UNITEXT 161

Damiano Rossello

A Cookbook with Probability One

With Financial Applications

Springer

UNITEXT

La Matematica per il 3+2

Volume 161

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Damiano Rossello
Department of Economics and Business
Università di Catania
Catania, Italy

ISSN 2038-5714 ISSN 2532-3318 (electronic)
UNITEXT
ISSN 2038-5722 ISSN 2038-5757 (electronic)
La Matematica per il 3+2
ISBN 978-3-031-54687-7 ISBN 978-3-031-54688-4 (eBook)
https://doi.org/10.1007/978-3-031-54688-4

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To Francesca and Sara

Preface

This book has been written for graduate students in economics with no prior exposition to probability theory out of undergraduate courses in Applied Statistics. It is intended to provide a self-contained account of probabilistic modelling mainly applied to finance. The fundamental concept of *random variable's distribution* is introduced beforehand with respect to the usual treatment of this subject in standard probability textbooks. All the analytic tools developed subsequently are illustrated through examples concerning the probability distribution of future stock prices and returns modeled as proper random variables. Given the intended audience in mind, the book tries to strike a balance between precise mathematical definitions and their applied knowledge. For example, events induced by random variables are implicitly defined as elements of a fixed sigma-algebra, and the expectation of a random variable is first defined separately for discrete and continuous distribution, then a unified treatment is offered to underline the Lebesgue approach to abstract integration. The corresponding 'symbolic manipulation' of integrals is easily established and illustrated through examples. Overall, the necessary measure-theoretic approach to the subject is underpinned without 'too tears': students learn the working aspect of the advanced theory, but some technical details are deferred to appropriate appendices within each section or postponed to the end of the book, containing review material from elementary real analysis and algebra courses. Indeed, to guarantee a double level of readability the book is divided into two parts. The probabilistic aspect of building a financial model such as the univariate random variables describing future asset prices or returns are illustrated from the scratch together with the corresponding distributions. The most important characteristics of these distributions are studied: moments, moment generating and characteristic functions, location-scale families, quantiles. Then their generalization to the multivariate case is presented: joint distribution functions and the corresponding vectorial first and second moments. Moreover, the fundamentals of stochastic dependence/independence are defined besides the treatment of Frechét bounds. The 'probability redux' stochastic processes are analyzed in term of their finite dimensional distributions, and typical examples are illustrated both in discrete and continuous time as candidate models for asset prices returns, and

loss variables. Convergence concepts as applied to the problem of evaluate point estimators or their asymptotic distribution are introduced. Special emphasis is put on non-parametric estimation of means, variances, correlation coefficients and risk measures, using the tool of empirical distribution function. Short chapters on copula functions and prediction constitute an important addendum, as well as a targeted section on risk measures and option pricing. The key definitions and theorems are stated without burdening too much the reader with unnecessary technical details, so it represents what I call a 'cookbook'. Since I don't sacrifice some mathematical rigor, the book can be accessed by those students with more mathematical preparation/sophistication, indeed Lebegue measure and integration together with the Fubini theorem and iterated integrals are used throughout. The appendices to each chapter and exercises (with complete solutions) also contain some proofs of results which are only stated in the main text and aim at making the book *quite* self-contained. The double level of reading should be attractive also for students of Mathematics. Another distinguished feature is the use of computer software (Excel, Matlab, R) to build numerical examples and simulations of some probabilistic objects. I provide the complete coding of each numerical example within the textbook. The treatment is not encyclopedic, rather it concentrates only on the relevant probabilistic tools although the reader will benefit from a *more quick learning* of topics that help a transition towards more advanced textbooks. The book itself has been developed in the last 17 years and has been evolved from lectures I taught for a graduate course on Probability for Finance (formerly called Calcolo delle Probabilita' per la Finanza) at the Department of Economics and Business, University of Catania. A word of caution: in each chapter a reference to Appendix concerns that contained at the end of the chapter. Other appendices are placed after the solutions to exercises, and are labelled with uppercase letters.

Catania, Italy Damiano Rossello

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Chapter 1
Probability, Events and Random Variables



1.1 The Need of Random Variables

Suppose today you want to make a guess about the S&P500's quote at the end of tomorrow. What is the chance to observe a value 35,501?[1] Indeed, today we don't know if 35,501 will be the true value tomorrow, thus we are faced something real called **uncertainty**. The word 'uncertainty' is a synonymous for **random experiment** , an empirical phenomenon or a given procedure whose outcome cannot be predicted with certainty. This can be interpreted as lack of information. A random experiment can be observed, but it hasn't deterministic regularity at all. It doesn't need to be performed, for example we can only think of tossing a coin.

Denoting S_T the S&P500's quote at time $T > 0$, then it behaves like a *variable* $X(\cdot)$ depending on some **scenarios** (viz. outcomes) ω collected in some universal set. A mathematical model for measuring numerically X based on the underlying random experiment should require at least the following (see Fig. 1.1):

- A **sample space** Ω, i.e. the set of all conceptually possible outcomes ω;
- A function $X : \Omega \to \mathbb{R}$, called a **random variable** (rv) .

Actually, X is not a variable, but in a sense it 'varies randomly': ending the random experiment, we only observe one realization of the rv, out of all possible numerical values it may take on.[2] Possible values of a rv by themselves are not interesting, we need to quantify how likely $X(\omega)$ is, for example, less than or equal to 35,501. Collecting those outcomes $\omega \in \Omega$ such that (s.t.) the condition $X(\omega) \leqslant 35,501$ holds, leads to consider subsets of the sample space called **events**. Extensively:

$$A = \{\omega \mid X(\omega) \leqslant a\}, \quad \omega \in \Omega, \qquad a \in \mathbb{R},$$

[1] It is a synonymous for price, since the S&P500 is nothing but a financial asset.

[2] X is sometimes called a *variate* and written x.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_1

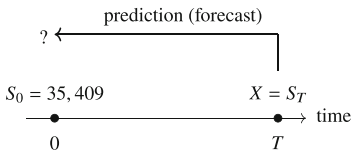


Fig. 1.1 The S&P500's price has a known value S_0 in $t = 0$, but its value S_T in $t = T$ is unknown

where $a = 35,501$. One can drop the symbol ω and writes more compactly $\{X \leqslant a\}$ which is *always* a subset of Ω. Other events can be considered, then a forecast about the future S&P500's quote involves:

$$A = \{X \geqslant a\} \text{ or } = \{X = a\} \text{ or } = \{a \leqslant X \leqslant b\}, \quad \text{etc.}, \quad a, b \in \mathbb{R}.$$

To evaluate how likely is to observe all such events we can assign them a **probability**, i.e. a numerical value between 0 and 1. For example, we can ask what is the probability that the future S&P500's quote is less than or equal to $a = 33,615$ or is greater than $b = 35,501$ and this writes:[3]

$$p = \mathsf{P}(X \leqslant a), \text{ or } p = \mathsf{P}(X > b) \quad \text{where} \quad p \in [0, 1].$$

In summary what we plan to study is based on the following:

- Dealing with phenomena we perceive as random.
- Modelling these phenomena as scenarios of some experiment.
- Outcomes are elements of a sample space $\Omega \neq \varnothing$.
- Subsets of Ω are called events.
- Events will be assigned a probability, a number between 0 and 1 that expresses how likely the event is to occur.

In particular, we will use a mathematical approach to the theory of probability firmly established by the Russian mathematician A.N. Kolmogorov in the 1930s. The core of this approach is a list of axioms about the behavior of probability expressed in terms of sets and functions. As we will see, using set theory and formal logic we are able to deduce anything about probability. Since we are interested in developing probabilistic modelling with finance in view, reference to financial econometrics is often called for:

- Financial econometrics is the econometrics of financial markets.
- Econometric theory is a framework needed for the empirical analysis in economics based on mathematical tools and principles, especially those of probability and theoretical statistics (quantitative laws of economics).

[3] We avoid the cumbersome notation $\mathsf{P}(\{X \leqslant a\})$, but notice that it would be correct since $\mathsf{P}(\cdot)$ is a function whose argument is an event $\{\cdots\} = A$.

- Financial econometrics is the bulk of models describing -in particular- financial time series such as prices, returns, interest rates, market indices, financial ratios, derivatives prices, etc.

Indeed, in the rest of the book we will apply the probabilistic framework especially to rvs modelling asset prices, besides we also treat rvs modelling asset returns.[4] Anyway, the probabilistic machinery we will present in the rest of this book is quite standard and suitable for random experiment different from those involving financial models. Often, we refer to time index $t = 0, 1, 2, \ldots, T$ or $t \in [0, \infty)$ to emphasize time *dependence* of relevant financial quantities. Only the initial price S_0 is a known quantity, while each subsequent S_t for $t > 0$ is a rv.

1.2 Events and Their Manipulation

To begin with, we introduce the formal framework of set theory to describe events.

Definition 1.1 An **event** is a subset $A \subset \Omega$ of the sample space. We say that the event A **occurs** if and only if (iff) $\omega \in A$.

We can think of ω as the effective result showed by the experiment.

Example 1.1 Rolling a dice once gives the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$. The subset $A = \{2, 4, 6\}$ can be thought of as the event $A = \{\text{even number shown}\}$. If the dice shows 3, then A does not occur.[5] We can use a 'verbal rule' to describe A.

Since $\Omega \subset \Omega$ and $\varnothing \subset \Omega$, the sample space itself and the empty set are events. Given events $A, B \subset \Omega$ we can build other sensible events such as: $A \cup B = \{\omega \,|\, \omega \in A \text{ or } \omega \in B\}$, $A \cap B = \{\omega \,|\, \omega \in A \text{ and } \omega \in B\}$, $A \setminus B = \{\omega \in A \,|\, \omega \notin B\}$ and $A^c = \{\omega \,|\, \omega \notin A\}$, see Fig. 1.2 and Table 1.1. Typically, in probability theory *relevant* events deserve a special description. In econometrics events are induced by rvs, i.e. they received a numerical description. We can write $\{X \in B\}$ for a suitable subset $B \subset \mathbb{R}$ of real numbers, to mean the collection of those scenarios $\omega \in \Omega$ s.t. $X(\omega)$ takes value on B. Therefore different choices of such B entail different events, see Table 1.2.

1.3 Probability Function

It is not mandatory to regard all subsets of Ω as events. Whatever the cardinality of Ω, the assignment of probabilities to events like $\{X \in B\}$ may cause inconsistencies.

[4] Returns have nice statistical properties than prices. We refer to closing daily prices.

[5] Observe that the sample space above need not to be described by a rv!

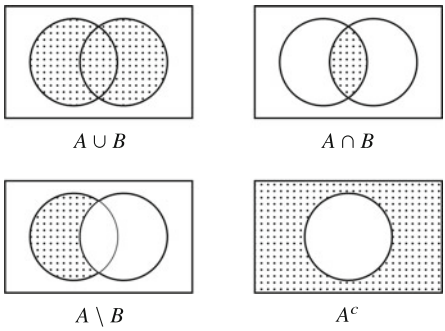


Fig. 1.2 Events can be depicted using Venn diagrams. Usually a rectangle represents the sample space Ω, while circles inside are used for smaller events. In particular: left circle $= A$, right circle $= B$. The result of unions, intersections, set difference and complements are filled properly

Table 1.1 Description of events as special subsets of Ω

Description: 'verbal rule'	Sets and subsets
Certain event	Ω (every ω is a member)
Impossible event	$\varnothing$ (no ω is a member)
At least one events occurs	$A \cup B$ (ωs lie in A or in B)
Both events occur	$A \cap B$ (ωs lie in A and in B)
Mutually disjoint or incompatible events	$A \cap B = \varnothing$
The event A occurs but not B	$A \setminus B = A \cap B^c$ (ωs never fall in B)
The event A implies the event B	$A \subset B$ (ωs in A are also in B)
The event A does not occur	A^c (ωs never fall in A)

Table 1.2 Numerical description of events induced by rvs

Subset of real numbers $B =$	Events ($a, b \in \mathbb{R}$ and $a < b$)
$\mathbb{R}$	$\{\omega \mid -\infty < X(\omega) < \infty\} = \Omega$
$\varnothing$	$\{\omega \mid X(\omega) \in \varnothing\} = \varnothing$
$(a, b]$	$\{\omega \mid a < X(\omega) \leqslant b\}$
$[a, b)$	$\{a \leqslant X < b\}$
$[a, b]$	$\{a \leqslant X \leqslant b\}$
(a, b)	$\{a < X < b\}$
$(-\infty, b]$	$\{X \leqslant b\}$
$(-\infty, b)$	$\{X < b\}$
$[a, \infty)$	$\{X \geqslant a\}$
(a, ∞)	$\{X > a\}$

Example 1.2 Consider the experiment involving the daily stock price with three possible tick movements ± 0.5, starting from an initial price $S_0 = 3.5$, see also Example 1.9. Any scenario in the sample space can be coded with u corresponding to an upward tick movement $+0.5$, and d corresponding to an downward tick movement -0.5. Thus, each scenario ω is of the form uuu,ddd and all other possible combination of labels u,d. If we fail to record the last tick movement, then the event

$$A = \{S_1 < 3.5\} = \{\text{ud}\mathbf{..}, \text{du}\mathbf{..}, \text{dd}\mathbf{..}, \text{dd}\mathbf{..}\} = \{\text{at least two downwards } -0.5\}$$

cannot be fully depicted. We can say A is not 'measurable', i.e. we cannot give a *yes or not answer* to the question 'Is $\omega \in A$?' since we have not full information about ω given the encoding based on u,d.

A sensible collection of events to which assign probabilities must be built upon the concept of occurrence and set operations as unions, complementation and intersections.

Definition 1.2 (Partial Information) Given a nonempty sample space Ω, a collection $\mathcal{F} \subset 2^{\Omega}$ of events is called a **σ-algebra** or **σ-field** if:

S1 $\varnothing \in \mathcal{F}$.
S2 $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$ (hence $\Omega \in \mathcal{F}$).
S3 If $A_1, A_2, \ldots$ belong to $\mathcal{F}$ also $A_1 \cup A_2 \cup \cdots = \cup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Usually $A = \{X \in B\}$ or $A_i = \{X \in B_i\}$, for a rv X and suitable subsets of real numbers B, B_i with $i = 1, 2, \ldots$

Remark 1.1 (Technical Issues)

- We write 2^{Ω} for the **power set** of Ω, the collection of all subsets of the sample space.
- A **countable set** contains an enumeration of elements. If there is a 1-to-1 correspondence between a set A and the integers $\mathbb{N} = \{1, 2, \ldots\}$, then A is countable and we write $A = \{\omega_1, \omega_2, \ldots\}$. If A is **finite** then there is 1-to-1 correspondence between A and $\{1, 2, \ldots, n\} \subset \mathbb{N}$. When A is neither finite nor countable, then it is **uncountable**, or uncountably infinite, and there is a 1-to-1 correspondence with the unit interval (0, 1) or with $\mathbb{R}$.
- A countable list of events $A_1, A_2, \ldots$ is a sequence written $(A_i)_{i \in \mathbb{N}}$ or $(A_i)_i$.
- For a finite sample space, i.e. $|\Omega| = n$ elements,[6] we always choose $\mathcal{F} = 2^{\Omega}$; the power set is also finite with cardinality $|2^{\Omega}| := 2^{|\Omega|} = 2^n$ elements.

Here some examples of more general event manipulation.

Example 1.3 $\cup_{i=1}^{\infty} A_i$ is the event {at least one of the A_i occurs}: indeed $\omega \in \cup_{i=1}^{\infty} A_i$ is the same to say $\omega \in A_i$ for <u>at least one</u> $i \in \mathbb{N}$. The special case is 'finite union', i.e. $A_1 \cup A_2 \cup \cdots \cup A_n$ and the verbal description is the same as above. Observe that we can write $A_1 \cup A_2 \cup \cdots \cup A_n \cup \varnothing \cup \cdots$

Example 1.4 What about intersections? $\cap_{i=1}^{\infty} A_i$ is the event {all the A_i's occur}, in fact $\omega \in \cap_{i=1}^{\infty} A_i$ is equivalent to say $\omega \in A_n$ <u>for all</u> $i \in \mathbb{N}$. The special case is finite intersection, i.e. $A_1 \cap A_2 \cap \cdots \cap A_n$

Example 1.5 The compound event $A = \{S_{t,1} \leqslant s_1\} \cap \cdots \cap \{S_{t,n} \leqslant s_n\}$, where $S_{t,i}$ are random stock prices for $i = 1, \ldots, n$, will play a central role in studying

[6] We write $|\Omega|$ to denote the cardinality of a set. Actually, two sets have the same cardinality iff there is a bijection (viz. a 1-to-1 correspondence) between them. This does not preclude that one can be a proper subset of the other.

stochastic dependence among several rvs. Clearly, for each scenario $\omega \in \Omega$ the event A describes the conjoint measurement of n asset prices at a future date t.

Hence, events from a σ-algebra can be manipulated using set operations, especially unions and intersections. Here some properties.

- Commutativity: $A \cup B = B \cup A$, and similarly for intersections.
- Associativity: $A \cup (B \cup C) = (A \cup B) \cup C$, and similarly for intersections.
- Distributivity: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- DeMorgan's Laws: $(A \cup B)^c = A^c \cap B^c$, and $(A \cap B)^c = A^c \cup B^c$.
- $(A^c)^c = A$.

The probability's assignment to events is then based on the following:

Definition 1.3 (Probability Function) Given a σ-algebra of events, we call **probability measure** the set function $\mathsf{P} : \mathcal{F} \to [0, 1]$ satisfying:

PA1 $\mathsf{P}(\varnothing) = 0$ and $\mathsf{P}(\Omega) = 1$

PA2 $\mathsf{P}\left(\cup_{i=1}^{\infty} A_i\right) = \mathsf{P}(A_1) + \mathsf{P}(A_2) + \cdots = \sum_{i=1}^{\infty} \mathsf{P}(A_i)$, for any pairwise disjoint sequence $(A_i)_{i \in \mathbb{N}} \subset \mathcal{F}$ of events, $A_i \cap A_j = \varnothing$ each $i \neq j$ (countable additivity or σ-additivity).

The couple $(\Omega, \mathcal{F})$ is the event space (or measurable space) and the triple $(\Omega, \mathcal{F}, \mathsf{P})$ is the **probability space** modelling a random experiment.

PA1 and PA2 are *axioms* for a probability measure. Thanks to Definitions 1.2 and 1.3 we can calculate probabilities for all the events in Table 1.1, also for intersections and differences of events (see Exercise 1.7). In summary, quantifying the likelihood of an event $A \in \mathcal{F}$ requires evaluation of $\mathsf{P}(\cdot)$ at A to get a number $\mathsf{P}(A) \in [0, 1]$ measuring *how easy* is that A occurs. The impossible event never occurs while the certain event always occurs, thus is sensible to assign them a probability zero and a probability 1, respectively. Thinking geometrically and using Venn diagrams to display Ω as a rectangle having area 1, $\mathsf{P}(A)$ measures the 'area' of the 'circle' representing A. Disjoint circles A_i inside Ω must have a union whose probability is given by adding each area computed separately. However, we must be careful in dealing with certain events behaving like $\varnothing$ and Ω. More precisely, if we 'add' the impossible event to any event $A \in \mathcal{F}$, then the probability is not altered:

$$\mathsf{P}(A) = \mathsf{P}(A) + 0 = \mathsf{P}(A \cup \varnothing).$$

This may happen for other events than $\varnothing$. We say that $B \in \mathcal{F}$ is **almost impossible** if $\mathsf{P}(A \cup B) = \mathsf{P}(A)$ whenever A and B are incompatible. This implies $\mathsf{P}(B) = 0$, which can be taken as the true definition of almost impossibility. We call $A \in \mathcal{F}$ an **almost sure** event if $\mathsf{P}(A) = 1$ and A is not necessarily equal to Ω. Almost impossible events are also called P-**null** or **negligible**. Obviously, if A is almost

sure its complement A^c is P-null, see Exercise 1.11. A property $S(\omega)$ depending on scenarios $\omega \in \Omega$ is said to hold **almost surely**, a.s. for short, if

$$\mathsf{P}(\{\omega \mid S(\omega) \text{ is false}\}) = 0.$$

Such a property holds almost everywhere in Ω except on a subset with zero probability.

Example 1.6 Consider the experiment of predicting the 1-day price S_1 of the S&P500, which can takes on the values

$$\{35,407.5,\ 35,408,\ 35,408.5,\ 35,409.5,\ 35,410,\ 35,410.5\},$$

starting from the initial value $35,409$. The statement 'only an integer price is shown' is a property $S(\omega)$ depending on the outcomes ω of this experiment. The set $A = \{\omega \mid S(\omega) \text{ is true}\}$ is equal to the event $\{35,408,\ 35,410\}$ while $A^c = \{\omega \mid S(\omega) \text{ is false}\}$ represents the event $\{35,407.5,\ 35,408.5,\ 35,409.5,\ 35,410.5\}$. If the probability assignment is based on experience and a subjective view (e.g. of a particular financial analyst), then it is allowed to assign $\mathsf{P}(A^c) = 0$ and the property $S(\omega)$ holds almost surely. In words: it is almost impossible that the S&P500 quote at the end of the day is fractional.

1.4 Frequency Justification and Discrete Probability Spaces

A random experiment can be repeated, maybe indefinitely. Let $n \in \mathbb{N}$ be the number of repetitions or trials of the same experiment: looking at daily stock price S_1. Suppose we observe stock prices over $n = 100$ consecutive days,[7] see Table 1.3. The relative frequency that $S_1 < 3.5$ is $\frac{21+17}{100} = 0.38$, and it could be thought of an approximation of the probability that the rv X modelling S_1 has value < 3.5, i.e. $\mathsf{P}(X < 3.5)$ which amounts to evaluate how easy is to occur $\{X < 3.5\}$. Moral: in a repeated experiment, writes $0 \leqslant f_n(A) \leqslant n$ for the number of occurrence of an event A, for example $A = \{X < 3.5\}$, in n trials and get the relative frequency $\frac{f_n(A)}{n}$. Thus, we believe that

$$\frac{f_n(A)}{n} \text{ converges to a real number } p \in [0,1] \text{ as } n \text{ becomes large,}$$

i.e. $\lim_{n\to\infty} \frac{f_n(A)}{n}$ would give $\mathsf{P}(A)$ or $\boxed{\frac{f_n(A)}{n} \approx \mathsf{P}(A)}$ for $n < \infty$ large enough. We trivially have:

[7] Clearly, we refer to the same stock quoted repeatedly. Do not confuse this with the experiment based on 3 consecutive tick movements over the same day.

Table 1.3 Observed daily prices can be though of as a drawn of $x_i = S_1(\omega)$ values of the rv S_1, for $i = 1, 2, 3, 4$, each corresponding to a different scenario $\omega \in \Omega$. Probabilities can be approximated by relative frequencies

Observed daily prices $S_1 =$	1.5	2.5	3.5	4.5
Frequency of occurrence	21	17	39	23

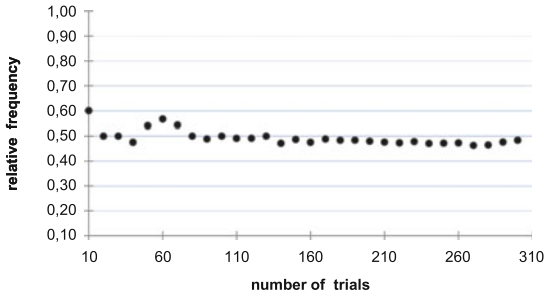


Fig. 1.3 Plot of relative frequencies of heads and tails against number of trials

- $\frac{f_n(\varnothing)}{n} = 0$ and $\frac{f_n(\Omega)}{n} = 1$, i.e. the relative frequencies of observing 'nothing' or 'everything';
- $\frac{f_n(A \text{ or } B)}{n} = \frac{f_n(A)}{n} + \frac{f_n(B)}{n}$ for disjoint events A and B, i.e. $f_n(A)$ is the number of time we observed A while $f_n(B)$ is the number of time we observed B, and they cannot be observed in the same trial.

Therefore, we expect that these properties transfer to the corresponding probabilities as the number of trials n becomes very large. In fact, this is the reason why Definition 1.3 holds.

Example 1.7 (Empirical Law of Large Number) We illustrate the behavior of heads and tails in the long run. Toss a *fair* coin $n = 10, 20, \ldots, 300$ times, and count the frequencies of observing heads. Compute $\frac{f_n(A)}{n}$, each n, and interpret the relative frequency as approximating the probability of the event $A = \{1\} = \{\text{H is shown}\}$. Here we encode H $= 1$ and use 0 for a tail, in a single coin flipping. We expect $\frac{f_{300}(A)}{300} \approx \frac{1}{2}$, i.e. the probability $\mathsf{P}(A)$ of getting a head in the random experiment of tossing a fair coin once, see Fig. 1.3 and Example 1.8. Now, there is a further justification of the definition and the properties of $\mathsf{P}(\cdot)$ introduced previously, called 'stabilization of relative frequencies'.

Example 1.8 (Simulating Heads and Tails in the Long Run...) To avoid tossing a fair coin 10 times, 20 times, etc. up to 300 times we suggest the following controlled experiment using your PC and Microsoft Excel 2007©. First, select the tab `Data` and then the tool `Data Analysis` from the `Tools Menu` also called `Ribbon`. Make sure to have installed the `Analysis ToolPak`. Otherwise: click the `Office Button` (left-up corner) and find the command to customize `Excel`. Then, in the `Add-Ins dialog box` check the box to the left of

`Analysis ToolPak`, and click `OK`. In the `Data Analysis dialog box` choose `Sampling`, then click `OK` to get the corresponding dialog box and do:

- `Number of Variables` $= 1$ (how many `Excel`'s columns to fill)
- `Number of Random Numbers` $= 300$
- `Distribution` $=$ Bernoulli
- `Parameters` $p = 0.5$
- `Output Options` `A1:A300` (what column to fill with random samples).

See Sect. 5.3 for the Bernoulli distribution.

Example 1.9 (Discrete Probabilities and Stock Price) Consider the sample space modelling 3 consecutive tick movements ± 0.5 starting from an initial price $S_0 = 3$, with corresponding rv $X = S_1$ for the 1-day price. We try to assign probabilities as follows: attach a probability weight

$$\mathsf{P}(\{\omega\}) \quad \text{for} \quad \omega = \text{uuu, uud, udu, duu, udd, dud, ddu, ddd.}$$

For example: $\mathsf{P}(X = 2.5) = \mathsf{P}(\{\text{udd, dud, ddu}\})$. But $\{X = 2.5\}$ is the disjoint union of three events (each consisting of a singleton):

$$\{\text{udd, udu, ddu}\} = \{\text{udd}\} \cup \{\text{dud}\} \cup \{\text{ddu}\},$$

thus

$$\mathsf{P}(X = 2.5) = \mathsf{P}(\{\text{udd}\}) + \mathsf{P}(\{\text{udu}\}) + \mathsf{P}(\{\text{ddu}\}) = \frac{3}{8} = 0.375.$$

Example 1.10 Assigning a probability weight as we did before is based on 'equally likely outcomes'. On the other hand, looking at the values X takes on we see that its range is the discrete set $\{1.5, 2.5, 3.5, 4.5\}$ and the assignment of probabilities is

$$\mathsf{P}(X = 4.5) = \mathsf{P}(X = 1.5) = \frac{1}{8}, \quad \mathsf{P}(X = 3.5) = \mathsf{P}(X = 2.5) = \frac{3}{8}.$$

In fact, the price $X(\omega) = 3.5$ can be reached in 3 ways, with $\omega = \text{uud, udu, duu}$; the same is true for the price $X(\omega) = 2.5$.

The above examples correspond to modelling **discrete uniform probability spaces**, where $|\Omega| = n$ and $\mathcal{F} = 2^{\Omega}$, i.e. we can always choose the power set as σ-algebra, see Example 1.12 in Sect. 1.6, which contains singletons $\{\omega\}$ as essential building blocks for probability computations. Here for each $A \in \mathcal{F}$ we have

$$\mathsf{P}(A) = \sum_{\omega \in A} \mathsf{P}(\{\omega\}), \tag{1.1}$$

where $|A| \leqslant |\Omega|$ is essential for P to be a probability measure.

Example 1.11 We compute the probability that the 1-day stock price is equal to 3.5 as that of the event $A = \{X = 3.5\}$ with cardinality $|A| = 3$, and

$$\mathsf{P}(A) = \sum_{\omega \in A} \mathsf{P}(\{\omega\}) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{|A|}{|\Omega|}.$$

Discrete uniform probabilities may fail to work with countably infinite sample spaces. Moreover, they are inadequate for not equally likely outcomes. However, there is a unifying treatment for the discrete case: given a finite Ω and its power set 2^Ω we have $(\Omega, 2^\Omega, \mathsf{P})$ called a **discrete probability space**, and P is in 1-to-1 correspondence with a sequence $(p_i)_{i \in \mathbb{N}}$ of real numbers s.t. $\sum_i p_i = 1$. Note that it may happen $p_i \neq p_j$ for $i \neq j$ (the space has no longer uniform probability weights). More precisely, if $\mathsf{P} : 2^\Omega \to [0, 1]$ is a probability measure then $p_i = \mathsf{P}(\{\omega_i\})$, for $i = 1, \ldots, n$, whenever $|\Omega| = n$ and indeed

$$\sum_{i=1}^{n} p_i = \sum_{i=1}^{n} \mathsf{P}(\{\omega_i\}) = \mathsf{P}\left(\cup_{i=1}^{n} \{\omega_i\}\right) = \mathsf{P}(\Omega) = 1.$$

It is worth noting $\{\omega_i\}$ are mutually disjoint events for which the probability measure is additive. Therefore,

$$\mathsf{P}(A) = \sum_{i \text{ s.t. } \omega_i \in A} p_i \tag{1.2}$$

which is nothing else formula (1.1) but without equally weights. On the other hand, assuming $\sum_{i=1}^{n} p_i = 1$ and defining the set function P on 2^Ω by formula (1.2) we can easily verify PA1 of Definition 1.3, Sect. 1.3, and to check countable additivity[8] we simply write

$$\begin{aligned} \mathsf{P}(A_1 \cup \cdots A_n \cup \varnothing \cup \varnothing \cup \cdots) &= \mathsf{P}\left(\cup_{i=1}^{n} A_i\right) \\ &= \sum_{j \text{ s.t. } \omega_j \in \cup_{i=1}^{n} A_i} \sum_{j=1}^{n} p_j = \sum_{j \text{ s.t. } \omega_j \in A_i} p_j \\ &= \sum_{i=1}^{n} \mathsf{P}(A_i), \end{aligned}$$

[8] For a finite sample space we have 2^Ω is also finite and then it cannot contain infinite sequences of events $A_1, A_2 \ldots$ Thus, property S3 of a σ-algebra (see Definition 1.2, Sect. 1.3) should be stated for finite sequences of mutually disjoint events $A_1, \ldots, A_2$, all belonging to 2^Ω, s.t. $\cup_{i=1}^{n} A_i \in 2^\Omega$. In this case the collection $\mathcal{F} = 2^\Omega$ is called an **algebra**, see Appendix in Chap. 2.

and observe that $\mathsf{P}(\{\omega_j\}) = p_j$. Further is true: we can extend this 1-to-1 correspondence to countably infinite sample spaces, keeping 2^Ω as their σ-algebra, under the usual condition $\sum_{i=1}^{\infty} p_i = 1$ and $\mathsf{P}(\{\omega_i\}) = p_i$, which is now an infinite sequence of positive real numbers, see Appendix C. Eventually, for an arbitrary sample space Ω forming $(\Omega, \mathcal{F}, \mathsf{P})$, a probability is said to be **discrete** provided that the finite or countable $C \subset \Omega$ satisfies $\mathsf{P}(C) = 1$. As byproduct, $\mathsf{P}(C^c) = 0$ and for every event $A \in \mathcal{F}$ we have

$$\mathsf{P}(A) = \mathsf{P}(A \cap C) = \sum_{\omega \in A \cap C} \mathsf{P}(\{\omega\}),$$

so that P is *concentrated* on C. From this more general setting we can deduce that for finite of countable sample spaces discussed above any P must be discrete. If we insist on using weights p_i to set up the probability of any other event $A \in 2^\Omega$, when the sample space is uncountable, then it can be shown that $p_i = 0$ and, moreover, there is no probability which assigns a value to every subset of Ω in such a way axioms PA1 and PA2 are fulfilled, see Appendix. On the other hand, it is possible to circumvent this and build a probability measure:

- restricting the σ-algebra 2^Ω to a more convenient class $\mathcal{F}$ of well representative events satisfying properties SA1-SA3 of Definition 1.2;
- defining the measure P without using countable summation.

The last item is directly linked to the question of what type of rv has to be used for random experiments with finite/countable scenarios and random experiment with uncountable scenarios, see Chap. 2.

Remark 1.2 Given a probability space where $\mathcal{F}$ contains all singletons $\{\omega\}$ for every $\omega \in \Omega$, we say that the outcome ω is an **atom** if $\mathsf{P}(\{\omega\}) > 0$. A probability measure is **diffusive** if it has no atoms, i.e. $\mathsf{P}(\{\omega\}) = 0$ for every scenario ω. A probability measure is said to be **purely atomic** if its atoms form a countable set H such that $\mathsf{P}(H^c) = 0$. Thus a (uniform) discrete probability measure is purely atomic.

1.5 Further Properties of a Probability Measure

Any probability measure P acting as a set function obeys to some arithmetic and ordering rules.

Proposition 1.1 *Given a probability space* $(\Omega, \mathcal{F}, \mathsf{P})$, *the measure* P *satisfies:*

Pr1 $\mathsf{P}(A^c) = 1 - \mathsf{P}(A)$, *for every* $A \in \mathcal{F}$
Pr2 *If* $A \subset B$, *then* $\mathsf{P}(A) \leqslant \mathsf{P}(B)$, *for every pair* $A, B \in \mathcal{F}$ *(monotonicity)*
Pr3 *If* $A \subset B$, *then* $\mathsf{P}(B \setminus A) = \mathsf{P}(B) - \mathsf{P}(A)$, *for every pair* $A, B \in \mathcal{F}$
Pr4 $\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(B) - \mathsf{P}(A \cap B)$, *for every pair* $A, B \in \mathcal{F}$.

Proving Proposition 1.1 is a constructive exercise in computing probabilities.

Proof First we check Pr1. Events $A \in \mathcal{F}$ and A^c are incompatible, thus $1 = \mathsf{P}(\Omega) = \mathsf{P}(A \cup A^c) = \mathsf{P}(A) + \mathsf{P}(A^c)$. To prove Pr2, let $B = A \cup (B \setminus A)$ be decomposed by two disjoint events, then $\mathsf{P}(B) = \mathsf{P}(A) + \mathsf{P}(B \setminus A)$ by countable additivity, thus subtracting the second term on the right-hand side we are done. Now, Pr3 easily follows from Pr2. Claim Pr4 is based on decomposing $A \cup B$ by the disjoint events A and $B \setminus (A \cap B)$, since the latter is such that $(A \cap B) \subset B$ and Pr3 applies. ■

Next we present an important property of P that will help in computing limiting probabilities, see Chap. 2, Sect. 2.1, and Appendix in Chap. 2.

Theorem 1.1 (Continuity of Probability) *Given a probability space* $(\Omega, \mathcal{F}, \mathsf{P})$ *and an increasing sequence of events* $A_1 \subset A_2 \subset \cdots$ *from* $\mathcal{F}$, *define* $A := \lim_{i\to\infty} A_i := \cup_{i=1}^{\infty} A_i$. *Then it holds*

$$\lim_{i\to\infty} \mathsf{P}(A_i) = \mathsf{P}(A).$$

The same result holds true for decreasing sequences of events $A_1 \supset A_2 \supset \cdots$ *from* $\mathcal{F}$, *and with* $A = \lim_{i\to\infty} A_i := \cap_{i=1}^{\infty} A_i$.

Property Pr2 does imply $\mathsf{P}(A_i) \leqslant \mathsf{P}(A_{i+1})$ is an increasing sequence of nonnegative reals, which is bounded above by $\mathsf{P}(A)$ for $A_i \subset A$ and then converges to the *limit* $\mathsf{P}(A)$. This justifies the interchange of limit and the probability measure, i.e. $\lim_{i\to\infty} \mathsf{P}(A_i) = \mathsf{P}(\lim_{i\to\infty} A_i) = \mathsf{P}(A)$ as for real-valued continuous function. The same reasoning applies to the case of a decreasing sequences of A_is. In some textbooks increasing events $A_1, A_2, \ldots$ and their union A are referred as $A_i \uparrow A$, using a notation similar to that of convergent sequences of real numbers. Similarly, decreasing events $A_1, A_2, \ldots$ and their intersection A are referred as $A_i \downarrow A$. Continuity of the probability measure can be generalized to some extent. In a σ-algebra $\mathcal{F}$ the σ reminds us that not only finite but also countable unions of events are possible (σ = 'sums'). Finite unions cannot capture tail events in experiment like tossing a coin infinitely many times or predicting the future stock price with infinitely many uptick and downtick movements. Let $(A_i)_{i\in\mathbb{N}}$ be a sequence in $\mathcal{F}$ and define:

$$\limsup A_i := \cap_{n\geqslant 1} \cup_{i\geqslant n} A_i$$
$$\liminf A_i := \cup_{n\geqslant 1} \cap_{i\geqslant n} A_i.$$

They are easily verified belonging to $\mathcal{F}$. The first is often denoted by $\{A_i \text{ i.o.}\}$ referred to as 'A_i infinitely often', since for a given $i \in \mathbb{N}$ the event $\cup_{i\geqslant n} A_i$ is interpreted as 'at least one of the A_i occurs' and

$$\cap_{n=1}^{\infty} \cup_{i=n}^{\infty} A_i = \left\{\omega | \forall\, n \;\; \omega \in \cup_{i=n}^{\infty} A_i\right\}$$

is the event that infinitely many of the A_i occur. The second limiting event above is often written $\{A_i \text{ ev}\}$ and referred to as 'A_i eventually' or almost always, indeed $\cap_{i \geqslant n} A_i$ stands for 'all the A_i's occur' and

$$\cup_{n=1}^{\infty} \cap_{i=n}^{\infty} A_i = \left\{\omega \mid \exists\, n \;\; \omega \in \cap_{i=n}^{\infty} A_i\right\},$$

is the event that all but a finite number of A_i occur.[9] For these tail events we have

$$\liminf A_i \subset \limsup A_i, \quad (\limsup A_i)^c = \liminf A_i^c, \quad \limsup A_i^c = (\liminf A_i)^c.$$

For example $\mathsf{P}(A_i \text{ i.o.}) = 1 - \mathsf{P}(A_i^c \text{ ev})$. The relation on the right-hand side is based on DeMorgan's laws. If $\liminf A_i = \limsup A_i = A$ we write $\lim_{i\to\infty} A_i = A$ and say that the sequence $(A_i)_{i\in\mathbb{N}}$, not supposed to be monotone, converges to the event A. This in fact generalizes Theorem 1.1 (see Appendix). We also have:

$$\mathsf{P}(\liminf A_i) \leqslant \liminf \mathsf{P}(A_i) \leqslant \limsup \mathsf{P}(A_i) \leqslant \mathsf{P}(\limsup A_i).$$

If a sequence $A_i \to A$ in the sense given above, then

$$\lim_{i\to\infty} \mathsf{P}(A_i) = \mathsf{P}(A),$$

since[10] $\liminf \mathsf{P}(A_i) = \limsup \mathsf{P}(A_i)$. With this machinery at hand we present the following:

Lemma 1.1 (Borel-Cantelli) *Let $A_1, A_2, \ldots \in \mathcal{F}$ any sequence of events on the probability space $(\Omega, \mathcal{F}, \mathsf{P})$.*

(i) If $\sum_{i=1}^{\infty} \mathsf{P}(A_i) < \infty$, then $\mathsf{P}(A_i \text{ i.o.}) = 0$.
(ii) If $\sum_{i=1}^{\infty} \mathsf{P}(A_i) = \infty$ and A_i are independent events, then $\mathsf{P}(A_i \text{ i.o.}) = 1$.

Proof Let $A = \limsup A_i$. (i) We have $A \subset \cup_{i=n}^{\infty} A_i$ and then $\mathsf{P}(A) \leqslant \sum_{i=n}^{\infty} \mathsf{P}(A_i)$ for all $i \in \mathbb{N}$, by monotonicity and subadditivity. Since the sum is taken over the tail of a convergent series, $\sum_{i=1}^{\infty} \mathsf{P}(A_i) < \infty$, then we know that $\lim_{i\to\infty} \mathsf{P}(A_i) = 0$ and then the sum also tends to zero as $i \to \infty$. (ii) Since $(\limsup A_i)^c = \liminf A_i^c$ we can write $A^c = \cup_{n=1}^{\infty} \cap_{i=n}^{\infty} A_i^c$ and thus:

$$\begin{aligned}\mathsf{P}(A^c) &\leqslant \sum_{n=1}^{\infty} \mathsf{P}\big(\cap_{i=n}^{\infty} A_i^c\big) = \sum_{n=1}^{\infty} \lim_{m\to\infty} \mathsf{P}\big(\cap_{i=n}^{m} A_i^c\big) \\ &= \sum_{n=1}^{\infty} \lim_{m\to\infty} \prod_{i=n}^{m} \{1 - \mathsf{P}(A_i)\} \leqslant \sum_{n=1} \lim_{m\to\infty} \prod_{i=n}^{m} \mathrm{e}^{-\mathsf{P}(A_i)}\end{aligned}$$

[9] Only a finite number of A_i does not occur.
[10] For a formal proof see e.g. [2, Th 4.1].

$$= \sum_{n=1}^{\infty} \mathrm{e}^{-\lim_{m\to\infty} \sum_{i=n}^{m} \mathsf{P}(A_i)} = \sum_{n=1}^{\infty} \mathrm{e}^{-\infty} = \sum_{n=1}^{\infty} 0 = 0,$$

provided $\sum_{i=1}^{\infty} \mathsf{P}(A_i) = \infty$. ■

In the proof above we use continuity of P, independence among complements and the following fact: for every $0 < x < 1$ it is true $g(x) = 1 - x \leqslant \mathrm{e}^{-x} = h(x)$. Just take the derivative of both functions g, h. The Borel-Cantelli Lemma asserts that if $(A_i)_{i\in\mathbb{N}}$ are independent then $\mathsf{P}(A_i \text{ i.o.})$ is either 0 or 1. See Appendix C for related concepts of lim inf, lim sup concerning real-valued sequences.

1.6 Further σ-algebra

Recall that for a random experiment with sample space Ω the corresponding σ-algebra $\mathcal{F}$ may be interpreted as 'partial information'.

Example 1.12 (Richer Information) The power set 2^{Ω} is indeed a σ-algebra: the impossible event and the certain event are contained in it (they are subsets of Ω). For every $A \in 2^{\Omega}$ we have $A^c \in 2^{\Omega}$, because every event A belongs to 2^{Ω} as well as its complement A^c. Given events $A_1, A_2, \ldots$ from 2^{Ω}, their union $\cup_{i=1}^{\infty} A_i$ is a subset of Ω and then it belongs to the power set. Here we have information on any event of interest: $A \in 2^{\Omega}$ tells us the event A can occur, but also its contrary A^c can occur; if each $A_1, A_2, \ldots \in 2^{\Omega}$ may occur, then the event {at least one of A_i occur} is conceivable.

Example 1.13 (Trivial σ-Algebra) The set $\{\varnothing, \Omega\}$ is a σ-algebra: the impossible event and the certain event are in here; their union is the whole sample space, then again belongs to it. The information provided by $\{\varnothing, \Omega\}$ is minimal (nothing or everything can happen).

Example 1.14 (σ-Algebra Generated By an Event) Given an event $A \subset \Omega$, the collection $\{\varnothing, A, A^c, \Omega\}$ is a σ-algebra (the verification is quite simple). It provides information restricted to A. It can be denoted by $\boxed{\sigma(A)}$, the **σ-algebra generated by the event A**, actually the smallest σ-algebra containing this event.

Example 1.15 The collection of all events $\{X \in B\}$, for a rv X and every suitable subsets B of real numbers, is a σ-algebra. It may be obtained by unions, intersections and complements of events $\{X \leqslant a\}$, for every $a \in \mathbb{R}$. The information it provides is based on the numerical description of the underlying random experiment, given by the rv X. Usually it is denoted by $\boxed{\sigma(X)}$, the **σ-algebra generated by the rv X**, actually the smallest σ-algebra containing events of the form $\{X \in B\}$.

We can assign different σ-algebras to a fixed experiment, depending on our needs.

Example 1.16 For two possible tick movements, consider the following collections of events:

$$\mathcal{F}_1 = \{\varnothing, \Omega, \{\text{uu}\}, \{\text{ud,du,dd}\}\} \quad \mathcal{F}_2 = \{\varnothing, \Omega, \{\text{uu,dd}\}, \{\text{ud,du}\}\}\,.$$

Properties S1–S3 of Definition 1.2 are easily verified, thus both are σ-algebra. Their intersection $\mathcal{F}_1 \cap \mathcal{F}_2 = \{\varnothing, \Omega\}$ is the trivial σ-algebra. Their union

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \{\varnothing, \Omega, \{\text{uu}\}, \{\text{ud,du,dd}\}, \{\text{uu,dd}\}, \{\text{ud,du}\}\}$$

is not a σ-algebra, for example $\{\text{uu}\} \cup \{\text{ud,du}\}$ does not belongs to it.

Example 1.17 Let $X_1, X_2, \ldots$ be rvs over the same probability space. They are indexed by natural numbers (viz. they form a sequence of rvs). Then $\sigma(X_1, \ldots, X_n)$ is the σ-algebra generated by the n-tuple $(X_1, \ldots, X_n)$, see Sect. 3.1, and contains those events A for which we can verify whether or not they have occurred, i.e., $\omega \in A$, by checking only numerical values of the n-tuple. More generally, if $(X_i)_{i\in I}$ is any indexed family of rvs, for a nonempty set I, then $\sigma(X_i, i \in I)$ is the σ-algebra generated by the whole family, see Chap. 9. Eventually, suppose that $S_1, S_2, \ldots$ is a sequence of rvs representing a stock price over subsequent days $t = 1, 2, \ldots$. The σ-algebra $\sigma_t := \sigma(S_{t+1}, S_{t+2}, \ldots)$ contributes to the **tail σ-algebra** $\cap_{t=1}^{\infty}\sigma_t$, see Exercise 1.13: the intersection of σ-algebras is again a σ-algebra. It can be shown that $\cap_{t=1}^{\infty}\sigma_t$ contains events like

$$\left\{\lim_{t\to\infty} S_t \text{ exists}\right\}, \quad \left\{\lim_{t\to\infty} t^{-1}\sum_{i=1}^{t} S_i \text{ exists}\right\}, \quad \left\{\sum_{t=1}^{\infty} S_t \text{ converge}\right\},$$

see [16, Def 4.10]. Knowing the values of the stock price at subsequent days tells us whether they will be from some day $t + 1$ on, and this also informs us about the limiting behavior of S_t for t large enough, about the convergence of partial sums of daily prices together with that of their arithmetic mean.

We have seen how a σ-algebra can be generated. In Example 1.14 we actually start from $\mathcal{G} = \{A, A^c\}$, add $\varnothing$ which extends $\mathcal{G}$ to $\{\varnothing, A, A^c\}$ where performing *all* unions we obtain $\sigma(A) \equiv \sigma(\mathcal{G})$. The class $\mathcal{G}$ form a partition of the sample space Ω,[11] and is called a **generator**.[12] We continue to call $\sigma(\mathcal{G})$ the σ-algebra generated by $\mathcal{G}$: it is the smallest σ-algebra containing that class and is equal to the intersection of all σ-algebras containing it, see Exercises 1.13 and 1.14 .

Remark 1.3 Any σ-algebra of events is a *π-system*, meaning that it is closed under finite intersection, i.e., for every $A, B \in \mathcal{F}$ we have $A \cap B \in \mathcal{F}$. Moreover, $\mathcal{F}$ is also

[11] I.e., their intersection is empty, and their union is the sample space.

[12] For finite sample space, Property S3 of a σ-algebra is expressed through finite union of event, see Appendix in Chap. 2.

closed under proper differences, i.e., $A, B \in \mathcal{F}$ s.t. $A \subset B$ implies $B \setminus A \in \mathcal{F}$, and it is closed under countable increasing union, i.e., $A_1 \subset A_2 \subset \cdots$, all belonging to $\mathcal{F}$, implies $A = \cup_i A_i \in \mathcal{F}$, see Exercise 1.15. These last two properties makes $\mathcal{F}$ a *d-system* or *Dynkin system* . Conversely, it can be shown that any collection of events which is both a π-system and a d-system, is also a σ-algebra.

The construction of a σ-algebra based on generators can be pushed further. We can consider classes of events $\mathcal{G}$ which are finite, $\{A_1, \ldots, A_n\}$, or countable, $\{A_1, A_2, \ldots\}$, and form a partition of the sample space. Hence, after adding $\varnothing$ we get $\sigma(\mathcal{G}) = \mathcal{F}$ and we can concentrate on minimal (viz. elementary or amenable) classes of events in such a way performing set operation yields the whole σ-algebra. A special case is $\mathcal{G} = \{\{\omega\} \mid \omega \in \Omega\}$, i.e., the collection of all elementary events (viz. singletons) which together with the impossible event $\varnothing$ generate the σ-algebra, i.e., $\sigma(\mathcal{G}) = \mathcal{F} = 2^{\Omega}$: every event $A \subset \Omega$ is a union of some elementary events, $A = \cup_{\omega \in A}\{\omega\}$, then $A \in \sigma(\mathcal{G})$.

Remark 1.4 A technical result from measure theory implies that uniqueness of a probability measure on an event space $(\Omega, \mathcal{F})$ can be tested on a generator $\mathcal{G}$ of $\mathcal{F}$, provided that it is a π-system, see also Appendix in Chap. 2.[13]

1.7 Introduction to Trees

Trees, especially binomial trees, play a relevant pedagogic role in probability theory. In the following we introduce them to illustrate the *wrong* way to build a model for the evolution of a stock price over a finite set of dates. The S&P500 is a market index whose value at a future time is unknown in advance. In finance other traded assets behave similarly: stock prices, interest rates, foreign exchange rates, derivatives prices. A simple additive model for the future price S_t can be built if we are interested in predicting a stock price at the end of $t = 1$ trading day. In actual markets, starting from the initial (spot) price we would observe upward or downward **tick** movements of ± 0.5, as many times as permitted by trading rules.

Example 1.18 Let $S_1(\omega)$ be the 1-day stock price: the sample space contains all possible tick movements. If we admit 3 upwards and/or downwards and label each outcomes $\omega \in \Omega$ accordingly, then

$$\Omega = \{\text{uuu, uud, udu, duu, udd, dud, ddu, ddd}\}.$$

[13] It can be also showed that a class of events $\mathcal{G}$ which is a π-system generates a σ-algebra $\sigma(\mathcal{G})$ equal to the smallest d-system containing the class, denoted $d(\mathcal{G})$.

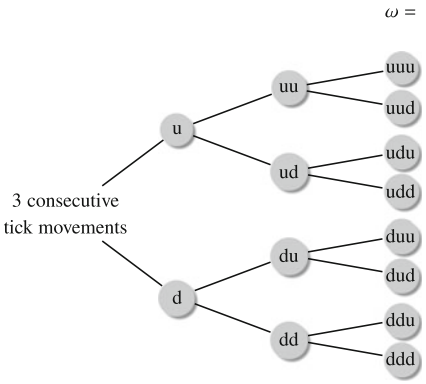


Fig. 1.4 Getting scenarios ω for the experiment 'three tick movements of a daily stock price' through a binomial tree. There is an obvious analogy with the experiment of tossing a coin three times, revealed by the corresponding coding (e.g. uuu can be replaced by HHH)

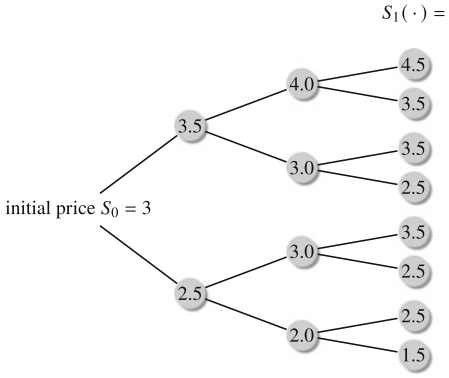


Fig. 1.5 The binomial tree is also good to figure out the underlying rv S_t

If we further assume a spot price $S_0 = 3$, then the daily price is given by the rv with values

$$S_1(\text{uuu}) = 4.5,\ S_1(\text{uud}) = S_1(\text{udu}) = S_1(\text{duu}) = 3.5,$$
$$S_1(\text{udd}) = S_1(\text{dud}) = S_1(\text{ddu}) = 2.5,\ S_1(\text{ddd}) = 1.5.$$

From Example 1.18, the correpsonding binomial tree is a diagram representing different paths that might be followed by the price within one day: each node represents either an upward or a downward price movement, see Figs. 1.4, 1.5 and 1.6. The very limitation of a tree is that it cannot be used if the number of scenarios is too large. Indeed, for the additive model this can lead to some inconsistencies.

Example 1.19 It is worth noting that 2 upwards and 1 downward entail the same value 3.5, and similarly 1 upward together with 2 downwards give the same value 2.5. If we assume a spot price $S_0 = 2$, then the daily price is given by a zero value $S_1(\text{ddd}) = 0$. A spot price $S_0 = 1$ gives a negative daily price $S_1(\text{ddd}) = -0.5$.

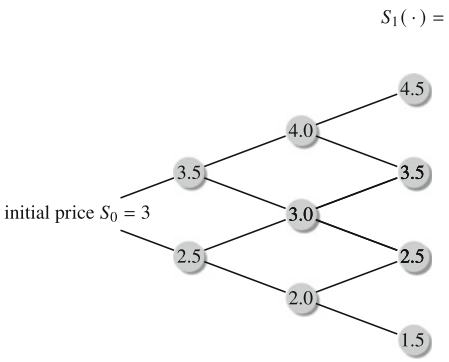


Fig. 1.6 Because of the symmetry in successive up and down stock price's movements the tree is **recombining**, a prelude of the so called *Markov property* we will study in the sequel

These are not real traded value!. Later, we introduce a *multiplicative* model for the future price which works, whatever the spot price is (see Sect. 5.4).

Labeling scenarios is not necessarily relevant.

Example 1.20 (Three Coin Tossing) The random experiment of flipping a fair coin three times produces an 8-elements sample space,

$$\Omega = \{\text{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}\},$$

where H stands for 'head' and T stands for 'tail', which shares the same structure of the sample space based on 3 possible up/down tick movements. In fact, we can use 0 and 1 instead of H and T, respectively.

We observe that the sample space underlying tick movements driving future stock prices is related to a σ-algebra built upon generating classes, see Sect. 1.6. More precisely, we have repeated single trials based on u or d up to three times to get the whole random experiment producing events like

$$A = \{\omega \mid \text{u is showed first}\}, \text{ or } B = \{\omega \mid \text{d is showed first}\} = A^c.$$

This is the same to handle

$$\Omega = \{\text{d}, u\} \times \{\text{d}, u\} \times \{\text{d}, u\} = \{\omega \mid \omega = (a_1, a_2, a_3),\ a_i = \text{u or d for } i = 1, 2, 3\}.$$

We can turn to the system $\mathcal{G} = \{A, B\}$ representing a decomposition of Ω, then adding $\varnothing$ we generate the class $\mathcal{F} = \{\varnothing, A, B, \Omega\}$ on which a probability measure can be defined if we are interested only in the first tick movements. This framework can be generalized by considering further tick movements yielding scenario $\omega = (a_1, a_2, \ldots)$, and events such as

$$C = \{\omega = (a_i)_{i\in\mathbb{N}} : a_1 = a_2 = \text{d}\},\quad D = \{\omega = (a_i)_{i\in\mathbb{N}} : a_1 = a_2 = \text{u}\},$$
$$E = \{\omega = (a_i)_{i\in\mathbb{N}} : a_1 = \text{d}, a_2 = \text{u}\},\quad F = \{\omega = (a_i)_{i\in\mathbb{N}} : a_1 = \text{u}, a_2 = \text{d}\}.$$

For example, C is the event of two consecutive tick movements. The collection $\mathcal{G} = \{C, D, E, F\}$ decomposes Ω and serves as a generator. Note that $C \cup E = A$ and $D \cup F = B$. Furthermore, we can perform double unions $C \cup D$, $C \cup F$, $E \cup F$, $D \cup E$ and triple unions such as $C \cup E \cup F = D^c$. This process end with the class

$$\mathcal{F} = \left\{\varnothing, \Omega, A, B, C, D, E, F, C^c, D^c, E^c, F^c, C \cup D, C \cup F, E \cup F, D \cup E\right\},$$

after we added the impossible event. Note how this is linked to the problem of defining a probability measure over a sample space when we pass to countably infinite tick movements, see Appendix.

1.8 Exercises

1.1 Let $S_0 = 3$ be the initial price of a traded asset. What the event $\{S_1 \leqslant a\}$ is, if we take $a = 3.5$?

1.2 Find the events $\{S_1 \geqslant a\}$, $\{S_1 < a\}$, $\{S_1 > a\}$ and $\{S_1 = a\}$, again using $a = 3.5$.

1.3 Given two real numbers $a < b$, write $\{X \leqslant b\}$ using $\{X \leqslant a\}$ and $\{a < X \leqslant b\}$.

1.4 Given a real number $a \in \mathbb{R}$, write $\{X \geqslant a\}$ as the disjoint union of two events.

1.5 Suppose the rv S_1 is a model for the daily stock price, over an uncountably infinite sample space Ω. Consider the event $A = \{1 < S_1 < 3\}$, then show that it equals $\cup_{i=1}^{\infty}\{1 < S_1 \leqslant 3 - \frac{1}{i}\}$.

1.6 For the rv S_1 modelling the 1-day price, consider the event $A = \{1 \leqslant S_1 \leqslant 3\}$, then show that it equals $\cap_{i=1}^{\infty}\{1 - \frac{1}{i} < S_1 \leqslant 3\}$.

1.7 Given the intersection of events $A = \cap_{i=1}^{\infty} A_i$ s.t. every $A_i \in \mathcal{F}$, show that $A \in \mathcal{F}$ too, i.e. we also have $\cap_{i=1}^{\infty} A_i$ is a member of the σ-algebra. Then, show that $A \setminus B \in \mathcal{F}$ whenever both $A, B \in \mathcal{F}$.

1.8 Show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, for events $A, B, C \in \mathcal{F}$. This in particular implies $A \cap (B \cup C) \in \mathcal{F}$.

1.9 Prove DeMorgan's Law $\left(\cup_{i=1}^{\infty} A_i\right)^c = \cap_{i=1}^{\infty} A_i^c$, for events $A_i \in \mathcal{F}$.

1.10 For events $A, B, C \in \mathcal{F}$ prove the following using a different method than the one given in a previous exercise:

- $A \cup (A \cap B) = A$
- $(A \cup B) \cap (A \cup C) = A \cup (B \cap C)$.

1.11 (P-Null Events) Let $\mathsf{P}(A \cap B) = \mathsf{P}(B)$, Show that this implies $\mathsf{P}(A) = 0$.

1.12 (Discrete Uniform Probabilities) Let Ω be a finite sample space with n outcomes, equally likely. Set $\mathcal{F} = 2^{\Omega}$ and as usual assign a probability weight $\mathsf{P}(\{\omega\}) = \frac{1}{n}$ to each elementary event $\{\omega\} \in \mathcal{F}$, for every $\omega \in \Omega$. Thus, define $\mathsf{P}(A) = \frac{|A|}{n}$ for any other event $A \in \mathcal{F}$. Verify that properties Pr1-Pr4 hold for this set function.

1.13 Let $\mathcal{G} = \{\mathcal{S}_i\}_{i \in I}$ be a collection of σ-algebras with respect to a sample space Ω. Show that the intersection $\mathcal{H} = \cap_{i \in I} \mathcal{S}_i$ is also a σ-algebra.

1.14 Let C be a collection of subsets of the sample space Ω. Prove that there exists the smallest σ-algebra with respect to Ω which contains C.

1.15 Show that a σ-algebra $\mathcal{F}$ of events contains proper difference of its events and countable increasing union of its events.

1.16 (Trace σ-Algebra) Let Ω be a sample space and let $\sigma_E := \{A \cap E \mid A \in \mathcal{F}\}$ be a collection containing all intersections between each event in $\mathcal{F}$ and a fixed event $E \subset \Omega$. Show that σ_E is a σ-algebra.

Appendix

Extending Probability Spaces

Assume a finite sample space $\Omega = \{\omega_0, \omega_1, \ldots, \omega_n\}$, where ω_i is the outcome consisting of i upwards and $n - i$ downwards price movements, for $i = 0, 1, 2, \ldots, n$. For example, for $n = 4$ we have:

$$\omega_2 \text{ corresponds to 2 downwards and 2 upwards.}$$

Observe that ω_2 can be given by $\binom{4}{2}$ different combination of 2 downwards and 2 upwards, so it is natural to assign a probability weight $\mathsf{P}(\{\omega_2\}) = \binom{4}{2}\frac{1}{2^4}$. More generally,[14] $\boxed{\mathsf{P}(\{\omega_i\}) = \binom{n}{i}\frac{1}{2^n}}$ where $\frac{1}{2}$ is the probability of a single upward or downward. We claim $\mathsf{P}(\Omega) = \sum_{i=0}^{n} \mathsf{P}(\{\omega_i\})$ is 1. Indeed:

$$\begin{aligned}\sum_{i=0}^{n} \mathsf{P}(\{\omega_i\}) &= \sum_{i=0}^{n} \binom{n}{i}\frac{1}{2^n} = \sum_{i=0}^{n} \binom{n}{i}\left(\frac{1}{2}\right)^i\left(\frac{1}{2}\right)^{n-i} \\ &= \left(\frac{1}{2} + \frac{1}{2}\right)^n = 1,\end{aligned}$$

[14] $\binom{n}{i} = \frac{n!}{i!(n-i)!} = \frac{(i+(n-i))!}{i!(n-i)!}$, i.e the number of permutations of two objects (1 upward and 1 downward) repeated i times and $n - i$ times, respectively.

where we used the binomial formula $(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i}$. As we have seen in Sect. 1.4, for finite Ω, formula $\mathsf{P}(A) = \frac{|A|}{|\Omega|}$ defines a finitely additive probability function. Equivalently $\mathsf{P}(A) = \sum_{\omega \in A} \mathsf{P}(\{\omega\})$, a finite sum of probability weights each given by $\mathsf{P}(\{\omega\}) = \frac{1}{n}$. This is a discrete uniform probability space, where the σ-algebra is the power set 2^{Ω}, which is finite too. One is tempted to extend this formula to infinite countable sample spaces $\Omega = \{\omega_1, \omega_2, \ldots\}$, through the usual set function

$$\mathsf{P}(A) = \sum_{\omega_j \in A} \mathsf{P}(\{\omega_j\}).$$

First, we need the additional requirement of a convergent series of real numbers, all lying in $[0, 1]$,

$$\sum_{\omega_j \in \Omega} \mathsf{P}(\{\omega_j\}) = 1$$

to meet axiom PA1, Definition 1.3, for the probability of the whole Ω. If the last convergence property holds, then the set function defined above is a probability measure. In particular, for pairwise disjoint events $A_1, A_2, \ldots \in \mathcal{F}$ countable additivity PA2 now reads:

$$\mathsf{P}(\cup_{i=1}^{\infty} A_i) = \sum_{\omega_j \in \cup_i A_i} \mathsf{P}(\{\omega_j\}) = \sum_{i=1}^{\infty} \sum_{\omega_j \in A_i} \mathsf{P}(\{\omega_j\}) = \sum_{i=1}^{\infty} \mathsf{P}(A_i).$$

Example 1.21 Consider the random experiment of forecasting the number of price changes or *jumps*. We don't care of upwards and downwards movements, only the number of jumps is of interest: $\Omega = \{\omega \,|\, \omega = 0, 1, 2, \ldots\}$. We associate the σ-algebra $\mathcal{F} = 2^{\Omega}$ of all events. Next, we assign the probability weight

$$\mathsf{P}(\{\omega\}) = \frac{\lambda^i \, \mathrm{e}^{-\lambda}}{i!}, \quad \text{with } i = \omega = 0, 1, 2, \ldots, \quad \text{and } \lambda > 0.$$

For example, $A \in \mathcal{F}$ given by $A = \{0, 1, 4\}$ is the event that no jumps or 1 jump or 4 jumps have occurred, and it corresponds to the union $\{0\} \cup \{1\} \cup \{4\}$ of 3 pairwise disjoint events with probabilities $\frac{\lambda^0 \mathrm{e}^{-\lambda}}{0!}$, $\frac{\lambda^1 \mathrm{e}^{-\lambda}}{1!}$, $\frac{\lambda^4 \mathrm{e}^{-\lambda}}{4!}$. With $\lambda = 2$ they are 0.135335, 0.270670 and 0.090223. For $n = 10$ the probability is 0.000038: indeed $\frac{\lambda^i \mathrm{e}^{-\lambda}}{i!}$ is very small for a large number n of jumps. It can be interpreted as the *probability of rare events*.

Example 1.22 Consider the pairwise disjoint events $A_1 = \{1\}$, $A_2 = \{3\}$, $A_3 = \{5\}, \ldots$ from $\mathcal{F}$. We have[15]

$$\mathsf{P}(\cup_{i=1}^{\infty} A_i) = \sum_{\omega \in \cup_i A_i} \mathsf{P}(\{\omega\}) = \sum_{i=1}^{\infty} \underbrace{\sum_{\omega \in A_i} \mathsf{P}(\{\omega\})}_{=\mathsf{P}(A_i)} = \sum_{i=1}^{\infty} \frac{\lambda^i \, \mathrm{e}^{-\lambda}}{i!}$$

$$= \mathrm{e}^{-\lambda} \sum_{i=1}^{\infty} \frac{\lambda^i}{i!} = \mathrm{e}^{-\lambda} (\mathrm{e}^{\lambda} - 1),$$

where we used the Taylor series expansion of the exponential function e^{λ} around $\lambda_0 = 0$, i.e. $\mathrm{e}^{\lambda} = \mathrm{e}^{\lambda_0} + \frac{\mathrm{e}^{\lambda_0}}{1!}(\lambda - \lambda_0)^1 + \frac{\mathrm{e}^{\lambda_0}}{2!}(\lambda - \lambda_0)^2 + \cdots = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$. For example, with $\lambda = 2$, using `Excel` we compute `=EXP(-2)*(EXP(2)-1)=0.864665`. For the whole Ω we have:

$$\mathsf{P}(\{0\}) + \mathsf{P}(\{1\}) + \mathsf{P}(\{2\}) + \cdots = \sum_{i=0}^{\infty} \frac{\lambda^i \, \mathrm{e}^{-\lambda}}{i!} = \mathrm{e}^{-\lambda} \mathrm{e}^{\lambda} = 1.$$

We can get a countable sample space in the limit.

Example 1.23 Assume the sample space consists of n possible sequences of upwards and downwards, starting from a given stock price S_0. Encoding each outcome $\omega \in \Omega$ as a finite sequence of symbols u,d we have that $|\Omega| = 2^n$: we allow ω being the sequence with all u or all d, otherwise we have different sequences containing repeated u and d with alternative order. In fact: each symbol u or d can or cannot belong (2 possible choices) to the sequence encoding a fixed ω and we have a total of $2 \times 2 \times \cdots \times 2 = 2^n$ elements.

Example 1.24 Now we assign a probability weight $\mathsf{P}(\{\omega\}) = \frac{1}{2^n}$ to each simple event $\{\omega\}$ and get the usual discrete uniform probability space. But things go wrong when the number of possible upwards and downwards becomes larger: $\lim_{n \to \infty} \frac{1}{2^n} = 0$. Hence, each simple event has a probability zero as well as every other event A, which is obviously made of pairwise disjoint unions of some singletons $\{\omega\}$. Conclusion: all events A from the corresponding σ-algebra have $\mathsf{P}(A) = 0$. Also for the certain event $\mathsf{P}(\Omega) = 0$, since we can choose a finite or countable sequence $(A_i)_{i \in \mathbb{N}} \subset \mathcal{F}$ of pairwise disjoint events s.t. their union is equal to the whole sample space, $\Omega = \cup_i A_i$. Therefore, $\mathsf{P}(\Omega) = \mathsf{P}(\cup_i A_i)$ and using axiom PA2 we have $\mathsf{P}(\cup_i A_i) = \sum_i \mathsf{P}(A_i) = 0$.

[15] The probability of the event {an odd number of jumps}.

Example 1.25 When infinite repetitions[16] of the random experiment giving two possibilities (as u or d) are allowed, attaching a probability weight to singletons $\{\omega\}$ leads to some inconsistencies. A possible remedy is the following. Replace the encoding u or d with 1 or 0, so each outcome ω is made of an infinite sequence of ones and zeros. This corresponds to the binary expansion of any real number contained in the unit interval [0, 1):

$$\Omega = \{\omega \,|\, \omega = a_1 a_2 \cdots, \text{ where } a_i = 0 \text{ or } a_i = 1\}.$$

Since it is well known that $x \in [0, 1)$ may be written $0.a_1 a_2 \ldots$ we have a 1-to-1 correspondence between the elements $\omega \in \Omega$ and the numbers $x \in [0, 1)$. After we identify Ω with the right-open unit interval, a reasonable probability measure obeys to the formula

$$\mathsf{P}(A) = \text{length of } A, \quad \text{where } A = [a, b] \text{ or } [a, b) \text{ or } (a, b] \text{ or } (a, b),$$

and the σ-algebra is obtained by union, intersections and complements of intervals with finite endpoints. This is called *continuous uniform probability space*.

As we have seen in Sect. 1.4, discrete probability measure are adequate to model random experiments with at most countably infinite outcomes but cannot be extended to events larger than a countable sample space keeping as building blocks probability weights. To get further a flavor of how a remedy is possible, observe that for countable Ω there is a 1-to-1 correspondence with $\mathbb{N}$, thus we can turn the probability space to $(\mathbb{N}, 2^{\mathbb{N}}, \mathsf{P})$, and replace $\mathsf{P}(\{\omega_i\})$ with $p_i := p(i)$ s.t. as usual $\sum_{i=1}^{\infty} p(i) = 1$. When the sample space is uncountably infinite, there is a 1-to-1 correspondence with the open unit interval (0, 1) and then with the closed unit interval [0, 1] or the whole $\mathbb{R}$.[17] Therefore, we may turn to $(\mathbb{R}, \mathcal{F}, \mathsf{P})$ and try to define the probability measure as $\mathsf{P}([a, b]) = \int_a^b p(x)\mathrm{d}x$ for some nonnegative (Riemann) integrable function $p : \mathbb{R} \to \mathbb{R}$ s.t. $\int_{-\infty}^{\infty} p(x)\mathrm{d}x = 1$, and reals $a < b$. Observe that $\mathcal{F}$ must be generated by a proper collection of subsets in $\mathbb{R}$, see Appendix in Chap. 2, while the family of finite intervals is not a σ-algebra.

Example 1.26 We want to model the future stock price S_t as a rv taking uncountably infinite values on a closed interval $[S_{\min}, S_{\max}]$, where $S_{\min}$ is the minimum quoted price in t and $S_{\max}$ is the maximum future price. We can no longer use a discrete probability to compute the likelihood of events involving such S_t. In fact, now Ω can be identified with the above interval and it can be shown $\mathsf{P}(S_t = a) = \mathsf{P}(a) = 0$ for every $a \in [S_{\min}, S_{\max}]$. Compare this with results from Sects. 2.1, 2.2 and 2.3.

[16] We have to explicitly mention the auxiliary hypotheses that repetitions are *independent*. But the meaning of this will be clear later, so we tacitly assume it, see Sect. 5.4 in Chap. 5.

[17] This is a well known fact from real analysis.

Additional Properties of a Probability Measure

The *Boole's Inequality* is another fundamental property of a probability measure.

Lemma 1.2 (Subadditivity) *Let* $(\Omega, \mathcal{F}, \mathsf{P})$ *be a probability space. Then* $\mathsf{P}(\cup_{i=1}^{\infty} A_i) \leqslant \sum_{i=1}^{\infty} \mathsf{P}(A_i)$, *for any sequence of events* $(A_i)_{i\in\mathbb{N}} \subset \mathcal{F}$ *not necessarily pairwise disjoint.*

Remark 1.5 (Arbitrary Events vs Pairwise Disjoint Ones) Let $(A_i)_{i\in\mathbb{N}}$ be any sequence of events in a σ-algebra. We don't know if the A_i's are pairwise disjoint. From this we can build a new sequence of mutually disjoint events simply by letting $B_1 = A_1$ and $B_n = A_n \setminus \left(\cup_{i=1}^{n-1} A_i\right)$, for every $n \in \mathbb{N}$. Indeed, $B_2 = A_2 \setminus A_1$, $B_3 = A_3 \setminus (A_2 \cup A_1)$ and so on. Clearly, any partial union belongs to $\mathcal{F}$ and so any set difference. We have $\cup_{i=1}^{\infty} B_i = \cup_{i=1}^{\infty} A_i$.

Proof Define the sequence $(B_i)_{i\in\mathbb{N}}$ as in the above remark. For every $n \in \mathbb{N}$ we have $B_n \subset A_n$ and $B_n \in \mathcal{F}$ as $A_n \setminus \left(\cup_{i=1}^{n-1} A_i\right) = A_n \cap \left(\cup_{i=1}^{n-1} A_i\right)^c$ and $\mathcal{F}$ is a σ-algebra. This implies $\mathsf{P}(B_n) \leqslant \mathsf{P}(A_n)$ and since $\cup_{i=1}^{\infty} B_i = \cup_{i=1}^{\infty} A_i$ we have $\mathsf{P}(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathsf{P}(B_i) \leqslant \sum_{i=1}^{\infty} \mathsf{P}(A_i)$, for the events B_i are pairwise disjoint. ■

We can generalize Property Pr4, Proposition 1.1 of a probability measure. Pick a sequence of events $A_1, \ldots, A_n \in \mathcal{F}$, then it is easy to prove the following *union-intersection formula*:

$$\mathsf{P}(A_1 \cup \cdots \cup A_n) = \sum_{k=1}^{n} (-1)^{k-1} \sum_{\{i_1,\ldots,i_k\}\subset\{1,\ldots,n\}} \mathsf{P}(A_{i_1} \cap \cdots \cap A_{i_k}) \tag{1.3}$$

where the inner sum ranges over all subsets of $k \leqslant n$ indices in $\{1, \ldots, n\}$. Indeed, formula (1.3) is equal to the **inclusion-exclusion formula**:

$$\begin{aligned} \mathsf{P}(\cup_{i=1}^{n} A_i) = &\sum_{i=1}^{n} \mathsf{P}(A_i) - \sum_{i<j\leqslant n} \mathsf{P}(A_i \cap A_j) \\ &+ \sum_{i<j<k\leqslant n} \mathsf{P}(A_i \cap A_j \cap A_k) + \cdots + (-1)^{n+1} \mathsf{P}(\cap_{i=1}^{n} A_i). \end{aligned} \tag{1.4}$$

For $n = 2$ formula (1.4) holds by Pr4 of Proposition 1.1. To further illustrate the issue we also have:

$$\begin{aligned} \mathsf{P}(A_1 \cup A_2 \cup A_3) = &\mathsf{P}(A_1) + \mathsf{P}(A_2) + \mathsf{P}(A_3) \\ &- \mathsf{P}(A_1 \cap A_2) - \mathsf{P}(A_1 \cap A_3) - \mathsf{P}(A_2 \cap A_3) \\ &+ \mathsf{P}(A_1 \cap A_2 \cap A_3). \end{aligned}$$

The first sum on the right entails the area in $A_1 \cap A_2 \cap A_3$. To avoid repetitions we should subtract the area in $A_1 \cap A_2$, $A_1 \cap A_3$ and $A_2 \cap A_3$. We add the area in $A_1 \cap A_2 \cap A_3$, since we have subtracted area(A_1), area(A_2) and area(A_3) twice. Now write $\mathsf{P}(\cup_{i=1}^{n+1} A_i) = \mathsf{P}\left(\cup_{i=1}^{n} A_i \cup A_{n+1}\right)$ as $= \mathsf{P}(\cup_{i=1}^{n} A_i) + \mathsf{P}(A_{n+1}) - \mathsf{P}\left((\cup_{i=1}^{n} A_i) \cap A_{n+1}\right)$, by Pr4 applied to the events $\cup_{i=1}^{n} A_i$ and A_{n+1}. Trick: we prove inclusion-exclusion formula for $n+1$. By induction, the formula holds for n. Again by induction, we also have

$$
\begin{aligned}
& \mathsf{P}\left((\cup_{i=1}^{n} A_i) \cap A_{n+1}\right) \\
\overset{\text{distrib. of } \cup \text{ and } \cap}{=} \; & \mathsf{P}\left(\cup_{i=1}^{n} (A_i \cap A_{n+1})\right) \\
= \; & \sum_{i=1}^{n} \mathsf{P}(A_i \cap A_{n+1}) - \sum_{i<j\leqslant n} \mathsf{P}(A_i \cap A_j \cap A_{n+1}) + \cdots \\
= \; & \sum_{i<j=n+1}^{n} \mathsf{P}(A_i \cap A_j) - \sum_{i<j<k=n+1} \mathsf{P}(A_i \cap A_j \cap A_k) + \cdots
\end{aligned}
$$

Inserting Eq. (1.4), given by the first induction hypotheses, and the equation above into the expression of $\mathsf{P}(\cup_{i=1}^{n+1} A_i)$ we have done.

Continuity of P

Proof of Theorem 1.1 Let $A_1 \subset A_2 \subset \cdots\ldots$ be increasing events from $\mathcal{F}$ s.t. $A := \cup_{i=1}^{\infty} A_i$. We can find a new sequence of pairwise disjoint events $B_1, B_2, \ldots$ such that $\cup_{j=1}^{\infty} B_j = \cup_{i=1}^{\infty} A_i = A$. Furthermore, by countable additivity

$$
\mathsf{P}(A) = \mathsf{P}(\cup_{j=1}^{\infty} B_j) = \sum_{j=1}^{\infty} \mathsf{P}(B_j) = \lim_{i\to\infty} \sum_{j=1}^{i} \mathsf{P}(B_j).
$$

The definition of $(B_j)_{j\in\mathbb{N}}$ entails $\sum_{j=1}^{i} \mathsf{P}(B_j) = \mathsf{P}(\cup_{j=1}^{i} B_j) = \mathsf{P}(A_i)$, then $\lim_{i\to\infty} \mathsf{P}(A_i) = \mathsf{P}(A)$ and the proof is complete. If we handle a decreasing sequence of events $A_1 \supset A_2 \supset \cdots$ from $\mathcal{F}$ s.t. $A := \cap_{i=1}^{\infty} A_i$, then it suffices to note that $A_1^c \subset A_2^c \subset \cdots$, with $A^c = \cup_{i=1}^{\infty} A_i^c$ and moreover by Pr1 it holds $\mathsf{P}(A_i^c) = 1 - \mathsf{P}(A_i)$. Hence we have that $\lim_{i\to\infty} \mathsf{P}(A_i^c) = 1 - \lim_{i\to\infty} \mathsf{P}(A_i)$ is equivalent to

$$
\begin{aligned}
\lim_{i\to\infty} \mathsf{P}(A_i) \quad &= \quad 1 - \lim_{i\to\infty} \mathsf{P}(A_i^c) \\
&\overset{\text{cont. of } \mathsf{P} \text{ for incr. seq.}}{=} 1 - \mathsf{P}(A^c) \\
&= \quad \mathsf{P}(A).
\end{aligned}
$$

■

Chapter 2
Distribution of Random Variables



2.1 Distribution Function

In predicting the possible values of a rv X, given a probability space $(\Omega, \mathcal{F}, \mathsf{P})$, neither Ω nor X are given *explicitly*, instead we need to know $\mathsf{P}(X \in B)$ for suitable subsets $B \subset \mathbb{R}$, called **the probability law** of X, or simply the law, also written $\mathsf{P}_X(B)$, and the choice $B = (-\infty, x]$ suffices[1] to compute it.

Definition 2.1 The *cumulative* distribution function (CDF) of a rv X is defined as

$$F_X(x) := \mathsf{P}(X \leqslant x), \tag{2.1}$$

for any $-\infty < x < \infty$.

For every real number $x \in \mathbb{R}$, formula (2.1) entails a numerical function $F_X : \mathbb{R} \to [0, 1]$. Any CDF has some typical features whatever the corresponding rv X is. In fact, there is a 1-to-1 correspondence between P and F_X:

Theorem 2.1 *Let* P *be a probability measure over* $(\Omega, \mathcal{F})$ *and let* $F_X : \mathbb{R} \to [0, 1]$ *be a function given by formula (2.1). Then we have:*

(a) F_X *is nondecreasing,* $x_1 < x_2$ *implies* $F_X(x_1) \leqslant F_X(x_2)$*;*
(b) F_X *is right-continuous,* $F_X(x) = F_X(x^+) := \lim_{h \downarrow 0} F_X(x + h)$*;*
(c) $\lim_{x \to \infty} F_X(x) := F_X(\infty) = 1$ *and* $\lim_{x \to -\infty} F_X(x) := F_X(-\infty) = 0$*;*
(d) $\mathsf{P}(a < X \leqslant b) = F_X(b) - F_X(a)$ *for* $a < b$*, with finite* $b \in \mathbb{R}$ *and* $a \geqslant -\infty$.

[1] Other suitable $B \subset \mathbb{R}$ can be deduced from $(-\infty, x]$ using unions, intersections and complements. The same construction can be repeated with finite intervals, see Appendix.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_2

Conversely, if a function $F_X : \mathbb{R} \to [0,1]$ *satisfies properties (a)–(c) then there exists a probability* P *over some event space* $(\Omega, \mathcal{F})$ *s.t.* $F_X(x)$ *is the CDF of a rv* X*, according to Definition 2.1, with law* $\mathsf{P}_X(B) = \mathsf{P}(X \in B)$ *for suitable* $B \subset \mathbb{R}$.

Remark 2.1 (One-Sided Limits) Recall the following characterization of a limit for a real-valued function given $h > 0$:

- right limit, $F(x^+) = \lim_{h \downarrow 0} F(x+h)$;
- left limit, $F(x^-) = \lim_{h \downarrow 0} F(x-h)$.

Notation $h \downarrow 0$ means that h approaches zero by decreasing values, also written $h \to 0^+$. In the current context $F \equiv F_X$.

Remark 2.2 (Sequential Characterization) A real-valued function is continuous at a point x_0 in its domain, written $\lim_{x \to x_0} F(x) = F(x_0)$ iff there exists a sequence of points $x_1, x_2, \ldots$ in its domain such that $x_i \neq x_0$ and which converges to x_0, written $\lim_{i \to \infty} x_i = x_0$, so we have $F(x_i) \to F(x_0)$ whenever $i \to \infty$. In words for values x_i close to x_0 one has $F(x_i)$ staying close to $F(x_0)$. Therefore, we have:

- $F(x^+)$ is $\lim_{i \to \infty} F(x_i) = F(x)$ for any decreasing sequence $x_1 \geqslant x_2 \geqslant \cdots$ with limit x, written $x_i \downarrow x$;
- $F(x^-)$ is $\lim_{i \to \infty} F(x_i) = F(x)$ for any increasing sequence $x_1 \leqslant x_2 \leqslant \cdots$ with limit x, written $x_i \uparrow x$.

By Remarks 2.1 and 2.2 one-sided limits of a CDF can be given equivalently in sequential form, which is useful in proving probabilities as listed in Table 2.1. Right-continuity of the CDF stated in item (b) of Theorem 2.1 reads $F_X(x) = F_X(x^+)$. Observe that $F_X(x^-) \leqslant F_X(x)$ since the CDF is nondecreasing, i.e. it is a monotone

Table 2.1 Computing probabilities via F_X

#	$B \subset \mathbb{R}$	Event	$\mathsf{P}(X \in B)$
1	$(-\infty, a]$	$\{X \leqslant a\}$	$\mathsf{P}(X \leqslant a) = F_X(a)$
2	$(-\infty, a)$	$\{X < a\}$	$\mathsf{P}(X < a) = F_X(a^-)$
3	$(a, b]$	$\{a < X \leqslant b\}$	$\mathsf{P}(a < X \leqslant b) = F_X(b) - F_X(a)$
4	$[a, b]$	$\{a \leqslant X \leqslant b\}$	$\mathsf{P}(a \leqslant X \leqslant b) = F_X(b) - F_X(a^-)$
5	$\{a\}$	$\{X = a\}$	$\mathsf{P}(X = a) = F_X(a) - F_X(a^-)$
6	(b, ∞)	$\{X > b\}$	$\mathsf{P}(X > b) = 1 - F_X(b)$
7	$[b, \infty)$	$\{X \geqslant b\}$	$\mathsf{P}(X \geqslant b) = 1 - F_X(b^-)$
8	$[a, b)$	$\{a \leqslant X < b\}$	$\mathsf{P}(a \leqslant X < b) = F_X(b^-) - F_X(a^-)$
9	(a, b)	$\{a < X < b\}$	$\mathsf{P}(a < X < b) = F_X(b^-) - F_X(a)$

real-valued function and then both one-sided limits exist and are finite.[2] In Appendix we give a formal derivation of probabilities in the third column of Table 2.1.

2.2 Discrete Random Variables

As we have seen in Chap. 1, for some random experiment is more appropriate to compute discrete probabilities over finite or countable sample spaces. For general Ω the same applies as well when numerical evaluation of scenarios $\omega \in \Omega$ results in a finite or countable list of numbers. It should be clear that the law P_X together with F_X are the only responsible for the behavior of a rv X, not its analytical construction.[3]

Definition 2.2 A rv X is called **discrete** if its image $X(\Omega) = \{x \in \mathbb{R} \,|\, \exists\ \omega \in \Omega \text{ s.t. } X(\omega) = x\}$ is a set C either finite, $\{x_1, \ldots, x_n\}$, or countable, $\{x_1, x_2, \ldots\}$. The CDF of a discrete rv is a *piecewise constant* function or step function with jump discontinuities at the values $x_1, x_2, \ldots$, and is constant in between:

- $F_X(x) = y_{i-1}$ for $x \in [x_{i-1}, x_i)$ and $i = 2, 3, \ldots$; obviously $y_{i-1} \in (0, 1]$;
- on $(-\infty, x_1)$ it is $\lim_{x \to -\infty} F_X(x) = 0$, and $F_X(x_n) = 1$ or more generally $\lim_{i \to \infty} F(x_i) = 1$ as $x_i \to \infty$.

The situation described in Definition 2.2 has been already analyzed in Sect. 1.4 and Appendix in Chap. 1. Now, identifying P with P_X we can safely handle as a 'sample space' the whole $\mathbb{R}$ and find an at most countable subset C s.t. $\mathsf{P}_X(C) = \mathsf{P}(X \in C) = 1$ and $\mathsf{P}_X(C^c) = 0$. This would implies $X : \Omega \to C$, i.e. the rv attains at most countable many values. The above definition has some consequences.

- Whatever the sample space Ω is, for a discrete rv the events $\{X = x_i\}$ form a partition: they are pairwise disjoint and their (finite or countable) union is just Ω.
- Each probability $\mathsf{P}(X = x_i)$ is called a mass for $i = 1, 2, \ldots$; actually $\mathsf{P}(X = x)$ is a **probability mass function** of the values $x \in \mathbb{R}$.
- The masses are *concentrated* at $x_1, x_2, \ldots$, while their value is zero otherwise, i.e. $\mathsf{P}(X = x) = 0$ for $x \neq x_i$.
- We must have $\sum_i \mathsf{P}(X = x_i) = 1$.
- The set C is also where F_X is strictly increasing, sometimes called the **support** of X and written $\mathcal{X}$ in some textbooks.
- The value of $F_X(x)$ for any $x \in \mathbb{R}$ is computed summing the masses: $\mathsf{P}(X \leqslant x) = F_X(x) = \sum_{i \text{ s.t. } x_i \leqslant x} \mathsf{P}(X = x_i)$.

[2] In the proof of item (b) in Theorem 2.1 the right-continuity $\lim_{i \to \infty} F_X(x + \frac{1}{i}) = F_X(x)$ is understood in the sequential form, with $x_i = x + \frac{1}{i}$ being a decreasing sequence converging to x as $i \in \mathbb{N}$ becomes larger. On the other hand, it is equivalent to have $F_X(x^+) = \lim_{h \downarrow 0} F_X(x + h)$.

[3] In fact two rvs defined over the same probability space, are regarded as equal iff their laws are equal, see Sect. 2.6.

- $\mathsf{P}(X = x_i) = F_X(x_i) - F_X(x_i^-) > 0$, where the left limit $F_X(x_i^-)$ is equal to the value that F_X assumes at the leftmost point x_{i-1} for $i = 2, 3, \ldots$

Remark 2.3 We will see in Sect. 2.3 how for some rvs we may define the support as the set of points $x \in \mathbb{R}$ s.t. $\mathsf{P}(x - \epsilon < X < x + \epsilon) > 0$ for every $\epsilon > 0$.

Example 2.1 Let a function $F : \mathbb{R} \to [0, 1]$ be defined as:

$$F(x) := \begin{cases} 0, \text{ if } x < -1 \\ \frac{1}{2}, \text{ if } -1 \leqslant x < 1 \\ 1, \text{ if } x \geqslant 1. \end{cases}$$

It can be showed that F is a CDF and then satisfies properties (a)–(d) of Theorem 2.1.

Indeed, Example 2.1 provides us with an useful exercise in proving $F \equiv F_X$ for some discrete rv X. We verify:

- F is nondecreasing, indeed it is piecewise constant with values $F(x) = 0$ if $x \in (-\infty, -1)$, $F(x) = \frac{1}{2}$ if $x \in [-1, 1)$ and $F(x) = 1$ if $x \in [1, \infty)$;
- it is right-continuous, $F(-1) = \lim_{h \downarrow 0} F(-1 + h) = \frac{1}{2}$ and $F(1) = \lim_{h \downarrow 0} F(1 + h) = 1$, being constant then continuous over the three intervals above;
- its limits for $x \to -\infty$ and $x \to \infty$ are 0 and 1, respectively, by construction;
- for every reals $a < b$ the probability that $a < X \leqslant b$ is computed by $F(b) - F(a)$, e.g., with $a = 0$ and $b = 2$ we have $F(2) - F(0) = 1 - \frac{1}{2} = \frac{1}{2}$.

By its truly definition, this CDF gives the probability $\mathsf{P}(X \leqslant x)$ *accumulated* at each $x \in \mathbb{R}$:

- for $x < -1$ the probability that $X \leqslant x$ is 0;
- for $x = -1$ this probability increases to $\frac{1}{2}$ and remains constant up to $x < 1$;
- for $x = 1$ the cumulative probability hits 1.

The probability changes only at the jump discontinuities of F_X, namely at $x = -1$ and at $x = 1$. By Theorem 2.1(d) we compute:

- $\mathsf{P}(-\infty < X \leqslant -1) = \mathsf{P}(\{-\infty < X < -1\} \cup \{X = -1\}) = \mathsf{P}(-\infty < X < -1) + \mathsf{P}(X = -1)$ as $F_X(-1) - F_X(-\infty) = F_X(-1)$;
- $\mathsf{P}(-1 < X \leqslant 1) = \mathsf{P}(\{-1 < X < 1\} \cup \{X = 1\}) = \mathsf{P}(-1 < X < 1) + \mathsf{P}(X = 1)$ as $F_X(1) - F_X(-1)$.

Putting things together, we must have $\mathsf{P}(-\infty < X < -1) = \mathsf{P}(-1 < X < 1) = 0$ and $\mathsf{P}(X = -1) = \mathsf{P}(X = 1) = \frac{1}{2}$, thus the sum of the latter probabilities is just one.

Remark 2.4 There is a zero probability that X takes values on the open interval $(-\infty, -1)$. To see this more formally, observe that $F_X(x) = \mathsf{P}(X \leqslant x) = 0$ for every $x < -1$: say there is an increasing sequence of zero-probability events $\{X \leqslant -1 - \frac{1}{i}\}$, for $i \in \mathbb{N}$, s.t. their union is

$$\{x < -1\} = \{-\infty < X < -1\},$$

then by continuity we have $\lim_{i\to\infty} \mathsf{P}(X \leqslant -1 - \frac{1}{i}) = \mathsf{P}(-\infty < X < -1) = 0$. Also, $\mathsf{P}(-\infty < X < -1) = F_X(-1^-)$ is just the left-limit of the CDF at $x = -1$. Moreover, there is a zero probability that X takes value between the jump discontinuities, i.e. on the open interval $(-1, 1)$.

As byproduct, the current CDF corresponds to a rv X with only two values -1 and 1 and no relevant values in the open intervals $(-\infty, -1)$, $(-1, 1)$ and $(1, \infty)$. Attached to these two values we have two probability masses $\mathsf{P}(X = -1)$ and $\mathsf{P}(X = 1)$ whose values are given by the jump discontinuities of the CDF, i.e. $F_X(-1) - F_X(-1^-)$ and $F_X(1) - F_X(1^-)$ respectively, such that they sum up to one, see Fig. 2.1.

Example 2.2 (Changing the Masses) Let X be the discrete rv introduced so far with values $C = \{-1, 1\}$ but different masses $\mathsf{P}(X = -1) = \frac{1}{3}$ and $\mathsf{P}(X = 1) = \frac{2}{3}$. If $B = (-5, 1)$ we compute $\mathsf{P}(X \in B) = \frac{1}{3}$, because for any $x \in B$ the only nonzero probability that $X = x$ is $\mathsf{P}(X = -1)$. Observe that $X = 1$ is not included for the interval B is open on the right.

Theorem 2.2 *For a probability space $(\Omega, \mathcal{F}, \mathsf{P})$ and a discrete rv X there exists a countable set $C = \{x_1, x_2, \ldots\} \subset \mathbb{R}$ s.t. the probability that X assumes a value here is just 1.*

By Theorem 2.2 we can change the masses over corresponding points from the support: it suffices to guarantee that all them sum up to one.

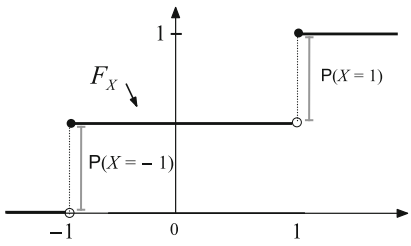


Fig. 2.1 CDF's graph of a discrete rv as a step function

2.3 Continuous Random Variables

The probability distribution of a discrete rv is characterized by a CDF which is not continuous: it is a nondecreasing step function with jump discontinuities at a finite or countable set of points where it increases by means of jumps given by probability masses. In fact, for a discrete rv X the CDF is not differentiable at each point x_i of its support where the derivative goes to infinity. To see this, first write

$$\lim_{h\to 0} \frac{F_X(x_i + \frac{h}{2}) - F_X(x_i - \frac{h}{2})}{h} = \frac{\lim_{h\to 0}\left[F_X(x_i + \frac{h}{2}) - F_X(x_i - \frac{h}{2})\right]}{\lim_{h\to 0} h}$$
$$= \frac{F_X(x_i) - F_X(x_i^-)}{0} = \frac{\mathsf{P}(X = x_i)}{0} = \infty.$$

Thus the above limit cannot provide the derivative of $F_X(x)$ at the point x_i. Moral: any discrete rv has CDF s.t. $F'_X(x) = 0$ except on its support, i.e. the finite or countable set of points where $F'_X(x)$ does not exist. When the CDF of a rv increases smoothly, then it can be the indefinite integral of its derivative.

Definition 2.3 A rv X i called **continuous** if the corresponding CDF is continuous i.e.

$$F_X(x) - F_X(x^-) = 0, \quad \text{for every } x \in \mathbb{R},$$

and there exists an integrable function $f_X : \mathbb{R} \to \mathbb{R}$ s.t. $f_X(x) \geqslant 0$ for all $x \in \mathbb{R}$, and $\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1$.

In Definition 2.3 the function f_X is called **density function** of the law P_X, or improperly of the rv X. It is not necessarily continuous, but due to the integrability condition[4] it is continuous or piecewise continuous.

Remark 2.5 By item 5 in Table 2.1, a possible jump of a CDF is just $F_X(x) - F_X(x^-) = \mathsf{P}(X = x)$. Moreover, it can be shown that a CDF has at most countably many discontinuity points. Therefore, for CDFs there are countably infinite $x \in \mathbb{R}$ where $\mathsf{P}(X = x) > 0$. A continuous CDF may be differentiable at $x \in \mathbb{R}$ where it has no jumps.

Proposition 2.1 (Absolute Continuity) *A rv X has a continuous probability distribution iff there exists a density function $f_X : \mathbb{R} \to \mathbb{R}$ which satisfies:*

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\mathrm{d}u, \quad x \in \mathbb{R}. \tag{2.2}$$

[4] In almost all cases, f_X is meant to be Riemann-integrable.

Thanks to this result we can compute probabilities via densities. In fact:

$$\int_{-\infty}^{b} f_X(u)\mathrm{d}u - \int_{-\infty}^{a} f_X(u)\mathrm{d}u \overset{(2.2)}{=} F_X(b) - F_X(a)$$
$$\overset{\text{Defn.on of } f_X}{=} \int_a^b f_X(u)\mathrm{d}u$$
$$\overset{\text{Th 2.1(d)}}{=} \mathsf{P}(a < X \leqslant b).$$

Loosely speaking, to compute $\mathsf{P}_X((a,b])$ it suffices to integrate the density over the corresponding interval. Moreover, since the support of a continuous rv (see Remark 2.3) is the uncountable set of points $x \in \mathbb{R}$ where $\mathsf{P}(x-\epsilon < X < x+\epsilon) > 0$ for all $\epsilon > 0$, the density is $f_X(x) > 0$ at these points. Henceforth, $\mathsf{P}(a \leqslant X \leqslant b) = \mathsf{P}(a \leqslant X < b) = \mathsf{P}(a < X < b)$ and all are equal to $\mathsf{P}(a < X \leqslant b)$. Indeed for a continuous probability distribution we always have $\mathsf{P}(X = x) = 0$ for every $x \in \mathbb{R}$ from the support: there is a zero probability that X takes on a single value. More is true: it can also be shown that for a continuous CDF which is continuously differentiable with the exception of a finite set of points, the law P_X has a density given by $f_X(x) = F_X'(x)$.[5] If the density function is additionally continuous then by the fundamental theorem of Calculus

$$F_X'(x) = \frac{\mathrm{d}}{\mathrm{d}x}\int_{-\infty}^{x} f_X(u)\mathrm{d}u = f_X(x). \tag{2.3}$$

Remark 2.6 if $f : [a,b] \to \mathbb{R}$ is continuous then the continuous function $F(x) := \int_a^x f(u)\mathrm{d}u$ is the antiderivative of f, i.e. $\frac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$ for $x \in [a,b]$ and $\int_a^b f(x)\mathrm{d}x = F(b) - F(a)$. Relation (2.3) is given in the improper form with $a \to -\infty$. It is worth recalling the improper Riemann integral $\int_{-\infty}^b f(x)\mathrm{d}x$ can be written $\lim_{a\to-\infty}\int_a^b f(x)\mathrm{d}x$, for $a < b$, and similarly $\int_{-\infty}^{\infty} f(x)\mathrm{d}x = \lim_{k\to\infty}\int_{-k}^{k} f(x)\mathrm{d}x$ for $k > 0$.

Hence, except for a finite number of points where F_X is not differentiable it is always the case that F_X can be recovered by f_X via (2.2), i.e. integration, and conversely f_X can be recovered by F_X through (2.3), i.e. differentiation. Observe that the density function alone does not yield a probability, since it could be the case $f_X(x) > 1$ for some $x \in \mathbb{R}$. For an interval with *infinitesimal* length $\mathrm{d}x$, say $[x - \frac{\mathrm{d}x}{2}, x + \frac{\mathrm{d}x}{2}]$, the probability that X takes on values in such a *small neighborhood* of x is just $f_X(x)\mathrm{d}x$. Since $\frac{\mathrm{d}F_X(x)}{\mathrm{d}x} = \lim_{h\to 0}\frac{F_X(x+h)-F_X(x)}{h}$ and $h \approx \mathrm{d}x$, we have $\frac{\mathrm{d}F_X(x)}{\mathrm{d}x}$ approximately given by $\frac{F_X(x+h)-F_X(x)}{\mathrm{d}x}$. Hence for the differential we have:

$$\mathrm{d}F_X(x) = f_X(x)\mathrm{d}x \approx F_X(x+\mathrm{d}x) - F_X(x) = \mathsf{P}(X \leqslant x+\mathrm{d}x) - \mathsf{P}(X \leqslant x),$$

[5] In this case $f_X(x)$ may assume arbitrary values on this finite exceptional set where $\nexists\, F_X'(x)$, and here we can let $f_X(x) = 0$.

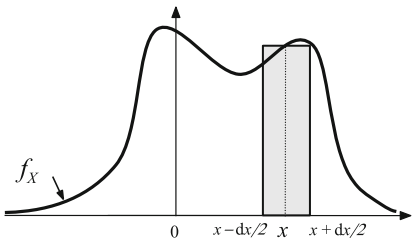


Fig. 2.2 Probability that X is close to x approximated by $f_X(x)\mathrm{d}x$, the area below the f_X's graph and between the dashed vertical lines. The total area below the f_X's graph amounts to 1

Table 2.2 Types of rvs in terms of their probability distribution

	Discrete	Absolutely continuous
F_X	Step function	Continuous
X's values	Finite or countable set $C = \{x_1, x_2, \ldots\}$	Interval(s) or $\mathbb{R}$
F_X increases	Jump $F_X(x_i) - F_X(x_i^-)$ for $x_i \in C$	Differential $\mathrm{d}F_X(x)$
F_X changes	Masses $\mathsf{P}(X = x_i)$ for $x_i \in C$	Density $f_X(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_X(x) \equiv F_X'(x)$
$\mathsf{P}(X \leqslant x)$	Finite or countable sum $\sum_{x_i \leqslant x} \mathsf{P}(X = x_i)$	'Uncountable sum' $\int_{-\infty}^{x} f_X(u)\mathrm{d}x$
$\mathsf{P}(X \in B)$	$\sum_{x_i \in B} \mathsf{P}(X = x_i)$	$\int_B f_X(x)\mathrm{d}x$

i.e., $\mathsf{P}_X((x + \frac{\mathrm{d}x}{2}])$ is approximately given by the X's law evaluated at $x \in \mathbb{R}$, and we ca formally write $\mathsf{P}_X(\mathrm{d}x) = \mathrm{d}F_X(x)$, where the left-hand side can be written $\mathrm{d}\mathsf{P}_X(x)$. The density function yields the instantaneous marginal change of the CDF, actually the slope at $x \in \mathbb{R}$ of the F_X's graph. Equivalently, $f_X(x)\mathrm{d}x$ is understood as the area below the f_X's graph, more or less the area of the rectangle with height $f_X(x)$ and base $\mathrm{d}x$, being x the center of $[x - \frac{\mathrm{d}x}{2}, x + \frac{\mathrm{d}x}{2}]$, see Fig. 2.2.

Example 2.3 Let $B = (-5, 1)$, and consider a continuous rv X with density $f_X(x) = \frac{1}{4}$ for $x \in [-2, 2]$ and $f_X(x) = 0$ otherwise. First, let us verify that

$$\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = \underbrace{\int_{-\infty}^{-2} f_X(x)\mathrm{d}x}_{=0} + \int_{-2}^{2} f_X(x)\mathrm{d}x + \underbrace{\int_{2}^{\infty} f_X(x)\mathrm{d}x}_{=0} = 1,$$

since $f_X(x)$ has primitive $\frac{1}{4}x$ (up to a constant) which integrates over $[-2, 2]$ to $\left[\frac{1}{4}x\right]_{-2}^{2} = \frac{1}{4}[2 - (-2)] = 1$. Thus, $\int_B f_X(x)\mathrm{d}x = \int_{-5}^{1} f_X(x)\mathrm{d}x$ and it equals

$$\int_{-5}^{-2} f_X(x)\mathrm{d}x + \int_{-2}^{1} f_X(x)\mathrm{d}x = 0 + \left[\frac{1}{4}x\right]_{-2}^{1} = \frac{3}{4} = \mathsf{P}(X \in B).$$

In Table 2.2, discrete and continuous rvs are compared.

2.4 Indicator Random Variables

Here we present a special type of discrete rv with support containing only two values. First, we need its deterministic version. For a given subset $B \subset \mathbb{R}$ the function

$$\mathbf{I}_B(x) := \begin{cases} 1, \text{ if } x \in B \\ 0, \text{ if } x \notin B \end{cases} \tag{2.4}$$

is called the *indicator* of the set B. Indicators can be useful for defining numerical functions changing over subsets B. For example, writing $\frac{1}{4}\mathbf{I}_{[-2,2]}(x)$ we get a function[6] equals to $\frac{1}{4}$ for every $-2 \leqslant x \leqslant 2$. When it is clear, the argument (x) can be omitted. The *random version* for a given event $A \subset \Omega$ is

$$\mathbf{I}_A(\omega) := \begin{cases} 1, \text{ if } \omega \in A \\ 0, \text{ if } \omega \notin A \end{cases} \tag{2.5}$$

called *indicator rv* . In fact, $\mathbf{I}_A$ is a rv iff $A \in \mathcal{F}$, see Exercise 2.3. Indicator rvs have the following properties, for any pair of events $A, B \in \mathcal{F}$:

1. $\mathbf{I}_{A\cup B} = \mathbf{I}_A + \mathbf{I}_B - \mathbf{I}_{A\cap B} = \max\{\mathbf{I}_A, \mathbf{I}_B\}$;
2. $\mathbf{I}_{A\cap B} = \mathbf{I}_A\mathbf{I}_B = \min\{\mathbf{I}_A, \mathbf{I}_B\}$;
3. $\mathbf{I}_{A\setminus B} = \max\{0, \mathbf{I}_A - \mathbf{I}_B\}$;
4. $\mathbf{I}_{A^c} = 1 - \mathbf{I}_A$;
5. $\mathbf{I}_A \leqslant \mathbf{I}_B$ iff $A \subset B$.

Their verification is based on definition (2.5). For example, property 1. is:

$$\begin{aligned} \mathbf{I}_{A\cup B}(\omega) = 1 &\iff \omega \in A \cup B \\ &\iff \omega \in A \text{ or } \omega \in B \text{ or } \omega \in A \cap B \end{aligned}$$

and provided A, B are compatible and $\mathbf{I}_A = \mathbf{I}_B = 1$ then also $\mathbf{I}_{A\cap B} = 1$ and the formula above holds; if only one of $\mathbf{I}_A, \mathbf{I}_B$ is 1 or A, B are disjoint, then $\mathbf{I}_{A\cap B} = 0$ and the formula still holds. Indicator rvs are useful in determining the structure of discrete rvs with finite values $\{x_1, \ldots, x_n\}$. Indeed, let $A_1, \ldots, A_n$ be events from $\mathcal{F}$ partitioning Ω, i.e. as usual $\Omega = A_1 \cup \cdots \cup A_n$ and $A_i \cap A_j = \varnothing$ for all $i \neq j$ ranging over $\{1, \ldots, n\}$. For a discrete rv with support $C = \{x_1, \ldots, x_n\}$ we can *safely* write

$$X(\omega) = x_1\mathbf{I}_{A_1}(\omega) + \cdots + x_n\mathbf{I}_{A_n}(\omega), \tag{2.6}$$

[6] Actually it is a density function.

whenever $X(\omega) = x_i$ for $\omega \in A_i$ and $i = 1, \ldots, n$. In fact by the disjointedness hypotheses we also have $\mathbf{I}_{A_i} = 1$, while $\mathbf{I}_{A_j} = 0$ for the remaining $n-1$ integers $j \neq i$ ranging over $\{1, \ldots, n\}$. For representing these disjoint events the obvious choice[7] is $A_i = \{X = x_i\}$, with positive masses $\mathsf{P}(A_i) > 0$. Discrete rvs as represented in (2.6) are usually called **simple rv** . A related issue is the representation of discrete probability distributions using indicators.

Example 2.4 (Discrete CDF via Indicators) Let X be a discrete rv modelling future stock prices, with values in the support $C = \{90, 150, 190\}$ denominated in euros. We set $x_1 = 90$, $x_2 = 150$ and $x_3 = 190$; the masses are $p_i := \mathsf{P}(X = x_i)$ for $i = 1, 2, 3$. Hence, the probability that the random price assumes value at most $x \in \mathbb{R}$ is

$$\mathsf{P}(X \leqslant x) = F_X(x) = p_1 \mathbf{I}_{[x_1,\infty)}(x) + p_2 \mathbf{I}_{[x_2,\infty)}(x) + p_3 \mathbf{I}_{[x_3,\infty)}(x). \tag{2.7}$$

Here each $\mathbf{I}_{[x_i,\infty)}(x)$ is 1 whenever $x \in [x_i, \infty)$, which is equivalent to $x_i \leqslant x$ for fixed x_i. For example, with $x = 155$ euro the future stock price is at most x with probability

$$\begin{aligned} F_X(155) &= p_1 \underbrace{\mathbf{I}_{[x_1,\infty)}(155)}_{=1} + p_2 \underbrace{\mathbf{I}_{[x_2,\infty)}(155)}_{=1} + p_3 \underbrace{\mathbf{I}_{[x_3,\infty)}(155)}_{=0} \\ &= \mathsf{P}(X = 90) + \mathsf{P}(X = 150). \end{aligned}$$

Each indicator of the type $\mathbf{I}_{[x_i,\infty)}(x)$ is sometimes written $\mathbf{I}_{\{x_i \leqslant x\}}$.

Example 2.5 The CDF of a discrete rv with $n \in \mathbb{N}$ values can be represented as

$$F_X(x) = \sum_{i=1}^{n} p_i \mathbf{I}_{\{x_i \leqslant x\}}.$$

Observe that this is the convex combination of n step functions $\mathbf{I}_{\{x_i \leqslant x\}}$, which are actually CDFs, with weights $p_i \geqslant 0$ s.t. $\sum_{i=1}^{n} p_i = 1$.

Remark 2.7 In real analysis we found the *unit step function* or *Heaviside function*:

$$u(x) := \begin{cases} 1, \text{ if } x \geqslant 0 \\ 0, \text{ if } x < 0. \end{cases} \tag{2.8}$$

As byproduct $\mathbf{I}_{\{x_i \leqslant x\}} = u(x - x_i)$ and we may write $\boxed{F_X(x) = \sum_{i=1}^{n} p_i u(x - x_i)}$ which is the way a discrete CDF is represented in some textbooks.

[7] We will see how nonnegative rvs can be approximated by discrete rvs in the form (2.6).

Example 2.6 From a different perspective, the indicator $\mathbf{I}_B(x)$ can be understood as a set function $\mathbf{I}_x(B)$, i.e. definition (2.4) is the same but now $x \in \mathbb{R}$ is fixed and what changes is the set B. A discrete probability distribution can be written also[8]

$$\mathsf{P}_X(B) := \sum_{i=1}^{n} p_i \mathbf{I}_{x_i}(B),$$

where $p_i \geqslant 0$ and $\sum_{i=1}^{n} p_i = 1$. For example, considering $B = [0, 155]$ and the previous rv X modelling future stock prices we can compute

$$\begin{aligned}\mathsf{P}(X \in B) = P_X(B) &= p_1 \underbrace{\mathbf{I}_{90}(B)}_{=1} + p_2 \underbrace{\mathbf{I}_{150}(B)}_{=1} + p_3 \underbrace{\mathbf{I}_{190}(B)}_{=0} \\ &= p_1 + p_2 = \mathsf{P}(X = 90) + \mathsf{P}(X = 150) \\ &= \sum_{x \in B} \mathsf{P}(X = x).\end{aligned}$$

In some textbook $\mathbf{I}_x(B)$ is written $\boxed{\delta_x(B)}$ and called *Dirac delta measure* .

We close this section with another application of indicators.

Example 2.7 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a discrete uniform probability space whit $\mathsf{P}(A) = \frac{|A|}{|\Omega|}$ for every event $A \in \mathcal{F} = 2^{\Omega}$, where Ω is finite. We can use the indicators to redefine the probability measure. If we let $\mathsf{P}(A) = \frac{\sum_{\omega \in \Omega} \mathbf{I}_A(\omega)}{\sum_{\omega \in \Omega} \mathbf{I}_\Omega(\omega)}$, then the ratio 'number of scenarios in A' to 'number of total scenarios' amounts to sum the indicators over all ω contained in A and Ω, respectively. The ratio has a frequency interpretation: in computing the likelihood of A to occur we divide the number of 'favorable' outcomes by the number of total outcomes.

2.5 Operations on Random Variables

It is possible to scale a rv and apply algebraic operations to two or more rvs, obtaining a new rv.

Theorem 2.3 *Let $(\Omega, \mathcal{F})$ be an event space and let X, Y be two rvs defined here. Then:*

1. *$X + Y$ is a rv*
2. *αX is a rv, for any $\alpha \in \mathbb{R}$*
3. *$\frac{X}{Y}$ is a rv, for any $\omega \in \Omega$ such that $Y(\omega) \neq 0$*

[8] See Exercise 2.7.

4. *the maximum and minimum between rvs,* $X \vee Y := \max\{X, Y\}$ *and* $X \wedge Y := \min\{X, Y\}$, *is again a rv.*

Theorem 2.3 can be proved by showing that $\{X + Y \leqslant x\}$ is an event in $\mathcal{F}$ for every $x \in \mathbb{R}$, and similarly for scaling and division. For the maximum and minimum of rvs, one shows that for every $x \in \mathbb{R}$ the event $\{X \vee Y \leqslant x\}$ equals $\{X \leqslant x\} \cap \{Y \leqslant x\}$ and the event $\{X \wedge Y \leqslant x\}$ equals $\{X \leqslant x\} \cup \{Y \leqslant x\}$. Theorem 2.3 yields the following consequences. Items 1 and 2 imply that a multiple integer of X is again a rv: nX for $n \in \mathbb{N}$. By item 3 with $X = 1$, reciprocal $\frac{1}{Y}$ is a rv. The constant function $X : \Omega \to \mathbb{R}$ with $X(\omega) = c$ for every scenarios ω and a fixed $c \in \mathbb{R}$ is a rv, usually called **degenerate**. By item 3, we can start from $Y = \frac{1}{Z}$, for a rv Z over the same event space, and then get multiplication XY. The linear combination $\alpha X + \beta Y$ is a special case based on items 1 and 2. Taking $\beta = 1 - \alpha$ for some $0 \leqslant \alpha \leqslant 1$, the *convex combination* $\alpha X + (1-\alpha)Y$ as a new rv does not necessarily share the same distribution as the rv with mixed CDF $\alpha F_X(x) + (1-\alpha)F_Y(x)$, i.e. in general

$$F_{\alpha X + (1-\alpha)Y}(x) \neq \alpha F_X(x) + (1-\alpha)F_Y(x), \quad \text{for all } x \in \mathbb{R} \text{ and } 0 \leqslant \alpha \leqslant 1.$$

Remark 2.8 (Mixed CDFs) Given two CDFs F_X, F_Y corresponding to rvs X, Y defined over the same probability space, the convex combination

$$\alpha F_X(x) + (1-\alpha)F_Y(x) \tag{2.9}$$

is a new CDF, for all $x \in \mathbb{R}$ and $0 \leqslant \alpha \leqslant 1$. In fact, we can check properties (a) to (d) of Theorem 2.1 in Sect. 2.1, see Exercise 2.13.

The negative $-X$, a special case of item 2, has interesting features for symmetric distributions (see Sect. 2.7). We can consider one more special case: the difference $X - Y$. Theorem 2.3 may be strengthen by considering more general transformations of rvs.

Scalar-to-scalar. Given a rv X we can always get a new rv through a suitable function[9] $g : \mathbb{R} \to \mathbb{R}$ s.t. $Y = g(X)$, i.e. for each scenario $\omega \in \Omega$ the scalar value $Y(\omega)$ is obtained by evaluating $g(\cdot)$ at the scalar value $X(\omega)$, see Table 2.3

Vector-to-scalar. Given two rvs[10] X, Y we can always get one new rv through a function $h : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ s.t. $Z = h(X, Y)$, i.e. for each scenario $\omega \in \Omega$ the scalar value $Z(\omega)$ is obtained by evaluating $h(\cdot, \cdot)$ at $(X(\omega), Y(\omega))$, a pair of two rvs, see Table 2.4.

Sums $h(X, Y) = X + Y$ can be obtained by linear combinations with $\alpha = 1$ and $\beta = 1$; if $\beta = -1$ we have differences $h(X, Y) = X - Y$. Convex combinations are special cases with $0 \leqslant \alpha \leqslant 1$ and $\beta = 1 - \alpha$. Scalar-to-scalar and vector-to-scalar

[9] Usually it is a continuous or even a differentiable function.

[10] Actually they form a pair (X, Y) which may be thought as a function from Ω to the product space $\mathbb{R} \times \mathbb{R}$, see Chap. 3.

Table 2.3 Elementary scalar-to-scalar transformations

Exponential	e^X or a^X for any $a > 0$
Logarithmic	$\log X$ or $\log_a X$ for any $a > 0, a \neq 1$ for all $\omega \in \Omega$ with $X(\omega) > 0$
Powers	X^n with $n \in \mathbb{N}$
Root	$X^{\frac{1}{n}} = \sqrt[n]{X}$ with $n \in \mathbb{N}$ for all $\omega \in \Omega$ with $X(\omega) \geqslant 0$
Linear	$a + bX$ for $a \in \mathbb{R}$ and $b \neq 0$

Table 2.4 Elementary vector-to-scalar transformations

Multiplication	XY
Division	$\frac{X}{Y}$ for all $\omega \in \Omega$ with $Y(\omega) \neq 0$
Linear combination	$\alpha X + \beta Y$ for $\alpha, \beta \in \mathbb{R}$
Maximum and minimum	$\max\{X, Y\}$, $\min\{X, Y\}$

transformation can be properly combined. Other scalar-to-scalar transformations are $\sin X$ or $\cos X$, which we will use in the sequel (see Chap. 4). Two further examples of vector-to-scalar transformation are the following.

- *Negative part of a rv*: take the maximum $X^- := \max\{0, -X\}$ where $Y = 0$ is a degenerate rv. It can be written $-\min\{0, X\}$ or $-X\mathbf{I}_{\{X<0\}}$.
- *Positive part of a rv*: take the maximum $X^+ := \max\{0, X\}$. It can be written $X\mathbf{I}_{\{X\geqslant 0\}}$.

Vector-to-scalar transformation can be in the form $Z = h(X_1, \ldots, X_n)$, for a suitable function $h(\cdot, \ldots, \cdot)$ and an n-dimensional vector of rvs $(X_1, \ldots, X_n)$.

Example 2.8 Asset prices and returns play important role in finance. Given a positive rv [11] $S_{t+1} > 0$ modelling the $(t+1)$-price, we known that $S_{t+1} - S_t$, for integers $t = 0, 1, 2, \ldots$, represents either a random profit or loss. Thus, $r_{t+1} = \frac{S_{t+1}-S_t}{S_t} = \frac{S_{t+1}}{S_t} - 1$ is the periodic random rate of return. In particular, $\frac{S_{t+1}}{S_t} = 1 + r_{t+1}$ is called *gross return*. For $h > 0$ small enough, $R_{t+h} = \ln\left(\frac{S_{t+h}}{S_t}\right)$ is the *log-return*. From well known properties of historical returns, it holds $R_{t+1} = \ln(1 + r_{t+1}) \approx r_{t+1}$, because for time window $[t, t+1]$ corresponding to one day, on week or at most one month the periodic return r_{t+1} is near to zero, and $\ln(x) \approx 0$ iff $x \approx 1$.

1. *Arithmetic Return.* Investing the total amount $€ V$ in $t = 0$ we eventually get

$$V\,(1 + r_1)(1 + r_2)\cdots(1 + r_T) = V\,h(r_1, r_2, \ldots, r_T),$$

by using compound interest for $T > 1$ years, e.g. $T = 365$ yields a 1-year value.

[11] Recall what this means: $S_{t+1}(\omega) > 0$ for every scenario $\omega \in \Omega$.

2. *Continuously Compounded Return.* For instantaneous rate of returns the future value is

$$V\,\mathrm{e}^{R_1}\mathrm{e}^{R_2}\cdots\mathrm{e}^{R_T} = V\,\mathrm{e}^{R_1+R_2+\cdots+R_T} = V\,h(R_1, R_2, \ldots, R_T),$$

where $R_{t+1} = \ln\left(\frac{S_{t+1}}{S_t}\right)$.

3. *Payoff of European options.* Let S_T be a rv representing the future price of a stock underlying an European call option with maturity $T > 0$ and *strike price* $K > 0$, a fixed amount known in advance. Call options are financial products giving the right (but not the obligation) to buy an asset at K in T, the time when they expire. The value in T of this derivative instrument is $\max\{0, S_T - K\} = (S_T - K)^+$, called *payoff*. The holder of an European call option[12] can choose between:

   - exercises it, whenever $S_T > K$, since buying the stock for K and sell it for S_T he earns a profit $S_T - K > 0$;
   - does not exercise it, if $S_T < K$, as he can get the stock at a cheaper price on the market, and in this case there is a zero profit.

   An European put option gives the right to sell the underlying stock at K, then its payoff is $\max\{K - S_T, 0\} = (K - S_T)^+$.

Example 2.9 In actuarial science, if $X > 0$ is a positive rv modelling a future loss, then the amount to be paid by the insurer when X is subject to a deductible $s > 0$ is $\max\{X - s, 0\} = (X - s)^+$, also known as *stop-loss function*.

Example 2.10 (Prices from Returns) Given the positive rv $S_t > 0$ representing future stock prices, we can recover prices from returns. Using log-returns R_t (see Example 2.8), the new rv

$$S_{t+1} = S_t\mathrm{e}^{R_{t+1}}$$

represents future prices one period ahead, and are well defined even if $R_{t+1}(\omega) < 0$ for some scenarios $\omega \in \Omega$. On the other hand, arithmetic returns would give

$$S_{t+1} = S_t(1 + r_{t+1}),$$

which works well only when $r_{t+1}(\omega) \geqslant -1$, for all scenarios. Observe what happen when we are in t: the rv S_t becomes degenerate, i.e. it could be considered as a constant function.

Example 2.11 (Portfolio Value) Portfolios are basket of different financial assets, e.g. stocks, bonds, derivatives, etc.[13] Assume we have $n \in \mathbb{N}$ assets in the portfolio,

[12] American options can be exercised before the maturity. Options are examples of *contingent claims*.

[13] These form the *asset universe*, where differences among assets are based on the degree of risk and other additional features. As a proxy of risk-free assets we can choose bank accounts or zero-coupon bonds.

with holdings h_i, for $i = 1, \ldots, n$, e.g., number of units held in asset i, as shares of a given stock, money invested in a bank account, etc. From this we can deduce some interesting rvs as the portfolio value

$$V_t = h_1 S_{1,t} + \cdots + h_n S_{n,t}, \quad \text{for } t = 0, 1, 2, \ldots, T,$$

and the arithmetic portfolio return in $t + 1$

$$r_{p,t+1} := \frac{V_{t+1} - V_t}{V_t} = \frac{\sum_{i=1}^n h_i S_{i,t+1} - \sum_{i=1}^n h_i S_{i,t}}{\sum_{i=1}^n h_i S_{i,t}} = \sum_{i=1}^n w_i r_{i,t+1},$$

which is a linear combination of arithmetic asset returns with weights $w_i = \frac{h_i S_{i,t}}{V_t}$, implying $\sum_{i=1}^n w_i = 1$. It is worth noting that each $h_i \in \mathbb{R}$ and it can be thought as a degenerate rv, and also $w_i \in \mathbb{R}$ are constant once we act on the time horizon $t, t + 1$ where $S_{i,t}$ are known.[14]

2.6 Comparing Random Variables

Two rvs as functions from Ω to $\mathbb{R}$ can be pointwise equal, i.e. $X(\omega) = Y(\omega)$ for every scenario $\omega \in \Omega$. But as we have seen, what matters for rvs is their probability distribution, thus a different kind of comparison can be set up, compare with Sect. 1.3, Chap. 1.

Definition 2.4 Two rvs X, Y over $(\Omega, \mathcal{F})$ are **almost surely equal**, abbreviated $X \overset{\text{a.s.}}{=} Y$ iff the event $\{X = Y\} \in \mathcal{F}$ has probability 1.

In some textbooks the above is also written $X = Y$ P-a.s. The fact that $\{X = Y\}$ is an event in $\mathcal{F}$, and then has assigned a probability, is due to Theorem 2.3 guaranteeing $\{X = Y\} = \{X - Y = 0\}$, i.e. $X - Y$ is a rv. Observe that $\mathsf{P}(X \neq Y) = 0$, where the events $\{X = Y\}$ and $\{X \neq Y\}$ form a partition of Ω. It could be the case that two rvs share the same probability law.

Definition 2.5 Two rvs X, Y defined over $(\Omega, \mathcal{F})$ are **equal in distribution**, abbreviated $X \overset{\text{d}}{=} Y$ iff

$$\mathsf{P}(X \in B) = \mathsf{P}(Y \in B),$$

for every subset $B \subset \mathbb{R}$ s.t. $\{X \in B\}$ and $\{Y \in B\}$ are in $\mathcal{F}$.

[14] When $h_i > 0$ we have a long position on asset i (buy); $h_i < 0$ means *short selling*, i.e. a trade in the market involving an asset which we do not hold in t, although it will be bought and delivered in $t + 1$.

As usual, choosing $B = (-\infty, x]$ we get the equivalent condition

$$F_X(x) = F_Y(x), \quad \text{for every } x \in \mathbb{R}.$$

Thus, for two rvs being equal in distribution the corresponding CDFs must be equal as numerical functions. When two rvs X and Y are equal in distribution all probability statements about them will be the same, but they can be different as functions.

Example 2.12 (Actually...Counterexample) Assume that X is discrete with support $C = \{-1, 1\}$ and masses $\mathsf{P}(X = -1) = \mathsf{P}(X = 1) = \frac{1}{2}$, interpreted as a bet giving € 1 in case of win otherwise yielding a loss of −€ 1, with equal probability. Let the rv $Y = -X$ given by the negative of the bet. We have Y as a discrete rv with masses $\mathsf{P}(Y = 1) = \mathsf{P}(Y = -1) = \frac{1}{2}$. Hence we conclude $X \overset{\mathrm{d}}{=} Y$, but X and Y are not equal because when $X = -1$ the other is $Y = 1$ and vice versa. Moreover, $\mathsf{P}(X = Y) = 0$ because $\{X = 1\} = \{Y = -1\}$ and $\{X = -1\} = \{Y = 1\}$ hence

$$\begin{aligned} 0 \leqslant \mathsf{P}(X = Y) &= \mathsf{P}\Big(\Big[\{X = 1\} \cap \{Y = 1\}\Big] \cup \Big[\{X = -1\} \cap \{Y = -1\}\Big]\Big) \\ &\leqslant \mathsf{P}\left(\{X = 1\} \cap \{Y = 1\}\right) + \mathsf{P}\left(\{X = -1\} \cap \{Y = -1\}\right) \\ &= \mathsf{P}(\varnothing) + \mathsf{P}(\varnothing) = 0. \end{aligned}$$

Other comparisons between two rvs involve inequalities, for example $X \leqslant Y$ pointwise or $X \leqslant Y$ P-a.s. A different kind of inequalities comes into play when Definition 2.5 does not hold. In fact, by considering a suitable class of CDFs of rvs defined over the same probability space we have:[15]

Definition 2.6 (Usual Stochastic Order) X is said to be smaller than Y in the **usual stochastic order**, written $X \leqslant_{\mathrm{st}} Y$, if

$$F_X(x) \geqslant F_Y(x), \quad \text{for every } x \in \mathbb{R}. \tag{2.10}$$

This is pointwise comparison of CDFs. Despite the inequality between CDFs should be counterintuitive, Definition 2.6 means that Y assumes large values with higher probability than X does. Since $F_X(x) \geqslant F_Y(x)$, we can choose $x_1 < x_2$ such that $\mathsf{P}(X \leqslant x_1) = \mathsf{P}(Y \leqslant x_2)$, then by increasing x_2 we get larger values of Y with more probability (recall that a CDF is nondecreasing). The usual stochastic order is also called *first order stochastic dominance*, and alternatively denoted $X \preccurlyeq_{\mathrm{FSD}} Y$. It can be shown that $X \leqslant_{\mathrm{st}} Y$ is equivalent to have $X \leqslant Y$ pointwise or $\mathsf{P}(X \leqslant Y) = 1$. If we define

$$\bar{F}_X(x) := 1 - F_X(x) = 1 - \mathsf{P}(X \leqslant x) = \mathsf{P}(X > x), \tag{2.11}$$

[15] Over this class of CDFs, namely $\mathcal{D}$, one can define a partial order, i.e. a binary relation $\preccurlyeq$ between couples of CDFs which is *reflexive*, $F_X \preccurlyeq F_X$ for all $F_X \in \mathcal{D}$, *antisymmetric*, if $F_X \preccurlyeq F_Y$ and $F_Y \preccurlyeq F_X$ then $F_X = F_Y$, *transitive*, if $F_X \preccurlyeq F_Y$ and $F_Y \preccurlyeq F_Z$ then $F_X \preccurlyeq F_Z$.

called *survival function*, then Definition 2.6 is equivalent to $\bar{F}_X(x) \leqslant \bar{F}_Y(x)$ for all real x. The usual stochastic order is only partial, for it could be the case $F_X(x_0) \geqslant F_Y(x_0)$ for some $x_0 \in \mathbb{R}$ and $F_X(x) < F_Y(x)$ for the remaining $x \in \mathbb{R}$, where x_0 is a crossing point of the two CDFs graphs.

Example 2.13 Let X be the strike price $K > 0$ of an European call option. It can be viewed as a degenerate rv $X(\omega) = K$ for every scenario $\omega \in \Omega$, and its probability distribution can be represented by a Dirac delta $\delta_K(B)$, for all suitable $B \subset \mathbb{R}$. Now, suppose we have two such options and corresponding strike prices K_1, K_2. Thus, degenerate rvs $X = K_1$ and $Y = K_2$ have distributions in usual stochastic order $\delta_{K_1} \preccurlyeq_{\text{FSD}} \delta_{K_2}$ iff $K_1 \leqslant K_2$, In fact, $\delta_{K_1}((-\infty, x]) = F_X(x)$ and $\delta_{K_2}((-\infty, x]) = F_Y(x)$, then $F_X(x) \geqslant F_Y(x)$ for all $x \in \mathbb{R}$. Indeed, this should be write $F_X(x) = \mathbf{I}_{\{K_1 \leqslant x\}}$ and similarly $F_Y(x) = \mathbf{I}_{\{K_2 \leqslant x\}}$, or also $F_X(x) = u(x - K_1)$ and $F_Y(x) = u(x - K_2)$, see Sect. 2.4.

Since $\leqslant_{\text{st}}$ is only a partial order, it could be the case that two rvs representing uncertain outcomes (viz. future values) of risky assets cannot be compared under this stochastic order. In such a situation, a risk-adverse decision maker would prefer the rv with lower variability. We say that X is less than Y in the *second order stochastic dominance*, written $X \preccurlyeq_{\text{SSD}} Y$, or **increasing concave order**, denoted $X \leqslant_{\text{icv}} Y$, if

$$\int_{-\infty}^{x} F_X(u)\mathrm{d}u \geqslant \int_{-\infty}^{x} F_Y(u)\mathrm{d}u \quad \text{for all } x \in \mathbb{R}. \tag{2.12}$$

For continuous rvs, this stochastic order has a geometric interpretation: when $\leqslant_{\text{st}}$ fails to hold and the CDFs' graphs have at least one crossing point, say $x_0 \in \mathbb{R}$, then condition (2.12) can be written

$$\int_{-\infty}^{x_0} (F_X(u) - F_Y(u))\mathrm{d}u \geqslant 0,$$

i.e., the F_X's graph is above that of F_Y up to x_0, hence the area between the two curves is positive up to this point. Moreover, the contribution of the negative area computed as

$$\int_{x_0}^{x} (F_X(u) - F_Y(u))\mathrm{d}u \leqslant 0, \quad \text{for every } x \geqslant x_0,$$

is negligible with respect to the first, see Fig. 2.3.

2.7 Symmetric Probability Laws

Let S_T be the rv representing a future stock price, given the initial price S_0. As we have seen, $X = S_T - S_0$ is a model for profits and losses, provided that we invest in the stock. A reasonable assumption should be assigning different probabilities to

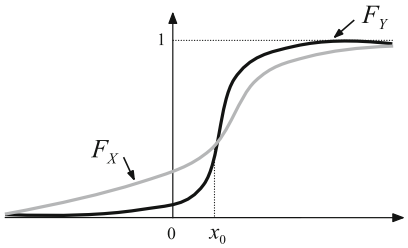


Fig. 2.3 F_X doe not always dominates F_Y, but the area on the left (+) is greater than that on the right (−)

X^+ and X^-, and indeed this is the typical situation for realized values of X over a fixed time horizon $[0, T]$.

Definition 2.7 A rv X is called **symmetric** (or symmetrically distributed) around $a \in \mathbb{R}$ if

$$\mathsf{P}(X \leqslant a - x) = \mathsf{P}(X \geqslant a + x), \quad \text{for every } x \in \mathbb{R}. \tag{2.13}$$

If the X's support is strictly contained in $\mathbb{R}$, then $a \in \mathbb{R}$ not necessarily belongs to it. Using the definition of CDF $F_X(\cdot) = \mathsf{P}(X \leqslant \cdot)$ and considering its left-limit $F_X(\cdot^-) = \mathsf{P}(X < \cdot) = 1 - \mathsf{P}(X \geqslant \cdot)$, for a symmetric X we have[16]

$$\begin{aligned} F_X(a - x) &= 1 - \mathsf{P}(X < a + x) = 1 - F_X((a + x)^-) \\ &\iff F_X(a - x) + F_X(a + x^-) = 1, \quad \text{for every } x \in \mathbb{R}. \end{aligned} \tag{2.14}$$

As byproduct, two conditions for symmetric probability distributions around a fixed point $a \in \mathbb{R}$ are:

- Discrete rvs, $\boxed{\mathsf{P}(X = a - x) = \mathsf{P}(X = a + x)}$, for all $x \in \mathbb{R}$;
- Absolutely Continuous rvs, $\boxed{f_X(a - x) = f_X(a + x)}$, for all $x \in \mathbb{R}$.

The second condition is obtained by the chain rule of differentiation: we differentiate $F_X(g(x))$ where $g(x) = a - x$, thus $\frac{\mathrm{d}}{\mathrm{d}x} F_X(g(x)) = F'_X(g(x))g'(x)$ with $g'(x) = -1$, and similarly for $g(x) = a + x$. Observe that $F_X(a + x^-) = F_X(a + x)$ for continuous distributions. Therefore, symmetry around zero is respectively:

- $\mathsf{P}(X = -x) = \mathsf{P}(X = x)$, for all $x \in \mathbb{R}$;
- $f_X(-x) = f_X(x)$, for all $x \in \mathbb{R}$, i.e. the density need not to be an even function.

For symmetric distributions the rvs $a - X$ and $X - a$ have the same probability law, $\boxed{a - X \overset{\mathrm{d}}{=} X - a}$. The precise reasoning is the following. Let X be symmetrically distributed around $a \in \mathbb{R}$, then condition (2.14) holds. Observe that $X - a$ has CDF

$$F_{X-a}(x) = \mathsf{P}(X - a \leqslant x) = \mathsf{P}(X \leqslant a + x) = F_X(a + x).$$

[16] Observe that $(a + x)^- = \lim_{h \downarrow 0}(a + x) - h = \lim_{h \downarrow 0} a + (x - h) = a + x^-$, for $h > 0$.

Now, $a - X$ has CDF

$$F_{a-X}(x) = \mathsf{P}(a - X \leqslant x) = \mathsf{P}(X \geqslant a - x)$$
$$= 1 - \mathsf{P}(X < a - x) = 1 - F_X((a - x)^-).$$

Since we found that $F_X((a-x)^-) = F_X(a - x^+)$ and the symmetry condition can be equivalently written $F_X(a - x^+) + F_X(a + x) = 1$, which implies $F_X(a + x) = 1 - F_X(a - x^+)$, we conclude $F_{a-X}(x) = F_X(a + x)$ for every $a \in \mathbb{R}$ and thus $a - X \overset{\mathrm{d}}{=} X - a$ as required, see Appendix.

Remark 2.9 (Stylized Facts Part 1) Since prices can be recovered from returns (see Sect. 2.5, Example 2.11), when we face the task of building a model for random stock prices some well known statistical properties of historical returns can be taken into account. Thus, being $R_{t+1} = \ln\left(\frac{S_{t+1}}{S_t}\right)$ the rv modelling log-returns over the time horizon $[t, t + 1]$, we have:

- The probability distribution $\mathsf{P}_{R_{t+1}}$ is not symmetric;
- $\mathsf{P}_{R_{t+1}}$ has fatter tails.

In financial applications, these two features are synonymous of not symmetrically distributed rvs. E.g., if $\mathsf{P}_{R_{t+1}}$ is absolutely continuous with density $f_{R_{t+1}}$, then the latter need not to be an even function.

2.8 Quantile Function

For a given rv X it could be useful to start with a probability level $c \in (0, 1)$ from the y-axis of the corresponding F_X's plot, and obtain possibly a value x in the support.

Definition 2.8 The ***c*th quantile** of the distribution P_X is

$$Q_X(c) := \inf\{x \in \mathbb{R} \mid F_X(x) \geqslant c\}, \quad \text{for every } 0 < c < 1. \tag{2.15}$$

By convention $Q_X(0) = -\infty$. The **quantile function** $Q : (0, 1) \to \mathbb{R}$ is given by formula (2.15) as the probability level c varies.

Sometimes $Q_X(c)$ is written q_c or $F_X^{\leftarrow}(c)$.

Remark 2.10 (Features of the Quantile Function) The behavior of $Q_X(c)$ depends upon the functional form of the CDF, see Figs. 2.4 and 2.5.

- When F_X is strictly increasing and continuous (i.e. 1-to-1) then $Q_X(c)$ is just the inverse function $F_X^{-1}(c)$ of the CDF and we have $Q_X(c) = x$ where x is the unique value in the X's support mapped to $c \in (0, 1)$ by F_X. In this case $\{x \in \mathbb{R} \mid F_X(x) \geqslant c\}$ is the same set as $\{x \in \mathbb{R} \mid x \geqslant F_X^{-1}(c)\}$.

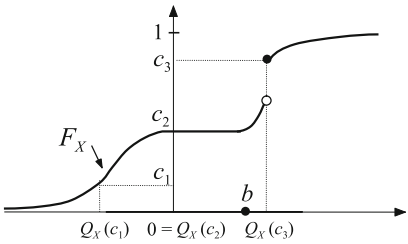


Fig. 2.4 CDF with a flat on $(0, b)$ and a jump

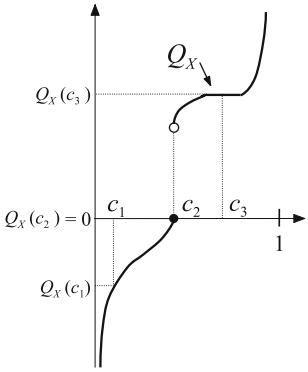


Fig. 2.5 Q_X's graph has a left-continuity point at the F_X's jump. The F_X's flat is now the Q_X's jump

- If the probability level $c \in (0, 1)$ is crossed by the horizontal line $F_X(x) = c$ whose intersection with the $F_X(x)$'s plot is a flat part of the graph, for any x in an open interval (a, b) with finite endpoints, then $Q_X(c)$ is the leftmost point of this interval.
- A value $c \in (0, 1)$ not corresponding to a value of the CDF (i.e. there is no $x \in \mathbb{R}$ s.t. $F_X(x) = c$) yields $Q_X(c)$ as the smallest value in the x-axis mapped to a probability level *at least* equal to c. This is verified when c is in between a jump of F_X.

CDFs and quantiles are linked by construction.

Proposition 2.2 *Let $Q_X(c)$ be the quantile function corresponding to a CDF $F_X(x)$. Then:*

1. $Q_X(F_X(x)) \leqslant x$, *for all* $x \in \mathbb{R}$
2. $F_X(Q_X(c)) \geqslant c$, *for all* $c \in (0, 1)$
3. $Q_X(c) \leqslant x$ *if and only if* $F_X(x) \geqslant c$
4. $Q_X(c)$ *is increasing.*[17]

Remark 2.11 It is worth noting that continuity of F_X implies Q_X is increasing, and vice versa. Moreover, F_X is strictly increasing iff Q_X is continuous; strictly

[17] This means: $c_1 < c_2$ implies $Q_X(c_1) \leqslant Q_X(c_2)$. If the latter inequality is strict, then quantile functions are strictly increasing.

monotonicity of F_X implies $Q_X(F_X(x)) = x$. Eventually, continuity of Q_X implies $F_x(Q_X(c)) = c$.

Moreover, as for $F_X(x)$ the quantile function $Q_X(c)$ determines the distribution of a rv X.

Lemma 2.1 *Let X and Y be two rvs over the same probability space $(\Omega, \mathcal{F}, \mathsf{P})$. Let $Q_X(c)$, $Q_Y(c)$ be their quantile functions. If $Q_X(c) = Q_Y(c)$ for every $c \in (0, 1)$, then X and Y have the same distribution, i.e. $X \stackrel{d}{=} Y$.*

Remark 2.12 There are some interesting interpretation of quantile functions when computed for certain values of $c \in (0, 1)$.

- When F_X is absolutely continuous with density f_X, then a large value $c \in (0, 1)$ of the probability level (e.g. $c = 0.95$ or 95%) with a corresponding positive large value $Q_X(c)$ gives raise to a long *right-tail* of the density. Similarly, a small value $c \in (0, 1)$ of the probability level (e.g. $c = 0.05$ or 5%) with a corresponding negative large value $Q_X(c)$ gives raise to a long *left-tail* of the density.
- The special case $c = \frac{1}{2}$ (50% of probability level) yields the *median* $Q_X(\frac{1}{2})$, also written $\mathsf{med}(X)$.
- $Q_X(\frac{1}{4})$ is called the *first quartile*, and $Q_X(\frac{3}{4})$ is the *third quartile*. Obviously, $\mathsf{med}(X)$ is the second quartile.

The median $\mathsf{med}(X)$ is the number $m \in \mathbb{R}$ satisfying

$$\mathsf{P}(X \leqslant m) \geqslant \tfrac{1}{2} \quad \text{and} \quad \mathsf{P}(X \geqslant m) \geqslant \tfrac{1}{2}. \tag{2.16}$$

Here $m = \mathsf{med}(X)$ is a measure of the center of the distribution P_X, since there is a 50% of probability that X will exceed m. Actually, the median represents a value that splits P_X into two halves. It is not necessarily unique, unless F_X is 1-to-1. Writing $\mathsf{P}(X \geqslant m) = 1 - \mathsf{P}(X < m) \geqslant \frac{1}{2}$ we get $\mathsf{P}(X < m) \leqslant \frac{1}{2}$ which leads to the equivalent condition

$$F_X(m-) \leqslant \tfrac{1}{2} \leqslant F_X(m). \tag{2.17}$$

If F_X is continuous it holds $F_X(m-) = F_X(m) = \frac{1}{2}$. As a result, the c-quantile given by Eq. (2.15) can be equivalently defined as

$$\mathsf{P}(X < Q_X(c)) \leqslant c \leqslant \mathsf{P}(X \leqslant Q_X(c)). \tag{2.18}$$

We conclude this section by noting that for any increasing function $f : \mathbb{R} \to \mathbb{R}$, event $\{X \leqslant x\}$, for a fixed $x \in \mathbb{R}$, is a strict subset of $\{f(X) \leqslant f(x)\}$. For example, let $\Omega = (0, 1]$ and consider the rv $X(\omega) = x$, for every $\omega \in \Omega$, then take $f(X(\omega)) = 2X(\omega)\mathbf{I}_{\{\omega < \frac{1}{4}\}} + \frac{1}{2}\mathbf{I}_{\{\frac{1}{4} \leqslant \omega < \frac{1}{2}\}} + X(\omega)\mathbf{I}_{\{\frac{1}{2} \leqslant \omega \leqslant 1\}}$, and observe that $\{X \leqslant \frac{1}{4}\} = (0, \frac{1}{4}]$, while $\{2X \leqslant \frac{1}{2}\} = (0, \frac{1}{2})$. Only when f is strictly increasing we have $\{X \leqslant x\} =$

$\{f(X) \leqslant f(x)\}$. Moreover, $\{X \leqslant x\}$ and $\{f(X) = f(x)\} \cap \{X > x\}$ are disjoint, then it is easily seen that

$$\mathsf{P}(f(X) \leqslant f(x)) = \mathsf{P}(X \leqslant x) + \mathsf{P}(\{f(X) = f(x)\} \cap \{X > x\}). \tag{2.19}$$

Now, for a CDF it holds $\mathsf{P}(\{F_X(X) = F_X(x)\} \cap \{X > x\}) = 0$ since the underlying event corresponds to a flat of the CFD's graph, then it must have zero probability. As a consequence, for any rv X the set of values $x \in \mathbb{R}$ s.t. $Q_X(F_X(x)) \neq x$ is again a flat and we have:

$$\mathsf{P}(Q_X(F_X(X)) = X) = 1, \tag{2.20}$$

in words $Q_X(F_X(X)) \overset{\text{a.s.}}{=} X$. Eventually:

Proposition 2.3 *Let X be a rv and $f : \mathbb{R} \to \mathbb{R}$ be an increasing and left-continuous function (or strictly increasing instead). Then, the new rv $f(X)$ has quantile function*

$$Q_{f(X)}(c) = f(Q_X(c)), \tag{2.21}$$

for every probability level $c \in (0, 1)$.

2.9 Mixed Random Variables and Distributions

In some financial and actuarial applications, we can define rvs having a probability law characterized by CDFs which are convex combination of discrete and absolutely continuous CDFs. The corresponding rvs are called **mixed rvs**. Before we show the exact construction, it is useful to know that a convex combination of two distinct probability measure is a probability measure too.

Remark 2.13 Given a random experiment modelled via $(\Omega, \mathcal{F})$, let us introduce two different probability functions $\mathsf{P}_1, \mathsf{P}_2$ together with a number $0 \leqslant \alpha \leqslant 1$. The set function

$$\mathsf{P}(A) := \alpha \mathsf{P}_1(A) + (1 - \alpha)\mathsf{P}_2(A), \quad \text{for each } A \in \mathcal{F}$$

is actually a probability function. We only have to show the above $\mathsf{P}(\cdot)$ satisfies Definition 1.3 of Sect. 1.3, i.e. it fulfills properties PA1 and PA2. In fact:

- $\mathsf{P}(\varnothing) := \alpha \mathsf{P}_1(\varnothing) + (1-\alpha)\mathsf{P}_2(\varnothing) = 0$, while $\mathsf{P}(\Omega) := \alpha \mathsf{P}_1(\Omega) + (1-\alpha)\mathsf{P}_2(\Omega)$ is just $\alpha + (1 - \alpha) = 1$;

- for pairwise sequence of events $A_1, A_2, \ldots$ in $\mathcal{F}$ we have

$$\begin{aligned}\mathsf{P}(\cup_{i=1}^{\infty} A_i) &= \alpha \mathsf{P}_1(\cup_{i=1}^{\infty} A_i) + (1-\alpha)\mathsf{P}_2(\cup_{i=1}^{\infty} A_i) \\ &= \alpha \sum_{i=1}^{\infty} \mathsf{P}_1(A_i) + \sum_{i=1}^{\infty} (1-\alpha)\mathsf{P}_2(A_i) \\ &= \sum_{i=1}^{\infty} \Big[\alpha \mathsf{P}_1(A_i) + (1-\alpha)\mathsf{P}_2(A_i)\Big] = \sum_{i=1}^{\infty} \mathsf{P}(A_i).\end{aligned}$$

Definition 2.9 A rv X is called **mixed** if its CDF has a finite[18] number of jump discontinuities at $x = x_1, \ldots, x_n$, where the masses are $p_i = \mathsf{P}(X = x_i)$ for $i = 1, \ldots, n$ with $\sum_{i=1}^{n} p_i \leqslant 1$. Then:

$$F_X(x) = \alpha F_{\mathrm{d}}(x) + (1-\alpha) F_{\mathrm{c}}(x), \quad 0 \leqslant \alpha \leqslant 1, \tag{2.22}$$

where F_{d} is the **discrete part** and F_{c} is the **continuous part**. The former can be given as a linear combination of unit step functions, $F_{\mathrm{d}}(x) = \sum_{i=1}^{n} \frac{p_i}{\alpha} u(x - x_i)$. The latter can be given through a density $f_{\mathrm{c}}(x) \geqslant 0$.

If $\alpha = 0$ the rv is continuous; if $\alpha = 1$ it is discrete. Since F_X corresponds 1-to-1 to P, then F_{d} and F_{c} correspond[19] 1-to-1 to P_{d} and P_{c} and $\mathsf{P} = \alpha \mathsf{P}_{\mathrm{d}} + (1-\alpha)\mathsf{P}_{\mathrm{c}}$. The following is crucial in the mixing procedure.

- The continuous part can be recovered as $F_{\mathrm{c}}(x) = \frac{F_X(x) - \sum_{i=1}^{n} p_i u(x - x_i)}{1 - \sum_{i=1}^{n} p_i}$.
- α in Definition 2.9 is equal to $\sum_{i=1}^{n} p_i$.
- The weights of the discrete part in Definition 2.9 are given by $\frac{p_i}{\alpha}$ s.t. $\sum_{i=1}^{n} \frac{p_i}{\alpha} = 1$.

Here the formal derivation. First, $F_X(x) - \sum_{i=1}^{n} p_i u(x - x_i)$ so we remove the jump discontinuities from the continuous part of the mixed CDF. The resulting function is not yet a CDF:

$$\begin{aligned}\lim_{x \to \infty} \Big[F_X(x) - \sum_{i=1}^{n} p_i u(x - x_i)\Big] &= \underbrace{\lim_{x \to \infty} F_X(x)}_{=1} - \sum_{i=1}^{n} p_i \Big[\underbrace{\lim_{x \to \infty} u(x - x_i)}_{=1}\Big] \\ &= 1 - \alpha \leqslant 1.\end{aligned}$$

[18] We can extend this definition to a countable number of discontinuities with minor modifications.

[19] Thus the law P_X is split into a discrete law $\mathsf{P}_{X,1}$ and an absolutely continuous law $\mathsf{P}_{X,2}$, to which correspond a set of masses and a density, respectively.

Thus we need to scale the above function dividing by $1 - \sum_{i=1}^{n} p_i = 1 - \alpha$. Rearranging the formula for $F_{\mathrm{c}}(x)$ we get

$$\boxed{F_X(x) = \sum_{i=1}^{n} p_i u(x - x_i) + (1 - \alpha) F_{\mathrm{c}}(x)},$$

and writing $\sum_{i=1}^{n} p_i u(x - x_i) = \alpha \sum_{i=1}^{n} \frac{p_i}{\alpha} u(x - x_i)$ we finally obtain expression (2.22). Now we introduce the generalized[20] *delta function* centered at $a \in \mathbb{R}$:

$$\delta_a(x) = \begin{cases} 0, & \text{if } x \neq a \\ \infty, & \text{if } x = a \end{cases}, \qquad \int_{-\infty}^{\infty} \delta_a(x) \mathrm{d}x = 1, \qquad \tfrac{\mathrm{d}}{\mathrm{d}x} u(x - a) = \delta_a(x).$$

Observe that we use the same symbol as for the Dirac delta measure, see Sect. 2.4. With this in mind we can *formally* differentiate the CDF and get the generalized density of the mixed distribution in formula (2.22):

$$\begin{aligned} f_X(x) = \tfrac{\mathrm{d}}{\mathrm{d}x} F_X(x) &= \alpha \tfrac{\mathrm{d}}{\mathrm{d}x} F_{\mathrm{d}}(x) + (1 - \alpha) \tfrac{\mathrm{d}}{\mathrm{d}x} F_{\mathrm{c}}(x) \\ &= \alpha \sum_{i=1}^{n} \frac{p_i}{\alpha} \underbrace{\tfrac{\mathrm{d}}{\mathrm{d}x} u(x - x_i)}_{=\delta_{x_i}(x)} + (1 - \alpha) f_{\mathrm{c}}(x) \\ &= \sum_{i=1}^{n} p_i \delta_{x_i}(x) + (1 - \alpha) f_{\mathrm{c}}(x) \end{aligned}$$

Hence, we may always write the expression

$$\boxed{f_X(x) = \sum_{i=1}^{n} p_i \delta_{x_i}(x) + (1 - \alpha) f_{\mathrm{c}}(x)}$$

as the 'density' of a mixed rv. In fact, by the defining properties of the delta function we have:

- $\sum_{i=1}^{n} p_i \delta_{x_i}(x) + (1 - \alpha) f_{\mathrm{c}}(x) \geqslant 0$ for every $x \in \mathbb{R}$;

[20] In real analysis, there could exist functions whose definition is not standard. These are called **generalized functions**. The rules of integration and differentiation are different, and require an excursion to functional analysis concerning vector spaces of *test functions* and the so called *distributions* (which are not defined in probabilistic terms).

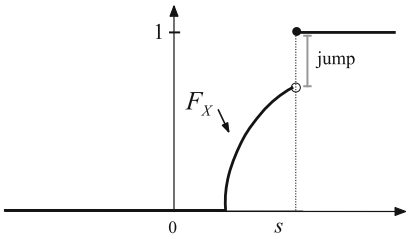


Fig. 2.6 CDF's graph of a mixed distribution

- $\int_{-\infty}^{\infty} \left[\sum_{i=1}^{n} p_i \delta_{x_i}(x) + (1-\alpha) f_c(x) \right] \mathrm{d}x$ is equal to

$$\sum_{i=1}^{n} p_i \left[\underbrace{\int_{-\infty}^{\infty} \delta_{x_i}(x)\mathrm{d}x}_{=1} \right] + (1-\alpha) \underbrace{\int_{-\infty}^{\infty} f_c(x)\mathrm{d}x}_{=1}$$

  which in turn is given by $\sum_{i=1}^{n} p_i + (1-\alpha) = \alpha + (1-\alpha) = 1$.

It is worth noting that the area under the graph of f_c over a subset excluding the set of points x_i where δ_{x_i} are concentrated is less than 1: in fact it is $1 - \sum_{i=1}^{n} p_i$. Each $p_i \delta_{x_i}(x)$ in the expression $\sum_{i=1}^{n} p_i \delta_{x_i}(x) + (1-\alpha) f_c(x)$ can be interpreted as the probability $p_i = \mathsf{P}(X = x_i)$ concentrated at $x_i \in C$ where the jump discontinuity of the corresponding CDF has infinity derivative $\frac{\mathrm{d}}{\mathrm{d}x} F_X(x)\big|_{x=x_i} = \delta_{x_i}(x_i) = \frac{\mathrm{d}}{\mathrm{d}x} u(x - x_i)\big|_{x=x_i} = \infty$.

Example 2.14 (Insurance Model) Suppose an individual has purchased an insurance contract against a random loss[21] for which the rv X represents the monetary reimbursement with *maximum* payment s denominated in euros. A candidate CDF for this rv is:

$$F_X(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1 - \mathrm{e}^{-\lambda x}, & \text{if } 0 \leqslant x < s \\ 1, & \text{if } x \geqslant s \end{cases}$$

where $\lambda > 0$ and $s > 0$ are parameters. Interpretation: Random losses are measured with positive values, see Fig. 2.6. Since the reimbursement X equals the loss occurred except when its amount is at least € s, then X is zero for negative losses while it assumes values in $[0, s)$ for positive losses less than € s. If losses exceed the latter level then $X = s$. It is easily verified that $F_X(x)$ is a CDF:

- it is non decreasing, $x_1 < x_2$ implies $F_X(x_1) \leqslant F_X(x_2)$ since the CDF is zero up to $x < 0$, is strictly increasing over $[0, s)$ and is 1 for $x \geqslant s$;
- it is continuous up to $x < s$ and has a jump discontinuity at $x = s$, hence it is right continuous;
- $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$ by construction.

[21] In case of an unpredictable loss, as for example due to a car accident.

We claim F_X corresponding to the mixed rv X is of the form (2.22) as in Definition (2.9). The only jump discontinuity is

$$\mathsf{P}(X = s) = F_X(s) - F_X(s^-) = 1 - \left(1 - \mathrm{e}^{-\lambda s}\right) = \mathrm{e}^{-\lambda s} = \alpha.$$

The CDF's continuous part $F_{\mathrm{c}}(x) = \frac{F_X(x) - \alpha u(x-s)}{1-\alpha}$ is given by

$$F_{\mathrm{c}}(x) = \begin{cases} 0, & \text{if } x < 0,\ u(x-s) = 0,\ F_X(x) = 0 \\ \frac{1-\mathrm{e}^{-\lambda x}}{1-\alpha}, & \text{if } 0 \leqslant x < s,\ u(x-s) = 0,\ F_X(x) = 1 - \mathrm{e}^{-\lambda x} \\ 1, & \text{if } x \geqslant s,\ u(x-s) = 1,\ F_X(x) = 1. \end{cases}$$

Now $f_{\mathrm{c}}(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_{\mathrm{c}}(x) = \frac{\lambda \mathrm{e}^{-\lambda x}}{1-\alpha}$, for $0 \leqslant x < s$, and the derivative (viz. the density) is zero otherwise. Hence we have:

$$F_X(x) = \alpha u(x-s) + (1-\alpha) F_{\mathrm{c}}(x) = \mathrm{e}^{-\lambda s} u(x-s) + \left(1 - \mathrm{e}^{-\lambda x}\right),$$

with generalized density $f_X(x) = \mathrm{e}^{-\lambda s} \delta_s(x) + \lambda \mathrm{e}^{-\lambda x}$.

More generally, a **mixture** is a rv with CDF given in the following form: $F_X = \sum_{i=1}^n \alpha_i F_{X_i}$, for rvs X_i and weights s.t. $\sum_{i=1}^n \alpha_i = 1$, where the CDFs F_{X_i} can be of different type.

Remark 2.14 In a far general setting (viz. measure theory) it can be shown that any probability law P_X of a rv (acting as a probability on $\mathbb{R}$) has a unique representation

$$\mathsf{P}_X = \alpha_1 \mathsf{P}_{\mathrm{d}} + \alpha_2 \mathsf{P}_{\mathrm{c}} + \alpha_3 \mathsf{P}_{\mathrm{s}}, \quad \alpha_1, \alpha_2, \alpha_3 \geqslant 0, \quad \alpha_1 + \alpha_2 + \alpha_3 = 1,$$

where the probability P_{s} is called **singular**, i.e. there exists a CDF $F_{\mathrm{s}}(x)$ such that its derivative $F'_{\mathrm{s}}(x) = 0$ except at those x forming a set of measure zero. Therefore, any discrete rv has a distribution which is singular, since the derivative of its CDF exists and is zero except at each jump discontinuity (whose set has measure zero, and where the derivative does not exist). However there are *singular continuous* probability distributions which corresponds 1-to-1 to a continuous CDF that is not discrete but nevertheless has no density function.

Remark 2.15 For a mixed rv X with countably many jumps over the set $\{x_1, x_2, \ldots\}$, we can succinctly write its law as:

$$\mathsf{P}_X = \alpha \sum_{i=1}^n \delta_{x_i} + (1-\alpha) \mathsf{P}_Y,$$

where Y is an (absolutely) continuous rv.[22] This is again a set function defined for suitable $B \subset \mathbb{R}$.

[22] P_Y has CDF s.t. $F'_Y(x) = f_Y(x)$, at all points $x \in \mathbb{R}$ except a finite number.

Another kind of mixture, say of a rv X's CDF, with mixing weights given by the values that another rv, say Y, might assume is defined as follows:

$$F_X(x) = \int_{-\infty}^{\infty} H(x; y)\mathrm{d}G(y), \tag{2.23}$$

where $H(x; y) = \mathsf{P}(X \leqslant x; Y = y)$ denotes the CDF[23] of X indexed by the real values of Y, and $G(y)$ is just the CDF of Y, called the *mixing distribution*, governing in a random way the weighting scheme which entails $F_X(x)$ as a CDF of the same type of $H(x; y)$. The integral in Eq. (2.23) is of the Stieltjes type. We can write down the density or mass function as

$$f_X(x) = \int_{-\infty}^{\infty} h(x; y)\mathrm{d}G(y), \tag{2.24}$$

where $h(x; y)$ is the corresponding density or mass function in the mixing scheme. In fact, integrating $f_X(x)$, i.e.,

$$\begin{aligned}\int_{-\infty}^{x} f_X(u) &= \int_{-\infty}^{x} \left[\int_{-\infty}^{\infty} h(u; y)\mathrm{d}G(y)\right]\mathrm{d}u \\ &= \int_{-\infty}^{\infty} \underbrace{\left[\int_{-\infty}^{x} h(u; y)\mathrm{d}u\right]}_{=H(x;y)}\mathrm{d}G(y).\end{aligned}$$

The exchange in the order of integration is permitted by Fubini's theorem, see Appendix in Chap. 3 and Appendix I. When F_X and H are both continuous we can write formulas (2.23) and (2.24) by replacing $\mathrm{d}G(y)$ with $g(y)\mathrm{d}y$, where $g(y)$ is the density function of the mixing CDF $G(y)$. In the discrete case we replace $g(y)\mathrm{d}y$ with $g(y)$ and the integral $\int_{-\infty}^{\infty}$ with the summation $\sum_{y\in\mathcal{Y}}$, where $\mathcal{Y}$ is the support of Y.

2.10 Exercises

2.1 (One-Sided Limits of a CDF) Let F_X be the CDF of a rv X. Prove the following:

(e) $\mathsf{P}(X > a) = 1 - F_X(a)$ for every $a \in \mathbb{R}$.
(f) $\mathsf{P}(X = a) = F_X(a) - F_X(a^-)$ for every $a \in \mathbb{R}$.

[23] It can be defined in term of conditional CDF, see Chap. 6.

2.2 (Continuous Distribution) Let $f_X(x)$ be a continuous density function. Consider the interval $[x - \frac{h}{2}, x + \frac{h}{2}]$ with length $h > 0$. Compute the probability $\mathsf{P}(X \in [x - \frac{h}{2}, x + \frac{h}{2}])$.

2.3 Show that the indicator $\mathbf{I}_A$ is a rv iff $A \in \mathcal{F}$.

2.4 Show that the composition $\mathbf{I}_B \circ X : \Omega \to \mathbb{R}$ is the indicator rv $\mathbf{I}_{\{X \in B\}}$, for a suitable $B \subset \mathbb{R}$ and a rv over $(\Omega, \mathcal{F})$.

2.5 (Borel Sets) Recall that the Borel σ-algebra is the smallest σ-algebra containing all the finite and infinite intervals of $\mathbb{R}$. Show that $\cup_{i=1}^{\infty}(a, b - \frac{1}{i}] = (a, b)$ and $\cap_{i=1}^{\infty}(a - \frac{1}{i}, b] = [a, b]$, for $a < b$, i.e. open and closed interval are Borel sets.

2.6 Let P_0 be defined by formula (2.26), and let $\mathcal{G}$ be the class containing all finite disjoint unions of $(a, b]$ together with $\varnothing$. Show that:

(1) $\mathbb{R} \in \mathcal{G}$;
(2) $\mathsf{P}_0(\varnothing) = 0$;
(3) $\mathsf{P}_0(\mathbb{R}) = 1$.

2.7 (Dirac Delta Measure and Probability) Recall that a Dirac delta measure can be used to model a probability distribution over $\mathbb{R}$ corresponding in a 1-to-1 fashion to the law P_X of a given rv X. Let $\delta_x(B)$ be the Dirac delta measure whose definition we recall for the reader's convenience:

$$\delta_x(B) := \begin{cases} 1, \text{ if } x \in B \\ 0, \text{ if } x \notin B \end{cases} \tag{2.25}$$

Equation (2.25) defines a function $\delta_x(\,\cdot\,) : \mathcal{B} \to \mathbb{R}$. Prove the following: $\mathsf{P}_X(B) := \sum_{i=1}^{\infty} p_i \delta_{x_i}(B)$ is the probability distribution of a discrete rv X with countable many values $\{x_1, x_2, \ldots\}$ and masses $\mathsf{P}(X = x_i) = p_i$.

Remark 2.16 We have seen that a rv $X(\omega) = k$, for some $k \in \mathbb{R}$, behaves like a constant. To see this more formally observe that the CDF is $F_X(x) = 1$ for every $x \geqslant k$, and $F_X(x) = 0$ otherwise. We have:

$$F_X(k^-) = \mathsf{P}(X < k) = \lim_{h \downarrow 0} \mathsf{P}(X \leqslant k - h) = 0,$$

while $F_X(k) = \mathsf{P}(X \leqslant k) = 1$. This implies

$$\mathsf{P}(X = k) = F_X(k) - F_X(k^-) = 1$$

and we can safely claim that X acts like a constant. Its probability distribution is characterized by a Dirac-delta $\delta_k(B)$, for any reasonable subset of real numbers $B \subset \mathbb{R}$.

2.8 (Positive and Negative Parts) Let X be a rv describing the daily rate of return of a traded stock. Find the expression of its magnitude.

Remark 2.17 Any simple rv as defined by Eq. (2.6) is bounded, i.e. we can find $k > 0$ s.t. $|X(\omega)| \leqslant k$, thus $-k \leqslant X(\omega) \leqslant k$ for all scenarios $\omega \in \Omega$.

2.9 Let $X > 0$ be a positive rv describing the future price of a stock or the future loss of an insurance claim. Prove that, for all $s, c \in \mathbb{R}$:

- $\max\{X, s\} \pm c = \max\{X \pm c, s \pm c\}$;
- $(X - s)^+ + c = \max\{X - s + c, c\}$;
- $(X - s)^+ - (s - X)^+ = X - s$.

2.10 Let F_X, F_Y be two CDFs s.t. $X \leqslant_{\text{st}} Y$. Show this is equivalent to $Q_X(F_Y(x)) \leqslant x$, for every $x \in \mathbb{R}$.

2.11 (Symmetric Distributions) Write the equivalent definition (2.14) of symmetry for discrete and continuous rvs.

2.12 Let X be a discrete rv with symmetric distribution around zero. Prove the new rv $Y = -X$ has the same distribution of X. Provide a prove also for continuous X.

2.13 (Mixed CDF) Show that the function

$$F(x) := \alpha F_X(x) + (1 - \alpha) F_Y(x), \quad \text{for every } x \in \mathbb{R}, \quad 0 \leqslant \alpha \leqslant 1$$

is a CDF, given two CDFs F_X, F_Y.

Appendix

Some Details on CDF's Right Continuity

It is worth recalling that continuity of a real-valued function might be equivalently given by $\epsilon - \delta$ definition or by sequential characterization. In the first case, for $F(x)$ being continuous at a point x_0 lying in its domain it is required that

$$\text{for each } \epsilon > 0 \text{ there exists } \delta > 0 \text{ s.t. } |x - x_0| < \delta \Longrightarrow |F(x) - F(x_0)| < \epsilon,$$

written $\lim_{x \to x_0} F(x) = F(x_0)$, i.e. for any x *near* to x_0 also $F(x)$ is *near* to $F(x_0)$. All this applies to $F \equiv F_X$. We see more formally why the sequential characterization of CDF's right-continuity is equivalent to the $\epsilon - \delta$ definition. Given a decreasing sequence $x_1 \geqslant x_2 \geqslant \cdots$ of values in the domain of F_X, s.t. $x_i \downarrow x$ for $i \to \infty$ and $x \neq x_i$, we know that $F_X(x^+) = \lim_{i \to \infty} F_X(x + \frac{1}{i}) = F_X(x)$ with $x_i = x + \frac{1}{i}$. Choosing another value y in the CDF's domain with $x < y < x_i$, then $F_X(x) \leqslant F_X(y) \leqslant F_X(x_i)$ because F_X is nondecreasing. Since for $x_i \downarrow x$ we have $F_X(x_i) \downarrow F_X(x)$, then $|F_X(x_i) - F_X(x)| \to 0$ and also $|F_X(y) - F_X(x)| \to 0$ provided that y decreases to x, i.e. $y \downarrow x$. This is the required $\epsilon - \delta$ definition of

right-continuity, alternatively given by setting $y = x + h$ for $h > 0$ and writing as usual $F_X(x) = \lim_{h \downarrow 0} F_X(x + h)$.

Five Properties of a CDF

Proof of Theorem 2.1 We prove the implication $\Longrightarrow$, the other *direction* being a much subtle result from measure theory.

(a) Take two real numbers $a < b$ and observe that $\{X \leqslant a\} \subset \{X \leqslant b\}$, thus $F_X(a) = \mathsf{P}(X \leqslant a)$ is less than or equal to $F_X(b) = \mathsf{P}(X \leqslant b)$ by the monotonicity of $\mathsf{P}(\cdot)$ for increasing events.

(b) Recall that $\mathsf{P}(\cdot)$ is continuous for a decreasing sequence of events with given intersection. Now take $A_i = \{X \leqslant x + \frac{1}{i}\}$ and observe that $\cap_{i=1}^{\infty}\{X \leqslant x + \frac{1}{i}\}$ is just $\{X \leqslant x\}$. Then

$$F_X(x) = \mathsf{P}(X \leqslant x) = \lim_{i \to \infty} \mathsf{P}(X \leqslant x + \tfrac{1}{i})$$
$$= \lim_{i \to \infty} F_X(x + \tfrac{1}{i}),$$

and since the CDF is nondecreasing the last limit exists and must be $F_X(x)$.

The following proof of the other implication can be skipped, and require the notion of trace σ-algebra (see Exercise 1.16 in Chap. 1). Thus, we are going to prove that a map $F : \mathbb{R} \to [0, 1]$ satisfying properties (a)–(c) of Theorem 2.1 is the CDF of the probability $\mathsf{P}_{\mathbf{X}}(B)$ induced by some rv X for every Borel set $B \in \mathcal{B}$, i.e., $F \equiv F_X$. Let X be a rv defined on a probability space $(\Omega, \mathcal{F}, \mathsf{P})$ where $\Omega = (0, 1]$, $\mathcal{F} = \mathcal{B}_{(0,1]}$ and $\mathsf{P} = m$, the Lebesgue measure restricted to the unit interval. For each scenario $\omega \in (0, 1]$ define the map $X(\omega) := \sup\{z \mid F(z) < \omega\}$. We first verify that X is indeed a rv. To any $x \notin \{z \mid F(z) < \omega\}$ will correspond $\omega \leqslant F(x)$. Taking the supremum requires $X(\omega) \leqslant x$. By assumption F is right-continuous at x implying that we can find $h > 0$ such that $F(x+h) \leqslant F(x) < \omega$ but also $x < x+h \leqslant X(\omega)$, since X is the supremum of the set $\{z \mid F(z) < \omega\}$, even if $z = x+h$. Putting things together we have

$$\{\omega \mid X(\omega) \leqslant x\} = \{\omega \mid \omega \leqslant F(x)\},$$

i.e. the inverse image $\{X \leqslant x\}$ is a Borel set in $\mathcal{B}_{(0,1]}$. Eventually we can write $\mathsf{P}(X \leqslant x) = m((0, F(x)]) = F(x)$ and the proof is complete. ■

Checking Probabilities in Table 2.1

Number # 1 is Definition 2.1 and # 3 is Theorem 2.1(d). To verify # 2 we consider the increasing sequence of events $\{X \leqslant a - \frac{1}{i}\}$ for $i = 1, 2, \ldots$ with union $\cup_{i=1}^{\infty}\{X \leqslant a - \frac{1}{i}\} = \{X < a\}$. By continuity of $\mathsf{P}(\cdot)$ we have

$$\mathsf{P}(X < a) = \lim_{i\to\infty} \mathsf{P}(X \leqslant a - \tfrac{1}{i}) = \lim_{i\to\infty} F_X(a - \tfrac{1}{i}) = F_X(a^-).$$

To prove # 5 we use the decreasing sequence of events $\{a - \frac{1}{i} < X \leqslant a\}$ with intersection $\cap_{i=1}^{\infty}\{a - \frac{1}{i} < X \leqslant a\} = \{a \leqslant X \leqslant a\} = \{X = a\}$. Again, using the continuity of $\mathsf{P}(\cdot)$ and Theorem 2.1(d) we have

$$\begin{aligned}\mathsf{P}(X = a) &= \lim_{i\to\infty} \mathsf{P}(a - \tfrac{1}{i} < X \leqslant a)\\ &= \lim_{i\to\infty} \left[F_X(a) - F_X(a - \tfrac{1}{i})\right] = F_X(a) - F_X(a^-).\end{aligned}$$

Now we can prove # 9 by considering $\{a < X < b\} = \{a < X \leqslant b\} \setminus \{X = b\}$ and then by Pr3 and the previous result we have

$$\mathsf{P}(a < X < b) = F_X(b) - F_X(a) - \left[F_X(b) - F_X(b^-)\right] = F_X(b^-) - F_X(a).$$

Number # 4 is based on the decreasing sequence of events $\{a - \frac{1}{i} < X \leqslant b\}$ with intersection $\cap_{i=1}^{\infty}\{a - \frac{1}{i} < X \leqslant b\} = \{a \leqslant X \leqslant b\}$ and the continuity of the probability measure:

$$\begin{aligned}\mathsf{P}(a \leqslant X \leqslant b) &= \lim_{i\to\infty} \mathsf{P}(a - \tfrac{1}{i} < X \leqslant b) = \lim_{i\to\infty} \left[F_X(b) - F_X(a - \tfrac{1}{i})\right]\\ &= F_X(b) - F_X(a^-).\end{aligned}$$

Number # 6 is (e) of Exercise 2.1. To prove # 7 it suffices to note that $\{X \geqslant b\} = \{X = b\} \cup \{X > b\}$, so applying $\mathsf{P}(\cdot)$ together with #5 and # 6 we have

$$\mathsf{P}(X \geqslant b) = \mathsf{P}(X = b) + \mathsf{P}(X > b) = F_X(b) - F_X(b^-) + 1 - F_X(b) = 1 - F_X(b^-).$$

Finally we prove # 8 considering the increasing sequence of events $\{a \leqslant X \leqslant b - \frac{1}{i}\}$ with union $\{a \leqslant X < b\}$, thus thanks to # 4:

$$\begin{aligned}\mathsf{P}(a \leqslant X < b) &= \lim_{i\to\infty} \mathsf{P}(a \leqslant X \leqslant b - \tfrac{1}{i}) = \lim_{i\to\infty} \left[F_X(b - \tfrac{1}{i}) - F_X(a^-)\right]\\ &= F_X(b^-) - F_X(a^-).\end{aligned}$$

Details on Discrete Distributions

Proof of Theorem 2.2 (Identification) We know that X is discrete if its support (the set where the CDF F_X is strictly increasing) is $C = \{x_1, x_2, \ldots\}$. Since by construction each fixed $\omega \in \Omega$ which belongs to $\{X = x_i\}$ cannot be in another $\{X = x_j\}$, for $i \neq j$, the sample space is partitioned by all these events:

$$\{X = x_1\} \cup \{X = x_2\} \cup \cdots = \Omega.$$

As byproduct we have (pairwise disjoint events)

$$1 = \mathsf{P}(\Omega) = \mathsf{P}\big(\cup_i \{X = x_i\}\big) = \mathsf{P}(X = x_1) + \mathsf{P}(X = x_2) + \cdots = \mathsf{P}(X \in C).$$

■

Details on Continuous Distributions

Proof of Proposition 2.1 If X has a continuous probability distribution, then as a consequence of Definition 2.1 its P_X is given in term of a density function s.t. $\mathsf{P}(-i \leqslant X \leqslant i) = \mathsf{P}(-i < X \leqslant i) = \int_{-i}^{i} f_X(x)\mathrm{d}x$ for some integer $i \in \mathbb{N}$. Thus, $\{-i \leqslant X \leqslant i\}$ is an increasing sequence of events and by continuity of $\mathsf{P}(\,\cdot\,)$ we have $\lim_{i\to\infty} \mathsf{P}(-i \leqslant X \leqslant i) = \mathsf{P}(-\infty \leqslant X \leqslant \infty) = \int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1$. Moreover, $F_X(x) = \mathsf{P}(X \leqslant x)$ can be written as $\mathsf{P}(-\infty < X \leqslant x)$ which in turn is equal to $\lim_{i\to\infty} \int_{-i}^{x} f_X(u)\mathrm{d}u$ hence (2.2) holds true. Conversely, if $\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1$ then by (2.2) the nonnegative function f_X delivery F_X as a CDF:

- it is nondecreasing, $F_X(x_1) \leqslant F_X(x_2)$ for reals $x_1 < x_2$;
- it is continuous, thus also right-continuous;
- $F_X(\infty) = 1$ and $F_X(-\infty) = \int_{-\infty}^{\infty} F_X(x)\mathrm{d}x = 0$;
- the probability of $\{a < X \leqslant b\}$ is $F_X(b) - F_X(a)$.

■

Probability Measure on $\mathbb{R}$

To begin with, let $F : \mathbb{R} \to [0, 1]$ be a nondecreasing, right-continuous functions for which it is not necessary that $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. Such an F is called **càdlàg**, the French acronym of **c**ontinue **à d**roite (continuous on the

right), **l**imité **à g**auche (with finite left-hand limit).[24] Next, collect all finite disjoint unions of intervals $(a, b]$ together with $\varnothing$ into a class $\mathcal{G}$, where $-\infty \leqslant a < b < \infty$ and by convention $(a, \infty) = (a, \infty]$. Thus a nonempty $A \in \mathcal{G}$ may be written as $\cup_{i=1}^{n}(a_i, b_i]$ with $(a_i, b_i] \cap (a_j, b_j] = \varnothing$ for every $i \neq j$. Now, define the set function $\mathsf{P}_0(A) : \mathcal{G} \to [0, 1]$ by

$$\mathsf{P}_0(A) := \sum_{k=1}^{n} F(b_i) - F(a_i), \tag{2.26}$$

which is **finitely additive**, i.e. $\mathsf{P}_0(\cup_{i=1}^{m} A_i) = \sum_{i=1}^{m} \mathsf{P}_0(A_i)$ for every $A_i \in \mathcal{G}$. It can be shown that P_0 in formula (2.26) is also countable additive on $\mathcal{G}$ (see Definition 1.3, axiom PA2, in Sect. 1.3). To get a probability measure we need P_0 to be extended on a σ-algebra $\mathcal{F}$ containing $\mathcal{G}$: it suffices to perform countable many unions, intersections, complements of sets A from $\mathcal{G}$ to get $\mathcal{F} = \sigma(\mathcal{G})$, the σ-algebra generated by the class $\mathcal{G}$. An appeal to the following result entails the desired extension denoted Q.

Theorem 2.4 (Carathéodory) *Suppose $\mathcal{G}$ is an algebra of subsets of $\mathbb{R}$ and $\mathsf{P}_0 : \mathcal{G} \to [0, 1]$ is a countable additive set function. Then, $\exists$ a measure Q on the σ-algebra $\sigma(\mathcal{G})$ generated by $\mathcal{G}$ such that they agree, $\mathsf{Q}(A) = \mathsf{P}_0(A)$, $\forall A \in \mathcal{G}$. If P_0 is σ-finite, then Q is its unique extension.*

It is worth noting $\mathcal{G}$ is defined to be an *algebra*, i.e. a collection of subsets from $\mathbb{R}$ such that properties (S1) and (S2) of a σ-algebra hold but property (S3) becomes:

(S3') For any *finite* sequence $A_1, A_2, \ldots, A_n$ of events belonging to $\mathcal{G}$ also $\cup_{i=1}^{n} A_i \in \mathcal{G}$.

In general, an algebra is smaller than a σ-algebra and any σ-algebra is also an algebra, but the converse is not true: an algebra may contain countable sequences of sets $(A_i)_{i \in \mathbb{N}}$ but there is not guarantee that their union $\cup_{i=1}^{\infty} A_i$ is in here. Theorem 2.4 is a general result from measure theory: in our application Q is understood as a probability. Here $\sigma(\mathcal{G})$ deserves the special name of **Borel σ-algebra**, usually denoted by $\mathcal{B}$; its members are called **Borel sets** . Observe that $\mathsf{P}_0(\varnothing) = \mathsf{Q}(\varnothing) = 0$.

Remark 2.18 For (a premeasure P_0 or) a measure Q, the term σ-finite means that $\mathsf{Q}(A_i) < \infty$ for any covering $\cup_{i=1}^{\infty} A_i = \Omega$, i.e. $A_i \subset \Omega$ for all $i \in \mathbb{N}$. Any probability Q is a *finite measure*, i.e. $\mathsf{Q}(\Omega) < \infty$, and this in turn implies σ-finiteness.

Remark 2.19 Our construction of a probability measure Q on the *real measurable space* $(\mathbb{R}, \mathcal{B})$ uses the concept of a generated σ-algebra $\sigma(\mathcal{G})$: given $\mathcal{G} \subset 2^{\mathbb{R}}$ the generated σ-algebra always exists and equals the intersection of all σ-algebras

[24] If in addition $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$ we call the càdlàg F a distribution function.

containing $\mathcal{G}$, i.e. $\sigma(\mathcal{G})$ is the smallest σ-algebra containing $\mathcal{G}$. If $\mathcal{G}$ and $\mathcal{D}$ are different families of subsets in $\mathbb{R}$ and $\mathcal{G} \subset \sigma(\mathcal{D})$, then $\sigma(\mathcal{G}) \subset \sigma(\mathcal{D})$.

For the Curious

The class $\mathcal{G}$ containing all finite disjoint unions of $(a, b]$ together with $\varnothing$ is an algebra but not a σ-algebra. More formally:

1. Every $A = \cup_{k=1}^{m}(a_k, b_k] \in \mathcal{G}$ is itself of the form $(\ \ ]$ iff $b_k = a_{k+1}$ for $k = 1, \ldots, m-1$. Moreover, A^c is $(-\infty, a_1] \cup (b_1, a_2] \cup \cdots \cup (b_{m-1}, a_m] \cup (b_m, \infty]$ and then belong to $\mathcal{G}$.
2. If $B = (c_1, d_1] \cup \cdots \cup (c_n, d_n] \in \mathcal{G}$, then $A \cup B \in \mathcal{G}$.
3. $A \cap B = \cup_{i=1}^{m} \cup_{j=1}^{n} (a_i, b_i] \cap (c_j, d_j]$ is made of finite disjoint unions, thus $A \cap B \in \mathcal{G}$; intersections can be $\varnothing$ or $(\ \ ]$.
4. Taking $A_n = (0, 1 - \frac{1}{n}]$, each $n \in \mathbb{N}$, we have $\cup_{n=1}^{\infty} A_n = (0, 1)$ is not in $\mathcal{G}$; $\cap_{n=1}^{\infty}(a - \frac{1}{n}, a] = \{a\}$ and neither singletons are in $\mathcal{G}$.

Items 1–3 give properties (S1),(S2), but 4 shows that $\mathcal{G}$ may not contain countable unions (and also intersections). Though, taking unions and complements of elements from $\mathcal{G}$, uncountably often, we generate $\mathcal{B}$. Observe that singletons $\{a\}$ belongs to $\mathcal{B}$ by item 4 above. The Borel σ-algebra can be generated in different ways.

- Open finite intervals generate $\mathcal{B}$, since taking $c = \frac{b-a}{2}$ we have $(a, b) = \cup_{i=1}^{\infty}(a, b - \frac{c}{i}]$ as a countable union of elements from $\mathcal{G}$.
- Open subsets[25] of $\mathbb{R}$ generate $\mathcal{B}$. Indeed, every open set can be written as countable union $A = \cup_{i=1}^{\infty}(a_i, b_i)$. Conversely, we have $(a, b] = \cap_{i=1}^{\infty}(a, b + \frac{1}{i})$. In fact, the standard definition of $\mathcal{B}$ is based on open sets like $(-\infty, a) \cup (b, \infty)$ which are not open finite intervals.
- Half-rays $(-\infty, x]$ generate $\mathcal{B}$ for every $x \in \mathbb{R}$.

Actually, $\mathcal{B}$ can be generated by closed sets, see Exercise 2.5. Now we show that the family $\mathcal{D}$ of half-rays together with $\varnothing$ generate the Borel σ-algebra. Clearly $(-\infty, x] = (x, \infty)^c$, for all $x \in \mathbb{R}$. Then intervals $(-\infty, x]$ are contained in $\mathcal{B}$, as complements of open finite intervals. By assumption they generate $\sigma(\mathcal{G})$ and then, see Remark 2.19, we have $\sigma(\mathcal{G}) \subset \mathcal{B}$. It suffices to show the reverse inclusion. Intervals

$$(x, y] = (-\infty, y] \cap (-\infty, x]^c \quad \text{for all } x < y$$

are contained in $\sigma(\mathcal{G})$. Moreover, for any open interval we can write $(a, b) = \cup_{i=1}^{\infty}(a, b - \frac{c}{i}]$ with $c = \frac{(b-a)}{2}$. It follows $(a, b) \in \sigma(\mathcal{G})$. But every open set in $\mathcal{B}$ is given by a countable union of open intervals (a, b), then the collection of open

[25] See Appendix.

subsets of $\mathbb{R}$ is contained in $\sigma(\mathcal{G})$. By Remark 2.19 we have $\mathcal{B} \subset \sigma(\mathcal{G})$ and this together with the above inclusion shows that $\mathcal{B} = \sigma(\mathcal{G})$.

Measurable Maps and Image Measures

In the main text we have defined a random function $X : \Omega \to \mathbb{R}$ to get a numerical representation, namely a random number $x = X(\omega)$, for all possible scenarios $\omega \in \Omega$. To compute $\mathsf{P}(A)$ for every $A \in \mathcal{F}$ we restrict ourselves to the following kind of random functions.

Definition 2.10 Let $(\Omega, \mathcal{F})$ be an event space and let $(\mathbb{R}, \mathcal{B})$ be the real measurable space. A random function $X : \Omega \to \mathbb{R}$ is a rv iff

$$X^{-1}(B) := \{\omega \in \Omega \mid X(\omega) \in B\} \in \mathcal{F}, \qquad \text{for all } B \in \mathcal{B}, \tag{2.27}$$

and we say X is a **measurable** map, or it is $\mathcal{F}$-measurable.

Remark 2.20 Definition 2.10 is still valid with $(\Omega, \mathcal{F})$ replaced by $(\mathbb{R}, \mathcal{B})$, i.e. any map $g : \mathbb{R} \to \mathbb{R}$ s.t. (2.27) holds is measurable, and we call it *Borel-measurable function*. E.g., $B = (x, y] \in \mathcal{B}$, with $x < y$ yields $g^{-1}((x, y]) = \{x \in \mathbb{R} \mid g(x) \in (x, y]\} \in \mathcal{B}$.

Every collection of scenarios $\omega \in \Omega$ forming an event $A \in \mathcal{F}$ is transformed by X into a 'measurable' object, i.e. a subset of $\mathbb{R}$ which belongs to the Borel σ-algebra:

$$\Omega \xrightarrow{\ X\ } \mathbb{R} \qquad \text{From scenarios to random numbers}$$

$$\mathcal{F} \xleftarrow{\ X^{-1}\ } \mathcal{B} \qquad \text{Pull back a Borel set to an event}$$

Now, for the reader's convenience we recall the following. For any random function $X : \Omega \to \mathbb{R}$ we have:

- $X(A) = \{b \in \mathbb{R} \mid b = X(\omega) \text{ for some } \omega \in \Omega\}$ is the **image** of $A \subset \Omega$ under X. Clearly $X(A) \subset \mathbb{R}$. The special case $X(\Omega)$ is called the **range** of X;
- $X^{-1}(B) = \{\omega \in \Omega \mid X(\omega) \in \mathbb{R}\}$ is the **inverse image** of $B \subset \mathbb{R}$; clearly $X^{-1}(B) \subset \Omega$;
- The inverse image is always defined as a set function $X^{-1} : 2^{\mathbb{R}} \to 2^{\Omega}$, the domain being the power set of $\mathbb{R}$. When X^{-1} maps every Borel set $B \in \mathcal{B} \subset 2^{\mathbb{R}}$ into an event $A \in \mathcal{F} \subset 2^{\Omega}$, then as a measurable function it is also a rv.

Remark 2.21 (Properties of Image and Inverse Image) Let $(A_i)_{i \in I} \subset 2^{\Omega}$, $(B_j)_{j \in J} \subset 2^{\mathbb{R}}$ be two families of subsets of Ω and $\mathbb{R}$, respectively. Then, define a random function $X : \Omega \to \mathbb{R}$. The image of X commutes with unions but not with intersections, while the inverse image commutes with unions, intersections and set complement:

1. $X(\cup_{i\in I} A_i) = \cup_{i\in I} X(A_i)$
2. $X(\cap_{i\in I} A_i) \subset \cap_{i\in I} X(A_i)$
3. $X^{-1}(\cup_{j\in J} B_i) = \cup_{j\in J} X^{-1}(B_j)$
4. $X^{-1}(\cap_{j\in J} B_i) = \cap_{j\in J} X^{-1}(B_j)$
5. $X^{-1}(B^c) = (X^{-1}(B))^c$.

In fact, not every mapping $X : \Omega \to \mathbb{R}$ is a rv.

Example 2.15 Toss a fair coin twice and let $\Omega = \{\text{HH, HT, TH, TT}\}$ coupled with the σ-algebra $\mathcal{F} = \{\varnothing, \{\text{HT, TH}\}, \{\text{HH, TT}\}, \{\text{HH, HT, TH, TT}\}\}$. You either win $\in 1$ if a head is showed twice or loose $\in 1$ if a tail is showed twice, otherwise you neither win nor loose. Define the map

$$X(\omega) = \begin{cases} 1, & \text{if} \quad \omega = \text{HH} \\ 0, & \text{if} \quad \omega = \text{HT} \quad \text{or} \quad \omega = \text{TH} \\ -1, & \text{if} \quad \omega = \text{TT}. \end{cases}$$

Choosing the Borel set $B = (-\infty, -1] \cup (-1, 0) \cup (1, \infty)$ we have $\{X \in B\} = \{\text{TT}\} \notin \mathcal{F}$. Similarly, $B = (-\infty, -1) \cup (-1, 0) \cup [1, \infty)$ gives $\{X \in B\} = \{\text{HH}\} \notin \mathcal{F}$. In both cases the choice of a Borel set in the image space $(\mathbb{R}, \mathcal{B})$ does not guarantee that the inverse image X^{-1} is in the σ-algebra $\mathcal{F}$ of events.

Example 2.16 (i) The identity map $\text{id} : \Omega \to \Omega$ defined by $\text{id}(\omega) = \omega$ is measurable, since for every $B \in \mathcal{F}$ we have $\text{id}^{-1}(B) = B \in \mathcal{F}$. (ii) Any map X between $(\Omega, \mathcal{F})$ and $(\mathbb{R}, \mathcal{S})$ such that $\mathcal{F} = 2^\Omega$ or $\mathcal{S} = \{\varnothing, \mathbb{R}\}$ is always measurable. In the former case, for every $B \in \mathcal{S}$ we have $X^{-1}(B) \in 2^\Omega$. In the latter case, either $X^{-1}(\varnothing) = \varnothing$ or $X^{-1}(\mathbb{R}) = \Omega$ by definition, and both inverse images are in $\mathcal{F}$.

Example 2.17 Let $(\Omega, \mathcal{F})$ be an event space. Define the map $X : \Omega \to \mathbb{R}$ by $X(\omega) = c$ for every $\omega \in \Omega$ and $c \in \mathbb{R}$ fixed. Then X is measurable, since for every $B \in \mathcal{B}$ we have either $c \in B$ or $c \in B^c$ so that $X^{-1}(B) = \{X \in B\}$ is in $\mathcal{F}$. The collection of its inverse images is the trivial σ-algebra $\{\varnothing, \Omega\}$. We have $X^{-1}(\{c\}) = \Omega$, with $\varnothing$ as its complement.

When a random function if indeed a rv X, then the abstract probability space $(\Omega, \mathcal{F}, \mathsf{P})$ is translated into a **real probability space** $(\mathbb{R}, \mathcal{B}, \mathsf{P}_X)$, where the law of X is actually a composed map:

$$\mathsf{P}_X(B) := \mathsf{P}(X^{-1}(B)) = \mathsf{Q}(B), \qquad \text{for all Borel sets } B \in \mathcal{B}. \tag{2.28}$$

Therefore, the set function P_X is a probability measure Q defined on $\mathcal{B}$, also called the **image measure** of P. The intuition behind this is depicted in Fig. 2.7 .

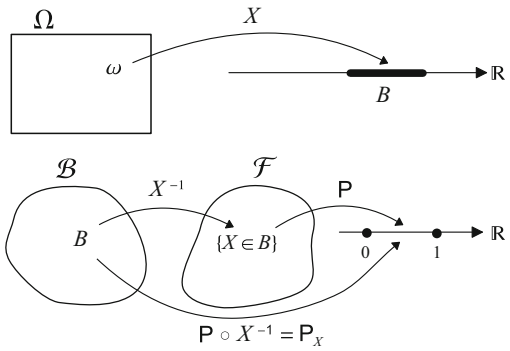


Fig. 2.7 A rv X as a measurable function: its image X entails $B \subset \mathbb{R}$, while its inverse image X^{-1} pullbacks it to $\mathcal{B}$, since it is a Borel set, and eventually the composed map $\mathsf{P} \circ X^{-1}$ delivers the law $\mathsf{P}_X(B)$

Distribution Function on $\mathbb{R}$

As in probability spaces $(\Omega, \mathcal{F}, \mathsf{P})$, the role of distribution function (DF) is similar to that of CDF but now for probabilities on $\mathbb{R}$.[26]

Theorem 2.5 *Let $F : \mathbb{R} \to [0, 1]$ be a DF. Then, there exists a unique probability measure Q on the event space $(\mathbb{R}, \mathcal{B})$ s.t.*

$$\mathsf{Q}((a, b]) = F(b) - F(a), \tag{2.29}$$

for all $-\infty \leqslant a < b < \infty$.

If Q is a probability measure on $(\mathbb{R}, \mathcal{B})$ we have also the converse: to Q there corresponds a DF $F : \mathbb{R} \to [0, 1]$ through:

$$F(x) = \mathsf{Q}((-\infty, x]), \quad \forall\, x \in \mathbb{R}. \tag{2.30}$$

By the above and Theorem 2.5 we conclude that there exists a 1-to-1 correspondence between probability measures Q and DFs on $(\mathbb{R}, \mathcal{B})$.

$$\begin{aligned} F \text{ is a DF} \implies & \text{formula } \mathsf{Q}((a, b]) := F(b) - F(a) \\ & \text{defines a probability measure on } (\mathbb{R}, \mathcal{B}) \\ \mathsf{Q} \text{ is a probability} \implies & F(x) := \mathsf{Q}((-\infty, x]) \text{ is a DF} \\ & \text{from } \mathbb{R} \text{ to } [0, 1] \end{aligned}$$

The ultimate connection between $(\Omega, \mathcal{F}, \mathsf{P})$ and probability spaces defined on $\mathbb{R}$ is given by considering $\mathbb{R}$ as the sample space, $\mathcal{F}$ replaced by $\mathcal{B}$ and $\mathsf{P} = \mathsf{Q} = \mathsf{P}_X$,

[26] The term DF replaces CDF when we consider probabilities defined over the event space $(\mathbb{R}, \mathcal{B})$. Thus, any DF satisfies properties (a)–(c) of Theorem 2.1.

i.e. the probability measure on $\mathbb{R}$ can be taken as the law of a fixed rv X so that $F \equiv F_X$. Moral: we can compute probabilities on $\mathbb{R}$ without explicitly mention any rv, but instead by working with Borel sets (what in the main text we have called *suitable* subsets $B \subset \mathbb{R}$).

Example 2.18 (Discrete Distribution on $\mathbb{R}$) Given a piecewise constant DF F let us consider the set $\{q_1, q_2, \ldots\}$ with $q_i := \mathsf{Q}(\{x_i\})$, for $i = 1, 2, \ldots$ made of probability weights given by the F's jumps:

$$\mathsf{Q}(\{x_i\}) = F(x_i) - F(x_{i-1}) > 0, \text{ with } F(x_{i-1}) = F(x_i-).$$

Indeed, using formula (2.29) we claim

$$\mathsf{Q}((a, b]) = \mathsf{Q}((a, b) \cup \{b\}) = \mathsf{Q}((a, b)) + \mathsf{Q}(\{b\}),$$

for $a = x_{i-1}$ and $b = x_i$. This is equivalent to have: $\mathsf{Q}((x_{i-1}, x_i)) = 0$ and $\sum_{i=1}^{\infty} q_i = 1$. On the other hand, $F(x) = \mathsf{Q}((-\infty, x]) = \sum_{x_i \leqslant x} q_i$.

Example 2.19 (Continuous Distribution on $\mathbb{R}$) Let F be a continuous DF, $F(x) - F(x-) = 0$ each $x \in \mathbb{R}$. Thus, given another real-valued integrable (in some sense) function f such that $f(x) \geqslant 0$, for $x \in \mathbb{R}$, we can build a probability measure by defining $F(x) = \int_{-\infty}^{x} f(u)\mathrm{d}u$, provided $\int_{-\infty}^{\infty} f(x)\mathrm{d}x = 1$. Thus,

$$\mathsf{Q}((a, b]) = F(b) - F(a) = \int_{-\infty}^{b} f(x)\mathrm{d}x - \int_{-\infty}^{a} f(x)\mathrm{d}x,$$

and F is said to be absolutely continuous. Now, we have $\{x\} = (-\infty, x] \setminus (-\infty, x)$ and $F(x-) = \lim_{i \to \infty} \mathsf{Q}((-\infty, x_i]) = \mathsf{Q}((-\infty, x))$, for $x_i \uparrow x$, and then

$$\begin{aligned} F(x) - F(x-) &= \mathsf{Q}((-\infty, x]) - \mathsf{Q}((-\infty, x)) \\ &= \mathsf{Q}(\{x\}) = 0 \end{aligned}$$

which is equivalent to $\mathsf{Q}(\{x\}) = \int_{x}^{x} f(u)\mathrm{d}u = 0$, and the corresponding probability Q put a zero weight on singletons.

Stieltjes Measure

Let P be a probability measure[27] on $(\mathbb{R}, \mathcal{B})$, Then, P determines a càdlàg function. First, define $F : \mathbb{R} \to \mathbb{R}$ by

$$F(x) = \begin{cases} \mathsf{P}((0, x]), & \text{if } x \geqslant 0 \\ -\mathsf{P}((x, 0]), & \text{if } x \leqslant 0. \end{cases}$$

[27] Or more generally a finite measure.

By Definition 1.3, item PA2, Chap. 1, we have:

$$F(x_1) \leqslant F(x_2) \Longleftrightarrow \mathsf{P}((0, x_1]) \leqslant \mathsf{P}((0, x_2])$$

when $0 \leqslant x_1 < x_2 \in \mathbb{R}$. If $x \geqslant 0$ and $x_i \downarrow x$, then intervals $(0, x_i]$ decrease to $(0, x]$, the case $x \leqslant 0$ being similar. Therefore, using continuity from above, see Appendix in Chap. 1, we see that

$$F(x_i) = \mathsf{P}((0, x_i]) \downarrow \mathsf{P}((0, x]) = F(x)$$

and then $\lim_i F(x_i) = F(x)$. Using continuity from below gives a similar result for $x \leqslant 0$. Thus F is both continuous from the right and non-decreasing, so it is càdlàg. Finally, since $(a, b] = (-\infty, b] \setminus (-\infty, a]$ we have

$$\mathsf{P}((a, b]) = F(b) - F(a), \qquad a < b \in \mathbb{R}, \tag{2.31}$$

for every interval with finite endpoints,[28] which is nothing else (2.29) in Theorem 2.5. On the other hand, a bounded càdlàg function $F : \mathbb{R} \to \mathbb{R}$ determines a measure on $(\mathbb{R}, \mathcal{B})$. Indeed, define a set function P by Eq. (2.31) and then extend it[29] to the Borel σ-algebra $\mathcal{B}$. If F and $\tilde{F}$ are two càdlàg functions such that $F = \tilde{F}$, then they define two set functions through (2.31) extending to a unique measure P on $\mathcal{B}$. But we should define $\tilde{F}(x) = F(x) + c$ for $c \in \mathbb{R}$, thus two càdlàg functions F and $\tilde{F}$ differing by a constant induce the same finite measure which assigns finite values to bounded intervals. This is also called a **Stieltjes measure**.

Absolutely Continuous Functions

The concept of absolutely continuous probability distribution has a counterpart in real analysis which we present here for the sake of completeness. A real-valued function $F : [a, b] \to \mathbb{R}$ is said to be **absolutely continuous** if for every $\epsilon > 0$ there exists $\delta > 0$ s.t. for every $n \in \mathbb{N}$ and all collections $(a_1, b_1), \ldots, (a_n, b_n)$ of disjoint open subintervals of $[a, b]$, of total length at most δ, i.e. $\sum_{i=1}^{n} |b_i - a_i| \leqslant \delta$, we have $\sum_{i=1}^{n} |F(b_i) - F(a_i)| \leqslant \epsilon$. This always implies $F(x)$ is continuous on $[a, b]$. A function $F : \mathbb{R} \to \mathbb{R}$ is said to be absolutely continuous if its restriction on $[a, b]$ is. If $F(x)$ has a bounded derivative $F'(x)$ for every x then $|F(x_2) - F(x_1)| \leqslant M(x_2 - x_1)$ for $x_1 < x_2$, where $M > 0$ is some constant depending on F. This is

[28] Using the one-dimensional Lebesgue measure, $\mathsf{P} \equiv m$, it holds $m((0, x]) = x$, for $x \geqslant 0$ and $-m((x, 0]) = -(-x)$ for $x \leqslant 0$, so we must have $F(x) = x$.

[29] The standard machine works: Carathéodory extension theorem together with Dynkin π-d theorem stating that a generator $\mathcal{G}$ which is a π-system entails $\sigma(\mathcal{G})$ equal to the generated d-system, $d(\mathcal{G})$. Actually, P should be defined on the class $\mathcal{G}$ of all finite disjoint unions of intervals $(a, b]$ which is a π-system generating $\mathcal{B}$.

called **Lipschitz condition**. Indeed, by the boundedness of the derivative and the mean value theorem we have:

$$|F'(x)| \leqslant M \iff \tfrac{|F(x_2)-F(x_1)|}{|x_2-x_1|} \leqslant M, \qquad \text{for some } x \in (x_1, x_2).$$

The above condition further implies

$$\sum_{i=1}^{n} |F(b_i) - F(a_i)| \leqslant M \sum_{i=1}^{n} |b_i - a_i| \leqslant \delta M,$$

where $\delta = \frac{\epsilon}{M}$. Thus the Lipschitz condition is sufficient for a function F to be absolutely continuous. It can be shown that for any such function there exists another function $f(x)$ such that the former can be defined by integration:

$$F(x) = F(a) + \int_a^x f(u)\mathrm{d}u.$$

As byproduct, any CDF of a continuous rv X, with law having density f_X, is an absolutely continuous function with $F = F_X, f = f_X$ and $a = -\infty$ so that $F_X(-\infty) = 0$. Since F_X represents the probability law P_X we also say, with slight abuse, that the law of X is absolutely continuous with density f_X,

More on Quantile Function

Proof of Proposition 2.2 (1) Define $L_c := \{x \mid F_X(x) \geqslant c\}$ and observe that $x \in L_{F_X(x)}$. It follows $Q_X(F_X(x)) = \inf L_{F_X(x)} \leqslant x$ and the claim is proved. (2) By definition any choice $x \in L_c$ implies $F_X(x) \geqslant c$. This holds for every sequence $(x_i)_{i\in\mathbb{N}} \subset L_c$ decreasing to $\inf L_c = Q_X(c)$. Then by right-continuity of F_X we have $\lim_{i\to\infty} F_X(x_i) = F_X(\lim_{i\to\infty} x_i) = F_X(Q_X(c))$ and by the sign rule we are done. (3) Assume $F_X(x) \geqslant c$. Then $x \in L_c$ and $Q_X(c) = \inf L_c \leqslant x$. On the other hand, F_X is nondecreasing and applying (2) gives $c \leqslant F_X(Q_X(c)) \leqslant F_X(x)$ proving the claim. (4) If $c_1 \leqslant c_2$ then $L_{c_2} \subset L_{c_1}$ by definition (draw a graph). This is equivalent to $\inf L_{c_1} \leqslant \inf L_{c_2}$ and we are done. ■

Proof of of Lemma 2.1 Pick $x \in \mathbb{R}$ and let $Q_Y(c) = Q_X(c)$. If $F_X(x) < F_Y(x)$, then by items 1. and 4. of Proposition 2.2 we have $Q(F_X(x)) \leqslant Q(F_Y(x)) \leqslant x$. But item 3. of the same Proposition gives $F_X(x) \leqslant F_Y(x)$, a contradiction. A similar argument shows that $F_X(x) > F_Y(x)$ is another contradiction. Thus we must have $F_X(x) = F_Y(x)$ for any $x \in \mathbb{R}$ and the lemma is proved. ■

Proof of Proposition 2.3 By Proposition 2.2, Item 3 we readily compute:

$$Q_{f(X)}(c) \leqslant x \iff F_{f(X)}(x) \geqslant c.$$

Consider $\sup\{z \mid f(z) \leqslant x\}$ and denote it z_0, for a fixed $x \in \mathbb{R}$. Since f is left-continuous (or strictly increasing) we readily have

$$f(z) \leqslant x \iff z \leqslant z_0.$$

As byproduct, $\{f(X) \leqslant x\} = \{X \leqslant z_0\}$ then

$$F_{f(X)}(x) \geqslant c \iff F_X(z_0) \geqslant c.$$

The two equivalences above, together with item 3 in Proposition 2.2, which gives $F_X(z_0) \geqslant c$ iff $Q_X(x) \leqslant z_0$, and

$$Q_X(c) \leqslant z_0 \iff f(Q_X(c)) \leqslant x, \quad \text{by the second equivalence above,}$$

complete the proof for an arbitrary $x \in \mathbb{R}$. ■

Uniform Probability Spaces Again

We have seen in Exercise 1.16, Chap. 1, that $\sigma_E := \{A \cap E \mid A \in \mathcal{F}\}$ is a σ-algebra on Ω called *trace-σ-algebra*. Now, consider the experiment of choosing at random a number in $\Omega = (0, 1]$ and endow the sample space with $\mathcal{B}_{(0,1]}$ defined as the Borel-trace-σ-algebra containing all Borel sets in $\mathcal{B}$ intersected with the unit interval $(0, 1]$.

Remark 2.22 Given a sample space Ω and a generator $\mathcal{G}$, let $\mathcal{G} \cap E := \{A \cap E \mid A \in \mathcal{G}\}$ contains all intersections between each event in $\mathcal{G}$ and a fixed event $E \subset \Omega$. It can be shown that $\sigma(\mathcal{G} \cap E) = \sigma(\mathcal{G}) \cap E =: \{A \cap E \mid A \in \sigma(\mathcal{G})\}$, i.e. the class of all intersections between each event A in the generated σ-algebra $\sigma(\mathcal{G})$ and a fixed event E is again a σ-algebra, which is generated by $\mathcal{G} \cap E$. Taking $\mathcal{G}$ as the collection of open (or left-open, right-closed) intervals in $\mathbb{R}$ and using the notation above, we have $\mathcal{B}_{(0,1]} = \mathcal{B} \cap (0, 1]$: a Borel set in $\mathbb{R}$ in the form $A \subset (0, 1]$ belongs to σ-algebra $\mathcal{B}_{(0,1]}$.

Next, consider the probability space $((0, 1], \mathcal{B}_{(0,1]}, \mathsf{P})$, where P is a probability measure[30] Then, we have a new formulation of a continuous uniform probability space where, in particular, the likelihood of finding a point into an interval is computed as its length, see Appendix in Chap. 1. Observe how this corresponds to $\mathsf{P}(I) = \int_I p(\omega)\mathrm{d}\omega$, where p is a density and I is any interval picked from $\mathcal{B}_{(0,1]}$. Actually, this probability measure is similar to the law P_X of a uniform rv X, see Chap. 5. We can extend this framework to the n-dimensional case.

[30] Actually, it is the Lebesgue measure restricted to the unit interval and the corresponding trace-σ-algebra.

Further Details on Indicators

Indicator rvs better describe the behavior of a sequence of events $A_1, A_2, \ldots, A_n, \ldots \in \mathcal{F}$, not necessarily monotonic. We have:

$$\mathbf{I}_{\limsup A_n} = \begin{cases} 1 & \text{se } \omega \in \limsup A_n \\ 0 & \text{se } \omega \notin \limsup A_n. \end{cases} \tag{2.32}$$

Indeed, $\mathbf{I}_{\limsup A_n} = 1$ if for every $n \in \mathbb{N}$ there exists $k \geqslant n$ such that $\omega \in A_k$. This is easily seen to be equivalent to:

$$\limsup \mathbf{I}_{A_n} = \inf_n \left\{ \sup_{k \geqslant n} \mathbf{I}_{A_k} \right\}. \tag{2.33}$$

Thus, if $k \geqslant n$ and $\omega \in A_k$, by (2.33) we have

$$\sup_{k \geqslant n} \mathbf{I}_{A_k} = \sup \left\{ \mathbf{I}_{A_n}, \mathbf{I}_{A_{n+1}}, \ldots \right\} = 1,$$

which implies $\inf_n \left\{ \sup_{k \geqslant n} \mathbf{I}_{A_k} \right\} = \inf_n \{1\} = 1$, for every $n \in \mathbb{N}$. We conclude that $\mathbf{I}_{\limsup A_n} = \limsup \mathbf{I}_{A_n}$ and analogously $\mathbf{I}_{\liminf A_n} = \liminf \mathbf{I}_{A_n}$. We are now in position to prove:

Lemma 2.2 *Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space and $(A_n)_{n \in \mathbb{N}} \subset \mathcal{F}$ be an arbitrary sequence of events. Then, $A = \liminf A_n = \limsup A_n = \lim_{n \to \infty} A_n$ is the limit event in $\mathcal{F}$ if and only if $\lim_{n \to \infty} \mathbf{I}_{A_n}(\omega) = \mathbf{I}_A(\omega)$, for all $\omega \in \Omega$.*

Proof

($\Rightarrow$) Assume $A = \liminf A_n = \limsup A_n = \lim A_n$. Then, $\mathbf{I}_{\liminf A_n} = \mathbf{I}_{\limsup A_n} = 1$ provided that $\omega \in A$, otherwise its value is zero. Since $\mathbf{I}_{\liminf A_n} = \liminf \mathbf{I}_{A_n}$ and $\mathbf{I}_{\limsup A_n} = \limsup \mathbf{I}_{A_n}$ we have $\liminf \mathbf{I}_{A_n} = \limsup \mathbf{I}_{A_n}$, which implies:

$$\lim \mathbf{I}_{A_n}(\omega) = \liminf \mathbf{I}_{A_n}(\omega) = \limsup \mathbf{I}_{A_n}(\omega) = \mathbf{I}_A(\omega), \qquad \text{for every} \quad \omega \in \Omega.$$

($\Leftarrow$) Suppose $\lim \mathbf{I}_{A_n} = \mathbf{I}_A$, for every $\omega \in \Omega$. Then, it follows $\liminf \mathbf{I}_{A_n} = \limsup \mathbf{I}_{A_n} = \mathbf{I}_A$. But we know that $\mathbf{I}_{\liminf A_n} = \liminf \mathbf{I}_{A_n}$ and $\mathbf{I}_{\limsup A_n} = \limsup \mathbf{I}_{A_n}$. Hence:

$$\mathbf{I}_{\liminf A_n} = \mathbf{I}_{\limsup A_n} = \mathbf{I}_A,$$

so that $\liminf A_n = \limsup A_n = A$ and we are done.

■

By lemma 2.2, convergence of a monotonic sequence of events is extended to more general cases. Another consequence is:

Theorem 2.6 *Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ be a sequence of events over a probability space $(\Omega, \mathcal{F}, \mathsf{P})$. If $(A_n)_{n\in\mathbb{N}}$ converges to the set A, then $A \in \mathcal{F}$ and $\lim_{n\to\infty} \mathsf{P}(A_n) = \mathsf{P}(A)$.*

Proof Assume $A_n \to A$ and Lemma 2.2. We have $\lim_{n\to\infty} \mathbf{I}_{A_n}(\omega) = \mathbf{I}_A(\omega)$ for every $\omega \in \Omega$. This is equivalent to $A = \liminf A_n = \limsup A_n$ and then $A \in \mathcal{F}$. Now, let $B_n = \cap_{k\geqslant n} A_n$ and $C_n = \cup_{k\geqslant n} A_n$. The two sequences $(B_n)_{n\in\mathbb{N}}$ and $(C_n)_{n\in\mathbb{N}}$ are increasing and decreasing, respectively. Then, $B_n \uparrow A$ and $C_n \downarrow A$ and

$$\lim_{n\to\infty} \mathsf{P}(B_n) = \lim_{n\to\infty} \mathsf{P}(C_n) = \mathsf{P}(A).$$

But $B_n \subset A_n \subset C_n$, then $\mathsf{P}(B_n) \leqslant \mathsf{P}(A_n) \leqslant \mathsf{P}(C_n)$. By the squeeze theorem for real sequences we eventually obtain $\lim_n \mathsf{P}(A_n) = \mathsf{P}(A)$ and the proof is completed. ■

Measurability of Rvs

Recall that for any Borel-measurable function $Y : \mathbb{R} \to \mathbb{R}$ the composition $Y(X(\omega))$ is again a rv for every $\omega \in \Omega$. A variant of this result is valid for continuous functions. Any continuous function $Y : \mathbb{R}^n \to \mathbb{R}$ is $(\mathcal{B}^n, \mathcal{B})$-measurable: for any open set in $\mathcal{B}$ the inverse image is again an open set in $\mathcal{B}^n$, the latter being the n-dimensional Bore σ-algebra generated by the open subsets of $\mathbb{R}^n$. Hence, when $n = 1$ and Y is continuous its inverse image $Y^{-1}(O)$ is an open subset of $\mathbb{R}$ for any open subset O and $X^{-1}(Y^{-1}(O))$ is actually an event in $\mathcal{F}$. This remains true when replacing O by any other Borel set,[31] since open sets generate the Borel σ-algebra. A slightly generalization is the following. Given a non empty set Ω, a map $\mathbf{X} : \Omega \to \mathbb{R}^n$ is a vector function, for it transforms a point $\omega \in \Omega$ into an Euclidean vector $\mathbf{X}(\omega) = (X_1(\omega), \ldots, X_n(\omega))$. If further we assume $\mathbf{X}$ is a $(\mathcal{F}, \mathcal{B}^n)$-measurable map, then it is nothing but a n-dimensional rv whose image space is $(\mathbb{R}^n, \mathcal{B}^n)$, see Chap. 3. An obvious question is when the measurability of each coordinate map X_i is implied by that of $\mathbf{X}$ and viceversa, for $i = 1, \ldots, n$. This is the content of the next:

Theorem 2.7 *Let $(\Omega, \mathcal{F})$ be a measurable space and let $X_1, \ldots, X_n$ be maps : $\Omega \to \mathbb{R}$. Define a vector function $\mathbf{X} : \Omega \to \mathbb{R}^n$. Each X_i is $(\mathcal{F}, \mathcal{B})$-measurable if and only if $\mathbf{X}$ is $(\mathcal{F}, \mathcal{B}^n)$-measurable.*

For the proof of Theorem 2.7 we need the following definition. We call $\pi_i : \mathbb{R}^n \to \mathbb{R}$ a **projection map** if $\pi_i(x_1, \ldots, x_n) = x_i$ for $i = 1, \ldots, n$ and $\mathbf{x} = (x_1, \ldots, x_n) \in$

[31] In general, any continuous function $G : \mathbb{R}^m \to \mathbb{R}^m$ is measurable.

$\mathbb{R}^n$. Thus π_i projects the vector $\mathbf{x}$ on the ith coordinate. It can be easily seen that π_i is a continuous function.

Proof of Theorem 2.7 ($\Rightarrow$) For any $\mathbf{b} \in \mathbb{R}^n$, we can write

$$\begin{aligned}\mathbf{X}^{-1}((-\infty, \mathbf{b}]) &= \{\omega \in \Omega : (X_1(\omega), \ldots, X_n(\omega)) \in (-\infty, \mathbf{b}]\} \\ &= \{\omega \in \Omega : X_1(\omega) \leqslant b_1, \ldots, X_n(\omega) \leqslant b_n\} \\ &= \{X_1 \leqslant b_1\} \cap \cdots \cap \{X_n \leqslant b_n\} = \cap_{i=1}^n X_i^{-1}((-\infty, b_i]).\end{aligned}$$

By assumption, each X_i is measurable and then $\mathbf{X}^{-1}((-\infty, \mathbf{b}])$ is in $\mathcal{F}$. This implies that $\mathbf{X}$ is measurable as the half-rays $(-\infty, \mathbf{b}]$ generate $\mathcal{B}^n$. ($\Leftarrow$) Assume that $\mathbf{X}$ is measurable. For each $i = 1, \ldots, n$ let π_i be the projection on the ith coordinate, which is a continuous function and then $(\mathcal{B}^n, \mathcal{B})$-measurable. Thus, the composition $X_i = \pi_i \circ \mathbf{X}$ is a measurable map. ■

The main application of these results is to arithmetic operations among rvs. We formulate the next theorem in n-dimensions, although its main application is in the case $n = 1$.

Theorem 2.8 *Let $(\Omega, \mathcal{F})$ be a measurable space on which we define a rv $Z : \Omega \to \mathbb{R}$ and two random vectors $\mathbf{X}, \mathbf{Y} : \Omega \to \mathbb{R}^n$. Then also the maps $\mathbf{X} + \mathbf{Y}$ and $Z\mathbf{Y}$ are rvs.*

Observe that the product $Z\mathbf{Y}$ between a one-dimensional rv Z and a random vector $\mathbf{X}$ is defined componentwise, $(ZY_1, \ldots, ZY_n)$. Recall that a vector in $\mathbb{R}^{n+1}$ can be written as $(\alpha, \mathbf{x})$ where $\alpha \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$.

Proof of Theorem 2.8 The map $\mathbf{h} : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ defined by $\mathbf{h}(\alpha, \mathbf{y}) = \alpha\mathbf{y} = (\alpha y_1, \ldots, \alpha y_n)$ is continuous and of course measurable. Define the map $\mathbf{G} : \Omega \to \mathbb{R} \times \mathbb{R}^n$ by $\mathbf{G}(\omega) = (Z(\omega), \mathbf{Y}(\omega)) = (Z(\omega), Y_1(\omega), \ldots, Y_n(\omega))$. By Theorem 2.7 it is measurable, hence the composed map $Z \cdot \mathbf{Y} = \mathbf{h} \circ \mathbf{G}$ is again measurable. Similar reasoning applies to $\mathbf{X} + \mathbf{Y}$. ■

Theorem 2.8 enable us to prove the measurability of arithmetic operations between rvs. Let $n = 1$ and replace $\mathbf{X}, \mathbf{Y}$ with two rvs $X, Y : \Omega \to \mathbb{R}$. Thus the sum $X + Y$ and the product ZY between rvs is again a rv. The latter specialize to the case of a constant map $Z = a$, whence the product of a rv Y by a constant is again a rv. Taking $a = -1$ we can combine sums and products to get $X - Y$ as a rv. Once more, dividing two rvs $\frac{X}{Z}$ gives a rv. Indeed, we can write $\frac{X}{Z}$ as the composition $X(h \circ Z)$, using the function $h : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ defined by $h(x) = \frac{1}{x}$. If we adopt the convention that diving by zero gives zero and let $h(0) = 0$, the only thing we have to show is that h is measurable. But in fact it is continuous on $\mathbb{R} \setminus \{0\}$. To prove this, first recall that $(\mathbb{R}, \tau)$ is a topological space. Then, for any open set $O \in \tau$ the proper difference $\tilde{O} = O \setminus \{0\}$ is also an open set[32] which lies in the

[32] Indeed $\tilde{O} = O \cap \{0\}^c$, but $\{0\}^c$ equals the open set $(-\infty, 0) \cup (0, \infty)$.

Borel σ-algebra $\mathcal{B}$. This implies that the disjoint union $O = \tilde{O} \cup (O \cap \{0\})$ is in $\mathcal{B}$, since $O \cap \{0\}$ is the intersection of two Borel sets. Furthermore, $h^{-1}(\tilde{O}) \in \mathcal{B}$ and $h^{-1}(\{0\}) = \{0\} \in \mathcal{B}$. Finally, by the preserving property of inverse images we have $h^{-1}(O) = h^{-1}(\tilde{O}) \cup h^{-1}(O \cap \{0\}) \in \mathcal{B}$ and this proves that h is measurable and thus $X(h \circ Z)$ is a rv.

Remark 2.23 When arithmetic operations among rvs are involved, the following rules are adopted.

- For XY we have:
  - $0 \times \pm\infty = \pm\infty \times 0 = 0$
  - $x \times \pm\infty = \pm\infty \times x = \pm\infty \qquad \text{if} \quad 0 < x < \infty$
  - $x \times \pm\infty = \pm\infty \times x = \mp\infty \qquad \text{if} \quad -\infty < x < 0$
  - $\pm\infty \times \pm\infty = \infty$ and $\pm\infty \mp \infty = -\infty$.
- $X \pm Y$ remains undefined in the case $\infty - \infty$.
- For $\frac{X}{Y}$ the same rules for products applies, but $\frac{\pm\infty}{\pm\infty}$, $\frac{\pm\infty}{0}$ remain undefined.

Limits of Rvs

We know that a sequence of rvs can converge pointwise, and we aim at showing that the limit is indeed a rv.

Proposition 2.4 *If a sequence $(X_n)_{n\in\mathbb{N}}$ of rvs is defined on the measurable space $(\Omega, \mathcal{F})$ and converges pointwise to the limit function $X : \Omega \to \mathbb{R}$, then X is a rv.*

This result is actually a corollary of more general limiting procedures. Before introduce them we give an example.

Example 2.20 The minimum between two rvs, $\min\{X(\omega), Y(\omega)\}$, is again a rv. To put things in a more general setting define $\inf\{X_1(\omega), X_2(\omega), \ldots\} =: \inf_n X_n$ for each outcome $\omega \in \Omega$. This is a map from Ω to $\mathbb{R}$ which value should be $-\infty$. If all terms of the sequence $(X_n)_{n\in\mathbb{N}}$ are strictly greater than a number $x \in \mathbb{R}$ for all ω, we have $X_n(\omega) > x$ for all $n \in \mathbb{N}$ and then also $\inf_n X_n(\omega) > x$. Otherwise, $\inf_n X_n \leqslant x$ where the infimum is taken for some terms in the sequence such that $X_n \leqslant x$. Thus we can write:

$$\omega \in \left\{ \inf_n X_n \leqslant x \right\} \iff \omega \in \{X_n \leqslant x\} \text{ for some } n \in \mathbb{N}$$
$$\iff \omega \in \cup_n \{X_n \leqslant x\}.$$

Since each $\{X_n \leqslant x\}$ is an event, the last countable union is an event too. We showed that the map $\inf_n X_n$ is $(\mathcal{F}, \mathcal{B})$-measurable and then is a rv. Similarly, $\sup_n X_n$ is a rv as $\{\sup_n X_n \geqslant x\}$ equals $\cup_n \{X_n \geqslant x\}$ which by is an event for every $x \in \mathbb{R}$.

Based on the preceding example we are in position to prove:

Theorem 2.9 *Let $X_1, X_2, \ldots$ be rvs defined on the same measurable space $(\Omega, \mathcal{F})$. Then the maps $\inf_n X_n$, $\sup_n X_n$, $\liminf_n X_n$, $\limsup_n X_n$ are also rvs.*

Proof By Example 2.20 the maps $\inf_n X_n$ e $\sup_n X_n$ are rvs. For each $\omega \in \Omega$ let us consider the limit inferior and the limit superior of the sequence $(X_n)_n$:

$$\liminf_n X_n = \sup_n \Big\{ \inf_{k \geqslant n} X_k \Big\}, \qquad \limsup X_n = \inf_n \Big\{ \sup_{k \geqslant n} X_k \Big\}.$$

If we let $Y_n = \inf_{k \geqslant n} X_k$ and $Z_n = \sup_{k \geqslant n} X_k$ then $\sup_n Y_n$ e $\inf_n Z_n$ are also rv and we are done. ■

Thanks to Theorem 2.9 we can prove that also $\max_{n \leqslant k} X_n$ and $\min_{n \leqslant k} X_n$ are rvs. Finally, we give:

Proof of Proposition 2.4 For every outcome $\omega \in \Omega$ the assumption of pointwise convergence enable us to take $\lim X_n(\omega) = X(\omega)$. But the sequence $X_1(\omega), X_2(\omega), \ldots$ of real numbers converges to the number $X(\omega)$ if and only if $\liminf_n X_n = \limsup_n X_n = X$. By Theorem 2.9 $\liminf_n X_n$ and $\limsup_n X_n$ are rvs and the proof is complete. ■

Observe that for $(X_n)_{n \in \mathbb{N}}$ be a sequence of rvs all defined on the same measurable space $(\Omega, \mathcal{F})$, the set $\{\sup_n X_n > x\}^c$ is the event $\cap_n \{X_n \leqslant x\}$ for every $x \in \mathbb{R}$. In fact, $\omega \in \{\sup_n X_n > x\}^c$ iff $\omega \notin \{\sup_n X_n > x\}$ and this implies $\omega \in \{\sup_n X_n \leqslant x\}$. More formally, write

$$\Big\{ \sup_n X_n > x \Big\}^c = \big(\cup_n \{X_n > x\} \big)^c = \cap_n \{X_n > x\}^c = \cap_n \{X_n \leqslant x\}$$

and we are done. Eventually, the event

$$\{\omega \in \Omega \mid X_n(\omega) \text{ conv. point.}\}$$

equals $\{X_n \to X\}$, where $X = \liminf_n X_n = \limsup_n X_n$. Hence, $\{X_n \to X\}^c = \{\liminf_n X_n < \limsup_n X_n\}$. Observe that for each $\omega \in \Omega$ we may find a rational number q that lies between the numbers $\underline{x} = \liminf_n X_n(\omega)$ and $\overline{x} = \limsup_n X_n(\omega)$. This induces the events $\{\liminf_n X_n \leqslant q < \limsup_n X_n\}$, which indeed form a countable collection depending upon rationals. Thus we can rewrite $\{X_n \to X\}^c$ as:

$$\cup_{q \in \mathbb{Q}} \{\liminf_n X_n \leqslant q < \limsup_n X_n\} = \cup_{q \in \mathbb{Q}} \big(\{\liminf_n X_n \leqslant q\} \cap \{\limsup_n X_n \leqslant q\}^c \big).$$

Then, the set $\{X_n \text{ conv. point.}\}$ is an event since both sets in the second equality are contained in $\mathcal{F}$.

Remark 2.24 We tacitly assumed the measurability of the limit X. Observe that

$$\{X_n \to X\} \Longleftrightarrow \{\omega \in \Omega \mid X_n(\omega) \text{ converge pointwise}\},$$

and its complement $\{X_n \to X\}^c = \{X_n \nrightarrow X\}$ is the set where convergence fails to hold. But this need not to be an event in $\mathcal{F}$ even if the convergence set belong to the σ-algebra. Loosely speaking, if we find a P-null event $A \in \mathcal{F}$ such that $\{X_n \nrightarrow X\} \subset A$ and assume the probability measure is complete, $\mathsf{P}(A) = 0$, this implies $\mathsf{P}(X_n \nrightarrow X) = 0$ too. This is a delicate issue which we turn around by assuming the necessary condition that also the limit function X is a random variable, i.e. it is measurable.

Chapter 3
Multidimensional Random Variables



3.1 Random Vectors

What is the probability that a pair of future asset prices $(S_{1,t+1}(\omega), S_{2,t+1}(\omega))$ will satisfy

$$a < S_{1,t+1} \leqslant b \text{ and } c < S_{2,t+1} \leqslant d,$$

for real $a < b$ and $c < d$, whatever time $t = 0, 1, 2, \ldots$? Situations like this require a *conjoint measurement* of quantities depending on the same sample space Ω. Despite we do not know how many and what are the scenarios underlying the random experiment, we need to predict both possible values $S_{1,t+1}$ and $S_{2,t+1}$ to gain useful information for financial decision making.

Example 3.1 Assume the rvs $S_{i,t+1}$ represent future prices of two distinct financial assets, for $i = 1, 2$. Further, assume we buy h_1 shares of asset # 1 and h_2 shares of asset # 2, i.e. h_i are holdings on the corresponding asset. Whence, we have the new rv

$$V_{t+1} := h_1 S_{1,t+1} + h_2 S_{2,t+1} = h(S_{1,t+1}, S_{2,t+1})$$

modelling future values of the portfolio comprising both assets, see Sect. 2.5, Example 2.5. Clearly, $V_{t+1}(\omega)$ is an univariate rv with a probability distribution $\mathsf{P}_{V_{t+1}}$ and a CDF $F_{V_{t+1}}$ both derived from the single distributions $\mathsf{P}_{S_{i,t+1}}$, $F_{S_{i,t+1}}$ and some mechanism aggregating them into a joint law governing the random pair of future prices.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_3

As Example 3.1 suggests, starting from a probability space $(\Omega, \mathcal{F}, \mathsf{P})$ now we need events in $\mathcal{F}$ that are induced by a function $\mathbf{X} : \Omega \to \mathbb{R}^n$, called **random vector**, s.t. for every scenario $\omega \in \Omega$ it gives a conjoint measurement

$$\mathbf{X}(\omega) = (X_1(\omega), \ldots, X_n(\omega)) = (x_1, \ldots, x_n) = \mathbf{x},$$

each coordinate being a univariate rv $X_i : \Omega \to \mathbb{R}$, for $i = 1, \ldots, n$. Events given by $\mathbf{X}$ are usually in the form

$$A = \left\{\omega \,|\, (X_1(\omega), \ldots, X_n(\omega)) \in B\right\} \equiv \{\mathbf{X} \in B\}$$

and B is a suitable subset of $\mathbb{R}^n$ s.t. $\{\mathbf{X} \in B\} \in \mathcal{F}$ and thus has a probability $\mathsf{P}(\mathbf{X} \in B)$.

Remark 3.1 The subset B_2 in Fig. 3.1 can be expressed as the Cartesian product of two intervals in $\mathbb{R}$. E.g., $B_1 = (a, b] \times (c, d]$ is the collection of those pairs (x_1, x_2) s.t. $a < x_1 \leqslant b$ and $c < x_2 \leqslant d$, to be meant as the possible pairs $(X_1(\omega), X_2(\omega))$ for some scenarios $\omega \in \Omega$ with $X_i(\omega) = x_i$ and $i = 1, 2$. The subset B_1 cannot be given as a Cartesian product, but in most application subsets of $\mathbb{R}^n$ for $n \geqslant 2$ representing events induced by a random vector are Cartesian products.

Remark 3.2 (Bi-Variate Rv) When $B \subset \mathbb{R}^2$ can be expressed as $B_1 \times B_2$, with each $B_1, B_2 \subset \mathbb{R}$ we have:

$$\begin{aligned}\{\mathbf{X} \in B\} &= \left\{\mathbf{X} \in B_1 \times B_2\right\} \\ &= \left\{(X_1, X_2) \in B_1 \times B_2\right\} \\ &= \left\{\omega \,|\, X_1(\omega) \in B_1, X_2(\omega) \in B_2\right\} \\ &= \{X_1 \in B_1\} \cap \{X_2 \in B_2\}.\end{aligned}$$

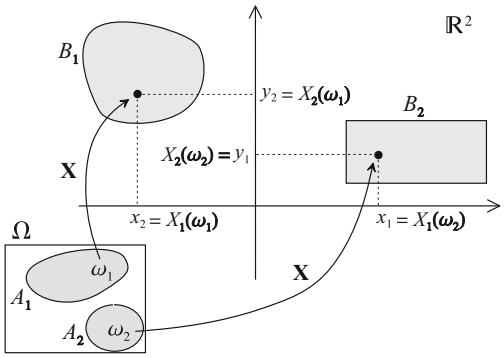


Fig. 3.1 Subsets B_1, B_2 of the Cartesian plane $\mathbb{R}^2$ depicting geometrically events $A_1 = \{\mathbf{X} \in B_1\}$ and $A_2 = \{\mathbf{X} \in B_2\}$

Thus, events like these are intersections of $\{X_i \in B_i\} \in \mathcal{F}$. Notice that the intersection also belongs to $\mathcal{F}$. In general, choosing $B_i = (-\infty, x_i]$ we have

$$\{X_1 \leqslant x_1\} \cap \cdots \cap \{X_n \leqslant x_n\}.$$

In some textbooks this is written $\{\mathbf{X} \leqslant \mathbf{x}\}$, where $X_i \leqslant x_i$ for all indices i and at least one inequality is strict. The σ-algebra generated by $\mathbf{X}$, denoted $\sigma(\mathbf{X})$, is the smallest collection of events $\{\mathbf{X} \in B\}$. We always have $\sigma(\mathbf{X}) \subset \mathcal{F}$.

For a random vector $\mathbf{X}$ over $(\Omega, \mathcal{F}, \mathsf{P})$ the probabilities $\mathsf{P}(\mathbf{X} \in B)$ for $B \subset \mathbb{R}^n$, s.t. the events $\{\mathbf{X} \in B\}$ are in $\mathcal{F}$, determines the law $\mathsf{P}_{\mathbf{X}}$. Viewed as a function of suitable n-dimensional sets, $\mathsf{P}_{\mathbf{X}}(B)$ translates the probability of $\{\mathbf{X} \in B\}$ into the probability of $B \subset \mathbb{R}^n$. For $n = 2$ this yields a geometric flavor often in terms of area of appropriate regions inside the Cartesian plane $\mathbb{R}^2$, see Fig. 3.1. Also in the current multidimensional setting there is a 1-to-1 correspondence between P and the CDF of the random vector $\mathbf{X}$ defined as

$$F_{\mathbf{X}}(x_1, \ldots, x_n) := \mathsf{P}(X_1 \leqslant x_1, \ldots, X_n \leqslant x_n), \tag{3.1}$$

called the **joint or multivariate CDF**. The event inside $\mathsf{P}(\cdot)$ is the shorthand notation for the intersection met in the fourth equality of the previous Remark (for $n \geqslant 2$). For $n = 2$, the bi-variate CDF $F_{\mathbf{X}} : \mathbb{R}^2 \to [0, 1]$ CDF satisfies the following properties:

1. is nondecreasing, i.e. for any $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ s.t. $\mathbf{x} \leqslant \mathbf{y}$ it follows $F_{\mathbf{X}}(\mathbf{x}) \leqslant F_{\mathbf{X}}(\mathbf{y})$, where the vector inequality means that $x_i \leqslant y_i$ for $i = 1, 2$ and at least one inequality is strict;
2. is right-continuous, i.e. $\lim_{h \downarrow 0} \left(\lim_{k \downarrow 0} F_{\mathbf{X}}(x_1 + h, x_2 + k)\right) = F_{\mathbf{X}}(x_1, x_2)$, for $h, k > 0$;
3. $\lim_{x_1, x_2 \to \infty} F_{\mathbf{X}}(x_1, x_2) := F_{\mathbf{X}}(\infty, \infty) = 1$ and $\lim_{x_i \to -\infty} F_{\mathbf{X}}(x_1, x_2) = 0$ for $i = 1, 2$;
4. $\mathsf{P}(a_1 < X_1 \leqslant b_1, a_2 < X_2 \leqslant b_2)$ is computed as

$$F_{\mathbf{X}}(b_1, b_2) - F_{\mathbf{X}}(a_1, b_2) - F_{\mathbf{X}}(b_1, a_2) + F_{\mathbf{X}}(a_1, a_2) \geqslant 0.$$

Properties 1–3 also hold for $n > 2$. Here a geometric interpretation based on identifying $\{\mathbf{X} \in B\}$ with regions in the Cartesian plane. Recall that $\mathbf{a} = (a_1, a_2) \leqslant (b_1, b_2) = \mathbf{b}$, and let $\mathbf{a}_1 = (a_1, b_2)$, $\mathbf{a}_2 = (b_1, a_2)$, see Fig. 3.2.

3.2 Discrete Random Vectors

Assume that $\mathbf{X} = (X, Y)$ and each of the component rvs has a countable support, $\mathcal{X} = \{x_1, x_2, \ldots\}$ and $\mathcal{Y} = \{y_1, y_2, \ldots\}$ respectively. The range of $\mathbf{X}$ is the set

$$\mathbf{X}(\Omega) = \{(x, y) \,|\, x \in \mathcal{X}, y \in \mathcal{Y}\},$$

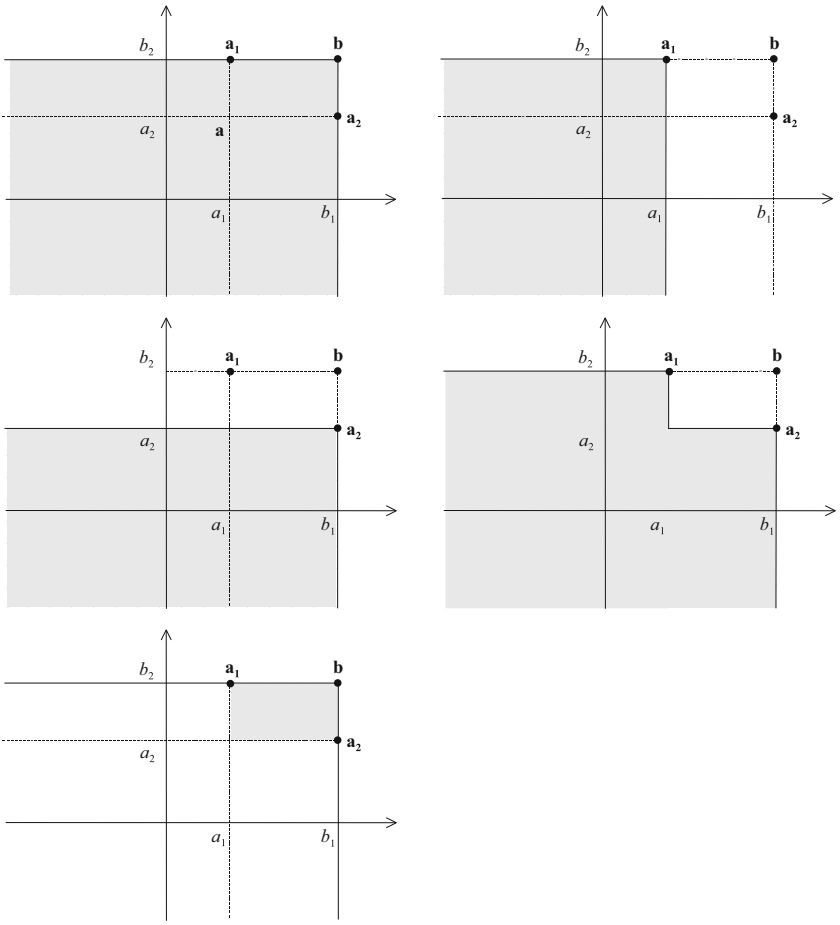


Fig. 3.2 Geometrical identification of event $\{a_1 < X_1 \leqslant b_1\} \cap \{a_2 < X_2 \leqslant b_2\}$: from the region on top-left subtract the union of regions on top-right and on middle-left, i.e., region on middle-right, then get the 'rectangle' $(\mathbf{a}, \mathbf{b}] = (a_1, b_1] \times (a_2, b_2]$ on bottom-left as the desired event

which equals the Cartesian product $\mathcal{X} \times \mathcal{Y}$ of two countable sets, and is itself a countable set.[1] Coordinates X and Y have their own discrete distributions given by masses $\mathsf{P}(X = x)$ and $\mathsf{P}(Y = y)$, and their CDFs $F_X(x)$ and $F_Y(y)$ are as usual step functions called **marginal distributions**, thus, $\mathbf{X}$ is said to have a **discrete distribution** with joint masses[2] $\mathsf{P}(X = x, Y = y)$. It holds:

$$\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \mathsf{P}(X = x, Y = y) = 1. \tag{3.2}$$

[1] There is a bijection $\mathbb{N} \times \mathbb{N} \to \mathcal{X} \times \mathcal{Y}$.

[2] The event inside P is $\{X = x\} \cap \{Y = y\}$.

Table 3.1 Joint masses of a bi-variate random vector

		$\mathcal{X}$		
		30	50	90
	100	0.1	0.3	0.1
$\mathcal{Y}$	110	0.1	0.2	0
	165	0	0.1	0.1

Indeed, for a fixed $x \in \mathcal{X}$ the events $\{X = x\} \cap \{Y = y_1\}, \{X = x\} \cap \{Y = y_2\}, \ldots$ form a partition of Ω. The same is true for $\{X = x_1\} \cap \{Y = y\}, \{X = x_2\} \cap \{Y = y\}, \ldots$ whenever $y \in \mathcal{Y}$ is fixed. Hence, the union of each of these sequence of events is the entire sample space. To compute the probability that $\mathbf{X}$ lies in some suitable subset $B \subset \mathcal{X} \times \mathcal{Y} \subset \mathbb{R}^2$ we simply sum $\sum_{(x,y)\in B} \sum \mathsf{P}(X = x, Y = y)$. Eventually, to recover the CDF of $\mathbf{X}$ from the joint masses we take:

$$F_{\mathbf{X}}(x_1, x_2) = \sum_{x \leqslant x_1} \sum_{y \leqslant x_2} \mathsf{P}(X = x, Y = y).$$

Note that $\mathsf{P}(\mathbf{X} \in \mathbf{X}(\Omega)^c) = 0$.

Example 3.2 To forecast the future asset prices of two stock indexes, let define the discrete random vector $\mathbf{X} = (X, Y)$ over $(\Omega, \mathcal{F})$ with $\mathcal{X} = \{30, 50, 90\}$ and $\mathcal{Y} = \{100, 110, 165\}$. Let us assign the probabilities

$$\mathsf{P}(X = 30) = 0.2,\ \mathsf{P}(X = 50) = 0.6,\ \mathsf{P}(X = 90) = 0.2$$
$$\mathsf{P}(Y = 100) = 0.5,\ \mathsf{P}(Y = 110) = 0.3,\ \mathsf{P}(Y = 165) = 0.2$$

to each random future quotes. The masses in the first line form the marginal distribution of X, and the second line corresponds to the marginal distribution of Y.

Example 3.3 Each entry in Table 3.1 is a joint mass $\mathsf{P}(X = x, Y = y)$ for every pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$.

- Adding the entries in each row we get the probability masses $\mathsf{P}(X = x)$. E.g., for $X = 30$ we get $\mathsf{P}(X = 30, Y = 100) + \mathsf{P}(X = 30, Y = 110) + \mathsf{P}(X = 30, Y = 165)$ is equal to $\mathsf{P}(\{X = 30, Y = 100\} \cup \{X = 30, Y = 110\} \cup \{X = 30, Y = 165\})$, which in turn gets $\mathsf{P}(X = 30) = 0.2$. This is because the union is based on pairwise disjoint events[3] and $\cup_{y\in\mathcal{Y}}\{X = 30, Y = y\} = \{X = 30\}$.
- Adding the entries in each column we get the probability masses $\mathsf{P}(Y = y)$, by the same reasoning as above.

[3] Recall that $\{X = 30, Y = y\}$ is the intersection $\{X = 30\} \cap \{Y = y\}$.

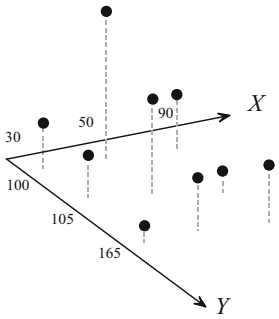


Fig. 3.3 Joint masses of the bi-variate random vector in Example 3.2. Plotting the couples (x, y) against probability masses $\mathsf{P}(X = x, Y = y)$ gives a finite number of points in the three-dimensional Euclidean space $\mathbb{R}^3$. Their heights are just the values of $\mathsf{P}(X = x, Y = y)$

- The double sum in (3.2) can be performed in whatever order: we can first sum the joint masses either as in item (a) or as in item (b), then summing again we get 1.

Choosing $B = (10, 55] \times (105, 170]$, to compute

$$\mathsf{P}(\mathbf{X} \in B) = \mathsf{P}((X, Y) \in B) = \mathsf{P}(10 < X \leqslant 55, 105 < Y \leqslant 170),$$

we simply take the double sum

$$\sum_{(x,y)\ \in B} \sum \mathsf{P}(X = x, Y = y) = \sum_{x=30,50} \sum_{y=110,165} \mathsf{P}(X = x, Y = y) = 0.4.$$

A graphical illustration is depicted in Fig. 3.3.

3.3 Continuous Random Vectors

A random vector $\mathbf{X} = (X, Y)$ with (absolutely) continuous components X, Y is said to be **continuous**. The marginal distributions are given by the densities $f_X(x)$ and $f_Y(x)$ s.t. their CDFs are $F_X(x) = \int_{-\infty}^{x} f_X(u)\mathrm{d}u$ and $F_Y(y) = \int_{-\infty}^{y} f_Y(u)\mathrm{d}u$. Vice versa, we also have $f_X(x) = F_X'(x)$ and $f_Y(y) = F_Y'(y)$. There is a **continuous joint distribution** if the multivariate CDF $F_{\mathbf{X}}$ is continuous and

$$\begin{aligned} F_{\mathbf{X}}(x, y) &= \int_{(-\infty,x]\times(-\infty,y]} f_{\mathbf{X}}(u, v)\mathrm{d}u\mathrm{d}v \\ &= \int_{-\infty}^{x}\int_{-\infty}^{y} f_{\mathbf{X}}(u, v)\mathrm{d}u\mathrm{d}v, \end{aligned} \tag{3.3}$$

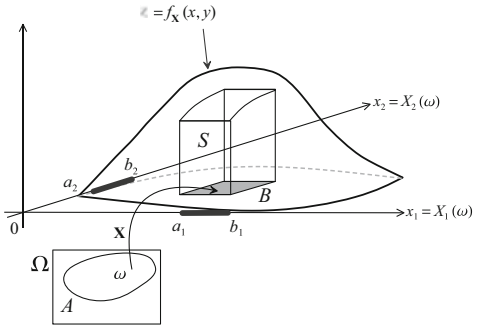


Fig. 3.4 The bell shaped surface is the density's graphic $\{(x, y, z) \in \mathbb{R}^3 \mid z = f_{\mathbf{X}}(x, y)\}$. The probability that $\mathbf{X}$ attains values in B equals the volume of the solid S, computed via double integration

where $f_{\mathbf{X}}(x, y) \geqslant 0$, for every $(x, y) \in \mathbb{R}^2$, is the **joint density function** of two variables, which must be integrable and satisfying[4]

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y) \mathrm{d}x \mathrm{d}y = 1.$$

Figure 3.4 depicts an example of a bi-variate joint density function. Given $f_{\mathbf{X}}$, we can compute the probability that a bi-variate random vector $\mathbf{X}$ takes values on the rectangle $B = (a_1, b_1] \times (a_2, b_2]$ as

$$\mathsf{P}(a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f_{\mathbf{X}}(x, y) \mathrm{d}x \mathrm{d}y. \tag{3.4}$$

The probability remains the same over rectangles with 'sides' $[a_i, b_i]$ or $[a_i, b_i)$ or (a_i, b_i), for $i = 1, 2$. In the discrete case we can recover[5] the marginal masses by the joint masses. Similarly, in the continuous case the two densities $f_X(x)$, $f_Y(y)$ satisfies:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y) \mathrm{d}x \mathrm{d}y = \int_{-\infty}^{\infty} f_X(x) \mathrm{d}x = \int_{-\infty}^{\infty} f_Y(y) \mathrm{d}y = 1.$$

The marginal densities are $f_X(x) = \int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y) \mathrm{d}y$ and $f_Y(y) = \int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y) \mathrm{d}x$. To compute $\mathsf{P}(\mathbf{X} \in B)$ over $B = (a_1, b_1] \times (a_2, b_2]$ we may first integrate the joint density over $(a_1, b_1]$ and then over $(a_2, b_2]$, i.e. $\int_{a_2}^{b_2} \left(\int_{a_1}^{b_1} f_{\mathbf{X}}(x, y) \mathrm{d}x \right) \mathrm{d}y$; we may as well reverse the order of integration. A geometric illustration is depicted in Fig. 3.5.

The joint CDF and density defined for bi-variate random vectors can be easily extended to the n-dimensional case. To get the joint density by the multivariate

[4] The first integral above is a double integral and is equal to two iterated integrals by Fubini's Theorem, see Appendix to this chapter and Appendix I.

[5] We need only to sum once, e.g. $\mathsf{P}(X = x) = \sum_{y \in \mathcal{Y}} \mathsf{P}(X = x, Y = y)$ for a fixed $x \in \mathcal{X}$.

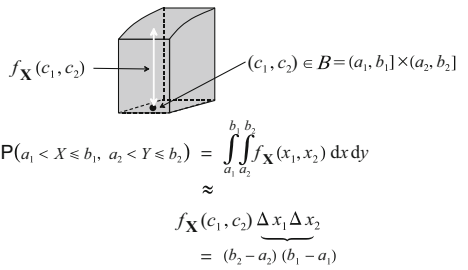


Fig. 3.5 The volume of S approximated by that of a parallelepiped with base given by the rectangle $B = (a_1, b_1] \times (a_2, b_2]$ and height given by the value $f_{\mathbf{X}}(c_1, c_2)$ of the joint density at some point (c_1, c_2) inside B

CDF we proceed as follow. Take $\mathbf{x} \in \mathbb{R}^n$ and define $\mathbf{e}_i \in \mathbb{R}^n$ as the n-dimensional vector with all entries zero except the ith which is 1. The partial derivative of[6] $F_{\mathbf{X}}$ with respect to the ith coordinate x_i is[7]

$$\frac{\partial F_{\mathbf{X}}}{\partial x_i}(\mathbf{x}) = \lim_{h \to 0} \frac{F_{\mathbf{X}}(\mathbf{x} + h\mathbf{e}_i) - F_{\mathbf{X}}(\mathbf{x})}{h}, \qquad h \in \mathbb{R}.$$

If $F_{\mathbf{X}}$ is differentiable, then there exist all partial derivatives.[8] The continuity of $f_{\mathbf{X}}(\mathbf{x})$ suffices to guarantee that it can be recovered from the multivariate CDF performing n partial differentiation:

$$\frac{\partial^n F_{\mathbf{X}}(x_1, \ldots, x_n)}{\partial x_1 \cdots \partial x_n} = f_{\mathbf{X}}(x_1, \ldots, x_n). \tag{3.5}$$

The order of partial derivatives is irrelevant. For example,

$$\frac{\partial^2 F_{\mathbf{X}}(x_1, x_2)}{\partial x_1 \partial x_2} = \tfrac{\partial}{\partial x_2}\big[\tfrac{\partial}{\partial x_1}[F_{\mathbf{X}}(x_1, x_2)]\big] = f_{\mathbf{X}}(x_1, x_2) = \frac{\partial^2 F_{\mathbf{X}}(x_1, x_2)}{\partial x_2 \partial x_1}.$$

If the joint CDF is continuous at $\mathbf{y} \in \mathbb{R}^n$ we could write

$$\lim_{x_1, \ldots, x_n \to y_1, \ldots, y_n} F_{\mathbf{X}}(x_1, \ldots, x_n) = F_{\mathbf{X}}(y_1, \ldots, y_n) \text{ iff } \lim_{\mathbf{x} \to \mathbf{y}} F_{\mathbf{X}}(\mathbf{x}) = F_{\mathbf{X}}(\mathbf{y}).$$

This means that we can made $F_{\mathbf{X}}(\mathbf{x})$ stay as close to $F_{\mathbf{X}}(\mathbf{y})$ as we like by restricting[9] $\mathbf{x}$ to be close enough to $\mathbf{y}$. Using the Euclidean distance

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}, \quad \text{for every } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

[6] Formally the derivative of the function $g(h) := \frac{F_{\mathbf{X}}(\mathbf{x} + h\mathbf{e}_i) - F_{\mathbf{X}}(\mathbf{x})}{h}$.

[7] Observe that $\mathbf{x} + h\mathbf{e}_i$ is the vector with all the entries equal to those of $\mathbf{x}$, except the ith which is $x_i + h$.

[8] Moreover, each $\frac{\partial F_{\mathbf{X}}}{\partial x_i}(\mathbf{x})$ is a continuous function of $\mathbf{x}$.

[9] The vector $\mathbf{x}$ can approach $\mathbf{y}$ over different directions.

we can rephrase the above definition as: for any $\epsilon > 0$ the choice $d(\mathbf{x}, \mathbf{y}) < \delta$, for $\delta > 0$, does imply $|F_{\mathbf{X}}(\mathbf{x}) - F_{\mathbf{X}}(\mathbf{y})| < \epsilon$. A continuous joint CDF $\boldsymbol{F}_{\mathbf{X}}$ is continuous in each variable separately. E.g., for $n = 2$ we have $g(\cdot) := F_{\mathbf{X}}(\cdot, x_2)$ is continuous as a function of the first variable only, for every fixed $x_2 \in \mathbb{R}$, and similarly for the second variable.[10]

Remark 3.3 For general multivariate CDF $F_{\mathbf{X}}(x_1, \ldots, x_n)$ that are (absolutely) continuous we have

$$f_{\mathbf{X}}(x_1, \ldots, x_n) = \frac{\partial^n F_{\mathbf{X}}}{\partial x_1 \cdots \partial x_n}(x_1, \ldots, x_n)$$

and $\mathrm{d}^n F_{\mathbf{X}}(x) = \partial^n F_{\mathbf{X}}(x) =: \partial_{x_1} \cdots \partial_{x_n} F_{\mathbf{X}}(x)$, where for $\mathbf{x} = (x_1, \ldots, x_n)$ the subscripts denotes the variable w.r.t. which we take partial differentiation. It is conventional to write $\mathrm{d}^n F_{\mathbf{X}}(x)$ as $\mathrm{d}F_{\mathbf{X}}(x)$ is no confusion arises w.r.t. partial differentiation and total differentiation. We adhere to this usage from now on.

With all these in mind, one obtains the first marginal density function $f_{X_1}(x_1)$ from the multivariate CDF of a random vector $\mathbf{X} = (X_1, \ldots, X_n)$ as follows:

$$\begin{aligned} f_{X_1}(x_1) &= \frac{\mathrm{d}F_{\mathbf{X}}}{\mathrm{d}x_1}(x_1, \infty, \ldots, \infty) \\ &= \frac{\mathrm{d}}{\mathrm{d}x_1} \int_{-\infty}^{x_1} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(u_1, u_2, \ldots, u_n) \mathrm{d}u_1 \mathrm{d}u_2 \cdots \mathrm{d}u_n \\ &= \underbrace{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty}}_{n-1 \text{ iterated integrals}} f_{\mathbf{X}}(x_1, u_2, \ldots, u_n) \mathrm{d}u_2 \cdots \mathrm{d}u_n. \end{aligned}$$

The second equality is due to the Leibniz's formula for differentiating integrals, see Appendix L. Interpretation: after we write

$$F_X(x_1, x_2, \ldots, x_n) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_n} f_{\mathbf{X}}(u_x, u_2, \ldots, u_n) \mathrm{d}u_1 \mathrm{d}u_2 \cdots \mathrm{d}u_n \tag{3.6}$$

and evaluate $x_2, \ldots, x_n$ at ∞, to get the marginal density (w.r.t. x_1) from the joint CDF we just differentiate this expression (w.r.t. x_1) so that the corresponding integral cancel out. Then, we integrate the joint density inside the remaining $n - 1$ iterated integrals ranging all over $u_i \in \mathbb{R}$, for $i = 2, \ldots, n$. Obtaining the other marginal densities is done similarly.

Remark 3.4 (Probability Zero Over Special Subsets) When $f_{\mathbf{X}}(x, y)$ is continuous as a function of two variables, we know that the marginal densities $f_X(x)$ and

[10] Warning: do not confuse $\mathrm{d}^n F_{\mathbf{X}}(x)$ with the total differential, e.g. in the bi-variate case $\mathrm{d}^2 F_{\mathbf{X}}$ is not $\frac{\partial F_{\mathbf{X}}}{\partial x}\mathrm{d}x + \frac{\partial F_{\mathbf{X}}}{\partial y}\mathrm{d}y$.

$f_Y(y)$ are continuous functions. Henceforth, $\mathsf{P}(X = x) = \mathsf{P}(Y = y) = 0$ and $\mathsf{P}(X = x, Y = y) = 0$ because $\{X = x\} \cap \{Y = y\}$ is a subset of zero-probability events $\{X = x\}, \{Y = y\}$ yielding $\mathsf{P}(X = x, Y = y) \leqslant \mathsf{P}(X = x) = \mathsf{P}(Y = y)$, see Proposition 1.1, Pr2, in Sect. 1.5 . As a generalization of the well know result in the one-dimensional case we have:

$$\begin{aligned}\mathsf{P}(a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2) &= \mathsf{P}(a_1 \leqslant X \leqslant b_1, a_2 \leqslant Y \leqslant b_2)\\ &= \mathsf{P}(a_1 \leqslant X < b_1, a_2 \leqslant Y < b_2)\\ &= \mathsf{P}(a_1 < X < b_1, a_2 < Y < b_2)\\ &= \int_{a_1}^{b_1}\int_{a_2}^{b_2} f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y.\end{aligned}$$

In fact, singletons $\{(x, y)\}$, segments, lines and planes are subsets of $\mathbb{R}^2$ where a bi-variate continuous density assigns a zero probability if properly integrated.

Remark 3.5 It is well known that $\mathsf{P}(x < X \leqslant x + \Delta x) \approx f_X(x)\Delta X$, i.e.

$$\mathsf{P}(x < X \leqslant x + \Delta x) = f_X(x)\Delta X + o(\Delta x)$$

for $\Delta x > 0$ and $\lim_{\Delta x \to 0} \frac{o(\Delta x)}{\Delta x} = 0$. In the bi-variate case the situation is more involved:

$$\mathsf{P}(x < X \leqslant x+\Delta x, y < Y \leqslant y+\Delta y) = f_X(x, y)\Delta x\Delta y+o((\Delta x, \Delta y)) \qquad (3.7)$$

where now $\lim_{\Delta x \to 0, \Delta y \to 0} \frac{o((\Delta x, \Delta y))}{\Delta x \Delta y} = 0$. Thus, dividing both sides of equation (3.7) by $\Delta x \Delta y$ and writing the expression of $\mathsf{P}(x < X \leqslant \Delta x, y < Y \leqslant \Delta y)$ in term of the bi-variate CDF we get alternatively that $f_{\mathbf{X}}(x, y) = \frac{\partial^2 F_{\mathbf{X}}}{\partial x \partial y}(x, y)$.

3.4 Stochastic Dependence/Independence

We know how joint masses or CDFs can generate univariate marginal masses or densities. The converse is not true in general, unless some stronger condition is imposed on the rvs forming the random vector $\mathbf{X}$, since knowledge of the marginal distributions is not *sufficient* in determining the joint distribution. However, there is a special condition that could do the job. When events $\{X \leqslant x\}$ and $\{Y \leqslant y\}$ can occur together, i.e. $\{X \leqslant x\} \cap \{Y \leqslant y\}$ is not necessarily the impossible event, it could be the case that they *do not influence* each other. This may be formalized by saying that their joint probability of occurrence factors:

$$\mathsf{P}(X \leqslant x, Y \leqslant y) = \mathsf{P}(X \leqslant x)\mathsf{P}(Y \leqslant y). \qquad (3.8)$$

The interpretation is straightforward: conditions $X \leqslant x$ and $Y \leqslant y$ are both verified with a probability computed through the the individual $\mathsf{P}(X \leqslant x)$ and $\mathsf{P}(Y \leqslant y)$,

and this sounds like $\{X \leqslant x\}$ does not affect $\{Y \leqslant y\}$, and vice versa regardless the common content of the resulting joint event.[11] Thus, although the joint behavior of X and Y is driven by a joint probability distribution the knowledge of X's values does not affect the knowledge of Y's values, provided condition (3.8) holds. Then there is no mutual information on the events induced by component rvs in $\mathbf{X}$. The opposite happens when for any pair of scenarios $\omega_1, \omega_2 \in \Omega$:

- $(X(\omega_1) - X(\omega_2))(Y(\omega_1) - Y(\omega_2)) \geqslant 0$;
- $(X(\omega_1) - X(\omega_2))(Y(\omega_1) - Y(\omega_2)) \leqslant 0$.

In the first case the rvs are called **comonotone** and they move in the same direction, while in the second case they are called **antimonotone** and move in opposite direction. This simply means they are dependent, and knowing events concerning X affects events induced by Y. If X, Y are neither comonotone nor antimonotone and the mutual information is not *maximal*, we can consider the following degree of mutual dependence: let $A = \{X \in B_1\}$, $B = \{Y \in B_2\}$ where $B_1, B_2 \subset \mathbb{R}$, then we always have:

$$\max\{0, \mathsf{P}(A) + \mathsf{P}(B) - 1\} \leqslant \mathsf{P}(A \cap B) \leqslant \min\{\mathsf{P}(A), \mathsf{P}(B)\}. \tag{3.9}$$

In fact, $A \cap B \subset A$ and $A \cap B \subset B$ so applying the probability function entails

$$\mathsf{P}(A \cap B) \leqslant \mathsf{P}(A) \text{ and } \mathsf{P}(A \cap B) \leqslant \mathsf{P}(B),$$

and we have the inequality in the right-hand side of (3.9). Moreover by Proposition 1.1, Pr4, we have

$$\mathsf{P}(A \cap B) = \mathsf{P}(A) + \mathsf{P}(B) - \mathsf{P}(A \cup B)$$

and replacing $\mathsf{P}(A \cup B)$ with $\mathsf{P}(\Omega) = 1$ we have $\mathsf{P}(A \cap B) \geqslant \mathsf{P}(A) + \mathsf{P}(B) - 1$, unless $\mathsf{P}(A) + \mathsf{P}(B) - 1 < 0$ thus we obtain the inequality in the left-hand side of (3.9). If $A \cap B \neq \varnothing$, and the two events may occur together, the joint probability is bounded above by one of the individual probability. Now, the following is crucial in the final understanding of possible dependence.

Theorem 3.1 (Conditional Probability) *Let* $(\Omega, \mathcal{F}, \mathsf{P})$ *a probability space. The set function* $\mathsf{P}(\cdot \mid B) : \mathcal{F} \to [0, 1]$ *defined by*

$$\mathsf{P}(A \mid B) := \frac{\mathsf{P}(A \cap B)}{\mathsf{P}(B)}, \qquad \textit{for all } A, B \in \mathcal{F}, \ \textit{s.t. } \mathsf{P}(B) > 0, \tag{3.10}$$

is a probability measure.

Since the idea underlying conditional probability is to suppose B is occurred and then use this assumption as 'new information' to update the value of $\mathsf{P}(A)$, whenever

[11] We emphasize how this concept is based on evaluating probability, instead of looking at the joint event as a (not necessarily empty) set of certain scenarios.

Table 3.2 Summary of dependence/independence

	$A \cap B = \varnothing$	$A \cap B \neq \varnothing$
$\mathsf{P}(A) > 0$ and $\mathsf{P}(B) > 0$	Dependent	Independent if $0 < \mathsf{P}(A \cap B) = \mathsf{P}(A)\mathsf{P}(B)$
		Dependent if $0 = \mathsf{P}(A \cap B)$ or $0 < \mathsf{P}(A \cap B) \neq \mathsf{P}(A)\mathsf{P}(B)$
$\mathsf{P}(A) = 0$ and/or $\mathsf{P}(B) = 0$	Independent	Independent if $0 = \mathsf{P}(A \cap B)$
		Dependent if $0 < \mathsf{P}(A \cap B)$

these events do not affect each other probabilities (though they are compatible) we have:

$$\mathsf{P}(A \mid B) = \mathsf{P}(A) \iff \mathsf{P}(A \cap B) = \mathsf{P}(A)\mathsf{P}(B). \tag{3.11}$$

So there is no influence of B on A but they may occur together and we say they are **independent** and write $A \perp\!\!\!\perp B$. Moreover, assuming $\mathsf{P}(A) > 0$ the conditioning can be reversed, $\mathsf{P}(B \mid A) := \frac{\mathsf{P}(A \cap B)}{\mathsf{P}(A)}$, and with no mutual influence we again have $\mathsf{P}(B \mid A) = \mathsf{P}(B)$ iff the joint probability factors. Hence, if the knowledge of one event does not affect the knowledge of the other both inequalities in (3.9) become equalities and we conclude that A, B are independent events, see Table 3.2 for a summary.

Remark 3.6 When two events are incompatible, $A \cap B = \varnothing$, then their joint probability is zero and independence holds iff *at least* one of the individual probability is zero too. But for $\mathsf{P}(A) = 0$ the condition $0 = \mathsf{P}(A \cap B) = \mathsf{P}(A)\mathsf{P}(B) = 0$ implies the conditional probability in (3.10) cannot be defined if in addition $\mathsf{P}(B) = 0$. The same is true for the reversed conditional probability with $\mathsf{P}(B) = 0$. In words, we can define independence without conditional probability, unless we require $\mathsf{P}(A \mid B) = \mathsf{P}(A)$ if $\mathsf{P}(B) = 0$ and the same for reversed conditioning.

For a random vector $\mathbf{X} = (X, Y)$ with two independent rvs,

$$\underbrace{\mathsf{P}(X \in B_1, Y \in B_2)}_{=\mathsf{P}_{\mathbf{X}}(B_1 \times B_2)} = \underbrace{\mathsf{P}(X \in B_1)}_{=\mathsf{P}_X(B_1)} \underbrace{\mathsf{P}(Y \in B_2)}_{=\mathsf{P}_Y(B_2)} \quad \text{for every } B_1, B_2 \subset \mathbb{R},$$

i.e. the joint distribution $\mathsf{P}_{\mathbf{X}}(B_1 \times B_2)$ describing X, Y as a pair $(X(\omega), Y(\omega)) = (x, y) \in B_1 \times B_2$ within a subset of the plane $\mathbb{R}^2$ given by a Cartesian product,[12] is just the product of the probabilities $\mathsf{P}_X(B_1)$ and $\mathsf{P}_Y(B_2)$ based on the marginal distributions. Thus the joint CDF factors too (see Eq. (3.8)):

$$F_{\mathbf{X}}(x, y) = F_X(x)F_Y(y), \quad \text{for every pair } (x, y) \in \mathbb{R}^2. \tag{3.12}$$

[12] E.g. $[a_1, b_1] \times [a_2, b_2]$.

Moral: the independence assumption enable us to build the joint distribution of a random vector starting from the marginals.

- If the bi-variate random vector $\mathbf{X}$ is continuous then using the marginal densities we get the joint density as $f_{\mathbf{X}}(x, y) = f_X(x) f_Y(y)$. Indeed by differentiating (3.12) we get

$$\frac{\partial^2 F_{\mathbf{X}}}{\partial x \partial y}(x, y) = \frac{\mathrm{d}F_X}{\mathrm{d}x}(x) \frac{\mathrm{d}F_Y}{\mathrm{d}y}(y),$$

  which is just the product of the marginal densities.
- For a discrete bi-variate random vector we easily have

$$\sum_{x_i \leqslant x} \sum_{y_j \leqslant y} \mathsf{P}(X = x_i, Y = y_j) = \sum_{x_i \leqslant x} \mathsf{P}(X = x_i) \sum_{y_j \leqslant y} \mathsf{P}(Y = y_j)$$

  and the joint masses are given by the product of the marginals $\mathsf{P}(X = x_i)$ and $\mathsf{P}(Y = y_j)$.

Any family of rvs $\{X_1, \ldots, X_n\}$ is said to be independent whenever their joint distribution factors,

$$\mathsf{P}(X_1 \in B_1, \ldots, X_n \in B_n) = \mathsf{P}(X_1 \in B_1) \cdots \mathsf{P}(X_n \in B_n), \tag{3.13}$$

for all suitable subsets $B_1, \ldots, B_n \subset \mathbb{R}$. This can be extended to more than a finite family of rvs. We may take an **indexed family** $(X_i)_{i \in I}$, where $I \neq \varnothing$ is a set of indices, and say that the whole family is independent iff any *finite* subfamily $\{X_{i_1}, \ldots, X_{i_n}\}$, for indices $i_1, \ldots, i_n \in I$, is independent in the sense of (3.13):

$$\boxed{X_i \perp\!\!\!\perp X_j, \quad \text{for every } i, j \in \{i_1, \ldots, i_n\}}.$$

If a family of rvs is independent, then any pair, triple, etc. of rvs taken from this family is also independent. The converse is not true, since we may take X_i, X_j as independent rvs for every $i \neq j$ and $i, j = 1, \ldots, n$ but the whole family will not be independent.

Remark 3.7 For a class of n rvs independence implies pairwise independence as follows: write equality (3.13), choose two rvs, say X_i, X_j for $i \neq j$, then choose $B_k = \mathbb{R}$ for all $k \neq i, j$ and recall that $\mathsf{P}(X_k \in \mathbb{R}) = 1$.

Moreover, independence is not transitive: if $\{X_1 \in B_1\}, \{X_2 \in B_2\}$ are independent, and $\{X_2 \in B_2\}, \{X_3 \in B_3\}$ are independent, we may have $\{X_1 \in B_1\}, \{X_3 \in B_3\}$ as dependent events. The pairs X_1, X_2 and X_2, X_3 of rvs are independent but the pair X_1, X_3 is not.

Remark 3.8 Two events $A, B \in \mathcal{F}$ are independent iff A and B^c are independent and also A^c and B^c are. In fact, $\mathsf{P}(B^c) = 1 - \mathsf{P}(B)$ and $\mathsf{P}(A) = \mathsf{P}(A \cap B) + \mathsf{P}(A \cap B^c)$ and it follows:

$$\begin{aligned}\mathsf{P}(A \cap B^c) &= \mathsf{P}(A) - \mathsf{P}(A \cap B) = \mathsf{P}(A) - \mathsf{P}(A)\mathsf{P}(B) \\ &= \mathsf{P}(A)(1 - \mathsf{P}(B)) = \mathsf{P}(A)\mathsf{P}(B^c).\end{aligned}$$

Since A, B are independent iff A, B^c are too, it follows that B^c, A are independent iff B^c, A^c are independent.

Now, let $A = \{X_1 \in B_1\}$, $B = \{X_2 \in B_2\}$ and $C = \{X_3 \in B_3\}$. We can assume $C = A^c$ and by Remark 3.8 independence of A and B is equivalent to independence of B and C. However A is not independent of its own complement C, since $\mathsf{P}(A \cap C) = \mathsf{P}(\varnothing) = 0$ which is not equal to $\mathsf{P}(A)\mathsf{P}(C)$ in general. The following is especially useful in the treatment of conditional distributions, see Chap. 6.

Lemma 3.1 (Sequential Conditioning) *Let $A_1, \ldots, A_n$ be events in $\mathcal{F}$. Then*

$$\begin{aligned}\mathsf{P}(A_1 \cap \cdots \cap A_n) &= \mathsf{P}(A_1) \\ &\quad \times \mathsf{P}(A_2 \mid A_1) \\ &\quad \times \mathsf{P}(A_3 \mid A_1 \cap A_2) \\ &\quad \vdots \\ &\quad \times \mathsf{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).\end{aligned} \tag{3.14}$$

Indeed, for $n = 2$ formula (3.14) reduces to formula (3.10). Next, assume the statement holds for some integer $n - 1 > 1$ and define $B = A_1 \cap \cdots \cap A_{n-1}$. We have:

$$\begin{aligned}\mathsf{P}(B \cap A_n) &= \mathsf{P}(B)\mathsf{P}(A_n \mid B) \\ &= \mathsf{P}(A_1 \cap \cdots \cap A_{n-1})\mathsf{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}) \\ &= \mathsf{P}(A_1)\mathsf{P}(A_2 \mid A_1) \\ &\quad \vdots \\ &\quad \times \mathsf{P}(A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2})\mathsf{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).\end{aligned}$$

This proves that the statement also holds for $n \in \mathbb{N}$. Independence among rvs is preserved under transformations: there is no reason why a rv $f(X)$ provides information on a rv $g(Y)$, whenever X, Y provides no information about each other, i.e., when they are independent.

Theorem 3.2 *Given two independent rvs X, Y over the same probability space, then $f(X)$ is independent of $g(Y)$, for any choice of suitable functions*[13] $f, g : \mathbb{R} \to \mathbb{R}$.

If X, Y are discrete rvs we can prove Theorem 3.2 easily. First, let $A = \{x \in \mathbb{R} \mid f(x) = u\}$ and $B = \{x \in \mathbb{R} \mid g(y) = v\}$, then write down the joint mass of the bi-variate random vector $\mathbf{Z} = (f(X), g(Y))$:

$$\begin{aligned}\mathsf{P}(f(X) = u, g(Y) = v) &= \mathsf{P}(X \in A, Y \in B)\\ &= \sum_{x\in A}\sum_{y\in B} \mathsf{P}(X = x, Y = y)\\ &= \sum_{x\in A} \mathsf{P}(X = x) \sum_{y\in B} \mathsf{P}(Y = y)\\ &= \mathsf{P}(X \in A)\mathsf{P}(Y \in B) = \mathsf{P}(f(X) = u)\mathsf{P}(g(Y) = v).\end{aligned}$$

Theorem 3.2 might be extended to $n > 2$, see Appendix. A note on terminology: n independent rvs $X_1, \ldots, X_n$ having the same distribution are denoted **IID**, the acronym of independent and identically distributed. Be warned that whether two rvs are independent has no relation with whether they share the same distribution: we can have any combination of independence/dependence and identical law/not identical law, out of four possible cases.

3.5 More on Discrete and Continuous Random Vectors

Given a random vector $\mathbf{X} = (X_1, \ldots, X_n)$, we recall the following.

- $\mathbf{X}$ is discrete iff there is an (at most) countable $C \subset \mathbb{R}^n$ s.t.

$$\mathsf{P}(\mathbf{X} \in C) = \mathsf{P}((X_1, \ldots, X_n) \in C) = 1,$$

  where C may be taken as the Cartesian product of n countable subsets $\mathcal{X}_i = \{x_{i1}, x_{i2}, \ldots\}$ for $i = 1, \ldots, n$ so that $(X_1, \ldots, X_n) \in C$ means $X_i = x_{ij}$ for $j = 1, 2, \ldots$ and fixed i. Thus the event $\{\mathbf{X} \in C\}$ is the disjoint union of intersection of events $\{X_i = x_{ij}\}$ for all indices i, j. Indeed, if its probability is 1 then also $\mathsf{P}(X_i \in \mathcal{X}_i) = 1$, compare with Theorem 2.2 Sect. 2.2.
- $\mathbf{X}$ is absolutely continuous iff there exists a joint density s.t.

$$\mathsf{P}(X_1 \leqslant x_1, \ldots, X_n \leqslant x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_{\mathbf{X}}(u_1, \ldots, u_n)\mathrm{d}u_1 \cdots \mathrm{d}u_n. \tag{3.15}$$

[13] The functions must be Borel-measurable.

Be warned that:

- if $\mathbf{X}$ is discrete in the sense above, then each rv X_i is discrete and the converse is also true;
- if $\mathbf{X}$ is absolutely continuous then each rv X_i is also absolutely continuous and each marginal density $f_{X_i}(x_i)$ is obtained by sending to ∞ all x_j in (3.15), for $j \neq i$, then differentiating w.r.t. x_i, since the left-hand side is just the multivariate CDF $F_{\mathbf{X}}(x_1, \ldots, x_n)$. But the converse 'each X_i is absolutely continuous then $\mathbf{X}$ is absolutely continuous too' is false in general unless the family $\{X_1, \ldots, X_n\}$ forming the random vector is independent.

For independent rvs X_i picked from $\mathbf{X}$ it holds[14]

$$F_{\mathbf{X}}(x_1, \ldots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n), \tag{3.16}$$

which extends formula (3.12), Sect. 3.4, to dimension $n > 2$.

Theorem 3.3 *For an absolutely continuous random vector $\mathbf{X} = (X_1, \ldots, X_n)$ the rvs $X_1, \ldots, X_n$ are independent iff the joint density factors,*

$$f_{\mathbf{X}}(x_1, \ldots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n). \tag{3.17}$$

Taking independent and absolutely continuous $X_1, \ldots, X_n$, multiplying their marginal densities we always obtain the joint density of an absolutely continuous random vector $\mathbf{X} = (X_1, \ldots, X_n)$. For the multidimensional setting we can summarize independence in terms of CDFs.

Theorem 3.4 *Let $F_i : \mathbb{R} \to [0, 1]$ be nondecreasing, right-continuous and s.t. $F_i(-\infty) = 0$ and $F_i(\infty) = 1$, for $i = 1, \ldots, n$. Then there exists a probability space $(\Omega, \mathcal{F}, \mathsf{P})$ and rvs $X_1, \ldots, X_n$ here defined that are independent with $F_i = F_{X_i}$. The probabilities on $\mathbb{R}$ to which they corresponds 1-to-1 are the X_i's laws and $\mathsf{P}_{X_i}((-\infty, \mathbf{x}]) = F_{X_i}(x_i)$. Moreover, there exists a random vector $\mathbf{X} = (X_1, \ldots, X_n)$ s.t. $F_{\mathbf{X}}(x_1, \ldots, x_n) = \prod_{i=1}^{n} F_{X_i}(x_i)$.*

More generally, let $(X_i)_{i \in I}$ be an indexed collection of rvs and consider a finite number of indices $i_1, \ldots, i_n \in I$. The joint distribution on $\mathbb{R}^n$

$$\mathsf{P}_{i_1 \ldots i_n}(B_1 \times \cdots \times B_n) := \mathsf{P}(X_{i_1} \in B_1, \ldots X_{i_n} \in B_n) \tag{3.18}$$

is said to be a **finite dimensional distribution** (hereafter FIDIS) of the collection for any $n \geqslant 2$. In term of joint CDF this reads:

$$F_{i_1 \ldots i_n}(x_1, \ldots, x_n) := \mathsf{P}(X_{i_1} \leqslant x_1, \ldots X_{i_n} \leqslant x_n),$$

[14] The sufficiency is obvious. For the necessity, the fact that the product of $F_{X_i}(x_i)$ equals a joint CDF is a more delicate issue of real analysis.

corresponding to the random vector $\mathbf{X} = (X_{i_1}, \ldots, X_{i_n})$ obtained from the indexed family. The family is independent iff any FIDIS factors and thus $F_{i_1 \ldots i_n}(x_1, \ldots, x_n) = \prod_{k=1}^{n} F_{X_{i_k}}(x_k)$, for any choice $i_1, \ldots, i_n$ with $n \geqslant 2$. We conclude this section with a classical result on obtaining the univariate density of a rv given by the sum (vector-to-scalar transformation) of two other rvs. Let X_1, X_2 be two independent and absolutely continuous rvs. Thus we know that $\mathbf{X} = (X_1, X_2)$ is an absolutely continuous random vector with joint density $f_{\mathbf{X}}(x_1, x_2)$, therefore we have:

Theorem 3.5 *The sum $X_1 + X_2$ of two independent and absolutely continuous rvs is an absolutely continuous rv with density*

$$f_{X_1+X_2}(u) = \int_{-\infty}^{\infty} f_{X_1}(u - x_2) f_{X_2}(x_2) \mathrm{d}x_2, \quad u \in \mathbb{R}. \tag{3.19}$$

Observe that $g(u) = f_{X_1+X_2}(u)$ is a density w.r.t. variable $u = x_1$ since we integrate out the other variable x_2.

Remark 3.9 It is worth noting that the **convolution formula** (3.19) for densities, also written $f_{X_1+X_2}(x) =: f_{X_1} * f_{X_2}(x)$, is commutative i.e. $f_{X_2} * f_{X_2} = f_{X_1} * f_{X_2}$. It is also associative.

3.6 Block Dependence

Let $\mathbf{X} = (X_1, X_2)$ and $\mathbf{Y} = (\tilde{X}_1, \tilde{X}_2)$ be two random vectors with CDFs $F_{\mathbf{X}}, F_{\mathbf{Y}}$. We say that $\mathbf{X}$ is independent of $\mathbf{Y}$, written $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$, if

$$F_{\mathbf{X},\mathbf{Y}}(a, b, c, d) = F_{\mathbf{X}}(a, b) F_{\mathbf{Y}}(c, d), \quad \text{for every } a, b, c, d \in \mathbb{R}, \tag{3.20}$$

where the left-hand side equals the join probability $\mathsf{P}(X_1 \leqslant a, X_2 \leqslant b, \tilde{X}_1 \leqslant c, \tilde{X}_2 \leqslant d)$. This **block independence** is sufficient for the independence of $\mathbf{X}$ from $\tilde{X}_1$, in the sense that

$$\begin{aligned} F_{\mathbf{X},\tilde{X}_1}(a, b, c) &= F_{\mathbf{X},\mathbf{Y}}(a, b, c, \infty) \\ &\overset{\text{ind. (3.20)}}{=} F_{\mathbf{X}}(a, b) F_{\mathbf{Y}}(c, \infty) \\ &= F_{\mathbf{X}}(a, b) F_{\tilde{X}_1}(c). \end{aligned}$$

The interesting point here is that this also implies X_1 is independent of $\tilde{X}_1$ and X_2 is independent of $\tilde{X}_1$. We show only the first statement is true, since the independence of X_2 and $\tilde{X}_1$ can be proved similarly. Therefore:

$$\begin{aligned} F_{(X_1,\tilde{X}_1)}(a,c) &= F_{\mathbf{X},\mathbf{Y}}(a,\infty,c,\infty) \\ &\overset{\text{ind. (3.20)}}{=} F_{\mathbf{X}}(a,\infty)F_{\mathbf{Y}}(c,\infty) \\ &= F_{X_1}(a)F_{\tilde{X}_1}(c). \end{aligned}$$

Hence, if two bi-variate random vectors $\mathbf{X}$, $\mathbf{Y}$ are independent then each component X_i of $\mathbf{X}$ is independent of each component $\tilde{X}_i$ of $\mathbf{Y}$, for $i = 1, 2$. It is worth noting that the components of $\mathbf{X}$ are not necessarily independent each other, and the same is true for the second random vector. If in addition to the independence of $\mathbf{X}$ from $\mathbf{Y}$ it also holds

$$\mathbf{X} \overset{\text{d}}{=} \mathbf{Y}, \quad \text{i.e. } F_{\mathbf{X}}(a,b) = F_{\mathbf{Y}}(a,b), \quad \text{for every } a, b \in \mathbb{R}. \tag{3.21}$$

then $F_{X_i}(x) = F_{\tilde{X}_i}(x)$ for $i = 1, 2$ and every $x \in \mathbb{R}$. In fact, using (3.21) and letting $a = x$ we have

$$F_{\mathbf{X}}(x,\infty) = F_{\mathbf{Y}}(x,\infty)$$

and for $b = x$

$$F_{\mathbf{X}}(\infty,x) = F_{\mathbf{Y}}(\infty,x).$$

As byproduct, we also have:

$$\left.\begin{matrix} F_{X_1}F_{\tilde{X}_2} \\ F_{\tilde{X}_1}F_{X_2} \end{matrix}\right\} = F_{X_1}F_{X_2}$$

3.7 Exercises

3.1 Let $\mathbf{a} = (a_1, a_2)$, $\mathbf{b} = (b_1, b_2)$, $\mathbf{a}_1 = (a_1, b_2)$ and $\mathbf{a}_2 = (a_2, b_1)$ be three vectors in $\mathbb{R}^2$. Furthermore, let $(-\infty, \mathbf{b}] := (-\infty, b_1] \times (-\infty, b_2]$, $(-\infty, \mathbf{a}_1] := (-\infty, a_1] \times (-\infty, b_2]$ and $(-\infty, \mathbf{a}_2] := (-\infty, b_1] \times (-\infty, a_2]$. Show that

$$(\mathbf{a}, \mathbf{b}] = (-\infty, \mathbf{b}] \setminus (-\infty, \mathbf{a}_1] \cup (-\infty, \mathbf{a}_2], \tag{3.22}$$

where the union is not disjoint.

3.2 Given $\mathbf{a}$, $\mathbf{b}$, $\mathbf{a}_1$ and $\mathbf{a}_2$ as in Exercise 3.1. Prove the following:

$$\begin{aligned}\mathsf{P}(\mathbf{a} < \mathbf{X} \leqslant \mathbf{b}) = {} & F_{\mathbf{X}}(b_1, b_2) - F_{\mathbf{X}}(a_1, b_2) \\ & - F_{\mathbf{X}}(b_1, a_2) + F_{\mathbf{X}}(a_1, a_2).\end{aligned}$$

3.3 Let $\mathbf{X} = (X_1, X_2)$ be a random vector and consider events $A_1 = \{X_1 + X_2 \leqslant 1\}$, $A_2 = \{X_1^2 + X_2^2 \leqslant 1\}$ and $A_3 = \{\max\{X_1, X_2\} \leqslant 1\}$. Plot the regions in the plane $\mathbb{R}^2$ corresponding to these events.

3.4 Show that for a bi-variate CDF it is true that $\mathsf{P}(a_1 < X \leqslant b_1, Y \leqslant y)$ equals $F_{\mathbf{X}}(b_1, y) - F_{\mathbf{X}}(a_1, y)$ and $\mathsf{P}(X \leqslant x, a_2 < Y \leqslant b_2)$ equals $F_{\mathbf{X}}(x, b_2) - F_{\mathbf{X}}(x, a_2)$.

3.5 Show that for a bi-variate CDF we have $F_{\mathbf{X}}(b_1, b_2) - F_{\mathbf{X}}(a_1, b_2) - F_{\mathbf{X}}(b_1, a_2) + F_{\mathbf{X}}(a_1, a_2) \geqslant 0$.

3.6 Let $\mathbf{X} = (X, Y)$ be a random vector with CDF $F_{\mathbf{X}}(x, y)$. It is worth noting that if the second order partial $\frac{\partial^2 F_{\mathbf{X}}}{\partial x \partial y}(x, y)$ exists and is continuous, expect at a finite number of points (x, y), then also the first order partials exist and are continuous, the CDF is differentiable and also continuous itself with the aforementioned exceptions. Verify the following:

(a) $f_{\mathbf{X}}(x, y) \geqslant 0$;
(b) $\mathsf{P}(a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y$;
(c) $F_{\mathbf{X}}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{\mathbf{X}}(u, v)\mathrm{d}u\mathrm{d}v$;
(d) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y = 1$.

3.7 (Further Details on Conditional Probability) Prove Theorem 3.1 and give the appropriate interpretation.

3.8 A financial analyst is faced with the problem of forecasting the future prices of two securities, labeled #1 and #2 respectively. He assume 9 outcomes based on upward and downward price movements together with the hypotheses that prices remain unchanged:

$$\Omega = \{\mathrm{dd}, \mathrm{ds}, \mathrm{du}, \mathrm{sd}, \mathrm{ss}, \mathrm{su}, \mathrm{ud}, \mathrm{us}, \mathrm{uu}\}.$$

Here 's' stands for 'same price'. If the analyst assigns the probabilities

- $\mathsf{P}(\{\mathrm{dd}\}) = \mathsf{P}(\{\mathrm{ds}\}) = \mathsf{P}(\{\mathrm{su}\}) = \mathsf{P}(\{\mathrm{ud}\}) = \mathsf{P}(\{\mathrm{uu}\}) = 0.1$
- $\mathsf{P}(\{\mathrm{du}\}) = \mathsf{P}(\{\mathrm{us}\}) = 0$; $\mathsf{P}(\{\mathrm{sd}\}) = 0.48$; $\mathsf{P}(\{\mathrm{ss}\}) = 0.02$,

what is the probability that price #1 will end down, under the condition that price #2 will be either the same or raised?

3.9 Let X, Y be rvs defined over the same probability space, and let $f(x)$, $g(y)$ two continuous functions of only x and y, respectively. Show that the rvs $U = f(X)$ and $V = g(Y)$ are independent, provided they are assumed to be continuous.

Appendix

Double Integration and Fubini's Theorem

The double integral $\iint_B \mathrm{d}F_{\mathbf{X}}(x, y)$ cannot in general be equal to iterated integrals, but can be interpreted geometrically as the weighting of area(B) using a bi-variate CDF s.t. $\mathrm{d}F_{\mathbf{X}}(x, y)$ is to be thought of as the limiting case of

$$\Delta F_{\mathbf{X}}(x, y) = F_{\mathbf{X}}(x+\Delta x, y+\Delta y) - F_{\mathbf{X}}(x+\Delta x, y) - F_{\mathbf{X}}(x, y+\Delta y) + F_{\mathbf{X}}(x, y) \geqslant 0.$$

In fact we have the following *version* of the Fubini's Theorem.

Theorem 3.6 ('Light' Fubini) *Let $F(x, y) = F_1(x)F_2(y)$, where F_1, F_2 are two bounded functions defined over $[a, b]$ and $[c, d]$, respectively, Then*

$$\begin{aligned} \iint_{[a,b]\times[c,d]} g(x, y)\mathrm{d}F(x, y) &= \int_c^d \left[\int_a^b g(x, y)F_1(x)\mathrm{d}x\right] F_2(y)\mathrm{d}y \quad (3.23) \\ &= \int_a^b \left[\int_c^d g(x, y)F_2(y)\mathrm{d}y\right] F_1(x)\mathrm{d}x, \end{aligned}$$

where the function $g(x, y)$ is bounded on $[a, b] \times [c, d]$.

Obviously, we can choose $F = F_{\mathbf{X}}$ and $F_1 = F_X$, $F_2 = F_Y$, i.e. we can refer to a joint CDF of a random vector $\mathbf{X} = (X, Y)$ with independent rvs X, Y having marginal CDFs F_X, F_Y. We can integrate first w.r.t. the variable x and then w.r.t. y or vice versa. We only need the following generalization of the Riemann double integral. Let $B_n := [-n, n] \times [-n, n]$ for any integer $n \in \mathbb{N}$ and write

$$\lim_{n\to\infty} \iint_{B_n} f(x, y)\mathrm{d}x\mathrm{d}y = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\mathrm{d}x\mathrm{d}y.$$

As byproduct, we can always define the **double improper Stieltjes integral**[15]

$$\boxed{\iint_{\mathbb{R}^2} g(x, y)\mathrm{d}F(x, y)}$$

as equal to $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x, y)\mathrm{d}F(x, y)$, which eventually can be computed by formula (3.23). Indeed, for $F_{\mathbf{X}}(x, y) \neq F_X(x)F_Y(y)$ we can consider the bi-variate

[15] For an overview of Stieltjes integration see Appendix in Chap. 4.

CDF function $F_{\mathbf{X}}(x, y) = F_{X \mid Y}(x, y)F_Y(y)$ with $F_{X \mid Y}(\cdot, y)$ being a CDF on $\mathbb{R}$ in the variable x for a fixed $y \in \mathbb{R}$ and compute:

$$\iint_{\mathbb{R}^2} g(x, y)\mathrm{d}F_{\mathbf{X}}(x, y) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} g(x, y)\mathrm{d}F_{X \mid Y}(x, y) \right] F_Y(y)\mathrm{d}y.$$

The same is true for $F_{\mathbf{X}}(x, y) = F_{Y \mid X}(x, y)F_X(x)$ with $F_{Y \mid X}(x, \cdot)$ being a CDF on $\mathbb{R}$ in the variable y for a fixed $x \in \mathbb{R}$:

$$\iint_{\mathbb{R}^2} g(x, y)\mathrm{d}F_{\mathbf{X}}(x, y) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} g(x, y)\mathrm{d}F_{Y \mid X}(x, y) \right] F_X(x)\mathrm{d}x.$$

Remark 3.10 Later we will recognize

$$F_{X \mid Y}(x, y) := \frac{F_{\mathbf{X}}(x,y)}{F_Y(y)}$$

and

$$F_{Y \mid X}(x, y) := \frac{F_{\mathbf{X}}(x,y)}{F_X(x)}$$

as the conditional CDF of one rv w.r.t. values assumed by the other, see Chap. 6.

For the sake of completeness we provide another version of the Fubini's Theorem.

Theorem 3.7 ('Weak' Fubini) *If $f : [a, b] \times [c, d] \to \mathbb{R}$ is continuous on $[a, b] \times [c, d]$, then the iterated integrals*

$$\int_c^d \int_a^b f(x, y)\mathrm{d}x\mathrm{d}y \quad \textit{and} \quad \int_a^b \int_c^d f(x, y)\mathrm{d}y\mathrm{d}x$$

exist and are equal.

1-to-1 Correspondence Between P and $F_{\mathbf{X}}$

Theorem 3.8 (Joint or Multivariate CDF) *Let P be some probability over $(\Omega, \mathcal{F})$ and let $F_{\mathbf{X}} : \mathbb{R}^n \to [0, 1]$ be a function given by formula (3.1). Then we have:*

1. *$F_{\mathbf{X}}$ is nondecreasing, i.e. for any $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ s.t. $\mathbf{x} \leqslant \mathbf{y}$ it follows $F_{\mathbf{X}}(\mathbf{x}) \leqslant F_{\mathbf{X}}(\mathbf{y})$, where the vector inequality means that $x_i \leqslant y_i$ for every $i = 1, \ldots, n$ and at least one inequality is strict;*
2. *$F_{\mathbf{X}}$ is right-continuous, i.e. for any sequence of vectors $(\mathbf{x}_m)_{m \in \mathbb{N}}$ s.t. $\mathbf{x}_m \downarrow \mathbf{x} \in \mathbb{R}^n$ it holds $F_{\mathbf{X}}(\mathbf{x}_m) \downarrow F_{\mathbf{X}}(\mathbf{x})$, where $\mathbf{x}_m \geqslant \mathbf{x}$ with $\mathbf{x}_m = (x_{1m}, \ldots, x_{nm})$, each $m \in \mathbb{N}$, and $x_{im} \downarrow x_i$ for $i = 1, \ldots, n$ where in general $n \neq m$;*

3. $\lim_{x_1,\ldots,x_n\to\infty} F_{\mathbf{X}}(x_1,\ldots,x_n) := F_{\mathbf{X}}(\infty,\ldots,\infty) = 1$ *and* $\lim_{x_i\to-\infty} F_{\mathbf{X}}(x_1,\ldots,x_n) = 0$ *for* $i = 1,\ldots,n$;
4. *for every pair of vectors* $\mathbf{a} = (a_1,\ldots,a_n) \leqslant (b_1,\ldots,b_n) = \mathbf{b}$, *i.e.* $a_i \leqslant b_i$ *for* $i = 1,\ldots,n$, *defining the* ***difference operator*** *as*

$$\begin{aligned}\Delta_{a_i,b_i} F(x_1,\ldots,x_n) &= F(x_1,\ldots,x_{i-1},b_i,x_{i+1},\ldots,x_n)\\ &\quad - F(x_1,\ldots,x_{i-1},a_i,x_{i+1},\ldots,x_n),\end{aligned}$$

entails $\Delta_{a_1,b_1}\cdots\Delta_{a_n,b_n} F_{\mathbf{X}}(x_1,\ldots,x_n) = \mathsf{P}(\mathbf{a} < \mathbf{X} \leqslant \mathbf{b})$.

Conversely, if a function $F_{\mathbf{X}} : \mathbb{R}^n \to [0,1]$ *satisfies properties (a)–(d) then there exists a probability* P *over some event space* $(\Omega,\mathcal{F})$ *s.t.* $F_{\mathbf{X}}$ *is the joint CDF of a random vector* $\mathbf{X}$, *i.e.* $\mathsf{P}_{\mathbf{X}}(B) = \mathsf{P}(\mathbf{X}\in B)$ *for a suitable* $B \subset \mathbb{R}^n$.

For the sake of intuition we illustrate properties 2,3 and 4 for the bi-dimensional case.

1. The characterization of right-continuity requires a sequence of vectors $\mathbf{x}_1 = (x_{11},x_{21}) \geqslant \mathbf{x}_2 = (x_{12},x_{22}) \geqslant \cdots \geqslant \mathbf{x}_m = (x_{1m},x_{2m})$ s.t. for each $i = 1,\ldots,n$ the corresponding real-valued sequence $x_{i1} \geqslant x_{i2} \geqslant \cdots \geqslant x_{im}$ converges (in decreasing order) to x_i, i.e. $\lim_{m\to\infty} x_{im} = x_i$, and $\mathbf{x} = (x_1,x_2)$. Thus, the vectors $\mathbf{x}_m = (x_{1m},x_{2m})$ 'approach the vector' $\mathbf{x}$ 'from the right' and correspondingly $F_{\mathbf{X}}(x_{1m},x_{2m})$ are decreasing real values tending to $F_{\mathbf{X}}(x_1,x_2)$. Alternatively we could write

$$\lim_{h\downarrow 0}\left(\lim_{k\downarrow 0} F_{\mathbf{X}}(x_1+h,x_2+k)\right) = F_{\mathbf{X}}(x_1,x_2), \quad h,k>0.$$

   Without loss of generality we can take $k = h$. Note that $d(\mathbf{x}_m,\mathbf{x}) \to 0$ as $m \to \infty$ or $d((x_1+h,x_2+h),(x_1,x_2)) \to 0$ as $h \downarrow 0$.
2. The limit

$$\lim_{x_1\to\infty}\left(\lim_{x_1\to\infty} F_{\mathbf{X}}(x_1,x_2)\right) = F_{\mathbf{X}}(\infty,\infty) = 1$$

   can be obtained using the sequential characterization. In fact, $x_i \to \infty$ is equivalent to $x_i + m \to \infty$ and $m \to \infty$, for $i = 1,2$. Thus:

$$\begin{aligned}&\lim_{m\to\infty} \mathsf{P}(X_1 \leqslant x_1+m, X_2 \leqslant x_2+m)\\ \overset{\text{contin. of }\mathsf{P}}{=}\ &\mathsf{P}\left(\lim_{m\to\infty}\{X_1 \leqslant x_1+m\}\cap\{X_2 \leqslant x_2+m\}\right)\\ =\ &\mathsf{P}(\Omega) = 1.\end{aligned}$$

With the same reasoning we get (which is true also for the variable x_2):

$$\lim_{m\to\infty} \mathsf{P}(X_1 \leqslant x_1 - m,\, X_2 \leqslant x_2) \overset{\text{contin. of } \mathsf{P}}{=} \mathsf{P}\left(\lim_{m\to\infty} \{X_1 \leqslant x_1 - m\} \cap \{X_2 \leqslant x_2\}\right)$$
$$= \quad \mathsf{P}\left(\varnothing \cap \{X_2 \leqslant x_2\}\right) = 0,$$

where the first limit on the left equals $\lim_{x_1\to-\infty} F_{\mathbf{X}}(x_1, x_2) = F_{\mathbf{X}}(-\infty, x_2) = 0$.

3. The difference operator acts iteratively.[16] Thus we have:

$$\begin{aligned} \Delta_{a_1,b_1}\Delta_{a_2,b_2} F_{\mathbf{X}}(x_1, x_2) &:= \Delta_{a_1,b_1}\left(\Delta_{a_2,b_2} F_{\mathbf{X}}(x_1, x_2)\right) \\ &= \Delta_{a_1,b_1}\left(F_{\mathbf{X}}(x_1, b_2) - F_{\mathbf{X}}(x_1, a_2)\right) \\ &= \Delta_{a_1,b_1} F_{\mathbf{X}}(x_1, b_2) - \Delta_{a_1,b_1} F_{\mathbf{X}}(x_1, a_2) \\ &= F_{\mathbf{X}}(b_1, b_2) - F_{\mathbf{X}}(a_1, b_2) \\ &- F_{\mathbf{X}}(b_1, a_2) + F_{\mathbf{X}}(a_1, a_2). \end{aligned} \tag{3.24}$$

It is worth noting that the event $\{\mathbf{a} < \mathbf{X} \leqslant \mathbf{b}\}$ means that the random vector takes values on the Cartesian product $(\mathbf{a}, \mathbf{b}] = (a_1, b_1] \times (a_2, b_2]$, i.e. $(X_1, X_2) \in (\mathbf{a}, \mathbf{b}]$ iff $a_i < X_i \leqslant b_i$ for $i = 1, 2$.

The necessary part ($\Longrightarrow$) of Theorem 3.8 for independent rvs is Theorem 3.4 in Sect. 3.5.

Remark 3.11 Property 1 of Theorem 3.8 can be restated by saying that $F_{\mathbf{X}}$ is **n-increasing**, i.e., $\Delta_{a_1,b_1} \cdots \Delta_{a_n,b_n} F_{\mathbf{X}}(x_1, \ldots, x_n) \geqslant 0$, for every rectangle $(\mathbf{a}, \mathbf{b}]$ with $\mathbf{a} = (a_1, \ldots, a_n)$ and $\mathbf{b} = (b_1, \ldots, b_n)$. Here the difference operator gives raise to the sum $S_0 - S_1 + S_2 - \cdots + (-1)^n S_n$, where S_i is itself a sum of all terms $F_{\mathbf{X}}(s_1, \ldots, s_n)$ with $s_k = a_k$ for exactly i integers from $\{1, \ldots, n\}$, and $s_k = b_k$ for the remaining $n - i$ integers; there are $\binom{n}{i}$ of such terms. In fact, this is just the $F_{\mathbf{X}}$-volume of the rectangle $(\mathbf{a}, \mathbf{b}]$ which can be equivalently computed as $\sum_{\mathbf{v}\in \mathrm{V}} \mathrm{sign}(\mathbf{v}) F_{\mathbf{X}}(\mathbf{v})$, where

$$\mathrm{sign}(\mathbf{v}) = \begin{cases} 1, \text{ if } v_i = a_i \text{ for an even number of indices} \\ -1, \text{ if } v_i = a_i \text{ for an odd number of indices,} \end{cases}$$

and $\mathrm{V} = \{a_1, b_1\} \times \cdots \times \{a_n, b_n\}$ is the set containing all vertices of $(\mathbf{a}, \mathbf{b}]$, each given as $\mathbf{v} = (v_1, \ldots, v_n)$. Observe that right-continuity (b) is equivalent to have $F_{\mathbf{X}}$ right-continuous in all variables.

[16] Δ_{a_i,b_i} acts on $F_{\mathbf{X}}$ by substituting the number b_i to the ith coordinate x_i in $F_{\mathbf{X}}(x_1, \ldots, x_n)$ then substituting the number a_i to the same coordinate and eventually taking the difference.

Some Details on Independence

Given independent events $A_1, \ldots, A_n \in \mathcal{F}$ we have $\sigma(A_1), \ldots, \sigma(A_n)$ are independent and also $\mathbf{I}_{A_1}, \ldots, \mathbf{I}_{A_n}$ are independent. These three conditions are equivalent. Observe that indexed classes of events $\mathcal{G}_i \subset \mathcal{F}$, for $i \in I$, are said independent iff $A_{i_1} \in \mathcal{G}_{i_1}, \ldots, A_{i_n} \in \mathcal{G}_{i_n}$ are independent for any finite choice of indices[17] $i_1, \ldots, i_n \in I$. For example, rvs $X_1, \ldots, X_n$ are independent iff $\sigma(X_1), \ldots, \sigma(X_n)$ are independent as families of events $\sigma(X_i) = \{\{X_i \in B_i\} \mid B_i \in \mathbb{R} \text{ suitable}\}$, see also Example 1.15 in Sects. 1.6 and 6.4. Now, a multidimensional version of Theorem 3.2 for classes of events induced by indexed families of rvs, can be proved based on the fact that certain independent classes $\mathcal{G}_i$ of events in $\mathcal{F}$, each containing intersection of all pairs $A, B \in \mathcal{G}_i$, generate σ-algebra $\sigma(\mathcal{G}_{i_1}, \ldots, \mathcal{G}_{i_n})$ that are independent for all choice of indices $i_1, \ldots, i_n$, and this yields rvs $Y = f(X_{i_1}, \ldots, X_{i_n})$ are independent because their generated σ-algebras $\sigma(Y)$ are. This is a more general form of block independence, see Sect. 3.6.

Remark 3.12 In the proof of Theorem 3.2 we used special sets A, B for the values of $f(X)$ and $g(Y)$. Actually, using the definition of inverse image (see Appendix in Chap. 2) we can consider two composed maps

$$\Omega \xrightarrow{X} \mathbb{R} \xrightarrow{f} \mathbb{R}, \quad \Omega \xrightarrow{Y} \mathbb{R} \xrightarrow{g} \mathbb{R},$$

and observing that X, Y and f, g are Borel-measurable, passing to their corresponding inverse images and composed mappings we have:

$$\{f(X) = u\} = X^{-1}(f^{-1}(\{u\})) = \{X \in f^{-1}(\{u\})\}, \ A = f^{-1}(\{u\}),$$

and

$$\{g(Y) = v\} = Y^{-1}(g^{-1}(\{v\})) = \{Y \in g^{-1}(\{v\})\}, \ B = g^{-1}(\{v\}),$$

then taking intersection $\{f(X) = u\} \cap \{g(Y) = v\}$ we get the same result. It is worth noting that

$$\mathcal{B} \xrightarrow{f^{-1}} \mathcal{B} \xrightarrow{X^{-1}} \Omega$$

has composed inverse image $\mathcal{B} \xrightarrow{(f \circ X)^{-1}} \Omega$, where $(f \circ X)^{-1} := X^{-1} \circ f^{-1}$.

We can formulate a result about independence also for different probability spaces.

[17] Independence of $\sigma(A_1), \ldots, \sigma(A_n)$ implies Independence of $A_1, \ldots, A_n$ trivially. Independence among $\sigma(A_i)$ and independence among $\mathbf{I}_{A_i}$ are equivalent since independence of rvs can be stated in terms of their generated σ-algebras. The fact that independence of events A_i implies independence of their generated σ-algebras is more subtle, and we omit the proof.

Theorem 3.9 *Let the rvs X, Y be defined over $(\Omega_1, \mathcal{F}_1)$ and over $(\Omega_2, \mathcal{F}_2)$, respectively. For every pair of Borel-measurable functions $f, g : \mathbb{R} \to \mathbb{R}$, the rvs $f(X), g(Y)$ are independent iff X, Y are.*

Proof Recall that a composition between a rv (which is measurable itself) and a Borel-measurable function if measurable. ($\Rightarrow$) Assuming $f(X), g(Y)$ are independent we simply let $f(x) = x$ and $g(y) = y$. ($\Leftarrow$) Assuming X, Y are independent we known that their generated σ-algebra $\sigma(X), \sigma(Y)$ are independent too. Now, let $U = f(X)$ and $V = g(Y)$, then

$$\{f(X) \in C\} = X^{-1}(f^{-1}(C)) = \{X \in A\}, \ A = f^{-1}(C),$$

and

$$\{g(Y) \in D\} = Y^{-1}(g^{-1}(D)) = \{Y \in B\}, \ B = g^{-1}(D).$$

Since $\sigma(U) \subset \sigma(X)$ and $\sigma(V) \subset \sigma(Y)$, events $\{X \in A\} \in \sigma(X)$ and $\{Y \in B\} \in \sigma(Y)$ are independent as well as $\sigma(X), \sigma(Y)$, we are done. ■

Conditional Independence

From Sect. 3.4 we see that two rvs X, Y can be:

- independent but not identically distributed;
- independent and identically distributed;
- dependent but not identically distributed;
- dependent and identically distributed.

For example, X being the lifetime of a randomly selected newborn and $Y = S_1$ the daily closing price of the S&P500 are independent but not identically distributed. If X represents tick movements ± 0.5 of the S&P500's closing price at the end of day 1, and Y represents tick movements of the same stock index at the end of day 2, then X, Y are IID. Moreover, let $X = \mathbf{I}_{\{S_1 > 0 \text{ at the end of day 3}\}}$ and let Y be the number of downward tick movements -0.5 in 3 consecutive trading days, then X, Y are dependent but they do not have the same distribution. Eventually, let X be the number of upward tick movements 0.5 in one trading day and let Y be the number of downward tick movements -0.5 over the same time horizon, then X, Y might safely have the same distribution (see Sect. 5.4) and they are clearly dependent. There is another form of dependence among rvs that we can state by requiring

$$\mathsf{P}(X \leqslant x, Y \leqslant y \mid Z = z) = \mathsf{P}(X \leqslant x, \mid Z = z)\mathsf{P}(Y \leqslant y \mid Z = z), \ \text{for all } x, y \in \mathbb{R}$$

for any pair of rvs X, Y defined over the same probability space, and any z in the support of a third rv Z. This is called *conditional independence*, and has nothing to

do with (unconditional) independence. The requirement above can be generalized by considering the notion of general conditional probability, see Sect. 6.4.

Proof of Theorem 3.5

Proof We simply write the CDF of the rv $X_1 + X_2$ and differentiating the proof is complete. Thus for the event $\{X_1 + X_2 \leqslant x\}$ we consider the subset given by

$$B = \{(x_1, x_2) \,|\, x_1 + x_2 \leqslant x\} \subset \mathbb{R}^2$$

involving the random vector $\mathbf{X} = (X_1, X_2)$:

$$\begin{aligned}
F_{X_1+X_2}(x) &= \mathsf{P}(X_1 + X_2 \leqslant x) \\
&= \int_B f_{\mathbf{X}}(x_1, x_2)\mathrm{d}x_1\mathrm{d}x_2 \overset{\text{Indep.}}{=} \int_B f_{X_1}(x_1) f_{X_2}(x_2)\mathrm{d}x_1\mathrm{d}x_2 \\
&\overset{\text{Fubini's Th}}{=} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{x-x_2} f_{X_1}(x_1)\mathrm{d}x_1 \right] f_{X_2}(x_2)\mathrm{d}x_2 \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{x} f_{X_1}(x_1 - x_2)\mathrm{d}x_1 \right] f_{X_2}(x_2)\mathrm{d}x_2 \\
&= \int_{-\infty}^{x} \left[\int_{-\infty}^{\infty} f_{X_1}(x_1 - x_2)\mathrm{d}x_2 \right] f_{X_2}(x_2)\mathrm{d}x_1,
\end{aligned}$$

so choosing $u = x_1$ and applying $\frac{\mathrm{d}}{\mathrm{d}x} F_{X_1+X_2}$ we are done. ■

Proof of Theorem 3.3

Proof Assume $X_1, \ldots, X_n$ are independent. By (3.16) and (3.15) we may write:

$$\begin{aligned}
&\underbrace{\int_{-\infty}^{x_1} f_{X_1}(u_1)\mathrm{d}u_1 \times \cdots \times \int_{-\infty}^{x_n} f_{X_n}(u_n)\mathrm{d}u_n}_{=\prod_{i=1}^n \int_{-\infty}^{x_i} f_{X_i}(u_i)\mathrm{d}u_i} \\
&= \prod_{i=1}^n \mathsf{P}(X_i \leqslant x_i) = \mathsf{P}(X_1 \leqslant x_1, \ldots, X_n \leqslant x_n) \\
&\overset{\text{abs. cont.}}{=} \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_{\mathbf{X}}(u_1, \ldots, u_n)\mathrm{d}u_1 \cdots \mathrm{d}u_n,
\end{aligned}$$

for every $x_1, \ldots, x_n$. Subtracting the former expression, which by Fubini's Theorem is written as $\int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} \left[\prod_{i=1}^n f_{X_i}(u_i) \right] \mathrm{d}u_1 \cdots \mathrm{d}u_n$, from the latter we get

$$\int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} \left[f_{\mathbf{X}}(u_1, \ldots, u_n) - \prod_{i=1}^n f_{X_i}(u_i) \right] \mathrm{d}u_1 \cdots \mathrm{d}u_n = 0.$$

Differentiating w.r.t. the variables $x_1, \ldots, x_n$ we are done. Conversely, if the joint density factors we have:

$$\begin{aligned} \mathsf{P}(X_1 \leqslant x_1, \ldots, X_n \leqslant x_n) &= \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} \left[\prod_{i=1}^n f_{X_i}(u_i) \right] \mathrm{d}u_1 \cdots \mathrm{d}u_n \\ &= \prod_{i=1}^n \int_{-\infty}^{x_i} f_{X_i}(u_i) \mathrm{d}u_i = \prod_{i=1}^n \mathsf{P}(X_i \leqslant x_i), \end{aligned}$$

which is independence by (3.16). ■

As a corollary we also get that independent rvs X_i which are absolutely continuous provide a random vector $\mathbf{X} = (X_1, \ldots, X_n)$ that is absolutely continuous. It suffices to start from $\mathsf{P}(X_1 \leqslant x_1, \ldots, X_n \leqslant x_n)$, writes it as $\prod_{i=1}^n \mathsf{P}(X_i \leqslant x_i) = \prod_{i=1}^n \int_{-\infty}^{x_i} f_{X_i}(u_i) \mathrm{d}u_i$ and then by Fubini's Theorem we get the iterated integrals

$$\int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} \left[\prod_{i=1}^n f_{X_i}(u_i) \right] \mathrm{d}u_1 \cdots \mathrm{d}u_n$$

showing that $\prod_{i=1}^n f_{X_i}(u_i)$ satisfies the property of a joint density $f_{\mathbf{X}}$.

Chapter 4
Moments and the Alike



4.1 A Motivating Starting Point

Let $(\Omega, \mathcal{F})$ be an event space representing the following random experiment: a gamble s.t. you win € 1000 with probability $p > 0$ and you loose € 900 with probability $1-p$. How much you wish to get a gain or you fear to suffer a loss, from this gamble on average? First, we can define the discrete rv X having distribution with masses $\mathsf{P}(X = 1000) = p$ and $\mathsf{P}(X = -900) = 1 - p$. Then, we can compute the forecasted win/loss as

$$\overline{X} = p1000 + (-900)(1 - p).$$

E.g., assuming $p = \frac{1}{2}$ you expect to win € $\overline{X} = 50$. A physical interpretation of $\overline{X}$ is that it represents the center of gravity of the numbers -900 and 1000 in $\mathbb{R}$, on which you put two probabilities determining $\mathsf{P}_X(B) = p\delta_{1000}(B) + (1 - p)\delta_{-900}(B)$ as the law of X, for every suitable subset[1] $B \subset \mathbb{R}$. Observe that $\overline{X}$ is a number lying between -900 and 1000, for every $p \in (0, 1)$, i.e. it is a convex combination with weights $1 - p$ and p.

The number $\overline{X}$ synthesizes possible values that the rv X takes on, using partial information given by its probability masses. In fact, interpreting

$$p = \mathsf{P}(X = 1000) \approx \frac{f_n(A)}{n} \quad \text{and} \quad 1 - p = \mathsf{P}(X = -900) \approx \frac{f_n(A^c)}{n}$$

as the relative frequencies of the events $A = \{X = 1000\}$ and $A^c = \{X = 1000\}$, respectively,[2] $\overline{X}$ deserves the name *average* and represents what one expects

[1] Recall: $\delta_x(B) = \mathbf{I}_B(x)$.

[2] Assuming the independent repetition $n \in \mathbb{N}$ times of the underlying random experiment.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
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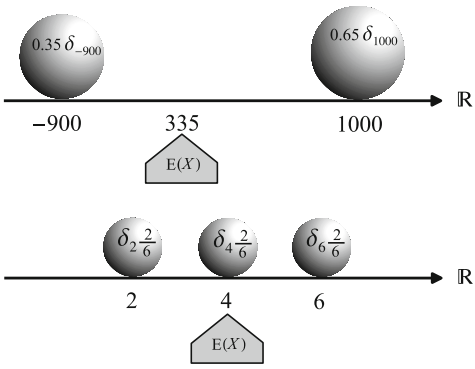


Fig. 4.1 (Top) The distribution P_X is discrete with total mass 1 and $\overline{X}$ being its center of gravity, with $p = 0.65$. (Bottom) Discrete uniform distribution with masses $p = \frac{2}{6}$

from the random experiment governing X. Figure 4.1 illustrates the physical interpretation of $\overline{X}$.

4.2 Expectation

Mathematical expectation can be defined for a rv as a predicted value with average probability. We will distinguish between discrete and continuous rvs.

Definition 4.1 Let X be a discrete rv with finite or countable values and probability masses $\mathsf{P}(X = x_i)$, for every x_i in the support C, and let $g : \mathbb{R} \to \mathbb{R}$ be a real-valued function s.t. $g(X)$ is a rv.[3] The **expectation** of $g(X)$ is

$$\mathsf{E}(g(X)) := \sum_{i \text{ s.t. } x_i \in C} g(x_i)\mathsf{P}(X = x_i), \tag{4.1}$$

provided that $g(X) \geqslant 0$ or the sum is[4] *absolutely convergent*, $\sum_i |g(x_i)|\mathsf{P}(X = x_i) < \infty$.

Using the identity function $g(x) = x$, expectation of discrete rvs satisfies the following properties.

1. If $X(\omega) = k$ for every scenario $\omega \in \Omega$, with $k \in \mathbb{R}$, then there is only $\{X = k\} = \Omega$ in the sum (4.1) and $\mathsf{E}(X) = k\mathsf{P}(X = k) = k$. Also, $\mathsf{P}(X = k) = 1$, i.e., $X = k$ P-a.s. does imply $\mathsf{E}(X) = k$.
2. If $a, b \in \mathbb{R}$ then $\mathsf{E}(a + b\,X) = a + b\,\mathsf{E}(X)$, i.e. *expectation is linear*, provided $\mathsf{E}(X)$ is well defined.

[3] This is especially true for continuous g.

[4] We use $|g(x_i)\mathsf{P}(X = x_i)| = |g(x_i)||\mathsf{P}(X = x_i)|$ and $|\mathsf{P}(X = x_i)| = \mathsf{P}(X = x_i)$.

3. If $X(\omega) \leqslant Y(\omega)$ for every scenario then $\mathsf{E}(X) \leqslant \mathsf{E}(Y)$, i.e. *expectation preserves ordering*, where Y is another discrete rv with well-defined $\mathsf{E}(Y)$.

For a discrete rv X there are interesting relationships between indicators and expectation.

- Let $g(x) = \mathbf{I}_B(x)$ be the indicator of a suitable subset $B \subset \mathbb{R}$. Thus $\mathsf{E}(\mathbf{I}_B(X)) = \mathsf{P}(X \in B)$. In fact, by Definition 4.1 the expectation on the left-hand side is $\sum_i \mathbf{I}_B(x_i)\mathsf{P}(X = x_i) = \sum_{x_i \in B} \mathsf{P}(X = x_i)$, and since the events $\{X = x_i\}$ are pairwise disjoint the last sum is equal to $\mathsf{P}\left(\cup_{x_i \in B}\{X = x_i\}\right)$ which is just $\mathsf{P}(X \in B)$.
- An indicator rv $X = \mathbf{I}_A$ with $A \in \mathcal{F}$ is discrete with support $C = \{0, 1\}$. Thus, $A = \{X = 1\}$ and $A^c = \{X = 0\}$ and we can write

$$\mathsf{E}(X) = 1 \times \mathsf{P}(X = 1) + 0 \times \mathsf{P}(X = 0)$$

  and finally obtain $\mathsf{E}(\mathbf{I}_A) = \mathsf{P}(A)$. Choosing $A = \{X \in B\}$ we obtain the result in the first item.

Example 4.1 Let X be a discrete rv with support $\{0, 1, 2, \ldots\}$ and mass function $\mathsf{P}(X = i) = \mathrm{e}^{\lambda}\frac{\lambda^i}{i!}$, with parameter $\lambda > 0$ (see Sect. 5.5). Taking the identity function $g(x) = x$, we have[5]

$$\mathsf{E}(X) = \sum_{i=1}^{\infty} i\mathrm{e}^{-\lambda}\frac{\lambda^i}{i!}.$$

Moreover, we have the following chain of equivalences:

$$\begin{aligned}
\mathsf{E}(X) &= \lambda\,\mathrm{e}^{-\lambda}\sum_{i=1}^{\infty} i\frac{\lambda^{i-1}}{i!} = \lambda\,\mathrm{e}^{-\lambda}\sum_{i=1}^{\infty}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda}\frac{\lambda^i}{i!}\right] \\
&= \lambda\,\mathrm{e}^{-\lambda}\,\frac{\mathrm{d}}{\mathrm{d}\lambda}\left[\sum_{i=0}^{\infty}\frac{\lambda^i}{i!}\right] = \lambda\,\mathrm{e}^{-\lambda}\,\frac{\mathrm{d}}{\mathrm{d}\lambda}\left[\mathrm{e}^{\lambda}\right] \\
&= \lambda\,\mathrm{e}^{-\lambda}\,\mathrm{e}^{\lambda} = \lambda.
\end{aligned}$$

Remark 4.1 In the previous chain of equivalences we used facts from real analysis.

- The exponential function $h(\lambda) = \mathrm{e}^{\lambda}$ has the Taylor series expansion[6] $h(\lambda) = \sum_{i=0}^{\infty}\frac{\lambda^i}{i!}$, around $\lambda = 0$.

[5] Observe that $\sum_{i=1}^{\infty}\left|i\mathrm{e}^{-\lambda}\frac{\lambda^i}{i!}\right| = \sum_{i=1}^{\infty} i\mathrm{e}^{-\lambda}\frac{\lambda^i}{i!}$, thus verifying $\mathsf{E}(X) = \lambda$ implies the sufficient convergent condition in Definition 4.1 is fulfilled.

[6] Recall the derivative $\frac{\mathrm{d}}{\mathrm{d}\lambda}\frac{\lambda^i}{i!} = i\frac{\lambda^{i-1}}{i!}$.

- The function $\sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$ is a power series converging for all $\lambda \in \mathbb{R}$, see Appendix H.
- For a convergent power series we can exchange the derivative with the sum and differentiate term by term, $\frac{\mathrm{d}}{\mathrm{d}\lambda}\left[\sum_{i=0}^{\infty} \frac{\lambda^i}{i!}\right] = \sum_{i=0}^{\infty}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda}\frac{\lambda^i}{i!}\right]$.

For a continuous rv with density function, the following characterization is an intuitive extension of the discrete case.

Definition 4.2 Let X be a continuous rv with probability density function $f_X(x)$ and let $g : \mathbb{R} \to \mathbb{R}$ be a real-valued function s.t. $g(X)$ is a rv. The expectation of $g(X)$ is

$$\mathsf{E}(g(X)) := \int_{-\infty}^{\infty} g(x) f_X(x)\,\mathrm{d}x, \tag{4.2}$$

provided that $g(X) \geqslant 0$ or the integral is[7] *absolutely convergent*, $\int_{-\infty}^{\infty} |g(x)| f_X(x)\,\mathrm{d}x < \infty$.

Similarly to the discrete case, choosing $g(x) = x$ we also have:

1. If $a, b \in \mathbb{R}$ then $\mathsf{E}(a + b\,X) = a + b\,\mathsf{E}(X)$, i.e. expectation is linear, provided $\mathsf{E}(X)$ is well defined according to Definition 4.2.
2. If $X(\omega) \leqslant Y(\omega)$ for every scenario then $\mathsf{E}(X) \leqslant \mathsf{E}(Y)$, i.e. expectation preserves ordering, where Y is another continuous rv with density $f_Y(y)$ and well-defined $\mathsf{E}(Y)$.

We can as well provide a unifying treatment of expectation stating the **change of variable formula** also known as *the rule (law) of the lazy (unconscious) statistician*[8]

$$\mathsf{E}(g(X)) = \int_{-\infty}^{\infty} g(x)\,\mathrm{d}F_X. \tag{4.3}$$

For $g(x) = x$, Definitions 4.1 and 4.2 can be deduced as follow:

- $F_X(x)$ is (absolutely) continuous and the density $f_X(x)$ is continuous $\Rightarrow \mathrm{d}F_X(x) = F'_X(x)\mathrm{d}x = f_X(x)\mathrm{d}x$ and $\mathsf{E}(X) = \int_{-\infty}^{\infty} x\, f_X(x)\mathrm{d}x$, is a Riemann integral;
- $F_X(x)$ is 'discrete' $\Rightarrow \mathrm{d}F_X(x) = F_X(x_i) - F_X(x_{i-1})$ and $\mathsf{E}(X) = \sum_i x_i \mathsf{P}(X = x_i)$.

The precise construction is based on measure theory, where eventually the change of variable formula is indeed stated as a theorem. We sketch it in several steps. Before

[7] We use $|g(x) f_X(x)| = |g(x)||f_X(x)|$ and $|f_X(x)| = f_X(x)$, since the density if nonnegative.

[8] The integral is of the Stieltjes type.

to start, we need the following concept stated informally. A function $g : \mathbb{R} \to \mathbb{R}$ is *measurable*[9] when $g^{-1}(B) := \{x \in \mathbb{R} \mid g(x) \in B\}$ is 'suitable'.[10] Thus, $x \in g^{-1}(B)$ iff $g(x) \in B$. Let $Y = g(X)$ and take a suitable $\tilde{B} \subset \mathbb{R}$:

$$\begin{aligned}\{Y \in \tilde{B}\} &= \{\omega \mid g(X(\omega)) \in \tilde{B}\} \\ &= \{\omega \mid X(\omega) \in g^{-1}(\tilde{B})\}.\end{aligned}$$

Since g is measurable and X is a rv then $B = g^{-1}(\tilde{B})$ is also suitable and $\{X \in B\} \in \mathcal{F}$, thus Y is a rv. But in general $\mathsf{P}_X \neq \mathsf{P}_Y = \mathsf{P}_{g(X)}$, or using CDFs $F_X \neq F_{g(X)}$. What the change of variable formula says is that we do not care about the new distribution in computing expectations of transformed rvs. In words, the expectation is the integral of a measurable function $g(x)$.

1. Suppose $X = x_1\mathbf{I}_{\{X\in B_1\}} + \cdots + x_n\mathbf{I}_{\{X\in B_n\}}$ with $\{X \in B_i\}$ partitioning Ω and B_i partitioning $\mathbb{R}$. Thus,[11] $g(X) = g(x_1)\mathbf{I}_{\{X\in B_1\}} + \cdots + g(x_n)\mathbf{I}_{\{X\in B_n\}}$ has expectation

$$\int g(x)\mathrm{d}F_X(x) := \sum_{i=1}^{n} g(x_i)\mathsf{P}(g(x_i) \in \tilde{B}_i), \quad \text{where } \tilde{B}_i = g(B_i).$$

2. If $g(x) \geqslant 0$ then $\int g(x)\mathrm{d}F_X(x) = \lim_{n\to\infty} \int g_n(x)\mathrm{d}F_X(x)$, where $0 \leqslant g_n(x) \leqslant g_{n+1}(x)$ is an increasing sequence s.t. $g_n(X(\omega))$ are discrete rvs with finite support converging pointwise to the rv $g(X(\omega))$.
3. For a general rv $g(X)$ we take $\int g(x)\mathrm{d}F_X(x) = \int g^+(x)\mathrm{d}F_X(x) - \int g^-(x)\mathrm{d}F_X(x)$, where $g^+(x) = \max\{0, g(x)\}$ is the positive part and $g^-(x) = \max\{0, -g(x)\}$ is the negative part. Hopefully, both are finite. If only one is $\pm\infty$ the expectation is infinite. If $g^+(x) = g^-(x) = \infty$ then the expectation is undefined.

Given this framework, change of variable in the discrete and continuous case is easily arranged.

- If X has finite or countable support, then $Y = g(X)$ is also discrete. Thus we write[12]

$$\mathsf{E}(g(X)) = \sum_i y_i \mathsf{P}(Y = y_i),$$

but in fact $X = x_i$ with $\mathsf{P}(X = x_i)$, then $Y = g(x_i)$ with the same probability. Moreover, $g(X) = y_i$ iff $g(X) \in \{y_i\}$ iff $X \in g^{-1}(\{y_i\})$.

[9] Actually Borel-measurable, see Appendix in Chap. 2.

[10] This means $g^{-1}(B) \in \mathcal{B}$ for every $B \in \mathcal{B}$.

[11] This is indeed a simple rv, see Sect. 2.4.

[12] Observe that $g^{-1}(\{a\}) = \{x \mid g(x) = a\}$.

- If X is continuous, then $\mathsf{P}(X = x) = 0$ and we can approximate $\mathsf{E}(g(X))$ as $\sum_i g(x_i) f_X(x_i) \mathrm{d}x_i$ over all subintervals $(x_i, x_i + \mathrm{d}x_i]$, for $\mathsf{P}(x_i < X \leqslant x_i + \mathrm{d}x_i) \approx f_X(x_i)\mathrm{d}x_i$ and in this case $g(X) \approx g(x_i)$ on each subinterval. In 'the limit' we get formula (4.2).

We can perform integration over subsets.

- For probabilities:
  - Given a rv X we have $\mathsf{P}(X \in B) = \int_B \mathrm{d}F_X(x)$, for every subset $B \subset \mathbb{R}$ such as intervals, half-rays, etc.
  - In particular, $\mathsf{P}(X \in B) = \int_B f_X(x)\mathrm{d}x$, provided that X is continuous with density $f_X(x)$.
  - Otherwise, $\mathsf{P}(X \in B) = \sum_{x \in B} \mathsf{P}(X = x)$, if X is discrete.
- For expectations: take the truncated version $X\mathbf{I}_A$ of a rv X, for any event $A \in \mathcal{F}$, then write $\mathsf{E}(X\mathbf{I}_A)$. Observe that $X\mathbf{I}_A = X$ for every $\omega \in A$.

We give a derivation of the first item. Recall $\mathsf{E}(\mathbf{I}_A) = \mathsf{P}(A)$ and apply it to $A = \{X \in B\} \in \mathcal{F}$, for $B \subset \mathbb{R}$. Then we have:

$$\mathsf{P}(X \in B) = \mathsf{E}(\mathbf{I}_{\{X \in B\}}) = \int_{-\infty}^{\infty} \mathbf{I}_B(x)\mathrm{d}F_X(x) = \int_B \mathrm{d}F_X(x).$$

Fact: $\mathbf{I}_B(X(\omega)) = \mathbf{I}_{\{X \in B\}}$, for a rv X and suitable $B \subset \mathbb{R}$, see Exercise 2.4 in Chap. 2. Whenever a rv is nonnegative, $X \geqslant 0$,[13] we compute its expectation as:

$$\boxed{\mathsf{E}(X) = \int_0^{\infty} [1 - F_X(x)]\mathrm{d}x}. \tag{4.4}$$

We could check formula (4.4) for discrete and continuous rvs, but instead we prove it for general nonnegative rvs:

$$\begin{aligned} \mathsf{E}(X) &= \int_0^{\infty} x \, \mathrm{d}F_X(x) \\ &= \int_0^{\infty} \left[\int_0^x \mathrm{d}u \right] \mathrm{d}F_X(x) \\ &= \int_0^{\infty} \left[\int_u^{\infty} \mathrm{d}F_X(x) \right] \mathrm{d}u \\ &= \int_0^{\infty} [1 - F_X(u)]\mathrm{d}u, \end{aligned} \tag{4.5}$$

[13] Observe that for any rv $X \geqslant 0$ we must have $F_X(0) = 0$.

which is formula (4.4) with $x = u$. This is especially useful whenever $X \geqslant 0$ represents the future price of a traded asset or a derivative instrument. Dropping the assumption $X \geqslant 0$, formula (4.4) generalizes to:

$$\boxed{\mathsf{E}(X) = \int_0^\infty [1 - F_X(x)]\mathrm{d}x - \int_{-\infty}^0 F_X(x)\mathrm{d}x}. \tag{4.6}$$

We give some details on formula (4.4) for nonnegative rvs.

- We used $x = \int_0^x \mathrm{d}u$ which is $u = x$ minus $u = 0$. It is worth noting that we integrate an ordinary Riemann integral over the open interval $(0, x)$.
- The iterated integral in the second equality of (4.5) is a *double integral* with respect to the variables u and x. The third equality is based on exchange the order of integration.[14]
- The underlying double integral is given for each pair (x, u) in the region $(0, \infty) \times (0, x)$, i.e. we must have $0 < u < x$. Thus the inner Stieltjes integral $\int_u^\infty \mathrm{d}F_X(x)$ is correctly evaluated for $x > u$ and yields $F_X(\infty) - F_X(u) = 1 - F_X(u)$.

Formally, the double integral involved in formula (4.5) is $\int_{(a,b]\times(c,d]} g(x, u)\mathrm{d}F_X(x)\,\mathrm{d}F_Y(u)$ which equals

$$\int_a^b \left[\int_c^d g(x, u)\mathrm{d}F_Y(u)\right]\mathrm{d}F_X(x) = \int_c^d \left[\int_a^b g(x, u)\mathrm{d}F_X(x)\right]\mathrm{d}F_Y(u),$$

namely a double integral of the bounded function $g(x, u)$ whatever is the order of integration. We used $g(x, u) = 1$ and $F_Y(u) = u$. We close this section by yet another notation for the expectation, which is useful in symbolic manipulations: $\mathsf{E}(X) = \int_\Omega X\mathrm{d}\mathsf{P}$.

Remark 4.2 By linearity, the expectation of a mixed rv X (see Sect. 2.9) is simply the sum of two expectations: one for the continuous part and one for the discrete part. Formally, given the generalized density $f_X(x) = \sum_{i=1}^n p_i \delta_{x_i}(x) + (1-\alpha) f_c(x)$ we can compute

$$\mathsf{E}(X) = \int f_X(x)\mathrm{d}x = \sum_{i=1}^n p_i x_i (1 - \alpha) \int x f_c(x)\mathrm{d}x,$$

where we have used $\int x\delta_z(x)\mathrm{d}x = z$ to compute

$$\int \sum_{i=1}^n p_i \delta_{x_i}(x)\mathrm{d}x = \sum_{i=1}^n \int p_i \delta_{x_i}(x)\mathrm{d}x = \sum_{i=1}^n p_i x_i.$$

[14] This is justified by the nonnegativity of the functions involved, guaranteeing an application of the Fubini's theorem.

4.3 Variance and Moments

For a rv X modelling profits/losses or rate of return, the spread of its values around its expectation can be interpreted as a risk.

Definition 4.3 Given a rv X with finite expectation and finite $\mathsf{E}(X^2)$, the **variance** is

$$\mathsf{V}(X) := \mathsf{E}\left((X - \mathsf{E}(X))^2\right) \tag{4.7}$$

often written σ_X^2. Denoting μ_X the expectation $\mathsf{E}(X)$, as usual we distinguish between two cases.

- Discrete rvs: $\sigma_X^2 = \sum_i (x_i - \mu_X)^2 \mathsf{P}(X = x_i)$
- Continuous rvs: $\sigma_X^2 = \int (x - \mu_X)^2 f_X(x)\mathrm{d}x$, where we write $\int$ instead of $\int_{-\infty}^{\infty}$.

Expanding the square in formula (4.7), i.e. $\mathsf{E}\big(X^2 - 2X\mathsf{E}(X) + (\mathsf{E}(X))^2\big)$, we get equivalently $\sigma_X^2 = \mathsf{E}(X^2) - (\mathsf{E}(X))^2$. If $\mathsf{E}(X)$ is not defined or is infinite, then $\mathsf{V}(X)$ cannot be defined either. Otherwise, $\mathsf{E}(X)$ is defined and finite and $\mathsf{V}(X)$ is also defined, and is infinite if and only if $\mathsf{E}(X^2) = \infty$. Another useful indicator is **standard deviation**: $\mathsf{SD}(X) := \sqrt{\mathsf{V}(X)} = \sigma_X$. Let $f(m) := \mathsf{E}((X - m)^2)$, for any $m \in \mathbb{R}$. Then, it is minimized by the expectation $m = \mu_X$, and the minimum is the variance, $\min_{m \in \mathbb{R}} f(m) = \mathsf{V}(X)$:

$$\begin{aligned} f(m) &= \mathsf{E}((X - \mu_X + \mu_X - m)^2) \\ &= \mathsf{E}((X - \mu_X)^2) + (\mu_X - m)^2 + 2(\mu_X - m)\underbrace{\mathsf{E}(X - \mu_X)}_{=0} \\ &= \mathsf{V}(X) + (\mu_X - m)^2 \geqslant \mathsf{V}(X) \end{aligned}$$

Then $f(m) \geqslant \mathsf{V}(X)$ and equality holds if and only if $m = \mu_X$. Taking powers of order greater than 2 in (4.7) we can generalize the variance.

Definition 4.4 Given a rv X with pth power X^p, the pth-**moment** is

$$\mathsf{E}(X^p), \quad \text{provided expectation is finite.} \tag{4.8}$$

- Discrete rvs: $\mathsf{E}(X^p) = \sum_i x_i^p \mathsf{P}(X = x_i)$
- Continuous rvs: $\mathsf{E}(X^p) = \int x^p f_X(x)\mathrm{d}x$.

For integer $p = n \in \mathbb{N}$ the corresponding nth-moment $\mathsf{E}(X^n)$ reduces to the expectation, for $n = 1$, or the variance for $n = 2$ provided that we consider pth-**central moments**, i.e. $\mathsf{E}((X - \mu_X)^p)$, for every $p > 0$.

- Discrete rvs: $\mathsf{E}((X-\mu_X)^p) = \sum_i (x_i-\mu_X)^p \mathsf{P}(X=x_i)$
- Continuous rvs: $\mathsf{E}((X-\mu_X)^p) = \int (x-\mu_X)^p f_X(x)\mathrm{d}x$.

We can treat transformed rvs with finite first moment as equally distributed, see Appendix in Chap. 2.

Lemma 4.1 *Let X, Y be rvs and $g : \mathbb{R} \to \mathbb{R}$ be a measurable function such that $g(X), g(Y)$ have finite expectation. We have $X \overset{d}{=} Y$ if and only if $\mathsf{E}(g(X)) = \mathsf{E}(g(Y))$.*

4.4 Moment Generating and Characteristic Functions

Some transformation of rvs can lead to useful tools for getting moments and finding the distribution of sums of rvs.[15]

Definition 4.5 Given a rv X and a real number $u \in \mathbb{R}$, the **moment generating function** (MGF) is

$$M_X(u) := \mathsf{E}(\mathrm{e}^{uX}), \quad \text{for all } u \in \mathbb{R} \text{ s.t. the expectation is finite.} \tag{4.9}$$

- Discrete rvs: $M_X(u) = \sum_i \mathrm{e}^{ux_i} \mathsf{P}(X=x_i)$
- Continuous rvs: $M_X(u) = \int \mathrm{e}^{ux} f_X(x)\mathrm{d}x$.

Fact: assuming $p \in \mathbb{N}$, if MGF exists and is finite in an open interval containing 0, then the pth-moment exists too and is given by the pth-derivative w.r.t. u evaluated at $u=0$, i.e.

$$\boxed{\mathsf{E}(X^p) = \frac{\mathrm{d}^p}{\mathrm{d}u^p} M_X(u)\Big|_{u=0} = \mathsf{E}\left(\frac{\mathrm{d}^p}{\mathrm{d}u^p}\mathrm{e}^{uX}\Big|_{u=0}\right)}.$$

Taking $p=2$ we have $M_X''(0) = \mathsf{E}\left(\frac{\mathrm{d}^2}{\mathrm{d}u^2}\mathrm{e}^{uX}\Big|_{u=0}\right) = \mathsf{E}\left(X^2\mathrm{e}^{uX}\Big|_{u=0}\right) = \mathsf{E}(X^2)$.

Theorem 4.1 *MGF posses the following properties.*

- *If $M_X(u)$ exists for all u in an open interval containing 0, then $M_{a+bX}(u) = \mathrm{e}^{au} M_X(b\,u)$, for all a, b.*
- *If $M_{X_i}(u)$ all exists and $X_1, \ldots, X_n$ are independent rvs, then $M_{\sum_{i=1}^n X_i}(u) = M_{X_1}(u) \cdots M_{X_n}(u) = \prod_{i=1}^n M_{X_i}(u)$.*
- *If $M_X(u) = M_Y(u)$ for all u in an open interval containing 0, then $X \overset{d}{=} Y$.*

[15] They can also be used in proving theorems.

In fact, for independent rvs expectation of their product factorizes, see Lemma 4.2 in Sect. 4.8.

Definition 4.6 Given a rv X and a real number $u \in \mathbb{R}$, the **characteristic function** (CHF) is

$$\phi_X(u) := \mathsf{E}(\mathrm{e}^{\mathrm{i}uX}) \equiv \mathsf{E}(\cos(uX)) + \mathrm{i}\mathsf{E}(\sin(uX)).^{16} \tag{4.10}$$

- Discrete rvs: $\phi_X(u) = \sum_i \mathrm{e}^{\mathrm{i}ux_i}\mathsf{P}(X = x_i)$
- Continuous rvs: $\phi_X(u) = \int \mathrm{e}^{\mathrm{i}ux} f_X(x)\mathrm{d}x$, also known as *Fourier transform*.

Fact: CHF always exists. Recall that the set of complex numbers $\mathbb{C}$ contains $z = a + \mathrm{i}b$ such that $a, b \in \mathbb{R}$ and $\mathrm{i} = \sqrt{-1}$. In fact: $\mathrm{i}^2 = -1, \mathrm{i}^3 = -\mathrm{i}$ and $\mathrm{i}^4 = 1$. The absolute value is $|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$, where $\bar{z} = a - \mathrm{i}b$ is the conjugate of z. Warning: the product of two complex numbers $z, w \in \mathbb{C}$ is given formally by $(a + \mathrm{i}b)(c + \mathrm{i}d)$. Similarly, we can define their sum.

Theorem 4.2 *CHF fulfills the following properties.*

- $|\phi_X(u)| \leqslant \phi_X(0) = 1$, *for every* $u \in \mathbb{R}$
- $\phi_{a+bX}(u) = \mathrm{e}^{\mathrm{i}au}\phi_X(b\,u)$, *for all* $a, b, u \in \mathbb{R}$
- *If* $\mathsf{E}(|X|^n) < \infty$, *then the nth-derivative of CHF exists and is given by*

$$\frac{\mathrm{d}^n}{\mathrm{d}u^n}\phi_X(u)\Big|_{u=0} = \mathrm{i}^n\mathsf{E}(X^n),$$

  for any integer $n = 0, 1, \ldots, p$.
- *If* $X_1, \ldots, X_n$ *are independent rvs, then* $\phi_{\sum_{i=1}^n X_i}(u) = \phi_{X_1}(u) \cdots \phi_{X_n}(u) = \prod_{i=1}^n \phi_{X_i}(u)$.
- *For rvs* X, Y *such that* $\phi_X(u) = \phi_Y(u)$, *for all reals* u, *then* $X \stackrel{d}{=} Y$.
- $\phi_X(u)$ *is a continuous function of* $u \in \mathbb{R}$.

We can recover the moment again by differentiation, i.e.,

$$\boxed{\mathsf{E}(X^n) = \frac{1}{\mathrm{i}^n}\frac{\mathrm{d}^n}{\mathrm{d}u^n}\phi_X(u)\Big|_{u=0}}.$$

Remark 4.3 Further details on CHF are worth to note.

- To any pair $(a, b) \in \mathbb{R} \times \mathbb{R}$ we can associate uniquely a complex number $z = a + \mathrm{i}b$. Moreover, (a, b) corresponds 1-to-1 to a point in the xy plane. Thus, $\phi_X(u)$ can be viewed as a vector-valued function $\mathbb{R} \to \mathbb{C}$ s.t. to each real number $u \mapsto (a, b)$ corresponds a point in the xy plane, where $a = \mathsf{E}(\cos(uX))$ and $b = \mathsf{E}(\sin(uX))$. Hence $\phi(u)$ is a complex number with modulus $|\phi(u)| =$

[16] De Moivre identity or **Euler's Formula**: $\mathrm{e}^{\mathrm{i}x} = \cos x + \mathrm{i}\sin x$, for any real number x.

$\sqrt{(\mathsf{E}(\cos(uX)))^2 + (\mathsf{E}(\sin(uX)))^2}$, and

$$\phi'_X(u) = \left(\tfrac{\mathrm{d}}{\mathrm{d}u}\mathsf{E}(\cos(uX)), \tfrac{\mathrm{d}}{\mathrm{d}u}\mathsf{E}(\sin(uX))\right), \quad \text{[vector of two derivatives]}.$$

We can interchange differentiation with $\mathsf{E}(\cdot)$.

- Assume X has density f_X which is symmetric around zero, i.e. $f_X(x) = f_X(-x)$, and the density is an even function. Then we can write

$$\phi_X(u) = \int \cos(ux) f_X(x)\mathrm{d}x + \mathrm{i}\int \sin(ux) f_X(x)\mathrm{d}x,$$

and since $\sin(ux)$ is an odd function the product $\sin(ux) f_X(x)$ is odd too and the second integral is zero. Fact: $\phi_X(u)$ reduces to a real-valued function iff X is symmetric around zero.

It can be shown that for CHF we have:[17]

$$|\phi_X(u)| \leqslant \mathsf{E}(|\mathrm{e}^{\mathrm{i}uX}|) = \mathsf{E}\left(\cos^2(uX) + \sin^2(uX)\right) = \mathsf{E}(1) = 1.$$

When X is absolutely continuous, the Fourier Transform can be denoted $\mathfrak{F}(f_X)$, which is actually a map acting on the set of densities functions whose image is the CHF. It can be shown that this map has an inverse called *inverse Fourier transform*,

$$f_X(x) = \mathfrak{F}^{-1}(\phi_X) := \tfrac{1}{2\pi}\int \mathrm{e}^{-\mathrm{i}ux}\phi_X(u)\mathrm{d}u,$$

which is nothing else the density. There exist commercial software (e.g. Matlab) which can compute the numerical inverse transform of a specified CHF to obtain the corresponding density.

4.5 Summary Statistics

To any rv X we can associate several numerical indices measuring how its distribution behaves.

Example 4.2 Let X be a degenerate rv, i.e. $X(\omega) = k$ for every scenario $\omega \in \Omega$, or $\mathsf{P}(X = k) = 1$, i.e. $X \overset{\text{a.s.}}{=} k$. Thus, $\mathsf{E}(X) = k$ and the new rv $a + bX$ obtained by a linear transformation, i.e. scaling of the values $x \in X(\Omega)$ by a factor $b \neq 0$ and shifting by a factor a, has a linearly transformed expectation $\mathsf{E}(a + bX) = a + bk$.

[17] See Jensen's inequality.

A number $m \in \mathbb{R}$ is said to be a **location parameter** of F_X if there is a high probability $\mathsf{P}(m - \epsilon < X < m + \epsilon)$ to 'locate' the values $x = X(\omega)$ near m, for a small $\epsilon > 0$; in addition the new rv $a + bX$ must have location $a + b\,m$. Fact: expectation $\mathsf{E}(X) = \mu_X$ and median $\mathsf{med}(X)$ are location parameters. In words, a location parameter shifts a distribution to the right or left without changing the distribution's shape or standard deviation. If P_X is symmetric around $a \in \mathbb{R}$ and $\mathsf{E}(X) < \infty$ then $a = \mathsf{E}(X) = \mathsf{med}(X)$. A **dispersion parameter** of F_X is a number $d \in \mathbb{R}$ measuring the *error* we made in predicting the X's values by using its location parameter. In words, it is a measure of how X fluctuates around its location. If X is a continuous rv, then a dispersion parameter quantifies the extent to which the density f_X is stretched or squeezed. Any dispersion parameter should be positive homogeneous.

Example 4.3 Pick $a, b \in \mathbb{R}$, such that $b \neq 0$. Let $m \in \mathbb{R}$ a location parameter. We define the **mean absolute deviation**

$$\mathsf{MAD}(X) := \mathsf{E}(|X - m|),$$

which is a dispersion parameter. We can show $\mathsf{MAD}(a + bX) = |b|\mathsf{MAD}(X)$. Similarly, standard deviation is a dispersion parameter for which $\mathsf{SD}(a + bX) = |b|\mathsf{SD}(X)$. In both cased we have positive homogeneity.

A **shape parameter** of F_X is a number $s \in \mathbb{R}$ which is neither a location nor a scale parameter, similar to a dispersion parameter. Recall that a rv Y is symmetrically distributed around $a \in \mathbb{R}$ if $a - Y \overset{\mathrm{d}}{=} Y - a$. Hence

$$\begin{aligned}\mathsf{E}((a - Y)^n) = \mathsf{E}((Y - a)^n) &= \mathsf{E}((-(a - Y))^n) \\ &= -\mathsf{E}((a - Y)^n), \quad \text{for integer } n = 1, 3, 5, \ldots\end{aligned}$$

implying $\mathsf{E}(Y^n) = 0$. Now, let $a = \frac{\mu_X}{\sigma_X}$ and $Y = \frac{X}{\sigma_X}$. The following shape parameters are the most relevant.

- **Skewness coefficient**: $\mathsf{sk}(X) := \frac{\mathsf{E}((X-\mu_X)^3)}{\sigma_X^3}$, a measure of symmetry. If $\mathsf{sk}(X) > 0$ we say X is positively skewed or right-skewed, while for $\mathsf{sk}(X) < 0$ we say X is negatively skewed or left-skewed. In words, $\mathsf{sk}(X) > 0$ implies a long right tail. The opposite holds for negative skewness. If X has a symmetric distribution then $\mathsf{sk}(X) = 0$.
- **Kurtosis**: $\mathsf{kur}(X) := \frac{\mathsf{E}((X-\mu_X)^4)}{\sigma_X^4}$. If $\mathsf{P}(|X| \geqslant a)$ is large, for $a > 0$, then the values $x = X(\omega)$ in the center and in the tails of the X's distribution are most likely to be observed and the kurtosis is high. Approximately, this is true for the X's values lying in intervals $(-\infty, \mu_x - 2\sigma_X)$, $(\mu_x - \sigma_X, \mu_x + \sigma_X)$ and $(\mu_x + 2\sigma_X, \infty)$.

The skewness is the standardized third central moment. Indeed, we can write

$$\mathsf{sk}(X) = \mathsf{E}\left(\left(\frac{X-\mu_X}{\sigma_X}\right)^3\right),$$

then the skewness does not depend on the location or the scale parameters (the units in which X is 'measured' do not affect $\mathsf{sk}(X)$). Although $\mathsf{sk}(X) > 0$ implies a long right tail compared to the left tail and $\mathsf{sk}(X) < 0$ implies a long left tail compared to the right tail, there exist asymmetric distributions whose odd central moments are all zero, and in general $\mathsf{sk}(X) = 0$ does not imply symmetry. Rvs having distributions with large kurtosis are more peaked in the center, have low shoulders, and heavy tails. Roughly, the X's values lying on the interval $(\mu_x - \sigma_X, \mu_x + \sigma_X)$ form the P_X's center; the X's values lying on the intervals $(-\infty, \mu_x - 2\sigma_X)$ and $(\mu_x + 2\sigma_X, \infty)$ form the tails; the remaining values form the P_X's shoulders. $\mathsf{sk}(X) = 0$ and $\mathsf{kur}(X) = 3$ are typical for $X \sim \mathrm{N}(\mu, \sigma^2)$. Therefore, $\mathsf{sk}(X) \neq 0$ and high $\mathsf{kur}(X) - 3$ suggest a deviation from the Gaussian model (in case of continuous distributions). $\mathsf{sk}(X) > 0$ implies high $\mathsf{P}(X \geqslant \mu_X) > 0$ and $\mathsf{sk}(X) < 0$ implies low $\mathsf{P}(X \leqslant \mu_X) > 0$. For densities of transformed continuous rvs, the effect of shifting their density's original location as well as modifying their variability is interesting in its own. Let X be continuous with density f_X, and let $Y = g(X)$ for strictly increasing $g(x)$. By monotonicity, g has an inverse function[18] $h(y)$. We have:

$$F_Y(y) = \mathsf{P}(Y \leqslant y) = \mathsf{P}(g(X) \leqslant y) = \mathsf{P}(X \leqslant h(y)) = F_X(h(y)).$$

Now differentiating[19] $F_Y(y) = F_X(h(y))$ we find

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y} F_X(h(y)) = f_X(h(y))h'(y).$$

In general,[20] for a monotone $g(x)$ we have

$$\boxed{f_Y(y) = f_X(h(y))|h'(y)|}. \tag{4.11}$$

Let $Y = a + bX$ for $a, b \in \mathbb{R}$ with $b \neq 0$. Then, $g(x) = a + bx$ has inverse $h(y) = \frac{y-a}{b}$ with $h'(y) = \frac{1}{b}$. We have $f_Y(y) = \frac{1}{|b|} f_X(\frac{y-a}{b})$. Defining a **scale**

[18] That is $h(g(x)) = x$, for any x in the domain of g and $y = g(x)$. Recall that if g is increasing (decreasing), then h is increasing (decreasing) too.

[19] Use the chain rule of differentiation.

[20] Consider $g(x)$ strictly decreasing so that $\mathsf{P}(Y \leqslant y) = \mathsf{P}(X \geqslant h(y)) = 1 - F_X(h(y))$, then differentiate.

parameter $\kappa \in \mathbb{R}$ as a multiple of the standard deviation we have:

- $m + X$ has density $f_X(x - m)$, where $m \in \mathbb{R}$ is a location parameter;
- κX has density $\frac{1}{\kappa} f_X(\frac{x}{\kappa})$, where $\kappa > 0$ is a scale parameter;
- fully, $m + \kappa X$ has density $\frac{1}{\kappa} f_X(\frac{x-m}{\kappa})$.

Fact: the cth-quantile $Q_Y(c)$ of $Y = g(X)$, with strictly increasing[21] $g(x)$, is $g(Q_X(c))$. Recall that for a composite function $v(s(x))$, where both v, s are invertible, we have that the inverse is $s^{-1}(v^{-1}(y))$. In particular, for linear functions: $Q_Y(c) = a + b\, Q_X(c)$, if $b > 0$, and $Q_Y(c) = a + b\, Q_X(1 - c)$, if $b < 0$. If the location parameter $a \in \mathbb{R}$ is shifted by $\lambda \in \mathbb{R}$, i.e. $a + \lambda$, then $\mathsf{E}(X)$, $\mathsf{med}(X)$, whenever they exist, are shifted too by λ. If the scale parameter is scaled, i.e. λb and $\lambda \neq 0$, then $\mathsf{V}(X)$ is scaled by λ^2.

Remark 4.4 (Proper Location-Scale Density) Given a density function $f_\bullet$ and $a \in \mathbb{R}$, $b > 0$, then Y is a rv with density function $\frac{1}{b} f_\bullet\left(\frac{y-a}{b}\right)$ iff there exists another rv X with density $f_\bullet(x)$ and $Y = a + bX$. Indeed, if $f_\bullet = f_X$ is a density we know that

$$f_Y(y) = f_X(h(y))h'(y) = \tfrac{1}{b} f_X\left(\frac{y-a}{b}\right),$$

since $y = g(x) = a + bx$ is increasing with inverse $x = h(y) = \frac{y-a}{b}$. Conversely, assuming the rv Y has density $f_\bullet\left(\frac{y-a}{b}\right)$ defining $h(y) = \frac{y-a}{b}$ and setting $X = h(Y)$ we can apply once again the above result on transformed rvs with densities:

$$f_X(x) = f_\bullet(g(x))g'(x) = \tfrac{1}{b} f_\bullet\left(\tfrac{(a+bx)-a}{b}\right) b,$$

since $h(y)$ is also increasing. Moreover:

$$a + bX = a + bh(Y) = a + b\left(\tfrac{Y-a}{b}\right) = Y.$$

4.6 Inequalities

Proposition 4.1 (Jensen's Inequality) *Let X be a rv with finite expectation and $g : \mathbb{R} \to \mathbb{R}$ be a convex function such that $\mathsf{E}(|g(X)|) < \infty$. Then*

$$g(\mathsf{E}(X)) \leqslant \mathsf{E}(g(X)). \tag{4.12}$$

The proof of Jensen's inequality is instructive from the geometrical point of view.

[21] It suffices to have g as increasing and left-continuous.

Proof of Jensen's Inequality Assume $g(x)$ is convex. Then, there exists a linear function $h(x) = ax + b$ such that its graph is the *support* line of g at the point $x = \mathsf{E}(X)$. This implies $h(x) \leqslant g(x)$ for every $x \in \mathbb{R}$ and also $h(\mathsf{E}(X)) = g(\mathsf{E}(X))$. Thus we have

$$\begin{aligned}\mathsf{E}(g(X)) &\geqslant \mathsf{E}(h(X)) = \mathsf{E}(aX + b) = a\mathsf{E}(X) + b \\ &= h(\mathsf{E}(X)) = g(\mathsf{E}(X)),\end{aligned}$$

and the proof is complete. ■

Remark 4.5 If g is concave, then $-g$ is convex and Jensen's inequality becomes:

$$-g(\mathsf{E}(X)) \leqslant \mathsf{E}(-g(X)) \Longleftrightarrow g(\mathsf{E}(X)) \geqslant \mathsf{E}(g(X)).$$

Jensen's inequality applied to $g(x) = |x|^p$ for $p > 1$ yields:

$$\mathsf{E}(|X|) \leqslant (\mathsf{E}(|X|^p))^{1/p}.$$

Here a list of useful inequalities involving expectation and other moments of order greater than one.

Theorem 4.3 (Cauchy-Schwarz) *Let X and Y be two rvs such that $\mathsf{E}(X^2)$ and $\mathsf{E}(Y^2)$ are both finite. Then*

$$|\mathsf{E}(XY)| \leqslant \mathsf{E}(|XY|) \leqslant \sqrt{\mathsf{E}(X^2)\mathsf{E}(Y^2)}. \tag{4.13}$$

Proof of Cauchy-Schwarz Inequalities By the Jensen's inequality we have $|\mathsf{E}(XY)| \leqslant \mathsf{E}(|XY|)$. Thus, choosing $p = q = 2$ and applying the Hölder's inequality below to $\mathsf{E}(|XY|)$ we have

$$\mathsf{E}(|XY|) \leqslant (\mathsf{E}(X^2))^{1/2}(\mathsf{E}(Y^2))^{1/2},$$

which together with the former inequality prove the result. ■

Cauchy-Schwarz inequalities may be reduced to $(\mathsf{E}(XY))^2 \leqslant \mathsf{E}(X^2)\mathsf{E}(Y^2)$, in fact they can be reduced to only one, considering either the left-hand side and the right-hand side or the innermost term and the right-hand side, each termed Cauchy-Schwarz inequality. The following is a generalization.

Theorem 4.4 (Hölder's Inequality) *Let $p \geqslant 1$ such that $\frac{1}{p} + \frac{1}{q} = 1$, and $\mathsf{E}(|X|^p)$ and $\mathsf{E}(|Y|^q)$ are both finite. Then*

$$\mathsf{E}(|XY|) \leqslant \sqrt[p]{\mathsf{E}(|X|^p)}\sqrt[q]{\mathsf{E}(|Y|^q)}. \tag{4.14}$$

If X and Y are rvs with finite expectation then $\mathsf{E}(|X+Y|) < \infty$: write the triangle inequality $|X + Y| \leqslant |X| + |Y|$ and take expectation of both sides. The following is a generalization.

Theorem 4.5 (Minkowski's Inequality) *Let $p \geqslant 1$ and X, Y be two rvs with $\mathsf{E}(|X|^p)$ and $\mathsf{E}(|Y|^p)$ both finite. Then $\mathsf{E}(|X+Y|^p) < \infty$ and*

$$\sqrt[p]{\mathsf{E}(|X+Y|^p)} \leqslant \sqrt[p]{\mathsf{E}(|X|^p)} + \sqrt[p]{\mathsf{E}(|Y|^p)}. \tag{4.15}$$

Proposition 4.2 (Chebyshev's Inequality) *Let X be a nonnegative rv on some probability space. Then for all $a > 0$ and every $p > 0$ we have*

$$\mathsf{P}(X \geqslant a) \leqslant \frac{\mathsf{E}(X^p)}{a^p}. \tag{4.16}$$

Here are some special cases.

- For $p = 1$ Proposition 4.2 gives **Markov's inequality**;
- For $p = 2$ and a general rv X:
  - Replacing X with $|X - \mu_X|$, Chebyshev's inequality yields $\mathsf{P}(|X - \mu_X| \geqslant a) \leqslant \frac{\mathsf{V}(X)}{a^2}$;
  - Taking $a = \lambda \mathsf{SD}(X)$, with $\lambda > 0$, the last item gives $\mathsf{P}(|X - \mu_X| \geqslant \lambda \mathsf{SD}(X)) \leqslant \frac{1}{\lambda^2}$.

The proof of Chedychev's inequality involves indicators.

Proof of Chebyshev's Inequality If $\mathsf{E}(X^p) = \infty$ the inequality is trivially satisfied. Thus, let us assume[22] $\mathsf{E}(X^p) < \infty$. By truncation, we can write $X^p = X^p \mathbf{I}_{\{X<a\}} + X^p \mathbf{I}_{\{X \geqslant a\}}$. Taking expectation we have:

$$\begin{aligned} \mathsf{E}(X^p) &= \mathsf{E}\big(X^p \mathbf{I}_{\{X<a\}}\big) + \mathsf{E}\big(X^p \mathbf{I}_{\{X \geqslant a\}}\big) \\ &\geqslant \mathsf{E}\big(X^p \mathbf{I}_{\{X \geqslant a\}}\big) \geqslant \mathsf{E}\big(a^p \mathbf{I}_{\{X \geqslant a\}}\big) = a^p \mathsf{P}(X \geqslant a), \end{aligned}$$

because $a^p \leqslant X^p$. ■

Observe that $X(\omega) \geqslant X(\omega)\mathbf{I}_{\{X \geqslant a\}}$: if $\omega \notin \{X \geqslant a\}$ then $X(\omega) \geqslant 0$. Interesting enough, we can group rvs based on finiteness of their moments. We write $X \in L^p$ whenever the pth- moment exists and is finite, i.e. $\mathsf{E}(|X|^p) < \infty$. For integers $n < m$ we have $L^m \subset L^n$. E.g.: $L^2 \subset L^1$, and $X \in L^2$ means $0 \leqslant \mathsf{V}(X) < \infty$; $X \in L^1$ means finite expectation. In words, if X has finite variance, then it has also a finite expectation but the converse is not true.

Remark 4.6 All these collections are understood as equivalence classes with respect to the equivalence relation given by equality P-a.s. among rsv. E.g., L^2 is that collection of $X \overset{\text{a.s.}}{=} Y$ with finite second moment, see Exercise 8.7 in Chap. 8.

[22] For every nonnegative rv $X \geqslant 0$ we always have $\mathsf{E}(X)$ exists, maybe infinite.

The collection of all rvs over a fixed $(\Omega, \mathcal{F}, \mathsf{P})$ is denoted L^0. For rvs $X \in L^2$ we have:

$$\boxed{\mathsf{V}(a + bX) = b^2 \mathsf{V}(X)}, \quad \text{for } a \in \mathbb{R} \text{ and } b \neq 0.$$

In fact: $a + bX \in L^2$, then we have $(a + bX - \mathsf{E}(a + bX))^2 = b^2(X - \mathsf{E}(X))^2$ and taking expectation of both sides we are done. Taking $b = 0$, the variance of a degenerate rv is zero.

Remark 4.7 There is a connection between L^p classes and inequalities involving expectation.

- For any pair of rvs X, Y with finite second moment, i.e. $X, Y \in L^2$, their product XY has finite expectation: it suffices to apply Hölder's inequality with $p = q = 2$.
- Given $X \in L^2$ we always have $(\mathsf{E}(X))^2 \leqslant \mathsf{E}(X^2)$: apply the Cauchy-Schwarz inequality with $Y = 1$. Thus $\mathsf{E}(X^2) - \mu_X^2 \geqslant 0$.
- Any linear combination $aX + bY$, of two rvs $X, Y \in L^2$, again belongs to L^2: given $x, y \in \mathbb{R}$ and for every $a, b \in \mathbb{R}$, take the quadratic $(ax - by)^2 \geqslant 0$. Add $(a^2x^2 + b^2y^2)$ to both sides of this inequality and consider $x = X(\omega)$, $y = Y(\omega)$ for every $\omega \in \Omega$. We have:

$$\begin{aligned} \mathsf{E}\big((a\,X + b\,Y)^2\big) &\leqslant \mathsf{E}\big(2\,a^2\,X^2 + 2\,b^2\,Y^2\big) \\ &= 2\,a^2 \mathsf{E}(X^2) + 2\,b^2 \mathsf{E}(Y^2) < \infty, \end{aligned}$$

  from the assumption $X, Y \in L^2$.

4.7 Covariance and Moments of Random Vectors

Stochastic dependence among rvs can be measured by proper numerical indices. In this section we summarize the most useful ones and limit ourselves to show some properties, postponing remaining details to Chap. 7.

Definition 4.7 Given two rvs $X, Y \in L^2$, their **covariance** is

$$\mathsf{cov}(X, Y) := \mathsf{E}\left((X - \mu_X)(Y - \mu_Y)\right) = \mathsf{E}(XY) - \mathsf{E}(X)\mathsf{E}(Y) =: \sigma_{X,Y}. \tag{4.17}$$

For $X, Y \in L^2$, the **Pearson correlation coefficient** is[23]

$$\rho_{X,Y} = \mathsf{corr}(X, Y) := \frac{\mathsf{cov}(X, Y)}{\sqrt{\mathsf{V}(X)\mathsf{V}(Y)}} = \frac{\mathsf{cov}(X, Y)}{\sigma_X \sigma_Y}. \tag{4.18}$$

Recall that $X, Y \in L^2$ does imply $XY \in L^1$, but for independent $X, Y \in L^1$ then $\mathsf{E}(XY) = \mathsf{E}(X)\mathsf{E}(Y) < \infty$, thus $XY \in L^1$. Factorization of expectation will be investigated later, see Theorem 4.8, Sect. 4.8. If $\mathsf{cov}(X, Y) > 0$, then X's values and Y's values are both above (below) their means, on average.

- $\mathsf{cov}(X, X) = \mathsf{V}(X)$ and $\mathsf{cov}(X, Y) = \mathsf{cov}(Y, X)$; also, $\mathsf{cov}(X, a) = 0$, for $a \in \mathbb{R}$.
- By Cauchy-Schwarz inequality, $\mathsf{E}\left((X - \mu_X)(Y - \mu_Y)\right) \leqslant \sqrt{\mathsf{E}((X - \mu_X)^2)\mathsf{E}((Y - \mu_Y)^2)}$, thus $\mathsf{cov}(X, Y)$ is finite provided[24] $X, Y \in L^2$.
- By the last item, $-1 \leqslant \mathsf{corr}(X, Y) \leqslant 1$.

Expectation can be defined for random vectors.

Definition 4.8 Given a random vector $\mathbf{X} = (X_1, \ldots, X_n)$, with $X_i \in L^1$ for $i = 1, \ldots, n$, the first moment or expectation is

$$\boldsymbol{\mu}_{\mathbf{X}} = \mathsf{E}(\mathbf{X}) := (\mathsf{E}(X_1), \ldots, \mathsf{E}(X_n)) \in \mathbb{R}^n.$$

For $n = 2$, assume $\mathbf{X}$ has a bi-variate density $f_{\mathbf{X}}(x_1, x_2)$. Thus, $\mathsf{E}(\mathbf{X}) = (\mathsf{E}(X_1), \mathsf{E}(X_2))$ is the location of the $f_{\mathbf{X}}$'s graph, i.e. the set of points in $\mathbb{R}^3$ with coordinates (x_1, x_2, z), where $z = f_{\mathbf{X}}(x_1, x_2)$, i.e. a three-dimensional surface. Expectation of coordinate rvs can be deduced from the multivariate distribution.

Example 4.4 (Univariate vs Multivariate Expectation) Let $\mathbf{X} = (X, Y)$ a bi-variate random vector with joint density $f_{\mathbf{X}}$. Then

$$\begin{aligned}\mathsf{E}(X) &= \int_{-\infty}^{\infty} x\, f_X(x)\mathrm{d}x = \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y)\mathrm{d}y\right]\mathrm{d}x \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y.\end{aligned}$$

Similarly we have:

$$\begin{aligned}\mathsf{E}(Y) &= \int_{-\infty}^{\infty} y\, f_Y(y)\mathrm{d}y = \int_{-\infty}^{\infty} y \left[\int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y)\mathrm{d}x\right]\mathrm{d}y \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y\, f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y.\end{aligned}$$

[23] It is a measure of **linear dependence** between X and Y.

[24] Recall that $|\mathsf{E}(Z)| \leqslant \mathsf{E}(|Z|)$.

Having defined the expectation of random vectors, we can go a step further.

Definition 4.9 Let $\mathbf{X} = (X_1, \ldots, X_n)$ be an n-dimensional random vector with $X_i \in L^2$, for $i = 1, \ldots, n$. The $n \times n$ variance-covariance matrix, or simply the covariance matrix, $\mathbf{C}$ of $\mathbf{X}$ is given by taking expectation of each entry inside the auxiliary $n \times n$ matrix

$$\begin{bmatrix} \{X_1 - \mathsf{E}(X_1)\}^2 & \ldots & \{X_1 - \mathsf{E}(X_1)\}\{X_n - \mathsf{E}(X_n)\} \\ \{X_2 - \mathsf{E}(X_2)\}\{X_1 - \mathsf{E}(X_1)\} & \ldots & \{X_2 - \mathsf{E}(X_2)\}\{X_n - \mathsf{E}(X_n)\} \\ \vdots & \ddots & \vdots \\ \{X_n - \mathsf{E}(X_n)\}\{X_1 - \mathsf{E}(X_1)\} & \ldots & \{X_n - \mathsf{E}(X_n)\}^2 \end{bmatrix}.$$

For example, with $n = 2$ we have:

$$\mathbf{C} = \begin{bmatrix} \mathsf{V}(X_1) & \mathsf{cov}(X_1, X_2) \\ \mathsf{cov}(X_2, X_1) & \mathsf{V}(X_2) \end{bmatrix}.$$

In general, for each entry of $\mathbf{C}$ we have $c_{ij} = \mathsf{cov}(X_i, X_j)$, for $i, j = 1, \ldots, n$. Since the n-dimensional random vector $\mathbf{X}$ can be viewed as a $1 \times n$ row vector, the $n \times n$ covariance matrix can be written as

$$\mathbf{C} = \mathsf{E}\Big((\mathbf{X} - \boldsymbol{\mu}_\mathbf{X})'(\mathbf{X} - \boldsymbol{\mu}_\mathbf{X})\Big)$$

where each entry c_{ij} equals the covariance $\mathsf{cov}(X_i, X_j)$ for each pair X_i and X_j inside $\mathbf{X}$. The symbol $'$ denotes matrix transposition. The matrix product $(\mathbf{X} - \boldsymbol{\mu}_\mathbf{X})'(\mathbf{X} - \boldsymbol{\mu}_\mathbf{X})$ is indeed

$$\begin{bmatrix} X_1 - \mathsf{E}(X_1) \\ \vdots \\ X_n - \mathsf{E}(X_n) \end{bmatrix} \text{ by } \Big[X_1 - \mathsf{E}(X_1), \ldots, X_n - \mathsf{E}(X_n)\Big],$$

yielding an $n \times n$ matrix. Now we provide a generalization of moments for random vectors.

Proposition 4.3 *Let* $\mathbf{X}$ *be a* n*-dimensional random (row) vector, and let* $\mathbf{A}$ *be a* $m \times n$ *with real valued entries* a_{ij}*. Then, the* m*-dimensional random vector* $\mathbf{Y} = \mathbf{A}\mathbf{X}'$ *admits:*

1. *first moment* $\mathsf{E}(\mathbf{Y}) = \mathbf{A}\boldsymbol{\mu}'_\mathbf{X} \in \mathbb{R}^m$*;*
2. *second moment* $\mathbf{C}_\mathbf{Y} = \mathbf{A}\mathbf{C}_\mathbf{X}\mathbf{A}'$*,*

where $\mathbf{C}_\mathbf{X}$ *is the* $n \times n$ *covariance matrix of* $\mathbf{X}$ *and* $\mathbf{C}_\mathbf{Y}$ *is the* $m \times m$ *covariance matrix of* $\mathbf{Y}$*.*

For example, given $\mathbf{X} = (X, Y)$ with first moment $\boldsymbol{\mu}_{\mathbf{X}} = \mathsf{E}(\mathbf{X}) = (\mu_X, \mu_Y)$, covariance matrix and deterministic matrix

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} \sigma_X^2 & \sigma_{X,Y} \\ \sigma_{X,Y} & \sigma_Y^2 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 2 & 2 \end{bmatrix},$$

respectively, we have

$$\mathbf{A}\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} \sigma_X^2 + 2\sigma_{X,Y} & \sigma_{X,Y} + 2\sigma_Y^2 \\ 3\sigma_X^2 + \sigma_{X,Y} & 3\sigma_{X,Y} + \sigma_Y^2 \\ 2\sigma_X^2 + 2\sigma_{X,Y} & 2\sigma_{X,Y} + 2\sigma_Y^2 \end{bmatrix}, \quad \text{and}$$

$$\mathbf{A}\mathbf{C}_{\mathbf{X}}\mathbf{A}' = \begin{bmatrix} \sigma_X^2 + 2\sigma_{X,Y} & \sigma_{X,Y} + 2\sigma_Y^2 \\ 3\sigma_X^2 + \sigma_{X,Y} & 3\sigma_{X,Y} + \sigma_Y^2 \\ 2\sigma_X^2 + 2\sigma_{X,Y} & 2\sigma_{X,Y} + 2\sigma_Y^2 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 2 \end{bmatrix},$$

which is a new 3×3 covariance matrix.

Example 4.5 (Portfolio Return) Let $\mathbf{S} = (S_{1,t}, \ldots, S_{n,t})$ be the n-dimensional row vector comprising stock prices $S_{i,t}$ at the end of a fixed time horizon. Assume $S_{i,t} \in L^2$, each $i = 1, \ldots, n$. Recall that

$$V_t = h(S_{1,t}, \ldots, S_{n,t}) := h_1 S_{1,t} + \cdots + h_n S_{n,t}$$

is the portfolio value in t with holdings h_i. Thus, the arithmetic portfolio return in $t+1$ is $r_{p,t+1} = \sum_{i=1}^n w_1 r_{i,t+1}$, where $r_{i,t+1}$ is the $t+1$-arithmetic return of stock ith, see Sect. 2.5, Chap. 2. In matrix notation:

$$\mathbf{r} = (r_{1,t+1}, \ldots, r_{n,t+1}), \qquad \mathbf{w} = (w_1, \ldots, w_n)$$

and $r_p = \mathbf{w}\mathbf{r}' = \sum_{i=1}^n w_i r_{i,t+1}$ yields a special application of Proposition 4.3, Item 1, with the matrix $\mathbf{A}$ replaced by the $1 \times n$ matrix $\mathbf{w}$ of portfolio weights and $\mathbf{X}$ replaced by $\mathbf{r}$:

$$\mathsf{E}(r_p) = \mathsf{E}(\mathbf{w} \cdot \mathbf{r}) = \mathbf{w} \cdot \boldsymbol{\mu}_{\mathbf{r}} \qquad \text{portfolio expected return.}$$

Observe that we used the 'dot'-notation for the inner product of vectors as n-tuples in $\mathbb{R}^n$.

Example 4.6 (Portfolio Variance) Writing $\mathbf{C}_{\mathbf{r}}$ for the $n \times n$ covariance matrix of portfolio return $\mathbf{r}$, we can define the portfolio *expected risk* as:

$$\begin{aligned} \mathsf{V}(r_p) &= \mathsf{V}(\mathbf{w} \cdot \mathbf{r}) \\ &= \mathbf{w}\mathbf{C}_{\mathbf{r}}\mathbf{w}' = (\mathbf{w}\mathbf{C}_{\mathbf{r}}) \cdot \mathbf{w}, \end{aligned}$$

a special case of Proposition 4.3, Item 2 with a scalar variance.

Example 4.7 (Portfolio Optimization) Given estimates of expected stock returns $\mathsf{E}(r_i)$, variances $\mathsf{V}(r_i)$ and covariances $\mathsf{cov}(r_i, r_j)$ we can solve

$$\min \left\{ \mathbf{w}\mathbf{C}_\mathbf{r}\mathbf{w}' = f(w_1, \ldots, w_n) \;\middle|\; \sum_{i=1}^{n} w_i = 1, \text{ and } \mathbf{w} \cdot \boldsymbol{\mu}_\mathbf{r} = c \right\}, \tag{4.19}$$

a constrained quadratic program, for a fixed level $c \in \mathbb{R}$ of portfolio expected return. In fact $f(w_1, \ldots, w_n)$ is a convex function of $n \in \mathbb{N}$ variables, and the constrain set $\{w \in \mathbb{R}^n \mid \sum_{i=1}^{n} w_i = 1, \ \mathbf{w}\boldsymbol{\mu}_\mathbf{r} = c\}$ is a polyhedral set.

Example 4.8 (Portfolio Performance) The portfolio optimization problem in Example 4.7 (actually minimizing the portfolio risk) can be equivalently formulated by replacing the objective function $\mathbf{w}\mathbf{C}_\mathbf{r}\mathbf{w}' = f(w_1, \ldots, w_n)$ with its square root, the portfolio standard deviation, $\sqrt{\mathbf{w}\mathbf{C}_\mathbf{r}\mathbf{w}}$. This is because the function $f(x) = \sqrt{x}$ is increasing for $x \geqslant 0$ and optimal solutions of the former optimization problem are the same as those of the new optimization problem. Under mild conditions, we can also refer to the alternative maximization:

$$\max \left\{ \frac{\mathbf{w} \cdot \boldsymbol{\mu}_\mathbf{r}}{\sqrt{\mathbf{w}\mathbf{C}_\mathbf{r}\mathbf{w}'}} \;\middle|\; \sum_{i=1}^{n} w_i = 1 \right\}. \tag{4.20}$$

Here $\frac{\mathbf{w} \cdot \boldsymbol{\mu}_\mathbf{r}}{\sqrt{\mathbf{w}\mathbf{C}_\mathbf{r}\mathbf{w}'}}$ represents a measure of portfolio performance (viz. risk-adjusted return). For a stock index with future random return $X \in L^2$ the Sharpe Ratio is $\frac{\mathsf{E}(X)}{\mathsf{SD}(X)}$, although X should measures the excess return w.r.t. a benchmark index's return.

We conclude this section with a list of useful formulae. Let $\sum_{i=1}^{n} a_i X_i$ and $\sum_{j=1}^{m} b_j Y_j$ be two linear combination of n and m rvs, respectively, with $a_i, b_j \in \mathbb{R}$. Then:

$$\mathsf{cov}\left(\sum_{i=1}^{n} a_i X_i, \sum_{j=1}^{m} b_j Y_j \right) = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i b_j \mathsf{cov}(X_i, Y_j). \tag{4.21}$$

Formula (4.21) implies:

$$\begin{aligned} \mathsf{V}\left(\sum_{i=1}^{n} a_i X_i \right) &= \mathsf{cov}\left(\sum_{i=1}^{n} a_i X_i, \sum_{j=1}^{n} a_j X_j \right) \\ &= \sum_{i=1}^{n} a_i^2 \mathsf{V}(X_i) + \sum_{i} \sum_{j \neq i} a_i a_j \mathsf{cov}(X_i, X_j). \end{aligned}$$

Further, we also have the following.

- $\mathsf{V}(X+Y) = \mathsf{V}(X) + \mathsf{V}(Y) + 2\mathsf{cov}(X,Y)$ and $\mathsf{V}(X-Y) = \mathsf{V}(X) + \mathsf{V}(Y) - 2\mathsf{cov}(X,Y)$
- $\mathsf{cov}(a+bX, c+dY) = b\,d\,\mathsf{cov}(X,Y)$, thus $\mathsf{cov}(a+X, c+Y) = \mathsf{cov}(X,Y)$.
- $\mathsf{corr}(a+bX, c+dY) = \frac{b\,d\,\mathsf{cov}(X,Y)}{\sqrt{b^2d^2\mathsf{V}(X)\mathsf{V}(Y)}} = \frac{b\,d\,\mathsf{cov}(X,Y)}{|b\,d|\sqrt{\mathsf{V}(X)\mathsf{V}(Y)}} = \frac{b\,d\,\mathsf{cov}(X,Y)}{|b\,d|\,\sigma_X\sigma_Y} = \frac{b\,d}{|b\,d|}\mathsf{corr}(X,Y)$.
- $\mathsf{corr}(X, a+b\,X) = \frac{b\,\mathsf{V}(X)}{\sqrt{\mathsf{V}(X)b^2\,\mathsf{V}(X)}} = \frac{b\,\mathsf{V}(X)}{|b|\sqrt{\mathsf{V}(X)\mathsf{V}(X)}} = \frac{b}{|b|} = \pm 1$.

4.8 Multivariate vs Univariate Expectation

For a function $g : \mathbb{R}^n \to \mathbb{R}$ of a random vector $\mathbf{X} = (X_1, \ldots, X_n)$ s.t. $g(\mathbf{X})$ is a rv inducing events $\{g(\mathbf{X}) \in B\} \in \mathcal{F}$, for suitable subsets $B \subset \mathbb{R}$, we have:

Theorem 4.6 *If* $\mathbf{X}$ *is absolutely continuous with joint density* $f_{\mathbf{X}}$ *and the rv* $g(X_1, \ldots, x_n) \in L^1$ *then*

$$\mathsf{E}(g(X_1, \ldots, X_n)) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(x_1, \ldots, x_x) f_{\mathbf{X}}(x_1, \ldots, x_x) \mathrm{d}x_1 \cdots \mathrm{d}x_n. \tag{4.22}$$

In general[25] Eq. (4.22) can be written using the CDF[26] $F_{\mathbf{X}}$:

$$\boxed{\mathsf{E}(g(X_1, \ldots, x_n)) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(x_1, \ldots, x_n) \mathrm{d}F_{\mathbf{X}}(x_1, \ldots, x_n)}. \tag{4.23}$$

In the bi-variate case the above becomes $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)\mathrm{d}F_{\mathbf{X}}(x,y)$ and we can write $\iint_{\mathbb{R}^2}$ instead of $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}$, or simply $\iint$. Dropping the continuity assumption of components in the random vector, and performing integration via Stieltjes integrals entails the formula for expectation of a function of two rvs.

Theorem 4.7 *If* $\mathbf{X} = (X, Y)$ *with independent components* X, Y, *and* $g : \mathbb{R}^2 \to \mathbb{R}$ *is integrable (e.g. continuous) then*

$$\begin{aligned} \mathsf{E}(g(X,Y)) &= \int \left[\int g(x,y)\mathrm{d}F_X(x) \right] \mathrm{d}F_Y(y) \\ &= \int \left[\int g(x,y)\mathrm{d}F_Y(y) \right] \mathrm{d}F_X(x). \end{aligned} \tag{4.24}$$

[25] It holds also in the case of a discrete random vector. Notice that Theorem 4.6 is based on Fubini's theorem yielding iterated integrals.

[26] Recall that $\mathrm{d}F_{\mathbf{X}}(x_1, \ldots, x_n)$ is not the CDF's total differential.

Applying Theorem 4.7 to independent rvs we get the factorization formula for expectation.

Theorem 4.8 *Assume X, Y are independent and $h, g : \mathbb{R} \to \mathbb{R}$ are integrable functions (e.g. continuous) s.t. $h(X) \in L^1$ and $g(Y) \in L^1$. Then*

$$\mathsf{E}(h(X)g(Y)) = \mathsf{E}(h(X))\mathsf{E}(g(Y)). \tag{4.25}$$

Indeed, $h(x) = g(x) = x$ entails $\mathsf{E}(XY) = \mathsf{E}(X)\mathsf{E}(Y)$ and the rvs X, Y are called **uncorrelated**. In fact: $\mathsf{cov}(X, Y) = \mathsf{E}(XY) - \mathsf{E}(X)\mathsf{E}(Y) = 0$, thus independent rvs are uncorrelated but the converse is not true. Here yet another application of Theorem 4.7 which extend the convolution formula Theorem 3.5 in Sect. 3.5, Chap. 3 to CDFs. Let X, Y be independent rvs. Then we have:

$$F_{X+Y}(u) = \int_{-\infty}^{\infty} F_X(u - y)\mathrm{d}F_Y(y), \quad u \in \mathbb{R}. \tag{4.26}$$

In fact, we simply apply formula (4.23) with $g(x, y) = \mathbf{I}_{\{x+y<u\}}$, where the indicator is 1 if the sum of values $x + y$ is $\leqslant u$, and get:

$$\begin{aligned}
\mathsf{P}(X + Y \leqslant u) &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \mathbf{I}_{\{x+y<u\}} \mathrm{d}F_X(x) \right] \mathrm{d}F_Y(y) \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \mathbf{I}_{\{x<u-y\}} \mathrm{d}F_X(x) \right] \mathrm{d}F_Y(y) \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{u-y} \mathrm{d}F_X(x) \right] \mathrm{d}F_Y(y) \\
&= \int_{-\infty}^{\infty} [F_X(u - y) - F_X(-\infty)] \, \mathrm{d}F_Y(y).
\end{aligned}$$

In Theorems 4.7 and 4.8 we assumed integrable functions g, h. Here a general version based on measurability.

Lemma 4.2 *Let X, Y be rvs on the same probability space. They are independent if and only if*

$$\mathsf{E}(h(X)g(Y)) = \mathsf{E}(h(X))\mathsf{E}(g(Y)),$$

for any pair of measurable functions h, g positive or bounded.

To close this section we summarize the main facts about expectation and variance that can be deduced from the above results.

- Let $X_1, \ldots, X_n$ be rvs over a common $(\Omega, \mathcal{F}, \mathsf{P})$. Then:

$$\mathsf{E}(X_1 + \cdots + X_n) = \mathsf{E}(X_1) + \cdots + \mathsf{E}(X_n),$$

  provided that $\mathsf{E}(X_i) \in L^1$ for[27] $i = 1, \ldots, n$.
- Adding independence among all X_i:

$$\mathsf{E}(X_1 \cdots X_n) = \mathsf{E}(X_1) \cdots \mathsf{E}(X_n)$$

  and

$$\mathsf{V}(X_1 + \cdots + X_n) = \mathsf{V}(X_1) + \cdots + \mathsf{V}(X_n),$$

- Given a rv X s.t. all the moments $\mathsf{E}(X^n)$ exists and are finite, i.e. $X \in L^n$ for $n \geqslant 2$, then we get the central moments as:

$$\mathsf{E}((X - \mu_X)^n) = \sum_{i=0}^{n} \binom{n}{i} \mathsf{E}(X^i)(-\mu_X)^{n-i}, \tag{4.27}$$

  just expand $(X - \mu_X)^n$ by the binomial theorem to get $\sum_{i=0}^{n} \binom{n}{k} X^i (-\mu_X)^{n-i}$, then take expectation.[28]

4.9 More on Transformation of Rvs

Recall that $\mathbb{R}^n$ is a normed space, and in particular a vector space (see Appendix G) and linear functions $\mathbf{T} : \mathbb{R}^n \to \mathbb{R}^n$ can be given a matrix representation. Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be a collection of points in $\mathbb{R}^n$ such that the jth coordinate of $\mathbf{e}_j$ is 1 while the others are zero, each $j = 1, \ldots, n$. This set is called the *standard basis* of $\mathbb{R}^n$, and we can write (uniquely) every point $\mathbf{x} \in \mathbb{R}^n$ as:

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Note that for each element of the basis it holds $\|\mathbf{e}_j\| = 1$. Then, the linear function maps $\mathbf{e}_j$ into $\mathbf{T}(\mathbf{e}_j) \in \mathbb{R}^n$, which can be represented as $\sum_{i=1}^{n} a_{ij} e_i$ for some

[27] All the expectations are admitted to be $+\infty$ or $-\infty$, but not both.

[28] This can be performed term by term in the summation since all the moments exist and are finite, thus we cannot have terms as $+\infty - \infty$.

constants $a_{1j}, \ldots, a_{nj}$. We may define a $n \times n$ matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

whose columns are just the coefficients of the representation of $\mathbf{T}(\mathbf{e}_j)$ for $j = 1, 2, \ldots, n$. It follows that $\mathbf{T}(\mathbf{x}) = \mathbf{A} \cdot \mathbf{x}'$, and we call $\mathbf{A}$ the *matrix representation* of the linear function $\mathbf{T}$ with respect to the standard basis. Conversely, given a $n \times n$ matrix $\mathbf{A} = [a_{ij}]$ we can define a linear function $\mathbf{T} : \mathbb{R}^n \to \mathbb{R}^n$ by the formula $\mathbf{T}(\mathbf{x}) = \mathbf{A} \cdot \mathbf{x}'$, for each $\mathbf{x} \in \mathbb{R}^n$. The determinant of $\mathbf{T}$ is simply $\det \mathbf{A}$.

Remark 4.8 If V, W are vector spaces (see Appendix G), then a linear function $T : V \to W$ is called a **linear operator** provided that $T(ax+by) = aT(x)+bT(y)$ for every $x, y \in V$ and $a, b \in \mathbb{R}$. If $W = \mathbb{R}$ we call T a **linear functional**. The matrix representation of a linear operator rests upon the finite-dimension of V and W, as in the simple case discussed above. Since $(S + T)(x) = S(x) + T(x)$ and $T(ax) = aT(x)$, the collection of all linear operators from a vector space V to another vector space W is a vector space itself. The subclass of all **bounded linear operators**, i.e. those $T : V \to W$ with the property $\sup\{\|T(x)\| : \|x\| = 1\} < \infty$, where the norms in V and W are not necessarily equivalent, is denoted by $L(V, W)$. Observe that if $T \in L(V, W)$, then there exists some $M \geqslant 0$ such that $\|T(x)\| \leqslant M\|x\|$, for all $x \in V$. Furthermore, it can be shown that any bounded linear operator is continuous, that is for any sequence $(x_n)_n \subset V$ such that $\lim_n x_n = x$ we have that $\lim_n T(x_n) = T(x)$. In fact, the two limits yields $\|x_n - x\| = 0$ and $\|T(x_n) - T(x)\|$ as $n \to \infty$.

If we let $M = n \cdot \max\{|a_{ij}| : i, j = 1, \ldots, n\}$ then $\|\mathbf{T}(\mathbf{x})\| = \|\mathbf{A}\mathbf{x}'\| \leqslant M\|\mathbf{x}\|$ and we see that the matrix representation defines a continuous linear operator.

Example 4.9 Let $\mathbf{X} = (X_1(\omega), X_2(\omega))$ be a two-dimensional random vector and

$$\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

be a 2×2 matrix with real entries. Then, $\mathbf{A}$ represents a linear transformation $\mathbf{T} : \mathbb{R}^2 \to \mathbb{R}^2$ which entails a composed map $\mathbf{T} \circ \mathbf{X} : \Omega \to \mathbb{R}^2$ defined by

$$\mathbf{T} \circ \mathbf{X}(\omega) := \mathbf{Y} = \mathbf{A}\mathbf{X}' = \big(4X_1(\omega) + X_2(\omega), 2X_1(\omega) + 3X_2(\omega)\big).$$

Thus any deterministic matrix $\mathbf{A}$ whose dimensions are compatible with that of the random vector $\mathbf{X}$ provides a new random vector $\mathbf{Y}$ as a consequence of Theorem 2.7.

Next we aim at extending the transformation formula for densities viewed in Sect. 4.5, to the n-dimensional case. Let $\mathbf{X}$ be a random vector with joint density

$f_{\mathbf{X}}(x_1, \ldots, x_n)$, and $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^n$ be a continuously differentiable and one-to-one function, i.e. it is differentiable (then continuous) and all its partial derivatives are continuous functions from $\mathbb{R}^n$ to $\mathbb{R}^n$. Such a map is called a C^1-differentiable function. The value of $\mathbf{g}$ at a point $\mathbf{x} = (x_1, \ldots, x_n)$ is the vector of functions $(g_1(\mathbf{x}), \ldots, g_n(\mathbf{x}))$, where $g_i : \mathbb{R}^n \to \mathbb{R}$. Recall that the derivative of $\mathbf{g}$ at $\mathbf{x}$ is represented by its **Jacobian matrix**,

$$\mathbf{J_g}(\mathbf{x}) = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial x_1} & \cdots & \frac{\partial g_n}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla g_1(\mathbf{x}) \\ \vdots \\ \nabla g_n(\mathbf{x}) \end{bmatrix},$$

where $\nabla g_i(\mathbf{x})$ denotes the gradient of each component function g_i at $\mathbf{x}$. We observe that the determinant of the Jacobian matrix, $\det \mathbf{J_g}(\mathbf{x})$, is a continuous function from $\mathbb{R}^n$ to $\mathbb{R}$ and is called simply the **Jacobian** of $\mathbf{g}$ at $\mathbf{x}$. If we assume that $\det \mathbf{J_g}(\mathbf{x}) \neq 0$ then the map $\mathbf{g}$ is called a **diffeomorphism**.[29] In this case $\mathbf{g}$ is invertible (on a neighborhood of $\mathbf{x}$)[30] and the Jacobian of the inverse function $\mathbf{g}^{-1}$ at $\mathbf{y} = \mathbf{g}(\mathbf{x})$ equals $\frac{1}{\det \mathbf{J_g}(\mathbf{x})}$. We have the following result:

Proposition 4.4 *Let $\mathbf{X}$ be a n-dimensional random vector with joint density $f_{\mathbf{X}}(\mathbf{x})$, and $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^n$ be a diffeomorphism. Then $\mathbf{Y} = \mathbf{g}(\mathbf{X})$ is a new n-dimensional random vector with density*

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{g}^{-1}(\mathbf{y}))|\det \mathbf{J}_{\mathbf{g}^{-1}}(\mathbf{y})| \tag{4.28}$$

if $\mathbf{y} \in \mathbf{Y}(\Omega)$, and zero otherwise.

Note that the Jacobian of the inverse function is taken in absolute value. Proposition 4.4 is based on the following theorem, which we present without proof (see for example [2]).

Theorem 4.9 (Jacobi's Transformation Formula) *Let V be an open set in $\mathbb{R}^n$ and $\mathbf{g} : V \to \mathbb{R}^n$ be a diffeomorphism. Then, for a measurable function $\mathbf{h} : \mathbf{g}(V) \to \mathbb{R}^n$ such that $\mathbf{h} \in \mathcal{L}^1(\mathbf{g}(V), \mathcal{B}^n, m^n)$, the composed function $\mathbf{h} \circ \mathbf{g}$ belongs to $\mathcal{L}^1(\mathbb{R}^n, \mathcal{B}^n, m^n)$ and*

$$\int_{\mathbf{g}(V)} \mathbf{h}(\mathbf{y})\mathrm{d}\mathbf{y} = \int_V \mathbf{h}(g(\mathbf{x}))|\det J_{\mathbf{g}}(\mathbf{x})|\mathrm{d}\mathbf{x}$$

holds.

[29] It follows that a C^1-differentiable function g with nonzero Jacobian is ever a bijection. The domain and range of a diffeomorphism can be restricted to open subsets V, W of $\mathbb{R}^n$.

[30] It is also a diffeomorphism.

Here, $\mathcal{B}^n$ is the n-dimensional Bore σ-algebra generated by the open subsets of $\mathbb{R}^n$, and m^n is the n-dimensional Lebesgue measure. In Theorem 4.9, $\mathbf{g}(V)$ is the image of V, i.e. $\{\mathbf{y} \in \mathbb{R}^n : \mathbf{y} = \mathbf{g}(\mathbf{x})\}$. The assumption $\mathbf{h} \in \mathcal{L}^1(\mathbf{g}(V), \mathcal{B}^n, m^n)$ means that $\mathbf{h}$ belongs to the space $\mathcal{L}^1(\mathbf{g}(V), \mathcal{B}^n, m^n)$ of functions defined on $\mathbf{g}(V)$ and having finite n-dimensional Lebesgue integral. This implies the integral of its transformation $\mathbf{h} \circ \mathbf{g}$ also exists, see Appendices M, N and O.[31] A word of caution. If we define a function $g : \mathbb{R}^n \to \mathbb{R}$ such that $g(\mathbf{x})$ takes value on the range of a random vector $\mathbf{X}$, i.e. $\mathbf{x} = \mathbf{X}(\omega)$, we return to the simple situation yet analyzed in the one-dimensional case. More precisely, the distribution of the rv $Y = g(\mathbf{X})$ can be recovered from the joint density function of $\mathbf{X}$, since we have $\{\omega \in \Omega : Y(\omega) \leqslant y\} = \{g(X_1, \ldots, X_n) \leqslant y\}$ and the set $\{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) \leqslant y\}$ equals $g^{-1}((-\infty, y])$ which is contained in the range of the random vector, $\mathbf{X}(\Omega)$. Hence it holds:

$$F_Y(y) = \int_{g^{-1}((-\infty, y])} f_{\mathbf{X}}(\mathbf{x}) \mathrm{d}\mathbf{x}.$$

A special case of Proposition 4.4 is when the diffeomorphism $\mathbf{g}$ is linear, represented by a non-singular $n \times n$ matrix $\mathbf{A}$. We have $\mathbf{Y} = \mathbf{A} \cdot \mathbf{X}'$ and its density function is clearly

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{A}^{-1} \cdot \mathbf{y}') \frac{1}{|\det \mathbf{A}|},$$

where $\mathbf{A}^{-1} = \mathbf{g}^{-1}$ is the inverse of $\mathbf{A}$ and its Jacobian $\mathbf{J}_{\mathbf{g}^{-1}} = \det \mathbf{A}^{-1}$ is the inverse of the determinant of $\mathbf{A}$.

4.10 Exercises

4.1 Let X be a discrete rv. Prove $\mathsf{E}(a + b\,X) = a + b\,\mathsf{E}(X)$.

4.2 Let X be a continuous rv with a density. Prove $\mathsf{E}(a + b\,X) = a + b\,\mathsf{E}(X)$.

4.3 (First order stochastic dominance) Let X, Y be two profit and loss rvs such that Y dominates X in the usual stochastic order, i.e. $F_X(x) \geqslant F_Y(x)$ also written $X \preccurlyeq_{\mathrm{FSD}} Y$. Show that $\mathsf{E}(X) \leqslant \mathsf{E}(Y)$, whenever the expectations are both finite.[32]

4.4 (Symmetric distributions) For any kind of rv X, which is symmetrical distributed around $a \in \mathbb{R}$, we have $a = \mathsf{med}(X)$. Moreover, if $\mathsf{E}(X) < \infty$ then $a = \mathsf{E}(X)$. Prove these.

[31] The appropriate space is now $\mathcal{L}^1(\mathbb{R}^n, \mathcal{B}^n, m^n)$.

[32] In fact, we have seen that when $X \preccurlyeq_{\mathrm{FSD}} Y$ then either $X \leqslant Y$ for each scenario or $\mathsf{P}(X \leqslant Y) = 1$. Moreover, the result holds for increasing $g(x)$ s.t. $g(X), g(Y)$ have finite expectations.

4.5 (Proper location-scale density) Given a continuous rv X with density $f_X(x)$ show that $\frac{1}{b} f_X\left(\frac{x-a}{b}\right)$ is again a density function, provided $a \in \mathbb{R}$ and $b > 0$.

4.6 Let $X \in L^1$, choose $d \in \mathbb{R}$ and let $g(d) = \mathsf{E}(|X - d|)$. Show that the median minimizes the MAD, i.e. $\min_{d\in\mathbb{R}} g(d) = g(m)$ where m is the median of F_X.

4.7 (L^p classes of rvs) Assume $\mathsf{E}(X^m)$ is finite and $0 < n < m$ (not necessarily integers) then $\mathsf{E}(X^n)$ is finite too.

4.8 Let $X \in L^2$ and set $g(X) = \frac{X-\mathsf{E}(X)}{\mathsf{SD}(X)}$. Show that linearity of expectation, $\mathsf{E}(a + bX) = a + b\mathsf{E}(X)$ and quasi-linearity of standard deviation $\mathsf{SD}(a + bX) = |b|\mathsf{SD}(X)$ are equivalent to $(g(a + bX))^2 = (g(X))^2$.

4.9 Given two rvs $X, Y \in L^2$ show that $XY \in L^1$ without using Hölder's inequality.

4.10 Let $g(x) = \mathbf{I}_{\{x\leqslant 1\}}$ and $F(x) = \mathbf{I}_{\{x\geqslant 1\}}$. Then, the former is bounded on every interval $(a, b]$ while the latter is nondecreasing. Compute the Stieltjes integral of $g(x)$ with respect to $F(x)$ for $a = 0$ and $b = 2$ using the intermediate point $x = 1$.

4.11 Assume $0 \leqslant \mathsf{E}(X) < \infty$, for a nonnegative rv X. Show that $n\mathsf{P}(X > n) \to 0$ for $n \to \infty$.

4.12 Let X be a nonnegative rv with $0 \leqslant \mathsf{E}(X) < \infty$, i.e. $X \in L^1$. Show that $\mathsf{P}(X < \infty) = 1$, i.e. X is P-a.s. finite.

4.13 Let X, Y be any pair of rvs for which the expectation exists. Show that[33] $\mathsf{E}(X + Y) = \mathsf{E}(X) + \mathsf{E}(Y)$.

4.14 $X + Y$ is a simple (viz. discrete) rv if X and Y are too. Prove the claim.

4.15 Show that $\mathsf{E}(X) \leqslant \mathsf{E}(Y)$, provided that $X \leqslant Y$ for every scenario and the expectations are both finite.

4.16 Let X be a rv with CDF $F_X(x)$ and quantile function $Q_X(c)$. Show that $\boxed{\mathsf{E}(X) = \int_0^1 Q_X(c)\mathrm{d}c}$

4.17 Show that:

1. Given a rv X $\boxed{X \overset{\text{a.s.}}{=} 0 \text{ implies } \mathsf{E}(X) = 0}$;
2. For two rvs $\boxed{X \overset{\text{a.s.}}{=} Y \text{ with finite expectations implies } \mathsf{E}(X) = \mathsf{E}(Y)}$; this holds for bounded or continuous $g : \mathbb{R} \to \mathbb{R}$ s.t. $g(X)$, $g(Y)$ have finite expectations too.
3. For a rv $\boxed{X \geqslant 0 \text{ with } \mathsf{E}(X) = 0 \text{ then } X \overset{\text{a.s.}}{=} 0}$.

[33] This holds with minor modification: $\mathsf{E}(a\,X + b\,Y) = a\,\mathsf{E}(X) + b\,\mathsf{E}(Y)$ for constants $a, b \in \mathbb{R}$.

4.18 Let X be a nonnegative discrete rv with countable values $X(\Omega) = \{x_1, x_2, \ldots\} = \mathcal{X}$, where $x_i \geqslant 0$ each $i \in \mathbb{N}$. Show that formula

$$\sum_{i=1}^{\infty} x_i \mathsf{P}(X = x_i), \quad \text{where events } \{X = x_i\} \text{ decompose } \Omega, \tag{4.29}$$

a special case of (4.1) with $g(x) = x$, correctly defines $\mathsf{E}(X)$.

4.19 Let $X_1, X_2, \ldots$ be a sequence of nonnegative rvs with $\mathsf{E}(X_i) < \infty$ each $i \in \mathbb{N}$. Show that

$$\mathsf{E}(X_1 + X_2 + \cdots) = \mathsf{E}(X_1) + \mathsf{E}(X_2) + \cdots,$$

linearity of expectation holds for countable many rvs $X_i \geqslant 0$.

4.20 Let $X, X_1, X_2, \ldots$ be nonnegative rvs on the same probability space. Assume $\lim_{n\to\infty} X_n = X$ pointwise. Show that: (i) $(X_n - X) \to 0$ pointwise; (ii) $(X - X_n)^+ \leqslant X$ and $(X_n - X)^- \leqslant X$. Prove the same relations hold P-a.s.

4.21 Prove Proposition 4.4.

4.22 Let $\mathbf{X} = (X_1, X_2)$ be a random vector where $X_i \sim \mathrm{N}(0, 1)$ are independent Gaussian variables. Compute the joint density function of the new random vector $\mathbf{Y} = \mathbf{g}(\mathbf{X})$, where $\mathbf{g} : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $\mathbf{g}(x_1, x_2) = (x_1 + x_2, x_1 - x_2)$.

4.23 Let X be a nonnegative rv taking values $x_i = i = 1, 2, \ldots$. Compute the expectation.

4.24 Let X be a rv representing a profit and loss or a rate of return with finite expectation. Compute $\mathsf{E}(X)$ using the integration by parts formula (4.38) with $a = -\infty$ and $b = \infty$.

4.25 Let X be a rv with CDF F_X and $\mathsf{E}(|X|) < \infty$. Show that $\int_{-\infty}^{x} F_X(u)\mathrm{d}u = \mathsf{E}\big((x - X)^+\big)$, where $(x - X)^+ := \max\{0, x - X\} = (x - X)\mathbf{I}_{\{X \leqslant x\}}$.

4.26 Let $X \geqslant 0$ be a nonnegative rv with cdf F_X and $\mathsf{E}(|X|) < \infty$, representing a loss. A *stop-loss insurance* offers the payment $(X - s)^+ := \max\{0, X - s\}$: the insurer retains a risk of monetary amount $s \in$ and let the *reinsurer* pays for the remainder (the insurer's loss stops at $s > 0$). Show that $\mathsf{E}\big((X - s)^+\big) = \int_s^{\infty}[1 - F_X(x)]\mathrm{d}x$.

Appendix

The Variance Can Be Zero

$\mathsf{V}(X)$ is zero if and only if X is a degenerate rv. Assume $X \overset{\text{a.s.}}{=} c$ and recall that $\mathsf{E}(X) = c$. But $X^2 = c^2 \mathbf{I}_{\{X=c\}}$ so $\mathsf{V}(X) = \mathsf{E}(X^2) - (\mathsf{E}(X))^2 = c^2 - c^2 = 0$. For the converse, assume $\mathsf{V}(X) = 0$ and apply Chebychev's inequality to the rv $|X - c|$ with $p = 2$ and $c = \mu_X \in \mathbb{R}$ being the expectation of X. Then

$$\mathsf{P}(|X - c| \geqslant a) \leqslant \frac{\mathsf{E}((X-c)^2)}{a^2} = \frac{\mathsf{V}(X)}{a^2} = 0,$$

and $\mathsf{P}(|X - c| \geqslant a)$ must be zero for every $a > 0$. Now $\{|X - c| \geqslant a\} \subset \{|X - c| > 0\}$ Hence:

$$1 = 1 - \mathsf{P}\big(|X - c| \geqslant a\big) \leqslant 1 - \mathsf{P}(|X - c| > 0)$$

which implies $1 = \mathsf{P}(|X - c| = 0) = \mathsf{P}(X = c)$ and we conclude that $X \overset{\text{a.s.}}{=} c$.

Hölder and Minkowski Inequalities

For all $x, y > 0$ it holds

$$x^{1/p} y^{1/q} \leqslant \frac{x}{p} + \frac{y}{q}, \tag{4.30}$$

where $p, q \in (1, \infty)$ are said to be **conjugate exponents**, i.e. they satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Now, $\ln(\,\cdot\,)$ is concave on $I = (0, \infty)$ and then:

$$\ln(\tfrac{x}{p} + \tfrac{y}{q}) \geqslant \tfrac{1}{p} \ln x + \tfrac{1}{q} \ln y = \ln\left(x^{1/p} y^{1/q}\right).$$

To get (4.30) take the exponential $\mathrm{e}^{(\,\cdot\,)}$ of both sides of the last inequality.

Proof of Hölder's Inequality Inequality (4.30) implies $|XY| \leqslant \frac{|X|^p}{p} + \frac{|Y|^q}{q}$, after we let $x^{1/p} = |X|$ and $y^{1/q} = |Y|$. Assume that $(\mathsf{E}(|X|^p))^{1/p}$ and $(\mathsf{E}(|Y|^q))^{1/q}$ are both equal to 1 so that also $\mathsf{E}(|X|^p) = \mathsf{E}(|Y|^q) = 1$. Then, taking expectation of both sides of the above inequality gives $\mathsf{E}(|XY|) \leqslant \frac{\mathsf{E}(|X|^p)}{p} + \frac{\mathsf{E}(|Y|^q)}{q} = \frac{1}{p} + \frac{1}{q} = 1$. Now, assume that neither $(\mathsf{E}(|X|^p))^{1/p}$ nor $(\mathsf{E}(|Y|^q))^{1/q}$ is zero. Hence, replacing

X and Y in the last inequality by $\frac{X}{(\mathsf{E}(|X|^p))^{1/p}}$ and $\frac{Y}{(\mathsf{E}(|Y|^q))^{1/q}}$ we have

$$\frac{1}{(\mathsf{E}(|X|^p))^{1/p}(\mathsf{E}(|Y|^q))^{1/q}} = \mathsf{E}(|XY|) \leqslant 1$$

and the theorem is proved. ■

Proof of Minkowski's Inequality We first prove that $\mathsf{E}(|X+Y|^p)$ is finite, for $p > 1$. Let $g(x) = (1+x)^p - 2^{p-1}(1+x^p)$ and note that $g(1) = 0$ and $\max_{x\in\mathbb{R}} g(x) = g(1)$. Then, $g(x) \leqslant 0$. Let $x = \frac{|Y|}{|X|}$ and rewrite the last inequality:

$$\frac{|X+Y|^p}{|X|^p} \leqslant 2^{p-1}\left(\frac{|X|^p+|Y|^p}{|X|^p}\right) \quad \text{if and only if}$$
$$|X+Y|^p \leqslant 2^{p-1}|X|^p + |Y|^p.$$

Taking expectation of both sides and using $\mathsf{E}(|X+Y|^p) < \infty$ we are done. Now, let $q = \frac{p}{p-1}$ so that $\frac{1}{p} + \frac{1}{q} = 1$. Then, apply twice the Hölder's inequality, first to X and $|X+Y|^{p-1}$ and then to Y and to $|X+Y|^{p-1}$, we get $\mathsf{E}(|X||X+Y|^{p-1}) \leqslant (\mathsf{E}(|X|^p))^{1/p}(\mathsf{E}(|X+Y|^p))^{1/q}$ and $\mathsf{E}(|Y||X+Y|^{p-1}) \leqslant (\mathsf{E}(|Y|^p))^{1/p}(\mathsf{E}(|X+Y|^p))^{1/q}$. But $|X+Y|^p = |X+Y||X+Y|^{p-1} \leqslant |X||X+Y|^{p-1} + |Y||X+Y|^{p-1}$, by the triangle inequality. Hence, we have

$$\mathsf{E}(|X+Y|^p) \leqslant (\mathsf{E}(|X+Y|))^{1/q}\left((\mathsf{E}(|X|^p))^{1/p}(\mathsf{E}(|Y|^p))^{1/p}\right)$$

and dividing by $(\mathsf{E}(|X+Y|))^{1/q}$ inequality (4.15) is proved. ■

The Covariance Matrix Is Semi-definite Positive

From formula (4.22), it follows the covariance matrix is **semi-definite positive**:

$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \mathsf{cov}(X_i, X_j) = \mathsf{cov}\left(\sum_{i=1}^{n} a_i X_i, \sum_{j=1}^{n} a_j X_j\right) = \mathsf{V}\left(\sum_{i=1}^{n} a_i X_i\right) \geqslant 0. \tag{4.31}$$

In matrix notation: $\mathbf{aCa'} \geqslant 0$, for every $\mathbf{a} \in \mathbb{R}^n$. Recall that every element of $\mathbb{R}^n$ is indeed a row vector, i.e. an $1 \times n$ matrix. Notation $\mathbf{a}'$ means transposition,[34] yielding a column vector, i.e. a $n \times 1$ matrix. Notation $\mathbf{aC}$ means matrix multiplication,

[34] For a $m \times n$ matrix $\mathbf{A}$, its transpose is the $n \times m$ matrix $\mathbf{A}'$.

yielding a $(1 \times n)(n \times n) = 1 \times n$ matrix (a row vector). Hence, $\mathbf{aCa}'$ is a 1×1 matrix, i.e. a scalar whose algebraic expression is given by (4.31).

Some Results on Measure-Theoretic Expectation

Recall that a simple rv is of the form $X = \sum_{i=1}^{n} x_i \mathbf{I}_{\{X \in B_i\}}$, with $\{X \in B_i\}$ partitioning Ω, i.e. a discrete rv with finite values by setting $\{X \in B_i\} = \{X = x_i\}$ for $B_i = \{x_i\}$. It can be show that sums of simple rvs, products of constants by simple rvs are also simple rvs, see Exercise 5.11. We have:

Lemma 4.3 (Limit of Increasing Simple Rvs) *Suppose $X \geqslant 0$ is a nonnegative rv. Then, there exists an increasing sequence $X_1 \leqslant X_2 \leqslant X_3 \leqslant \cdots$ of simple rvs which are nonnegative and converge pointwise to X.*

Remark 4.9 We can define a measurable map $X_n : \Omega \to \mathbb{R}$ for every $n \in \mathbb{N}$, then obtaining a sequence $(X_n)_{n\in\mathbb{N}}$ of rvs. For each $\omega \in \Omega$ the set

$$\{X_1(\omega), X_2(\omega), \ldots\} = \{x_1, x_2, \ldots\}$$

is an infinite sequence of real numbers. If $\lim_{n\to\infty} x_n = x$, then the number x corresponds to scenario $\omega \in \Omega$. It might happen that for any other scenario the above limit exists but in general it is not equal to the same number x. Thus, we have a function that to each ω attaches a number given by a limit. In this case the sequence $(X_n)_{n\in\mathbb{N}}$ is said to **converge pointwise** to the limit function $X : \Omega \to \mathbb{R}$. Namely, $\forall\, \omega \in \Omega$ and $\forall\, \epsilon > 0$, there exists a positive integer N such that $n \geqslant N$ implies $|X_n(\omega) - X(\omega)| < \epsilon$. It can be shown that the random function X is also a rv.

To prove Lemma 4.3, we provide an increasing sequences of discrete rvs for the pointwise convergence:

$$X_n := \sum_{i=1}^{n2^n} \tfrac{i-1}{2^n} \mathbf{I}_{\left\{\frac{i-1}{2^n} < X \leqslant \frac{i}{2^n}\right\}} + n\mathbf{I}_{\{X>n\}}. \tag{4.32}$$

It approximates X from below. It is worth noting that we have $n2^n$ terms in the sum and a $(n2^n + 1)$th term outside it. To figure out the behavior of such a sequence, we display the first two terms:

$$\begin{aligned} X_1 &= \sum_{i=1}^{2} \tfrac{i-1}{2} \mathbf{I}_{\left\{\frac{i-1}{2} < X \leqslant \frac{i}{2}\right\}} + \mathbf{I}_{\{X>1\}} \\ &= 0 + \tfrac{1}{2}\mathbf{I}_{\left\{\frac{1}{2} < X \leqslant 1\right\}} + \mathbf{I}_{\{X>1\}} \end{aligned}$$

$$X_2 = \sum_{i=1}^{2\times 2^2} \tfrac{i-1}{2^2}\mathbf{I}_{\left\{\frac{i-1}{2^2} < X \leqslant \frac{i}{2^2}\right\}} + 2\mathbf{I}_{\{X>2\}} = \begin{cases} 0 & \text{if } i = 1 \text{ and } X \text{ is on } (0, \frac{1}{4}] \\ \frac{1}{4} & \text{if } i = 2 \text{ and } X \text{ is on } (\frac{1}{4}, \frac{1}{2}] \\ \frac{1}{2} & \text{if } i = 3 \text{ and } X \text{ is on } (\frac{1}{2}, \frac{3}{4}] \\ \vdots & \vdots \\ \frac{7}{4} & \text{if } i = 8 \text{ and } X \text{ is on } (\frac{7}{4}, 2] \\ 2 & \text{at stage } 2 \times 2^2 + 1 = 9 \text{ and } X \text{ is on } (2, \infty). \end{cases}$$

This construction yields the following consequences.

- For fixed $n \in \mathbb{N}$ the above nonnegative simple rv is based on a $n2^n + 1$-fold partition of the positive half-axe $[0, \infty)$, into $n2^n$ intervals $(\frac{i-1}{2^n}, \frac{i}{2^n}]$ of equal length $\frac{1}{2^n}$, for $i = 1, \ldots, n2^n$, and a last interval $(\frac{n2^n}{2^n}, \infty)$.
- For every scenario ω, the $X_n(\omega)$'s values are $\frac{i-1}{2^n}$ for $i = 1, \ldots, n2^n + 1$ i.e. they are the left-hand points of $(\frac{i-1}{2^n}, \frac{i}{2^n}]$ over the first $n2^n$ intervals, with the addition of the value $\frac{n2^n}{2^n} = n$ over the last interval.
- Incrementing n bisects each interval, in particular passing from n to $n + 1$ yields $(\frac{i-1}{2^n}, \frac{i}{2^n}]$ divided into two halves of length $\frac{\frac{1}{2^n}}{2} = \frac{1}{2^{n+1}}$.
- $X_{n+1}(\omega) = \frac{i-1}{2^n} = X_n(\omega)$ or $X_{n+1}(\omega) = \frac{i}{2^n} - \frac{1}{2^{n+1}} = \frac{2i-1}{2^{n+1}}$, for every scenario.
- Since $\frac{i-1}{2^n} = \frac{2(i-1)}{2^{n+1}}$ we have $X_n(\omega) \leqslant X_{n+1}(\omega)$, for every scenario; moreover $X_n(\omega) \leqslant X(\omega)$ for every ω.
- For $n \to \infty$ each interval collapses to a single value $x = X(\omega)$.

Thus we have proved $0 \leqslant X_1 \leqslant X_2 \leqslant \cdots \leqslant X$ and $\lim_{n\to\infty} X_n = X$ pointwise. A general definition of expectation is valid for every kind of rv X and is based on the notion of abstract Lebesgue integral, see Appendix N.

1. Let X_n be a simple rv for any $n \in \mathbb{N}$. Then, we define $\mathsf{E}(X_n) = \sum_{i=1}^{m} x_{in}\mathsf{P}(X_n \in B_{im})$ and denote it $\int_\Omega X_n \mathrm{d}\mathsf{P}$, for $m \in \mathbb{N}$.
2. For $X \geqslant 0$, we use Lemma 4.3 to produce a sequence $(X_n)_{n\in\mathbb{N}}$ which approximates X from below, then define $\mathsf{E}(X)$ as $\lim_{n\to\infty} \mathsf{E}(X_n)$.

Up to this point, we need the following result for interchanging expectation and limits.

Theorem 4.10 (Monotone Convergence Theorem (MCT)) *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of nonnegative rvs and let X be any rv such that $0 \leqslant X_1 \leqslant X_2 \leqslant \cdots \leqslant X$. If the sequence converges pointwise to X, then*

$$\lim_{n\to\infty} \mathsf{E}(X_n) = \mathsf{E}(X).$$

Comparing with Lemma 4.3, we do not necessarily use simple rvs.

Proof Assuming pointwise convergence $X_n \to X$, we have X is a rv. Furthermore, $\mathsf{E}(X_1) \leqslant \mathsf{E}(X_2) \leqslant \cdots \leqslant \mathsf{E}(X)$ by assumption. Hence, $\lim_n \mathsf{E}(X_n) \leqslant \mathsf{E}(X)$. It suffices to show the converse inequality. The case $\mathsf{E}(X_1) = \infty$ is trivial, then we assume $\mathsf{E}(X_1) < \infty$. By formula (4.33) we can approximate $\mathsf{E}(X)$ by the expectation of a nonnegative simple rv $Z = \sum_{i=1}^m z_i \mathbf{I}_{A_i}$ such that $Z \leqslant X$. The latter condition means that $z_i \leqslant X(\omega)$ for every $\omega \in A_i$. Let us show that $\lim_n \mathsf{E}(X_n) \geqslant \sum_{i=1}^m z_i \mathsf{P}(A_i)$. Pick $\epsilon > 0$, define $A_{in} = \{\omega \in A_i \mid X_n(\omega) > z_i - \epsilon\}$ and observe that $A_{in} \uparrow A_i$ as $n \to \infty$. Furthermore, $\mathsf{E}(X_n) \geqslant \sum_{i=1}^m (z_i - \epsilon)\mathsf{P}(A_{in})$. By continuity from below we have

$$\lim_{n\to\infty} (z_i - \epsilon)\mathsf{P}(A_{in}) = (z_i - \epsilon)\mathsf{P}(A_i),$$

which implies $\lim_{n\to\infty} \mathsf{E}(X_n) \geqslant \sum_{i=1}^m (z_i - \epsilon)\mathsf{P}(A_i) = \sum_{i=1}^m z_i \mathsf{P}(A_i) - \epsilon$. Since $\epsilon > 0$ is arbitrary it follows $\lim_n \mathsf{E}(X_n) \geqslant \sum_{i=1}^m z_i \mathsf{P}(A_i)$ and we are done. ■

Remark 4.10 In the proof of Theorem 4.10 we used the following fact. Let $a, b \in \mathbb{R}$ such that $a \leqslant b + \epsilon$, and $\epsilon > 0$, then $a \leqslant b$. The proof runs by contradiction: assume $a \leqslant b + \epsilon$ and $\neg(a \leqslant b) = a > b$. But choosing $\epsilon = \frac{a-b}{2} > 0$ we arrive to $a > b + \epsilon$, which contradicts the assumption.

Remark 4.11 Recall that any increasing sequence of real numbers may have a finite supremum, which is the limit of the sequence. Thus:

$$\mathsf{E}(X) := \sup\left\{\mathsf{E}(Z) \colon 0 \leqslant Z \leqslant X, \;\; Z \text{ simple}\right\} \quad \text{identify } Z \equiv X_n. \tag{4.33}$$

Here, the increasing sequence of real numbers is $(\mathsf{E}(X_n))_{n\in\mathbb{N}}$ and its supremum entails $\mathsf{E}(X)$ for $X \geqslant 0$. The set $\left\{\mathsf{E}(Z) \colon 0 \leqslant Z \leqslant X, \;\; Z \text{ simple}\right\}$ is of the form $[0, x)$ or $[0, x]$, where $x = \infty$ is allowed.

Now, step 2 reads:

$$\lim_{n\to\infty} \mathsf{E}(X_n) = \mathsf{E}\left(\lim_{n\to\infty} X_n\right) = \mathsf{E}(X),$$

i.e. expectation and limits can be exchanged.

3. For a rv X not necessarily $\geqslant 0$, we decompose $X = X^+ - X^-$ and by Lemma 4.3 we always find two sequences of nonnegative rvs $(X_n^+)_{n\in\mathbb{N}}$ and $(X_n^-)_{n\in\mathbb{N}}$ s.t. $\lim_{n\to\infty} X_n^+ = X^+$ and $\lim_{n\to\infty} X_n^- = X^-$, pointwise. If both expectations $\mathsf{E}(X^+)$ and $\mathsf{E}(X^-)$ are finite we can define:

$$\int_\Omega X \mathrm{d}\mathsf{P} = \mathsf{E}(X) := \mathsf{E}(X^+) - \mathsf{E}(X^-) = \int_\Omega X^+ \mathrm{d}\mathsf{P} - \int_\Omega X^- \mathrm{d}\mathsf{P} < \infty. \tag{4.34}$$

When the expectation of either X^+ or X^- is infinite we have $\mathsf{E}(X) = \pm\infty$; when both equal infinity we no longer can define $\mathsf{E}(X)$.

Remark 4.12 We have seen that formulae (4.1) and (4.2) in Sect. 4.2, for expected values of discrete and continuous rvs use sufficient conditions of absolute convergence (either sum or integral). In fact, in the current general setting we have $|X| = X^+ + X^-$, then:

- define $\mathsf{E}(|X|) = \mathsf{E}(X^+) + \mathsf{E}(X^-)$ by formula (4.34) and assume it is finite. From $X^+ \leqslant |X|$ and $X^- \leqslant |X|$ it follows both $\mathsf{E}(X^+)$, $\mathsf{E}(X^-)$ are finite so that $\mathsf{E}(X) < \infty$.
- On the other hand, assume both $\mathsf{E}(X^+)$, $\mathsf{E}(X^-)$ are finite. Then summing and using linearity we get $\mathsf{E}(|X|) < \infty$.

Moral: absolute convergence $\mathsf{E}(|X|) < \infty$ is equivalent to existence of $\mathsf{E}(X) < \infty$.

What we have learned is summarized below.

- For fixed $n \in \mathbb{N}$, expectation $\mathsf{E}(X_n)$ is the Lebesgue integral $\sum_{i=1}^{n} x_i \, \mathsf{P}(A_i)$, for a simple rv $X_n = \sum_{i=1}^{m} x_{i,n} \mathbf{I}_{A_{in}}$ where $A_{in} = \{X_n \in B_{in}\}$ are events decomposing Ω.
- Steps 2 and 3 yield a limiting procedure corresponding to a more refined 'version' of a general rv X with more values in its support, and consequently a 'refined partition' of Ω through an increasing number of events A_i. As byproduct, an integration procedure is carried on Ω: the **abstract expectation** $\int_\Omega X \mathrm{d}\mathsf{P}$ is the Lebesgue integral of the measurable function X with respect to the measure P.

We have the extended change of variable formula

$$\int_\Omega X(\omega)\mathrm{d}\mathsf{P} = \int_{\mathbb{R}} x \mathrm{d}\mathsf{P}_X = \int_{-\infty}^{\infty} x \mathrm{d}F_X(x), \tag{4.35}$$

which also holds for transformation $g(X)$. The precise statement follows.[35]

Theorem 4.11 *Let X be a rv with distribution P_X and let $g : \mathbb{R} \to \mathbb{R}$ be a Borel-measurable function. Then we have*

$$\mathsf{E}(g(X)) = \int_{\mathbb{R}} g(x)\mathrm{d}\mathsf{P}_X, \tag{4.36}$$

provided that both sides are well-defined.

By 1-to-1 correspondence between P and CDF F_X, via the law P_X, formula (4.36) extends formula (4.35).

Proof First we check formula (4.36) for indicators $g(X) = \mathbf{I}_B(X)$, where $B \in \mathcal{B}$. We have $\int_\Omega \mathbf{I}_B(X)\mathrm{d}\mathsf{P} = \int_\Omega \mathbf{I}_{\{X \in B\}}\mathrm{d}\mathsf{P} = \mathsf{E}(\mathbf{I}_{\{X \in B\}}) = \mathsf{P}(X \in B) = \mathsf{P}_X(B) = \int_{\mathbb{R}} \mathbf{I}_B(x)\mathrm{d}\mathsf{P}_X$. By linearity, formula (4.36) holds for simple rvs $g(X)$. If $g(X)$ is

[35] Recall that $g \circ X$ is a rv, provided X is a rv and g is Borel-measurable.

any rv, then we apply MCT to both $g(X)^+$ and $g(X)^-$ as follows:

$$\begin{aligned}\int_\Omega g(X)\mathrm{d}\mathsf{P} &= \int_\Omega \lim_n (g_n^+(X) - g_n^+(X))\mathrm{d}\mathsf{P} \\ &= \lim_n \int_\Omega g_n^+(X)\mathrm{d}\mathsf{P} - \lim_n \int_\Omega g_n^-(X)\mathrm{d}\mathsf{P} \\ &= \lim_n \int_{\mathbb{R}} g_n^+(x)\mathrm{d}\mathsf{P}_X - \lim_n \int_{\mathbb{R}} g_n^-(x)\mathrm{d}\mathsf{P}_X \\ &= \int_{\mathbb{R}} \lim_n (g_n^+(x) - g_n^+(x))\mathrm{d}\mathsf{P}_X \\ &= \int_{\mathbb{R}} (g^+(x) - g^-(x))\mathrm{d}\mathsf{P}_X = \int_{\mathbb{R}} g(x)\mathrm{d}\mathsf{P}_X,\end{aligned}$$

where $g_n^+(X) \uparrow g^+(X)$, $g_n^-(X) \uparrow g^-(X)$ pointwise as $n \to \infty$ and $g_n^+(X)$, $g_n^-(X)$ are simple rvs. ■

There is a connection between the aforementioned approach to expectation and Stieltjes integration. Choosing $(X_n)_{n\in\mathbb{N}}$ in MCT as defined by (4.32), we have

$$\begin{aligned}\mathsf{E}(X_n) &= \mathsf{E}\left(\sum_{i=1}^{n2^n} \tfrac{i-1}{2^n}\mathbf{I}_{\left\{\frac{i-1}{2^n} < X \leqslant \frac{i}{2^n}\right\}} + n\mathbf{I}_{\{X>n\}}\right) \\ &= \sum_{i=1}^{n2^n} \tfrac{i-1}{2^n}\mathsf{P}\left(\tfrac{i-1}{2^n} < X \leqslant \tfrac{i}{2^n}\right) + n\mathsf{P}(X > n).\end{aligned}$$

Observe that $\mathsf{P}\left(\frac{i-1}{2^n} < X \leqslant \frac{i}{2^n}\right) = \mathsf{P}\left(X_n = \frac{i-1}{2^n}\right)$, and recall it equals $F_X(\frac{i}{2^n}) - F_X(\frac{i-1}{2^n})$, for every $i = 1, \ldots, n2^n$ and each $n \in \mathbb{N}$. Eventually, by MCT:

$$\begin{aligned}\mathsf{E}(X) = \lim_{n\to\infty} \mathsf{E}(X_n) = \lim_{n\to\infty} \sum_{i=1}^{n2^n} \tfrac{i-1}{2^n}\left[F_X(\tfrac{i}{2^n}) - F_X(\tfrac{i-1}{2^n})\right] \\ + \lim_{n\to\infty} n\,\mathsf{P}(X > n).\end{aligned}$$

Now, we have 2 cases:

- If $\mathsf{P}(X = \infty) > 0$, then the limit is $\mathsf{E}(X) = \infty$;
- If $\mathsf{P}(X < \infty) = 1$, then the limit $\mathsf{E}(X)$ is finite and writes

$$\mathsf{E}(X) = \lim_{n\to\infty} \sum_{i=1}^{n2^n} \tfrac{i-1}{2^n}\left[F_X(\tfrac{i}{2^n}) - F_X(\tfrac{i-1}{2^n})\right] := \int_0^\infty x\,\mathrm{d}F_X(x), \tag{4.37}$$

where se used $\lim_{n\to\infty} n\,\mathsf{P}(X > n) = 0$, see Exercise 5.10.

The Stieltjes integral in (4.37) can be formally given by $\lim_{k\to\infty}\int_0^k g(x)\mathrm{d}F(x)$, for $k > 0$. Here we recall typical properties of the integral.

- (Linearity in the integrand) $\int_0^\infty [g(x) + h(x)]\mathrm{d}F_X(x) = \int_0^\infty g(x)\mathrm{d}F_X(x) + \int_0^\infty h(x)\mathrm{d}F_X(x)$, provided that the latter integral is also finite and g, h are two suitable transformation of the rv X. In particular, $\int_0^\infty kg(x)\mathrm{d}F_X(x) = k\int_0^\infty g(x)\mathrm{d}F_X(x)$, for $k \in \mathbb{R}$.
- (Linearity in the measure) For mixed CDFs $F_1(x) + (1-\alpha)F_2(x)$, with $0 \leqslant \alpha \leqslant 1$, we have

$$\int_0^\infty g(x)\mathrm{d}\left(\alpha F_{X_1}(x) + (1-\alpha)F_{X_2}(x)\right) = \alpha\int_0^\infty g(x)\mathrm{d}F_{X_1}(x) + (1-\alpha)\int_0^\infty g(x)\mathrm{d}F_{X_2}(x),$$

  for two rvs X_1, X_2.
- If $g(x) \leqslant h(x)$ for every x, then $\int_0^\infty g(x)\mathrm{d}F_X(x) \leqslant \int_0^\infty h(x)\mathrm{d}F_X(x)$.
- $\left|\int_0^\infty g(x)\mathrm{d}F_X(x)\right| \leqslant \int_a^b |g(x)|\mathrm{d}F_X(x)$.
- We can partition $(0,\infty)$ in $(0, y] \cup (y, \infty)$ and get $\int_0^y g(x)\mathrm{d}F_X(x) + \int_y^\infty g(x)\mathrm{d}F_X(x)$, for $0 < y < \infty$.

All above properties can be extended to $\int_{-\infty}^\infty g(x)\mathrm{d}F_X(x) = \lim_{k\to\infty}\int_{-k}^k g(x)\mathrm{d}F_X(x)$. Stieltjes integration over finite intervals is indeed based on *left-open/right-closed* intervals, i.e, computing $\int_a^b g(x)\mathrm{d}F_X(x)$, for real $a < b$, equals $\int_{(a,b]} g(x)\mathrm{d}F_X(x)$. In fact, F_X is increasing right-continuous. For $g(x) = 1$, each $x \in \mathbb{R}$, we have:

$$\int_{(a,b]} \mathrm{d}F_X(x) = F_X(b) - F_X(a) \quad \text{and} \quad \int_{(a,b)} \mathrm{d}F_X(x) = F_X(b^-) - F_X(a).$$

Given two CDFs $F_X(x)$ and $F_Y(x)$, corresponding to rvs X, Y, we have the following **integration by parts formula**:

$$\boxed{F_X(b)F_Y(b) - F_X(a)F_Y(a) = \int_a^b F_X(x^-)\mathrm{d}F_Y(x) + \int_a^b F_Y(x)\mathrm{d}F_X(x),} \tag{4.38}$$

see [12, Prop 4.5]. As byproduct we also have

$$F_X(x)F_Y(x) = \int_{-\infty}^x F_X(u^-)\mathrm{d}F_Y(u) + \int_{-\infty}^x F_Y(u)\mathrm{d}F_X(u).$$

For continuous CDFs, formula (4.38) reads

$$\int_a^b F_X(x)\mathrm{d}F_Y(x) = F_X(b)F_Y(b) - F_X(a)F_Y(a) - \int_a^b F_Y(x)\mathrm{d}F_X(x).$$

For $F_Y(x) = 1$, we may as well apply integration by parts formula (4.38) to get:

$$F_X(b) - F_X(a) = \int_{(a,b]} \mathrm{d}F_X(x) \equiv \mathsf{P}(a < X \leqslant b),$$

where $\mathrm{d}F_Y(x) = 0$. Moral: Integration by parts may serve for integrating over subsets.

Proof of Lemma 4.2

Proof ($\Rightarrow$) Let $g(x) = \mathbf{I}_{B_1}(x)$ and $h(y) = \mathbf{I}_{B_2}(y)$ be two indicator functions, for some Borel sets $B_1, B_2 \subset \mathbb{R}$. Thus, $g(X) = \mathbf{I}_{\{X\in B_1\}}, h(Y) = \mathbf{I}_{\{Y\in B_2\}}$ and $\mathbf{I}_{\{X\in B_1\}\cap\{Y\in B_2\}} = \mathbf{I}_{\{X\in B_1\}}\mathbf{I}_{\{Y\in B_2\}}$. By assumption we have $\mathsf{P}(\{X \in B_1\} \cap \{Y \in B_2\}) = \mathsf{P}(X \in B_1)\mathsf{P}(Y \in B_2)$ and this is equivalent to $\mathsf{E}(\mathbf{I}_{\{X\in B_1\}\cap\{Y\in B_2\}}) = \mathsf{E}(\mathbf{I}_{\{X\in B_1\}}\mathbf{I}_{\{Y\in B_2\}}) = \mathsf{E}(\mathbf{I}_{\{X\in B_1\}})\mathsf{E}(\mathbf{I}_{\{Y\in B_2\}})$. We may find increasing sequences g_n, h_n of nonnegative step (bounded) functions converging pointwise to positive functions g and h, respectively. The above result for indicators extend to $g_n(X)$ and $h_n(Y)$ by linearity, while MCT does the job:

$$\begin{aligned}\mathsf{E}(g(X)h(Y)) &= \mathsf{E}(\lim_n g_n(X)h_n(Y)) = \lim_n \mathsf{E}(g_n(X)h_n(Y))\\ &= \lim_n \big(\mathsf{E}(g_n(X))\mathsf{E}(h_n(Y))\big) = \mathsf{E}(g(X))\mathsf{E}(h(Y)).\end{aligned}$$

If g, h is any pair of bounded functions, then we may always apply the above to their positive and negative parts. ($\Leftarrow$) It suffices to choose $g(x) = \mathbf{I}_{B_1}(x)$ and $h(y) = \mathbf{I}_{B_2}(y)$. ■

Proof of Lemma 4.1

Proof ($\Rightarrow$) $X \stackrel{\mathrm{d}}{=} Y$ means that $\mathsf{P}_X(B) = \mathsf{P}_Y(B)$ for every Borel set $B \subset \mathbb{R}$. If $X = \mathbf{I}_A, Y = \mathbf{I}_{A'}$ for $A, A' \in \mathcal{F}$, then $g(X) = X$ and $g(Y) = Y$. Since $\{X \in B\} = A, \{Y \in B\} = A'$, we have that $\mathsf{E}(\mathbf{I}_A) = \mathsf{P}(A) = \mathsf{P}_X(B)$ and $\mathsf{E}(\mathbf{I}_{A'}) = \mathsf{P}(A') = \mathsf{P}_X(B)$, hence $\mathsf{E}(g(X)) = \mathsf{E}(g(Y))$. The same argument extends to nonnegative simple rvs X, Y by linearity. We may apply the Monotone Convergence Theorem to $X, Y \geqslant 0$, or for general rvs to their positive and negative parts. ($\Leftarrow$) If $\mathsf{E}(g(X)) =$

$\mathsf{E}(g(Y))$ for every Borel function g, then in particular we may choose $g = \mathbf{I}_B$ with $B \in \mathcal{B}$. We have $\mathbf{I}_B(\,\cdot\,) = \mathbf{I}_{\{\,\cdot\, \in B\}}$ for both rvs Thus, $\mathsf{E}[\mathbf{I}_{\{\,\cdot\, \in B\}}] = \mathsf{P}(\,\cdot\, \in B)$, and their respective distributions coincide. ■

Limit and Expectation Again

There are two further results on interchanging limits and expectation, other than Theorem 4.10, which can be useful in some proofs. Before to present the results, be warned that Theorem 4.10 can be stated[36] for an increasing sequence $(X_n)_{n\in\mathbb{N}}$ which converges P-a.s. to a limit X, i.e., given $A = \{\lim_{n\to\infty} X_n = X\}$ the convergence fails on A^c with probability $\mathsf{P}(A^c) = 0$. To see why, we note $X_n \overset{\text{a.s.}}{=} X_n\mathbf{I}_A$ since $\{X_n = X_n\mathbf{I}_A\}$ is exactly the set A and the same is true for X. By Remark 3, after the Solution to Exercise 4.17, 3, we have $\mathsf{E}(X_n) = \mathsf{E}(X_n\mathbf{I}_A)$ and $\mathsf{E}(X) = \mathsf{E}(X\mathbf{I}_A)$, since the expectation does not change by adjusting measurable functions on a P-null set. Now we have that $X_n \overset{\text{a.s.}}{=} X$ implies $X_n\mathbf{I}_A \to X\mathbf{I}_A$ for every $\omega \in A$ and $\lim_n \mathsf{E}(X_n\mathbf{I}_A) = \mathsf{E}(X\mathbf{I}_A)$ easily follows by Theorem 4.10.

Lemma 4.4 (Fatou) *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of nonnegative rvs. Then, $\mathsf{E}\big(\liminf X_n\big) \leqslant \liminf \mathsf{E}(X_n)$ where both sides may be infinite.*

Proof Define the random variable $Y_k = \inf_{n\geqslant k} X_n$ and $Y = \lim_k Y_k = \liminf X_n$, where clearly each Y_k is nonnegative. For $n \geqslant k$ it holds $X_n \geqslant Y_k$ and by monotonicity of expectation also $\mathsf{E}(X_n) \geqslant \mathsf{E}(Y_k)$. This implies $\mathsf{E}(Y_k) \leqslant \inf_{n\geqslant k} \mathsf{E}(X_n)$ But $Y_k \uparrow Y$ and by the monotone convergence theorem it follows $\liminf \mathsf{E}(X_n) \geqslant \liminf \mathsf{E}(Y_k) = \mathsf{E}(Y)$, which proves the lemma. ■

Again, we have the a.s.-version of Fatou's lemma where the sequence $(X_n)_{n\in\mathbb{N}}$ is s.t. $X_n \geqslant 0$ except on a P-null set A^c, for all $n \in \mathbb{N}$. The proof runs as above with the only exception that we replace X_n with $X_n\mathbf{I}_A$ and change all the expectations involved on A^c, without affecting their values.

Theorem 4.12 (Lebesgue's Dominated Convergence) *Let $X, X_1, X_2, \ldots$ be rvs such that $X_n \to X$ pointwise. If the sequence $(X_n)_{n\in\mathbb{N}}$ is dominated by a random variable $Y \in L^1$, i.e., $|X_n| \leqslant Y$ for every $n \in \mathbb{N}$, then $\lim_n \mathsf{E}(|X_n - X|) = 0$ and in particular $\lim_n \mathsf{E}(X_n) = \mathsf{E}(X)$.*

Proof By assumption we have $Y + X_n \geqslant 0$ and applying the Fatou's lemma to the sequence $(Y + X_n)_{n\in\mathbb{N}}$ it holds: $\mathsf{E}(Y) + \mathsf{E}(X) = \mathsf{E}(Y + X) \leqslant \liminf \mathsf{E}(Y + X_n) = \mathsf{E}(Y) + \liminf \mathsf{E}(X_n)$. Using $\mathsf{E}(Y) < \infty$ we can rule out this expectation from both sides obtaining $\mathsf{E}(X) \leqslant \liminf \mathsf{E}(X_n)$. From real analysis we know that $-\liminf(-x_n) = \limsup x_n$, for any sequence $(x_n)_{n\in\mathbb{N}} \subset \mathbb{R}$. Applying once more the Fatou's lemma to the sequence $(Y - X_n)_{n\in\mathbb{N}}$ and following the same reasoning

[36] This is true also for Lemma 4.3.

as above we have $\mathsf{E}(X) \geqslant \limsup \mathsf{E}(X_n)$, since $\liminf \mathsf{E}(-X_n) = \limsup \mathsf{E}(X_n)$. Moreover, from $\limsup \mathsf{E}(X_n) \geqslant \liminf \mathsf{E}(X_n)$ we must have $\limsup \mathsf{E}(X_n) = \liminf \mathsf{E}(X_n) = \mathsf{E}(X)$, which is equivalent to $\mathsf{E}(|X_n - X|) = 0$ and the proof is complete. ■

In Theorem 4.12 we can again weaken the hypotheses by requiring that $|X_n| \leqslant Y$ holds almost surely and $\mathsf{P}(\lim_{n\to\infty} X_n = X) = 1$, i.e. $\lim_{n\to\infty} X_n \overset{\text{a.s.}}{=} X$.

Chapter 5
Special Distributions



5.1 Summary of Univariate Distributions

Special models of probability distributions are available for application, especially in financial modelling. A brief summary of those used in this book is showed in Tables 5.1 and 5.2.

Usually, notations are correctly referred to CDFs and one can write $X \sim F_X$, read 'X is distributed as F_X', where F_X is the CDF corresponding to the masses or densities in Tables 5.1 and 5.2. Any distribution listed above is a *family* of distributions indexed by the corresponding parameters.

5.2 Discrete Uniform

From Tables 5.1 and 5.2, we see how notations for the uniform distributions are similar, but be careful with the discrete case where

$$\mathrm{U}\{a,\ldots,b\} = \begin{cases} 0 & \text{if } x_i < a \\ \frac{x_i - a + 1}{b - a} & \text{if } a \leqslant x_i \leqslant b \\ 1 & \text{if } x_i > b, \end{cases}$$

denotes the CDF of the discrete uniform and

$$\mathsf{P}(X = x_i) = \frac{\mathbf{I}_{\{a < x_i < b\}}}{b - a + 1}$$

is the probability mass function, with $a = \min C$ and $b = \max C$, with $a < b$.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_5

Table 5.1 Discrete univariate distributions

Name	Notation	Mass $\mathsf{P}(X = x_i) =$	Parameters
Uniform	$\mathrm{U}\{a, \ldots, b\}$	$\frac{1}{n}$, for $x_i = i \in C \subset \mathbb{Z}$	$\lvert C\rvert = n \in \mathbb{N}$
Bernoulli	$\mathrm{Ber}(p)$	p, for $x_1 = 1$, or $1 - p$ for $x_2 = 0$	$0 < p < 1$
Binomial	$\mathrm{Bin}(n, p)$	$\binom{n}{i} p^i (1-p)^{n-i}$, for $x_i = i \in \{0, 1, 2, \ldots, n\}$	$0 < p < 1$, $n \in \mathbb{N}$
Poisson	$\mathrm{Po}(\lambda)$	$\mathrm{e}^{-\lambda} \frac{\lambda^{x_i}}{x_i!}$, for $x_i = i \in \{0, 1, 2, \ldots\}$	$\lambda > 0$

Table 5.2 Continuous univariate distributions

Name	Notation	Density $f_X(x) =$	Parameters
Uniform	$\mathrm{U}(a, b)$	$\frac{1}{b-a}$ for $a \leqslant x \leqslant b$, zero otherwise	$a, b \in \mathbb{R}$, $a < b$
Gaussian or Normal	$\mathrm{N}(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-(x-\mu)/2\sigma^2}$, for $x \in \mathbb{R}$	$\mu \in \mathbb{R}$, $\sigma > 0$
Exponential	$\mathrm{Exp}(\lambda)$	$\lambda \mathrm{e}^{-\lambda x}$, for $x \geqslant 0$, zero otherwise	$\lambda > 0$
Gamma	$\mathrm{Gamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha x^{\alpha-1} \mathrm{e}^{-\lambda x}}{\Gamma(\alpha)}$, for $x > 0$, zero otherwise	$\alpha > 0$, $\lambda > 0$
Student-t	$\mathrm{T}(n)$	$\frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-(\frac{n+1}{2})}$, for $x \in \mathbb{R}$	$n > 1$, degree of freedom

Example 5.1 Let $C = \{-2, 1, 0, 1\}$ where $X \sim \mathrm{U}\{-2, \ldots, 1\}$ is distributed with equally likely probability $\mathsf{P}(X = x_i) = \frac{1}{|C|} = \frac{1}{4}$, for $x_i \in C$. Then, choosing (at random) $x_i = 0$ we have

$$\mathsf{P}(X = 0) = \frac{\mathbf{I}_{\{-2<0<1\}}}{1 - (-2) + 1} = \frac{1}{4}.$$

The law of a discrete uniform rv can be though of as a probability on $(\mathbb{R}, \mathcal{B})$, see Appendix in Chap. 2, using the Dirac delta measure introduced in Sect. 2.4, Chap. 2 (see also Exercise 2.7). It suffices to define probability the $\mathsf{P} \equiv \mathsf{P}_X$ by a linear combination of several Dirac δ measures concentrated at each point of a countable set[1] $C = \{x_1, , \ldots, x_n\}$. If we assign a weight $p_i \geqslant 0$ to each $x_i \in C$ and require $\sum_{i=1}^n p_i = 1$, then the set function $\mathsf{P}(B) := \sum_{i=1}^n p_i \delta_{x_i}(B)$ is a discrete probability measure for every $B \in \mathcal{B}$ provided $\mathsf{P}(C) = 1$. If $p_i = \frac{1}{|C|}$ we recover the masses above, otherwise we give raise to a discrete non-uniform rv.

[1] Recall that the Borel σ-algebra contains all singletons $\{x\}$ with $x \in \mathbb{R}$.

5.3 Bernoulli

The Bernoulli distribution is used in random experiments where only two results are possible, and they are interpreted as 'success' and 'failure', called *Bernoulli trial*. For example, a tick movement of a daily stock price, as in Sect. 1.7, gives raise to such an experiment provided that the rv[2] S_1 is obtained using a Bernoulli rv $X \sim \text{Ber}(p)$, and then scaling it as follows:

$$S_1 = 0.5\mathbf{I}_{\{X=1\}} - 0.5\mathbf{I}_{\{X=0\}},$$

where $p = \mathsf{P}(X = 1)$ is the *success probability* to be assigned. Alternatively, we can transform the Bernoulli rv using $g(x) = -(\frac{1}{2} - x)$. For $p = \frac{1}{2}$ the X's distribution is symmetric with the same masses $\mathsf{P} = p$, indeed it is a special case of the discrete uniform with only two values in its support. Since $X \sim \text{Ber}(p)$ is discrete, then its expected value is easily computed:

$$\mathsf{E}(X) = 1 \times \mathsf{P}(X = 1) + 0 \times \mathsf{P}(X = 0) = p. \tag{5.1}$$

The variance is:

$$\begin{aligned} \mathsf{V}(X) &= (1 - \mathsf{E}(X))^2 \times \mathsf{P}(X = 1) + (0 - \mathsf{E}(X))^2 \times \mathsf{P}(X = 0) \\ &= (1 - p)^2 p + p^2(1 - p) = p(1 - p). \end{aligned} \tag{5.2}$$

Example 5.2 The expectation of the first day stock price is

$$\mathsf{E}(S_1) = 0.5\mathsf{E}(\mathbf{I}_{\{X=1\}}) - 0.5\mathsf{E}(\mathbf{I}_{\{X=0\}}) = 0.5p - 0.5(1 - p) = p - \frac{1}{2},$$

or alternatively

$$\mathsf{E}(S_1) = \mathsf{E}\left(-\left(\frac{1}{2} - X\right)\right) = -\frac{1}{2} + \mathsf{E}(X),$$

which entails the same result. The variance is:

$$\begin{aligned} \mathsf{V}(S_1) &= (0.5 - (p - \tfrac{1}{2}))^2 p + (-0.5 - (p - \tfrac{1}{2}))^2 (1 - p) \\ &= (1 - p)^2 p + p^2(1 - p) = p(1 - p). \end{aligned}$$

It is no surprisingly that $\mathsf{V}(S_1) = \mathsf{V}(X)$: the rv S_1 is obtained by the Bernoulli rv X by a linear transformation which affects only the location of $\text{Ber}(p)$ not its dispersion.

[2] We confine ourselves to the first trading day.

Remark 5.1 We can use $X \sim \text{Ber}(\frac{1}{2})$ to create new discrete rvs with two values $a, b \in \mathbb{R}$ as follows:

$$Y = aX + b(1 - X).$$

For example, if S_1 has values ± 0.5 with equal probability, then we can use $a = \frac{1}{2}, b = -\frac{1}{2}$ which is the same as $g(X) = -(\frac{1}{2} - X)$.

As we did in Sect. 5.2, considering $(\mathbb{R}, \mathcal{B})$, the *Bernoulli measure* can be defined as $\mathsf{P} := p\delta_0 + (1-p)\delta_1$, where $0 < p < 1$ is interpreted as the 'success probability' in one trial. This corresponds to the law P_X of $X \sim \text{Ber}(p)$. The Bernoulli measure is easily verified to be a probability. It is countably additive for any pairwise disjoint sequence $(A_i)_{i\in\mathbb{N}} \subset \mathcal{B}$:

$$\begin{aligned}\mathsf{P}(\cup_i A_i) &= p\delta_0(\cup_i A_i) + (1-p)\delta_1(\cup_i A_i) \\ &= p\sum_i \delta_0(A_i) + (1-p)\sum_i \delta_1(A_i) = \sum_i \mathsf{P}(A_i).\end{aligned}$$

Observe that $\mathsf{P}(A) = p\delta_0(A) + (1-p)\delta_1(A)$ yields p whenever $0 \in A$, otherwise is equal to $1 - p$. Normalization is easily established:

$$\mathsf{P}(\mathbb{R}) = \mathsf{P}(\{0,1\} \cup \mathbb{R} \setminus \{0,1\}) = 1.$$

Remark 5.2 It is worth noting that $\mathbf{I}_A \sim \text{Ber}(p)$, whenever $p = \mathsf{P}(A)$ for $A \in \mathcal{F}$.

The MGF of a Bernoulli rv is

$$\begin{aligned}M_X(u) &= \mathsf{E}(\mathrm{e}^{uX}) = \mathrm{e}^{u\,0}(1-p) + \mathrm{e}^u p \\ &= (1-p) + p\mathrm{e}^u.\end{aligned}$$

5.4 Binomial

Let Ω consists of $n \in \mathbb{N}$ independent Bernoulli trials with the same success probability p. There are different sequences of successes and failures in n trials, and we can let $q = 1 - p$ be the *failure* probability. Letting X be the rv representing the number of successes entails $C = \{0, 1, 2, \ldots, n\}$ as its support. To figure out possible scenarios, we also let $A_i = \{\text{success at trial } i\}$ and $B_i = \{\text{failure at trial } i\}$ be events of either a single success or failure, for $i = 1, 2, \ldots, n$. Clearly, for a fixed i we have A_i and B_i as incompatible, while $A_i \cap B_j = \{\text{success at trial } i, \text{ failure at trial } j\}$ are compatible for $i \neq j$ and also independent. As byproduct, the probability of any specific sequence of i successes and $n - i$ failures is simply given by $p^i(1-p)^{n-i}$. For fixed i, and then $n - i$, there are $\frac{n!}{i!(n-i)!} = \binom{n}{i}$ such sequences: they are the number of combinations of two objects,

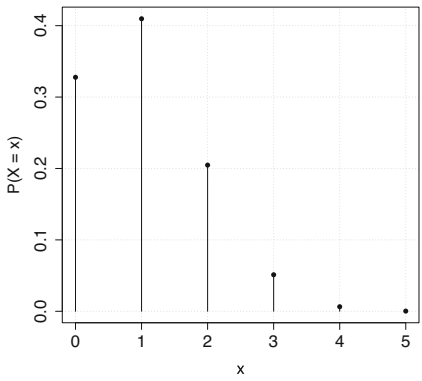


Fig. 5.1 Mass function of a Bin(6, 0.3)

namely one success and one failure, the former repeated exactly i times and the latter repeated $n-i$ times. Hence, we have to sum $p^i(1-p)^{n-i}$ across incompatible events each given by the union of A_i and B_i showing the same number i of successes together with $n-i$ failures, but with different position in the whole trials, using additivity of P. For example, with $n=4$ and $i=2$ we can have A_1, B_2, A_3, B_4 or B_1, A_2, B_3, A_4 with the same number of successes and failures but different position in the sequence, hence $A_1 \cap B_2 \cap A_3 \cap B_4$ and $B_1 \cap A_2 \cap B_3 \cap A_4$ are incompatible. This yields the expression of $\mathsf{P}(X = x_i)$ in row 3 of Table 5.1, where $x_i = i \in C$. By the binomial theorem it is easily verified: $\sum_{i=0}^{n} \binom{n}{i} p^i(1-p)^{n-i} = 1$. Observe that $\mathsf{P}(X = x_i) = 0$ if $x_i \notin C$. For $p = \frac{1}{2}$ and even n the binomial distribution is symmetric around[3] $\frac{n}{2}$, since $X \sim \text{Bin}(n, p)$ implies $n - X \sim \text{Bin}(n, q)$, see Exercise 5.1, but $p = q$ and then, letting $x_i = i$,

$$\mathsf{P}(X = i) = \binom{n}{i} p^i q^{n-i} = \binom{n}{i} q^i p^{n-i} = \mathsf{P}(n - X = i),$$

which entails $\mathsf{P}(X = i) = \mathsf{P}(X = n - i)$ so for $i = \frac{n}{2} + j$ we have

$$\mathsf{P}\left(X = \frac{n}{2} + j\right) = \mathsf{P}\left(X = \frac{n}{2} - j\right), \quad \text{for all } j \in \mathbb{N}.$$

In Fig. 5.1 a typical plot of the binomial mass function is showed: when the success and failure probabilities are different, there is more probability weight put on smaller values in the support. As a result, for $p < 0.5$ the distribution has positive skewness with a log right-tail. Up to this point it is worth noting that a binomial rv can be alternatively defined as the sum of n independent Bernoulli rvs $X_i \sim \text{Ber}(p)$, for $i = 1, \ldots, n$:

$$X = X_1 + \cdots + X_n. \tag{5.3}$$

[3] For even n it holds $\frac{n}{2} = m$ is also even.

Since $\mathsf{E}(X_i) = p$ we have

$$\mathsf{E}(X) = \mathsf{E}(X_1) + \cdots + \mathsf{E}(X_n) = np, \tag{5.4}$$

using linearity of expectation and also

$$\mathsf{V}(X) = \mathsf{V}(X_1) + \cdots + \mathsf{V}(X_n) = np(1-p), \tag{5.5}$$

using independence and $\mathsf{V}(X_i) = p(1-p)$, each i. We see that the sum of two independent binomial rvs with the same success probability is again binomial. Let $X \sim \mathrm{Bin}(n_1, p)$ and $Y \sim \mathrm{Bin}(n_2, p)$, then $Z = X + Y$ is the sum of $m = n_1 + n_2$ IID $X_i \sim \mathrm{Ber}(p)$, for $i = 1, \ldots, m$, thus $Z \sim \mathrm{Bin}(m, p)$, see Exercise 5.2 for a different derivation. The MGF of $X \sim \mathrm{Bin}(n, p)$ is

$$\begin{aligned} M_X(u) = \mathsf{E}\big(\mathrm{e}^{uX}\big) &= \mathsf{E}\big(\mathrm{e}^{u(X_1+\cdots+X_n)}\big) \\ &= \mathsf{E}\big(\mathrm{e}^{uX_1}\big) \cdots \mathsf{E}\big(\mathrm{e}^{uX_n}\big) \\ &= (p\mathrm{e}^u + (1-p))^n, \end{aligned}$$

where the second line is due to the property of the exponential function and to the independence among the Bernoulli rvs, while the last line stems from

$$\mathsf{E}\big(\mathrm{e}^{uX_i}\big) = p\mathrm{e}^u + (1-p)\mathrm{e}^0.$$

The MGF exists for all $u \in \mathbb{R}$, which obviously is the largest open interval containing zero. In `R` we can generate random numbers from a given distribution, obtain the density or mass function for given quantiles, and get the quantile for given probability levels. Here an example:

Program Code

```
S=20; rbinom(S,size=10,prob=0.5)
q=seq(0:10)
p=seq(from = 0.01,to = 0.99, length = 50)
dbinom(q,size=10,prob=0.5)
pbinom(q,size=10,prob=0.5)
qbinom(p,size=10,prob=0.5)
```

The first `R` command sets the variable `S` to the number of simulation used in the second command, which yields twenty simulated values drawn from $X \sim \mathrm{Bin}(10, 0.5)$. The third commands creates a list of ten quantiles of X, actually the whole support $\{0, 1, 2, \ldots, 10\}$, while the fourth command entails a list of fifty probability values from 0.01 to 0.99. Eventually, the fifth and sixth commands yield the mass function and CDF, respectively. Clearly each command can be

used with the first argument, `s`, `q` and `p` in scalar form. We can also set apart parameter, e.g., `rbinom(x,n,p)` can be used after we specify `x <- 2`, `n <- 5` and `p <- 1/2`. We close this section with a classical application to finance. Recall from Example 1.2 in Sect. 1.3 the model for the daily stock price through tick movements ± 0.5: there is a fallacy due to possible negative value for very large tick movements. A plausible stochastic model for the future evolution of stock closing prices over subsequent trading days can be built using the binomial rv. Let the closing price at day $n \in \mathbb{N}$ be defined by

$$S_n = S_0 \mathrm{u}^X \mathrm{d}^{n-X}, \quad X \sim \mathrm{Bin}(n, p), \tag{5.6}$$

where $0 < \mathrm{d} < 1 < \mathrm{u}$ are interpreted as the percentages of downward and upward movements, respectively, starting from the initial stock price S_0. This is a special case of Example 2.8, item 1 in Sect. 2.5, which in the current setting becomes $S_0(1+r_1)\cdots(1+r_n)$, with arithmetic returns rvs $r_{t+1} = \frac{S_{t+1}}{S_t} - 1$, for $t = 0, 1, \ldots, n-1$, and n source of randomness.[4] The special case arises letting $\mathrm{u} = 1 + r_{t+1}$ for some t and $\mathrm{d} = 1 + r_{t+1}$ for the remaining $n - t$, and considering one source of randomness given by the binomial rv X which counts the number of upward movements of the stock price. Events $\{S_n = \mathrm{u}^i \mathrm{d}^{n-i}\}$ and $\{X = i\}$ are the same by construction, then

$$\mathsf{P}(S_n = \mathrm{u}^i \mathrm{d}^{n-i}) = \mathsf{P}(X = i), \quad \text{for } i = 0, 1, 2, \ldots, n.$$

More is true: by (5.3) the expression for the nth day stock price is:

$$\begin{aligned} S_n &= S_0 \mathrm{u}^{X_1 + \cdots + X_n} \mathrm{d}^{n - (X_1 + \cdots + X_n)} \\ &= S_0 \mathrm{u}^{X_1} \cdots \mathrm{u}^{X_n} \mathrm{d}^{n - X_1} \cdots \mathrm{d}^{n - X_n} \\ &= S_0 f(X_1) \cdots f(X_n) g(X_1) \cdots g(X_n). \end{aligned} \tag{5.7}$$

For $n = 1$ we have $S_1 = S_0 \mathrm{u}^{X_1} \mathrm{d}^{1 - X_1}$, so the stock price at the end of day 1 may take on values in $\{S_0\mathrm{u}, S_0\mathrm{d}\}$, and S_1 is independent of S_0. At the end of day 2 we have $S_2 = S_0 \mathrm{u}^{X_1 + X_2} \mathrm{d}^{1 - (X_1 - X_2)}$ with values in $\{S_0\mathrm{u}^2, S_0\mathrm{ud}, S_0\mathrm{d}^2\}$, and since (5.7) holds we compute

$$\begin{aligned} S_2 &= S_0 \mathrm{u}^{X_1} \mathrm{u}^{X_2} \mathrm{d}^{1 - X_1} \mathrm{d}^{1 - X_2} \\ &= S_1 \mathrm{u}^{X_2} \mathrm{d}^{1 - X_2}. \end{aligned}$$

This recursive structure still holds for $t = 3, 4, \ldots, n$. There are some interesting consequences:

[4] This corresponds to a vector-to-scalar transformation $S_0 h(r_1, r_2, \ldots, r_n)$.

1. the stock price at day $t+1$ is dependent only on the previous stock price,

$$\boxed{S_{t+1} = S_t \mathrm{u}^{X_{t+1}} \mathrm{d}^{1-X_{t+1}}}$$

   for all $t = 1, 2, \ldots, n-1$;
2. gross returns $\frac{S_1}{S_0}, \frac{S_2}{S_1}, \ldots, \frac{S_n}{S_{n-1}}$ are IID rvs with distribution

$$\frac{S_{t+1}}{S_t} = \begin{cases} \mathrm{u} \text{ with mass } p \\ \mathrm{d} \text{ with mass } q, \end{cases} \tag{5.8}$$

   by the independence assumption about the Bernoulli rvs $X_1, \ldots, X_n$ underlying the model, see Theorem 3.2, Sect. 3.4;
3. log-returns $\ln(\frac{S_{t+1}}{S_t})$ as well as arithmetic returns $r_{t+1} = \frac{S_{t+1}}{S_t} - 1$ are IID rvs with the same distribution as for gross returns.

Property 1 is a form of market efficiency: there is no predictive information in past stock prices up to day $t-1$. Observe that the probability that S_t take on a specific value depends upon uptick and downtick movements of the daily stock price given by gross returns i.e., it might be computed by multiplying Bernoulli probabilities of success p and of failure q over t consecutive trading days, for $t = 1, \ldots, n$. On the other hand, the conditional probability

$$\mathsf{P}(S_{t+1} = \mathrm{u}^i \mathrm{d}^{t+1-i} \mid S_t = \mathrm{u}^{i-1} \mathrm{d}^{t-i+1}), \quad \text{for all } i = 0, 1, 2, \ldots, t+1, \text{ and } t = 1, \ldots, n-1 \tag{5.9}$$

is p or q. For example, $\mathsf{P}(S_3 = \mathrm{u}^3 \mid S_2 = \mathrm{u}^2) = p$, while $\mathsf{P}(S_3 = \mathrm{u}^2\mathrm{d} \mid S_2 = \mathrm{ud}) = q$. Do not confuse the parameter $u \in \mathbb{R}$ in the definition of MGF with the uptick movement u in the binomial stock price model.

5.5 Poisson

A rv X with a Poisson distribution has a countably infinite support $C = \{0, 1, 2, \ldots\}$. Each mass is of the form $\mathsf{P}(X = i) = \mathrm{e}^{-\lambda} \frac{\lambda^i}{i!}$, for $x_i = i \in C$, see Appendix in Chap. 1 for a first encounter with this distribution; see also Sect. 4.2. In fact, they are truly masses:

$$\begin{aligned} \sum_{i=0}^{\infty} \frac{\mathrm{e}^{-\lambda} \lambda^i}{i!} &= \mathrm{e}^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} \\ &= \mathrm{e}^{-\lambda} \mathrm{e}^{\lambda} = 1, \end{aligned}$$

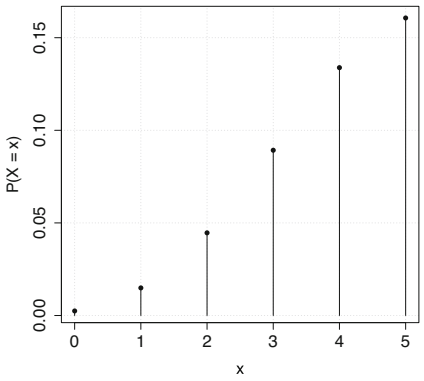


Fig. 5.2 Density of a Po(6)

since the exponential function $g(x) = \mathrm{e}^x$ has the Taylor series expansion $g(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, each $n \in \mathbb{N}$. Figure 5.2 shows a plot of the Poisson mass function, which has typical positive skewness. Just as we did in Sect. 5.3, the law $\mathsf{P} \equiv \mathsf{P}_X$ of a Poisson rv X as a probability on $(\mathbb{R}, \mathcal{B})$ can be written using a countable linear combination of Dirac delta measures concentrated at C with weights $p_i = \mathrm{e}^{-\lambda} \dfrac{\lambda^{x_i}}{x_i!}$, it suffices to define $\mathsf{P} := \sum_{i=1}^{\infty} p_i \delta_i$. Thus, for any $B \in \mathcal{B}$ we can compute $\mathsf{P}(B)$ as $\sum_{i=1}^{\infty} p_i \delta_i(B)$. In fact, this yields a probability measure satisfying axioms PA1 and PA2 of Definition 1.3, Sect. 1.3: normalization amounts to choose $B = \mathbb{R}$, and using the three equivalence above; countable additivity is

$$\begin{aligned} \sum_{n=1}^{\infty} \mathsf{P}(A_n) &= \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \tfrac{\mathrm{e}^{-\lambda} \lambda^i}{i!} \delta_i(A_n) \\ &= \sum_{i=1}^{\infty} \tfrac{\mathrm{e}^{-\lambda} \lambda^i}{i!} \delta_i(\cup_n A_n) = \mathsf{P}(\cup_n A_n), \end{aligned}$$

for a pairwise disjoint sequence $(A_n)_{n \in \mathbb{N}} \subset \mathcal{B}$. The expectation of a Poisson rv is computed as follows:

$$\begin{aligned} \mathsf{E}(X) &= \sum_{i=1}^{\infty} i\, \mathrm{e}^{-\lambda} \frac{\lambda^i}{i!} = \mathrm{e}^{-\lambda} \sum_{i=1}^{\infty} i\, \frac{\lambda^i}{i!} \\ &= \lambda \mathrm{e}^{-\lambda} \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} = \lambda \mathrm{e}^{-\lambda} \mathrm{e}^{-\lambda} = \lambda, \end{aligned} \tag{5.10}$$

where we dropped the $i = 0$ term since it was 0 and used the Taylor series expansion for e^{λ} starting from $i = 1$. Surprisingly enough, the variance is

$$\mathsf{V}(X) = \lambda, \tag{5.11}$$

see Exercise 5.4. The MGF of a Poisson rv is

$$\begin{aligned} M_X(u) = \mathsf{E}\big(\mathrm{e}^{uX}\big) &= \sum_{i=0}^{\infty} \mathrm{e}^{u\,i}\,\mathrm{e}^{-\lambda}\,\frac{\lambda^i}{i!} \\ &= \mathrm{e}^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda \mathrm{e}^u)^i}{i!} \\ &= \mathrm{e}^{-\lambda}\mathrm{e}^{\lambda \mathrm{e}^u} = \exp(\lambda(\mathrm{e}^u - 1)), \end{aligned}$$

where we used the Taylor series expansion of $\exp(\lambda \mathrm{e}^u)$. The MGF exists for every $u \in \mathbb{R}$, thus also for $u = 0$. The sum of two independent Poisson rvs $X \sim \mathrm{Po}(\lambda_1)$ and $Y \sim \mathrm{Po}(\lambda_2)$ is just $X + Y \sim \mathrm{Po}(\lambda_1 + \lambda_2)$. Indeed, using the MGF and independence we have:

$$\begin{aligned} M_{X+Y}(u) &= M_X(u) M_Y(u) = \mathsf{E}\big(\mathrm{e}^{uX}\big)\mathsf{E}\big(\mathrm{e}^{uY}\big) \\ &= \exp(\lambda_1(\mathrm{e}^u - 1)) \exp(\lambda_2(\mathrm{e}^u - 1)) \\ &= \exp((\lambda_1 + \lambda_2)(\mathrm{e}^u - 1)). \end{aligned}$$

The Poisson and the binomial distribution are related each other: if $n \in \mathbb{N}$ is large enough and the success probability is very small, with np finite, bounded and nonzero, then we can approximate the latter distribution with the former.

Proposition 5.1 *Let $X \sim \mathrm{Bin}(n, p)$ and send n to ∞, taking $p \to 0$ s.t. $\lambda = np$ remains fixed. Then the X's mass function converges to that of $Y \sim \mathrm{Po}(\lambda)$.*

Example 5.3 Suppose an insurance company shall receive a certain number of claims in a fixed time interval, say $(0, t]$, for $t > 0$. Let X be the rv modelling the number of claims, with support $C = \{0, 1, 2, \ldots, n\}$ for a positive integer n. Thus, we partition $(0, t]$ into n subinterval of equal length $\frac{t}{n}$, so for n large enough each subinterval is small and we might assume that the company will receive only one claim in it. Further, we can assume that the probability for such a claim is proportional to the length of each subinterval $(\frac{(i-1)t}{n}, \frac{i\,t}{n}]$, for $i = 1, \ldots, n$. We set $p = \frac{\lambda t}{n}$, with $\lambda > 0$, the probability that a Bernoulli rv X_i takes value 1 if there is just one claim in $(\frac{(i-1)t}{n}, \frac{i\,t}{n}]$. We also assume that claim arrivals over each subinterval are independent each other. So we have $X = X_1 + \cdots + X_n$ for IID Bernoulli $X_i \sim \mathrm{Ber}(p)$, thus we approximate the number j of claims during $(0, t]$ by the binomial rv $X \sim \mathrm{Bin}(n, \frac{\lambda t}{n})$, where $\frac{\lambda t}{n}$ is the expected number of claims over each subinterval $(\frac{(i-1)t}{n}, \frac{i\,t}{n}]$, i.e., $\mathsf{E}(X_i) = p$. By Proposition 5.1 we have $\lim_{n\to\infty} \mathsf{P}(X = j) = \mathrm{e}^{-\lambda}\frac{\lambda^i}{i!}$, where $\lambda = \frac{np}{t}$ is kept fixed and clearly $p \to 0$.

The following `R` commands are similar to those used for the binomial distribution in Sect. 5.3, but now without the size and success probability parameters which are replaced by λ.

Program Code

```
S=20; rpois(S,lambda=0.1)
q=seq(0:10)
p=seq(from=0.01,to=0.99, lenght=50)
dpois(q,lambda=0.1)
ppois(q,lambda=0.1)
qunif(p,lambda=0.1)
```

5.6 Continuous Uniform

In Example 1.25 (see Appendix in Chap. 1) we have encountered a continuous uniform probability law. In the current setting it is nothing else the uniform distribution $X \sim \mathrm{U}(a, b)$ with $a = 0$ and $b = 1$. In general, a uniformly distributed rv X has constant density $f_X(x)\frac{1}{b-a}$ for all $x \in [a, b]$ and $f_X(x) = 0$ outside the interval. Its CDF is

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{b-a}\mathrm{d}u = \begin{cases} 0, & \text{if } x < a \\ \frac{x-a}{b-a}, & \text{if } a \leqslant x \leqslant b \\ 1, & \text{if } x > b, \end{cases}$$

which is easily obtained from

$$\int_{-\infty}^{x} f_X(u)\mathrm{d}u = \underbrace{\int_{-\infty}^{a} f_X(u)\,\mathrm{d}u}_{=0} + \int_{a}^{x} f_X(u)\mathrm{d}u.$$

In Fig. 5.3 the plots of the CDF and the density are depicted. Observe that the CDF is continuous but has not derivative at the points $x = a$ and $x = b$, where the corresponding density has two jump discontinuities.[5] The importance of uniform distributed rvs is due to the following couple of results.

Theorem 5.1 (Quantile Transform) *Let $U \sim U(0, 1)$ be a uniform rv, and Y be a rv with CDF $F_Y(y)$ and quantile function $Q_Y(c)$, Then, $Y \overset{d}{=} Q_Y(U)$, i.e.* $\mathsf{P}(Q_Y(U) \leqslant y) = F_Y(y)$.

Theorem 5.2 (Probability Transform) *If a rv Y has continuous CDF, then*

$$F_Y(Y) \sim U(0, 1),$$

i.e. $F_Y(Y)$ is a rv with a uniform distribution over $(0, 1)$.

[5] The plots are generated by R commands: that of f_X should not display the vertical bar at abscissas $x = 0$ and $x = 1$.

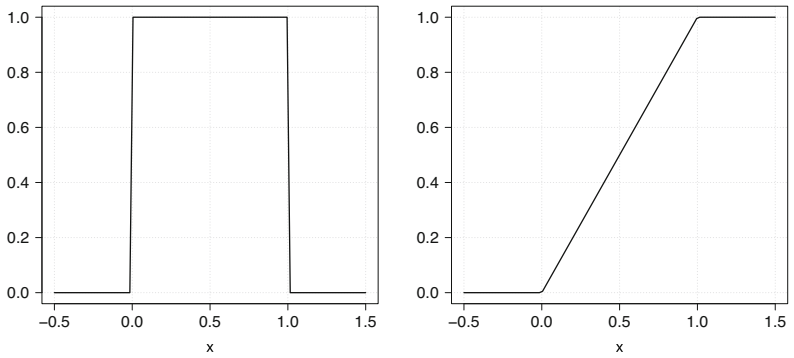


Fig. 5.3 Left: density of a U(0, 1). Right: CDF of a U(0, 1)

Theorem 5.2 provides the so called *inverse transform method* for simulating rvs with a prescribed distribution.

Example 5.4 Let X be a discrete rv with masses

$$\mathsf{P}(X = 3) = 0.3, \quad \mathsf{P}(X = 4) = 0.4, \quad \mathsf{P}(X = 5) = 0.3.$$

Its CDF is

$$F(x) = \begin{cases} 0 & \text{if } x < 3 \\ 0.3 & \text{if } 3 \leqslant x < 4 \\ 0.7 & \text{if } 4 \leqslant x < 5 \\ 1 & \text{if } x \geqslant 5, \end{cases}$$

the quantile function is

$$Q_X(u) = \begin{cases} 3 & \text{if } 0 < u \leqslant 0.3 \\ 4 & \text{if } 0.3 < u \leqslant 0.7 \\ 5 & \text{if } u > 0.7. \end{cases}$$

Hence, by Theorem 5.1 $Q_X(U)$, with uniform $U \sim \mathrm{U}(0, 1)$, yields values $u = U(\omega)$ in the support $\{3, 4, 5\}$ of X, for every scenario $\omega \in \Omega$.

Therefore, to get values from the support of a rv we can do following:

- assign a rv X with prescribed CDF
- randomly generate numbers u from the uniform distribution in (0,1)
- obtain X's values x after applying the X's quantile function to u.

If a rv Y is continuous we might as well plug it into its own CDF, then the resulting rv has U(0, 1).

Example 5.5 Let Y be a discrete rv with support $C = \{0, 1, 2, \ldots, n\}$ and mass function $\mathsf{P}(Y = i) =: p_i$, for $i \in C$. Thus, divide the unit interval (0, 1) into $n + 1$ open subintervals s.t. each has length equal to p_i. Then, we can define Y taking value $i \in C$ whenever a uniformly distributed rv $U \sim \mathrm{U}(0, 1)$ takes value in the ith subinterval of length p_i. Since the law of U is continuous uniform and

$$\{Y = i\} = \{U \text{ falls into interval of length } p_i\},$$

then being the law of U continuous uniform, i.e., characterized by length of intervals, we arrive to $\mathsf{P}(Y = i)$ is exactly p_i, for $i \in C$.

Applying a location-scale transformation to $U \sim \mathrm{U}(0, 1)$ we can change the unit interval: $X = a + (b - a)U \sim \mathrm{U}(a, b)$, for real numbers $a < b$. In fact, the support of U is the unit interval (0, 1) which has length 1, and we can pass to the support (a, b) by using the scaling factor $b-a$, i.e., $(b-a)U \sim \mathrm{U}(0, b-a)$. Thus, changing the left end point from 0 to a we need to shift all the support by adding just a, then getting the resulting distribution of X. This enable us to randomly generate numbers from any interval (a, b), for example using the Excel function `RAND()*(b-a)+a`, or the `R` command `runif(n,a,b)` where `n` is the number of simulations. Other `R` commands are: `dunif(x,a,b)` and `punif(x,a,b)` evaluating the density and the CDF, respectively, at `x`. To draw more values from the uniform over [0, 1] we refer to the following `R` commands:

Program Code

```
S=20; a <- 0; b <- 1; runif(S,a,b)
q=seq(0:10)
p=seq(from=0.01,to=0.99, lenght=50)
dunif(q,a,b)
punif(q,a,b)
qunif(p,a,b)
```

where `qunif(p,a,b)` yields the uniform quantile function evaluated at `p`. Observe that the first arguments `S`, `q` and `p` can be vectors of values, i.e., numerical arrays. The expectation of a continuous uniform rv is easy to compute:

$$\begin{aligned}\mathsf{E}(X) = \int_{-\infty}^{\infty} x f(x)\mathrm{d}x &= \frac{1}{b-a}\int_a^b x\mathrm{d}x \\ &= \frac{1}{b-a}\left[\tfrac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{b+a}{2}.\end{aligned}$$

To compute the variance we first write the second moment

$$\mathsf{E}(X^2) = \int_a^b x^2 \frac{1}{b-a} \mathrm{d}x = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^2 + ab + a^2}{3},$$

then we have

$$\mathsf{V}(X) = \mathsf{E}(X^2) - (\mathsf{E}(X))^2 = \frac{b^2 + ab + a^2}{3} - \frac{(b+a)^2}{4} = \frac{(b-a)^2}{12}.$$

The MGF of $X \sim \mathrm{U}(a, b)$ is computed below:

$$\begin{aligned} M_X(u) = \mathsf{E}(\mathrm{e}^{uX}) &= \frac{1}{b-a} \int_a^b \mathrm{e}^{ux} \mathrm{d}x \\ &= \frac{\mathrm{e}^{ub} - \mathrm{e}^{ua}}{u(b-a)}. \end{aligned}$$

For example, choosing $a = 0, b = 1$ we get $M_X(u) = \frac{\mathrm{e}^u - 1}{u}$ as the MGF of $U \sim \mathrm{U}(0, 1)$. Therefore, the MGF exists for all $u \in \mathbb{R}$ and this implies that the CHF can be obtained by replacing u with $\mathrm{i}u$.

Uniforms over the Same Interval
Let $U \sim \mathrm{U}(0, 1)$ and let $X = 1 - U$. We show that $X \sim \mathrm{U}(0, 1)$. Indeed, write

$$\begin{aligned} M_X(u) = \mathsf{E}(\mathrm{e}^{u(1-U)}) \overset{t=-u}{=} \mathsf{E}(\mathrm{e}^u \mathrm{e}^{tU}) &= \mathrm{e}^u \underbrace{\mathsf{E}(\mathrm{e}^{tU})}_{=M_U(t)} \\ &= \mathrm{e}^u \frac{\mathrm{e}^t - 1}{t} = \frac{\mathrm{e}^u - 1}{u}, \end{aligned}$$

where the term on the right-hand side of the latter equality is the MGF of $U \sim \mathrm{U}(0, 1)$.

Uniform rvs have also some interest to disprove a converse of the statement 'continuous bi-variate densities have continuous marginals'.

Example 5.6 (Actually... Counterexample) Let $X_1 \sim \mathrm{U}(0, 1)$ and define another rv $X_2(\omega) = X_1(\omega)$ for every scenario $\omega \in \Omega$. Obviously, $X_2 \overset{\mathrm{d}}{=} X_1$ so both rvs are absolutely continuous with uniform marginals. On the other hand, $\mathbf{X} = (X_1, X_2)$

does not posses a bi-variate density and then it is not an absolutely continuous random vector. In fact we have

$$\begin{aligned} F_{\mathbf{X}}(x_1, x_2) &= \mathsf{P}(X_1 \leqslant x_1, X_2 \leqslant x_2) = \mathsf{P}(X_1 \leqslant x_1, X_1 \leqslant x_2) \\ &= \mathsf{P}(X_1 \leqslant \min\{x_1, x_2\}) = \int_0^{\min\{x_1, x_2\}} \mathrm{d}x_1 \\ &\neq \int_0^{x_1} \int_0^{x_2} f_{\mathbf{X}}(u_1, u_2) \mathrm{d}u_1 \mathrm{d}u_2, \end{aligned}$$

for some joint density $f_{\mathbf{X}}(x_1, x_2)$. We used that $\{X_1 \leqslant x_1\} \cap \{X_1 \leqslant x_2\}$ is reduced to the event $\{X_1 \leqslant \cdot\}$ evaluated at the smaller value between x_1, x_2. Moreover, $X_1 = X_2$ has uniform density $f_{X_1}(x) = \mathbf{I}_{[0,1]}(x)$.

On the other hand, a bi-variate uniform distribution over the rectangle $[0, 1] \times [0, 1]$, called the *unit square*, is characterized by a joint density

$$f_{\mathbf{X}}(x, y) = \begin{cases} 1 & \text{if } (x, y) \in [0, 1] \times [0, 1], \\ 0 & \text{otherwise} \end{cases}$$

of a random vector $\mathbf{X} = (X, Y)$ where both coordinate rvs have a uniform distribution over the unit interval. In fact, for the X's marginal we have

$$f_X(x) = \int_0^1 f_{\mathbf{X}}(x, y) \mathrm{d}y = \int_0^1 1 \, \mathrm{d}y = 1,$$

and similarly for f_Y. Moreover, for this kind of uniform bi-variate distribution X, Y are independent:

$$\begin{aligned} 1 &= \int_0^1 \int_0^1 f_{\mathbf{X}}(x, y) \mathrm{d}x \mathrm{d}y = \int_0^1 \int_0^1 1 \, \mathrm{d}x \mathrm{d}y \\ &= \int_0^1 1 \, \mathrm{d}x \int_0^1 1 \, \mathrm{d}y = 1 \times 1 \\ &= \int_0^1 f_X(x) \mathrm{d}x \int_0^1 f_Y(y) \mathrm{d}y. \end{aligned}$$

Thus, the joint density factors and the distribution of Y conditional on $X = x$ is univariate uniform over the unit interval, whatever x is, see Chap. 6 for further details on conditional distributions.

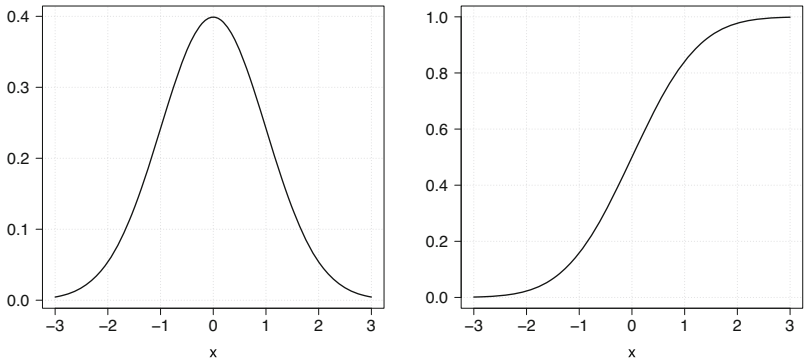


Fig. 5.4 Left: density of a standard normal. Right: CDF of a standard normal

5.7 Gaussian

A Gaussian rv X has the well known density with bell-shaped graph, centered at μ, which is a location parameter, and whose values are dispersed around it at a rate σ. The Normal distribution is ubiquitous in mathematical statistics since many random phenomena can be modelled in such a way some aggregation procedure comes into play, especially through the Central Limit Theorem we will discuss later. Moreover, it is plausible to model stock returns over fixed time periods as being symmetrical distributed according the Gaussian law, under *normal* market conditions.[6] The Gaussian density $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}$ is indeed symmetric around μ since $f_X(x') = f_X(x)$, whenever $x' + x = 2\mu$, i.e. when the points x and x' have the same distance from the localization parameter, and the bell-shaped graph of f_X has a maximum at $(\mu, f_X(\mu))$. In words, the function $g(x) = f_X(x+\mu)$ takes value $g(x-\mu) = f_X(x)$ for $m > 0$, the same values of the density at $x - m\mu$ which leads to a horizontal shift of the f_X's graph on the left. If $\mu = 0$ and $\sigma = 1$, X is said to have a *standard normal distribution*, see Fig. 5.4. It can be shown that such a function f_X is a truly density, see Exercise 5.9. We show that μ and σ are the expectation and standard deviation of X, respectively. First, write the expectation with the change of variable $x = (x-\mu) + \mu$:

$$\mathsf{E}(X) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (x-\mu) \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}} \mathrm{d}x + \mu \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}} \mathrm{d}x.$$

[6] Whereas this assumption is far for being true, as statistical analysis of historical returns shows, viz. stylized facts.

Let $y = x - \mu$ and $\mathrm{d}y = \mathrm{d}x$. The first integrand on the left-hand side, say $h(y) = y\mathrm{e}^{-\frac{y^2}{2\sigma^2}}$, is an odd function, $h(-y) = -h(y)$, so its integral is zero. The other integral refers to the Gaussian density and then is 1, thus we have $\mathsf{E}(X) = \mu$. Second, using again $\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1$ we have:

$$\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}\mathrm{d}x = \sigma\sqrt{2\pi}.$$

Hence, differentiate with respect to σ both sides we get:[7]

$$\int_{-\infty}^{\infty} \frac{(x-\mu)^2}{\sigma^3}\mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}\mathrm{d}x = \sqrt{2\pi}.$$

Multiplying by $\frac{\sigma^2}{\sqrt{2\pi}}$ we get the expression

$$\int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}\mathrm{d}x = \sigma^2,$$

which is nothing else $\mathsf{E}((X-\mu)^2) = \mathsf{V}(X)$. For the standard normal distribution we can let $X = \mu + \sigma Z$, where $Z \sim \mathrm{N}(0,1)$ and X turn to be $\sim \mathrm{N}(\mu, \sigma^2)$. In fact, this is a proper location-scale transform, see Sect. 4.5 in Chap. 4, and the two densities f_X and f_Z are related by $\frac{1}{\sigma} f_X\left(\frac{x-\mu}{\sigma}\right) = f_Z(x)$ so $\frac{X-\mu}{\sigma} \sim \mathrm{N}(0,1)$. The Gaussian MGF is computed after that of the standard normal. Thus, let $Z \sim \mathrm{N}(0,1)$ and compute:

$$\begin{aligned} M_Z(u) &= \int_{-\infty}^{\infty} \mathrm{e}^{ux}\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2}\mathrm{d}x = \mathrm{e}^{t^2/2}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-(x-t)^2/2}\mathrm{d}x \qquad (5.12)\\ &= \mathrm{e}^{t^2/2}, \end{aligned}$$

where the last integral refers to the density of a $\mathrm{N}(t,1)$ rv and then is equal to 1. The MGF exists for every $u \in \mathbb{R}$. Now, by the proper location-scale transform we obtain the MGF of a Gaussian $X \sim \mathrm{N}(\mu, \sigma^2)$ by recalling Theorem 4.1 in Sect. 4.4:

$$M_{\mu+\sigma Z}(u) = \mathrm{e}^{\mu u} M_Z(\sigma u) = \mathrm{e}^{\mu u + \frac{\sigma^2 u^2}{2}}.$$

[7] Let $h(x,\mu,\sigma) = \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}$ and use Leibniz's formula (see Appendix L) to differentiate: $\frac{\mathrm{d}}{\mathrm{d}\sigma}\int_{-\infty}^{\infty} h(x,\mu,\sigma)\mathrm{d}x$ is equal to $\int_{-\infty}^{\infty}\frac{\partial}{\partial\sigma}h(x,\mu,\sigma)\mathrm{d}x$.

There is a remarkable formula for the central moments of a Gaussian distribution:

$$\mathsf{E}((X-\mu)^p) = \begin{cases} \frac{(2p)!}{2^p p!}\sigma^{2p} & \text{for } p = 2, 4, 6, \ldots \\ 0 & \text{for } p = 1, 3, 5, \ldots, \end{cases} \tag{5.13}$$

see Exercise 5.14. Thus, for a standard normal $Z = \frac{X-\mu}{\sigma}$ we also found that

$$\mathsf{E}\left(\left(\frac{X-\mu}{\sigma}\right)^4\right) - 3 = \mathsf{E}(Z^4) - 3 = 3 - 3 = 0,$$

which confirms how the excess kurtosis of a standard normal is zero. The standard normal distribution is symmetric around $\mu = 0$ and this transfers to symmetry of tail areas, $F_Z(x) = 1 - F_Z(-x)$, where F_Z is the CDF function:

$$F_Z(x) = \int_{-\infty}^{-x} f_Z(t)\mathrm{d}t = \int_x^{\infty} f_Z(t)\mathrm{d}t = 1 - \int_{-\infty}^{x} = 1 - F_Z(x).$$

It follows that $Z \stackrel{\mathrm{d}}{=} -Z$:

$$\mathsf{P}(-Z \leqslant x) = \mathsf{P}(Z \geqslant -x) = 1 - F_Z(-x) = F_Z(x).$$

If we take the sum of $X \sim \mathrm{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathrm{N}(\mu_2, \sigma_2^2)$, which are independent, then

$$\begin{aligned} M_{X+Y}(u) &= M_X(u) M_Y(u) \\ &= \exp\left(\mu_1 u + \frac{\sigma_1^2 u^2}{2}\right) \exp\left(\mu_2 u + \frac{\sigma_2^2 u^2}{2}\right) \\ &= \exp\left((\mu_1+\mu_2)u + \frac{(\sigma_1^2+\sigma_2^2)u^2}{2}\right) = W, \end{aligned}$$

hence $W \sim \mathrm{N}(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$. Taking instead the sum of two dependent Gaussian rvs, the resulting rv can be Gaussian whenever the joint density is of particular form. In fact, a random vector $\mathbf{X} = (X_1, \ldots, X_n)$ is said to be Gaussian distributed if its components X_i are continuous rvs with joint density

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}(\det \boldsymbol{\Sigma})^{1/2}} \exp\left(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})\,\boldsymbol{\Sigma}^{-1}\,(\mathbf{x}-\boldsymbol{\mu})'\right), \quad \mathbf{x} \in \mathbb{R}^n, \tag{5.14}$$

where $\mathbf{\Sigma}$ is a $n \times n$ symmetric and *positive semi-definite* matrix,[8] $\boldsymbol{\mu} \in \mathbb{R}^n$. A prime $'$ stands for matrix transposition[9] and $\det \mathbf{\Sigma}$ denotes the determinant of $\mathbf{\Sigma}$. We write $\mathbf{X} \sim \mathrm{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$. For $n = 2$, a bi-variate Gaussian random vector has joint density given by (5.14), where:

- the inverse of $\mathbf{\Sigma}$ is

$$\mathbf{\Sigma}^{-1} = \frac{1}{c_{11}c_{22-} - c_{12}^2} \begin{bmatrix} c_{22} & -c_{12} \\ -c_{21} & c_{11} \end{bmatrix},$$

  for $\det \mathbf{\Sigma} = c_{11}c_{22} - c_{12}^2 \neq 0$, with $c_{12} = c_{21}$;
- $c_{11} = \mathsf{V}(X_1) = \sigma_1^2$ and $c_{22} = \mathsf{V}(X_2) = \sigma_2^2$;
- $c_{12} = \mathsf{cov}(X_1, X_2)$;
- $\mathsf{corr}(X_1, X_2) = \frac{c_{12}}{\sigma_1 \sigma_2} = \rho$, the Pearson correlation coefficient.

Indeed, let $\boldsymbol{\mu} = (\mu_1, \mu_2)$ and perform the matrix multiplication $(\mathbf{x} - \boldsymbol{\mu})\, \mathbf{\Sigma}^{-1}\, (\mathbf{x} - \boldsymbol{\mu})'$ to get

$$\begin{aligned} f(x_1, x_2) &= a_1(x_1 - \mu_1)^2 - 2a_2(x_1 - \mu_1)(x_2 - \mu_2) \\ &\quad + a_3(x_2 - \mu_2)^2 \geqslant 0, \quad \text{for every } (x_1, x_2) \in \mathbb{R}^2, \end{aligned}$$

i.e. a quadratic form with $a_1 = \frac{c_{11}}{\det \mathbf{\Sigma}}$, $a_2 = \frac{c_{12}}{\det \mathbf{\Sigma}}$, and $a_3 = \frac{c_{22}}{\det \mathbf{\Sigma}}$. The density is:

$$f_{\mathbf{X}}(x_1, x_2) = \frac{1}{2\pi (c_{11}c_{22} - c_{12}^2)^{1/2}} \mathrm{e}^{f(x_1, x_2)}$$

which we rewrite

$$\begin{aligned} f_{\mathbf{X}}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\Bigg(- \frac{1}{2\,(1-\rho^2)} \Bigg(\frac{(x_1 - \mu_1)^2}{\sigma_1^2} \\ - 2\rho \frac{x_1 - \mu_1}{\sigma_1} \frac{x_2 - \mu_2}{\sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \Bigg) \Bigg). \end{aligned}$$

Essentially, in the bi-dimensional case such a joint distribution has five parameters: $\mu_1 \in \mathbb{R}$, $\mu_2 \in \mathbb{R}$, $\sigma_1 > 0$, $\sigma_1 > 0$, and $-1 < \rho < 1$. The relevant fact here is:

Theorem 5.3 *If* $\mathbf{X} = (X_1, X_2) \sim \mathrm{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$, *then* $X_i \sim \mathrm{N}(\mu_i, \sigma_i^2)$ *for* $i = 1, 2$.

For a proof of this see Exercise 5.11. There is an analogous of Theorem 5.3 for n-dimensional Gaussian vectors. It is worth noting how for bi-variate Gaussian distributions $\rho = 0$ is equivalent to independence of X_1 and X_2. More generally, let

[8] It is invertible.

[9] We assume that any $\mathbf{x} \in \mathbb{R}^n$ is represented by a row vector.

$\boldsymbol{\Sigma} = \mathbf{I}$, where $\mathbf{I}$ is the $n \times n$ identity matrix, and $\boldsymbol{\mu}$ is the null vector, then $\mathbf{I}^{-1} = \mathbf{I}$ with $\det \mathbf{I} = 1$ and expression (5.14) reduces to

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}) &= \frac{1}{(2\pi)^{n/2}} \mathrm{e}^{\frac{\mathbf{x}\mathbf{x}'}{2}} \\ &= \frac{1}{\sqrt{2\pi} \cdots \sqrt{2\pi}} \mathrm{e}^{-x^2/2} \cdots \mathrm{e}^{-x^2/2}, \end{aligned}$$

which means that the joint density factors and the component rvs are independent. If $\mathbf{X} = (X_1, X_2)$ is bi-variate normal and $a_1, a_2 \in \mathbb{R}$, then

$$a_1 X_1 + a_2 X_2 \sim \mathrm{N}\left(a_1\mu_1 + a_2\mu_2,\ a_1\sigma_1^2 + a_2\sigma_2^2 + 2a_1a_2\rho\sigma_1\sigma_2\right).$$

In particular if X_1, X_2 are independent (and then $\rho = 0$) the dispersion parameter simplifies to $a_1\sigma_1^2 + a_2\sigma_2^2$. This is a consequence of Theorem 5.3 and the use of MGF $M_{a_1X_1+a_2X_2}(u)$. As for bi-variate uniform distributions seen in Sect. 5.6, bi-variate Gaussian random vectors lead to marginal distribution of one component rv conditional on the other, see Chap. 6. By Proposition 4.3 in Sect. 4.7, for a given deterministic $m \times n$ matrix $\mathbf{A}$ and a Gaussian random vector $\mathbf{X} \sim \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ we have

$$\mathbf{A}\mathbf{X}' \sim \mathrm{N}(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}').$$

Other univariate limiting distributions might be obtained from the univariate Gaussian, and this will be treated in Chap. 8. We highlight a connection between the binomial and the Gaussian distribution which could be useful in modelling future stock prices and related returns. Let $X_n \sim \mathrm{Bin}(n, p)$ and set $\frac{X_n - np}{\sqrt{np(1-p)}}$. Then, it can be showed that for n large enough the distribution of X_n is approximately standard normal: the trick is to consider X_n as the sum of IID $X_i \sim \mathrm{Ber}(p)$. Another way to express this is writing

$$X_n \quad \text{is approximately distributed as} \quad \mathrm{N}(np, np(1-p)).$$

The `R` commands to handle a Gaussian distribution are reported below.

Program Code

```
S=20; mu <- 0; sigma <- 1
rnorm(S,mu,sigma)
q=seq(0:10)
p=seq(from=0.01,to=0.99, lenght=50)
dnorm(q,mu,sigma)
pnorm(q,mu,sigma)
qnorm(p,mu,sigma)
```

Recall that `rnorm(x,mu,sigma)` yields a realization `x` drawn from $\mathrm{N}(\mu, \sigma^2)$, `dnorm(x,mu,sigma)` evaluates the Gaussian density at `x`, `pnorm(x,mu,sigma)` evaluates the Gaussian CDF up to `x` and `qnorm(p,mu,sigma)` entails the Gaussian quantile function evaluated at the probability level `p`. Obviously, when `x` and `p` are vectors, i.e, lists of values, these commands produce lists of output. Note that `sigma` is the standard deviation not the variance. For the bi-variate Gaussian, if we know the values of all the five parameters, then we can define the covariance matrix as

```
Sigma=matrix(c(s12,s1*s2*rho,s22,s1*s2*rho),nrow=2,ncol=2)
```

where `s12` is the variance σ_1^2 of the first component rv of the Gaussian vector, `s22` is the variance σ_2^2 of the second component, while `s1*s2*rho` is the covariance $\mathsf{cov}(X_1, X_2)$ with standard deviations σ_1, σ_2 and correlation coefficient ρ. To get the inverse of `Sigma` we type `solve(Sigma)`. Another rv whose distribution is related to the Gaussian is the *log-normal*: let $X \sim \mathrm{N}(\mu, \sigma^2)$ and let $Y = \ln X$, then we have the CDF

$$\begin{aligned} F_Y(y) = \mathsf{P}(Y \leqslant y) &= \mathsf{P}\left(\mathrm{e}^X \leqslant y\right) \\ &= \mathsf{P}(X \leqslant \ln y) \\ &= \int_{-\infty}^{\ln y} \frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}} \mathrm{d}x, \end{aligned} \tag{5.15}$$

see Sect. 4.5 in Chap. 4. Differentiating the CDF we get

$$\begin{aligned} f_Y(y) &= \frac{\mathrm{d}}{\mathrm{d}y} h(g(y)) = \frac{\mathrm{d}}{\mathrm{d}y} F_Y(y) \\ &= F_Y' \frac{\mathrm{d}}{\mathrm{d}y} (\ln y) \\ &= \frac{1}{y\sqrt{2\pi}\sigma} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right) \mathbf{I}_{\{y>0\}}. \end{aligned} \tag{5.16}$$

Note that we add the restriction $\mathbf{I}_{\{y>0\}}$ since Y cannot assume negative values. Thus, Y is log-normal with parameters μ, σ^2 because $\ln(Y) \sim \mathrm{N}(\mu, \sigma^2)$. The log-normal rv is used in models of future stock prices taking advantage of its positive values. The expectation and variance are

$$\mathsf{E}(Y) = \mathrm{e}^{\mu+\sigma^2/2} \tag{5.17}$$

$$\mathsf{V}(Y) = \mathrm{e}^{2\mu+\sigma^2}(\mathrm{e}^{\sigma^2} - 1), \tag{5.18}$$

see Exercise 5.12.

Remark 5.3 Let $Y = \ln X$ be a log-normal rv and assume $X \sim \mathrm{N}(0, 1)$ is standard normal. The density function is

$$f_Y(x) = \frac{1}{x} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-(\ln x)^2/2}, \qquad x > 0,$$

so the MGF writes

$$\frac{1}{\sqrt{2\pi}} \int_0^\infty \mathrm{e}^{u\,x - (\ln x)^2/2 - \ln x} \mathrm{d}x.$$

Now, for any $u > 0$ we recall from a basic calculus course that $\frac{u\,x}{\ln x} \to \infty$ as $x \to \infty$, then

$$\frac{u\,x - (\ln x)^2/2 - \ln x}{u\,x} = 1 - (\ln x)^2/2u\,x - \ln x/u\,x \to 1, \quad \text{as } x \to \infty,$$

and the exponent in the integrand of the integral is asymptotic to $u\,x$, i.e., $u\,x - (\ln x)^2/2 - \ln x \sim u\,x$ as $x \to \infty$. This implies that $\mathsf{E}(\mathrm{e}^{u\,X}) = \infty$ and actually the MGF does not exist. Notwithstanding, for some $p \in \mathbb{N}$ write

$$\mathsf{E}(X^p) = \int_{-\infty}^\infty \mathrm{e}^{pt} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2} \mathrm{d}t = \mathrm{e}^{p^2/2} < \infty,$$

where we use the change of variable $\ln x = t$ with $\mathrm{d} \ln x = \frac{1}{x} = \mathrm{d}t$ and $x^p = \mathrm{e}^{p \ln x} = \mathrm{e}^{pt}$. Hence, all the p-th moments of the log-normal distribution do exist and are finite.

5.8 Exponential

The exponential distribution characterizes rvs with nonnegative real values. Historically, it models waiting of a success in continuous time, where the rate of successes for unit time is governed by a parameter $\lambda > 0$. The density function reported in Table 5.2 has CDF given by

$$F_X(x) = \int_{-\infty}^x \lambda\, \mathrm{e}^{-\lambda z}\, \mathbf{I}_{\{z \geqslant 0\}} \mathrm{d}z = \int_0^x \lambda\, \mathrm{e}^{-\lambda s} \mathrm{d}z = 1 - \mathrm{e}^{-\lambda x}, \tag{5.19}$$

see also Fig. 5.5.

A financial application is when $X \sim \mathrm{Exp}(\lambda)$ represents the value of a callable bond before its drawing. Thus $\mathsf{P}(X > t) = \mathrm{e}^{-\lambda t}$ is the probability that the bond will have a life of at least $t > 0$ years. The expectation of $X \sim \mathrm{Exp}(\lambda)$ is $\mathsf{E}(X) = \frac{1}{\lambda}$ and

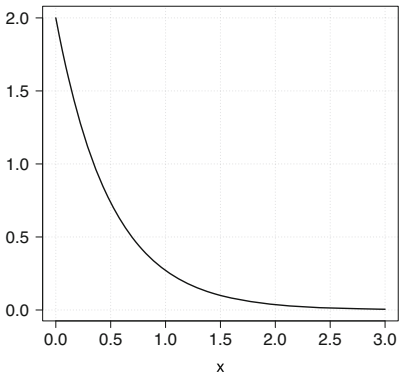


Fig. 5.5 Density of a Exp(2)

the variance is $\mathsf{V}(X) = \frac{1}{\lambda^2}$, see Exercise 5.13. The MGF of $X \sim \text{Exp}(\lambda)$ is derived below assuming that $u < \lambda$:

$$\begin{aligned} M_X(u) &= \int_0^\infty \mathrm{e}^{u\,x} \lambda \mathrm{e}^{-\lambda x} \mathrm{d}x \\ &= \lambda \left[\frac{\mathrm{e}^{(u-\lambda)x}}{u-\lambda} \right]_0^\infty \\ &= \frac{\lambda}{\lambda - u}. \end{aligned}$$

The above constraint $u < \lambda$ is needed since for $u \geqslant \lambda$ the integral does diverge.

Remark 5.4 From Example 2.14 in Sect. 2.9, we recognize

$$F_Y(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1 - \mathrm{e}^{-\lambda x}, & \text{if } 0 \leqslant x < s \\ 1, & \text{if } x \geqslant s, \end{cases}$$

with $\lambda > 0$ and $s > 0$, a CDF similar to that of an exponential rv. Indeed, this is the *truncated exponential* CDF, a mixed rv. This is because the Y's distribution has a continuous part given by the exponential density truncated at s, while the remaining discrete part is the point mass at $\mathrm{e}^{-\lambda s}$. Therefore, we can compute its MGF using

the change of variable formula to adapt the expectation of Y to the transform $\mathrm{e}^{u\,Y}$ via the original mixed distribution:

$$\begin{aligned} M_Y(u) &= \underbrace{\int_0^s \mathrm{e}^{u\,x}\lambda \mathrm{e}^{-\lambda s}\mathrm{d}x}_{\mathsf{E}(\,\cdot\,)\text{ of cont. part}} + \mathrm{e}^{u\,s}\mathrm{e}^{-\lambda s} \\ &\overset{u\neq\lambda}{=} \lambda\left[\frac{\mathrm{e}^{(u-\lambda)x}}{u-\lambda}\right]_0^s + \mathrm{e}^{-s(\lambda-u)} \\ &= \frac{\lambda}{\lambda-u} - \frac{u}{\lambda-u}\mathrm{e}^{-s(\lambda-u)}. \end{aligned}$$

For $u = \lambda$ we have $M_Y(u) = \lambda s + 1$.

An interesting property of the exponential distribution is stated as follows:

$$\begin{aligned} \mathsf{P}(X \geqslant x + y \,|\, X \geqslant x) &= \frac{\mathsf{P}(X \geqslant x + y)}{\mathsf{P}(X \geqslant x)} \\ &= \frac{\mathrm{e}^{-\lambda(x+y)}}{\mathrm{e}^{-\lambda x}} = \mathrm{e}^{-\lambda y} = \mathsf{P}(X \geqslant y), \end{aligned}$$

where we used $\{X \geqslant x + y\} \cap \{X \geqslant x\} = \{X \geqslant x + y\}$, because $x + y \geqslant x$, in the definition of the conditional probability, while the last line follows from the definition of the CDF. This is called *memoryless property*, if x, y are interpreted as waiting times. There is an interesting relationship between the exponential and the Poisson rvs, which can be easily derived by the definition of their probability distributions (viz. the CDF or density, and the mass). For $X \sim \mathrm{Exp}(\lambda)$ and $\mathrm{Po}(\lambda x)$ we have

$$\begin{aligned} \mathsf{P}(X > x) &= 1 - F_X(x) = \mathrm{e}^{-\lambda x} \\ &= \mathrm{e}^{\lambda x}\frac{\lambda^0}{0!} = \mathsf{P}(Y = 0). \end{aligned}$$

From this we also deduce $\mathsf{P}(X \leqslant x) = \mathsf{P}(Y > 0)$. There are `R` commands for handling the exponential distribution similar to those viewed for the aforementioned distributions, namely `rexp(x,lambda)`, `dexp(x,lambda)`, `pexp(x,lambda)` and `qexp(x,lambda)`.

5.9 Gamma

A rv $X > 0$ with gamma distribution generalizes the exponential law. The definition of its density requires the *gamma function*, which in turn generalizes the factorial function early defined only for positive integers:

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} \mathrm{e}^{-x} \mathrm{d}x, \quad \alpha > 0. \tag{5.20}$$

Note that for $\alpha = 1$ we have

$$\Gamma(1) = \int_0^\infty \mathrm{e}^{-x} \mathrm{d}x = 1. \tag{5.21}$$

Integrating by parts for $\alpha > 0$ we also have

$$\Gamma(\alpha+1) = \int_0^\infty x^{\alpha} \mathrm{e}^{-x} \mathrm{d}x = \underbrace{\left[-x^{\alpha} \mathrm{e}^{-x}\right]_0^\infty}_{=0} + \alpha \int_0^\infty x^{\alpha-1} \mathrm{e}^{-x} \mathrm{d}x = \alpha\Gamma(\alpha). \tag{5.22}$$

Hence, by induction it holds $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$. To get the density function of $X \sim \text{Gamma}(\alpha, 1)$ we simply divide both sides of (5.21) by $\Gamma(\alpha)$, then

$$1 = \int_0^\infty \frac{x^{\alpha-1} \mathrm{e}^{-x}}{\Gamma(\alpha)} \mathrm{d}x$$

and $f_X(x) = \frac{x^{\alpha-1} \mathrm{e}^{-x}}{\Gamma(\alpha)} \mathbf{I}_{\{x>0\}}$ is a truly density. With a little extra effort we obtain the density of $Y = \frac{X}{\lambda} \sim \text{Gamma}(\alpha, \lambda)$ reported in Table 5.2, where $\lambda > 0$ is the additional parameter, see Fig. 5.6.

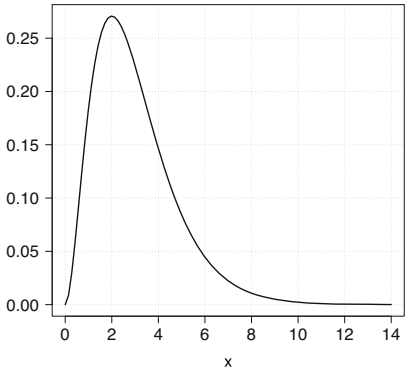


Fig. 5.6 Density of a Gamma(3, 1)

Remark 5.5 Any function $g(x) = k\, x^{\alpha-1}\mathrm{e}^{-x\lambda}\mathbf{I}_{\{x>0\}}$, for $k > 0$, is a density and $k = \frac{\lambda^\alpha}{\Gamma(\alpha)}$, see Exercise 5.15. On the other hand, by formula (4.11) in Sect. 4.5 applied to the increasing transformation $Y = \frac{X}{\lambda}$ we have

$$f_Y(y) = \frac{1}{\Gamma(\alpha)}(\lambda\, y)^\alpha \mathrm{e}^{-\lambda\, y} y^{-1}, \quad y > 0,$$

which is the density of Gamma(α, λ), see Exercise 5.16.

It is worth noting that for $\alpha = 1$ the gamma distribution $Y \sim$ Gamma$(1, \lambda)$ has the same density as that of the Exp(λ). The expectation and variance of a gamma rv $X \sim$ Gamma$(\alpha, 1)$ are $\mathsf{E}(X) = \mathsf{V}(X) = \alpha$, see Exercise 5.16. We deduce that $Y = \frac{X}{\lambda} \sim$ Gamma(α, λ) has expectation

$$\mathsf{E}(Y) = \frac{1}{\lambda}\mathsf{E}(X) = \frac{\alpha}{\lambda},$$

and variance

$$\mathsf{V}(Y) = \frac{1}{\lambda}\mathsf{V}(X) = \frac{\alpha}{\lambda^2}.$$

The MGF of Y is easily derived:

$$\begin{aligned} M_Y(u) &= \int_0^\infty \mathrm{e}^{ux}\frac{\lambda^\alpha x^{\alpha-1}\mathrm{e}^{-\lambda x}}{\Gamma(\alpha)}\mathrm{d}x \\ &= \left(\frac{\lambda}{\lambda-u}\right)^\alpha \int_0^\infty \underbrace{\frac{(\lambda-u)^\alpha x^{\alpha-1}\mathrm{e}^{-(\lambda-u)x}}{\Gamma(\alpha)}}_{=\Gamma(\alpha,\lambda-u)}\mathrm{d}x \\ &= \left(\frac{\lambda}{\lambda-u}\right)^\alpha, \quad u < \lambda. \end{aligned}$$

The condition $u < \lambda$ ensures that the MGF exists for u in an neighborhood of zero. Incidentally, we recognize that $M_Y(u)$ is also equal to the MGF

$$M_{X_1+\cdots+X_n}(u) = M_{X_1}\cdots M_{X_n}, \quad \text{for IID } X_i \sim \mathrm{Exp}(\lambda).$$

Interesting enough, there is a connection also between gamma and Gaussian distribution. Indeed, let $V \sim \mathrm{N}(0, \sigma^2)$ and consider the transform $W = V^2$. Then, W has a density $f_W(x)\mathbf{I}_{\{x>0\}}$ which is nothing else $\Gamma(\frac{1}{2}, \frac{1}{2\sigma^2})$, see Exercise 5.18. Observe that this implies $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ for the gamma function.

5.10 Student-*t*

To set up a rv with Student-*t* distribution we first need the *Chi-square distribution* defined as $K = \sum_{i=1}^{n} Z_i^2$, where $Z_i \sim \mathrm{N}(0, 1)$ are IID. We write $K \sim \chi_n^2$ and call n the degree of freedom. By Exercise 5.18, we see that $Z_1^2 \sim \mathrm{Gamma}(\frac{1}{2}, \frac{1}{2})$ and thus χ_n^2 is nothing else the $\mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$, the sum of n IID $\mathrm{Gamma}(\frac{1}{2}, \frac{1}{2})$. The expectation and variance are obtained by knowing their counterpart in the Gamma distribution, or by considering the sum of IID squared standard normals: $\mathsf{E}(K) = n\mathsf{E}(Z_1^2) = n$ and

$$\begin{aligned}\mathsf{V}(K) = n\mathsf{V}(Z_1^2) &= n\left(\mathsf{E}(Z_1^4) - (\mathsf{E}(Z_1^2))^2\right) \\ &= n(3-1) = 2n.\end{aligned}$$

To obtain the MGF, we simply set the parameters in $\mathrm{Gamma}(\alpha, \lambda)$ equal to $\frac{n}{2}$ and $\frac{1}{2}$, thus for $u < \frac{1}{2}$ we have

$$M_K(u) = \left(\frac{1}{2-u}\right)^{n/2},$$

from which the CHF can be get by replacing u with $\mathrm{i}u$. Now, the Student-*t* rv $T \sim \mathrm{T}(n)$ is build using a standard normal $Z \sim \mathrm{N}(0, 1)$ and a Chi-square $K \sim \chi_n^2$ through

$$T = \frac{Z}{\sqrt{\frac{K}{n}}},$$

provided that Z is independent of K. We say that T has a Student-*t* distribution with n degree of freedom. The Student-*t* distribution is symmetric, i.e., $T \sim \mathrm{T}(n)$ is the same as $-T \sim \mathrm{T}(n)$, indeed $-T = \frac{-Z}{\sqrt{K/n}}$, where $-Z \sim \mathrm{N}(0, 1)$, see Fig. 5.7. Moreover, as $n \to \infty$ the Student-*t* distribution approximate the standard normal, see Sect. 8.2 in Chap. 8.

5.11 Exercises

5.1 Let $X \sim \mathrm{Bin}(n, p)$ with failure probability $q = 1 - p$. Show that $n - X \sim \mathrm{Bin}(n, q)$, for $n \geqslant 2$.

5.2 Prove in an alternative way that $X + Y \sim \mathrm{Bin}(n_1 + n_2, p)$, provided $X \sim \mathrm{Bin}(n_1, p)$ and $Y \sim \mathrm{Bin}(n_2, p)$.

5.3 Prove in an alternative way that $\mathsf{E}(X) = np$ for $X \sim \mathrm{Bin}(n, p)$.

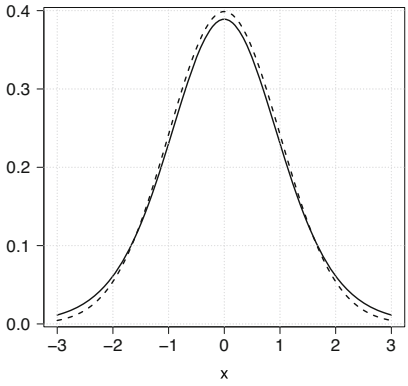


Fig. 5.7 Density of a Student-t with $n = 10$ degree of freedom (bold) vs standard normal (dashed)

5.4 Derive formula (5.11) for the variance of $X \sim \mathrm{Po}(\lambda)$.

5.5 Show that the sum of two independent Poisson rvs $X \sim \mathrm{Po}(\lambda_1)$ and $Y \sim \mathrm{Po}(\lambda_2)$ is again Poisson $X + Y \sim \mathrm{Po}(\lambda_1 + \lambda_2)$, using a different method than that of MGF.

5.6 Given two independent Poisson rvs $X \sim \mathrm{Po}(\lambda_1)$ and $Y \sim \mathrm{Po}(\lambda_2)$, show that the distribution of X conditioned on $X + Y = n$, for $n \in \mathbb{N}$ and $X + Y \sim \mathrm{Po}(\lambda_1 + \lambda_2)$, is the binomial $\mathrm{Bin}\big(n, \frac{\lambda_1}{\lambda_1+\lambda_2}\big)$.

5.7 Let $X_n \sim \mathrm{Bin}(n, p_n)$, for every positive integer $n \geqslant 1$, where $p_n = \frac{\lambda}{n}$ is s.t. $\lambda = np_n > 0$ is constant. Prove that $M_{X_n}(u) \to M_X(u)$, for $n \to \infty$, where $X \sim \mathrm{Po}(\lambda)$ and $u \in \mathbb{R}$ is taken from some interval containing zero.

5.8 Let $U \sim \mathrm{U}(a, b)$, and take $(c, d) \subset (a, b)$. Prove that conditional on $U \in (c, d)$ the distribution of U is $U \sim \mathrm{U}(c, d)$.

5.9 Let $X \sim \mathrm{N}(\mu, \sigma^2)$. Show that $\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1$, where f_X is the density.

5.10 Let $X \sim \mathrm{N}(\mu, \sigma^2)$ and let $Y = a + bX$. Prove that $Y \sim \mathrm{N}(a + b\mu, b^2\sigma^2)$.

5.11 Let $\mathbf{X} \sim \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $n = 2$. Prove Theorem 5.3.

5.12 Derive the expectation and variance of a log-normal rv.

5.13 Derive the expressions of the expectation and variance for an exponential rv with parameter $\lambda > 0$.

5.14 (Central moments of a Gaussian rv) Prove that $\mathsf{E}((X - \mu)^p) = \frac{(2p)!}{2^p p!}\sigma^{2p}$, for even $p \in \mathbb{N}$, when $X \sim \mathrm{N}(\mu, \sigma^2)$.

5.15 Assume $g(x) = k\, x^{\alpha-1}\mathrm{e}^{-x\lambda}\mathbf{I}_{\{x>0\}}$ is a density. Show that $k = \frac{\lambda^\alpha}{\Gamma(\alpha)}$.

5.16 Using the formula $f_Y(y) = f_X(h(y))|h'(y)|$, for $h(y) = \lambda\, y$, show that $Y = \frac{X}{\lambda} \sim \mathrm{Gamma}(\alpha, \lambda)$ provided that $X \sim \mathrm{Gamma}(\alpha, 1)$.

5.17 Prove that $\mathsf{E}(X) = \mathsf{V}(X) = \alpha$ for $X \sim \mathrm{Gamma}(\alpha, 1)$.

5.18 Let $V \sim \mathrm{N}(0, \sigma^2)$ and let $W = g(V) = V^2$. Show that the density of W is supported on $(0, \infty)$ and equal that of $\Gamma(\frac{1}{2}, \frac{1}{2\sigma^2})$.

Appendix

Binomial vs Poisson

Proof of Proposition 5.1 Using $\lambda = np$ we compute

$$\begin{aligned}
\mathsf{P}(X = i) &= \binom{n}{i} p^i (1-p)^{n-i} \\
&= \frac{n(n-1)(n-2)\cdots(n-i+1)}{i!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-i} \\
&= \frac{\lambda^i}{i!} \frac{n(n-1)(n-2)\cdots(n-i+1)}{n^i} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-i},
\end{aligned}$$

where letting $n \to \infty$, for fixed i, we get $\frac{n(n-1)(n-2)\cdots(n-i+1)}{n^i} \to 1$, $\left(1 - \frac{\lambda}{n}\right)^n \to \mathrm{e}^{\lambda}$ and $\left(1 - \frac{\lambda}{n}\right)^{-i} \to 1$, concluding the proof. ■

Continuous Uniform and Inversion

To prove Theorem 5.1 simply apply Proposition 2.2, item 1, with $u = c \in (0, 1)$, i.e. $F_Y(y) \geqslant u$ iff $Q_Y(u) \leqslant y$. Then,

$$\mathsf{P}(Q_Y(U) \leqslant y) = \mathsf{P}(U \leqslant F_Y(y)) = F_Y(y).$$

For Theorem 5.2 we have:

$$\begin{aligned}
\mathsf{P}(F_Y(Y) \leqslant u) &= \mathsf{P}(Q_Y(F_Y(Y)) \leqslant Q_Y(u)) \\
&= \mathsf{P}(Y \leqslant Q_Y(u))
\end{aligned}$$

where the first equality follows because for a continuous CDF F_Y we always have Q_Y is strictly increasing; the second equality follows from the fact that $Q_Y(F_Y(y)) \leqslant y$, but the values y where $Q_Y(F_Y(y)) \neq y$ form a flat of the F_Y's graph, then they have zero probability and $Q_Y(F_Y(Y)) \overset{\text{a.s.}}{=} Y$; the fourth equality is due to the assumption that F_Y is continuous from which it follows that the composition of F_Y and its quantile Q_Y is the identity map, i.e. $F_Y(Q_Y(u)) = u$, see Sect. 2.8, Remark 2.11, and Eq. 2.20.

Chapter 6
Conditioning



6.1 Conditional Distributions

Recall that $\sigma(X)$ is the smallest σ-algebra containing events $\{X \in B\} \in \mathcal{F}$, for every subset $B \subset \mathbb{R}$ and a fixed probability space $(\Omega, \mathcal{F}, \mathsf{P})$. Thus, by knowing the numerical values $X(\omega)$ we may identify $\{X \in B\}$ without knowing the scenarios $\omega \in \Omega$: the sub-σ-algebra $\sigma(X) \subset \mathcal{F}$ represents incomplete information about the underlying experiment, and this is equivalent to observe $\mathbf{I}_{\{X \in B\}}$. How can we assess the law P_X taking into account other partial information, i.e. assuming a fixed event $A \in \mathcal{F}$ occurred, provided $\mathsf{P}(A) > 0$? The answer is:

$$\mathsf{P}_{X|A}(B) := \mathsf{P}(X \in B \mid A) = \frac{\mathsf{P}\big(\{X \in B\} \cap A\big)}{\mathsf{P}(A)}, \tag{6.1}$$

which is nothing but a conditional probability given A, called the **conditional distribution** of X. Choosing $B = (-\infty, x]$ we get the conditional cumulative distribution function (CCDF):

$$F_{X \mid A}(x) := \frac{\mathsf{P}\big(\{X \leqslant x\} \cap A\big)}{\mathsf{P}(A)}. \tag{6.2}$$

We can specify the model of conditional distribution depending on the type of rv. If X is discrete, the **conditional masses** are:

$$p_{X \mid A}(x) := \frac{\mathsf{P}\big(\{X = x\} \cap A\big)}{\mathsf{P}(A)}, \quad x \text{ in the support } \mathcal{X}. \tag{6.3}$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_6

If $F_{X\,|\,A}$ is differentiable w.r.t. x we have the **conditional density**:

$$f_{X\,|\,A}(x) := \frac{\mathrm{d}}{\mathrm{d}x} F_X(x \mid A), \quad \text{for almost all } x \in \mathbb{R}. \tag{6.4}$$

Whenever X admits conditional density $f_{X\,|\,A}(x)$ we can always set:

$$F_{X\,|\,A}(x) = \int_{-\infty}^{x} f_{X\,|\,A}(u)\mathrm{d}u.$$

Thus we also have

$$\mathsf{P}_{X\,|\,A}(B) = \int_B f_{X\,|\,A}(x)\mathrm{d}x,$$

as we did in the unconditional setting.

Example 6.1 Let $A = \{a < X \leqslant b\}$ in the conditional CDF $F_X(x \mid A)$, for reals $a < b$, s.t. its probability is positive. Now

$$\{X \leqslant x\} \cap \{a < X \leqslant b\} = \begin{cases} \varnothing, & \text{if } x \leqslant a \\ \{a < X \leqslant x\}, & \text{if } a < x \leqslant b \\ \{a < X \leqslant b\}, & \text{if } x > b. \end{cases}$$

Applying $\mathsf{P}(\,\cdot\,)$ to the above events and dividing by $\mathsf{P}(a < X \leqslant b)$ we get:

$$F_{X\,|\,a<X\leqslant b}(x) = \begin{cases} \frac{\mathsf{P}(\varnothing)}{\mathsf{P}(a<X\leqslant b)} = 0, & \text{if } x \leqslant a \\ \frac{\mathsf{P}(a<X\leqslant x)}{\mathsf{P}(a<X\leqslant b)} = \frac{F_X(x)-F_X(a)}{F_X(b)-F_X(a)}, & \text{if } a < x \leqslant b \\ \frac{\mathsf{P}(a<X\leqslant b)}{\mathsf{P}(a<X\leqslant b)} = 1, & \text{if } x > b. \end{cases}$$

Differentiating w.r.t. x we get the conditional density

$$f_{X\,|\,a<X\leqslant b}(x) = \frac{f_X(x)}{\int_a^b f_X(x)\mathrm{d}x}, \quad \text{for } a < x \leqslant b, \tag{6.5}$$

whose value is zero for $x \leqslant a$ and $x > b$.

In Example 6.1, the conditioning event $A = \{a < X \leqslant b\}$ is reported without brackets inside notation $F_{X\,|\,\bullet}(x)$ for the CCDF. From now on, we keep this notation for distributions conditioned to events involving rvs. A special case arises when $A = \{Y \in C\}$ has positive probability, where Y is another rv defined on the same probability space and $C \subset \mathbb{R}$ is suitable. The conditional probability

$$\mathsf{P}(X \in B \mid Y \in C) = \frac{\mathsf{P}(X \in B, Y \in C)}{\mathsf{P}(Y \in C)} \tag{6.6}$$

is clearly based on the joint distribution of the random vector $\mathbf{X} = (X, Y)$. For half-rays $(-\infty, \mathbf{x}] = B \times C$, where $B = (-\infty, x]$ and $C = (-\infty, y]$, we can rewrite the CCDF as

$$F_{X \mid Y=y}(x) := \begin{cases} \frac{F_{\mathbf{X}}(x,y)}{F_Y(y)}, & \text{for every } y \in \mathbb{R} \text{ s.t. } F_Y(y) \neq 0 \\ 0, & \text{otherwise,} \end{cases} \tag{6.7}$$

where $F_Y(y)$ is the marginal CDF of Y. A simpler notation is $F_{X \mid Y}(x)$. We can revert conditioning and define $F_{Y \mid X=x}(y)$ in a similar way. If the random vector has absolutely continuous joint CDF $F_{\mathbf{X}}$, with joint density $f_{\mathbf{X}}$, then besides the marginals densities $f_X(x)$, $f_Y(y)$, which we call **unconditional densities**, we can define two further univariate densities:

$$f_{X \mid Y=y}(x) := \frac{f_{\mathbf{X}}(x, y)}{f_Y(y)}, \quad \text{for every } y \in \mathbb{R} \text{ s.t. } f_Y(y) \neq 0, \tag{6.8}$$

$$f_{Y \mid X=x}(y) := \frac{f_{\mathbf{X}}(x, y)}{f_X(x)}, \quad \text{for every } x \in \mathbb{R} \text{ s.t. } f_X(x) \neq 0. \tag{6.9}$$

Notation can be simplified to $f_{X \mid Y}$ and $f_{Y \mid X}$. They are truly densities functions, see Exercise 6.2. We give an intuitive interpretation of $f_{X \mid Y}$:

- assume $\Delta x, \Delta y$ are small changes in x, y;
- compute $f_{\mathbf{X}}(x, y)\Delta x \Delta y \approx \mathsf{P}\big(x \leqslant X \leqslant x + \Delta x, y \leqslant Y \leqslant y + \Delta y\big)$;
- compute $f_Y(y)\Delta y \approx \mathsf{P}\big(y \leqslant Y \leqslant y + \Delta y\big)$ and assume it positive;
- get $f_{X \mid Y=y}(x)\Delta z \approx \dfrac{\mathsf{P}\big(x \leqslant X \leqslant x + \Delta x, y \leqslant Y \leqslant y + \Delta y\big)}{\mathsf{P}\big(y \leqslant Y \leqslant y + \Delta y\big)} \approx \mathsf{P}\big(x \leqslant X \leqslant x + \Delta x \mid Y \approx y\big)$.

Conditional densities cannot be interpreted literally, because $\mathsf{P}(Y = y) = 0$ for absolutely continuous rvs. For discrete components in $\mathbf{X}$, with supports $\mathcal{X}$, $\mathcal{Y}$, the corresponding conditional masses are:

$$\begin{aligned} p_{X \mid Y}(x) &:= \mathsf{P}(X = x \mid Y = y) \\ &= \frac{\mathsf{P}(X = x, Y = y)}{\mathsf{P}(Y = y)} \quad \text{for every } x \in \mathcal{X}, \text{ and } y \in \mathcal{Y} \text{ s.t. } \mathsf{P}(Y = y) > 0. \end{aligned} \tag{6.10}$$

Example 6.2 (Market Efficiency) Let S_{t+1} denotes one random stock price to be predicted by investors at time t. Assuming they are *rational*, to make the best possible forecast investors should use all available *information* in t. We can model this as another rv denoted Y_t, and further assume $\mathbf{X} = (S_{t+1}, Y_t)$ has a joint distribution. Now, consider a third rv M representing the whole information available in the market where this single stock is traded; the joint distribution of $\mathbf{X}$ is considered to be known. Hence, the market is said to be *informationally efficient* provided

$$\mathsf{P}(S_{t+1} \in B \mid Y_t \in C) = \mathsf{P}(S_{t+1} \in B \mid M \in C), \quad \text{for all suitable } B, C \subset \mathbb{R},$$

where $\mathsf{P}(Y_t \in C), \mathsf{P}(M \in C)$ are both positive. In terms of CCDF the above reads:

$$F_{S_{t+1} \mid Y_t = y}(x) = F_{S_{t+1} \mid M = y}(x),$$

for every $x \in \mathbb{R}$ and every $y \in \mathbb{R}$ s.t. $F_{Y_t}(y)$ and $F_M(y)$ are both positive.

To appreciate how conditional distributions are related to conditional probability we further need the following classical results.

Theorem 6.1 *Let $\Omega = \cup_{i \in I} A_i$ be decomposed through a finite, $I = \{1, \ldots, n\}$, or a countable, $I = \mathbb{N}$, partition of pairwise disjoint events $A_i \in \mathcal{F}$. Then for every $B \in \mathcal{F}$:*

$$\textbf{law of total probability} \quad \mathsf{P}(B) = \sum_{i \in I} \mathsf{P}(B \mid A_i)\, \mathsf{P}(A_i) \tag{6.11}$$

$$\textbf{Bayes' formula} \quad \mathsf{P}(A_i \mid B) = \frac{\mathsf{P}(B \mid A_i)\mathsf{P}(A_i)}{\sum_{i \in I} \mathsf{P}(B \mid A_i)\mathsf{P}(A_i)}, \quad \textit{for } \mathsf{P}(B) > 0. \tag{6.12}$$

Proof Let $B = B \cap \Omega = B \cap (\cup_i A_i) = \cup_i (B \cap A_i)$. Since $(A_i)_{i \in I}$ are pairwise disjoint, so are the events $(B \cap A_i)$. Then $\mathsf{P}(B) = \mathsf{P}(\cup_i (B \cap A_i)) = \sum_i \mathsf{P}(B \cap A_i) = \sum_i \mathsf{P}(B \mid A_i)\mathsf{P}(A_i)$. Next, by the law of total probability we can write $\mathsf{P}(B) = \sum_i \mathsf{P}(B \mid A_i)\mathsf{P}(A_i)$. This implies $\frac{\mathsf{P}(B \mid A_i)\mathsf{P}(A_i)}{\sum_i \mathsf{P}(B \mid A_i)\mathsf{P}(A_i)} = \frac{\mathsf{P}(B \cap A_i)}{\mathsf{P}(B)} = \mathsf{P}(A_i \mid B)$. ■

With $A_i = A$, Bayes' formula yields $\mathsf{P}(B)\mathsf{P}(A \mid B) = \mathsf{P}(B \mid A)\mathsf{P}(A)$.

6.2 Conditional Expectation Given an Event

Expectation as defined in Chap. 4 can be interpreted as average prediction of a rv's values without additional information, called *unconditional expectation*. Using conditional distribution we may define the expected value with partially resolution of uncertainty. Following the route developed in Sect. 6.1, we start by defining **conditional expectation** as:

$$\mathsf{E}(X \mid A) = \int_{-\infty}^{\infty} x \, \mathrm{d}F_{X \mid A}(x) = \int_{\mathbb{R}} x \, \mathrm{d}\mathsf{P}_{X \mid A}, \quad \text{whenever } \mathsf{P}(A) > 0. \tag{6.13}$$

Immediately, we get versions for discrete and continuous rvs, respectively:

$$\sum_i x_i \, p_{X \mid A}(x_i), \tag{6.14}$$

$$\int_{-\infty}^{\infty} x \, f_{X \mid A}(x) \mathrm{d}x. \tag{6.15}$$

Example 6.3 Let $X \sim \mathrm{N}(0, \sigma^2)$ a Gaussian rv and choose $A = \{X > 0\}$. Recall that X has density $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{x^2}{2\sigma^2}}$. Based on (6.5) from Example 6.1 in Sect. 6.1, with $a = 0$ and $b = \infty$, we compute:[1]

$$f_{X \mid X>0}(x) = \frac{f_X(x)}{\int_0^\infty f_X(x)\mathrm{d}x} = 2\frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{x^2}{2\sigma^2}}, \quad \text{for } 0 < x \leqslant \infty.$$

The conditional density is zero for $x \leqslant 0$. As byproduct,

$$\mathsf{E}(X \mid X > 0) = 2\frac{1}{\sqrt{2\pi}\sigma} \int_0^\infty x \, \mathrm{e}^{-\frac{x^2}{2\sigma^2}} \mathrm{d}x,$$

because the conditional expectation is based on values x for which the conditional density is nonzero. Inside the conditional expectation $\mathsf{E}(X \mid \bullet)$, the conditioning event is denoted again without brackets.

Example 6.4 Let $X \sim \mathrm{N}(\mu, \sigma^2)$ a Gaussian rv and choose $A = \{X \leqslant b\}$, so that from (6.5) with $a = -\infty$ we get the conditional density $f_{X \mid X \leqslant b}(x) = \frac{f_X(x)}{\int_{-\infty}^b f_X(x)\mathrm{d}x}$. Instead of working out the explicit expression of this density we compute the conditional expectation

$$\mathsf{E}(X \mid X \leqslant b) = \int_{-\infty}^b x \, f_{X \mid X \leqslant b}(x)\mathrm{d}x = \frac{\int_{-\infty}^b x \, f_X(x)\mathrm{d}x}{\int_{-\infty}^b f_X(x)\mathrm{d}x}. \tag{6.16}$$

Since it can be shown that $X = \mu + \sigma Z$, for a Standard Normal $Z \sim \mathrm{N}(0, 1)$ we rewrite:

$$\begin{aligned} \mathsf{E}(X \mid X \leqslant b) &= \mathsf{E}(\mu + \sigma Z \mid \mu + \sigma Z \leqslant b) \\ &= \mu + \sigma \mathsf{E}\left(Z \mid Z \leqslant \tfrac{b-\mu}{\sigma}\right). \end{aligned}$$

[1] For Gaussian density with location parameter zero, integrating over $[0, \infty)$ yields $\frac{1}{2}$ by symmetry.

Next we evaluate the latter conditional expectation with $k = \frac{b-\mu}{\sigma}$, using (6.16). From the unconditional density $f_Z(z) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{z^2}{2}}$ we see that

$$\frac{\mathrm{d}}{\mathrm{d}z} f_Z(z) = -z \, \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{z^2}{2}}.$$

Hence by computing the integral in the numerator of (6.16) for the Standard Normal we get:

$$\begin{aligned} \int_{-\infty}^{k} z \, f_Z(z) \mathrm{d}z &= -\int_{-\infty}^{k} -z \, f_Z(z) \mathrm{d}z \\ &= -\left[\frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{z^2}{2}} \right]_{-\infty}^{k} = -f_Z(k) + \lim_{z \to -\infty} f_Z(z) = -f_Z(k). \end{aligned}$$

Finally we obtain $\mathsf{E}(Z \mid Z \leqslant k) = \frac{\int_{-\infty}^{k} z \, f_Z(z) \mathrm{d}z}{\int_{-\infty}^{c} f_Z(z) \mathrm{d}z} = \frac{-f_Z(k)}{F_Z(k)}$ and thus

$$\mathsf{E}(X \mid X \leqslant b) = \mu - \sigma \, \frac{f_Z(k)}{F_Z(k)}. \tag{6.17}$$

Example 6.5 Let us identify X, from Example 6.4, with the arithmetic rate of return r_{t+1} of a portfolio. Then, let $b = Q_{r_{t+1}}(c)$ be the $F_{r_{t+1}}$'s quantile, with $0 < c < 1$. Observe that for c small enough (e.g. $c = 0.01$) the rv r_{t+1} attains negative values and the choice for b can be interpreted as measuring percentage losses from investing in the portfolio, with probability at most c. In risk management, an average measure for market risk associated to a negative value of returns (viz. percentage losses) can be defined as follows:

$$\begin{aligned} -\mathsf{E}(r_{t+1} \mid r_{t+1} \leqslant Q_{r_{t+1}}(c)) &= -\mu + \sigma \, \frac{f_Z(Q_{r_{t+1}}(c))}{F_Z(Q_{r_{t+1}}(c))} \\ &= -\mu + \sigma \, \frac{f_Z(Q_{r_{t+1}}(c))}{c}. \end{aligned} \tag{6.18}$$

Warning: we have to let $k = \frac{Q_{r_{t+1}}(c) - \mu}{\sigma}$ and recall that quantiles are linear for positive linear transformation, $Q_{\mu+\sigma Z}(c) = \mu + \sigma \, Q_Z(c)$. Thus, $k = Q_Z(c)$ and the latter expression on the right-hand side of (6.18) is due to

$$\int_{-\infty}^{Q_Z(c)} f_Z(z) \mathrm{d}z = F_Z(Q_{r_{t+1}}(c)) = c,$$

for absolutely continuous distributions.

Example 6.4 suggests that $\mathsf{E}(X \mid X \leqslant b)$ can be generalized to any event $A \in \mathcal{F}$ than $\{X \leqslant b\}$. Indeed it holds:

$$\mathsf{E}(X \mid A) = \frac{\mathsf{E}(X\mathbf{I}_A)}{\mathsf{P}(A)}, \quad \text{if } \mathsf{P}(A) > 0. \tag{6.19}$$

This formula can be proved directly, when X is a discrete rv with conditional masses given by (6.3) and with countable support. The expected value of X conditional on the event A having positive probability is:

$$\begin{aligned}\mathsf{E}(X \mid A) &= \sum_{i=1}^{\infty} x_i \; p(x_i \mid A) \\ &= \sum_{i=1}^{\infty} x_i \; \frac{\mathsf{P}(\{X = x_i\} \cap A)}{\mathsf{P}(A)} \\ &= \frac{1}{\mathsf{P}(A)} \sum_{i=1}^{\infty} x_i \; \mathsf{P}(\{X = x_i\} \cap A) = \frac{\mathsf{E}(X\mathbf{I}_A)}{\mathsf{P}(A)}.\end{aligned}$$

The last line follows because the corresponding sum is the expectation for every scenario $\omega \in A$, i.e., it is the average of values $x_i = X(\omega)$ for scenario ω s.t. probability is $\mathsf{P}(\{X = x_i\} \cap A)$. For general rvs, see Exercise 6.5. It is worth noting that formula (6.19) cab also be written

$$\frac{1}{\mathsf{P}(A)} \int_{\Omega} X\mathbf{I}_A \mathrm{d}\mathsf{P} = \frac{1}{\mathsf{P}(A)} \int_{A} X \mathrm{d}\mathsf{P},$$

using the definition of Lebesgue integral involving expectation, see Appendix N, which emphasizes the averaging procedure restricted to a subset of $X(\omega)$'s values given by scenarios $\omega \in A$, with additional normalization by a factor $\frac{1}{\mathsf{P}(A)}$.

6.3 Conditional Expectation Given a Rv

As we did with conditional distributions, it is sensible to extend the expectation conditioned on an event to conditioning on a rv. We specialize to discrete rvs. Take a rv $X \in L^1$ and another discrete rv Y with support $\mathcal{Y} = \{y_1, y_2, \ldots\}$. Trying to forecast X conditioned on Y taking values on its support, amounts to find which event $\{Y = y_j\}$ will occur. We should use formula (6.19) in Sect. 6.2 and, provided $\mathsf{P}(Y = y_i) > 0$ for all $y_i \in \mathcal{Y}$, the conditional expectation of X given Y is a *second*

discrete rv $\boxed{Z = \mathsf{E}(X \mid Y)}$ s.t.

$$Z(\omega) := \frac{\mathsf{E}(X\mathbf{I}_{\{Y=y_i\}})}{\mathsf{P}(Y = y_i)}, \quad \text{provided } Y(\omega) = y_i \text{ for any } i = 1, 2, \ldots \tag{6.20}$$

A shorthand notation for Z is $\mathsf{E}(X \mid Y = y_i)$, whenever in formula (6.20) it is evaluated at those scenarios $\omega \in \Omega$ s.t. $Y(\omega)$ equals $y_i \in \mathcal{Y}$. In fact, $f(Y) = \mathsf{E}(X \mid Y)$ is a function[2] representing a 'version' of X obtained by averaging the values $f(y_j)$ with weights $\mathsf{P}_{X \mid Y=y_j}$, and which takes on constant values $\frac{\mathsf{E}(X)}{\mathsf{P}(Y=y_j)}$ over events $\{Y = y_j\}$. Lemma 6.1 below is a crucial ingredient in the current setting.

Lemma 6.1 (Doob-Dynkin) *Let Y be a rv and consider $\sigma(Y)$, its generated σ-algebra. Then, each rv Z which induces event $\{Z \in B\} \in \sigma(Y)$ is given as $Z = f(Y)$, for a suitable function $f : \mathbb{R} \to \mathbb{R}$ and every suitable[3] $B \subset \mathbb{R}$.*

It is plausible to define expectation conditioned on sub-σ-algebras for $X \in L^1$, extending definition (6.20) by declaring $\mathsf{E}(X \mid Y) \equiv \mathsf{E}(X \mid \sigma(Y))$, i.e., $\mathsf{E}(X \mid Y)$ depends on a σ-algebra rather than on Y's values.[4] In fact, by Exercises 6.11 and 6.12 we are able to define $Z = \mathsf{E}(X \mid Y)$ also when Y is no longer discrete[5] in such a way:

- the conditional expectation is a $\sigma(Y)$-measurable rv, in words events $\{Z \in B\}$ are in here,[6] and $Z = f(Y)$ for a suitable function;
- for every event $A \in \sigma(Y)$ the unconditional expectation of $Z\mathbf{I}_A$ coincides with that of $X\mathbf{I}_A$.

It can be shown that for $X, Y \in L^1$ in $\mathbf{X} = (X, Y)$ with joint density, the conditional expectation can be computed as

$$\mathsf{E}(X \mid Y) = \frac{\int_{-\infty}^{\infty} x f_{\mathbf{X}}(x, Y)\mathrm{d}x}{\int_{-\infty}^{\infty} f_{\mathbf{X}}(x, Y)\mathrm{d}x}, \tag{6.21}$$

see Exercise 6.15.

[2] Actually, a Borel measurable function, i.e., the composed map $\Omega \xrightarrow{Y} \mathbb{R} \xrightarrow{f} \mathbb{R}$ is a rv, see Appendix in Chap. 2.

[3] Again, we should say Z is $\sigma(Y)$-measurable and f is a Borel measurable function.

[4] Observe that $\sigma(Y) \subset \mathcal{F}$ and Z does not carries over more information than $\mathcal{F}$.

[5] Do not forget the condition $X \in L^1$, which can be rephrased as X being *integrable*.

[6] It can be shown $Z = \mathsf{E}(X \mid Y)$ induces events in $\sigma(Y)$ iff $\sigma(Z) \subset \sigma(Y)$.

6.4 Conditional Expectation Given a Sub-σ-algebra

The framework developed in Sect. 6.3 enable us to introduce a far general definition of conditional expectation.

Theorem 6.2 *Let X be a rv defined on a probability space $(\Omega, \mathcal{F}, \mathsf{P})$, assume $X \in L^1$ and let $\mathcal{A} \subset \mathcal{F}$ be a sub-σ-algebra. Then, there exists the conditional expectation of X given $\mathcal{A}$ which is the rv* $\boxed{Z = \mathsf{E}(X \mid \mathcal{A})}$ *satisfying:*

1. *events induced by Z are in $\mathcal{A}$, i.e., Z is $\mathcal{A}$-measurable;*
2. *Z has finite expectation, i.e. $Z \in L^1$;*
3. *for any $A \in \mathcal{A}$ it holds $\mathsf{E}(Z\mathbf{I}_A) = \mathsf{E}(X\mathbf{I}_A)$.*

If conditions 1–3 are satisfied by another rv Z', then $Z \overset{a.s.}{=} Z'$.

In fact, if Z and Z' are two rvs inducing events in $\mathcal{A}$ s.t.

$$\mathsf{E}(Z\mathbf{I}_A) = \mathsf{E}(X\mathbf{I}_A) = \mathsf{E}(Z'\mathbf{I}_A), \quad \text{for all } A \in \mathcal{A} \subset \mathcal{F},$$

then $\mathsf{E}((Z - Z')\mathbf{I}_A) = 0$ and $Z \overset{\text{a.s.}}{=} Z'$, see Exercise 6.13. Existence asserted in Theorem 6.2 can be proved by invoke another classical result. Before to state it, let us consider the problem of changing the probability measure P in the event space $(\Omega, \mathcal{F})$, using another probability measure. To some extent, this can be interpreted as different subjective opinions about events in $\mathcal{F}$.

Definition 6.1 A probability measure $\tilde{\mathsf{P}}$ is said to be absolutely continuous w.r.t. probability P, written $\tilde{\mathsf{P}} \ll \mathsf{P}$, if $\mathsf{P}(A) = 0$ implies $\tilde{\mathsf{P}}(A) = 0$ for any event $A \in \mathcal{F}$. They are equivalent probability measures, written $\tilde{\mathsf{P}} \sim \mathsf{P}$, whenever $\tilde{\mathsf{P}} \ll \mathsf{P}$ and $\mathsf{P} \ll \tilde{\mathsf{P}}$.

The two probabilities agree on impossible events but put different weights on the remaining ones. Observe that also $\mathsf{P}(A^c) = 1$ implies $\tilde{\mathsf{P}}(A^c) = 1$. Hence, impossible and certain events under P are evaluated the same under $\tilde{\mathsf{P}}$, but the converse is not true. Now unconditional expectation can be computed with two different probabilities, and to distinguish between them we can write E_{P} and $\mathsf{E}_{\tilde{\mathsf{P}}}$. Surprisingly enough, there is a simple way of getting $\tilde{\mathsf{P}}$ as absolutely continuous w.r.t. P. Take $X \geqslant 0$ be a nonnegative rv over $(\Omega, \mathcal{F}, \mathsf{P})$ and define the set function $\tilde{\mathsf{P}} : \mathcal{F} \to [0, \infty]$ by

$$\tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A), \quad \text{with } \mathsf{E}_{\mathsf{P}}(X) = 1. \tag{6.22}$$

As a result, $\tilde{\mathsf{P}} \ll \mathsf{P}$, see Exercise 6.16.

Example 6.6 Assuming no arbitrage, the price in $t = 0$ of an European call option with maturity in $t = T$ is

$$C_0 = \mathrm{e}^{-r\,T}\mathsf{E}_{\tilde{\mathsf{P}}}((S_T - K)^+),$$

where $K > 0$ is the strike price in $t = T$ and S_T is the underlying stock price; r is the continuously compounded risk-free rate. Under additional hypotheses, the financial *theory of no arbitrage* ensures the existence of a new probability measure $\tilde{\mathsf{P}}$, called **risk-neutral**, which is equivalent to the *market* probability measure P, in such a way pricing European call options reduces to compute the discounted value of the option random payoff $\mathrm{e}^{-rT}(S_T - K)^+$ using the risk-free return instead of the true random return of the option itself.

Example 6.7 To appreciate how $\tilde{\mathsf{P}} \ll \mathsf{P}$ are related each other through formula (6.22) we rewrite it using Lebesgue integration:

$$\tilde{\mathsf{P}}(A) = \underbrace{\int_\Omega X\mathbf{I}_A \mathrm{d}\mathsf{P}}_{=\mathsf{E}_\mathsf{P}(X\mathbf{I}_A)} = \boxed{\int_A X \mathrm{d}\mathsf{P}} = \underbrace{\int_\Omega \mathbf{I}_A \mathrm{d}\tilde{\mathsf{P}}}_{=\mathsf{E}_{\tilde{\mathsf{P}}}(\mathbf{I}_A)} = \boxed{\int_A \mathrm{d}\tilde{\mathsf{P}}}.$$

Comparing the second and the fourth integrals yields

$$X\mathrm{d}\mathsf{P} = \mathrm{d}\tilde{\mathsf{P}} \iff X = \frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}},$$

where X is called the **Radon-Nikodým derivative** or density of $\tilde{\mathsf{P}}$ w.r.t. P. It may be interpreted as the rv given by $\lim_{A\downarrow\{\omega\}} \frac{\tilde{\mathsf{P}}(A)}{\mathsf{P}(A)}$, for finer partitions of the event $A \in \mathcal{F}$.

Remark 6.1 For two equivalent probability measures $\tilde{\mathsf{P}} \sim \mathsf{P}$, we can recover the Radon-Nikodým derivative of P w.r.t. $\tilde{\mathsf{P}}$ simply by inverting, i.e. $\frac{\mathrm{d}\mathsf{P}}{\mathrm{d}\tilde{\mathsf{P}}} = \left(\frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}}\right)^{-1}$.

What happen if we start from two probability measures such that $\tilde{\mathsf{P}} \ll \mathsf{P}$ and wish to find a nonnegative rv acting as the Radon-Nikodým derivative of $\tilde{\mathsf{P}}$ w.r.t. P? The answer to this question is the converse of formula (6.22) which we present without proof.

Theorem 6.3 (Radon-Nikodým) *Let $\tilde{\mathsf{P}}$ be a probability measure absolutely continuous with respect to P over the event space $(\Omega, \mathcal{F})$. Then, there exists a rv $X \geqslant 0$ such that $\mathsf{E}_\mathsf{P}(X) = 1$ and $\tilde{\mathsf{P}}(A) = \mathsf{E}_\mathsf{P}(X\mathbf{I}_A)$, for every $A \in \mathcal{F}$. The same holds when the hypotheses are replaced by $\tilde{\mathsf{P}} \sim \mathsf{P}$ and $X \geqslant 0$ a.s.*

To verify the existence of $Z = \mathsf{E}(X \,|\, \mathcal{A})$ we can now use Theorem 6.3, see Exercise 6.17. Next, we immediately see that, similar to the special case of $\mathcal{A} = \sigma(Y)$, there is no additional information rather than $\mathcal{A}$, i.e., $\sigma(Z) \subset \mathcal{A}$. Observe that Theorem 6.2, item 3, implies $\mathsf{E}(\mathsf{E}(X \,|\, \mathcal{A})) = \mathsf{E}(X)$ provided $A = \Omega$. Moreover, we can condition on 'extreme' sub-σ-algebras contained in the information set $\mathcal{A}$, namely $\mathsf{E}(X \,|\, \{\varnothing, \Omega\}) = \mathsf{E}(X)$ and $\mathsf{E}(X \,|\, \mathcal{F}) = X$, see Exercise 6.21. The latter highlights that the greater $\mathcal{A}$ the larger is the set of rvs inducing events in $\mathcal{A}$ and representing $\mathsf{E}(X \,|\, \mathcal{A})$.

Remark 6.2 Conditional expectation $\mathsf{E}(X \mid Y_1, \ldots, Y_n)$ can be defined as $\mathsf{E}(X \mid \sigma(\mathbf{Y}))$, for a random vector $\mathbf{Y} = (Y_1, \ldots, Y_n)$ with discrete components Y_i. Indeed, $\sigma(\mathbf{Y}) \equiv \sigma(Y_1, \ldots, Y_n)$ is the smallest σ-algebra containing events $\cup_{j \in I}\{\mathbf{Y} = \mathbf{c}_j\}$ for some $I \subset \mathbb{N}$. Note that

$$\{\mathbf{Y} = \mathbf{c}_j\} = \cap_{i=1}^{n}\{Y_i = c_{ij}\}, \quad \mathbf{c}_j = (c_{1j}, \ldots, c_{nj}).$$

If the random vector is no longer discrete then $\sigma(Y_1, \ldots, Y_n)$ is the smallest σ-algebra containing events $\{Y_i \leqslant y_i\}$ for all $i = 1, \ldots, n$, i.e., $\cap_i\{Y_i \leqslant y_i\} = \{\mathbf{Y} \leqslant \mathbf{y}\} \in \sigma(\mathbf{Y})$ for some $\mathbf{y} \in \mathbb{R}^n$.

The general conditional expectation fulfills some interesting conditions.

Proposition 6.1 *For rvs $X, Y \in L^1$ and sub-σ-algebras $\mathcal{H} \subset \mathcal{A} \subset \mathcal{F}$ the following properties are valid.*

1. *Linearity.* $\mathsf{E}(\alpha X + \beta Y \mid \mathcal{A}) \overset{a.s.}{=} \alpha\mathsf{E}(X \mid \mathcal{A}) + \beta\mathsf{E}(Y \mid \mathcal{A})$, *for* $\alpha, \beta \in \mathbb{R}$.
2. *Positivity. If $X \geqslant 0$, then $\mathsf{E}(X \mid \mathcal{A}) \geqslant 0$ with probability 1.*
3. *Law of iterated expectations or tower property.* $\mathsf{E}\big(\mathsf{E}(X \mid \mathcal{A}) \mid \mathcal{H}\big) \overset{a.s.}{=} \mathsf{E}\big(\mathsf{E}(X \mid \mathcal{H}) \mid \mathcal{A}\big) \overset{a.s.}{=} \mathsf{E}(X \mid \mathcal{H})$
4. *Taking out what is known. If Y induces events in $\mathcal{A}$ and $XY \in L^1$ then* $\mathsf{E}(XY \mid \mathcal{A}) \overset{a.s.}{=} Y\,\mathsf{E}(X \mid \mathcal{A})$.
5. *If X and $\mathbf{I}_A$ are independent for any $A \in \mathcal{A}$, then* $\mathsf{E}(X \mid \mathcal{A}) = \mathsf{E}(X)$.

Remark 6.3 Sub-σ-algebra $\mathcal{H}$ in item 3 is said to be **coarser** than $\mathcal{A}$, which is said to be **finer** than $\mathcal{H}$. Thus, the 'coarser rules out the finer': conditioning X on the information set $\mathcal{H}$ is the same as conditioning first on $\mathcal{A}$. On the other hand, $Z = \mathsf{E}(X \mid \mathcal{H})$ induces events in $\mathcal{A}$ and the information in $\mathcal{A}$ is enough to determine $\mathsf{E}(Z \mid \mathcal{A})$, regardless a second *forecast* based on the finer information set. Observe that

$$\mathsf{E}(Z \mid \mathcal{A}) \overset{\text{a.s.}}{=} Z, \qquad \text{for any rv } Z \text{ inducing events in } \mathcal{A}. \tag{6.23}$$

This follows from Property 4. of Proposition 6.1 if we let $Y = Z$ and $X = 1$. Property 4. says that for a rv Y inducing events in $\mathcal{A}$, the expectation of XY conditional on the information $\mathcal{A}$ treats Y as a *constant*. Property 5. is equivalently stated by saying that the rv X and the sub-σ-algebra $\mathcal{A}$ are *independent*, in the sense that $\sigma(X)$ and $\mathcal{A}$ are independent.

We clarify the precise meaning of the statement about independence in the previous remark by stating equivalently that $\sigma(X)$ and $\mathcal{A}$ are independent. Let $\sigma(\mathbf{I}_{A_i})$ be the σ-algebra generated by the indicator rv $\mathbf{I}_{A_i}$, for some event A_i. Indicators $\mathbf{I}_{A_1}, \mathbf{I}_{A_2}, \ldots$ are independent iff their generated σ-algebras $\sigma(\mathbf{I}_{A_1}), \sigma(\mathbf{I}_{A_2}), \ldots$ are

independent. To see why, first by independence we have:

$$\begin{aligned}\mathsf{P}(A_i \cap A_j) &= \mathsf{E}(\mathbf{I}_{A_i \cap A_j}) = \mathsf{E}(\mathbf{I}_{A_i})\mathsf{E}(\mathbf{I}_{A_j}) \\ &= \mathsf{P}(A_i)\mathsf{P}(A_j) \quad \text{for all } i \neq j.\end{aligned}$$

Second, A_i and A_j are independent. Now, $\sigma(\mathbf{I}_{A_i}) = \sigma(A_i)$ and recall from Example 1.14, Sect. 1.6, that also $\sigma(A_i) = \{\varnothing, A_i, A_i^c, \Omega\}$, for $i = 1, 2, \ldots$. We conclude that the families $\sigma(A_i)$ and $\sigma(A_j)$ contain mutually independent events, see Sect. 3.4. We can write $\sigma(A_i) \perp\!\!\!\perp \sigma(A_j)$, for all possible choice of indices i, j. Henceforth, independence between two rvs X, Y can be stated in one of the following equivalent forms:

- $\{X \in B\}$ and $\{Y \in C\}$ are independent events, for any pair of suitable sets $B, C \subset \mathbb{R}$.
- $\sigma(X)$ and $\sigma(Y)$ are independent σ-algebras.
- $\mathbf{I}_{\{X \in B\}}$ and $\mathbf{I}_{\{Y \in C\}}$ are independent indicators.

We may pass from conditional probability to a notion of *stochastic* probability. In fact, let $X = \mathbf{I}_A$ and define $Z = \mathsf{E}(\mathbf{I}_A \,|\, \mathcal{A})$, with respect to the sub-σ-algebra $\mathcal{A} \subset \mathcal{F}$, as $Z : \Omega \to [0, 1]$. This is a rv called **regular conditional probability**, denoted $\mathsf{P}(A \,|\, \mathcal{A})$, which assigns probabilities to events $A \in \mathcal{A}$ under the *hypotheses* that other events in $\mathcal{A}$ occurred, i.e., what matters is the information set $\mathcal{A}$. In particular, for every $A, B \in \mathcal{A}$ we have $\mathsf{P}(A \cap B) = \mathsf{E}(\mathbf{I}_A \mathbf{I}_B) = \mathsf{E}\,(\mathsf{P}(A \,|\, \mathcal{A})\mathbf{I}_B)$, which is $\mathsf{P}(A \,|\, B)\mathsf{P}(B)$. If $B = \Omega$ then $\mathsf{E}(\mathsf{P}(A \,|\, \mathcal{A}) \,|\, \mathcal{A}) = \mathsf{E}(\mathbf{I}_A) = \mathsf{P}(A)$. For a more detailed treatment of regular conditional probability see for example [2] and [16] and Appendix. Eventually, we have a further formula based on Theorem 6.2. Let X be a rv independent from the sub-σ-algebra $\mathcal{A} \subset \mathcal{F}$, and let Y be another rv inducing events in $\mathcal{A}$. Given the random vector $\mathbf{X} = (X, Y)$, define a function $h : \mathbb{R}^2 \to \mathbb{R}$ s.t. the rv $Z = h(X, Y)$ has finite expectation. Then define another function $\psi(x) = \mathsf{E}(h(x, Y))$ given by the expectation of the rv $h(x, Y)$, which is obtained from Z by keeping X as a constant. Hence:

$$\mathsf{E}\big(h(X, Y) \,|\, \mathcal{A}\big) = \psi(X). \tag{6.24}$$

Under the assumptions above, the conditional expectation of a function of two rvs becomes *unconditional*, thus we get $\psi(X) = \mathsf{E}(h(X, Y))$.

Example 6.8 Considering the same setting of Example 6.6, under no arbitrage, the price in $t > 0$ of an European call option with maturity in $t = T$ is the rv

$$C_t = \mathrm{e}^{-r\,(T-t)}\mathsf{E}_{\tilde{\mathsf{P}}}((S_T - K)^+ \,|\, \mathcal{F}_t),$$

where now we add the effect of conditioning based on information $\mathcal{F}_t \subset \mathcal{F}$, to be interpreted as the sub-σ-algebra containing all the relevant information about the price of any contingent claim from the perspective of risk-neutral investors.

6.5 Exercises

6.1 Verify that $F_X(x \mid A)$ is a CDF.

6.2 Verify that $f_{X \mid Y}(x)$ is a density.

6.3 Let $X = \mathbf{I}_B$ for some event $B \in \mathcal{F}$. Compute $\mathsf{E}(X \mid A)$ with $\mathsf{P}(A) > 0$.

6.4 Let $X \in L^1$ be a discrete rv with support $\mathcal{X} = \{x_1, x_2, \ldots\}$ and masses $\mathsf{P}(X = x_i) > 0$. Write a formula to relate the unconditional expectation of X to its conditional expectation $\mathsf{E}(X \mid A)$ with $\mathsf{P}(A) > 0$.

6.5 Prove formula (6.19) for general rvs.

6.6 Let $X \in L^1$ be a nonnegative rv, and $u > 0$ s.t. $A_u := \{X \geqslant u\}$. Show that

$$\mathsf{P}(A_u) \leqslant \frac{1}{u}\mathsf{E}(X\mathbf{I}_{A_u}) \leqslant \frac{1}{u}\mathsf{E}(X).$$

6.7 If $Y = c$ for every scenario, show that $\mathsf{E}(X \mid Y)$ is also constant and equals $\mathsf{E}(X) = c$. Show that for $X = \mathbf{I}_A$ and $Y = \mathbf{I}_B$ we have $\mathsf{E}(\mathbf{I}_A \mid \mathbf{I}_B) = \mathsf{P}(A \mid B)$ for $\omega \in B$.

6.8 Evaluate $\mathsf{E}(X \mid Y)$ at $Y = y_j$, when both X and Y are discrete rvs with countably many values. Justify a similar expression when conditioning on a discrete random vector.

6.9 For discrete rvs X, Y, prove $\mathsf{E}(\mathsf{E}(X \mid Y)) = \mathsf{E}(X)$.

6.10 For absolutely continuous rvs X, Y, prove $\mathsf{E}(\mathsf{E}(X \mid Y)) = \mathsf{E}(X)$.

6.11 (Role of independence) Show that $\mathsf{E}(X \mid Y) = \mathsf{E}(X)$, provided that X and Y are independent, by examining three cases:

1. X and Y are both discrete;
2. X and Y are both continuous;
3. X is any rv and only Y is discrete.

6.12 Assume $X \in L^1$ and Y is a discrete rv. Show that:

1. $\mathsf{E}(X \mid Y)$ induces events in $\sigma(Y)$, i.e., it is $\sigma(Y)$-measurable (see also Appendix in Chap. 2);
2. for every $A \in \sigma(Y)$ it holds $\mathsf{E}(X\mathbf{I}_A) = \mathsf{E}(\mathsf{E}(X \mid Y)\mathbf{I}_A)$.

6.13 Let $\mathcal{A} \subset \mathcal{F}$ be a sub-σ-algebra. Show that if X induces events in $\mathcal{A}$, i.e., it is $\mathcal{A}$-measurable, and $\mathsf{E}(X\mathbf{I}_A) = 0$, for all $A \in \mathcal{A}$, then $X \overset{\text{a.s.}}{=} 0$. Recall what the last sentence means: $\mathsf{P}(X = 0) = 1$.

6.14 Prove that the existence of $\mathsf{E}(X \mid Y)$ is verified up to P-null sets

6.15 Show that $\mathsf{E}(X \mid Y) = \frac{\int_{-\infty}^{\infty} x f_X(x,Y)\mathrm{d}x}{\int_{-\infty}^{\infty} f_X(x,Y)\mathrm{d}x}$, for rvs $X, Y \in L^1$ with joint density.

6.16 (Expectation vs probability) Show that formula (6.22) gives a probability measure $\hat{\mathsf{P}}$ absolutely continuous w.r.t. P.

6.17 Prove existence of general conditional expectation in Theorem 6.2.

6.18 Let $X, Y \geqslant 0$ be two rvs over a probability space $(\Omega, \mathcal{F}, \mathsf{P})$, where $\mathsf{E}_{\mathsf{P}}(Y) = 1$, i.e. Y is a Radon-Nikodým derivative $\frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}}$. If $\tilde{\mathsf{P}}(A) = \mathsf{E}(Y\mathbf{I}_A)$, for any $A \in \mathcal{F}$, show that:[7]

$$\mathsf{E}_{\tilde{\mathsf{P}}}(X) = \mathsf{E}_{\mathsf{P}}(XY). \tag{6.25}$$

6.19 Let $X \geqslant 0$ be a rv and take $Y > 0$ be another rv with $\mathsf{E}_{\mathsf{P}}(Y) = 1$, both defined over the same probability space. Prove that

$$\mathsf{E}_{\mathsf{P}}(X) = \mathsf{E}_{\tilde{\mathsf{P}}}(\tfrac{X}{Y}). \tag{6.26}$$

6.20 Let $X \geqslant 0$ a rv with $\mathsf{E}_{\mathsf{P}}(X) = 1$. If $\mathsf{P}(A) = 0$ for some event $A \in \mathcal{F}$, show that $\tilde{\mathsf{P}}(A) = 0$ and then $\tilde{\mathsf{P}} \ll \mathsf{P}$.

6.21 Show that $\mathsf{E}(X \mid \mathcal{A}) = \mathsf{E}(X)$, if $\mathcal{A} = \{\varnothing, \Omega\}$, while it equals X for $\mathcal{A} = \mathcal{F}$.

6.22 Prove item 3 of Theorem 6.2, i.e. $Z \in L^1$ for $Z = \mathsf{E}(X \mid \mathscr{A})$.

6.23 Prove tower property, i.e., item 3 of Proposition 6.1.

Appendix

Existence of Conditional Distribution and Joint Law

Any conditional probability like that in (6.6) cannot be defined in most cases where there are events with zero probability, namely $\mathsf{P}(Y \in C) = 0$. The latter is especially true if $\{Y \in C\} = \{Y = y\}$ and the rv Y has absolutely continuous CDF F_Y. To guarantee the existence of conditional probability even for P-null event, the following theorem is of some importance.

Theorem 6.4 *Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space and $\mathbf{X} = (X, Y)$ be a random vector with joint law $\mathsf{P}_{\mathbf{X}}(\,\cdot\,)$ defined over those subsets $D \subset \mathbb{R}^2$ s.t. $\{\mathbf{X} \in D\} \in \mathcal{F}$.*

1. *There exists a conditional distribution $\mathsf{P}_{X\,|\,Y}(B \mid y)$ which equals*[8] $\mathsf{P}(X \in B \mid Y = y)$*, for any $B \subset \mathbb{R}$ s.t. $\{X \in B\} \in \mathcal{F}$;*
2. *$\mathsf{P}_{X\,|\,Y}(B \mid \cdot)$ is a univariate probability measure on $\mathbb{R}$ for every $y \in \mathbb{R}$;*
3. *Provided $\mathsf{E}(g(X, Y)) < \infty$ it holds*[9] $\mathsf{E}(g(X, Y) \mid Y = y) = \int_{\mathbb{R}} g(x, y)\mathrm{d}\mathsf{P}_{X\,|\,Y}$*, for a suitable function $g(x, y)$;*
4. *$\mathsf{P}_{\mathbf{X}}(D) = \mathsf{P}(X \in B, Y \in C) = \int_C \mathsf{P}_{X\,|\,Y}(B \mid y)\mathrm{d}\mathsf{P}_Y$, where $D = B \times C$.*

[7] This can be extended to general rvs.

[8] This relation holds P_Y-a.s. in $y \in \mathbb{R}$.

[9] Again, this holds P_Y-a.s. in $y \in \mathbb{R}$.

The conditional distribution whose existence is postulated by Theorem 6.4 is the regular conditional probability introduced in Sect. 6.4. Here we give a further heuristic justification:

$$\mathsf{P}_{X\,|\,Y}(B\,|\,y) = \lim_{h\downarrow 0} \frac{\mathsf{P}(X \in B,\, Y \in (y-h,\, y+h))}{\mathsf{P}(Y \in (y-h,\, y+h))}, \quad \text{for } h > 0. \tag{6.27}$$

This can be understood as the *derivative* of one measure, $\mathsf{Q}(C) := \mathsf{P}(X \in B, Y \in C)$, w.r.t. another, $\mathsf{P}_Y(C) := \mathsf{P}(Y \in C)$. Indeed both are probability measures on $\mathbb{R}$ and moreover $0 \leqslant \mathsf{Q}(C) \leqslant \mathsf{P}_Y(C)$, i.e. $\mathsf{Q} \ll \mathsf{P}_Y$ and we can invoke the Radon-Nikodỳm theorem recognizing the limit above as $\frac{\mathrm{d}\mathsf{Q}}{\mathrm{d}\mathsf{P}_Y} = f(y)$ and writing

$$\int_C f(y)\mathrm{d}\mathsf{P}_Y.$$

Compare $f(y)$ with the integrand in item (d) of Theorem 6.4.[10] The limit in the previous discussion cannot be conceived in the usual sense since it may be $\mathsf{P}(Y = y) = 0$. The measure-theoretic device (Radon-Nikodỳm derivative) enable us to avoid imposing restrictive conditions on P and (X, Y).

Remark 6.4 Assume $\mathbf{X} = (X, Y)$ has a joint density $f_{\mathbf{X}}(x, y)$. Further, assume that there exists $h > 0$ s.t. $f_{\mathbf{X}}(x, y)$ and the marginal $f_Y(y)$ are both continuous in the second variable over $[y-h, y+h]$, and also $f_Y(y) > 0$. Recast the limit in (6.27) as

$$\lim_{h\downarrow 0} \frac{\int_B \int_{y-h}^{y+h} f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y}{\int_{b-h}^{b+h} f_Y(y)\mathrm{d}y}.$$

Now, we apply the mean value theorem[11] to $\int_{y-h}^{y+h} f_{\mathbf{X}}(x, y)\mathrm{d}y$ and $\int_{y-h}^{y+h} f_Y(y)\mathrm{d}y$, both as functions of y. This yields that the limit above can be rewritten as

$$\lim_{h\downarrow 0} \frac{\int_B 2h\, f_{\mathbf{X}}(x, y_1)\mathrm{d}x}{2h\, f_Y(y_2)},$$

where $y_1, y_2 \in [y-h, y+h]$ and in the limit they coincide with y. So, the $2h$ in the numerator and denominator cancel each other and we end with the desired conditional probability $\mathsf{P}_{X\,|\,Y}(B\,|\,y)$.

[10] Be warned: we can find another Radon-Nikodỳm derivative s.t. $\tilde{f}(y) \neq f(y)$ for those $y \in \mathbb{R}$ forming a P_Y-null set.

[11] Given a continuous $g(x)$ over $[a, b]$, then there exists $c \in [a, b]$ s.t. $\int_a^b g(x)\mathrm{d}x = g(c)(b-a)$.

Chapter 7
Regression, Prediction and More Dependence



7.1 Prediction

How is a random experiment affected by collateral *random disturbances*? Let $X_1, \ldots, X_n$ be rvs over $(\Omega, \mathcal{F}, \mathsf{P})$ and assume Y is the rv of interest. Our aim is to tackle the problem of predicting $Y(\omega)$ given $X_i(\omega)$ for every $i = 1, \ldots, n$ and for all scenarios ω in the sample space. We will see how this problem can lead to find some *functional* dependence among Y and $X_1, \ldots, X_n$.

Definition 7.1 Let $\mathbf{X} = (X_1, \ldots, X_n)$ be *explanatory* rvs. Then a **predictor** for Y based on $\mathbf{X}$ is a suitable function $f(\mathbf{X})$. The **prediction error** is the rv $\varepsilon = Y - f(\mathbf{X})$.

To compare different predictors we use the mean-square prediction error:

$$\mathsf{E}\,(Y - f(\mathbf{X}))^2\,, \quad Y, f(\mathbf{X}) \in L^2.$$

Say, a predictor f_1 is better than a predictor f_2 (in the mean-square sense) iff

$$\mathsf{E}\,(Y - f_1(\mathbf{X}))^2 \leqslant \mathsf{E}\,(Y - f_2(\mathbf{X}))^2\,,$$

i.e. provided that ε is minimized with f_1.

Example 7.1 The rv $f(X_1, \ldots, X_n) = \mathsf{E}(Y \mid X_1, \ldots X_n)$ is called the **regression function** of Y on $\mathbf{X} = (X_1, \ldots, X_n)$. Its value at $X_i = x_i$, for $i = 1, \ldots, n$, is given by

$$f(x_1, \ldots, x_n) = \mathsf{E}(Y \mid X_1 = x_1, \ldots, X_n = x_n).$$

For $n = 1$ we have $f(X) = \mathsf{E}(Y \mid X)$, with value $f(x) = \mathsf{E}(Y \mid X = x)$.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_7

Example 7.2 (Linear Prediction) The function $f(X_1, \ldots, X_n) = a_0 + a_1 X_1 + \cdots + a_n X_n$, where $a_0, a_1, \ldots, a_n$ are all constants, is a linear predictor of Y based on $\mathbf{X} = (X_1, \ldots, X_n)$. For $n = 1$ we have $f(X) = a + bX$.

Some properties of conditional expectation are useful in prediction, then we summarize them here. Observe that they can be deduced from those satisfied by general conditional expectation, see Sect. 6.4, Theorem 6.2, and Proposition 6.1. (all rvs are in L^1).

1. $\mathsf{E}(c \mid X) = c \in \mathbb{R}$.
2. $\mathsf{E}(Y + Z \mid X) = \mathsf{E}(Y \mid X) + \mathsf{E}(Z \mid X)$.
3. $\mathsf{E}(cY \mid X) = c\mathsf{E}(Y \mid X)$.
4. $\mathsf{E}(Y \mid X) = \mathsf{E}(Y)$ iff X and Y are independent.
5. $\mathsf{E}(ZY \mid X) = Z\mathsf{E}(Y \mid X)$ iff Z and X are independent.
6. $\mathsf{E}(h(X)Y \mid X) = h(X)\mathsf{E}(Y \mid X)$, taking out what is known.
7. $\mathsf{E}(\mathsf{E}(Y \mid X)) = \mathsf{E}(Y)$, or more generally $\mathsf{E}(\mathsf{E}(h(Y) \mid X)) = \mathsf{E}(h(Y))$ for a suitable function h.
8. $\mathsf{E}(f(X) \mid X) = h(X)$, for a suitable function h.
9. $\mathsf{E}\,(\mathsf{E}(Y \mid X, Z) \mid X) = \mathsf{E}(Y \mid X)$.

In item 4 we have no information from X, and $\mathsf{E}(Y \mid X)$ becomes unconditional; in item 7 we have complete information from X, which we understood as complete dependence. Recall that $\mathsf{E}(Y \mid X)$ is a rv that equals $\mathsf{E}(Y \mid X = x)$ whenever $X = x$. In item (8) we used $\mathsf{E}(\cdot \mid X, Z) = \mathsf{E}(\cdot \mid \sigma(X, Z))$ and $\mathsf{E}(\cdot \mid X) = \mathsf{E}(\cdot \mid \sigma(X))$, and since $\sigma(X) \subset \sigma(X, Z)$ the tower property of general conditional expectation does the job, see Proposition 6.1 in Sect. 6.4. It is worth noting how $\mathsf{E}(Y \mid X)$ is based on reverse conditioning w.r.t. that treated in Sect. 6.3, which is now expressed as dependence between Y and X, i.e., $Y = f(X)$, see also Lemma 6.1.

Remark 7.1 (Projection) For any suitable[1] function $h(\cdot)$ we have $Y - \mathsf{E}(Y \mid X)$ is uncorrelated with $h(X)$. Indeed, in $\mathsf{cov}(Y - \mathsf{E}(Y \mid X), h(X))$ let $Z = \mathsf{E}(Y \mid X)$ then:

$$\begin{aligned}\mathsf{E}((Y - Z)h(X)) - \underbrace{\mathsf{E}(Y - Z)}_{=0 \text{ by (2), (3) and (7)}} \mathsf{E}(h(X)) &= \mathsf{E}(Yh(X)) - \mathsf{E}(h(X)Z) \\ &\overset{\text{by (6)}}{=} \mathsf{E}(Yh(X)) - \underbrace{\mathsf{E}(\mathsf{E}(h(X)Y \mid X))}_{=\mathsf{E}(Yh(X))} \\ &= 0\end{aligned}$$

Up to this point, it is useful to have a conditional version of the variance.

[1] Recall: this means Borel measurable.

Definition 7.2 Given two rvs $X, Y \in L^2$, the conditional variance of Y based on X is the rv

$$\mathsf{V}(Y \mid X) := \mathsf{E}\left(\{Y - \mathsf{E}(Y \mid X)\}^2 \mid X\right), \tag{7.1}$$

i.e., the variance of Y given X. A specific value is given by:

$$\mathsf{V}(Y \mid X = x), \quad x \in \mathbb{R}.$$

The conditional variance can be written:

$$\mathsf{V}(Y \mid X) = \mathsf{E}(Y^2 \mid X) - (\mathsf{E}(Y \mid X))^2 . \tag{7.2}$$

$\mathsf{V}(Y \mid X)$ is also a function of X, i.e. it is a rv.

Proposition 7.1 (Law of Total Variance) *For $Y \in L^2$ we have*

$$\mathsf{V}(Y) = \mathsf{V}(\mathsf{E}(Y \mid X)) + \mathsf{E}(\mathsf{V}(Y \mid X)). \tag{7.3}$$

$\mathsf{V}(Y)$ decomposes into the sum of:

- $\mathsf{V}(\mathsf{E}(Y \mid X))$ which explains the variability of Y in term of its average dependence on X (explained variation);
- $\mathsf{E}(\mathsf{V}(Y \mid X))$ accounts for other sources of variability (unexplained variation).

If $Y = f(X)$ for a suitable function f, then

$$\mathsf{E}(Y \mid X) = \mathsf{E}(f(X) \mid X) = f(X) = Y$$

thus

$$\mathsf{V}(Y \mid X) = \mathsf{E}\left((Y - Y)^2 \mid X\right) = 0.$$

Hence, in case of strong dependence the variance of $\mathsf{E}(Y \mid X)$ is the same as $\mathsf{V}(Y)$, and X is a good predictor of Y. In case of independence, $\mathsf{E}(Y \mid X) = \mathsf{E}(Y)$ and the conditional variance equals $\mathsf{V}(\mathsf{E}(Y)) = 0$. Observing that, from $L^2 \subset L^1$, Theorem 6.2 in Sect. 6.4 can be stated for square integrable rvs, and we have that the expectation of one rv Y conditioned on another rv X (recall to choose $\mathcal{A} = \sigma(X)$) represents a form of functional dependence which is the best in the mean-square sense.

Theorem 7.1 (Least-Square Estimate) *For a rv $Y \in L^2$ it holds*

$$\mathsf{E}\left(\{Y - \mathsf{E}(Y \mid \mathcal{A})\}^2\right) \leqslant \mathsf{E}\left((Y - Z)^2\right), \tag{7.4}$$

for any rv $Z \in L^2$ which induces events in $\mathcal{A} \subset \mathcal{F}$, i.e., which is $\mathcal{A}$-measurable.

Thanks to the Law of Total Variance (Proposition 7.1) we can write:

$$\begin{aligned}\mathsf{V}(Y) &\overset{\text{Def. of } \sigma_Y^2}{=} \mathsf{E}(\{Y - \mathsf{E}(Y)\}^2) \\ &\overset{\text{Def. (7.1)}}{=} \mathsf{E}\left(\{\mathsf{E}(Y \mid X) - \mathsf{E}(Y)\}^2\right) + \mathsf{E}\left(\mathsf{V}(Y \mid X)\right) \\ &\geqslant \mathsf{E}(\mathsf{V}(Y \mid X)).\end{aligned}$$

The above rewrites:

$$\begin{aligned}\mathsf{E}(\{Y - h(X)\}^2) &= \mathsf{E}\left(\{f(X) - h(X)\}^2\right) + \mathsf{E}\left(\mathsf{V}(Y \mid X)\right) \\ &\geqslant \mathsf{E}(\mathsf{V}(Y \mid X)),\end{aligned}$$

with $h(X) = \mathsf{E}(Y)$ being *another* predictor[2] of Y and $f(X) = \mathsf{E}(Y \mid X)$ the regression function of Y on X. When $h(X) \overset{\text{a.s.}}{=} f(X)$, we have equality above and minimal expected quadratic prediction error: $f(X) = \mathsf{E}(Y \mid X)$ is the best predictor of Y based on X, for rvs $Y \in L^2$.

7.2 Linear Prediction and Dependence

Although $f(X) = \mathsf{E}(Y \mid X)$ is the best predictor of $Y \in L^2$ in the mean square sense, computing the conditional expectation requires the joint distribution of (X, Y). When knowledge of the regression function can be difficult, we turn to the **best linear predictor**: $f(X) = a + bX$ for some $b \in \mathbb{R}$ and $a \neq 0$.

Proposition 7.2 *Let two rvs $X, Y \in L^2$ be given, with $\mathsf{E}(X) = \mu_X, \mathsf{V}(X) = \sigma_X^2$ and $\mathsf{E}(Y) = \mu_Y, \sigma_Y^2 = \mathsf{V}(Y)$, and with Pearson correlation coefficient $\rho = \frac{\mathsf{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{X,Y}}{\sigma_X \sigma_Y}$. Then, the best linear predictor $f(X) = a + bX$ of Y is that minimizing $\mathsf{E}\left(\{Y - f(X)\}^2\right)$, where parameters $a, b \in \mathbb{R}$ are s.t.*

$$f(X) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X). \tag{7.5}$$

In words, if Y depends linearly on X then Y is closely approximated by the straight line $Y = f(X) = a + bX$ with (minimizing) parameters $a = \mu_Y - \rho \frac{\sigma_Y}{\sigma_X}\mu_X$ and $b = \rho \frac{\sigma_Y}{\sigma_X}$. Moreover, computing the variance of Y we get:

$$\mathsf{V}(a + b) = \rho^2 \frac{\sigma_Y^2}{\sigma_X^2} \mathsf{V}(X) = \rho^2 \sigma_Y^2.$$

[2] It is the constant function $h(X)$ whose value is the unconditional expectation of X.

Since $\rho \in [-1, 1]$ we have $\mathsf{V}(a + bX) \leqslant \mathsf{V}(Y)$ and $\frac{\mathsf{V}(a+bX)}{\mathsf{V}(Y)} = \rho^2$, called **coefficient of determination**, which measures the variation of Y explained by a linear relationship with X. A perfect linear dependence is when $\rho = 1$. Here, $y = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ is the **regression line**, and $\rho\frac{\sigma_Y}{\sigma_X}$ is the **regression coefficient**, for every scenario ω s.t. $Y(\omega) = y$ and $X(\omega) = x$. The quantity $\sigma_Y^2(1 - \rho^2)$, the minimum attained by $\mathsf{E}(\{Y - f(X)\}^2)$, is called **residual variance**, see Exercise 7.3 and its solution. Prediction of X based on Y, via the best linear predictor (in the mean-square sense) gives the regression line:

$$x = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y).$$

This rewrites:

$$y = \mu_Y + \frac{1}{\rho}\frac{\sigma_Y}{\sigma_X}(x - \mu_X), \quad \rho \neq 0.$$

The two regression lines coincides when the slope is the same:

$$\rho\frac{\sigma_Y}{\sigma_X} = \frac{1}{\rho}\frac{\sigma_Y}{\sigma_X} \quad \text{iff} \quad |\rho| = 1.$$

Remark 7.2 From the perspective of conditional expectation, the best linear prediction is when $f(X) = \mathsf{E}(Y \mid X) = a + bX$. This is equivalent to have $Y = a + bX + \epsilon$, where ϵ is the prediction error s.t. $\mathsf{E}(\epsilon \mid X) = 0$. In fact by linearity:

$$\mathsf{E}(Y \mid X) = \mathsf{E}(a \mid X) + \mathsf{E}(bX \mid X) + \mathsf{E}(\epsilon \mid X) = a + bX.$$

Conversely, suppose $\mathsf{E}(Y \mid X) = a + bX$ and let $\epsilon = Y - (a + bX)$. Then, $Y = a + bX + \epsilon$ and

$$\mathsf{E}(\epsilon \mid X) = \mathsf{E}(Y \mid X) - \mathsf{E}(a + bX \mid X) = \mathsf{E}(Y \mid X) - (a + bX) = 0.$$

Note $\mathsf{E}(\epsilon) = \mathsf{E}(\mathsf{E}(\epsilon \mid X)) = \mathsf{E}(0) = 0$. Also that $\mathsf{E}(\epsilon X)$ equals

$$\mathsf{E}(\mathsf{E}(\epsilon X \mid X)) = \mathsf{E}(X\mathsf{E}(\epsilon \mid X)) = \mathsf{E}(0) = 0,$$

i.e. the explanatory variable X and the error ϵ are uncorrelated. Taking the covariance of both sides in $Y = a + bX + \epsilon$ with X we get $a = \mathsf{E}(Y) - \frac{\mathsf{cov}(X,Y)}{\mathsf{V}(X)}\mathsf{E}(X)$ and $b = \frac{\mathsf{cov}(X,Y)}{\mathsf{V}(X)}$. Observe that in all the above calculations we used properties (1),(2) and (8) of conditional expectation.

Example 7.3 $X^* = \mathsf{E}(X_t \mid \mathcal{F}_s)$ is the best predictor of a future price X_t for any date $t > s \geqslant 0$. The information $\mathcal{F}_s$ available at a past date if fully reflected in the

price X_t, i.e. is embodied optimally (in the mean square sense). This is a form of market efficiency given in term of *rational expectations*. $\mathsf{E}(X_t \mid \mathcal{F}_s)$ is a prediction of the price X_t at time t, given all the available information up to time $s < t$. Used as a forecast, $\mathsf{E}(X_t \mid \mathcal{F}_s)$ is the best predictor in the mean square sense, i.e.

$$\mathsf{E}\left(\{X_t - X^*\}^2\right)$$

is minimal for $X^* = \mathsf{E}(X_t \mid \mathcal{F}_s)$, provided that $X_t \in L^2$ for all t.

7.3 Nonlinear Dependence

The Pearson correlation coefficient is a scalar measure of linear dependence between two rvs, corresponding to the case $Y = a + bX$ or $Y \overset{\text{a.s.}}{=} a + b$, for $a \in \mathbb{R}$, $b \neq 0$ and $X, Y \in L^2$.

Remark 7.3 Two uncorrelated rvs $X, Y \in L^2$ are note necessarily independent. E.g., take $X \sim \mathrm{N}(0, 1)$ and let $Y = g(X) = X^2$. Now $\mathsf{E}(XY) = \mathsf{E}(X^3) = 0$ because the Normal X is symmetric (the odd moments are all zero), then $\mathsf{cov}(X, Y) = \mathsf{E}(X^3) - \mathsf{E}(X)\mathsf{E}(Y) = 0$, since $\mathsf{E}(X) = 0$, and they are uncorrelated although *strongly* dependent: the X's values yield perfect prediction of the Y's values. For rvs $X, Y \in L^2$ to be uncorrelated it suffices to have at least $\mathsf{E}(XY) = 0$ and $\mathsf{E}(X) = 0$, or $\mathsf{E}(XY) = 0$ and $\mathsf{E}(Y) = 0$.

Example 7.4 Let Ω be a finite sample space with $|\Omega| = 7$ and assume the discrete uniform distribution with $\mathsf{P}(\{\omega\}) = \frac{1}{7}$, for $\omega \in \Omega$. Define two rvs $X = \sum_{i=1}^{7}(\frac{1}{2} + \frac{(-1)^i}{2})\mathbf{I}_{A_i}$ and $Y = \mathbf{I}_{\{X=0\}} - \sum_{i \text{ even}}(-1)^{i+1}\mathbf{I}_{A_i}$, where $A_i = \{\omega\}$, each $\omega \in \Omega$, is a partition of Ω. We have $XY = 0$ and $\mathsf{E}(Y) = 0$ so $\mathsf{cov}(X, Y) = 0$ but $\{X = 1\} \cap \{Y = 1\} = \varnothing$, then $\mathsf{P}(X = 1, Y = 1) = 0$ while $\mathsf{P}(X = 1)\mathsf{P}(Y = 1) = \frac{3}{7} \cdot \frac{2}{7} \neq 0$. Thus, X and Y are not independent. Observe that $\mathbf{I}_{\{X=0\}} = \sum_{i \text{ even}} \mathbf{I}_{A_i}$.

Perfect linear dependence is when $|\mathsf{corr}(X, Y)| = 1$, similar to the concepts of antimonotone and comonotone rvs. Indeed, taking an increasing linear function $g : \mathbb{R} \to \mathbb{R}$ and considering $\mathsf{corr}(g(X), g(Y))$, then correlation is unchanged. But for *nonlinear* increasing transformations this does not apply. Moreover, correlation is not suitable when marginal distributions are heavy-tailed which means that possibly variances are not finite, i.e. $X, Y \notin L^2$.

Just to Remind
We summarize the framework of linear dependence.

- Pearson correlation is invariant only under linear increasing transform, i.e. $a + b\,X$ and $c + d\,Y$ with positive b, d. In fact we showed

$$\mathsf{corr}(a + b\,X, c + d\,Y) = \mathrm{sign}(bd)\mathsf{corr}(X, Y)$$

  which equals $\mathsf{corr}(X, Y)$ iff $\mathrm{sign}(bd) = 1$.
- Consider the linear relationship $Y \overset{\text{a.s.}}{=} a + b\,X$ and take $a = 0$ and $b = \pm 1$. Observe $Y \overset{\text{a.s.}}{=} \mathrm{sign}(b)X$ implies $Y \overset{\text{d}}{=} \mathrm{sign}(b)X$. Thus

$$\mathsf{cov}(X, Y) = \mathsf{cov}(X, \mathrm{sign}(b)X) = \mathrm{sign}(b)\mathsf{V}(X) \Longrightarrow \mathsf{corr}(X, Y) = \mathrm{sign}(b),$$

  so we have $Y \overset{\text{a.s.}}{=} X$ corresponds to **maximal linear dependence** with X, Y being comonotone, provided $\mathrm{sign}(b) = 1$, and $Y \overset{\text{a.s.}}{=} -X$ corresponds to **minimal linear dependence** with X, Y being antimonotone, provided $\mathrm{sign}(b) = -1$.

It is not always possible to build a bi-variate CDF $F_{\mathbf{X}}$ for the random vector $\mathbf{X} = (X, Y)$ with margins F_X, F_Y and assigned correlation, unless $\mathbf{X}$ is, e.g., a Gaussian random vector. In fact, linear and nonlinear dependence are related by Theorem 7.2 below.

Theorem 7.2 *Assume* $\mathbf{X} = (X_1, X_2)$ *has joint CDF* $F_{\mathbf{X}}$ *and marginal CDFs* F_{X_1}, F_{X_2} *with finite variances* $\mathsf{V}(X_1), \mathsf{V}(X_2) < \infty$.

(a) If X_1, X_2 *are comonotone, then* $\mathsf{corr}(X_1, X_2) > 0$. *If they are antimonotone, then* $\mathsf{corr}(X_1, X_2) < 0$.

(b) The correlation $\mathsf{corr}(X_1, X_2)$ *is maximal if* X_1, X_2 *are comonotone, in particular* $X_1 \overset{d}{=} X_2$ *yields* $\mathsf{corr}(X_1, X_2) = 1$. *The correlation* $\mathsf{corr}(X_1, X_2)$ *is minimal if* X_1, X_2 *are antimonotone, in particular* $X_1 \overset{d}{=} -X_2$ *yields* $\mathsf{corr}(X_1, X_2) = -1$.

The proof of Theorem 7.2, see Appendix, is partly based on the following well known formula for covariances.

Lemma 7.1 (Hoeffding) *If* $\mathbf{X} = (X_1, X_2)$ *has joint CDF* $F_{\mathbf{X}}$ *and marginal CDFs* F_{X_1}, F_{X_2}, *then*

$$\mathsf{cov}(X_1, X_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left[F_{\mathbf{X}}(x_1, x_2) - F_{X_1}(x_1)F_{X_2}(x_2)\right]\mathrm{d}x_1\mathrm{d}x_2. \tag{7.6}$$

The proof of Hoeffding's lemma is in Appendix. More concept and results on nonlinear dependence, than those presented in Sect. 7.4, are needed for the proof of Theorem 7.2.

Remark 7.4 Assume that $F_X(x)$ is the CDF of a degenerate rv with a point mass at $x = a$, and $F_Y(y)$ is another CDF corresponding to a different not necessarily degenerate rv. Then we have:

$$\min\{F_X(x), F_Y(y)\} = F_X(x)F_Y(y). \tag{7.7}$$

In fact, for $x < a$ we have $F_X(x) = 0$, for $x \geqslant a$ we have $F_X(x) = 1$ while for every $y \in \mathbb{R}$ we have $0 \leqslant F_Y(y) \leqslant 1$, and the product of CDFs equals their pointwise minimum. If at least one of the rvs corresponding to the marginal CDFs is not degenerate we also have

$$F_X(x)F_Y(y) < \min\{F_X(x), F_Y(y)\} \tag{7.8}$$

for every pair $(x, y) \in \mathbb{R}^2$.

7.4 Stochastic Dependence... Again

Assume $\mathbf{X} = (X, Y)$ is a random vector with components interpreted as *risk factors*. For example, X, Y can be thought as future stock prices, profits and losses, rates of return, contracts held by an insurance company. Thus the joint distribution of $\mathbf{X}$ provides probabilistic information on the dependence between X and Y, especially if they form a portfolio whose possible changes (viz. the loss incurred by the portfolio) are affected by their joint movements. Based on the analysis developed in Sect. 3.4, the two risk factors are dependent whenever their joint probability or CDF does not factor and assumes value within the interval[3] $[a, b]$ where a, b are values attained by

$$G(x, y) = \max\{0, F_X(x) + F_Y(y) - 1\} \tag{7.9}$$

$$H(x, y) = \min\{F_X(x), F_Y(y)\}, \tag{7.10}$$

respectively for $(x, y) \in \mathbb{R}^2$, which are proved to be CDFs, see Exercise 7.10. $F_{\mathbf{X}}(x, y)$ is bounded above by the joint CDF $H(x, y)$ describing positive dependence between the marginals $F_X(x), F_Y(y)$, and it is bounded below by the joint CDF $G(x, y)$ describing negative dependence. These are called **Fréchet bounds**. Now, we apply Theorem 5.1 to the risk factors getting the new random vectors

[3] Recall to set $A = \{X \leqslant x\}$ and $B = \{Y \leqslant y\}$ in Eq. (3.9).

$$\mathbf{X}^{\mathrm{c}} = (Q_X(U), Q_Y(U)) \text{ and } \mathbf{X}_{\mathrm{c}} = (Q_X(U), Q_Y(1-U)), \quad U \sim \mathrm{U}(0,1).$$

The former is called *comonotonic* the latter *antimonotonic*.

Theorem 7.3 *Let $\mathbf{X} = (X, Y)$ be two risk factors. Then:*

1. *The **upper Fréchet bound** $H(x, y)$ is the joint CDF of $\mathbf{X}^c$. Equivalently, there exists a rv Z and two nondecreasing functions f_1, f_2 s.t. $\mathbf{X}$ has the same joint distribution as the random vector $(f_1(Z), f_2(Z))$. Moreover X and Y are comonotone, i.e. $F_{\mathbf{X}} = H$.*
2. *The **lower Fréchet bound** $G(x, y)$ is the joint CDF of $\mathbf{X}_c$. Equivalently, there exists a rv Z, a nondecreasing function f_1 and a nonincreasing function f_2 (or vice versa) s.t. $\mathbf{X}$ has the same joint distribution as the random vector $(f_1(Z), f_2(Z))$. Moreover X and Y are antimonotone, i.e. $F_{\mathbf{X}} = G$.*

It is worth noting that $Q_X(U) \overset{\mathrm{d}}{=} X$ and also $Q_Y(U) \overset{\mathrm{d}}{=} Q_Y(1-U) \overset{\mathrm{d}}{=} Y$ but equality in distribution does not mean equality among the components of $\mathbf{X}$ and the components of $\mathbf{X}^{\mathrm{c}}, \mathbf{X}_{\mathrm{c}}$. Theorem 7.3, Item 1, is still valid for $n > 2$ and corresponding CDFs in $\mathbb{R}^n$, but Item 2 no longer holds for $n > 2$. The general version of the upper Fréchet bound is

$$H(x_1, \ldots, x_n) = \min\{F_{X_1}(x_1), \ldots, F_{X_n}(x_n)\}, \quad \mathbf{X} = (X_1, \ldots, X_n), \tag{7.11}$$

while the right form of the lower Fréchet bound is

$$G(x_1, \ldots, x_n) = \max\left\{0, \sum_{i=1}^{n} F_{X_i}(x_i) + 1 - n\right\}. \tag{7.12}$$

Remark 7.5 For comonotonic rvs X, Y there is a single source Z determining their *riskiness* as component of $\mathbf{X} = (X, Y)$: they are perfect positively dependent being both increasing functions of Z. Applying the same reasoning as above, two risk factors in $\mathbf{X} = (X, Y)$ are perfect negatively dependent in the antimonotonic case.

Remark 7.6 Observe that for a uniform rv $U \sim \mathrm{U}(0, 1)$ we have $F_U(x) = \mathsf{P}(U \leqslant x) = x$, which is obvious from the truly definition of its CDF: essentially, it is the identity function restricted to the unit interval.

For more details on dependence concepts as applied to risk management, see [11, Sec 7.2] and [13, Chps 1,2]. In risk management or credit risk, the probability distribution of portfolio losses $L = V_t - V_0$ is crucial. Recall that $V_t = \sum_{i=1}^{n} h_i S_{i,t}$ is a vector-to-scalar transformation, thus the V_t's distribution hardly depends on the joint distribution of $\mathbf{S} = (S_{1,t}, \ldots, S_{n,t})$, i.e., future asset prices, given the vector of portfolio holdings $h = (h_1, \ldots, h_n)$ s.t. the initial endowment is $V_0 = \sum_{i=1}^{n} h_i S_{i,0}$, which is assumed to be constant. The components in $\mathbf{S}$ are risk factors, since securities are exposed to fluctuations in the market. However, if $r_{p,t} = \sum_{i=1}^{n} w_i r_{i,t}$

is the portfolio return with weights[4] $w_i = \frac{h_i S_{i,0}}{V_t}$, where $r_{i,t} = \frac{S_{i,t} - S_{i,0}}{S_{i,0}}$, then it is equivalent to say that the random vector of risk factors is $\mathbf{r} = (r_{i,t}, \ldots, r_{n,t})$, since $V_t = V_0(1 + r_{p,t})$. Therefore, what really matters for L is the joint distribution $F_{\mathbf{r}}(x_1, \ldots, x_n)$, which contains relevant information about individual risk factors as well as their dependence structure. We know that only for independent marginals $F_{r_{i,t}}$ one may set up the joint CDF by multiplication. Assuming returns have symmetric distributions given by $F_{r_{i,t}}$, with no fat-tails, and assuming further linear dependence then multivariate Gaussian modelling of $\mathbf{r}$ comes in hand.[5] But empirical evidence shows different patterns of returns. Fortunately enough, there is a way to get the joint distribution of risk factors starting from the marginals and this take advantage of being easier to estimate univariate CDFs from time series of returns than to postulate a particular form of the dependence among risk factors.

Definition 7.3 A function $C : [0, 1] \times [0, 1] \to [0, 1]$ is a **bi-variate copula** if the following properties hold:

(1) $C(u_1, u_2) = 0$, if $u_1 = u_2 = 0$;
(2) $C(1, u_2) = u_2$, and $C(u_1, 1) = u_1$;
(3) $C(b_1, b_2) - C(a_1, b_2) - C(b_1, a_2) + C(a_1, a_2) \geqslant 0$, for any $0 \leqslant a_1 < b_1 \leqslant 1$ and $0 \leqslant a_2 < b_2 \leqslant 1$.

It is worth noting that $C(u_1, u_2)$ is non-decreasing in each components. Moral: $C(u_1, u_2) = \mathsf{P}(U_1 \leqslant u_1, U_2 \leqslant u_2)$ is a CDF having uniform $U_i \sim \mathrm{U}(0, 1)$ marginals, for $i = 1, 2$, which assigns a nonnegative probability to rectangles inside the unit cube $[0, 1] \times [0, 1]$, i.e., $\mathsf{P}(a_1 \leqslant U_1 \leqslant b_1, a_2 \leqslant U_2 \leqslant b_2) \geqslant 0$. Copula functions, marginal CDFs and joint CDF are related each other, as stated in:

Theorem 7.4 (Sklar) *Assume the bi-variate vector $\mathbf{X} = (X_1, X_2)$ has joint CDF $F_{\mathbf{X}}$, with marginals F_{X_1} and F_{X_2}. Then, there exists a copula function $C : [0, 1] \times [0, 1] \to [0, 1]$ s.t. for every $(x_1, x_2) \in \mathbb{R}^2$ we have*

$$F_{\mathbf{X}}(x_1, x_2) = C(F_{X_1}(x_1), F_{X_2}(x_2)). \tag{7.13}$$

If $F_{\mathbf{X}}$ has continuous marginals, then the copula C is unique. Conversely, if C is a copula and F_{X_1}, F_{X_2} are univariate CDFs, then the function of two variable given in (7.13) yields a joint CDF with marginals F_{X_1}, F_{X_2}.

[4] No-leverage is implicitly assumed: $\sum_{i=1}^{n} w_i = 1$

[5] Recall that in the multivariate Gaussian setting, each component rv is also Gaussian.

See Appendix for the n-dimensional version of copula and Sklar's theorem. Under the continuity hypotheses we have $F_{X_1}(X_1) = U_1 \sim U(0,1)$ and $F_{X_2}(X_2) = U_2 \sim U(0,1)$, thus assuming C as the joint CDF of (U_1, U_2) we have

$$\begin{aligned} C(u_1, u_2) &= \mathsf{P}(F_{X_1}(X_1) \leqslant u_1, F_{X_2}(X_2) \leqslant u_2) \\ &= \mathsf{P}(X_1 \leqslant Q_{X_1}(u_1), X_2 \leqslant Q_{X_2}(u_2)) \\ &= F_{\mathbf{X}}(Q_{X_1}(u_1), Q_{X_2}(u_2)), \end{aligned}$$

which is nothing else the only-if part of Sklar's theorem proof. Coming back to risk management, portfolio value V_t with two weights $\mathbf{w} = (w_i, w_2)$ gives raise to random loss $L = V_0 r_{p,t}$ where random return $r_{p,t} = \mathbf{w} \cdot \mathbf{r}$ is based on the random vector $\mathbf{r} = (r_{1,t}, r_{2,t})$ of two risk factors. If marginal CDFs $F_{r_{1,t}}, F_{r_{2,t}}$ and their dependence structure given by copula $C(u_1, u_2)$ are available, then by Sklar's theorem the latter is the bi-variate CDF C of $(F_{r_{1,t}}(r_{1,t}), F_{r_{2,t}}(r_{2,t}))$, and we can say that C is the copula of $F_{\mathbf{r}}$ or $\mathbf{r}$. Perfect positive/negative dependence[6] can be reproduced by copula functions since we always have:

$$G(u_1, u_2) \leqslant C(u_1, u_2) \leqslant H(u_1, u_2), \quad \text{for all } (u_1, u_2) \in [0,1] \times [0,1],$$

where G, H are the bi-variate Fréchet bounds in (7.9) and (7.10), respectively. Hence, H is the copula of $\mathbf{U} = (U_1, U_2)$ with $U_1 = U_2$, while G for $U_2 = 1 - U_1$. They can be used when the corresponding random vector $\mathbf{X} = (X_1, X_2)$ has marginals F_{X_1}, F_{X_2} for which we want to reproduce either a perfect positive dependence or a perfect negative dependence structure.

Proposition 7.3 *Let $\mathbf{X} = (X_1, X_2)$ be a random vector with continuous marginals F_{X_1}, F_{X_2}, copula function $C(u_1, u_2)$, and let $f_i : \mathbb{R} \to \mathbb{R}$ be two strictly increasing functions, for $i = 1, 2$. Then the random vector $(f_1(X_1), f_2(X_2))$ also has copula functions $C(u_1, u_2)$.*

Proposition 7.3 is analogous to Proposition 2.3 in Sect. 2.8, indeed copula functions represent stochastic dependence on a quantile scale.

Example 7.5 Assuming continuity of the CDFs, letting $X_1 = f_1(X_1)$ and $X_2 \overset{\text{a.s.}}{=} f_2(X_1)$ and applying Proposition 7.3 we have that copulas of $\mathbf{X} = (X_1, X_1)$ and $\mathbf{U} = (U_1, U_1)$ are the same, and since the copula of the former random vector is the CDF of the latter (with $U_i = F_{X_1}(X_1)$, for $i = 1, 2$), we get the copula H.

Example 7.6 (Independence Copula) For a vector of two random returns $\mathbf{r} = (r_{1,t}, r_{2,t})$ which can be assumed to be independent, their copula function is

$$C(u_1, u_2) = u_1 u_2. \tag{7.14}$$

[6] In risk management, comonotonicity entails undiversifiable risks and must be handled with care in capital requirements based on risk.

In fact, by assumption $F_{\mathbf{r}}(x_1, x_2) = F_{r_{1,t}}(x_1)F_{r_{2,t}}(x_2)$, and by Sklar's theorem the joint CDF of returns must be equal to $C(F_{r_{1,t}}(x_1), F_{r_{2,t}}(x_2))$ which in turn is (7.14).

Example 7.7 (Gaussian Copula) Consider two random returns $\mathbf{r} = (r_{1,t}, r_{2,t})$ with bi-variate Gaussian distribution $\mathbf{r} \sim \mathrm{N}(\boldsymbol{\mu}, \mathbf{C}_{\mathbf{r}})$, where $\boldsymbol{\mu} = (\mu_1, \mu_2)$, and the covariance matrix is

$$\mathbf{C}_{\mathbf{r}} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{1,2} & \sigma_2^2 \end{bmatrix},$$

with covariance $\sigma_{1,2} = \mathsf{cov}(r_{1,t}, r_{2,t})$. By Sklar's theorem, $\mathbf{r}$'s copula is $C(F_{r_{1,t}}(x_1), F_{r_{2,t}}(x_2))$ and equals $F_{\mathbf{r}}(x_1, x_2)$. On the other hand, standardizing returns, $X_i = f_i(r_{i,t}) = \frac{r_{i,t}-\mu_i}{\sigma_i}$ for $i = 1, 2$, we also have the random vector $\mathbf{X} = (X_1, X_2)$ distributed as $\mathrm{N}(\mathbf{0}, \mathbf{C}_{\mathbf{X}})$, where $\mathbf{0} = (0, 0)$ and

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},$$

with Pearson correlation coefficient $\rho = \mathsf{corr}(r_{1,t}, r_{2,t})$. Hence, we have the Gaussian copula $C^{\mathrm{Ga}}(F_{X_i}(x_1), F_{X_2}(x_2))$ as the CDF $F_{\mathbf{X}}(x_1, x_2)$, and by Proposition 7.3 it holds $C = C^{\mathrm{Ga}}$ since f_i is strictly increasing for $i = 1, 2$. We can do more, by writing the explicit expression of the Gaussian copula as follows:

$$\begin{aligned} C^{\mathrm{Ga}}(u_1, u_2) &= F_{\mathbf{X}}(Q_{X_1}(u_1), Q_{X_1}(u_2)) \\ &= \int_{-\infty}^{Q_{X_1}(u_1)} \int_{-\infty}^{Q_{X_2}(u_2)} \frac{1}{2\pi\sqrt{(1-\rho^2)}} \\ &\quad \times \exp\left(\frac{-(y_1^2 - 2\rho y_1 y_2 + y_2^2)}{2(1-\rho^2)}\right) \mathrm{d}y_1 \mathrm{d}y_2. \end{aligned} \tag{7.15}$$

Example 7.7 suggests that under the continuity assumption we can find the density of C. Indeed, assuming $F_{\mathbf{X}}$ is an absolutely continuous bi-variate CDF with strictly increasing, continuous marginals F_{X_1}, F_{X_2}, we can differentiate the copula function $C(u_1, u_2) = F_{\mathbf{X}}(Q_{X_1}(u_1), Q_{X_2}(u_2))$ to get

$$c(u_1, u_2) = \frac{\partial C(u_1, u_2)}{\partial u_1 \partial u_2}.$$

Observing that $Q_{X_i} \equiv F_{X_i}^{-1}$, i.e., quantiles are now inverse functions of CDFs, we can relate the copula density to the bi-variate density of $\mathbf{X}$ as

$$\begin{aligned} c(F_{X_1}(x_1), F_{X_2}(x_2)) &= \frac{f_{\mathbf{X}}(x_1, x_2)}{f_{X_1}(x_1) f_{x_2}(x_2)} \iff f_{\mathbf{X}}(x_1, x_2) \\ &= c(F_{X_1}(x_1), F_{X_2}(x_2)) f_{X_1}(x_1) f_{x_2}(x_2), \end{aligned} \tag{7.16}$$

see [11, Ch 7]. Using `R` or `Matlab` or `Excel` we can:

- simulate a random vector $\mathbf{X}$ according to a specified parametric bi-variate distribution $F_{\mathbf{X}}$, i.e. we get pseudo random numbers x_1, x_2;
- we let $u_i = F_{X_i}(x_i)$ for $i = 1, 2$ and get the vector $\mathbf{u} = (u_1, u_2)$, where u_i are drawn from uniform marginal $U \sim \mathrm{U}(0, 1)$ and thanks to Sklar's theorem we have the dependence structure given by copula $C(u_1, u_2)$;
- we let $x_i = Q_{X_i}(u_i)$ and get the vector $\mathbf{x} = (x_1, x_2)$, where now x_i are drawn from the marginal CDFs F_{X_i} and we get the required dependence structure given by the bi-variate CDF $F_{\mathbf{X}}$.

For in depth analysis of copula and dependence see [11, Ch 7] and [13, Ch 1]. See also Appendix.

7.5 Exercises

7.1 Prove formula (7.3) in Proposition 7.1.

7.2 Prove Theorem 7.1.

7.3 Prove Proposition 7.2.

7.4 Prove $|\mathsf{corr}(X, Y)| \leqslant 1$ and $\mathsf{corr}(a + bX, c + dY) = \mathsf{corr}(X, Y)$, provided $b, d > 0$.

7.5 Prove $\mathsf{V}(a + bX) = b^2\mathsf{V}(X)$ by:

(i) first showing that the degenerate rv $Y = a$ and the rv $Z = bX$ are independent;
(ii) then showing that for two independent rvs Y, Z it holds $\mathsf{V}(Y + Z) = \mathsf{V}(Y) + \mathsf{V}(Z)$.

7.6 Let $Y = a + bX$. Compute the covariance and the Pearson correlation of the pair X, Y.

7.7 Given two independent random vectors s.t. $(X_1, X_2) = \mathbf{X} \stackrel{\mathrm{d}}{=} \mathbf{Y} = (\tilde{X}_1, \tilde{X}_2)$ show that $2\,\mathsf{cov}(X_1, X_2) = \mathsf{E}\left[(X_1 - \tilde{X}_1)(X_2 - \tilde{X}_2)\right]$.

7.8 Let $Y \stackrel{\text{a.s.}}{=} X$ and show this implies $Y \stackrel{\mathrm{d}}{=} X$ so that it can be generalized to $Y \stackrel{\text{a.s.}}{=} g(X)$ implying $Y \stackrel{\mathrm{d}}{=} g(X)$ for a suitable function $g(x)$.

7.9 Let $Y \stackrel{\text{a.s.}}{=} X$ and show this implies $\mathsf{corr}(X, Y) = 1$.

7.10 Show that $G(x, y) = \max\{0, F_X(x) + F_Y(y) - 1\}$ and $H(x, y) = \min\{F_X(x), F_Y(y)\}$ are joint CDFs.

Appendix

Proof of Hoeffding's Lemma

Proof Let $\mathbf{X} = (X_1, X_2)$ and take another block independent random vector $\mathbf{Y} = (\tilde{X}_1, \tilde{X}_2)$ and assume they have the same distribution, $\mathbf{X} \stackrel{\mathrm{d}}{=} \mathbf{Y}$. Now, consider the formula

$$\int_{-\infty}^{\infty} \left[\mathbf{I}_{\{b \leqslant x\}} - \mathbf{I}_{\{a \leqslant x\}}\right] \mathrm{d}x = \int_b^x \mathrm{d}x - \int_a^x \mathrm{d}x = a - b.$$

Use this formula twice for the pairs $X_1, \tilde{X}_1$ and $X_2, \tilde{X}_2$ then multiply and get

$$\begin{aligned}(X_1 - \tilde{X}_1)(X_2 - \tilde{X}_2) = &\int_{-\infty}^{\infty} \left[\mathbf{I}_{\{\tilde{X}_1 \leqslant x_1\}} - \mathbf{I}_{\{X_1 \leqslant x_1\}}\right] \mathrm{d}x_1 \\ &\times \int_{-\infty}^{\infty} \left[\mathbf{I}_{\{\tilde{X}_2 \leqslant x_2\}} - \mathbf{I}_{\{X_2 \leqslant x_2\}}\right] \mathrm{d}x_2.\end{aligned}$$

The above product is actually a double integral w.r.t. the Lebesgue product measure in $\mathbb{R}^2$, thus we can rewrite it as:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left[\mathbf{I}_{\{\tilde{X}_1 \leqslant x_1\}} - \mathbf{I}_{\{X_1 \leqslant x_1\}}\right] \left[\mathbf{I}_{\{\tilde{X}_2 \leqslant x_2\}} - \mathbf{I}_{\{X_2 \leqslant x_2\}}\right] \mathrm{d}x_1 \mathrm{d}x_2.$$

Recall that $2\,\mathsf{cov}(X_1, X_2)$ equals the expectation of $(X_1 - \tilde{X}_1)(X_2 - \tilde{X}_2)$, so taking expectation of the iterated integral above we get:[7]

$$\begin{aligned}&\mathsf{E}\left(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left[\mathbf{I}_{\{\tilde{X}_1 \leqslant x_1\}} - \mathbf{I}_{\{X_1 \leqslant x_1\}}\right] \left[\mathbf{I}_{\{\tilde{X}_2 \leqslant x_2\}} - \mathbf{I}_{\{X_2 \leqslant x_2\}}\right] \mathrm{d}x_1 \mathrm{d}x_2\right) = \\ &\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathsf{E}\left[\mathbf{I}_{\{\tilde{X}_1 \leqslant x_1\}} - \mathbf{I}_{\{X_1 \leqslant x_1\}}\right] \left[\mathbf{I}_{\{\tilde{X}_2 \leqslant x_2\}} - \mathbf{I}_{\{X_2 \leqslant x_2\}}\right] \mathrm{d}x_1 \mathrm{d}x_2.\end{aligned}$$

Now performing the product between brackets inside the expectation yields:

$$\mathbf{I}_{\{\tilde{X}_1 \leqslant x_1\} \cap \{\tilde{X}_2 \leqslant x_2\}} - \mathbf{I}_{\{\tilde{X}_1 \leqslant x_1\} \cap \{X_2 \leqslant x_2\}} - \mathbf{I}_{\{X_1 \leqslant x_1\} \cap \{\tilde{X}_2 \leqslant x_2\}} + \mathbf{I}_{\{X_1 \leqslant x_1\} \cap \{X_2 \leqslant x_2\}}.$$

[7] This results in a new double integral w.r.t. to $\mathsf{P} \times (m \times m)$. By Fubini's theorem we get three iterated integrals $\int_\Omega \int_{\mathbb{R}} \int_{\mathbb{R}}$ that can be interchanged.

Taking expectation we get:

1. $\mathsf{P}(\tilde{X}_1 \leqslant x_1, \tilde{X}_2 \leqslant x_2)$ which is the same distribution as $\mathsf{P}(X_1 \leqslant x_1, X_2 \leqslant x_2)$, and these two CDFs do not factor;
2. $\mathsf{P}(\tilde{X}_1 \leqslant x_1, X_2 \leqslant x_2)$ factors by independence, and the same is true for the other cross product of indicators;
3. $\mathsf{P}(X_i \leqslant x_i) = \mathsf{P}(\tilde{X}_i \leqslant x_i)$ for $i = 1, 2$, by block independence.

As byproduct we finally have the desired formula (7.6). ■

Proof of Theorem 7.2

Proof To prove item (a) we apply Hoeffding's Lemma 7.1 and Theorem 7.3. First, if X_1, X_2 are comonotone then their joint CDF is $F_{\mathbf{X}}(x_1, x_2) = \min\{F_{X_1}(x_1), F_{X_2}(x_2)\}$ and by formula (7.6) we get:

$$\mathsf{cov}(X_1, X_2) = \iint \big[\min\{F_{X_1}(x_1), F_{X_2}(x_2)\} - F_{X_1}(x_1)F_{X_2}(x_2)\big]\mathrm{d}x_1\mathrm{d}x_2.$$

This is positive if at least one of the rvs is not degenerate, but this is the case since we assumed both $\mathsf{V}(X_1), \mathsf{V}(X_2)$ are strictly positive. The statement for antimonotone rvs follows by $F_{\mathbf{X}}(x_1, x_2) = \max\{0, F_{X_1}(x_1) + F_{X_2}(x_2) - 1\}$ and replacing X_2 with $-X_2$. To prove item (b) we note that

$$\begin{aligned}\mathsf{cov}(X_1, X_2) &= \iint \big[F_{\mathbf{X}}(x_1, x_2) - F_{X_1}(x_1)F_{X_2}(x_2)\big]\mathrm{d}x_1\mathrm{d}x_2 \\ &\leqslant \iint \big[\min\{F_{X_1}(x_1), F_{X_2}(x_2)\} - F_{X_1}(x_1)F_{X_2}(x_2)\big]\mathrm{d}x_1\mathrm{d}x_2,\end{aligned}$$

while the covariance is $\geqslant$ of $\iint \big[\max\{0, F_{X_1}(x_1)+F_{X_2}(x_2)-1\} - F_{X_1}(x_1)F_{X_2}(x_2)\big]$ $\mathrm{d}x_1\mathrm{d}x_2$. ■

In the proof of the antimonotone case we used the following fact. The function

$$g(x_1, x_2) = \max\{0, F_{X_1}(x_1) + F_{X_2}(x_2) - 1\} - F_{X_1}(x_1)F_{X_2}(x_2)$$

is negative if the maximum is zero. Then, assume the latter is positive and observe that

$$\begin{aligned}F_{X_1} + F_{X_2} - 1 - F_{X_1}F_{X_2} &= F_{X_1}(1 - F_{X_2}) - (1 - F_{X_2}) \\ &= (1 - F_{X_2})(F_{X_1} - 1) = -(1 - F_{X_2})(1 - F_{X_1}) \\ &= -\bar{F}_{X_2}\bar{F}_{X_1} < 0,\end{aligned}$$

where $\bar{F}_{X_i}(x_i) = 1 - F_{X_i}(x_i)$ is the survival function.

More on Copula Functions

The n-dimensional version of Sklar's theorem 7.4 requires only minor changes. Before to state it, we need to extend Definition 7.3.

Definition 7.4 An n-dimensional copula function $C : [0,1]^n \to [0,1]$ is a map from the unit hypercube $[0,1]^n := [0,1] \times \cdots \times [0,1]$ to the unit interval, fulfilling the following properties:

1. $C(u_1, \ldots, u_n) = 0$ whenever $u_i = 0$, each i;
2. C evaluated at $\mathbf{1}_i$, the vector of ones except the ith coordinate which is u_i, yields $C(\mathbf{1}_i) = u_i$ for $i = 1, \ldots, n$;
3. for every pair of vectors $\mathbf{a}, \mathbf{b} \in [0,1]^n$ forming a rectangle $(\mathbf{a}, \mathbf{b}]$ we have C as n-increasing,

$$\Delta_{a_1,b_1} \cdots \Delta_{a_n,b_n} C(u_1, \ldots, u_n) \geqslant 0,$$

for all rectangle, see Appendix in Chap. 3.

Hence, C is the joint CDF of a random vector $(U_1, \ldots, U_n)$ with uniform marginals, i.e. $U_i \sim \mathrm{U}(0,1)$ for $i = 1, \ldots, n$. As usual, Item 3 entails $\mathsf{P}(a_1 \leqslant U_1 \leqslant b_1, \ldots, a_n \leqslant U_n \leqslant b_n) \geqslant 0$.

Theorem 7.5 (Sklar) *Assume the random vector $\mathbf{X} = (X_1, \ldots, X_n)$ has multivariate CDF $F_{\mathbf{X}}$, with marginals $F_{X_1}, \ldots, F_{X_2}$. Then, there exists a copula function $C : [0,1]^n \to [0,1]$ s.t. for every $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ we have*

$$F_{\mathbf{X}}(x_1, \ldots, x_n) = C(F_{X_1}(x_1), \ldots, F_{X_n}(x_n)). \tag{7.17}$$

If $F_{\mathbf{X}}$ has continuous marginals, then the copula C is unique. Conversely, if C is a copula and $F_{X_1}, \ldots, F_{X_n}$ are univariate CDFs, then the mapping given in (7.17) yields a multivariate CDF with those marginals.

The proof of Theorem 7.5 is simplified in the 'if part' by assuming continuous marginals, while the 'only if part' is a simple extension to n-dimensions of the corresponding proof given for the bi-variate case, see the paragraph immediately after the statement of Theorem 7.4.

Proof ($\Longrightarrow$) Assume $\mathbf{X}$ has CDF $F_{\mathbf{X}}$ with continuous marginals $F_{X_1}, \ldots, F_{X_n}$. Thus, let $U_i = F_{X_i}(X_i)$, for $i = 1, \ldots, n$ so by Theorem 5.2 and Remark 2.11, Sect. 2.8 we have:

$$Q_{X_i}(U_i) = Q_{X_i}(F_{X_i}(X_i)) \overset{\text{a.s.}}{=} X_i, \quad \text{for all } i \in \{1, \ldots, n\}.$$

Thus, if C is the CDF of the uniform random vector $\mathbf{U} = (U_1, \ldots, U_n)$, then by Definition 7.4 and Property 3, Proposition 2.2, for every $\mathbf{x} \in \mathbb{R}^n$ we compute

$$\begin{aligned} F_{\mathbf{X}}(x_1, \ldots, x_n) &= \mathsf{P}(X_1 \leqslant x_1, \ldots, X_n \leqslant x_n) \\ &= \mathsf{P}(Q_{X_1}(U_1), \ldots, Q_{X_n}(U_n)) \\ &= \mathsf{P}(U_1 \leqslant F_{X_1}(x_1), \ldots, U_n \leqslant F_{X_n}(x_n)) \\ &= C(F_{X_1}(x_1), \ldots, F_{X_n}(x_n)), \end{aligned}$$

therefore C exists. Using again Remark 2.11, for $x_i = Q_{X_i}(u_i)$, each i, we get $F_{X_i}(x_i) = Q_{X_i}(u_i) = u_i$ and

$$F_{\mathbf{X}}(Q_{X_1}(u_1), \ldots, Q_{X_n}(u_n)) = C(u_1, \ldots, u_n).$$

We conclude that the copula C is given in terms of the multivariate CDF and its marginal quantiles, which entails uniqueness. ■

Eventually, we provide the following:

Proof of Proposition 7.3 The pair $\mathbf{Y} = (f_1(X_1), f_2(X_2))$ is a new random vector, and since f_i are strictly increasing,[8] each i, the coordinate $f_i(X_i)$ are rvs which implies $\mathbf{Y}$ is indeed a random vector. From the last paragraph in the ('only if part') proof of Theorem 7.5, we can rewrite the expression of C by observing that $\{X_i \leqslant x_i\} = \{f_i(X_i) \leqslant f_i(x_i)\}$:

$$\begin{aligned} C(u_1, \ldots, u_n) &= \mathsf{P}(X_1 \leqslant Q_{X_1}(u_1), \ldots, X_n \leqslant Q_{X_n}(u_n)) \\ &= \mathsf{P}(f_1(X_1) \leqslant f_1(Q_{X_1}(u_1)), \ldots, f_n(X_n) \\ &\qquad \leqslant f_n(Q_{X_n}(u_n))) \\ &\underset{\text{Proposition 2.3}}{=} \mathsf{P}\left(f_1(X_1) \leqslant Q_{f_1(X_1)}(u_1), \ldots, f_n(X_n)\right. \\ &\qquad \left. \leqslant Q_{f_n(X_n)}(u_n)\right). \end{aligned}$$

This shows that C as given above is the unique copula of $\mathbf{Y}$, and we are done. ■

[8] They are Borel-measurable.

Chapter 8
Convergence Concepts



8.1 Type of Convergence

Sequences of rvs are important in probability since they reveal interesting behavior essential for mathematical statistics, especially the theory of large sample.[1] The reason is that sequences of rvs or their partial sums enable us to connect patterns of real world data with named distributions. Suppose that $X_1, \ldots, X_n$ are IID rvs representing possible values of a *population* rv X with the same distribution. If we want to infer information on the X's law, then we might use realizations from each X_i, whose collection is termed a **random sample**. Suppose, we try to model the unknown distribution of a 1 day stock return X using $n > 2$ historical daily returns $x_1, \ldots, x_n$ collected within a past time horizon, say $[t_{m-1}, t_m]$ for $m \in \mathbb{N}$. Anytime we change the time horizon switching to another not overlapping time interval, say $[t_{m-3}, t_{m-2}]$, s.t. we keep collecting $n > 2$ historical returns $y_1, \ldots, y_n$ different from the previous ones, then we thought of x_i, y_i as two different realizations of rvs X_i, for $i = 1, \ldots, n$. In words, the ith rv X_i represents all possible historical returns $x_i, y_i, \ldots$, not necessarily with the same value, for fixed i. Moreover, we need to known what happen when the number n, called **sample size**, will increase and we gather more information about X via its copies X_i. In situation like this, we are interested in study the limiting behavior of $X_1, \ldots, X_n$ for $n \to \infty$, excluding the trivial pointwise convergence discussed in Appendix to Chap. 4.

Example 8.1 Let $X_1, \ldots, X_n$ a random sample of n IID daily observations drawn from the price S_t of the S&P500, say $t = 1$ day, acting as the population rv. We should believe that some model of rv with a specific parametric distribution well describe S_t, but we decide to handle the Gaussian the $\mathrm{N}(\mu, \sigma^2)$, under normal market conditions, by considering the log-normal return $Y = \ln(\frac{S_t}{S_0})$, where S_0

[1] Synonymous for this are *asymptotic theory* or *stochastic limit theory*.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_8

is the initial known quote of the S&P500, i.e., $Y \sim \mathrm{N}(\mu, \sigma^2)$ iff $S_t = S_0 \mathrm{e}^Y$ has the log-normal distribution with the same parameters. Doing this, we aim at recover the S_t's unknown distribution via the inverse transform $\exp(\ln(\frac{S_t}{S_0}))$. We gather actual data $x_1, \ldots, x_n$ for a fixed time horizon, and obtain the corresponding time series of log-returns $\ln(\frac{x_2}{x_1}), \ln(\frac{x_3}{x_2}), \ldots, \ln(\frac{x_n}{x_{n-1}})$ whose population counterpart is the random sample $Y_1, \ldots, Y_n$. To infer values of μ and σ^2 we can consider the **sample mean** $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^n Y_i$ and the **sample variance** $\bar{V}_n = \frac{1}{n}\sum_{i=1}^n (Y_i - \bar{Y}_n)^2$. They are **statistics**, meaning that a function $T(Y_1, \ldots, Y_n)$ has been defined and a possible value is obtained by evaluating $T(y_1, \ldots, y_n)$, called **estimate**, at a particular sample of historical log-returns. Clearly, we wish increasing n in such a way estimates of the sample mean and variance become more accurate and approach the true (unknown) values of μ and σ^2.

The question of approximating unknown statistics[2] requires defining some convergence concepts for rvs.

Example 8.2 Consider the sample mean $\bar{Z} = \frac{1}{n}\sum_{i=1}^n \frac{Y_i - \mu}{\sigma}$ and variance $\frac{1}{n}\sum_{i=1}^n (\frac{Y_i - \mu}{\sigma} - \bar{Z})^2$ obtained from those in Example 8.1 by standardizing (recall that we treat random samples, i.e., IID rvs having the same distribution and then the same population's parameters). Note that we average over IID rvs which are standard normal distributed. In `R` we have simulated two samples `z1` of length `n1` and `z2` of length `n2`, from a standard normal distribution; we have calculated their sample mean `m1`, `m2` and variance `v1`,`v2`, see the program code below. Visualizing their respective values via the last four commands, we have that the sample mean decreases from -0.08834072 to -0.007981814 and approaches $\mu = 0$, besides the sample variance also decreases from 1.171443 to 0.9927758 approaching $\sigma^2 = 1$.

Program Code

```
n1 <- 10^2; n2 <- 10^5
mu <- 0; sigma <- 1
z1 = rnorm(n1,mu,sigma); z2 = rnorm(n2,mu,sigma)
m1 = mean(z1); m2 = mean(z2); v1 = var(z1); v2 = var(z2)
m1 #..
v1 #..
m2 #..
v2 #..
```

Convergence of rvs can happen in different ways, in contrast to convergence of real-valued sequences.

Example 8.3 Given a sequence $X_1, X_2, \ldots$ of IID rvs, where each $X_n \sim \mathrm{N}(\mu, \frac{\sigma^2}{i})$ for $n = 1, 2, \ldots$. Statements like '$X_n \to 0$ as $n \to \infty$' requires a careful

[2] Other forms of the function T can be considered.

examination of the limiting behavior, since for continuous rvs $\mathsf{P}(X_n = 0) = 0$ and we are likely to rule out that increasing the sample size n will lead to have any X_n more concentrated around zero, almost in the probabilistic sense.

Let $X_1, X_2, \ldots$ be a sequence of rvs and let X be another rv, whose CDFs are F_{X_n} for $n = 1, 2, \ldots$ and F_X, respectively. We define the following:

1. **Convergence in distribution**: $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$, at every x at which the limiting CDF F_X is continuous,[3] written $X_n \xrightarrow{\mathrm{d}} X$.
2. **Convergence in probability**: $\lim_{n\to\infty} \mathsf{P}(|X_n - X| > \epsilon) = 0$, for every $\epsilon > 0$, written $X_n \xrightarrow{\mathsf{P}} X$.
3. **Convergence almost sure (a.s.)**: $\mathsf{P}(\lim_{n\to\infty} X_n = X) = 1$, written $X_n \xrightarrow{\text{a.s.}} X$.
4. **Convergence in mean square**: $\lim_{n\to\infty} \mathsf{E}((X_n - X)^2) = 0$, written $X_n \xrightarrow{\text{ms}} X$.

Convergence a.s. is the strongest one and convergence in probability is stronger than convergence in distribution. In a sequence converging in distribution, each X_n can be defined over a different probability space than X. Convergence in probability and a.s. make sense only if all the rvs involved are defined over the same probability space. In is worth noting that convergence in probability can be restated as:

- $\lim_{n\to\infty} \mathsf{P}(|X_n - X| \leqslant \epsilon) = 1$, for every $\epsilon > 0$;
- $|X_n - X| \xrightarrow{\mathsf{P}} 0$.
- $\mathsf{P}(|X_n - X| > \epsilon) < \delta$, for all $n \geqslant m$, any $\epsilon > 0$ and $\delta > 0$ with integer $m := m(\delta)$ depending on δ.

Convergence in mean square can be generalized to convergence in pth moment $\lim_{n\to\infty} \mathsf{E}(|X_n - X|^p) = 0$, written $X_n \xrightarrow{L^p} X$. A special case is $\lim_{n\to\infty} \mathsf{E}(|X_n - X|) = 0$, written $X_n \xrightarrow{L^1} X$ and called $\boldsymbol{L^1}$ **convergence**. Observe that L^p convergence needs $(X_n)_{n\in\mathbb{N}} \subset L^p$ and $X \in L^p$. Further, we note that

$$
\begin{aligned}
|\mathsf{E}(X_n - X)|^p \leqslant \mathsf{E}(|X_n - X|^p) \quad & \text{by Jensen's inequality} \\
& \text{since } f(x) = |x|^p \text{ is convex and} \\
|\mathsf{E}(|X_n| - \mathsf{E}(|X|))|^p \leqslant \mathsf{E}(|X_n - X|^p) \quad & \text{since } ||x_n| - |x|| \leqslant |x_n - x|,
\end{aligned}
$$

whence L^p convergence does imply convergence of moments, $\lim_{n\to\infty} \mathsf{E}(X_n) = \mathsf{E}(X)$, and absolute moments, $\lim_{n\to\infty} \mathsf{E}(|X_n|) = \mathsf{E}(|X|)$, which can be of some importance when handling L^p spaces of rvs. In all types of convergence, a special case arises when the limiting rv is degenerate, $X = c$, with a point mass at c and

[3] These are called *continuity points*. The set containing continuity points is dense in $\mathbb{R}$.

$\mathsf{P}(X = c) = 1$. Convergence in distribution is also called **weak convergence**, and it can be rephrased as $\mathsf{P}_{X_n}(B) \to \mathsf{P}_X(B)$, as $n \to \infty$, for every suitable (Borel) set whose boundary[4] ∂B is s.t. $\mathsf{P}_X(\partial B) = 0$.

Example 8.4 Proposition 5.1 in Sect. 5.5 asserts that the sequence $(X_n)_{n\in\mathbb{N}}$ with $X_n \sim \text{Bin}(n, \frac{\lambda}{n})$ converges in distribution to the $\text{Po}(\lambda)$, i.e., using that both the distribution of each X_n and the limiting distribution of X are discrete, we have $\mathsf{P}(X_n = x) \to \mathsf{P}(X = x)$ for every point $x \in \{0, 1, 2, \ldots\}$ where both laws have atoms.

Convergence in probability means that events $\{|X_n - X| > \epsilon\}$ describing the 'error' made by approximating X with elements X_n in the sequence, exceeding a fixed threshold ϵ, have smaller probability as n increases. Here, the asserted convergence pertains to sequence of real numbers $\mathsf{P}(|X_1 - X| > \epsilon), \mathsf{P}(|X_2 - X| > \epsilon), \ldots$.

Example 8.5 Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of rvs converging in distribution to the point mass distribution $F_X(x) = \mathbf{I}_{\{x\geqslant c\}}$. The event $\{|X_n - c| > \epsilon\}$ is equal to the disjoint union $\{X_n < c - \epsilon\} \cup \{X_n > c + \epsilon\}$, and $\{X_n < c - \epsilon\} \subset \{X_n \leqslant c - \epsilon\}$. Then, we have:

$$\begin{aligned}
\mathsf{P}(|X_n - c| > \epsilon) &= \mathsf{P}(X_n < c - \epsilon) + \mathsf{P}(X_n > c + \epsilon) \\
&\leqslant \mathsf{P}(X_n \leqslant c - \epsilon) + \mathsf{P}(X_n > c + \epsilon) \\
&= F_{X_n}(c - \epsilon) + 1 - F_{X_n}(c + \epsilon) \\
&\overset{n\to\infty}{\to} F_X(c - \epsilon) + 1 - F_X(c + \epsilon) \\
&= 0 + 1 - 1 = 0.
\end{aligned}$$

This shows that $X_n \xrightarrow{\mathrm{d}} c$, where $\mathsf{P}(X = c) = 1$, does imply $X_n \xrightarrow{\mathsf{P}} X$.

Typically, convergence types are related each other via the 'right' implication.

Theorem 8.1 *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of rvs and let X be a rv. Then*

(i) $X_n \xrightarrow{a.s.} X \Longrightarrow X_n \xrightarrow{\mathsf{P}} X$;
(ii) $X_n \xrightarrow{\mathsf{P}} X \Longrightarrow X_n \xrightarrow{d} X$;
(iii) $X_n \xrightarrow{ms} X \Longrightarrow X_n \xrightarrow{\mathsf{P}} X$;
(iv) $X_n \xrightarrow{ms} X \Longrightarrow X_n \xrightarrow{L^1} X$;
(v) $X_n \xrightarrow{L^1} X \Longrightarrow X_n \xrightarrow{\mathsf{P}} X$.

Exercise 8.5 shows that the reverse implication of item (ii) holds for a degenerate rv $X = c$. Convergence types can be generalized to sequences of random vectors

[4] The boundary of any $B \subset \mathbb{R}$ is the set difference $\bar{B} \setminus \mathring{B}$ between the closure and the interior of B. Equivalently (in $\mathbb{R}$ and more general metric spaces), it contains limit points of sequence in B^c.

$(\mathbf{X}_n)_{n\in\mathbb{N}}$. Convergence in distribution is for multivariate CDFs and the continuity points $\mathbf{x}$ of the limiting multivariate CDF. For convergence in probability, the absolute value $|X_n - X|$ is replaced by the Euclidean distance $d(\mathbf{X}_n, \mathbf{X}) = \|\mathbf{X}_n, \mathbf{X}\|$. For convergence in mean square expectation is understood as the vector of scalar expectations. Be careful that for $\mathbf{X}_n = (X_{1n}, \ldots, X_{nn})$ convergence in probability, in mean square and in L^1 is equivalent to convergence of components $(X_{in})_{n\in\mathbb{N}}$ in the same sense, for $i = 1, \ldots, n$.

Remark 8.1 For random vectors $\mathbf{X}_n = (X_{1n}, \ldots, X_{nn})$ convergence $\mathbf{X}_n \xrightarrow{\mathrm{d}} \mathbf{X}$ is not equivalent to convergence $X_{in} \xrightarrow{\mathrm{d}} X_i$ of components X_{in},X_i in $\mathbf{X}_n, \mathbf{X}$, respectively. Indeed, marginal CDFs $F_{X_{in}}$, F_{X_i} do not determine joint CDFs $F_{\mathbf{X}_n}$, $F_{\mathbf{X}}$ due to possible stochastic dependence, see Sect. 3.4. What is true is that convergence in distribution $\mathbf{X}_n \xrightarrow{\mathrm{d}} \mathbf{X}$ implies convergence of the components $X_{in} \xrightarrow{\mathrm{d}} X_i$, but the converse is not necessarily true, see Appendix to Chap. 8 for a partial converse.

There are two partial converse of Theorem 8.1: the first to items (i) and the second to items (iii),(v) in the general version $X_n \xrightarrow{L^p} X$, whose special cases are for $p = 1, 2$, respectively.[5]

Theorem 8.2 *Assume $X_n \xrightarrow{\mathsf{P}} X$ and suppose $|X_n| \leqslant Y$, for a rv $Y \in L^p$. Then:*

(a) there exists a subsequence $(X_{n_k})_{k\in\mathbb{N}}$ of rvs s.t. $X_{n_k} \xrightarrow{a.s.} X$, for $k \to \infty$;
(b) $|X| \in L^p$ and $X_n \xrightarrow{L^p} X$.

Convergences in distribution, probability and a.s. are preserved under continuous transformations.

Theorem 8.3 (Continuous Mapping) *Let $g : \mathbb{R} \to \mathbb{R}$ be a function which is continuous within a suitable $B \subset \mathbb{R}$ s.t. $\mathsf{P}(X \in B) = 1$. Then we have:*

1. If $X_n \xrightarrow{d} X$, $\Longrightarrow g(X_n) \xrightarrow{d} g(X)$.
2. If $X_n \xrightarrow{\mathsf{P}} X$, $\Longrightarrow g(X_n) \xrightarrow{\mathsf{P}} g(X)$.
3. If $X_n \xrightarrow{a.s.} X$, $\Longrightarrow g(X_n) \xrightarrow{a.s.} g(X)$.

Some convergence are preserved under arithmetic operations.

Theorem 8.4 *Let X_n, X Y_n, Y be rvs and $c \in \mathbb{R}$. Then we have:*

(a) If $X_n \xrightarrow{\mathsf{P}} X$ and $Y_n \xrightarrow{\mathsf{P}} Y$, $\Longrightarrow X_n + Y_n \xrightarrow{\mathsf{P}} X + Y$ and $X_n Y_n \xrightarrow{\mathsf{P}} XY$;
(b) If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, $\Longrightarrow X_n + Y_n \xrightarrow{d} X + c$ and $X_n Y_n \xrightarrow{d} Xc$;
(c) If $X_n \xrightarrow{L^p} X$ and $Y_n \xrightarrow{L^p} Y$, $\Longrightarrow X_n + Y_n \xrightarrow{L^p} X + Y$, for $p = 1, 2$;
(d) If $X_n \xrightarrow{ms} X$ and $Y_n \xrightarrow{L^p} Y$, $\Longrightarrow X_n Y_n \xrightarrow{L^1} XY$, for $p = 1, 2$.

[5] For a proof see [8, Th. 17.3, 17.4].

Part (b) is known[6] as *Slutsky's lemma*, and provided $c \neq 0$ we also have $\frac{X_n}{Y_n} \xrightarrow{\mathrm{d}} \frac{X}{c}$.

Remark 8.2 Recall that $L^2 \subset L^1$ and $X_n Y_n$ need not belong to L^2 even if X_n, Y_n do, for every $n \in \mathbb{N}$. On the other hand, if $X_n Y_n \in L^2$, then the product does belong to L^1 so $X_n \xrightarrow{L^1} X$ and $Y_n \xrightarrow{L^1} Y$ is implied by L^2 convergence of factors.

Convergence in distribution can be checked using CHF $\phi_\bullet(u) = \mathsf{E}(\mathrm{e}^{\mathrm{i}u\bullet})$.

Theorem 8.5 (Levy's Continuity Theorem) *Let X_n, X be rvs with CHFs $\phi_{X_n}(u), \phi_X(u)$. Then $X_n \xrightarrow{d} X$ iff $\lim_{n\to\infty} \phi_{X_n}(u) = \phi_X(u)$ for all $u \in \mathbb{R}$. If $\lim_{n\to\infty} \phi_{X_n}(u) = f(u)$ for every $u \in \mathbb{R}$, where $f(u)$ is continuous at zero, then $f(u)$ is the CHF $\phi_X(u)$ of a rv X for which $X_n \xrightarrow{d} X$.*

In the second part of Levy's continuity theorem, it suffices to have a sequence of CHFs converging pointwise to a function that it is the CHF of some rv, to which the sequence of rvs underlying the CHFs will converge in distribution. Note that all functions $u \mapsto \phi_\bullet(u)$ are continuous and bounded. Theorem 8.5 is the reason why the fifth item of Theorem 4.2 in Sect. 4.4 asserts that the distribution of rvs depends upon their CHFs. More generally, convergence in distribution can be equivalently stated as

$$\lim_{n\to\infty} \mathsf{E}(g(X_n)) = \mathsf{E}(g(X)),$$

for every bounded continuous function $g : \mathbb{R} \to \mathbb{R}$, and as

$$\lim_{n\to\infty} \mathsf{E}(g(\mathbf{X}_n)) = \mathsf{E}(g(\mathbf{X})),$$

for every bounded continuous function $g : \mathbb{R}^n \to \mathbb{R}$, see Appendix to Chap. 8.

8.2 The Law of Large Numbers and the Central Limit Theorem

The celebrated law of large number (LLN) is based on theorems stating conditions under which sample moments converge to their population counterparts. Here a version with weaker hypotheses.

[6] The version for random vector requires $\mathbf{c} \in \mathbb{R}^n$.

Theorem 8.6 (Weak Law of Large Numbers (WLLN)) *Let $Y_1, Y_2, \ldots$ be a sequence of IID rvs, each $X_i \in L^2$, with $\mathsf{E}(Y_1) = \mu$ and $\mathsf{V}(Y_1) = \sigma^2$. Then*

$$\lim_{n\to\infty} \mathsf{P}\Big(\Big|\frac{1}{n}(Y_1 + \cdots + Y_n) - \mu\Big| \geqslant \epsilon\Big) = 0,$$

for every $\epsilon > 0$.

Thus, WLLN asserts that $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^n Y_i \xrightarrow{\mathsf{P}} \mu$, i.e., the sample mean converges in probability to the population expectation μ.

Proof For the sample mean we have $\mathsf{E}(\bar{Y}_n) = n\mu$ and, by independence, $\mathsf{V}(\bar{Y}_n) = \frac{\sigma^2}{n}$. Then, by Chebyshev's inequality we have

$$\mathsf{P}\Big(\,|\bar{Y}_n - \mu| > \epsilon\Big) \leqslant \frac{\sigma^2}{\epsilon^2 n},$$

and letting $n \to \infty$ the proof is complete. ■

Remark 8.3 Observe that $\mathsf{E}((\bar{Y}_n - \mu)^2) = \mathsf{E}((\bar{Y}_n)^2) - 2\mu\mathsf{E}(\bar{Y}_n) + \mu^2$, and since $\mu = \mathsf{E}(\bar{Y}_n)$ we recover the variance $\mathsf{V}(\bar{Y}_n)$ of the sample mean.

Another version is:

Theorem 8.7 (Strong Law of Large Numbers (SLLN)) *Let $Y_1, Y_2, \ldots$ be a sequence of IID rvs, each $X_i \in L^1$, with $\mathsf{E}(Y_1) = \mu$. Then $\bar{Y}_n \xrightarrow{a.s.} \mu$.*

Theorem 8.7 does not require finite second moments and the involved convergence is a.s. For a proof see [10, Th 7.7], and for yet other versions see [2].

Example 8.6 Let $K_n = \sum_{i=1}^n Z_i^2$ for IID $Z_i^2 \sim \mathrm{N}(0,1)$, thus by the SLLN $K_n/n \xrightarrow{\text{a.s.}} \mathsf{E}(Z_1^2) = 1$. Thus, let $Z \sim \mathrm{N}(0,1)$ be independent of all Z_i and let $T_n = \frac{Z}{\sqrt{Z_n/n}}$, for every $n \in \mathbb{N}$. By construction, $T_n \sim T(n)$ and since the denominator goes to 1 for n large enough $T_n \xrightarrow{\text{d}} Z$.

The LLN suggests how the sample mean approximates the population counterpart for n large enough, but probability statements about $\bar{Y}_n$ need another important result.

Theorem 8.8 (Central Limit Theorem (CLT)) *Let $Y_1, Y_2, \ldots$ be a sequence of IID rvs, each $X_i \in L^2$, with $\mathsf{E}(Y_1) = \mu$ and $\mathsf{V}(Y_1) = \sigma^2$. Then*

$$\lim_{n\to\infty} \mathsf{P}\Big(\frac{Y_1 + \cdots + Y_n - n\mu}{\sqrt{n}\sigma} \leqslant x\Big) = \mathrm{N}(0,1). \tag{8.1}$$

It is worth noting that $\sum_{i=1}^{n} Y_i = n\bar{Y}_n$, then:

$$\frac{\sum_{i=1}^{n} Y_i - \mathsf{E}(\sum_{i=1}^{n} Y_i)}{\sqrt{\mathsf{V}(\sum_{i=1}^{n} Y_i)}} = \frac{n\bar{Y}_n - n\mu}{\sqrt{n^2\mathsf{V}(\bar{Y}_n)}} = \frac{n\bar{Y}_n - n\mu}{\sqrt{n}\sigma}$$

$$= \frac{\sqrt{n}(\bar{Y}_n - \mu)}{\sigma}.$$

Hence, the CLT asserts that the standardized sample mean converges in distribution to a standard normal,[7] namely $\frac{\sqrt{n}(\bar{Y}_n-\mu)}{\sigma} \xrightarrow{\mathrm{d}} \mathrm{N}(0,1)$. By the property of the Gaussian distribution, we deduce the following approximation which are equivalent to statement of Theorem 8.8 (viz. the necessary part) for n large enough: $\sqrt{n}(\bar{Y}_n - \mu) \approx \mathrm{N}(0, \sigma^2)$, or $\bar{Y}_n - \mu \approx \mathrm{N}(0, \frac{\sigma^2}{n})$, or also $\bar{Y}_n \approx \mathrm{N}(\mu, \frac{\sigma^2}{n})$. For stronger version of the CLT see [8, 10, Sec 7.3] or [2]. We can prove Theorem 8.8 using the CHF as a device, see Appendix in Chap. 8. Whenever the CLT holds, we might be interested in proving convergence in distribution for some transforms. The following result helps us in such a situation.

Theorem 8.9 (Delta Method) *With the same notation as in Theorem 8.8, assume that* $\frac{\sqrt{n}(\bar{Y}_n-\mu)}{\sigma} \xrightarrow{d} \mathrm{N}(0,1)$. *Let* $f : \mathbb{R} \to \mathbb{R}$ *be a differentiable function s.t.* $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\big|_{x=\mu} =: f'(\mu) \neq 0$. *Then:*

$$\frac{\sqrt{n}(f(\bar{Y}_n) - f(\mu))}{|f'(\mu)|\sigma} \xrightarrow{d} \mathrm{N}(0,1).$$

For the multivariate extension see [15, Ch 3].

Example 8.7 Let $R_1, \ldots, R_n$ be IID daily log-returns over n trading days, with $R_t \in L^2$, for $t = 1, \ldots, n$, and $\mathsf{E}(R_t) = \mu$, $\mathsf{V}(R_t) = \sigma^2$. We have $\frac{\sqrt{n}(\bar{S}_n-\mu)}{\sigma} \xrightarrow{\mathrm{d}} \mathrm{N}(0,1)$, where $\bar{S}_n = \frac{1}{n}\sum_{t=1}^{n} S_t$ is the sample mean return. If we pass to $f(\bar{S}_n) = \mathrm{e}^{\bar{S}_n}$, i.e. to the accumulation factor over the horizon $[0, n]$, with average log-return per day, then $\bar{S}_n \approx \mathrm{N}(\mu, \frac{\sigma^2}{n})$ and as a consequence $f(\bar{S}_n) \approx \mathrm{N}(f(\mu), (f'(\mu))^2\frac{\sigma^2}{n})$. The latter give $\approx \mathrm{N}(\mathrm{e}^{\mu}, \mathrm{e}^{2\mu}\frac{\sigma^2}{n})$.

There is a multivariate version of the CLT, for a proof see [2].

Theorem 8.10 (Multivariate CLT) *Let* $\mathbf{Y}_1, \mathbf{Y}_2, \ldots$ *be a sequence of IID random vectors, with* $\mathsf{E}(\mathbf{Y}_i) = (\mathsf{E}(Y_{i1}), \ldots, \mathsf{E}(Y_{in})) = \boldsymbol{\mu} = (\mu_1, \ldots, \mu_n)$, *each* i, *and* $n \times n$ *covariance matrix* $\boldsymbol{\Sigma} = [c_{i,j}]$, *where* $c_{ii} = \mathsf{V}(Y_{ik}) = \sigma_i^2$ *and* $c_{ij} = \mathsf{cov}(Y_{ik}, Y_{jk}) =$

[7] Be warned that probability statements about the sample mean can be approximated by the standard normal CDF. This by no means implies that the rv $\bar{Y}_n$ can be approximated by $Z \sim \mathrm{N}(0,1)$.

σ_{ij}, for every $k = 1, \ldots, n$ and each pair i, j. Let $\bar{\mathbf{Y}}_m = (\bar{Y}_1, \ldots, \bar{Y}_m)$, where each component is $\bar{Y}_j = \frac{1}{m}\sum_{j=1}^{m} Y_{ji}$. Then:

$$\lim_{m\to\infty} \mathsf{P}\Big(\sqrt{m}(\bar{\mathbf{Y}}_m - \boldsymbol{\mu}) \leqslant x\Big) = \mathrm{N}(\mathbf{0}, \boldsymbol{\Sigma}). \tag{8.2}$$

8.3 Application 1: Point and CDF Estimation

Suppose we drawn a random sample $R_1, \ldots, R_n$ of daily log-returns from a population's law P_X, i.e., R_i are IID and each is a copy of X (viz. same distribution). If we further assume $X \sim \mathrm{N}(\mu, \sigma^2)$, then the sample mean $\bar{R}_n = \frac{1}{n}\sum_{i=1}^{n} R_i$ and the sample variance $\bar{V}_n = \frac{1}{n}\sum_{i=1}^{n}(R_i - \bar{R}_n)^2$ could be considered as approximation of μ and σ^2, respectively. How good is this approximation? It is worth noting that the question remains unchanged if we drop the hypotheses of a parametric population law. In any case, we know that a statistics $T(R_1, \ldots, R_n)$ as a function of the random sample can be used to infer the true values of μ, σ^2, by specifying its functional form. The sample mean and variance are two such specification of $T(\cdot)$. We can collect n historical daily log-returns $c_1, \ldots, c_n$ and evaluate $T(\cdot)$ at the n-tuple $(c_1, \ldots, c_n)$ getting an estimated value of $\bar{R}_n$, $\bar{V}_n$ or whatever the statistics is. We call $T_n := T(R_1, \ldots, R_n)$ a **point estimator** of some parameter, say θ concerning a fixed parametric distribution, and this is the assumption we made together with $T_n \in L^2$. Observe that any random sample has its own parametric probability law $\mathsf{P}_{\mathbf{R}}$, where $\mathbf{R} = (R_1, \ldots, R_n)$. Turning to the CDF we write $F_{\mathbf{R}}(\mathbf{x}, \theta)$ to emphasize the parametric dependence. Since the daily log-returns are IID, we have that the CDF is equivalently given by $F_{R_1}(x_1, \theta) \cdots F_{R_n}(x_n, \theta)$.

Example 8.8 The function $T(R_1, \ldots, R_n) = \prod_{i=1}^{n} f(R_i, \theta)$ expresses the probability of observing R_i when the true value of the parameter if just θ, for IID R_i each with continuous density function $f(x_i, \theta)$. It is an estimator called *likelihood function*. The maximum likelihood estimator is the value θ maximizing $\prod_{i=1}^{n} f(R_i, \theta)$, written

$$\theta_n = \arg\max_{\theta\in\mathbb{R}} \prod_{i=1}^{n} f(R_i, \theta)$$

which is again an estimator as a function of $R_1, \ldots, R_n$. Usually, under mild assumptions, it is consistent, $\theta_n \xrightarrow{\mathsf{P}} \theta$. There is a related version, $\ln\theta_n$, called *log-likelihood function* which is maximized at the same value[8] of θ. This is used

[8] If a function attains a global maximum at a point, then any monotone transformation attains a maximum at the same point.

because, depending on the particular form of the densities involved, calculations with $\ln \theta_n$ can be easier.

We say that an estimator is *unbiased* if $\mathsf{E}(T_n) = \theta$, and we can let[9] $\mathsf{bias}(T_n) := \mathsf{E}(T_n) - \theta$ be the *bias* of the estimator. A point estimator T_n is consistent at θ if $T_n \xrightarrow{\mathsf{P}} \theta$. Also the estimator has its own probability law called *sampling distribution*. The standard deviation $\sqrt{\mathsf{V}(T_n)}$ is often used as measure of the error made in approximating θ with T_n, called *standard error*.

Example 8.9 Assume $X_i \sim \mathrm{Ber}(p)$ are IID, for $i = 1, 2, \ldots$. For an n-dimensional random sample we set $T_n := T(X_1, \ldots, X_n) = \bar{X}_n$, the sample mean. Then we have $\mathsf{E}(\bar{X}_n) = \frac{1}{n}\sum_{i=1}^n \mathsf{E}(X_i) = p = \theta$, then T_n is an unbiased estimator of the success probability, i.e. $\mathrm{bias}(T_n) = 0$. The standard error is $\sqrt{\mathsf{V}(T_n)} = \sqrt{\frac{p(1-p)}{n}}$. Switching to IID $X_i \sim \mathrm{N}(\mu, \sigma^2)$ we instead have $\mathsf{E}(\bar{X}_n) = \mu = \theta_1$, and the sample mean of a Gaussian sample is unbiased, besides the sample variance $\bar{V}_n = \frac{1}{n}\sum_{i=1}^n (X_i - \mu)^2$ has expectation $\mathsf{E}(\bar{V}_n) = n^{-1}\sum_{i=1}^n \mathsf{E}((X_i - \mu))^2 = \sigma^2 = \theta_2$ and then it is unbiased too. The variance of the sample mean is $\mathsf{V}(\bar{X}_n) = \frac{\sigma^2}{n}$. Observe that in this case we have a bi-variate parameter $\boldsymbol{\theta} = (\theta_1, \theta_2)$.

Defining $\mathsf{MSE} := \mathsf{E}((T_n - \theta)^2)$ as the *mean squared error*, we have a further measure of the quality of an estimator.

Lemma 8.1 $\mathsf{MSE} = (bias(T_n))^2 - \mathsf{V}(T_n)$. *Moreover, if* $bias(T_n) \to 0$ *and* $\sqrt{\mathsf{V}(T_n)} \to 0$, *as* $n \to \infty$, *then* $T_n \xrightarrow{\mathsf{P}} \theta$.

Hence, the less a point estimator T_n is biased and the less the standard error is, the more T_n is a good approximation of the unknown parameter θ, i.e., it is a consistent estimator. If we drop the assumption of a parametric population law, then the inference is said *nonparametric*. Thus, without assuming the population daily log-return X follows some parametric distribution, we can use *row* estimates of its unknown expectation and variance given by $\bar{c} = \frac{1}{n}\sum_{i=1}^n c_i$ and $\bar{v} = \frac{1}{n}\sum_{i=1}^n (c_i - \bar{c})^2$. The interesting thing now is that we can consider the whole unknown CDF as a 'parameter'. We define

$$\mathbb{F}_n(\omega, x) = \frac{1}{n}\sum_{i=1}^n \mathbf{I}_{\{R_i \leqslant x\}}(\omega), \tag{8.3}$$

to be the **empirical distribution function**, where we emphasize its dependence on both scenarios ω and rvs R_i. By varying the sample size n, the empirical distribution function will depend on n too, indeed it can be considered an indexed family of rvs known as *empirical process*.[10] Since R_i are IID, the indicators $\mathbf{I}_{\{R_i \leqslant x\}}$ are

[9] Expectation is taken with respect to the law P_X, or equivalently the CDF F_X. Clearly, in the discrete and continuous case we turn to the corresponding mass function or density.

[10] It fits the definition of a stochastic process, see Sect. 9.1.

IID $\text{Ber}(F_X(x))$, for every $x \in \mathbb{R}$. This is obvious by definition: if $\{R_i \leqslant x\}$ occurs, with probability $\mathsf{P}(R_i \leqslant x) = F_X(x)$, then the indicator takes value 1, otherwise is zero. Thus, $\mathbb{F}_n(\omega, x)$ can be used to estimate the true CDF $F_X(x)$ when it is unknown. If we collect a sample of historical daily log-returns $c_1, \ldots, c_n$, the empirical distribution function as a nonparametric estimator can be evaluated at the sample as $\bar{\mathbb{F}}_n = \frac{1}{n}\sum_{i=1}^n \mathbf{I}_{\{c_i \leqslant x\}}$ corresponding to a particular scenario ω. This is nothing else a sample mean, thus collecting more and more data the sample size n grows and by the WLLN we have $\mathbb{F}_n(\omega, x) \xrightarrow{\mathsf{P}} F_X(x)$. Clearly we have:

$$\begin{aligned}
\mathsf{E}(\mathbb{F}_n(\omega, x)) &= F_X(x), \\
\mathsf{V}(\mathbb{F}_n(\omega, x)) &= \mathsf{V}\left(\frac{1}{n}\sum_{i=1}^n \mathbf{I}_{\{R_i \leqslant x\}}(\omega)\right) \\
&= \frac{1}{n^2} n \mathsf{V}(\mathbf{I}_{\{R_i \leqslant x\}}(\omega)) = \frac{F_X(x)(1 - F_X(x))}{n}.
\end{aligned}$$

Hence, the empirical distribution function is unbiased, $\mathsf{E}(\mathbb{F}_n(\omega, x) - F_X(x)) = 0$, and by Lemma 8.1 its MSE equals $\mathsf{V}(\mathbb{F}_n(\omega, x))$.

Theorem 8.11 (Glivenko-Cantelli) *Let $R_1, R_2, \ldots$ be IID with common CDF F_X. Then $\sup_x |\mathbb{F}_n - F_X| \xrightarrow{a.s.} 0$, as $n \to \infty$.*

The Glivenko-Cantelli theorem says that for larger and larger samples the empirical distribution function converges to the true unknown CDF, and the convergence is uniform (does not depend on x). By Theorem 8.1, part (i), the convergence is also in probability. There is a relation between convergence in distribution and related quantiles of the distributions involved. It can be shown that $R_n \xrightarrow{\text{d}} X$ is equivalent to have $\lim_{n\to\infty} Q_{R_n}(c) = Q_X(c)$ at every point $c \in (0, 1)$ at which $Q_X(c)$ is continuous, see [15, Lemma 21.2]. There is an alternative definition to that of a statistics: a *statistical functional* $T(F_X)$ is a function acting on the set of CDFs.[11] Its role is analogous to $T(R_1, \ldots, R_n)$, but now we refer directly to the unknown CDF. Two examples are the expectation $\mu = T(F_X) = \int x \text{d}F_X(x)$ and variance $\int (x - \mu)^2 \text{d}F_X(x)$. Thus, we can recover some unknown parameter as $\theta = T(F_X)$. The plug-in estimator of θ is obtained by literally 'plugging in' the empirical distribution function, $T(\mathbb{F}_n)$. We drop the argument in $\mathbb{F}_n(\omega, x)$. It is possible to build a statistical functional by specifying the functional form of T. When we choose the integral against $\text{d}F_X(x)$ we say that $T(F_X) = \int g(x)\text{d}F_X(x)$ is a *linear functional*.[12] Interpreting $\mathbb{F}_n$ as the CDF that puts equal probability masses $\frac{1}{n}$ at each data point R_i we can consider the plug-in estimator of a linear statistical

[11] This set must include the empirical distribution functions and every point mass distribution.

[12] The term stems from functional analysis, studying vector spaces such as the set of all CDFs. In fact, $T(\alpha F_X + \beta F_Y) = \alpha T(F_X) + \beta T(F_Y)$, whenever $T(\cdot)$ is specified by an integral.

functional $T(F_X)$ as

$$T(\mathbb{F}_n) = \int g(x)\mathrm{d}\mathbb{F}_n = \frac{1}{n}\sum_{i=1}^{n} g(R_i).$$

Further examples of statistical functional can be found in chapters 5, 12, 20 and 22 of [15]. For any sample of log-returns, and in particular for IID rvs, we define

$$\begin{aligned} R_{(1)} &= \min\{R_1, \ldots, R_n\}, \\ R_{(2)} &= \text{second smallest of } R_1, \ldots, R_n, \\ &\vdots \\ R_{(n-1)} &= \text{second largest of } R_1, \ldots, R_n, \\ R_{(n)} &= \max\{R_1, \ldots, R_n\}. \end{aligned}$$

We call $R_{(i)}$ the ith *order statistics*, since it is a function of the sample. Note that by construction $R_{(1)} \leqslant \cdots, \leqslant R_{(n)}$. Using this we can define the *empirical quantile function* $\mathbb{F}_n^{-1}(c) = R_{(i)}$ for $c \in (\frac{i-1}{n}, \frac{i}{n}]$. In the IID case, it is easy to see that

$$\begin{aligned} F_{R_{(n)}}(x) &= \mathsf{P}(\max\{R_1, \ldots, R_n\} \leqslant x) \\ &= \mathsf{P}(R_1 \leqslant x, \ldots, R_n \leqslant x) \\ &= \mathsf{P}(R_1 \leqslant x) \cdots \mathsf{P}(R_n \leqslant x) = F_X(x), \end{aligned}$$

where again F_X is the unknown CDF s.t. $R_1, \ldots, R_n \sim F_X$. The first equality holds because the event $\{\max\{R_1, \ldots, R_n\} \leqslant x\}$ equals $\cap_{i=1}^{n}\{R_i \leqslant x\}$ and the others are due to independence. Similarly we get:

$$\begin{aligned} F_{R_{(1)}}(x) &= 1 - \mathsf{P}(\min\{R_1, \ldots, R_n\} > x) \\ &= 1 - \mathsf{P}(R_1 > x, \ldots, R_n > x) \\ &= 1 - (1 - F_X(x))^n. \end{aligned}$$

Then it can be shown that

$$\mathsf{P}(R_{(i)} \leqslant x) = \sum_{j=i}^{n} \binom{n}{j} F_X(x)^j (1 - F_X(x))^{n-j}.$$

A similar approach can be used for approximating an unknown density function. Let $X_1, X_2, \ldots$ IID distributed according to F_X, which admits the density f_X. Fix an interval $[a, b]$ with finite endpoints and split it into subintervals $I_1, \ldots, I_m$, then

$\sum_{i=1}^{n} \mathbf{I}_{\{X_i \in I_j\}}$ counts how many observations lie in I_j among the first n elements $X_1, \ldots, X_n$ of the sample, for $j = 1, \ldots, m$, and the rv

$$Y_n^j := \frac{1}{n} \sum_{i=1}^{n} \mathbf{I}_{\{X_i \in I_j\}},$$

expresses this in percentage. Assuming all subintervals have equal length, an *histogram* is a plot of m bar charts each having height proportional to the Y_n^j insisting on I_j. By the SLLN, as $n \to \infty$ we have

$$Y_n^j \xrightarrow{\text{a.s.}} \mathsf{E}(\mathbf{I}_{\{X_i \in I_j\}}) = \mathsf{P}(X_i \in I_j) = \int_{I_j} f_X(x)\mathrm{d}x,$$

which explains why we use histograms to approximate unknown density: for n large enough, each subinterval I_j shrinks and the corresponding bar approximating the value of $f_X(x)$ at $x \in I_j$. Another similar application refers to the *quantile plot*. Given IID $X_1, \ldots, X_n \sim F_X$ and for $B_x = (-\infty, x]$, with $\mathsf{E}(\mathbf{I}_{B_x}(X_1)) = F_X(x)$, we can see that

$$\frac{1}{n} \sum_{i=1}^{n} \mathbf{I}_{B_x}(X_i) \xrightarrow{\text{a.s.}} \mathsf{E}(\mathbf{I}_{B_x}(X_1)),$$

by the SLLN. Considering the ordered statistics $X_{(1)} \leqslant \cdots \leqslant X_{(n)}$ and the empirical distribution function $\mathbb{F}_n$ it is clear that

$$\mathbb{F}_n(\omega, x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{I}_{B_x}(X_i),$$

and $\mathbb{F}_n(\omega, x) = 0$ if $x < X_{(1)}$, it is $\mathbb{F}_n(\omega, x) = 1$ if $X_{(n)} \leqslant x$ and it is in between

$$\mathbb{F}_n(\omega, x) = \frac{i}{n}, \quad \text{if } x \in [X_{(i)}, X_{(i+1)}), \ \ i = 1, \ldots, n-1.$$

Thus, if we collect data $X_, \ldots, X_n$ and ask ourselves if they are generated by, say, $Z \sim \mathrm{N}(0, 1)$, since $\mathbb{F}_n = \frac{i}{n}$ at $X_{(i)}$ then the true $\frac{i}{n}$ and the theoretical counterpart $F_Z(X_{(i)})$ are near each other. Thus, starting from the quantile transform we have $Q_Z(\frac{i}{n})$ and $X_{(i)}$ tied together, and drawing a straight line with a (unit) slope of $\frac{\pi}{4}$ the pairs $(Q_Z(\frac{i}{n}), X_{(i)})$ lie 'along' the line. The resulting graphic is known as the *q-q plot*. Roughly speaking, given n historical returns $r_1, \ldots, r_n$ we can standardize them, $x_i = \frac{r_i - \bar{r}}{\bar{s}}$ for $i = 1, \ldots, n$, where $\bar{r} = n^{-1} \sum_{i=1}^{n} r_i$ is the sample mean and $\bar{s} = \sqrt{n^{-1} \sum_{i=1}^{n} (r_i - \bar{r})^2}$ is the sample standard deviation, then we arrange in increasing order $x_{(1)} < x_{(2)} < \cdots x_{(n)}$ and get the sample counterpart of the ordered statistics $X_{(1)}, \ldots, X_{(n)}$. Thus, we divide the x-axis into n bins expecting

one value $x_{(i)}$ in the ith bin provided that it is closest to the quantile $Q_Z(\frac{i}{n})$ of a standard normal. As byproduct, the smallest standardized return $x_{(1)}$ should be closed to $Q_Z(\frac{1}{n})$, the second smallest $x_{(2)}$ should be closed to $Q_Z(\frac{2}{n})$, and so on. For large sample size n, if the q-q plot[13] displays points lying approximately on a straight line of unit slope, then we can assume that historical returns are drawn from $Z \sim \mathrm{N}(0, 1)$. The `matlab` command `qqplot(x)` can be used to produce a q-q plot of input data contained in the vector `x`.

8.4 Application 2: Monte Carlo Simulation

Let $S_1, \ldots, S_n$ be daily stock prices and set $\theta = h(S_1, \ldots, S_n) = n^{-1} \sum_{t=1}^{n} S_t$, nothing else the sample mean of the rvs, s.t. $h(S_1, \ldots, S_n) \in L^1$. We can estimate θ by the following algorithm:

Input: $m \leftarrow$ integer
for $i = 1$ **to** m **do**
  simulate a vector $\mathbf{S}_i$ of n stock prices ;
  set $h(i)$ as the function $h(\mathbf{S}_i)$;
  compute the arithmetic average of $h(i)$
end

This kind of algorithm is at the heart of the so called *Monte Carlo methods*. It is worth noting that given the random vector $\mathbf{S}_i$ of n simulated daily stock prices we have $h(i) = h(\mathbf{S}_i)$ by the fourth step in the above code, then the latter step provides us with

$$\theta_n := \frac{h(1) + \cdots + h(m)}{m},$$

actually an average of averages. We can use θ_n as an estimator of θ, and indeed it is unbiased

$$\mathsf{E}(\theta_n) = \frac{\mathsf{E}\left(\sum_{i=1}^{m} h(\mathbf{S}_i)\right)}{m} = \frac{m\theta}{m} = \theta,$$

and by SLLN it is also consistent, $\theta_n \xrightarrow{\text{a.s.}} \theta$ as $n \to \infty$. If we specialize to the case of a random sample of stock prices, i.e., the IID hypotheses is considered, then Monte Carlo simulation can be used to estimate the unknown population mean using

[13] We plot ordered historical returns on the y-axis against the population standard normal quantiles on the x-axis.

the sample average,

$$\mu = \mathsf{E}(X) = \int_{-\infty}^{\infty} x \, \mathrm{d}F_X(x) \approx \frac{1}{n} \sum_{i=1}^{n} c_i,$$

where $c_1, \ldots, c_n$ is a realization of the sample and we have assumed[14] that $S_1, \ldots, S_n \sim F_X$. We can invoke Theorem 5.1, Sect. 5.6 for $X = Q_X(U)$ with $U \sim \mathrm{U}(0, 1)$, so that the sample mean can be approximated by $n^{-1} \sum_{t=1}^{n} Q_X(u_i)$, where $Q_X(u_i) = c_i$ and $u_1, \ldots, u_n$ are uniform random number on $[0, 1]$. This is possible because of:

Lemma 8.2 *Let* $h : [0, 1] \to \mathbb{R}$ *be continuous*[15] *s.t.* $\int_0^1 |h(x)| \mathrm{d}x < \infty$, *and consider IID uniform* $U_i \sim \mathrm{U}(0, 1)$, *for* $i = 1, 2, \ldots$. *Then* $n^{-1} \sum_{i=1}^{n} h(U_i) \xrightarrow{a.s.} \int_0^1 h(x) \mathrm{d}x$.

In fact, we have $\mathsf{E}(h(U_1)) = \int_0^1 h(x) \mathrm{d}x$ and considering $Y_i = h(U_i)$ with $\mathsf{E}(|Y_i|) = \int_0^1 |h(x)| \mathrm{d}x$ we apply SLLN, Theorem 8.7, to IID Y_i.

Example 8.10 (Option Pricing) Continuous time arbitrage theory provide us with the following formula for the price of a European call option:

$$C_0 = \mathrm{e}^{-rT} \mathsf{E}\left(\max\left\{S_0 \mathrm{e}^{(r-\sigma^2/2)T + \sigma\sqrt{T}Z} - K, 0\right\}\right),$$

where S_0 is the initial stock price of one share paying no dividend over the horizon $[0, T]$, K is the strike price, $S_0 \mathrm{e}^{(r-\sigma^2/2)T+\sigma\sqrt{T}Z}$ is the T-price of the stock, and r is the known continuously compounded return of a risk-free asset, see Sect. 2.5 in Chap. 2. Here, $Z \sim \mathrm{N}(0, 1)$. Estimate of σ^2 might be the sample variance $\hat{\sigma}^2$ of historical daily log-returns, then knowing the value S_0 and the strike price K, we can approximate T with the number of trading days in $[0, T]$ expressed in annual terms and run the simulation of n uniform random numbers $U_i \sim \mathrm{U}(0, 1)$ to compute $Z = Q_Z(U_i)$ and then the Monte Carlo estimate of C_0 will be:

$$\frac{1}{n} \mathrm{e}^{-rT} \sum_{i=1}^{n} \mathsf{E}\left(\max\left\{S_0 \mathrm{e}^{(r-\hat{\sigma}^2/2)T + \hat{\sigma}\sqrt{T}Q_Z(u_i)} - K, 0\right\}\right),$$

where $u_1, \ldots, u_n$ are the actual realizations of the uniform random sample.

[14] We also assume $X \in L^1$.

[15] This can be weakened by requiring $h(x)$ being Borel measurable.

Program Code

```
rand('state',0);
n=1000; T=0.5; S0=100; mu=0.15; sigma=0.2; K=92; r=0.01
U=rand(1,n); Z=norminv(U,0,1);
ST = S0 * exp((mu - (sigma^2)/2)*T + sigma*sqrt(T)*rand(1,n));
if ST>=K;
   C0=exp(-t*T)mean(ST - K*ones(1,n))
else C0=0;
end;
```

The code above is written in `Matlab`: command `rand('state',0)` initializes the uniform random number generator; the number of simulation of $Z \sim \mathrm{N}(0,1)$, the horizon, parameters μ, σ and the strike price are set in the second line; `U=rand(1,n)` generates 1000 uniform random numbers and `Z=norminv(U,0,1)` yields an array (row vector) of 1000 standard normal values. The function `ones(1,n)` yields an array (row vector) or 1000 ones. The fourth line sets the formula for the log-normal distributed stock price,[16] and the rest of the code provides a conditional calculation of the European call price.

8.5 Application to Log-Normal Stock Price Model

Assume you are interested in a single stock traded in the market. You are requested to build a sensible model of the S_T's probability distribution over the time horizon $[0, T]$. For each trading day, log-returns are good approximation of arithmetic returns, see Sect. 2.5. In fact, taking the Taylor series expansion[17] of $\ln(1+x)$ around $x = 0$ we have

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{2}{3!}x^3 - \cdots = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^n}{n},$$

[16] The horizon $T = 0.5$ is corresponds to 6 months, and the parameters μ, σ are expressed in annual terms.

[17] Equivalently, we can consider the Taylor series expansion of e^x near $x = 0$ up to the first-order term

$$\mathrm{e}^x = x + 1 + o(x),$$

where $o(x) \to 0$ as $x \to 0$ faster than x.

and observe that when $x \approx 0$, then powers $\frac{x^n}{n}$ vanishes rapidly for $n > 1$, which is the case for historical daily returns x. Recalling that $r_T = \frac{S_T - S_0}{S_0}$, we have:

$$\ln(1 + r_T) = \ln\left(\frac{S_T}{S_0}\right) \approx r_T.$$

We are interested in improving this approximation for high-frequency returns, when the following question arises: Fix a time-horizon $[0, T]$ and suppose you might invest a monetary amount in $t = 0$ and hold the investment until $t = T$. Is this a good strategy? An answer to this question requires probability statements about S_T, i.e., we need to recover the distribution of the final stock price. Our strategy based on taking higher frequency returns than daily ones, starts firstly by decomposing $[0, T]$ into $n \in \mathbb{N}$ subintervals of equal length $\Delta = \frac{T}{n}$ and then consider:

$$\begin{aligned} g_T = \frac{S_T}{S_0} &= \frac{S_{t_n}}{S_{t_{n-1}}} \frac{S_{t_{n-1}}}{S_{t_{n-2}}} \cdots \frac{S_{t_1}}{S_{t_0}} \\ &= g_{t_n} g_{t_{n-1}} \cdots g_{t_1}, \end{aligned}$$

where $t_i = i\Delta$ for $i = 0, 1, 2, \ldots, n$, $t_0 = 0$ and $t_n = T$. Next, we write

$$\begin{aligned} \ln(g_T) &= \ln\left(g_{t_n} g_{t_{n-1}} \cdots g_{t_1}\right) \\ &= \ln(g_{t_n}) + \ln(g_{t_{n-1}}) + \cdots + \ln(g_{t_1}) \\ &= \ln\left(\frac{S_T}{S_0}\right) \approx r_T \end{aligned}$$

which implies $S_T = S_0 \mathrm{e}^{R_T} \approx S_0(1 + r_T)$, where $R_T = \ln\left(\frac{S_T}{S_0}\right)$ is the log-return at T. Now, sending n to ∞ implies $\Delta \to 0$ since

$$\lim_{n\to\infty} \Delta = \lim_{n\to\infty} \frac{T}{n} = 0, \qquad T \text{ fixed.}$$

But $T = n\Delta$, then if both $n \to \infty$ and $\Delta \to 0$ we have that n and $\frac{1}{\Delta}$ are of the same order. Moreover, assume $\mathsf{E}(\ln(g_{t_i})) = \mu\Delta$ and $\mathsf{V}(\ln(g_{t_i})) = \sigma^2\Delta$, where $\mu \in \mathbb{R}$ and $\sigma > 0$. These represent the continuously compound expected return and volatility of a log-return as proxies for r_{t_i}, each $i = 1, \ldots, n$. From stylized facts about daily log-returns, the latter assumptions are meant as proportional rules for instantaneous returns over each subintervals. The expected return and volatility over the whole

time horizon are given respectively by:

$$\mathsf{E}(\ln(g_T)) = \mathsf{E}\left(\sum_{i=1}^{n} \ln(g_{t_i})\right) \sum_{i=1}^{n} \mathsf{E}(\ln(g_{t_i}))$$

$$\mathsf{V}(\ln(g_T)) = \mathsf{V}\left(\sum_{i=1}^{n} \ln(g_{t_i})\right) \sum_{i=1}^{n} \mathsf{V}(\ln(g_{t_i})).$$

The latter equation is based on the additional assumption that periodic log-returns are IID rvs, again based on stylized facts suggesting little auto-correlation among historical daily log-returns as a proxy of independence and their frequency histograms: they are almost similar through successive time windows. By standardization

$$\frac{\ln(g_T) - \mathsf{E}(\ln(g_T))}{\sqrt{\mathsf{V}(\ln(g_T))}} = \frac{\sum_{i=1}^{n} \ln(g_{t_i}) - \mu T}{\sigma\sqrt{T}},$$

as $n \to \infty$ log-returns are much more frequently observed within $[0, T]$ and by CLT the sample mean log-return $\bar{g}_n = n^{-1} \sum_{i=1}^{n} \ln(g_{t_i})$ is s.t. $\sqrt{T}\bar{g}_n \approx \mathrm{N}(\mu T, \sigma\sqrt{T})$, thus its CDF can be approximated by that of a Gaussian $\mu T + \sigma\sqrt{T}Z \sim \mathrm{N}(\mu T, \sigma^2 T)$ with standard normal $Z \sim \mathrm{N}(0, 1)$. As byproduct, a reasonable model for the stock price at the horizon is:

$$\begin{aligned} \ln(g_T) &= \ln\left(\frac{S_T}{S_0}\right) = \mu T + \sigma\sqrt{T}Z \\ &\iff \\ S_T &= S_0 \mathrm{e}^{\mu T + \sigma\sqrt{T}Z}, \end{aligned}$$

then S_T is a log-normal rv. If further assumptions are made, in particular the market is arbitrage-free, we can[18] take $\mu = r - \frac{\sigma^2}{2}$, where r is a risk-free rate.

8.6 Exercises

8.1 Prove (i) of Theorem 8.1.

8.2 Prove (iii) of Theorem 8.1.

8.3 Prove (iv) of Theorem 8.1.

[18] After a proper change of the probability measure P, replaced by the so called *equivalent martingale measure*.

8.4 Prove (v) of Theorem 8.1.

8.5 Write the likelihood function of a Gaussian sample $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$, and verify that it is maximized at the sample mean $\bar{R}_n$ and the sample standard deviation $\sqrt{\bar{V}_n}$.

8.6 Prove Lemma 8.1.

8.7 For $1 \leqslant p < \infty$, prove that in $\mathcal{L}^p$ the relation $X \mathcal{R} Y$ defined as $X \overset{\text{a.s.}}{=} Y$, for every $X, Y \in \mathcal{L}^p$, is indeed an equivalence relation, with quotient space L^p.

Appendix

Details on Theorem 8.1

Proof of Theorem 8.1, (ii) Let $g : \mathbb{R} \to \mathbb{R}$ be bounded and continuous, so that $g(X_n) \xrightarrow{\mathsf{P}} g(X)$ by the continuous mapping Theorem 8.3. By Theorem 8.2, part (b), we also have $\mathsf{E}(g(X_n)) \xrightarrow{L^1} \mathsf{E}(g(X))$ which in turn implies convergence of expectations, and this together with Portmanteau lemma 8.3 (see below), entails the desired result. ■

Portmanteau Lemma

We present a useful result on convergence in distribution without proof.

Lemma 8.3 (Portmanteau (Partial)) *Let X_n, X be rvs. The following are equivalent.*

(a) $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$ *at every continuity point* $x \in \mathbb{R}$*;*
(b) $\lim_{n\to\infty} \mathsf{E}(g(X_n)) = \mathsf{E}(g(X))$*, for every bounded continuous functions* g*;*
(c) $\lim_{n\to\infty} \mathsf{P}(X_n \in B) = \mathsf{P}(X \in B)$*, for every Borel set* $B \subset \mathbb{R}$ *s.t.* $\mathsf{P}(X \in \partial B) = 0$*, with* ∂B *being the boundary of* B*.*

This formulation of the lemma is incomplete, since we must add four equivalent provisos which are omitted here see [15, Th 2.2], items (iii)–(vi). For equivalence (a)⇔(b) see also [10, Th 5.8].

Convergence in Distribution: Special Cases

Convergence in distribution can be specialized to cases when all rvs X_n, X are either continuous with densities $f_{X_n}(x)$, $f_X(x)$ or discrete with masses $\mathsf{P}(X_n = x)$, $\mathsf{P}(X = x)$. In the first case:

$$\begin{aligned}\lim_{n\to\infty} \mathsf{E}(g(X_n)) &= \lim_{n\to\infty} \int g(x) f_{X_n}(x)\mathrm{d}x \\ &= \int g(x) \lim_{n\to\infty} f_{X_n}(x)\mathrm{d}x \\ &= \int g(x) f(x)\mathrm{d}x = \mathsf{E}(g(X)),\end{aligned}$$

where for the second equality to hold we must assume $f_{X_n}(x) \leqslant g(x)$, each $n \in \mathbb{N}$ and bounded continuous $g(x)$, together with $\int g(x)\mathrm{d}x < \infty$ and $\lim_{n\to\infty} f_{X_n}(x) = f_X(x)$ pointwise,[19] so we can apply a version of the Lebesgue's dominated convergence Theorem 4.12 for real-valued measurable functions, see Appendix to Chap. 4. But this is nothing else convergence in distribution $X_n \xrightarrow{\mathrm{d}} X$, see Portmanteau lemma 8.3, equivalence (a)⇔(b). In fact, this result can be extended to random vectors $\mathbf{X}_n, \mathbf{X}$ having joint densities $f_{\mathbf{X}_n}(\mathbf{x})$, $f_{\mathbf{X}}(\mathbf{x})$, s.t. $\lim_{n\to\infty} f_{\mathbf{X}_n}(\mathbf{x}) = f_{\mathbf{X}}(\mathbf{x})$ pointwise:[20] this again implies convergence in distribution $\mathbf{X}_n \xrightarrow{\mathrm{d}} \mathbf{X}$ of the joint CDFs. In the second case, one can prove $\lim_{n\to\infty} \mathsf{P}(X_n = x) = \mathsf{P}(X = x)$ is equivalent to $X_n \xrightarrow{\mathrm{d}} X$, for every x in the supports of each X_n and of X, see [8, Th 18.9] for a proof. Finally we present a partial converse of the implication

$$\mathbf{X}_n \xrightarrow{\mathrm{d}} \mathbf{X} \Longrightarrow X_{in} \xrightarrow{\mathrm{d}} X_i, \quad \text{for } i = 1, \ldots, n$$

for random vectors $\mathbf{X}_n = (X_{1n}, \ldots, X_{nn})$ and $\mathbf{X} = (X_1, \ldots, X_n)$ In fact, convergence in distribution for random vectors is equivalent to

$$\sum_{i=1}^{n} a_i X_{in} \xrightarrow{\mathrm{d}} \sum_{i=1}^{n} a_i X_i,$$

[19] This can be weakened to $f_{X_n}(x) \to f_X(x)$, excepts on a set of x of Lebesgue measure zero.

[20] This can also be weakened by requiring convergence on a set of $\mathbf{x}$ with n-dimensional Lebesgue measure zero.

for every choice of $\mathbf{a} = (a_1, \ldots, a_n)$, since defining the linear function

$$h(\mathbf{x}) = g(\mathbf{a} \cdot \mathbf{x}) = g\left(\sum_{i=1}^{n} a_i x_i\right)$$

we clearly have

$$\lim_{n\to\infty} \mathsf{E}(h(\mathbf{X}_n)) = \mathsf{E}(h(\mathbf{X})),$$

which is convergence in distribution due to $h(\cdot)$ being bounded and continuous on $\mathbb{R}^n$. For the converse see below Complements on CHF.

Further Details on Convergence in Distribution

Convergence in probability does not imply convergence of expectations, but there is one exception which requires an additional property. We say that a sequence $(X_n)_{n\in\mathbb{N}} \subset L^1$ is *uniformly integrable* if

$$\lim_{M\to\infty} \sup \mathsf{E}\left(|X_n|\mathbf{I}_{\{|X_n|>M\}}\right) = 0. \tag{8.4}$$

By Lebesgue's dominated convergence theorem 4.12 (see Appendix in Chap. 4) it is true[21]

$$\lim_{M\to\infty} \mathsf{E}(|X_n|\mathbf{I}_{\{|X_n|>M\}}) = 0.$$

Condition $\mathsf{E}(|X_n|\mathbf{I}_{\{|X_n|>M\}}) < \delta$ holds for all $\delta > 0$ and some constant M, thus, in (8.4) we further require it holds for the same[22] M for all X_n. It can be shown that uniform integrability is equivalent to the condition $\sup_n \mathsf{E}(|X_n|) < \infty$, see [10, Th 5.16], part i). Now, if $X_n \xrightarrow{\mathsf{P}} c$ and $(X_n)_{n\in\mathbb{N}} \subset L^1$ is uniformly integrable then $\lim_{n\to\infty} \mathsf{E}(X_n) = c$, for a constant $c \in \mathbb{R}$, see [10, Theorem 5.17] in the degenerate case $X = c$. A similar situation occurs for convergence in distribution. Intuitively, $\mathsf{P}(|X_n| > M) < \epsilon$ holds for all $\epsilon > 0$ and some constant M and rvs X_n are called *tight*. In condition (8.4) we impose choosing the same M for every $n \in \mathbb{N}$, and the sequence is said *uniformly tight*.[23] On the other hand, convergence in distribution

[21] Every truncated rv $|X_n|\mathbf{I}_{\{|X_n|>M|\}}$ is dominated by X_n, each n, and converge to zero.

[22] Recall that a sequence of functions $(f_n(x))_{n\in\mathbb{N}}$, defined over the same domain, say D, converges uniformly to a limit function $f(x)$ iff $\sup_{x\in D} |f_n(x) - f(x)| \to 0$ as $n \to \infty$, i.e., $\sup_{x\in D} |f_n(x) - f(x)| < \delta$ for all $\delta > 0$ and some $N \in \mathbb{N}$ s.t. $n > N$.

[23] Note that the complimentary event $\{|X_n| \leqslant \epsilon\}$ is defined over a compact subset $[-\epsilon, \epsilon]$.

$X_n \xrightarrow{\mathrm{d}} X$ does imply $(X_n)_{n\in\mathbb{N}}$ is uniformly tight.[24] In fact, a pathological behavior happens when $X_n \xrightarrow{\mathrm{d}} X$, and all X_n are continuous rvs but X is not. More generally, it could be the case that a sequence of CDFs $F_{X_n}(x)$ converges pointwise to a limit $F_X(x)$ which is not a properly CDF: it is nondecreasing, right-continuous but $F_X(\infty) < 1$ and $F_X(-\infty) > 0$. This is called a *defective* CDF, and can be right discontinuous at some points x where $F_X(x) - F_X(-x) > 0$. In the latter case, these form sets of isolated points $\{x\}$ which have to be excluded from the convergence since $\mathsf{P}(X \in \partial\{x\}) > 0$, where we considered $\partial\{x\} = \{x\}$. The following lemma fixes this, see [15, Lemma 2.5] or [10, Th 6.32].

Lemma 8.4 (Helly) *Let $(F_{X_n}(x))_{n\in\mathbb{N}}$ be a sequence of CDFs. Then there is a subsequence s.t. $\lim_{k\to\infty} F_{X_{n_k}}(x) = F_X(x)$, at each continuity point x of a possibly defective CDF F_X.*

When the limit rv X has a continuous CDF F_X then convergence in distribution is at every $x \in \mathbb{R}$, and then it is uniform convergence of CDFs F_{X_n}, i.e., $\sup_x |F_{X_n}(x) - F_X(x)| \to 0$ as $n \to \infty$, see [15, Lemma 2.11]. There is a more general version of Lemma 8.4, referred as *Helly's selection principle*, see for example [8, Th 18.6]. More complete accounts of convergence concepts are [1] and [6].

Further Convergence Relationships

Theorem 8.12 *Let X_n, X and Y_n, Y be rvs. Then:*

(i) if $X_n \xrightarrow{d} X$ and $|X_n - Y_n| \xrightarrow{\mathsf{P}} 0$, then $Y_n \xrightarrow{d} X$;
(ii) if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{\mathsf{P}} c$, then $(X_n, Y_n) \xrightarrow{d} (X, c)$, where $c \in \mathbb{R}$;
(iii) if $X_n \xrightarrow{\mathsf{P}} X$ and $Y_n \xrightarrow{\mathsf{P}} Y$, then $(X_n, Y_n) \xrightarrow{\mathsf{P}} (X, Y)$.

Proof We only prove (ii) and (iii). For a proof of (i) see [15, Th 2.7]. To check (ii), first we recall that convergence in probability for random vectors in $\mathbb{R}^n$ needs replacing absolute value in the condition $|X_n - X| \xrightarrow{\mathsf{P}} 0$ for univariate rvs, with the Euclidean distance $d((X_n, Y_n), (X_n, c)) = \sqrt{(X_n - X_n)^2 + (Y_n - c)^2}$, written for bi-variate random vectors (X_n, Y_n), each n, and (Y_n, c). Thus, $d((X_n, Y_n), (X_n, c)) = |Y_n - c|$ which converges to zero in probability. According to (i) the proof amounts to show $(X_n, c) \xrightarrow{\mathrm{d}} (X, c)$. In fact, for every bounded continuous function $g(x, y)$, the mapping $x \mapsto g(x, c)$ is also continuous and bounded. Then, provided $X_n \xrightarrow{\mathrm{d}} X$ we get $g(X_n, c) \xrightarrow{\mathrm{d}} g(X, c)$ and by

[24] See Prohorov's theorem [15, Th 2.4]. part (i). More is true, if the sequence of rvs is uniformly tight, then there exists a subsequence converging in distribution, which is Theorem [15, Th 2.4]. part (ii).

Theorem 8.3, part (b), we are done. To prove (iii) we simply invoke the inequality $d((X_n, Y_n), (X, Y)) \leqslant d(X_n, X) + d(Y_n, Y)$. ■

It is worth noting that Slutsky's lemma can be deduced by Theorem 8.12 in the following sense. By part (ii), if $X_n \xrightarrow{\mathrm{d}} X$ and $Y_n \xrightarrow{\mathsf{P}} c$ we have $(X_n, Y_n) \xrightarrow{\mathrm{d}} (X, c)$, and by the continuous mapping Theorem 8.3 we also have $g(X_n, Y_n) \xrightarrow{\mathrm{d}} g(X, c)$ for a function $g(x, y)$ which is continuous at every point in $\mathbb{R} \times \{c\}$ where the random vector (X, c) takes values. Choosing $g(x, y) = x + y$ or $g(x, y) = xy$ entails part (b) of Theorem 8.4.

Complements on CHF

Any CHF $\phi_X(u) = \mathsf{E}(\mathrm{e}^{\mathrm{i}uX})$ is uniformly continuous:

$$\begin{aligned} |\phi_X(u+h) - \phi_X(u)| &= |\mathsf{E}(\mathrm{e}^{(\mathrm{i}+h)uX}) - \mathsf{E}(\mathrm{e}^{\mathrm{i}uX})| \\ &= |\mathsf{E}(\mathrm{e}^{\mathrm{i}uX}(\mathrm{e}^{\mathrm{i}hX} - 1))| \\ &\leqslant \mathsf{E}(|\mathrm{e}^{\mathrm{i}uX}||\mathrm{e}^{\mathrm{i}hX} - 1|) = \mathsf{E}(|\mathrm{e}^{\mathrm{i}hX} - 1|), \end{aligned}$$

so $\lim_{h\to 0} \mathsf{E}(|\mathrm{e}^{\mathrm{i}hX} - 1|) = 0$ and by the Lebesgue's dominated convergence Theorem 4.12 the convergence is for all u. Moreover, from Euler's formula we can take conjugate as $\mathrm{e}^{-\mathrm{i}x} = \overline{\mathrm{e}^{\mathrm{i}x}} = \cos x - i \sin x$ then:

$$\begin{aligned} \phi_X(-u) &= \mathsf{E}(\mathrm{e}^{-\mathrm{i}uX}) = \mathsf{E}(\overline{\mathrm{e}^{\mathrm{i}uX}}) \\ &= \mathsf{E}(\cos(uX)) - \mathrm{i}\mathsf{E}(\sin(uX)) = \overline{\mathsf{E}(\mathrm{e}^{\mathrm{i}uX})} = \overline{\phi_X(u)}. \end{aligned}$$

In this respect, we also have $\phi_{-X}(u) = \mathsf{E}(\mathrm{e}^{-\mathrm{i}uX}) = \phi_X(-u)$ which in turn yields the conjugate CHF $\overline{\phi_X(u)}$. Moreover, the CHF determines the distribution, i.e., $\phi_X(u) = \phi_Y(u)$ for all u implies $X \overset{\mathrm{d}}{=} Y$, since

$$\mathsf{P}(x < X < y) = \lim_{k\to\infty} \int_{-k}^{k} \frac{\mathrm{e}^{-\mathrm{i}ux} - \mathrm{e}^{-\mathrm{i}ux}}{\mathrm{i}u} \phi_X(u)\mathrm{d}u, \ \ x, y \text{ con. points of } F_X,$$

see for example [10, Th 6.5], and then

$$F_X(x) - F_X(y) = F_Y(x) - F_Y(y)$$

so that taking the limits as $y \to -\infty$ we have $F_X(x) = F_Y(x)$ for every continuity point[25] x. We have seen in Sect. 4.4 that for an absolutely continuous rv X its symmetric density around zero can be recovered by the inverse Fourier transform. In general a sufficient condition for the inverse to exist is absolute integrability of the CHF:

$$\int |\phi_X(u)| \mathrm{d}u < \infty.$$

If this is fulfilled then X is an absolutely continuous rv with continuous and bounded density $f_X(x) = \frac{1}{2\pi} \int \mathrm{e}^{-\mathrm{i}ux} \phi_X(u) \mathrm{d}u$, see [10, Th 6.7]. For example, for a standard normal $X \sim (0, 1)$ we have $\phi(u) = \mathrm{e}^{-u^2/2}$ which is absolutely integrable since

$$\int |\mathrm{e}^{-u^2/2}| \mathrm{d}u = \int \mathrm{e}^{-u^2/2} = \sqrt{2\pi},$$

and the inverse Fourier transform can be obtained analytically by complex contour integration.

Remark 8.4 For an absolutely continuous rv X, its CHF is the *Fourier transform* $\phi_X(u) = \mathfrak{F}(f_X))$ of the density function. More generally, we can handle CDFs:

$$\phi_X(u) = \int \mathrm{e}^{\mathrm{i}ux} \mathrm{d}F_X(x).$$

Thus we have $\phi_X(u) = \mathfrak{F}(F_X))$ is a transform acting on the space of CDFs (with value onto the complex numbers $\mathbb{C}$).

For discrete rvs X the mass function can be recovered via

$$\mathsf{P}(X = x) = \lim_{k \to \infty} \frac{1}{2k} \int_{-k}^{k} \mathrm{e}^{-\mathrm{i}ux} \phi_X(u) \mathrm{d}u,$$

see [10, Prop 6.9]. If moments $\mathsf{E}(X^p)$ exist, they can be obtained by the pth derivative of the CHF $\phi_X(u)$ evaluated at zero, see [10, Th 6.11]. For example, for $X \in L^2$ we have $\mathsf{E}(X) = -i \frac{\mathrm{d}}{\mathrm{d}u} \phi_X(0)$ and $\mathsf{E}(X^2) = -\frac{\mathrm{d}^2}{\mathrm{d}u^2} \phi_X(0)$. It is also true that assuming the pth derivative $\frac{\mathrm{d}^p}{\mathrm{d}u^p} \phi_X(0)$ exists for even p, then all even moments exist since $\mathsf{E}(|X|^p)$ is finite, see [10, Th 6.12]. Turning things around, if $\mathsf{E}(|X|^p) < \infty$ for some integer $p \geqslant 1$, we can take the Taylor series expansion of the CHF:

$$\phi_X(u) = \sum_{i=0}^{p} \frac{(\mathrm{i}u)^i}{i!} \mathsf{E}(X^i) + o(|u|^p), \tag{8.5}$$

[25] This gives $F_X(x) = F_Y(x)$ also for all $x \in \mathbb{R}$, since CDFs have at most countably many discontinuous points.

where $o(|u|^p) \to 0$ as $u \to 0$. When we are sure that the MGF $M_X(u) = \mathsf{E}(\mathrm{e}^{uX})$ exists, then it is possible to recover it by $\phi_X(u) = M_X(iu)$, i.e., $M_X(u) = \phi_X(-iu)$. For a random vector $\mathbf{X} = (X_1, \ldots, X_n)$ and a deterministic vector $\mathbf{u} = u_1, \ldots, u_n$, the CHF is defined as

$$\phi_{\mathbf{X}}(\mathbf{u}) = \mathsf{E}(\mathrm{e}^{\mathrm{i}\mathbf{X}\cdot\mathbf{u}}) = \mathsf{E}\left(\exp\left(\mathrm{i}\sum_{j=1}^{n} X_j u_j\right)\right). \tag{8.6}$$

The multivariate CHF share the same properties than those of the univariate counterpart. We list them below.

- $\phi_{\mathbf{X}}(\mathbf{u})$ is uniformly continuous.
- $\phi_{\mathbf{X}}(-\mathbf{u}) = \overline{\phi_{\mathbf{X}}(\mathbf{u})} = \phi_{-\mathbf{X}}(\mathbf{u})$.
- For independent $\mathbf{X}_1, \ldots, \mathbf{X}_m$ it holds $\phi_{\mathbf{X}_1+\cdots+\mathbf{X}_m}(\mathbf{u}) = \phi_{\mathbf{X}_1}(\mathbf{u}) \cdots \phi_{\mathbf{X}_m}(\mathbf{u})$
- $\mathbf{X} \overset{\mathrm{d}}{=} \mathbf{Y}$ iff $\phi_{\mathbf{X}}(\mathbf{u}) = \phi_{\mathbf{Y}}(\mathbf{u})$ for every $\mathbf{u} \in \mathbb{R}^n$.
- The components X_i of $\mathbf{X}$ are independent iff $\phi_{\mathbf{X}}(\mathbf{u}) = \prod_{i=1}^{n} \phi_{X_i}(u_i)$.
- (Levy's Continuity theorem) Let $\mathbf{X}_n, \mathbf{X}$ be random vectors with CHFs $\phi_{\mathbf{X}_n}(\mathbf{u}), \phi_{\mathbf{X}}(\mathbf{u})$. Then $\mathbf{X}_n \overset{\mathrm{d}}{\to} \mathbf{X}$ iff $\lim_{n\to\infty} \phi_{\mathbf{X}_n}(\mathbf{u}) = \phi_{\mathbf{X}}(\mathbf{u})$, at every $\mathbf{u} \in \mathbb{R}^n$. If $\lim_{n\to\infty} \phi_{\mathbf{X}_n}(\mathbf{u}) = f(\mathbf{u})$, for all $\mathbf{u} \in \mathbb{R}^n$, where $f(\mathbf{u})$ is continuous at $\mathbf{0} \in \mathbb{R}^n$, then $f(\mathbf{u}) = \phi_{\mathbf{X}}(\mathbf{u})$ and $\mathbf{X}_n \overset{\mathrm{d}}{\to} \mathbf{X}$.

To prove that convergence of a linear combination of components in each $\mathbf{X}_n$ do converges in distribution to a linear combination of components in $\mathbf{X}$, with the same coefficients, define the rvs $Y_n = \mathbf{X}_n \cdot \mathbf{u}$ and $Y = \mathbf{X} \cdot \mathbf{u}$, actually linear combinations of X_{jn} and X_j with coefficients u_j, respectively, for $j = 1, \ldots, n$. If $\mathbf{X}_n \cdot \mathbf{u} \overset{\mathrm{d}}{\to} \mathbf{X} \cdot \mathbf{u}$ we have $\phi_{Y_n}(v) \to \phi_Y(v)$ for an arbitrary $v \in \mathbb{R}$ as $n \to \infty$. Thus choosing $v = 1$ we have $\phi_{\mathbf{X}_n}(\mathbf{u}) \to \phi_{\mathbf{X}}(\mathbf{u})$ as $n \to \infty$ and $\mathbf{X}_n \overset{\mathrm{d}}{\to} \mathbf{X}$ is established.

Complements on MGF

When a rv X admits MGF, then all its moments exist finite since from basic calculus courses we know that $\lim_{x\to\infty} \frac{x^{\mathrm{e}}}{\mathrm{e}^x} = 0$ and then $\mathrm{e}^{|ux|}$ is greater than any polynomial of finite degree in ux, as $|x| \to \infty$. Next, a useful application of the Lebesgue's dominated convergence theorem 4.12 enable us computing moments via the Taylor series expansion of the MGF around zero. The starting point is to replace the sequence of rvs assumed in Theorem 4.12 with any indexed family of rvs $(X_u)_{u\in I}$ defined on the same probability space and such that $\lim_{u\to 0} X_u(\omega) = X_0(\omega)$ pointwise, where $I \subset \mathbb{R}$ is any nonempty set. This implies the pointwise convergence $X_{u_n}(\omega) \to X_0(\omega)$ for any subsequence $u_1 < u_2 < \cdots$ of indexes with

$u_k \downarrow 0$ as $n \to \infty$. Thus, Theorem 4.12 guarantees that $\lim \mathsf{E}(X_{u_n}) = \mathsf{E}(X_0)$ and consequently $\lim_{u\to 0} \mathsf{E}(X_u) = \mathsf{E}(X_0)$. We have:

Theorem 8.13 (Differentiating Inside the Expectation) *Let $(X_u)_{u\in(a,b)}$ be an indexed family of rvs defined on the same probability space. Assume that the partial derivative $\frac{\partial}{\partial u} X_u(\omega)$ exists, for every $\omega \in \Omega$ and every $u \in (a, b)$. Furthermore, let $Y \in L^1$ be a random variable such that $|\frac{\partial}{\partial u} X_u| \leqslant Y$ holds for every scenario ω and every u. Then, $\frac{\partial}{\partial u} X_u(\omega)$ is a random variable and the function $u \mapsto \mathsf{E}(X_u)$ has derivative $\mathsf{E}(\frac{\partial}{\partial u} X_u)$, for every $a < u < b$.*

Observe that the indexed family in the above theorem can be though of as a collection of functions $X(u, \omega)$ of two variables, see Chap. 9.

Proof Let $u_k = u + \frac{1}{k}$ and take $u_k \downarrow u$ as $k \to \infty$. First, we show that $\frac{\partial}{\partial u} X_u$ is a random variable. The Newton quotient $\frac{X_{u_k}(\omega) - X_u(\omega)}{1/n}$ is a rv. This is nothing but the partial derivative claimed. Now, write $f'(u) = \lim_{h\to 0} \frac{f(u+h)-f(u)}{h}$ and observe that $|\frac{f(u+h)-f(u)}{h}| \leqslant |\frac{\partial}{\partial u} X_u| \leqslant Y$ by assumption. Then the Lebesgue's dominated convergence theorem gives:

$$\lim_{h\to 0} \frac{f(u+h) - f(u)}{h} = \lim_{h\to 0} \mathsf{E}\Big(\frac{X_{u+h} - X_u}{h}\Big) = \mathsf{E}\Big(\lim_{h\to 0} \frac{X_{u+h} - X_u}{h}\Big)$$
$$= \mathsf{E}\left(\frac{\partial}{\partial u} X_u\right).$$

This limit is finite, since the rvs $\frac{\partial}{\partial u} X_u$ are dominated by Y. Taking expectation we have $\mathsf{E}(|\frac{\partial}{\partial u} X_u|) \leqslant \mathsf{E}(Y) < \infty$. ■

Theorem 8.13 enable us to differentiate inside the expectation as $\frac{\mathrm{d}}{\mathrm{d}u}\mathsf{E}(X(u, \omega)) = \mathsf{E}(\frac{\partial}{\partial u} X_u)$. As a consequence we are in position to define the MGF $M_X(u) = \mathsf{E}(\mathrm{e}^{uX})$ under the condition that $t \neq 0$ is chosen so that the expectation exists and is finite. The assumption $X \in L^1$ is in general not sufficient to apply the change of variable formula to the transformation e^{uX}, since we must ensure that $\mathrm{e}^{uX} \in L^1$ depending on the values $u \in \mathbb{R}$. Indeed, e^{uX} is nonnegative and its expectation always exists, although it may be ∞. For the finiteness we must impose $\mathsf{E}(\mathrm{e}^{uX}) < \infty$ for $u < \delta$, and $\delta > 0$. If X is itself nonnegative, then the finiteness of its MGF holds for all $u \leqslant 0$, since e^{uX} is bounded and $M_X(-u)$ does exist and is called the *Laplace transform* of F_X, or of X. Using the Taylor series expansion of e^{ux} around zero and using the linearity of expectation we have:

$$M_X(u) = \mathsf{E}\left(\sum_{p=0}^{\infty} \frac{(uX)^p}{p!}\right) = \sum_{p=0}^{\infty} \frac{u^p}{p!}\mathsf{E}(X^p).$$

see for example [2]. Indeed, if $M_X(u)$ exists we may apply recursively Theorem 8.13 and get $\left(\frac{\mathrm{d}}{\mathrm{d}u}\right)^p M_X(u) = \mathsf{E}\left(\left(\frac{\partial}{\partial u}\right)^p \mathrm{e}^{uX}\right) = \mathsf{E}(X^p \mathrm{e}^{uX})$ and set $u = 0$. There is a version of MGF for random vectors.

Theorem 8.14 *Let* $\mathbf{X}, \mathbf{Y}$ *be n-dimensional random vectors with MGFs* $M_{\mathbf{X}}$ *and* $M_{\mathbf{Y}}$.

(i) If $\mathbf{A}$ *is an* $m \times n$ *matrix with real entries and* $\mathbf{b} \in \mathbb{R}^m$, *then*

$$M_{\mathbf{AX}'+\mathbf{b}}(\mathbf{u}) = \mathrm{e}^{\mathbf{u}\cdot\mathbf{b}} M_{\mathbf{X}}(\mathbf{AX}'),$$

for $\mathbf{u} \in \mathbb{R}^m$ *s.t.* $\|\mathbf{u}\| < \delta$, *with* $\delta > 0$ *depending on* $\mathbf{A}$.

(ii) $\mathbf{X} \stackrel{d}{=} \mathbf{Y}$ *if and only if there exists some* $\delta > 0$ *such that* $M_{\mathbf{X}}(\mathbf{u})$ *equals* $M_{\mathbf{Y}}(\mathbf{u})$, *for all* $\mathbf{u} \in \mathbb{R}^n$ *with* $\|\mathbf{u}\| < \delta$.

Property (ii) of Theorem 8.14 specializes to the one-dimensional case $n = 1$, but indeed it requires that the two rvs have finite moments of all orders that do not grow too fast, or equivalently it is required that P_X has tails of *exponential order*

$$\mathsf{P}(|X| > x) \leqslant M\mathrm{e}^{-\alpha x},$$

for $M, \alpha > 0$, see [2]. Recall that $\|\mathbf{x}\| = \sqrt{\sum_{i=1}^n x_i^2}$ for vectors $\mathbf{x} = (x_1, \ldots, x_n)$, considered as row vectors for which a prime $'$ denotes transposition. Also we used the dot product between two vectors in $\mathbb{R}_n$, i.e., $\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i$. For two rvs X, Y taken from a bi-variate random vector $\mathbf{X} = (X, Y)$, their joint MGF is

$$M_{\mathbf{X}}(\mathbf{u}) = \mathsf{E}(\mathrm{e}^{u_1 X + u_2 Y}) = \iint \mathrm{e}^{u_1 x + u_2 y} \mathrm{d}F_{\mathbf{X}}(x, y),$$

where $\mathbf{u} = u_1, u_2$. It specializes to the case of a continuous bi-variate random vector with a joint density, replacing $\mathrm{d}F_{\mathbf{X}}(x, y)$ with $f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y$. The *joint moment* $\mathsf{E}(X^n Y^m)$ can be recovered by the joint MGF as

$$\mathsf{E}(X^n Y^m) = \frac{\partial^{n+m}}{\partial^n u_i \partial^m u_2} M_{\mathbf{X}}(u_1, u_2)\Big|_{u_1 = u_2 = 0}.$$

Whatever the joint MGF does not exists, we can obtain the joint CHF in Theorem 8.14 by formally replacing $\mathbf{u}$ with $i\mathbf{u}$ and dropping the proviso $\|\mathbf{u}\| < \delta$ in part (ii). In the bi-variate case $\mathbf{X} = (X, Y)$ we can write:

$$\phi_{\mathbf{X}}(\mathbf{u}) = \mathsf{E}(\mathrm{e}^{\mathrm{i}u_1 X + \mathrm{i}u_2 Y}) = \iint \mathrm{e}^{\mathrm{i}u_1 x + \mathrm{i}u_2 y} \mathrm{d}F_{\mathbf{X}}(x, y),$$

and we can recover the *marginal CHFs* as

$$\phi_X(u_1) = \phi_{\mathbf{X}}(u_1, 0), \quad \phi_Y(u_2) = \phi_{\mathbf{X}}(0, u_2).$$

Proof of Theorem 8.8

Proof Let $\phi_{X_i}(u)$ be the CHF of each term in the sequence. Thanks to the hypotheses we have that $\phi_{X_i}(u)$ is differentiable at least once at $u = 0$, then $\frac{\mathrm{d}}{\mathrm{d}u}\phi_{X_i}(u)\big|_{u=0} = \mathrm{i}\mu$. Using the Taylor series expansion (8.5), we can write $\phi_{X_i}(u) = 1 + \mathrm{i}\mu u + o(t)$. Thus, the CHF of the sample mean is

$$\phi_{\bar{Y}_n}(u) = (\phi_{Y_i}(t/n))^n = \left(1 + \frac{\mathrm{i}\mu u}{n} + o(n^{-1})\right)^n,$$

where $f(n) = o(n^{-1})$ means that $\lim_{n\to\infty} nf(n) = 0$. As byproduct, the limit of the latter expression, as $n \to \infty$, is $\mathrm{e}^{\mathrm{i}\mu u}$ which together with the Levy's Continuity Theorem 8.5 entails the result. ■

Proof of Theorem 8.11

We present a short proof based on [15, Th 19.1].

Proof Let $\|\mathbb{F}_n - F_X\|_\infty := \sup_x |\mathbb{F}_n(\omega, x) - F_X(x)|$. By SLLN, Theorem 8.7, we have $\mathbb{F}_n(\omega, x) \xrightarrow{\text{a.s.}} F_X(x)$ at all $x \in \mathbb{R}$. A.s. convergence holds also for left-limits, i.e., $\mathbb{F}_n(\omega, x-) \xrightarrow{\text{a.s.}} F_X(x-)$. Take $\epsilon > 0$ and partition points $-\infty = x_0 < x_1 < \cdots < x_k = \infty$ of $\mathbb{R}$, s.t. the left-limit of $F_X(x)$ at any x_i does not exceed the right-limit at x_{i-1} more than ϵ, i.e., $F_X(x_i-) - F_X(x_{i-1}) < \epsilon$, for all $i = 0, 1, \ldots, k$. As byproduct, for $x \in [x_{i-1}, x_i)$ we can take $|\mathbb{F}_n(\omega, x) - F_X(x)|$ within the closed interval

$$[\mathbb{F}_n(\omega, x_{i-1}) - F_X(x_{i-1}) - \epsilon, \mathbb{F}_n(\omega, x_i-) - F_X(x_i-) + \epsilon].$$

Hence, the convergence of $\mathbb{F}_n(\omega, x)$ and $\mathbb{F}_n(\omega, x-)$ is uniform for $x \in \{x_1, \ldots, x_k\}$, then we conclude that $\limsup \|\mathbb{F}_n - F_X\|_\infty \leqslant \epsilon$, a.s., for all ϵ and $\|\mathbb{F}_n - F_X\|_\infty \xrightarrow{\text{a.s.}} 0$ as desired. ■

Further Details on L^p Spaces

The collection $\mathcal{L}^0$ of all rvs defined over the same probability space has subspaces $\mathcal{L}^p$ where $\mathsf{E}(|X|^p) < \infty$, for $1 \leqslant p < \infty$, see Sect. 4.6, Chap. 4. In fact, L^p is the

quotient space of $\mathcal{L}^p$ with respect to the equivalence relation[26] $X\mathcal{R}Y$ iff $X \overset{\text{a.s.}}{=} Y$, for all $X, Y \in \mathcal{L}^p$, see Exercise 8.7. Hence, $\|X\|_p := (\mathsf{E}(|X|^p))^{1/p}$ is finite whenever $X \in \mathcal{L}^p$, and we can define $\|X - Y\|_p$ as a pseudo-metric on $\mathcal{L}^p$ which turn to be a metric if we restrict ourselves to the quotient space L^p. It is worth noting that $\|X\|_p$ is a norm on L^p, see Appendix G, since:

- $\|X\|_p = 0$ iff $X = 0$, for $X \in L^p$;
- $\|X + Y\|_p \leqslant \|X\|_p + \|Y\|_p$, by Minkowski's inequality, see Theorem 4.5, Sect. 4.6, for $X, Y \in L^p$;
- $\|\lambda X\|_p = |\lambda| \|X\|_p$, for $\lambda \in \mathbb{R}$ and $X \in L^p$.

Remark 8.5 The first item should be written $\|X\|_p = 0$ *then* $X = 0$ for $X \in \mathcal{L}^p$, whenever $X \overset{\text{a.s.}}{=} 0$. This is ruled out passing to the quotient space, and considering equivalence classes of rvs which are equal P-a.s.

Therefore, this norm induces the metric $d(X, Y) = \|X - Y\|_p$ which drives convergence in L^p spaces. For a sequence $(X_n)_{n\in\mathbb{N}} \subset L^p$ and a rv $X \in L^p$ it holds:

$$X_n \xrightarrow{L^p} X \Longrightarrow X_n \xrightarrow{L^q} X, \quad \text{for every } 1 \leqslant q \leqslant p.$$

This is a consequence of the (Lyapunov's) inequality $\|X - Y\|_q \leqslant \|X - Y\|_p$, which in turn is due to the Hölder's inequality, see Theorem 4.4, Sect. 4.6, by setting $Y = 1$ and replacing p with $\frac{p}{q}$ in such a way $\mathsf{E}((|X|^q))^{1/q} \leqslant \mathsf{E}((|X|^p))^{1/p}$. Clearly L^p convergence implies convergence in probability since $X_n \xrightarrow{L^p} X$ yields $X_n \xrightarrow{L^1} X$ and this in turn gives $X_n \xrightarrow{\mathsf{P}} X$ by Theorem 8.1. A special subspace is L^2, which is endowed with the inner product

$$\langle X, Y\rangle := \mathsf{E}(XY), \quad \text{for all } X, Y \in L^2.$$

This is well defined since by the Cauchy-Schwarz inequality in Theorem 4.3 we have

$$|\mathsf{E}(XY)| \leqslant (\mathsf{E}(X^2))^{1/2} (\mathsf{E}(Y^2))^{1/2} < \infty,$$

for $X, Y \in L^2$. Note that $\langle X, Y\rangle \geqslant 0$ and equality occurs[27] iff $X = 0$. Observe that the inner product is *bilinear*, i.e., linear in each component,[28]

$$\langle aX + bZ, Y\rangle = a\langle X, Z\rangle + b\langle Z, Y\rangle$$

[26] $\mathcal{R}$ is reflexive, symmetric and transitive.

[27] We have that $\langle X, Y\rangle = 0$ is equivalent to $X \overset{\text{a.s.}}{=} 0$ which implies $X = 0$ when the quotient space L^2 replaces $\mathcal{L}^2$.

[28] Recall that L^2 is indeed a vector space.

and

$$\langle X, aY + bZ\rangle = a\langle X, Y\rangle + b\langle X, Z\rangle.$$

The inner product and the norm are connected by $\langle X, X\rangle^{1/2} = \|X\|_2$, and this turns L^2 to be a *normed space*. It can be showed that Cauchy sequences $(X_n)_{n\in\mathbb{N}} \subset L^2$ converge to a limit rv $X \in L^2$, which means that the latter is an *Hilbert space*.[29] Two rvs $X, Y \in L^2$ are said to be *orthogonal* if $\langle X, Y\rangle = 0$ and in this case $\|X + Y\|_2^2 = \|X\|_2^2 + \|Y\|_2^2$, i.e., $\mathsf{E}((X + Y)^2) = \mathsf{E}(X^2) + \mathsf{E}(Y^2)$. The important feature of L^2 as an Hilbert space is that we can always find the *projection* of $X \in L^2$ as the unique $Y \in L^2$ s.t. the latter is closest to X, see [8, Ch 22] for a brief yet complete review of all these concepts.

Remark 8.6 Convergence in L^0 spaces cannot be expressed by the norm $\|\cdot\|_p$ used in smaller L^p spaces. Apart convergence in distribution, L^0 can be metrized by $d(X, Y) = \mathsf{E}\left(\frac{|X-Y|}{1+|X-Y|}\right)$. Thus, we can write $X_n \to X$ in L^0 iff $\lim_{n\to\infty} \mathsf{E}\left(\frac{|X_n - X|}{1 + |X_n - X|}\right) = 0$, for every $(X_n)_{n\in\mathbb{N}} \subset L^0$ and $X \in L^0$. This turns to be equivalent to convergence in probability, see [8, Th 17.1]. A further equivalent metric is $d(X, Y) = \mathsf{E}(\min\{1, |X - Y|\})$.

[29] Cauchy sequences of rvs in L^2 are s.t. $\|X_n - X_m\|_2 \to 0$ as $n, m \to \infty$. The fact that for such sequences it holds $X_n \xrightarrow{\text{ms}} X$ and $X \in L^2$ is called *completeness* of L^2.

Chapter 9
Introduction to Stochastic Processes



9.1 Definition and Preliminary Classification

Any collection of observations $x_t \in \mathbb{R}$ recorded at fixed times t is called a **times series**. For example, x_t may represent a stock price at end of the day t or a log-return. In this case the time series is indexed by discrete time indices and we can call it historical prices or returns. Since nowadays electronic trading is almost continuous in time, we can have a continuous-time series over some time interval $[0, T]$. Our aim is to build a model to describe the time evolution of such quantities in the future, trying to fit some probabilistic framework to the past behavior of a time series. From now on, we assume $(\Omega, \mathcal{F}, \mathsf{P})$ is a fixed probability space.

Definition 9.1 A stochastic process denoted by $X = (X_t)_{t \in I}$, or $X = \{X_t \mid t \in I\}$, is an indexed family of rvs where $I \neq \varnothing$ is the index set.

Equivalently, a stochastic process can be thought of as a function of two variables $X : I \times \Omega \to \mathbb{R}$, where $X(\cdot, \omega)$ is a rv for each t and $X(t, \cdot)$ is a function called **sample path** of **trajectory**, that maps each scenario $\omega \in \Omega$ into a curve in the space $I \times \mathbb{R}$. When $I = [0, \infty)$, a sample path is the graph of a function in the Cartesian plane with abscissa $t \geqslant 0$. Usually t is understood as 'time', whence a stochastic process is a model of the random evolution of a system in time, e.g. $(S_t)_{t \geqslant 0}$ is referred to a stock price process, $(R_t)_{t \geqslant 0}$ is referred to a log-return process, and so on, see Fig. 9.1. The value X_t is the *state* of the process at time t, then the *state space* is the set of all X_t. When the time is indexed by $[0, \infty)$ we have a **continuous-time stochastic process**, while for $I = \mathbb{N}$ we have a **discrete-time stochastic process**. The law of each rv X_t is called the *unconditional distribution*, see Fig. 9.2. Each X_t can be discrete, (absolutely) continuous or mixed. We are also interested in the conditional distribution, e.g. $\mathsf{P}(X_t \leqslant x \mid X_s \leqslant y)$, for times $0 \leqslant s < t$.

We can ask ourselves how a process X could be at some point in time $t > 0$ compared to its past history at $0 \leqslant s < t$. We endow $(\Omega, \mathcal{F})$ with an increasing

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_9

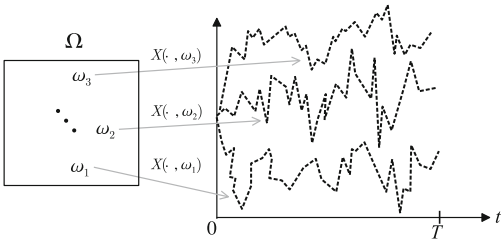


Fig. 9.1 Three trajectories of a process X as graphs of functions in $t \in [0, T]$

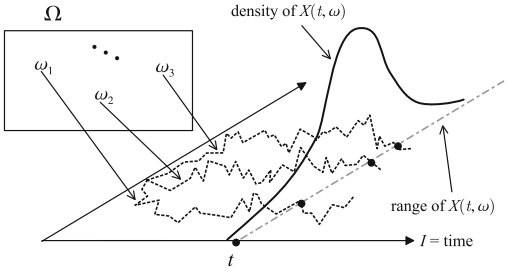


Fig. 9.2 One unconditional continuous distribution at time t

collection $\{\mathcal{F}_t\}_{t\geqslant 0}$ of sub-*sigma*-algebras of $\mathcal{F}$,

$$\mathcal{F}_0 \subset \cdots \subset \mathcal{F}_s \subset \mathcal{F}_t \subset \cdots \subset \mathcal{F} \qquad \text{for all } \ s < t,$$

called **filtration**.[1] The quadruple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathsf{P})$ is said to be a **filtered probability space**. We interpret $\{\mathcal{F}_t\}$ as a flow of information that elapses in time. We can set $\mathcal{F}_t = \sigma(X_s, s \leqslant t)$, meant to be the smallest σ-algebra containing events $\{X_t \in B\}$ for all $t \in [0, \infty)$ and all suitable sets $B \subset \mathbb{R}$. This is the σ-algebra generated by the process X, called the **natural filtration**, such that X_ts are $\mathcal{F}_t$-measurable for all t. X is **adapted** to the filtration $\{\mathcal{F}_t\}$ if X_t is $\mathcal{F}_t$-measurable rv each t; X is clearly adapted to its natural filtration. To verify that A is in $\mathcal{F}_t$ one should observes the value $x = X_t(\omega)$ instead of ω, i.e. $\mathcal{F}_t$ contains all the information about X_t.

Example 9.1 Consider the random experiment of observing the exchange rate Yen/Euro in the next nine days. We can model this quantity through a stochastic process $X = (X_t)_{0\leqslant t\leqslant 9}$ with respect to its natural filtration. If we know that $X_t = 150$ at some $t \in [0, 9]$ we can decide if $A = \{X_t \leqslant 150\}$ is occurred, for $A \in \sigma(X_s, s \leqslant 9)$.

Remark 9.1 We always have $\sigma(X_s) \subset \sigma(X_s, s \leqslant t)$, since $\sigma(X_s)$ contains only information on the values of the process at time $s \geqslant 0$, given by the rv X_s.

[1] It is required that $\mathcal{F}_0$ contains all impossible events.

Remark 9.2 For a process $X = (X_t)_{t\geqslant 0}$ its generated σ-algebras can be collected together by varying t as follows:

$$\sigma(X) := \sigma\left(\cup_{t\in[0,\infty)}\{X_t \in B\}\right) = \sigma\left(X_t \in B, t \geqslant 0,\ B \subset \mathbb{R}\right),$$

where B are suitable subsets of reals (actually Borel sets, i.e. $B \in \mathcal{B}$). This is the smallest σ-algebra that makes all X_t's true rvs (i.e. simultaneously measurable functions) and is equivalently generated by sets of the form

$$\{\omega \in \Omega \mid (X_{t_1}(\omega), \ldots, X_{t_n}(\omega)) \in B\}, \quad B \subset \mathbb{R}^n, \quad t_i \in [0,\infty), \text{ distinct}, \quad i \in \mathbb{N},$$

where $(X_{t_1}, \ldots, X_{t_n})$ is a random vector collecting a finite number of rvs from X. These sets are sometimes called *cylinder sets*. Here the suitable $B \subset \mathbb{R}^n$ are as usual Borel sets, picked from the n-dimensional Borel σ-algebra $\mathcal{B}^n$, which is generated by Cartesian products $B = B_1 \times \cdots \times B_n$, referred to as *measurable rectangles*, where $B_i \in \mathcal{B}$ each i is a Borel set.

9.2 Filtration and Distribution of a Process

The probability distribution of a process $X = (X_t)_{t\geqslant 0}$ requires the knowledge of probabilities $\mathsf{P}(X \in \tilde{B})$ for some set $\tilde{B}$ belonging to the collection

$$\mathbb{R}^{[0,\infty)} = \{\omega(t) \mid \ \omega : [0,\infty) \to \mathbb{R}\} \tag{9.1}$$

of all functions $\omega(t)$ from the index set $[0,\infty)$ to $\mathbb{R}$. The symbol ω better describes the association of a trajectory to each point in the sample space Ω. Things are easier if we are legitimate to use the FIDIS, see (3.18) in Sect. 3.5, Chap. 3:

$$\mathsf{P}_{t_1,\ldots,t_n}(B) = \mathsf{P}\big((X_{t_1}, \ldots, X_{t_n}) \in B\big), \qquad B \subset \mathbb{R}^n \tag{9.2}$$

with respect to the random vector $\mathbf{X} = (X_{t_1}, \ldots, X_{t_n})$ and all choices of distinct time indexes $t_1, \ldots, t_n$. Note that this random vector is a subset of the stochastic process, containing only a finite number of its component rvs, thus any FIDIS expresses the joint probability at finite points in time when the process is represented by events $\{X_{t_1} \in B_1, \ldots, X_{t_n} \in B_n\}$ provided that further $B = B_1 \times \cdots \times B_n$.

Remark 9.3 For a discrete-times process the probability law is based on a countable collection $X = (X_1, X_2, \ldots)$ of rvs. Thus we can figure out its joint distribution as $\mathsf{P}(X_1 \in B_1, X_2 \in B_2, \ldots)$, using the joint event $\cap_{t=1}^{\infty}\{X_t \in B_t\}$ which belongs to $\mathcal{F}$ since it is made of a countable intersection of events, provided that $\{X_t \in B_t\} \in \mathcal{F}_t$ for each time t. Problems arise when X is a continuous-time process, since it is no more a countable collection of rvs. For example, the uncountable intersection $A = \cap_{t\geqslant 0}\{X_t \in B_t\}$ is an event but it is no necessarily true that $A \in \mathcal{F}$, and then to some $\mathcal{F}_t$.

We say that FIDIS of X are *consistent* if

$$\mathsf{P}_{t_1,\dots,t_n}(B_1 \times \cdots \times B_{n-1} \times \mathbb{R}) = \mathsf{P}_{t_1,\dots,t_{n-1}}(B_1 \times \cdots \times B_{n-1}),$$

for any choice of distinct times $t_1, \dots, t_n$ and $n \in \mathbb{N}$. The probability that $(X_{t_1}, \dots, X_{t_{n-1}})$ takes values on the rectangle $B_1 \times \cdots \times B_{n-1}$ can be obtained by that of $(X_{t_1}, \dots, X_{t_{n-1}}, X_{t_n})$ by varying the nth coordinate X_{t_n} anywhere in $\mathbb{R}$, so that it does not matter at all. FIDIS can be equivalently defined via the joint distribution functions

$$F_{t_1,\dots,t_n}(x_1, \dots, x_n) = \mathsf{P}(X_{t_1} \leqslant x_1, \dots X_{t_n} \leqslant x_n),$$

for any choice of time indexes t_i. Then FIDIS are collections

$$\left\{F_{\mathbf{t}}(\cdot) \,|\, \mathbf{t} \subset I\right\} := \left\{F_{t_1,\dots,t_n}(x_1, \dots, x_n) \,|\, \text{all } (t_1, \dots, t_n) \subset I\right\}$$

with $(x_1, \dots, x_n) = \mathbf{x} \in \mathbb{R}^n$. We present a simplified version of a key result about existence of a stochastic process with prescribed FIDIS.

Theorem 9.1 (Kolmogorov's Theorem) *The FIDIS* $\left\{F_{\mathbf{t}}(\cdot) \,|\, \mathbf{t} \subset I\right\}$ *are the CDFs of some stochastic process iff for every* $n = 1, 2, \dots$, $\mathbf{t} \subset I$ *and integer* $i \in \{1, \dots, n\}$ *we have*

$$\lim_{x_i \to \infty} F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}(i)}(\mathbf{x}(i)),$$

where $\mathbf{t}(i) = \mathbf{t} \setminus \{t_i\}$ *and* $\mathbf{x}(i) = \mathbf{x} \setminus \{x_i\} \in \mathbb{R}^{n-1}$.

Notation $\mathbf{t}(i)$ is adapted from [4, Th 1.2.1]. Actually, what the Kolmogorov's theorem says is more involved than it may appear. Given that probability measures $\mathsf{P}_{t_1,\dots,t_n}$ on $\mathbb{R}^n$ are collected by varying the t_i's together with $n \in \mathbb{N}$, and each $\mathsf{P}_{t_1,\dots,t_n}$ is consistent then we can find a process X having such collection as its FIDIS. This means that there exists a probability space $(\mathbb{R}^{[0,\infty)}, \mathcal{F}^{[0,\infty)}, \mathsf{P})$ and rvs $(X_t)_{t \geqslant 0}$ s.t. for each n

$$\mathsf{P}((X_{t_1}, \dots, X_{t_n}) \in B) = \mathsf{P}_{t_1,\dots,t_n}(B), \quad B \in \mathcal{B}^n,$$

where $\mathcal{F}^{[0,\infty)}$ is the σ-algebra generated by the family containing events $\{X_t \in B_t\}$ with $t \geqslant 0$ and Borel sets $B_t \subset \mathbb{R}$. On the other hand, given $\mathbb{R}^{[0,\infty)}, \mathcal{F}^{[0,\infty)}$ as above we can define P starting from events of the form $\{(X_{t_1}, \dots, X_{t_n}) \in B\}$, for n-dimensional Borel sets $B \subset \mathbb{R}^n$ and distinct $t_1, \dots, t_n$, and then we can extend to the whole σ-algebra $\mathcal{F}^{[0,\infty)}$. Thanks to these necessary and sufficient conditions we can model a process using FIDIS usually taken with respect to $B = B_1 \times \cdots \times B_n$. Two more general versions of Theorem 9.1 can be found, for example, in [5, Th 4.7 and Th 4.18]. Really, Theorem 9.1 is an *extension theorem* from measure theory as applied to probability, see also [3, Th 2.18]. Let $X = (X_t)_{t \geqslant 0}$ be a process

defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathsf{P})$. Since the distribution of X is specified by its FIDIS, we can write $\mathsf{P}_{t_1,\dots,t_n}(B_1 \times \cdots \times B_n)$ using the multiplication formula (see Lemma 3.1, Sect. 3.4):

$$\begin{aligned}\mathsf{P}(X_{t_1} \in B_1, \dots, X_{t_n} \in B_n) &= \mathsf{P}(X_{t_1} \in B_1) \\ &\times \mathsf{P}(X_{t_2} \in B_2 \,|\, X_{t_1} \in B_1) \\ &\times \mathsf{P}(X_{t_3} \in B_3 \,|\, X_{t_2} \in B_2, X_{t_1} \in B_1) \\ &\vdots \\ &\times \mathsf{P}(X_{t_n} \in B_n \,|\, X_{t_{n-1}} \in B_{n-1}, \dots, X_{t_1} \in B_1),\end{aligned}$$

for any choice of distinct $t_1, \dots, t_n$ and suitable sets $B_i \subset \mathbb{R}$, each $i = 1, \dots, n$. Choosing $B_i = (-\infty, x_i]$ we can turn to the joint CDF $F(x_1, \dots, x_n)$ and get:

- the marginal distribution $F_1(x_1) = \mathsf{P}(X_{t_1} \leqslant x_1)$ setting $F(x_1, \infty, \dots, \infty)$, at t_1;
- the conditional distribution $F_2(x_2 \,|\, x_1) = \dfrac{\mathsf{P}(X_{t_1} \leqslant x_1, X_{t_2} \leqslant x_2)}{\mathsf{P}(X_{t_1} \leqslant x_1)}$, setting $\dfrac{F(x_1, x_2, \infty, \dots, \infty)}{F(x_1, \infty, \dots, \infty)} = \dfrac{F_{1,2}(x_1, x_2)}{F_1(x_1)}$, at t_2;
- the conditional distribution $F_3(x_3 \,|\, x_1, x_2) = \dfrac{\mathsf{P}(X_{t_1} \leqslant x_1, X_{t_2} \leqslant x_2, X_{t_3} \leqslant x_3)}{\mathsf{P}(X_{t_1} \leqslant x_1, X_{t_2} \leqslant x_2)}$, where we set $\dfrac{F(x_1, x_2, x_3, \infty, \dots, \infty)}{F(x_1, x_2, \infty, \dots, \infty)} = \dfrac{F_{1,2,3}(x_1, x_2, x_3)}{F_{1,2}(x_1, x_2)}$, at t_3;
- $\vdots$
- the conditional distribution $F_n(x_n \,|\, x_1, x_2, \dots, x_{n-1}) = \dfrac{\mathsf{P}(X_{t_1} \leqslant x_1, X_{t_2} \leqslant x_2, \dots, X_{t_n} \leqslant x_n)}{\mathsf{P}(X_{t_1} \leqslant x_1, X_{t_2} \leqslant x_2, \dots, X_{t_{n-1}} \leqslant x_{n-1})}$, setting $\dfrac{F(x_1, x_2, \dots, x_n)}{F(x_1, x_2, \dots, x_{n-1}, \infty)} = \dfrac{F(x_1, x_2, \dots, x_n)}{F_{1,2,\dots,n-1}(x_1, x_2, \dots, n-1)}$, at t_n.

Interpreting each X_{t_i} as a stock price or a log-return, sequential conditioning suggests how the dependence structure in the law of a stochastic process is crucial as well the past history of prices and returns up to some point in time.

9.3 Distributional Characteristics

Given an adapted process $X = (X_t)_{t \geqslant 0}$, if $X_t \in L^1$, i.e., any component rv has finite expectation each t, we can define the **mean function**

$$\mathsf{E}(t) := \mathsf{E}(X_t) < \infty, \quad \text{for all } t > 0, \tag{9.3}$$

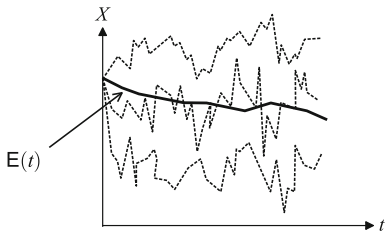


Fig. 9.3 Mean function as an 'average' path

see Fig. 9.3. If $X \in L^2$, i.e., any component rv has finite second moment, we can define the **auto-covariance function**

$$\mathsf{cov}(s,t) = \mathsf{E}((X_s - \mathsf{E}(s))(X_t - \mathsf{E}(t))), \quad \text{for all } 0 \leqslant s \leqslant t. \tag{9.4}$$

If $s = t$ we have $\mathsf{cov}(t,t) = \mathsf{V}(X_t)$. The mean function represents the 'trend' in the sample paths of X; the auto-covariance tells us how each pair of rvs X_s and X_t are (linearly) correlated in time.

A process $X = (X_t)_{t \geqslant 0}$ is said to be **strictly stationary** if for any pair of n-dimensional random vectors taken from X we have

$$(X_{t_1}, \ldots, X_{t_n}) \stackrel{\mathrm{d}}{=} (X_{t_1+h}, \ldots, X_{t_n+h}),$$

whatever are $t_1, \ldots, t_n$ and their shifted version obtained by $h > 0$. The probability laws that govern the behavior of the process do not change over time. The statistical properties of a stationary process are identical wherever we look at the data. For example, $X = (X_t)_{t \geqslant 0}$ is strictly stationary iff X_t's are IID rvs. A process X is said to be **weakly stationary** provided that:

1. $\mathsf{E}(t) = c$, for $c \in \mathbb{R}$ and every t.
2. $\mathsf{cov}(s, s+h) = \mathsf{cov}(t, t+h)$ for every $s < t$, i.e. the auto-covariance function only depends upon a temporal shift h.

A special case of weakly stationary process is when for every $s < t$ we have

$$X_t - X_s \stackrel{\mathrm{d}}{=} X_{t+h} - X_{s+h}, \qquad \forall\, h > 0. \tag{9.5}$$

Then, X is said to have **stationary increments**. If for any choice of time indexes $t_1, t_2, \ldots, t_n$

$$X_{t_2} - X_{t_1}, \ldots, X_{t_n} - X_{t_{n-1}} \tag{9.6}$$

are independent rvs then X is said to have **independent increments**. Observe that:

- strict stationarity implies weak stationarity iff X has finite second moments;
- weak stationarity implies constant **variance function** $\mathsf{V}(t) := \mathsf{V}(X_t)$ each t.

The auto-covariance function of a weakly stationary process is s.t.

$$\mathsf{cov}(t, t+h) := \gamma(h) = \gamma(-h).$$

The **auto-correlation function** is

$$\mathsf{corr}(s,t) = \frac{\mathsf{cov}(s,t)}{\sqrt{\mathsf{V}(s)\mathsf{V}(t)}}, \quad 0 \leqslant s < t.$$

Thus for weakly stationary process we also have:

$$\rho(h) := \mathsf{corr}(t, t+h) = \frac{\mathsf{cov}(t, t+h)}{\mathsf{V}(t)} = \frac{\gamma(h)}{\gamma(0)}.$$

9.4 Special Kinds of Processes

Kolmogorov's Theorem 9.1 enable us to use FIDIS and related distributional characteristics in order to identify special kinds of processes employed in application.

9.4.1 White Noise Processes

Let X be a weakly stationary process at time, $t = 1, 2, \ldots$. If $\mathsf{E}(t) = 0$ for all t and

$$\rho(h) = \mathsf{cov}(t, t+h) = \gamma(h) = \begin{cases} 0, \text{ if } h \neq 0 \\ \sigma^2, \text{ if } h = 0 \end{cases},$$

where $\sigma > 0$, then X is called **white noise**. A version with nonzero mean function is obtained by letting $\mathsf{E}(t) = \mu$ for $\mu \in \mathbb{R}$. We need only two parameters to model such a process, since all X_t's have the same expectation and variance $\mathsf{V}(X_t) = \sigma^2 = \mathsf{cov}(t,t)$. We have $\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{0}{\sigma^2} = 0$ for $h \neq 0$ and equals $\rho(h) = \frac{\gamma(0)}{\gamma(0)} = \frac{\sigma^2}{\sigma^2} = 1$ for $h = 0$, thus a white noise is made by uncorrelated rvs. We may further assume serial independence among X_ts, but recall that no correlation does not imply independence. From now on we will refer to a white noise $X = (X_t)_{t \in \mathbb{N}}$ with IID rvs X_t each with $\mathsf{E}(X_t) = 0$ and $\mathsf{V}(X_t) = \sigma^2$.

Remark 9.4 The term *uncorrelated process* is used when the autocorrelation function $\rho(h) = \mathsf{corr}(t, t+h)$ is 1 for $h = 0$ and 0 for $h > 0$, whether it is stationary or not. The best linear forecast of X_{t+1}, see Sect. 7.2, is its unconditional expectation.

9.4.2 Gaussian Processes

Recall the multivariate Gaussian distribution $\mathbf{X} \sim \mathbf{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu} = (\mathsf{E}(X_1), \ldots, \mathsf{E}(X_n)) = (\mu_1, \ldots, \mu_n)$ and $\boldsymbol{\Sigma}$ is the (positive semi-definite) $n \times n$ covariance matrix, with entries $c_{ii} = \mathsf{V}(X_i) = \sigma_i^2$ and $c_{ij} = \mathsf{cov}(X_i, X_j) = \rho_{i,j}\sigma_i\sigma_j$ for $i \neq j$ and $i, j \in \{1, \ldots, n\}$, with $\rho_{i,j}$ being the Pearson correlation coefficient between pairs X_i, X_j. Thus, $X = (X_t)_{t\in I}$ is said to be a **Gaussian process** if for any choice of distinct times $t_1, \ldots, t_n$ its FIDIS are multivariate Gaussian, i.e. $\mathbf{X} = (X_{t_1}, \ldots, X_{t_n})$ is a Gaussian vector. From Theorem 5.3 in Sect. 5.7 we easily deduce:

Theorem 9.2 *For a Gaussian random vector* $\mathbf{X} = (X_{t_1}, \ldots, X_{t_n})$ *the following are equivalent:*

1. *each* $X_{t_i} \sim \mathrm{N}(\mu_i, \sigma_i^2)$*;*
2. $Y = \sum_{i=1}^n a_i X_{t_i}$ *is a Gaussian rv, for every* $a_1, \ldots, a_n \in \mathbb{R}$.

As byproduct, a Gaussian process is strictly stationary iff it is weakly stationary. In fact, Assuming $X = (X_t)_{t\in I}$ is a strictly stationary Gaussian process, then both $(X_{t_1}, \ldots, X_{t_n})$ and $(X_{t_1+h}, \ldots, X_{t_n+h})$ are $\mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for $h > 0$, thus by Theorem 9.2 we have

$$\begin{aligned}\mathsf{E}(X_{t_i}) = \mathsf{E}(X_{t_i+h}) = \mu_i, \text{ and } \mathsf{V}(X_{t_i}) = \mathsf{V}(X_{t_i+h})\\ = \sigma_i^2 \quad \text{for every } h > 0, t_i, i \in \mathbb{N},\end{aligned}$$

and then

$$(X_{t_i}, X_{t_i+h}) \sim \mathrm{N}\left(\begin{bmatrix}\mu_i\\ \mu_i\end{bmatrix}, \begin{bmatrix}\sigma_i^2 & \rho\sigma_i\sigma_j\\ \rho\sigma_i\sigma_j & \sigma_i^2\end{bmatrix}\right)$$

which implies $\mathsf{cov}(t_i, t_i + h) = \mathsf{cov}(t_j, t_j + h)$ for $t_i < t_j$. For the converse, assuming $X = (X_t)_{t\in I}$ is a weakly stationary Gaussian process then $\mathsf{E}(X_{t_i}) = \mu_i$ for every $t_i, i \in \mathbb{N}$ and also $\mathsf{cov}(t_i, t_i + h) = \mathsf{cov}(t_j, t_j + h)$ for every $h > 0$ and $t_i < t_j$ which implies $\mathsf{V}(X_{t_i}) = \mathsf{V}(X_{t_i+h}) = \sigma_i^2$. Since both $(X_{t_1}, \ldots, X_{t_n})$ and $(X_{t_1+h}, \ldots, X_{t_n+h})$ are $\mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ it follows $X = (X_t)_{t\in I}$ is a strictly stationary Gaussian process.

Example 9.2 Let $R = (R_t)_{t\in\mathbb{N}}$ a discrete-time white noise process with mean function $\mathsf{E}(t) = \mu$, modelling subsequent log-returns R_t. Because of the lack of serial correlation, forecasting the future movements of R from its past realizations is not possible. Assuming R is in addition a Gaussian process, then we have:

$$\mathsf{E}(R_{t+n} \mid R_1, R_2, \ldots, R_n) = \mu, \quad \text{for all } t \geqslant 1.$$

Possible deviations of the R's trajectories from μ are difficult to predict. This is because the future realization R_{t+n} is independent from the past realizations $R_1, R_2, \ldots, R_n$ and the best forecast is just μ, see Sect. 7.2, Chap. 7.

When handling Gaussian processes further properties of multivariate Gaussian distributions are useful. First, yet another consequence of Theorem 5.3 is:

Corollary 9.1 *For a bi-variate Gaussian vector* $\mathbf{X} = (X_1, X_2)$ *the parameter* ρ *is the correlation coefficient of* X_1 *and* X_2.

Proof First recall:

- $\mathsf{E}(\mathsf{E}(X \mid Y)) = \mathsf{E}(X)$
- $\mathsf{E}(X\,Y \mid Y) = Y\,\mathsf{E}(X \mid Y)$.

Now, assume (X_1, X_2) is bi-variate Gaussian and let Z_1, Z_2 be two standard normal rvs. Thus:

$$\begin{aligned}\mathsf{cov}(Z_1, Z_2) &= \mathsf{E}(Z_1 Z_2) = \mathsf{E}\left(\mathsf{E}(Z_1 Z_2 \mid Z_1)\right)\\ &= \mathsf{E}\left(Z_1 \mathsf{E}(Z_2 \mid Z_1)\right) = \rho \mathsf{E}(Z_1^2) = \rho,\end{aligned}$$

where we used the fact (to be proved later) that $\mathsf{E}(Z_2 \mid Z_1) = \rho Z_1$. Since Z_1, Z_2 are standard normal rvs also $\rho = \mathsf{corr}(Z_1, Z_2)$. Now letting $X_i = \mu_i + \sigma_i Z_i$ for $i = 1, 2$ we have $X_i \sim \mathrm{N}(\mu_i, \sigma_i^2)$ as required. Moreover:

$$\mathsf{cov}(X_1, X_2) = \mathsf{cov}(X_1 - \mu_1, X_2 - \mu_2) = \mathsf{cov}(\sigma_1 Z_1, \sigma_2 Z_2)$$

and eventually $\mathsf{corr}(X_1, X_2) = \mathsf{corr}(\sigma_1 Z_1, \sigma_2 Z_2)$ which equals $\mathsf{corr}(Z_1, Z_2) = \rho$. ■

Proposition 9.1 *If* (X_1, X_2) *is bi-variate Gaussian, then for fixed* $x \in \mathbb{R}$ *the conditional distribution of* X_2 *given* $X_1 = x$ *is univariate Gaussian:*

$$X_2 \mid X_1 \sim \mathrm{N}\left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1),\ \sigma_2^2(1 - \rho^2)\right).$$

Proof Let $f_{X_2 \mid X_1}(x_2) = \frac{f_{(X_1,X_2)}(x_1,x_2)}{f_{X_1}(x_1)}$ be the conditional density of X_2 given X_1. Performing the division we get the expression of a Gaussian density with location parameter $\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1)$ and dispersion parameter $\sigma_2^2(1 - \rho^2)$. ■

Remark 9.5 Proposition 9.1 yields

$$\mathsf{E}(X_2 \mid X_1) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(X_1 - \mu_1),$$

i.e. the location parameter above is a conditional expectation depending on the values of X_1. In particular, for two standard normal rvs X_1, X_2 we get:

$$\mathsf{E}(X_2 \mid X_1) = 0 + \rho \frac{1}{1}(X_1 - 0) = \rho X_1.$$

9.4.3 Random Walk Processes

A discrete time process $W = (W_t)_{t \in \mathbb{N}}$ is called a **symmetric random walk** if it can be represented by the recursive equation

$$W_{t+1} = W_t + Z_{t+1}, \quad t = 0, 1, 2, \ldots \tag{9.7}$$

where $W_0 = Z_0 = 0$ and $Z_1, Z_2, \ldots$ are IID rvs with $\mathsf{P}(Z_t = 1) = \mathsf{P}(Z_t = -1) = \frac{1}{2}$, mean function $\mathsf{E}(Z_t) = 0$ for $t = 1, 2, \ldots$ and variance function $\mathsf{V}(Z_t) = \mathsf{E}(Z_t^2) = 1$ for $t = 1, 2, \ldots$ Of course we have $W_t = Z_1 + \cdots + Z_t$. Note that

$$\mathsf{E}(t) = \mathsf{E}(W_t) = \mathsf{E}(Z_1 + \cdots + Z_t) = 0$$

and $\mathsf{V}(t) = \mathsf{V}(W_t) = \mathsf{V}(Z_1 + \cdots + Z_t) = t$. The random walk has independent and stationary increments, but it is not weakly stationary.

Example 9.3 Let $R = (R_t)_{t \in \mathbb{N}}$ be a discrete-time process for the log-returns. Recall that $R_{t+1} = \log\left(\frac{S_{t+1}}{S_t}\right)$, where S_t is the stock price at $t = 0, 1, 2, \ldots$. Thus, $R_{t+1} = \ln(S_{t+1}) - \ln(S_t)$ and the stochastic model for the log-price differences might be a *random walk with drift*:

$$\ln(S_{t+1}) = \mu + \ln(S_t) + Z_{t+1},$$

where $\mu = \mathsf{E}(\ln(S_{t+1}) - \ln(S_t))$ and as usual $Z = (Z_t)_{t \in \mathbb{N}}$ is a white noise.

The sample paths of a random walk can be represented by a binomial tree, see Fig. 9.4. We can interpret $W_t = Z_1 + \cdots + Z_t$ as a repeated bet: we either win $Z_t = 1$ € at time $t = 1, 2, \ldots$ if, for example, the log-price in Example 9.3 moved up from $t - 1$ to t, or we loose $Z_t = -1$ € if the log-price moved down. Thus, W_t is the accumulated fortune up to time t. The tree in Fig. 9.4 may recombine, see Fig. 9.5.

It is worth noting that each $Z_t = X - (1 - X)$ with Bernoulli rv $X \sim \text{Ber}(\frac{1}{2})$, thus Z_t has the same Bernoulli distribution. Then, the FIDIS of a random walk and related conditional distributions are binomial and, for example, we have $\binom{t}{m} p^m (1 - p)^{t-m}$ with $m = 0, 1, \ldots, t$ and $t = 1, 2, 3$. Here m denotes the number of successes, i.e., the number of upward movements (viz. $Z_t = 1$) along the binomial tree. So, we can compute $\mathsf{P}(W_3 = 1 \mid W_2 = 2, W_1 = 1) = 1 - p$ and

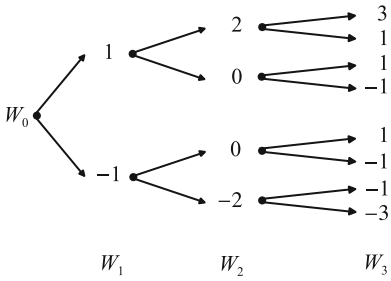


Fig. 9.4 Binomial tree for the random walk $W = (W_t)_{t=1,2,3}$. Each rv Z_t contributes with its support $\{-1, 1\}$ at each node

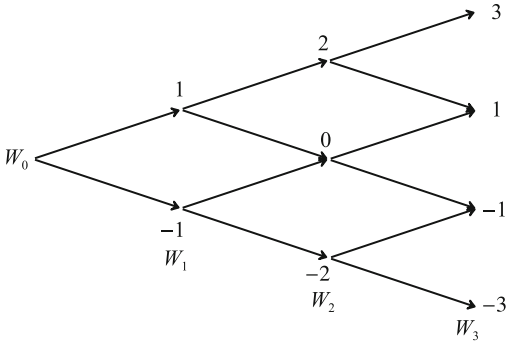


Fig. 9.5 Recombined binomial tree for the random walk $W = (W_t)_{t=1,2,3}$

$\mathsf{P}(W_3 = 1) = 3p^2(1-p)$, and $W_3 \sim \text{Bin}(t, p)$, with $t = 3$ and $p = \frac{1}{2}$. A typical plot of the W's path is depicted in Fig. 9.6. The behavior of a symmetric random walk without drift can be simulated by the following `R` code:

Program Code

```
npath <- 1; p <- 0.5; nsteps <- 30
set.seed(45568223)
RW <- matrix(NA,nrow=nsteps,ncol=npath)
for(j in 1:npath){
RW[1,j] <- 0
for(i in 2:nsteps)
RW[i,j] <- ifelse(runif(1) < p,RW[i-1,j]+1,RW[i-1,j]-1)
}
matplot(RW,ylim=c(range(RW)),type="l",lty=1,lwd=2,
xlab="time step",ylab="W(t)",panel.first=grid(),las=1)
points(1:nsteps,RW[,1],pch=16,col="black")
```

When $p \neq \frac{1}{2}$ we no longer have a symmetric random walk. The number of simulated values W_t per path is $t =$ `nsteps` set to 30. The for-next loop aims at fill the matrix `RW`, using the uniform random number generator `runif(1)`: comparing it with the success probability `p`, in case of a smaller value each entry `RW[i,j]` is incremented by 1, otherwise is reduced by -1. We used a matrix since by specifying `npath`

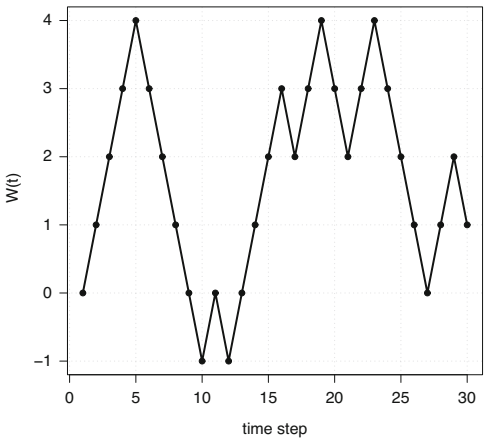


Fig. 9.6 Sample path of a symmetric random walk

greater than 1, we can simulate more sample paths at once. Note that we used `NA` in the matrix specification since no available data are loaded into `RW`, because we fill it with simulated values. Clearly, incrementing `npath` we pass from an array vector to a `nsteps` × `npath` matrix. The command `matplot` yields a plot of the path with the following arguments: `type="l"` connects the dots given by `pch=16`; `lty=1` draws a solid line; `lwd=2` makes the line thicker; `col="black"` entails a black solid line; `panel.first=grid()` adds grid lines.

9.4.4 Martingale Process

A process $X = (X_t)_{t \in I}$, is said to be a **martingale** if it satisfies:

1. X is adapted to the filtration $\{\mathcal{F}_t\}$.
2. For every $t \geqslant 0$ the rv X_t has finite expectation.
3. $\mathsf{E}(X_t \,|\, \mathcal{F}_s) \overset{\text{a.s.}}{=} X_s$, for every $0 \leqslant s < t$.

Property 3 is usually taken as the definition of a martingale. If we try to forecast the future values of X using conditional expectation, then the best we get is the present value X_s and we do not know how the process will evolve (there is no trend). Replacing $=$ in property 3 with $\leqslant$ or $\geqslant$ we have that X is a *supermartingale* or *submartingale* respectively: the process is increasing or decreasing on average. A process is a martingale iff it is both a supermartingale and a submartingale. We can apply all the properties of conditional expectation to a martingale, see Sect. 6.4 in Chap. 6. For example:

$$\mathsf{E}(X_t \,|\, \mathcal{F}_s) = \mathsf{E}\big(\mathsf{E}(X_t \,|\, \mathcal{F}_s) \,|\, \mathcal{F}_t\big)$$

for $s < t$, since $\mathcal{F}_s \subset \mathcal{F}_t$. If we condition on the σ-algebra $\sigma(X_s)$ then:

$$\mathsf{E}(X_t \,|\, \sigma(X_s)) := \mathsf{E}(X_t \,|\, X_s) = X_s, \qquad \text{for all } \ 0 \leqslant s < t$$

or equivalently

$$\mathsf{E}(X_t - X_s \,|\, X_s) = 0, \qquad \text{for all } \ 0 \leqslant s < t.$$

Observe that $\mathsf{E}(X_s \,|\, \mathcal{F}_s) = X_s \mathsf{E}(1 \,|\, \mathcal{F}_s)$ since X_s is $\mathcal{F}_s$-measurable.

Example 9.4 Let $X = (X_t)_{t \in \mathbb{N}}$ be a discrete-time stochastic process. It is a martingale if and only if for all integer t we have that X_t has finite expectation and with probability 1

$$\mathsf{E}(X_{t+1} \,|\, \mathcal{F}_t) = X_t,$$

that is on average the value of X_{t+1} is the same as that of X_t. Since $\mathsf{E}(X_{t+1} \,|\, \mathcal{F}_t)$ is a $\mathcal{F}_t$-measurable rv, by the Law of Iterated Expectations and the fact that $\sigma(X_0, X_1, \ldots, X_t)$ is a sub-σ-algebra of $\mathcal{F}_t$ we have

$$\begin{aligned} \mathsf{E}(X_{t+1} \,|\, X_0, X_1, \ldots, X_t) &= \mathsf{E}\big(\mathsf{E}(X_{t+1} \,|\, \mathcal{F}_t) \,|\, \sigma(X_0, X_1, \ldots, X_t)\big) \\ &= \mathsf{E}(X_t \,|\, X_0, X_1, \ldots, X_t) = X_t, \end{aligned}$$

and the two definition are equivalent.

By differentiating a discrete-time martingale we get an uncorrelated process.

Example 9.5 Let $X = (X_t)_{t \in \mathbb{N}}$ be the discrete-time martingale of Example 9.4. Thus, using the information provided by its own history, i.e., the σ-algebra $\sigma(X_{t-1}, X_{t-2} \ldots)$ we get

$$\mathsf{E}(X_t \,|\, X_{t-1}, X_{t-2}, \ldots) = X_t.$$

Observe that we can define a discrete-time process with $\mathbb{Z}$ as the time index set, so negative integers can be interpreted as past dates. Now, taking the difference $D_t = X_t - X_{t-1}$ we have that the process $D = (D_t)_{t \in \mathbb{Z}}$ is a *martingale difference* with

$$\mathsf{E}(D_t \,|\, D_{t-1}, D_{t-2}, \ldots) = 0,$$

and zero mean function, $\mathsf{E}(t) = 0$. It follows that D is also an uncorrelated process, and if we add stationarity it also becomes a white noise.

There is a property slightly similar to that characterizing a martingale: $X = (X_t)_{t\in I}$ is said to be a **Markov process** if for $s < t$ it satisfies

$$\mathsf{P}(X_t \in B \,|\, \mathcal{F}_s) = \mathsf{P}(X_t \in B \,|\, X_s),$$

where we used the general conditional probability for any Borel set $B \subset \mathbb{R}$. Equivalently we have:

$$\mathsf{P}(X_{t_{n+1}} \in B \,|\, X_{t_1}, \ldots, X_{t_n}) = \mathsf{P}(X_{t_{n+1}} \in B \,|\, X_{t_n}).$$

Another equivalent characterization is:

$$\mathsf{E}(g(X_t) \,|\, \mathcal{F}_s) = \mathsf{E}(g(X_t) \,|\, X_s), \quad g \text{ is measurable and } \mathsf{E}(g(X_t)) < \infty.$$

Only the information at time $s < t$ matter in forecasting the future value X_t. The past is forgotten. It can be shown that any process with independent increment is also of the Markov type. Taking $g(x) = x$ we get $\mathsf{E}(X_t \,|\, \mathcal{F}_s) = \mathsf{E}(X_t \,|\, X_s)$. Do not confuse this with the martingale property. By the equivalent characterization it follows

$$\mathsf{E}(g(X_t) \,|\, \mathcal{F}_s) = \mathsf{E}(g(X_t) \,|\, X_s), \quad g \text{ is measurable and } \mathsf{E}(g(X_t)) < \infty$$

and we get

$$\mathsf{E}(g(X_t) \,|\, X_s) = f(X_s), \quad 0 \leqslant s < t.$$

A Markov process need not to be a martingale, since this only happens when $g(X_t) = X_t$ and $f(X_s) = X_s$. A full account of martingale theory is [12], while a brief though precise analysis of discrete time martingale is [16].

9.4.5 *Autoregressive and GARCH Processes*

Let $X = (X_t)_{t\in\mathbb{N}}$ be a white noise with zero mean function $\mathsf{E}(t) = 0$. We can construct another discrete-time process $Y = (Y_t)_{t\in\mathbb{N}}$ by letting

$$Y_t - \mu = \theta(Y_{t-1} - \mu) + X_t, \quad \text{for all } t \in \mathbb{N} \text{ and } \theta, \mu \in \mathbb{R}. \tag{9.8}$$

The process Y is called **autoregressive** or **AR(1) process**; we can choose $X_t \sim \mathrm{N}(0, \sigma^2)$ with $\mathsf{cov}(X_s, X_t) = 0$ for all $s < t$. We only confine ourselves to first order AR processes, meaning that the stochastic difference equation (9.8) involves only two subsequent rvs Y_t, Y_{t-1} with time lag 1. For AR of greater integer order see [4]. The term $\theta(Y_{t-1} - \mu)$ records the past deviation from the mean, affecting the current behavior of Y. If θ has large absolute value the 'feedback' of the past into

the future is large too. If $\theta = 0$ we have $Y_t = \mu + X_t$ which is a white noise. If Y_t is the rate of return of a financial asset with $t = 1$ day, then X_t can be though of as the effect on the asset's price of new information revealed at time t. But this information cannot be anticipated, so X_t should be independent of X_{t-1}. It can be shown that if $|\theta| < 1$ then Y is weakly stationary, see [4], and we made this assumption from now on. The expression for Y can be written as

$$Y_t = (1-\theta)\mu + \theta Y_{t-1} + X_t, \tag{9.9}$$

which is a regression equation. By weak stationarity the mean function (viz. unconditional expectation) is constant, say $c \in \mathbb{R}$, and from Eq. (9.9):

$$\begin{aligned} c = \mathsf{E}(Y_t) &= (1-\theta)\mu + \theta\mathsf{E}(Y_t) \\ &= \mu - \theta\mu + \theta c \\ &\iff \\ c(1-\theta) &= \mu(1-\theta). \end{aligned}$$

Let $a = (1-\theta)\mu$, then recalling what a best linear prediction means we have:

- $\mathsf{E}(Y_t \mid Y_{t-1}) = a + \theta Y_{t-1}$ (one period forecast)[2]
- $\mathsf{V}(Y_t \mid Y_{t-1}) = \mathsf{V}(X_t) = \sigma^2$
- $\mathsf{E}(Y_t) = \frac{a}{1-\theta}$
- $\mathsf{V}(Y_t) = \frac{\sigma^2}{1-\theta^2}$.

The first item is trivial, for the other two see Appendix in Chap. 9. Yet another equivalent formulation is

$$Y_t = \mu + \sum_{h=0}^{\infty} \theta X_{t-h},$$

obtained by recursive substitutions in (9.8), which holds for times $t \in \mathbb{Z}$. A process related to AR(1) is the **GARCH(1,1)** which is used in modelling the volatility of asset prices, namely a measure of price variability over some time horizon. Before to explain it in detail, we complete the partial list of stylized facts about asset returns introduced in Sect. 2.7, Remark 2.9.

1. Daily returns have little correlation, $\mathsf{corr}(r_t, r_{t-h})$, whatever the time lag $h = 1, 2, \ldots$.
2. The unconditional probability distribution P_{r_t} is not Gaussian.
3. P_{r_t} is not symmetric (it is skewed), i.e., market exhibits very large drawdown but not equally large drawup.

[2] Parameters a, θ must be estimated.

4. Standard deviation $\mathsf{SD}(r_t)$ completely dominates the mean $\mathsf{E}(r_t)$, at short time horizon (usually days).
5. Squared returns r_t^2 as measures of variability displays positive correlation with its own past, i.e. $\mathsf{corr}(r_{t+h}^2, r_t^2) > 0$.
6. Leverage effects: equities (stocks and indices) display negative correlation between variance and mean returns.
7. Correlation between assets returns $\mathsf{corr}(r_t^{\text{asset \#1}}, r_t^{\text{asset \#2}})$ varies in t: it increases in highly volatile down markets or during market crashes.
8. P_{R_t} has fatter tails, even after standardizing returns by a time-varying volatility measure.
9. The more increases horizon, the more P_{r_t} aggregates as a Gaussian distribution.

Note that all the above facts apply as well to daily log-returns. As a result, the price volatility varies as time elapses and clusters in period of high and low market movements. In particular, item 5 suggests that squared returns are positively autocorrelated, thus if we want to model returns only one discrete-time process is not enough: we need a second discrete-time process for the volatility. In its simplest form we have:

$$r_t = \mu + \sigma_t \epsilon_t, \quad \epsilon_t \text{ IID } \mathrm{N}(0,1) \tag{9.10}$$

$$\sigma_t^2 = \alpha_1 + \alpha_2 (r_{t-1} - \mu)^2 + \alpha_3 \sigma_{t-1}^2 \quad \alpha_1, \alpha_2, \alpha_3 \geqslant 0. \tag{9.11}$$

The nonnegative constrains ensure that σ_t^2 is never negative, as it should be. It is further assumed that $\alpha_2 + \alpha_3 > 1$ and $\alpha_1 > 0$, in order the GARCH(1,1) being strictly stationary, see [4].

9.5 Brownian Motion Process

A standard **Wiener process** or **Brownian motion** $B = (B_t)_{t \geqslant 0}$ is a continuous-time process that satisfies the following properties:

1. $B_0 = 0$.
2. B has independent increments.
3. $B_t - B_s \sim \mathrm{N}(0, t-s)$, for $0 \leqslant s < t$.
4. $\mathsf{E}(B_t \mid \mathcal{F}_s) = B_s$, for $0 \leqslant s < t$
5. For any $\omega \in \Omega$, the sample path $t \mapsto B_t(\omega)$ is a continuous and nowhere differentiable function.

A Brownian motion is a Gaussian process, a stationary process with stationary increments, a martingale and a (strong) Markov process, see [12, Ch III, §3]. Intuitively, this means that the increments $B_{t_n} - B_{t_{n-1}}$ do not depend on the position

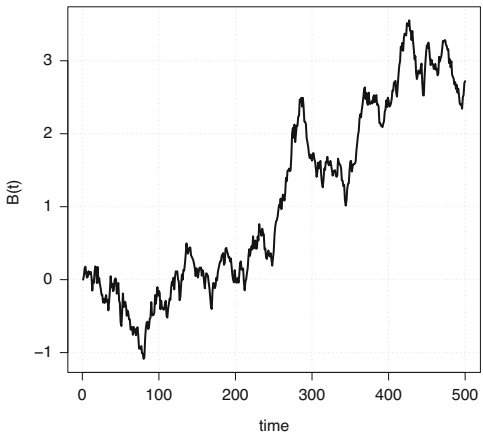


Fig. 9.7 Sample path of a standard Brownian motion

at t_{n-1} over the time interval $[t_{n-1}, t_n]$. It follows that

$$\begin{aligned} &\mathsf{P}\left(B_{t_n} \leqslant x_n \mid B_{t_i} = x_i,\ 0 \leqslant i \leqslant n-1\right) \\ &= \mathsf{P}\left(B_{t_n} \leqslant x_n - x_{n-1}\right) \\ &= \int_{-\infty}^{x_n - x_{n-1}} \frac{1}{\sqrt{2\pi(t_n - t_{n-1})}} \exp\left(-\frac{u^2}{2(t_n - t_{n-1})}\right) \mathrm{d}u. \end{aligned}$$

The behavior of a standard Brownian motion can be analyzed via simulation in R:

Program Code

```
nsteps <- 600
set.seed(8221056)
BM <- matrix(NA,nrow=nsteps,ncol=1)
for(i in 2:nsteps){
BM[i,j] <- BM[i-1,j] + sqrt(0.05)*rnorm(1)
}
matplot(BM,ylim=c(range(BM)),type="l",lty=1,lwd=2,
xlab="time",ylab="B(t)",panel.first=grid(),las=1,col="black")
```

For more details on this code see Sect. 9.4.3. Here `sqrt(0.05)` guarantees a time interval of size 0.05 for the `nsteps` increments of the simulated path. Figure 9.7 shows one typical sample path of a standard Brownian motion.

An additional property of a standard Brownian motion is the *reflection principle*: the first time B_t hits $a \in \mathbb{R}$, called the *hitting time*[3] $T_a := \inf\{t \geqslant 0 \mid B_t = a\}$, it is equally likely that B moves either up or down, so each path is reflected about $B_{T_a} =$

[3] Note that it is a rv.

a with equal distribution of the path after the point (a, T_a) in the Cartesian plane. It is interesting how the symmetric random walk we have seen in Sect. 9.4.3 can be used to get the Brownian motion using appropriate asymptotic stochastic limits. First, by CLT the scaled version converges in distribution to a normal standard: $\frac{W_t}{\sqrt{t}} \xrightarrow{\mathrm{d}} \mathrm{N}(0,1)$, as integer times $t \to \infty$. On the other hand, considering the time horizon $[0, t]$ with $t > 0$ fixed, we could increase the number of rvs Z_t in the definition of W being aware that t is a natural number. Thus, we can define

$$W_t := c(Z_1 + \cdots + Z_{\lfloor t/h \rfloor}), \tag{9.12}$$

for a constant $c \in \mathbb{R}$ and $h > 0$ s.t. $n = \lfloor t/h \rfloor$ is the number of subintervals of $[0, t]$ with equal length h, where $\lfloor \cdot \rfloor$ is the floor function giving the greater positive integer $\leqslant \frac{t}{h}$. Thus, for example, if $t = 4.4$ and $h = 0.5$ we get $\frac{t}{h} = 8.8$ and $\lfloor t/h \rfloor = 8$. If $\frac{t}{h}$ is not integer, we can interpolate W_s, W_u, for $s < t < u$, whenever $\frac{s}{h}$, $\frac{u}{h}$ are positive integer:

$$W_t = \alpha W_s + (1 - \alpha) W_u, \qquad \alpha = \tfrac{u-t}{u-s} \in [0, 1].$$

Substituting the expressions of W_s, W_u given by (9.12) and of α, then rearranging we get W_t. The expectation and variance are:

$$\mathsf{E}(W_t) = 0, \quad \mathsf{V}(W_t) = c^2 \lfloor t/h \rfloor.$$

By setting $c = \sqrt{h}$ and letting $n \to \infty$ we have

$$\lim_{n\to\infty} \mathsf{E}(W_t) = 0, \quad \lim_{n\to\infty} \mathsf{V}(W_t) = t.$$

The second equation follows since $t/h - \lfloor t/h \rfloor$ is the decimal part of $\frac{t}{h}$ and to recover the integer one we can write

$$\lfloor \tfrac{t}{h} \rfloor = \tfrac{t}{h} - \sum_{k=1}^{\infty} c_k 10^{-k}, \quad c_k \in \{1, 2, \ldots, 9\}.$$

Now, $n \to \infty$ is equivalent to taking the limit

$$\lim_{h\to 0} h \left(\frac{t}{h} - \sum_{k=1}^{\infty} c_k 10^{-k} \right) = t$$

and we are done. Eventually, by CLT and standardization we get:

$$\frac{W_t}{\sqrt{h \lfloor t/h \rfloor}} \xrightarrow{\mathrm{d}} \mathrm{N}(0,1), \quad \text{as } n \to \infty$$

which together with $h\lfloor \frac{t}{h} \rfloor \to t$ implies $\frac{W_t}{\sqrt{t}} \sim \mathrm{N}(0,1)$ or equivalently $W_t \sim \mathrm{N}(0,t)$. It is apparent how the limit rv W_t and B_t have the same distribution. This construction suggests why properties 1–4 of a Brownian motion hold. In particular, properties 2 and 3 depends on (9.12), since the scaled symmetric random walk is made of IID rvs. It follows that the MGF

$$\begin{aligned} M_{B_t - B_s}(u) &= \mathsf{E}(\mathrm{e}^{u(B_t - B_s)}) \\ &= \mathsf{E}(\mathrm{e}^{uB_t})\mathsf{E}(\mathrm{e}^{u(-B_s)}) \\ &= M_{B_t}(u) M_{-B_s}(u) = \mathrm{e}^{u^2(t-s)} \end{aligned}$$

it clearly that of a Gaussian rv, see Eq. (5.12) in Sect. 5.7. Moreover, it has been tacitly assumed that B is adapted to the filtration $\{\mathcal{F}_t\}$, and again the construction via the symmetric random walk yields that the rv $B_t - B_s$, for $s < t$, is s.t. its generated σ-algebra $\sigma(B_t - B_s)$ is independent from $\mathcal{F}_s$, i.e., the possible trajectories of B from $t = 0$ up to $s > 0$ do not affect the values of $B_t - B_s$. Form this and the properties of the general conditional expectation it follows:

$$\begin{aligned} \mathsf{E}(B_t \,|\, \mathcal{F}_s) &= \mathsf{E}((B_t - B_s) + B_s \,|\, \mathcal{F}_s) \\ &= \mathsf{E}(B_t - B_s \,|\, \mathcal{F}_s) + \mathsf{E}(B_s \,|\, \mathcal{F}_s) \\ &= \underbrace{\mathsf{E}(B_t - B_s)}_{=0} + B_s. \end{aligned}$$

Since for $s < t$ we have

$$\begin{aligned} \mathsf{E}\Big(\big((B_t - B_s) + B_s\big)B_s\Big) &= \mathsf{E}\big((B_t - B_s)B_s\big) + \mathsf{E}(B_s^2) \\ &= \underbrace{\mathsf{E}(B_t - B_s)\mathsf{E}(B_s)}_{=0} + s \\ &= s, \end{aligned}$$

and similarly for the reverse direction $t < s$, it holds:

$$\mathsf{cov}(s,t) := \mathsf{E}(B_s\, B_t) = \min\{s,t\}.$$

A proof of property 5 is in [2]. More detailed accounts of Brownian motion are [9] and [12]. Some processes can be derived by a standard Brownian motion. The first is Brownian motion starting at $x_0 \in \mathbb{R}$, which can be obtained as $x_0 + B_t$ for every $t \geqslant 0$. Scaled Brownian motions are B_{ct} and $\sqrt{c}B_t$, for all $t \geqslant 0$. A *Brownian motion with drift* is the process given by

$$\tilde{B}_t := \mu t + \sigma B_t, \quad \text{for all } t \geqslant 0, \tag{9.13}$$

where $\mu \in \mathbb{R}$ is called the *drift* and $\sigma > 0$ is called the *diffusion parameter*. This process can be used as a model for continuous-time log-returns, where μ can be thought of as the mean instantaneous compounded return, and σ can be interpreted as the instantaneous volatility of the underlying price process. In fact, the price process can be modelled as a *geometric Brownian motion*:

$$S_t := S_0 \mathrm{e}^{\mu t + \sigma B_t} = S_0 \mathrm{e}^{\tilde{B}_t}, \quad \text{for all } t \geqslant 0, \text{ and constant } S_0 \in \mathbb{R}. \tag{9.14}$$

Clearly, S_0 can be interpreted as the initial stock price and $\ln(\frac{S_t}{S_0}) = \tilde{B}_t$. Since $\tilde{B}_t \sim \mathrm{N}(\mu t, \sigma^2 t)$ it follows that S_t is log-normal with the same parameters, i.e., its natural logarithm is $\mathrm{N}(\mu t, \sigma^2 t)$, while $\mathsf{E}(S_t) = S_0 \mathrm{e}^{\mu t + \sigma^2 2t/2}$ and $\mathsf{V}(S_t) = S_0^2 \mathrm{e}^{2\mu t + \sigma^2 t}(\mathrm{e}^{\sigma^2 t} - 1)$, see Exercise 9.3.

9.6 Exercises

9.1 Prove that $B_{c\,t}$ and $\sqrt{c} B_t$ yield a Brownian motion, for all $t \geqslant 0$.

9.2 Compute the probability that a standard Brownian motion is below $b \in \mathbb{R}$ at $t = 4$, conditional on $B_1 = a$, for $a \in \mathbb{R}$.

9.3 Find the distribution of a Brownian motion with drift and of a geometric Brownian motion.

9.4 Given a standard Brownian motion, find the distribution of the hitting time $T_a = \inf\{t \geqslant 0 \mid B_T = a\}$, for a constant $a > 0$.

9.5 Given a standard Brownian motion, find the CDF of the *running maximum* $\bar{B}_t := \sup_{u \in [0,t]} B_u$, i.e., compute $F_{\bar{B}_t}(a)$ for $a, t > 0$.

Appendix

Details on AR(1) and GARCH(1,1) Processes

We show that $\mathsf{V}(Y_t \mid Y_{t-1}) = \mathsf{V}(X_t) = \sigma^2$. Note that by construction $f(Y_{t-1}) = a + \theta Y_{t-1}$ and X_t are independent thus we have:

$$\begin{aligned}
\mathsf{V}(Y_t \mid Y_{t-1}) &= \mathsf{V}(a + \theta Y_{t-1} \mid Y_{t-1}) + \mathsf{V}(X_t \mid Y_{t-1}) \\
&= \mathsf{E}((a + \theta Y_{t-1})^2 \mid Y_{t-1}) - \{\mathsf{E}(a + \theta Y_{t-1} \mid Y_{t-1})\}^2 + \mathsf{V}(X_t) \\
&= (a + \theta Y_{t-1})^2 - (a + \theta Y_{t-1})^2 + \mathsf{V}(X_t) = \mathsf{V}(X_t).
\end{aligned}$$

Next we show $\mathsf{E}(Y_t) = \frac{a}{1-\theta}$ and $\mathsf{V}(Y_t) = \frac{\sigma^2}{1-\theta^2}$. If Y is weakly stationary, then $\mathsf{E}(Y_t) = c$, $\mathsf{V}(Y_t) = \gamma(0)$ and $\mathsf{cov}(Y_t, Y_{t+h}) = \gamma(h)$, for integer $h \in \mathbb{N}$. Moreover,

$$\mathsf{E}(Y_t) = a + \theta \mathsf{E}(Y_{t-1}) \text{ iff } c = a + \theta c \text{ iff } c = \frac{a}{1-\theta}.$$

Note what this implies: the mean function of Y_t exists iff $\theta \neq 1$, and is zero iff $a = 0$. For the converse, assume $|\theta| < 1$. Now, by repeated substitutions (as time goes to ∞):

$$\begin{aligned} Y_t &= a + \theta Y_{t-1} + X_t \\ &= a + \theta(a + \theta Y_{t-2} + X_{t-1}) + X_t \\ &\vdots \\ &= a(1 + \theta + \theta^2 + \cdots) + (X_t + \theta X_{t-1} + \theta^2 X_{t-2} + \cdots) \\ &= a \sum_{i=0}^{\infty} \theta^i + \sum_{i=0}^{\infty} \theta^i X_{t-i}. \end{aligned}$$

Then $\sum_{i=0}^{\infty} \theta^i = \frac{1}{1-\theta}$ and

$$\mathsf{E}(Y_t) = \mathsf{E}\left(a \sum_{i=0}^{\infty} \theta^i\right) + \mathsf{E}\left(\sum_{i=0}^{\infty} \theta^i X_{t-i}\right) = \frac{a}{1-\theta} + 0.$$

Moreover we see that:

$$\begin{aligned} \mathsf{V}(Y_t) &= \mathsf{V}\left(\frac{a}{1-\theta} + \sum_{i=0}^{\infty} \theta^i X_{t-i}\right) = \mathsf{V}\left(\sum_{i=0}^{\infty} \theta^i X_{t-i}\right) \\ &\overset{\text{indep.}}{=} \sum_{i=0}^{\infty} \mathsf{V}(\theta^i X_{t-i}) = \sigma^2(1 + \theta^2 + \theta^4 + \cdots) \\ &= \frac{\sigma^2}{1-\theta^2}, \quad \text{since also } |\theta^2| < 1. \end{aligned}$$

Further, the autocovariance function is:[4]

$$\begin{aligned}
\mathsf{cov}(Y_t, Y_{t-h}) &= \mathsf{E}\left(\left(\sum_{i=0}^{\infty}\theta^i X_{t-i}\right)\cdot\left(\sum_{j=0}^{\infty}\theta^i X_{t-j-h}\right)\right) \\
&= \sum_{i=0}^{\infty}\sum_{j=0}^{\infty}\theta^i\theta^j\mathsf{E}(X_{t-i}X_{t-j-h}) \\
&\overset{k=j+h}{=} \sum_{i=0}^{\infty}\sum_{k=h}^{\infty}\theta^i\theta^j\mathsf{E}(X_{t-i}X_{t-k}) \\
&= \sigma^2\sum_{i=0}^{\infty}\theta^i\theta^{i+h} = \frac{\sigma^2\theta^h}{1-\theta^2} = \theta^h\mathsf{V}(Y_t).
\end{aligned}$$

As a consequence, the autocorrelation function is $\rho(h) = \theta^h$, for $h = 0, 1, 2, \ldots$. Notice that all the autocovariance are finite by the Cauchy-Schwarz inequality. For GARCH(1,1) Eq. (9.11) can be rewritten

$$\sigma_t^2 = \alpha_1 + (\alpha_2 X_{t-1}^2 + \alpha_3)\sigma_{t-1}^2,$$

from which its expectation is

$$\mathsf{E}(\sigma_t^2) = \alpha_1 + (\alpha_2 + \alpha_3)\mathsf{E}(\sigma_{t-1}^2),$$

by independence of X_{t-1} and σ_{t-1}^2. It is possible to generalize the representation of GARCH(1,1), if we consider the distribution of return r_t conditional on its past history $\sigma(r_{t-1}, r_{t-2}, \ldots)$ as $\mathrm{N}(\mu, \sigma_t^2)$, written

$$r_t \,|\, r_{t-1}, r_{t-2}, \ldots \sim \mathrm{N}(\mu, \sigma_t^2),$$

with volatility given by

$$\sigma_t^2 = \alpha_1 + \alpha_2(r_{t-1} - \mu)^2 + \alpha_3\sigma_{t-1}^2.$$

It can be shown that the unconditional variance $\mathsf{V}(\sigma_t^2)$ is finite, the correlation $(r_t, r_{t-h}) = 0$ for every lag $h > 0$, and the correlation $\mathsf{corr}((r_t-\mu)^2, (r_{t-h}-\mu)^2) > 0$ for every lag $h > 0$. By standardization we also have:

$$\frac{r_t - \mu}{\sigma_t} \,|\, r_{t-1}, r_{t-2}, \ldots \sim N(0, 1).$$

[4] We used that $a\sum_{i=0}^{\infty}\theta^i$ is a constant.

Donsker's Invariance Principle

We give some details on one possible construction of a standard Brownian motion, for which we gave an heuristic explanation in Sect. 9.5. Let $Z_1, Z_2, \ldots$ be a sequence of IID rvs with $\mathsf{E}(Z_i) = 0$ and $\mathsf{V}(Z_i) = 1$, for $i = 1, 2, \ldots$ Let $S_k = Z_1 + \cdots Z_k$, for $k \geqslant 1$ and set $S_0 = 0$. Next, define

$$S_t^{(n)} := \frac{S_{k-1}}{\sqrt{n}} + n\left(t - \frac{k-1}{n}\right)\frac{S_k - S_{k-1}}{\sqrt{n}},$$

for $t \in [\frac{k-1}{n}, \frac{k}{n}]$ and $1 \leqslant k \leqslant n$. This means that $S_t^{(n)} = \frac{S_k}{\sqrt{n}}$ at points $t = \frac{k}{n}$, for $0 \leqslant k \leqslant n$, and the mapping $t \mapsto S_t^{(n)}$ is linearly interpolated in $[\frac{k-1}{n}, \frac{k}{n}]$. Thus we have:

Theorem 9.3 *The stochastic process $S^{(n)} = (S_t^{(n)})_{t \geqslant 0}$ converge in distribution to the standard Brownian motion, as $n \to \infty$.*

Remark 9.6 They are the FIDIS of $S^{(n)}$ that converge to those of the standard Brownian motion. This is the notion of convergence in distribution for stochastic processes.

Chapter 10
Introduction to Market Risk Measures and Option Pricing



10.1 Value-at-Risk

Let $X = V_T - V_0$ be a rv describing the profits or losses (P&L) of a single asset or portfolio, with future traded value V_T and initial value V_0. For a single asset we have $V_t = S_t$ for $t = 0, T$ and the actual rv is S_T, e.g. a future stock price. In case of a portfolio it is possible that V_T depends on several underlying assets, i.e., $V_T = f(T, S_{1,T}, \ldots, S_{n,T})$, where $S_{i,T}$ is the T-value of the ith asset or risk factor. Financial institutions (e.g. insurance companies, banks, etc.) can use internal value-at-risk (VaR) models to asses some measures of risk since the 1st Basel Capital Accord in 1998, see `http://www.bis.org`.

Definition 10.1 Given a P&L rv X, an horizon $T > 0$ representing the near future, value-at-risk at probability level c is given by

$$\mathrm{VaR}_{c,T}(X) := -Q_X(c), \quad \text{for } c \in (0, 1). \tag{10.1}$$

It is a measure quantifying the loss due to portfolio changes over $[0, T]$, s.t. one is confident at $100(1-c)\%$ to undertake an *actual loss* not exceeding $\mathrm{VaR}_{c,T}(X) \equiv K$ €, while there is only a $100c\%$ chance of a larger loss. Usually $T = 1$ day or 1 week and $c = 0.01$ or $c = 0.05$, equivalently 1% or 5%. In some textbooks the loss rv X is defined as $V_0 - V_T = -(V_T - V_0)$, thus an actual loss in T is represented by some scenario $\omega \in \Omega$ s.t. $X(\omega) \geqslant 0$. The definition of X used in Eq. (10.1) gives $X(\omega) \leqslant 0$ so we need to take its negative to report a meaningful value. Hence with the convention $X = V_0 - V_T$ the value-at-risk in Definition 10.1 need to be stated without the minus sign:

$$\mathrm{VaR}_{c,T}(X) := Q_X(c), \quad \text{for } c \in (0, 1). \tag{10.2}$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4_10

The interpretation changes accordingly and becomes more *transparent*. For example, the same value-at-risk of $K \in$ must be reported now with a probability $c = 0.99$ or 0.95 (equivalently 99% or 95%) and quantifies the loss given portfolio changes over the same horizon; 1% or 5% of the time one could experience an actual loss exceeding $K \in$ (not acceptable), i.e. $\mathsf{P}(X > \mathrm{VaR}_{c,T}(X)) = 1 - c$. The latter formulation is equivalent to $\mathsf{P}(X \geqslant \mathrm{VaR}_{c,T}(X)) \leqslant 1 - c$. For absolutely continuous distributions we have

$$c = \int_{-\infty}^{Q_X(c)} f_X(x)\mathrm{d}x = F_X(Q_X(c)), \quad \text{for every } c \in (0, 1). \tag{10.3}$$

VaR is one possible choice for a market risk measure needed in risk management. Financial markets witnessed increasing risks over the last few decades, especially because equity returns, interest rates as well as foreign exchange rates become more volatile, and volumes traded in derivatives market are nowadays huge. Financial risks can be listed (not exhaustively) as:

- *Market Risk*: changes in quoted prices.
- *Credit Risk*: counterparts fail to meet contractual obligations.
- *Liquidity Risk*: investments (in general positions) cannot be bought or sold quickly *enough* to prevent (or minimize) losses.

Senior management should asses the risks that junior managers or traders are assuming (on behalf of firms). The economic interpretation of VaR is that it yields the amount of extra capital a firm[1] needs as cushion for possible losses with $100c\%$ probability of occurrence (e.g. in case of bankruptcy). This *economic capital* is interpreted as a risk-free investment which added to the financial exposure makes it acceptable to an external regulator. Hence, a negative VaR would imply that the firm may give back a part of its capital to shareholders, or alternatively it might accept more risk. In fact, another equivalent definition is:

$$\mathrm{VaR}_{c,T}(X) := \inf\left\{m \in \mathbb{R} \,\middle|\, \mathsf{P}(X + m < 0) \leqslant c\right\}, \quad c \in (0, 1). \tag{10.4}$$

Remark 10.1 Consider the rv $X = V_T - V_0$ as in Definition 10.1, Eq. (10.1). Its negative $-X = -(V_T - V_0) = V_0 - V_T$, corresponds to the rv used in the alternative definition of VaR in Eq. (10.2). Now we give a unified definition[2] of VaR corresponding to these two rvs and the same probability threshold $c \in (0, 1)$. Assuming the CDF F_X's graph has no flats, it is easy to verify

$$\mathrm{VaR}_{c,T}(X) = -Q_X(c) = Q_{-X}(1 - c).$$

[1] We have mentioned banks and insurance companies, i.e. financial firms. But VaR can be used by non-financial firms to quantify risks associated to their financial exposure.

[2] Is crucial to understand that we start with $X = V_T - V_0$ and interpret its negative value as possible losses, with $-X$ being the same losses reported as positive monetary amounts.

The latter equality can be deduced by observing the plots of the density. In case of flats, the situation is slightly more involved, see Appendix.

For a better understanding of Eq. (10.4) we observe that any CDF provides us with two definitions of inverses:

$$Q_X^-(c) := \sup\left\{x \in \mathbb{R} \,\middle|\, F_X(x) < c\right\}, \quad \textbf{left-inverse} \tag{10.5}$$

$$Q_X^+(c) := \inf\left\{x \in \mathbb{R} \,\middle|\, F_X(x) > c\right\}, \quad \textbf{right-inverse}. \tag{10.6}$$

It can be easily seen that

- $\sup\left\{x \in \mathbb{R} \,\middle|\, \mathsf{P}(X \leqslant x) < c\right\} = \inf\left\{x \in \mathbb{R} \,\middle|\, \mathsf{P}(X \leqslant x) \geqslant c\right\}$,
- $\inf\left\{x \in \mathbb{R} \,\middle|\, \mathsf{P}(X \leqslant x) > c\right\} = \sup\left\{x \in \mathbb{R} \,\middle|\, \mathsf{P}(X \leqslant x) \leqslant c\right\}$

and the two supremum remains the same by using $\mathsf{P}(X < x) = F_X(x^-)$. Actually, a quantile $Q_X(c)$ is any function from $(0, 1)$ to $\mathbb{R}$ s.t.

$$Q_X^-(c) \leqslant Q_X(c) \leqslant Q_X^+(c).$$

We usually set $Q_X(c) = Q_X^-(c)$, see Definition 2.8 in Sect. 2.8, but $Q_X(c) = Q_X^+(c)$ is also possible. Consequence: we might define VaR using right-inverses or left-inverses as well. In fact, quantiles are, for any CDF F_X, all the values in the following intervals:

- closed, $\left[Q_X^-(c), Q_X^+(c)\right]$, if $\mathsf{P}\big(X = Q_X^+(c)\big) = 0$,
- right-open, $\left[Q_X^-(c), Q_X^+(c)\right)$, if $\mathsf{P}\big(X = Q_X^+(c)\big) > 0$.

But observe that $Q_X^-(c) = Q_X^+(c) = Q_X(c)$ if and only if F_X has no flat. Thus, VaR can be further rewritten as

$$\mathrm{VaR}_{c,T}(X) = -Q_X^+(c) = Q_{-X}^-(1-c), \quad \text{fro any } \; c \in (0, 1), \tag{10.7}$$

see Exercise 10.2. The economic definition of VaR is then based on Eq. (10.7) relating lower and upper quantiles, see also Appendix. Thus, using

$$\mathrm{VaR}_{c,T}(X) := \inf\left\{m \in \mathbb{R} \,\middle|\, \mathsf{P}(X + m < 0) \leqslant c\right\}$$

one can additionally see that:

1. if $X \geqslant 0$, then $\mathrm{VaR}_{c,T}(X) \leqslant 0$ and VaR is correctly interpreted as the capital amount m that added to the risky position X makes it acceptable;
2. if $X \leqslant Y$, then $\mathrm{VaR}_{c,T}(X) \geqslant \mathrm{VaR}_{c,T}(Y)$, i.e. for a riskier position X the VaR is greater;
3. if $\lambda \geqslant 0$, then $\mathrm{VaR}_{c,T}(\lambda X) = \lambda \mathrm{VaR}_{c,T}(X)$, i.e. if a position is scaled (e.g. with $\lambda = 2$ it is doubled) the VaR is scaled too;

4. $\mathrm{VaR}_{c,T}(X+k) = \mathrm{VaR}_{c,T}(X) - k$, for any $k \in \mathbb{R}$ interpreted as additional risk-free cash added to the position (making it safer).

In item 2, $X \leqslant Y$ means that increasing the payoff profile of the position the risk is reduced. This in related to the usual stochastic order, see Definition 2.6, Sect. 2.6. Indeed, we have:

- if $X(\omega) \leqslant Y(\omega)$ for every scenario ω then $\{X \leqslant x\} \supset \{Y \leqslant x\}$ for any $x \in \mathbb{R}$. Indeed, reversing the set inclusion, $\{X \leqslant x\} \subset \{Y \leqslant x\}$, we would have that there exists a scenario $\omega \in \{Y \leqslant x\}$ which belongs to $\{X > x\}$, but this implies $x < X(\omega) \leqslant Y(\omega)$ contradicting $Y(\omega) \leqslant x$;
- as byproduct $\Omega = \{X \leqslant Y\} \supset \{X \leqslant x\} \supset \{Y \leqslant x\}$;
- applying the probability measure we have $1 \geqslant F_X(x) \geqslant F_Y(x)$, i.e., the usual stochastic order $X \leqslant_{\mathrm{st}} Y$.

Remark 10.2 If $F_{\tilde{X}}(x) \geqslant F_{\tilde{Y}}(x)$ for all $x \in \mathbb{R}$ is the usual stochastic order between the pair of rvs $\tilde{X}, \tilde{Y}$, then set $X = Q_{\tilde{X}}(c)$ and $Y = Q_{\tilde{Y}}(c)$ with $U \sim \mathrm{U}(0,1)$: we have $Q_{\tilde{X}}(c) \leqslant Q_{\tilde{Y}}(c)$ for every $c \in (0,1)$ and as byproduct $X \leqslant Y$ with probability 1. Therefore, $X \leqslant Y$ P-a.s. implies $X \leqslant Y$.

In the context of risk management, the usual stochastic order means that the P&L X is more risky than the P&L Y iff the VaR associated to the former is greater than the VaR associated to the latter. This is because $F_X(x) \geqslant F_Y(x)$ for all $x \in \mathbb{R}$ is equivalent to $Q_X(c) \leqslant Q_Y(c)$ for every $c \in (0,1)$. Using arithmetic returns and assuming $T = 1$, we can write

$$X = V_1 - V_0 = V_0 r, \quad \text{for } r = \frac{V_1 - V_0}{V_0},$$

thus Definition 10.1 can be restated as

$$\mathrm{VaR}_{c,1}(X) = -V_0 Q_r(c), \quad \text{for } c \in (0,1).$$

Now the VaR of a P&L is reduced to that of the corresponding arithmetic return, then we can let $\mathrm{VaR}_{c,1}(R) = -(1 + Q_r(c))$. Observe that in this daily basis we can also use log-returns, but be warned that this is true for a single asset, since portfolio log-returns are not equal to the linear combination of individual log-returns of the underlying components with given portfolio weights. Indeed, let $r_{1,t}, r_{2,t}$ be two arithmetic returns and let w_1, w_2 be the corresponding portfolio weights, see Example 2.11, Sect. 2.5, and recall that $w_1 + w_2 = 1$ and $w_1, w_2 \in [0,1]$. Thus, we have

$$\begin{aligned} \ln(1 + w_1 r_{1,t} + w_2 r_{2,t}) &= \ln(w_1 + w_2 + w_1 r_{1,t} + w_2 r_{2,t}) \\ &= \ln\left(w_1(1 + r_{1,t}) + w_2(1 + r_{2,t})\right) \\ &= \ln\left(w_1 R_{1,t} + w_2 R_{2,t}\right) \end{aligned}$$

from which it is apparent that the portfolio log-return (viz. the last term on the right-hand side) is different from portfolio arithmetic return $w_1 r_{1,t} + w_2 r_{2,t}$, where $1 + r_{i,t} = R_{1,t} = \frac{S_{i,t}}{S_{i,t-1}}$ for $i = 1, 2$ and any time $t = 1, 2, \ldots$. It only holds that for daily or intradaily arithmetic returns $r_{i,t} \approx \ln(R_{i,t})$, see Sect. 8.5. If $r \sim \mathrm{N}(\mu, \sigma^2)$ we have the following simple formula for daily VaR

$$\mathrm{VaR}_{c,1}(X) = -V_0 Q_{\mu+\sigma Z}(c) = -V_0(\mu + \sigma Q_Z(c)),$$

where $Z \sim \mathrm{N}(0, 1)$.

10.2 Coherent Risk Measures

One can think of a risk measure as a mapping $\varrho : L \to \mathbb{R}$. Possible choices for the domain are L^1 or L^2. Sometimes we can use L^∞ defined as the collection of rvs X s.t.

$$\inf\{a \geqslant 0 \,|\, \mathsf{P}(|X| > a) = 0\} = \inf\{a \geqslant 0 \,|\, \mathsf{P}(-a \leqslant X \leqslant a) = 1\} < \infty,$$

i.e. those rvs which are bounded with probability 1. It is advisable that ϱ satisfies desirable properties (axioms), to meet some requirements called for regulatory authorities and supervisory agencies. The following is a partial list of properties.

Definition 10.2 (Coherence) A risk measure $\varrho : L^p \to \mathbb{R}$ satisfying

- *monotonicity*, $X \leqslant Y$ implies $\varrho(X) \geqslant \varrho(Y)$,
- *positive homogeneity*, $\varrho(\lambda X) = \lambda \varrho(X)$, for $\lambda \geqslant 0$,
- *cash (translation) invariance*, if $m \in \mathbb{R}$ then $\varrho(X + m) = \varrho(X) - m$,
- *subadditivity*, $\varrho(X + Y) \leqslant \varrho(X) + \varrho(Y)$,

is called **coherent**.

The first three properties are the same as those in items 2, 3 and 4 satisfied by VaR, see Sect. 10.1. If ϱ satisfies monotonicity and cash invariance, then it is called a *monetary risk measure*. Interpreting X as P&L generated by some trading strategy, the former property means that the risk of a loss can be cut off by passing to a greater Y with increased payoff. Cash invariance means that $\varrho(X)$ can be interpreted as a capital requirement, needed to cushion the position underlying X and make it acceptable.[3] Subadditivity means that merging two separate positions X and Y cannot increase risk. VaR is not subadditive in general.

Remark 10.3 For a monetary risk measure ϱ, positive homogeneity implies *normalization*: $\varrho(0) = 0$. Indeed, for $\lambda > 0$ and $\varrho(0) = \varrho(\lambda\, 0)$ we have $\lambda \varrho(0) = \varrho(0)$ iff $\varrho(0) = 0$. In the case $\varrho(0) = \varrho(0\, X) = 0$, the latter is equal to $0\, \varrho(X)$ which in

[3] This is the typical point of view of a supervising agency.

turn is equal to zero. Now, a monetary risk measure is *convex* whenever

$$\varrho(\lambda X + (1-\lambda)Y) \leqslant \lambda\varrho(X) + (1-\lambda)\varrho(Y), \quad \text{for all } \lambda \in [0,1],$$

which can be interpreted as diversification (viz. the position $\lambda X + (1-\lambda)Y$, with a percentage λ invested in X and the remaining $(1-\lambda)$ invested in Y) reduces the risk. A convex risk measure is coherent if it is in addition positive homogeneous, since it can be shown that convexity together with positive homogeneity imply subadditivity, see [7, Ch 4]. On the other hand, any coherent risk measure is convex.

Further desirable properties are the following:

- instead of subadditivity, given the dependence structure of (X, Y) we can have additivity, $\varrho(X+Y) = \varrho(X) + \varrho(Y)$ provided X, Y are comonotonic. Putting this kind of P&L together is not an hedge against each other (riskiness never decreases);
- If $(X_n)_{n\in\mathbb{N}}$ is a sequence of P&L's converging in distribution to the P&L X, then

$$\lim_{n\to\infty} \varrho(X_n) = \varrho(X).$$

The following results characterize coherent risk measures using either probability laws or their Radon-Nikodým derivatives with respect to equivalent probability measures, see Theorem 6.3, Sect. 6.4.

Theorem 10.1 (Robust Representation of Coherent Risk Measures) *Any coherent risk measure is* ***scenario-based****, i.e. it can be represented as*

$$\varrho(X) = -\inf\{\mathsf{E}_{\mathsf{Q}}(X) \,|\, \mathsf{Q} \in \mathcal{P}\} = \sup\{\mathsf{E}_{\mathsf{Q}}(-X) \,|\, \mathsf{Q} \in \mathcal{P}\}, \tag{10.8}$$

where $\mathcal{P}$ is a (convex) set of probability measures and E_{Q} means expectation taken w.r.t. Q.

Theorem 10.2 (Dual Representation of Coherent Risk Measures) *Any coherent risk measure being scenario-based may be represented as*

$$\varrho(X) = -\inf\{\mathsf{E}_{\mathsf{P}}(XZ) \,|\, Z \geqslant 0,\ \mathsf{E}_{\mathsf{P}}(Z) = 1\} \tag{10.9}$$

where Z is the Radon-Nikodým derivative $\frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}}$ of probability measures $\tilde{\mathsf{P}}$ equivalent to the fixed P.

Based on the concept of coherent risk measure and since VaR does not account[4] for the magnitude of losses greater than it (beyond the c-level) the following founds alternative application.

[4] Moreover, VaR can discourages diversification.

Definition 10.3 Given a P&L rv $X \in L^1$ the **expected shortfall** for the horizon $[0, T]$ is defined as

$$\mathsf{ES}_c(X) := -\frac{1}{c}\int_0^c \mathrm{VaR}_{c,T}(X)\mathrm{d}c, \quad \text{for } c \in (0,1),$$

which can be equivalently given by

$$\mathsf{ES}_c(X) := -\frac{\mathrm{VaR}_{c,T}(X)}{c}\left(c - \mathsf{P}(X \leqslant -\mathrm{VaR}_{c,T}(X))\right) - \frac{1}{c}\mathsf{E}\left(X\mathbf{I}_{\{X\leqslant -\mathrm{VaR}_{c,T}(X)\}}\right). \tag{10.10}$$

Assuming the P&L X in Definition 10.3 being absolutely continuous, then Eq. (10.10) reduces to

$$\mathsf{ES}_c(X) = -\mathsf{E}(X \mid X \leqslant -\mathrm{VaR}_{c,T}(X)) = -\frac{\mathsf{E}\left(X\mathbf{I}_{\{X\leqslant -\mathrm{VaR}_{c,T}(X)\}}\right)}{\mathsf{P}(X \leqslant -\mathrm{VaR}_{c,T}(X))}.$$

Thus, we can interpret ES as the mean value of the P&L variable conditional on it (short)fall below a threshold given by the $100c\%$ VaR, which is taken with negative sign as representing the c-th left-quantile of the X's distribution. Observe that $\mathsf{P}(X \leqslant -\mathrm{VaR}_{c,T}(X)) = F_X(-\mathrm{VaR}_{c,T}(X)) = c$, for continuous distributions, and $-\mathrm{VaR}_{c,T}(X) = Q_X(c)$. For an in depth account of monetary risk measures see [7, Ch 4]. A last risk measure we present is based on the following definition. The *expectile* of $X \in L^2$, at a probability level $c \in (0,1)$, is

$$e_X(c) = \underset{z\in\mathbb{R}}{\mathrm{argmin}}\,\{\mathsf{E}(S(z,X))\}, \tag{10.11}$$

based on the asymmetric quadratic score function

$$S(z,X) := |c - \mathbf{I}_{\{X-z<0\}}|(X-z)^2.$$

The scoring function is equal to $c(X-z)^2$ whenever $\mathbf{I}_{\{X\geqslant z\}} = 1$, while it is $(1-c)(X-z)^2$ if $\mathbf{I}_{\{X<z\}} = 1$. The mean scoring function can be written at once as $\mathsf{E}\left(c((X-z)^2)^+ - (1-c)((X-z)^2)^-\right)$. The minimum is attained at $z = e_X(c)$, by differentiating inside the expectation w.r.t. z. We used the definition of absolute value together with that of indicator function and positive/negative part of a rv. E.g., if $z - \mathbf{I}_{\{X<z\}} \geqslant 0$ it must hold $\mathbf{I}_{\{X\geqslant z\}} = 1$. Expectiles are asymmetric generalization of the mean and also quadratic versions of the usual quantiles. The corresponding first order condition and then the equivalent definition of c-expectile which holds also for $X \in L^1$ is,

$$c\,\mathsf{E}\,(X - e_X(c))^+ - (1-c)\mathsf{E}\,(X - e_X(c))^- = 0 \tag{10.12}$$

giving $e_X(c)$ as a unique minimizer, see [11, Sec 8.2.2].

Remark 10.4 A scoring function $S : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ fulfills the following properties:

- $S(x, y) \geqslant 0$, and for $x = y$ it is zero;
- $S(x, y)$ is increasing for $x > y$ and decreasing for $x < y$;
- $S(x, y)$ is continuous in the first component x.

It can be shown that the quantile function $Q_X(c)$ can be associated to the scoring function $S(z, X) = |c - \mathbf{I}_{\{X - z \leqslant 0\}}||X - z|$, see [11, Sec 9.3.3].

Proposition 10.1 (Properties of Expectiles) *Let $X \in L^1$ be a random return with CDF F_X and $c \in (0, 1)$. Then:*

(a) Translation equivariance and homogeneity, $e_{aX+b}(c) = a\, e_X(c) + b$, for every $a \geqslant 0$ and $b \in \mathbb{R}$;
(b) Monotonicity, $X \leqslant Y$ P-a.s. implies $e_X(c) \leqslant e_Y(c)$, for all $c \in (0, 1)$.
(c) Strict Monotonicity, $X \leqslant Y$ P-a.s. and $\mathsf{P}(X < Y) > 0$ implies $e_X(c) < e_Y(c)$, for all $c \in (0, 1)$.
(d) Parameter continuity and monotonicity, $e_X(\,\cdot\,) : (0, 1) \to (\mathit{ess}\inf X, \mathit{ess}\sup X)$ is continuous and strictly increasing, except when X is degenerate and then $e_X(c)$ is constant for every $c \in (0, 1)$; the function $c \mapsto e_X(c)$ maps the unit interval $(0, 1)$ onto $\{x \in \mathbb{R} \mid 0 < F_X(x) < 1\}$.
(e) $\lim_{c \downarrow 0} e_X(c) = \mathit{ess}\inf X$ and $\lim_{c \uparrow 1} e_X(c) = \mathit{ess}\sup X$.
(f) Superadditivity and subadditivity, for another $Y \in L^1$, if $c \in (0, \frac{1}{2}]$, then $e_{X+Y}(c) \geqslant e_X(c) + e_Y(c)$; if $c \in [\frac{1}{2}, 1)$, then $e_{X+Y}(c) \leqslant e_X(c) + e_Y(c)$.
(g) Additivity for positively linear dependent returns, if $Y = aX + b$ with $a > 0$ and $b \in \mathbb{R}$, i.e. $\mathsf{corr}(X, Y) = 1$, then $e_{X+Y}(c) = e_X(c) + e_Y(c)$.
(h) Upper and lower expectiles, $e_X(c) = -e_{-X}(1 - c)$, for all $c \in (0, 1)$;
(i) Law invariance, for random returns $X, Y \in L^1$ sharing the same distribution, i.e. $X \overset{d}{=} Y$, is equivalent to have $e_X(c) = e_Y(c)$ for all $c \in (0, 1)$.

From Eqs. (10.11) and (10.12) it is clear that $e_X(\frac{1}{2}) = \mathsf{E}(X)$.

Remark 10.5 For a relation among random returns, the suffix P-a.s. is understood with probability 1. For example, $X \leqslant Y$ P-a.s. stands for $\mathsf{P}(X \leqslant Y) = 1$. Given $Y(\omega) = c$ for every $\omega \in \Omega$, we recall that the *essential supremum* of a random return is defined as $\inf\{c \in \mathbb{R} \mid X \leqslant c \ \mathsf{P}\text{-a.s.}\}$, denoted $\operatorname{ess\,sup} X$. Similarly, the *essential infimum* is defined as $\sup\{c \in \mathbb{R} \mid X \geqslant c \ \mathsf{P}\text{-a.s.}\}$, denoted $\operatorname{ess\,inf} X$. We have $X \leqslant \operatorname{ess\,sup} X$ P-a.s. and $X \geqslant \operatorname{ess\,inf} X$ P-a.s., provided that the essential supremum and the essential infimum are both finite. Otherwise, it could be the case that $\operatorname{ess\,sup} X = \infty$ or $\operatorname{ess\,inf} X = -\infty$.

A further consequence of properties (a), (g) and (h) is that

$$e_{aX+b}(c) = ae_X(1-c) + b,$$

for $b < 0$. This together with (a) gives that expectiles are location and scale equivariant. When X is a P&L, the functional $-e_X(c)$ is a coherent risk measure for $c \leqslant \frac{1}{2}$, also called *expectile VaR*. It is worth noting that the expectile VaR is a continuous function of the probability level $c \in (0,1)$ also for discrete X, while VaR is not since it is based on quantile which takes finite values for typical discrete distribution. It is possible to associate to each CDF F_X a derived CDF through expectile $e_X(c)$. Indeed,

$$\tilde{F}_X(x) = \frac{\mathsf{E}(X-x)^-}{\mathsf{E}(|X-x|)}, \tag{10.13}$$

is actually a new CDF, continuous and strictly increasing on $\{t \in \mathbb{R} \mid 0 < F_X(x) < 1\}$. It is worth noticing that $\mathsf{E}(X-x)^- \leqslant \mathsf{E}(|X-x|)$, then for all $x \in \mathbb{R}$ the ratio defining $\tilde{F}_X(x)$ is less than or equal to 1. The point now is that $e_X(c)$ is the c-quantile of the derived CDF $\tilde{F}_X(c)$, i.e

$$\tilde{F}_X(c) = e_X^{-1}(x), \text{ for all } x \in (\operatorname{ess\,inf} X, \operatorname{ess\,sup} X)$$
$$\Longleftrightarrow$$
$$\tilde{F}_X^{-1}(c) = e_X(c), \text{ for all } c \in (0,1).$$

Observe that $\mathsf{E}(|X-x|)$ equals $\mathsf{E}(X-x)^+ + \mathsf{E}(X-x)^-$, and the derived CDF in (10.13) can be rearranged as

$$\tilde{F}_X(x) = \frac{\mathsf{E}(X-x)^+ - \mathsf{E}(X) + x}{2\mathsf{E}(X-x)^+ - \mathsf{E}(X) + x},$$

where $\mathsf{E}(X-x)^+$ is the *stop-loss transform* (decreasing and convex in x).

Remark 10.6 For those $x \in \mathbb{R}$ such that $\mathsf{E}(X-x)^+ > 0$, the stop-loss transform is strictly decreasing with $\lim_{x\to-\infty} \mathsf{E}(X-x)^+ - \mathsf{E}(X) + x = 0$ and $\lim_{x\to\infty} \mathsf{E}(X-x)^+ = 0$. Observe that $\mathsf{E}(X-x)^+ = 0$ for such $x \in \mathbb{R}$ with $\mathsf{P}(X > x) = 0$ and $\mathsf{E}(X-x)^+ - \mathsf{E}(X) + x = 0$ for such $x \in \mathbb{R}$ with $\mathsf{P}(X < x) = 0$. Yet other equivalent expressions for the stop-loss transform are

$$\mathsf{E}(X-x)^+ = \mathsf{E}(X)^+ - \int_0^x [1 - F_X(u)]\mathrm{d}u = \int_x^\infty [1 - F_X(u)]\mathrm{d}u.$$

Derived CDFs should be continuous although the original ones are not, see Figs. 10.1 and 10.2.

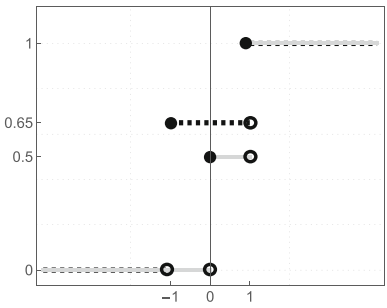


Fig. 10.1 Comparing F_X, F_Y: the rv X has law $\mathsf{P}(X = -1) = 0.65$ and $\mathsf{P}(X = 1) = 0.35$ (dashed, black); $Y \sim \text{Ber}(0.5)$ (solid, gray)

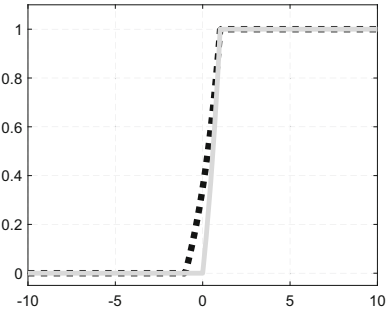


Fig. 10.2 Comparing derived CDFs: $\tilde{F}_X$ (dashed, black) and $\tilde{F}_Y$ (solid, gray)

10.3 Working Examples

Assume to invest only in one stock, say the S&P500 market index, with spot price $S_0 = 35,409$, and assume the log-normal model for the $T = 1$ day price $S_1 = S_0 \mathrm{e}^{\mu+\sigma Z}$ with Standard Normal $Z \sim \mathrm{N}(0, 1)$. In addition we let $\mu = 0.005$ and $\sigma = 0.02$ and consider to hold 500 shares of the underlying index. To compute $\mathrm{VaR}_{c,1}(X)$ we handle the P&L rv

$$X = 500\,(S_1 - S_0) = 500\,S_0 r,$$

where $r = \frac{S_1 - S_0}{S_0}$ so $S_1 = S_0(1 + r)$. To compute a daily VaR at level $c = 0.01$ or $c = 0.05$ we only need to obtain the negative of the quantile $Q_Z(c)$ and compute

$$\mathrm{VaR}_{c,1}(X) = -500\,S_0\,(\mu + \sigma\,Q_Z(c)).$$

Here the `Matlab` code for inverting the CDF of Z, i.e. $Q_Z(c) = F_Z^{-1}(c)$, then computing VaR with two different c-levels.

Program Code

```
z=norminv([0.01,0.05],0,1);
mu=[0.005  0.005]; sigma=0.02;
VaR=-500*35409*(mu + sigma*z)
VaR =
735214.02      493903.72
```

The command `VaR = -500*35409*(mu + sigma*z)` is not followed by a semicolon, then `Matlab` displays the two computed values showed in the last line. Buying 500 shares in the S&P500 with spot price $S_0 = 35{,}409$ we hold a portfolio value 17,704,500.00 \$. Holding this portfolio we may suffer a loss greater than

$$\mathrm{VaR}_{0.01,1}(X) = 735{,}214.02\,\$,$$

1% of times within 1 day, or

$$\mathrm{VaR}_{0.05,1}(X) = 493{,}903.72\,\$,$$

5% of times. We are confident to suffer smaller losses $100(1-c)\% = 99\%$ or $100(1-c)\% = 95\%$ of times, during the next day. The two VaR figures also quantify the amount of economic capital to set aside for handling unfavorable scenarios. Under the log-normal hypotheses we can compute a VaR figure for more than 1 day, using the same value μ, σ referred as the *1-day drift* and *1-day volatility*. Here the 'recipe':

$$\mathrm{VaR}_{c,T}(X) = -\#\text{ shares } S_0\left(\mu T + \sigma\sqrt{T}\,Q_Z(c)\right)$$

where $T =$ number of days within the horizon. Dropping the assumption that P&Ls are of the form $X = -\#\text{ shares } S_0\big(\mu + \sigma\mathrm{N}(0,1)\big)$, we can resort to nonparametric estimation of VaR. Once we have at disposal n data referring to returns, being historical values or simulated via Monte Carlo method, we consider them as realizations of an IID sample:

$$x_1, \ldots, x_n = X_1(\omega), \ldots, X_n(\omega).$$

We use the sample for computing the c-VaR without assuming any parametric probability distribution as follows.

1. Order returns from the smallest to the greatest
2. Compute $\lceil cn \rceil$, the greatest integer $\geqslant cn$
3. Write $x_{(\lceil cn \rceil)}$ for the corresponding ordered value within the sample
4. Compute the daily estimated $\widehat{\mathrm{VaR}_{c,1}(X)} = -\#\text{ shares } S_0 x_{(\lceil cn \rceil)}$.

Here the number $\lceil cn \rceil$ approximates the ith position of the value x_i in the sample to which corresponds a probability c for the theoretical quantile $Q_Z(c)$. This is especially true if we assume a continuous CDF F_Z of the population Z and recalling the empirical CDF

$$\mathbb{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{I}_{\{Z_i \leqslant x\}}, \quad x \in \mathbb{R}. \tag{10.14}$$

It measures the random proportion of the data points in the sample being $\leqslant x$. We can estimate it through the observed data, $\hat{\mathbb{F}}_n(x) = \frac{\text{number of data } x_i \leqslant x}{n}$, see Sect. 8.3. Considering the ordered statistics $Z_{(1)} < \cdots < Z_{(n)}$ we arrive to:

$$\widehat{Q_Z(c)} = \mathbb{F}_n^{-1}(c) = Z_{(i)} \Longleftrightarrow \mathbb{F}_n(Z_{(i)}) = \frac{i}{n}, \tag{10.15}$$

for $c \in (\frac{i-1}{n}, \frac{i}{n}]$ and $i = 1, \ldots, n$. To illustrate this, we start from a dataset comprising 1119 values s_i of the daily S&P500's quotes, from 31 December 2012 to 12 June 2017. Then we get daily log-returns $x_{i-1} = \log(s_i) - \log(s_{i-1})$ for $i = 2, 3, \ldots, 1118$, which approximate arithmetic returns $\frac{s_i - s_{i-1}}{s_{i-1}}$. Considering again to invest in 500 shares of this market index with initial price $S_0 = 35{,}409$, we obtain the estimated VaR at $c = 0.01$ level using `Matlab`:

Program Code

```
x=[0.0251  -0.0021  ... -0.0010];
y=prctile(x,0.01);
VaR=-500*35409*y
VaR =
711720.90
```

By the first command, historical log-returns are collected into the array `x`, which is a 1118×1 vector.[5] The function `prctile(x,0.01)` yields the empirical quantile -0.0402 corresponding to the $\lceil cn \rceil = 11$th ordered statistic $Z_{(11)}$ with $c = 0.01$ and $n = 1118$. It could be considered as the value obtained by inverting the empirical CDF through $\mathbb{F}_{1118}^{-1}(0.01)$.

10.4 Option Pricing

In Sect. 2.5, Example 2.8, we have seen that an European call option is a financial contract giving the right, but not the obligation, to buy an underlying asset at strike price $K > 0$ at maturity $T > 0$. The value in T, called payoff, is $C_T = \max\{0, S_T - K\} = (S_T - K)^+$, and we can ask what the fair price at some date $0 \leqslant t < T$ should be since it is clear how it depends on the whole path followed by the stochastic process $S = (S_t)_{t \geqslant 0}$ modelling the underlying. Any such a contract whose value in $[0, T]$ depends on S is called *contingent claim*. Observe that the payoff $(x - K)^+$ as a function of $x = S_T$ is zero for $x - K < 0$ (out-of-the money) and is equal to $x - K$ otherwise (in-the money), thus the graph of $y = (x - K)^+ = \mathbf{I}_{\{x - K \geqslant 0\}}$ is piecewise linear convex. Our aim is to determine C_T, and our first assumption is

[5] We have displayed only the first two and the last.

that the payoff is *replicable*, i.e. $C_T \overset{\text{a.s.}}{=} V_T$ for some portfolio $\mathbf{h}$, as the following example explains.

Example 10.1 Assume $-1 < r$ is the continuously compounded return of a risk-free asset. In $t = 0$ we form a portfolio $\mathbf{h} = (h^1, h^2)$, where the holding $h^1 = -K\mathrm{e}^{-rT}$ is given by short selling[6] and the corresponding position is risk-free, while $h^2 = 1$ is holding given by buying a corresponding share of the risky security. The initial value of the portfolio is[7]

$$V_0 = h^1 D_0 + h^2 S_0 = -K\mathrm{e}^{-rT} + S_0,$$

where $D_0 = 1$, without loss of generality, is the risk-free initial price and S_0 is the initial price of the risky security. We keep $\mathbf{h}$ unchanged until T, thus the final value of the portfolio is

$$V_T = h^1 D_T + h^2 S_T = -K\mathrm{e}^{-rT} D_0 \mathrm{e}^{rT} + h^2 S_T = -K + S_T.$$

This trading strategy has succeeded in create a final monetary value equal to the in-the money option payoff. If we choose $\mathbf{h} = (0, 0)$ then we replicate the out-of-the money option payoff. Here we have tacitly assumed that the time evolution of the risk-free price is deterministic, and this is can be described by the ordinary first order differential equation $\frac{\mathrm{d}}{\mathrm{d}t} D_t = D_t \mathrm{e}^{rt}$, equivalently written as $\mathrm{d}D_t = D_t \mathrm{e}^{rt} \mathrm{d}t$. Yet another formulation is $D_t = D_0 + \int_0^t D_u \mathrm{e}^{ru} \mathrm{d}u$, in fact the solution in both cases[8] is $D_t = D_0 \mathrm{e}^{rt}$.

Next, we assume that the market is *free of arbitrage*, since simultaneously buying a given asset where it is traded at a lower price, and selling it where it is traded at a higher price yields an *arbitrage profit* (at zero initial cost) which is followed by the action of many 'arbitrageurs' on a large scale. This will force that market, i.e., the price will raise in the first market and will fall in the second, thus we arrive at the same price. Then, a mathematical model of a financial market should not allow arbitrage. We refer to a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathsf{P})$, where $\{\mathcal{F}_t\} = \sigma(B_s, s \leqslant t)$ is the natural filtration of a standard Brownian motion, whose role will be clarified soon. Armed with the notion of replicating portfolio, we choose as components a risk-free asset with price process $(D_t)_{t \geqslant 0}$, and a risky security[9] with price process $(S_t)_{t \geqslant 0}$. Further, the replicating portfolio $\mathbf{h} = (h_t^1, h_t^2)$

[6] This means that in $t = 0$ we sell the asset with delivery in $t = T$, which is equivalent to borrow money.

[7] We further assume to invest all our wealth in the portfolio.

[8] In deterministic calculus, differentials and integral equations are equivalent.

[9] Actually we refer to a stock paying no dividend.

has components $(h_t^i)_{t\geqslant 0}$ which are adapted to $\{\mathcal{F}_t\}$, for $i = 1, 2$. This means that the portfolio value process $(V_t)_{t\geqslant 0}$ with

$$V_t = h_t^1 D_t + h_t^2 S_t, \tag{10.16}$$

is adjusted continuously in a non anticipated way. We require that V is *self-financing*, i.e., the changes in V_t depends upon trading in the risk-free and the risk security, but not upon either injection or withdraw of additional money. More precisely, given $0 \leqslant t_{i-1} \leqslant t_i \leqslant T$, where T is the maturity of the contingent claim, for $i \geqslant 1$,

$$V_{t_{i-1}} = h^1_{t_{i-1}} D_{t_{i-1}} + h^2_{t_{i-1}} S_{t_{i-1}} = h^1_{t_i} D_{t_{i-1}} + h^2_{t_i} S_{t_{i-1}}.$$

which means that the investor readjusts his positions from $h^i_{t_{n-1}}$ to $h^i_{t_n}$ without bringing or consuming any wealth, i.e., the wealth before a trading and that after are equal. As byproduct, the net trading gain from price changes is

$$\begin{aligned} V_{t_n} - V_0 &= \sum_{i=1}^{n} \left(V_{t_i} - V_{t_{i-1}}\right) \\ &= \sum_{i=1}^{n} \left(h^1_{t_i}(D_{t_i} - D_{t_{i-1}}) + h^2_{t_i}(S_{t_i} - S_{t_{i-1}})\right) \\ &\quad [\text{use } V_{t_{i-1}} = h^1_{t_i} D_{t_{i-1}} + h^2_{t_i} S_{t_{i-1}}] \\ &= \sum_{i=1}^{n} h^1_{t_i} \Delta D_{t_i} + \sum_{i=1}^{n} h^2_{t_i} \Delta S_{t_i}, \end{aligned}$$

where $\Delta D_{t_i} = D_{t_i} - D_{t_{i-1}}$ for $i = 1, \ldots, n$, and similarly for S. We are interested in the behavior of $\Delta V_{t_i} = V_{t_i} - V_{t_{i-1}}$ for finer partitions $0 \leqslant t_0 < t_1 < \cdots < t_n \leqslant T$ of the horizon $[0, T]$, in the limit. What we expect is to get something like $V_t = V_0 + \int_0^t h_u^1 \mathrm{d}D_u + \int_0^t h_u^2 \mathrm{d}S_u$. While the integral $\int_0^t h_u^1 \mathrm{d}D_u$ poses no problem, due to the assumption of the deterministic dynamics of the risk-free rate, what is the meaning of $\int_0^t h_u^2 \mathrm{d}S_u$? We try to give an answer firstly by recalling that a reasonable model for the risky price process could be that introduced in Sect. 8.5, the log-normal stock price model and a first attempt should be letting $S_t = S_0 \mathrm{e}^{\mu t + \sigma B_t}$ with B_t being a standard Brownian motion. In order to develop a unifying approach, we propose to specify $S = (S_t)_{t\geqslant 0}$ as a process based on B via the theory of *stochastic integration*. We limit ourselves to a very short excursion to this theory tailored to our current setting. The dynamics of the risky security is specified by the **stochastic differential equation** (SDE) equation

$$\mathrm{d}S_t = \alpha S_t \mathrm{d}t + \sigma S_t \mathrm{d}B_t, \tag{10.17}$$

whose solution[10] is $S_t = S_0 \mathrm{e}^{(\alpha-\sigma^2/2)t+\sigma B_t}$, corresponding to the log-normal model for the stock price, see Exercise 10.3. Hence, the portfolio value's dynamics can be specified by the diffusive equation

$$\mathrm{d}V_t = h_t^1 \mathrm{d}D_t + h_t^2 \mathrm{d}S_t, \tag{10.18}$$

due to the self-financing assumption. The above SDE is actually

$$V_t = V_0 + \int_0^t h_s^1 \mathrm{d}D_s + \int_0^t h_s^2 \mathrm{d}B_s, \tag{10.19}$$

where the first integral on the right-hand side is of the Riemann type, while the second is an Itô stochastic integral, see Appendix. In fact, $S_t = f(t, B_t)$ so the stock price is adapted to the natural filtration generated by the Brownian motion and in addition $\sigma(S_t) \subset \sigma(B_t)$, for every $t \in [0, T]$. Recall that the holdings processes $h^1 = (h_t^1)_{t\geqslant 0}$ and $h^2 = (h_t^2)_{t\geqslant 0}$ are also adapted to the above filtration. Thanks to the replicating hypotheses we have $V_T = (S_T - K)^+$, thus the pricing problem can be tackled by handling the hedge portfolio instead of the original European call option payoff C_T. This is guaranteed by the following:

Lemma 10.1 *Let $C = (C_t)_{t\in[0,T]}$ be the process for the contingent claim. Assume that the portfolio with price process $V = (V_t)_{t\in[0,T]}$ over the horizon replicates the claim and there are no arbitrage opportunity. Then, we have hedging, i.e., $C_t = V_t$ for $t \in [0, T]$.*

Proof Assume $C_T = V_T$ and $C_t < V_t$. We show that arbitrage exists, so we found a contradiction. Sell the portfolio $\mathbf{h}$ underlying V_t and buy the claim at price $-C_t$. Thus, invest the amount $V_t - C_t$ at the risk-free instantaneous rate of return $r > -1$ for $T - t$. At the horizon T we get the amount $(V_t - C_t)\mathrm{e}^{r(T-t)} > 0$; the delivery of V_T stems from the claim. This is actually an arbitrage. ■

Lemma 10.1 enable us to formally define an arbitrage portfolio.

Definition 10.4 A self-financing portfolio $\mathbf{h} = (h^1, h^2)$, where h_1, h^2 are adapted processes, is an arbitrage whenever:

- $V_t \leqslant 0$
- $\mathsf{P}(V_T \geqslant 0) = 1$ and $\mathsf{P}(V_T > 0) > 0$.

Observe that the last item above implies $\mathsf{E}(V_T) \geqslant 0$. Now we turn to discounted values:

- $\tilde{S}_t = f(t, S_t) = \mathrm{e}^{-rt} S_t$, for the stock price;
- $\tilde{V}_t = f(t, V_t) = \mathrm{e}^{-rt} V_t$, for the replicating portfolio.

It is worth noting that $\tilde{S}_0 = S_0$ and $\tilde{V}_0 = V_0$.

[10] A strong solution to this type of SDE is called *diffusive process*, see [9] and [12].

For the former we apply Itô's formula (see Appendix) and get:

$$\begin{aligned}
\mathrm{d}\tilde{S}_t = \mathrm{d}\left(\mathrm{e}^{-rt}S_t\right) &= \mathrm{e}^{-rt}\underbrace{(\alpha S_t\mathrm{d}t + \sigma S_t \mathrm{d}B_t)}_{=\mathrm{d}S_t} - r\mathrm{e}^{-rt}S_t\mathrm{d}t \\
&= \alpha\tilde{S}_t\mathrm{d}t + \sigma\tilde{S}_t\mathrm{d}B_t - r\tilde{S}_t\mathrm{d}t \\
&= \tilde{S}_t\mathrm{d}\left((\alpha - r)t + \sigma B_t\right).
\end{aligned}$$

Observe that we used $\frac{\partial f}{\partial S_t} = \mathrm{e}^{-rt}$; $\frac{\partial f}{\partial t} = -rS_t\mathrm{e}^{-rt}$; $\frac{\partial^2 f}{\partial S_t^2} = 0$. Now, let $\tilde{B}_t = B_t + \frac{\alpha - r}{\sigma}t$ and rewrite the last line above as $\mathrm{d}\tilde{S}_t = \sigma\tilde{S}_t\mathrm{d}\tilde{B}_t$. The following result is crucial.

Theorem 10.3 (Girsanov) *Let a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathsf{P})$ where a Brownian motion is defined with $\mathcal{F}_t = \mathcal{F}_t = \sigma(B_s, s \leqslant t)$ for every $t \in [0, T]$. Then:*

- *$M_t = \mathrm{e}^{-qB_t - \frac{1}{2}q^2 t}$ is an $\mathcal{F}_t$-martingale, for $q \in \mathbb{R}$;*
- *$\tilde{\mathsf{P}}(A) = \int_A M_T\mathrm{d}\mathsf{P}$ is a probability measure equivalent to the original one, i.e., $\tilde{\mathsf{P}} \sim \mathsf{P}$, with $M_T \geqslant 0$ and $\mathsf{E}_{\mathsf{P}}(M_T) = 1$, for every event $A \in \mathcal{F} = \mathcal{F}_T$;*
- *$\tilde{B}_t = B_t + q\,t$ is again a Brownian motion w.r.t. the filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \tilde{\mathsf{P}})$.*

Therefore, under the new measure $\tilde{\mathsf{P}}$, for $q = \frac{\alpha - r}{\sigma}$ we have that the discounted price process $\tilde{S} = (\tilde{S}_t)_{t\in[0,T]}$ is a martingale (its diffusive equation has no more drift). We can conclude that the SDE $\mathrm{d}\tilde{S}_t = \sigma\tilde{S}_t\mathrm{d}\tilde{B}_t$ has solution

$$\tilde{S}_t = \tilde{S}_0\mathrm{e}^{-\frac{\sigma^2}{2}t + \sigma\tilde{B}_t} \iff \boxed{S_t = S_0\mathrm{e}^{(r - \sigma^2/2)t + \sigma\tilde{B}_t}},$$

and the stock price process in the new filtered probability space has a *risk-neutral* drift $r > -1$. For the discounted portfolio value process we have a similar caveat:

$$\begin{aligned}
\mathrm{d}\tilde{V}_t &= -r\mathrm{e}^{-rt}V_t\mathrm{d}t + \mathrm{e}^{-rt}\mathrm{d}V_t \\
&= -r\mathrm{e}^{-rt}(h_t^1 D_t + h_t^2 S_t)\mathrm{d}t + \mathrm{e}^{-rt}\left(h_t^1\mathrm{d}D_t + h_t^2\mathrm{d}S_t\right) \\
&= h_t^2\left(-r\mathrm{e}^{-rt}S_t\mathrm{d}t + \mathrm{e}^{-rt}\mathrm{d}S_t\right) \\
&= h_t^2\mathrm{d}\tilde{S}_t, \quad \left[\text{see first line of } \mathrm{d}\tilde{S}_t\text{'s equation}\right]
\end{aligned}$$

where we have used the self-financing condition together with the definition of the portfolio value process V_t for $t \geqslant 0$ and $D_t = D_0\mathrm{e}^{rt}$. This SDE in integral form is

$$\tilde{V}_t = \tilde{V}_0 + \int_0^t h_u^2\tilde{S}_u\mathrm{d}\tilde{B}_u,$$

where we known that $\tilde{S}$ is a martingale and now also $\int_0^t h_u^2 \tilde{S}_u \mathrm{d}\tilde{B}_u$ is, by the martingale representation theorem, see [9] and [12] where it can be found a more general version of the Girsanov theorem.

Remark 10.7 We applied Itô's Formula to $\tilde{V}_t = f(t, V_t) = \mathrm{e}^{-rt} V_t$ in the form

$$\mathrm{d}f(t, V_t) = \frac{\partial f}{\partial t}\mathrm{d}t + \frac{\partial f}{\partial V_t}\mathrm{d}V_t + \frac{1}{2}\frac{\partial^2 f}{\partial V_t^2}(\mathrm{d}V_t)^2$$

given by equation (10.23), with partials

- $\frac{\partial f}{\partial V_t} = \mathrm{e}^{-rt}$
- $\frac{\partial f}{\partial t} = -r\mathrm{e}^{-rt} V_t$
- $\frac{\partial^2 f}{\partial V_t^2} = 0.$

NB. The term $h_t^1 \left(\mathrm{d}\mathrm{e}^{-rt} D_t + \mathrm{e}^{-rt}\mathrm{d}D_t\right)$ vanished because (recall that D_t is not stochastic):

$$\begin{aligned}\mathrm{d}\left(\mathrm{e}^{-rt} D_t + \mathrm{e}^{-rt}\mathrm{d}D_t\right) &= \mathrm{d}\left(\mathrm{e}^{-rt} D_t\right) \\ &= \mathrm{d}\left(\mathrm{e}^{-rt} D_0 \mathrm{e}^{rt}\right) = \mathrm{d}D_0 = 0.\end{aligned}$$

We arrive to the following pricing formula:

$$\begin{aligned}\tilde{V}_t &= \mathsf{E}_{\tilde{\mathsf{P}}}\left(\tilde{V}_T \,|\, \mathcal{F}_t\right) \\ &\Longleftrightarrow \\ \mathrm{e}^{r(T-t)} V_t &= \mathsf{E}_{\tilde{\mathsf{P}}}\left(V_T \,|\, \mathcal{F}_t\right).\end{aligned}$$

To get a closed form formula for the European call option, we recall that by the replicating hypotheses and no arbitrage we have $V_t = C_t$ and $V_T = (S_T - K)^+$. Plugging in the expression for the log-normal stock price process with risk-neutral drift we have:

$$\begin{aligned}C_t &= \mathsf{E}_{\tilde{\mathsf{P}}}\left(\mathrm{e}^{-r(T-t)}\left(S_t \mathrm{e}^{(r-\sigma^2/2)(T-t)+\sigma(\tilde{B}_T - \tilde{B}_t)} - K\right)^+ \,|\, \mathcal{F}_t\right) \\ &= \mathrm{e}^{-r(T-t)} \int_{-\infty}^{\infty} \left(y\mathrm{e}^{(r-\sigma^2/2)(T-t)+\sigma(z\sqrt{T-t})} - K\right)^+ f_Z(z)\mathrm{d}z.\end{aligned}$$

The second equality, with $S_t \overset{\text{a.s.}}{=} y$, follows[11] since S_t is a measurable function of B_t and then it is $\mathcal{F}_t$-measurable, the increment $\tilde{B}_T - \tilde{B}_t$ is independent from $\mathcal{F}_t$, then all these imply that the above expectation is indeed unconditional and computed

[11] Actually, we have also used a property of the Itô stochastic integral, see Exercise 10.4

(using the change of variable formula) via the standard normal density $f_Z(z)$ for $Z \sim \mathrm{N}(0,1)$, which is the formula in Example 6.8, Chap. 6. The fact that here we have an unconditional expectation is due to formula (6.24) which we report here for the reader convenience:

$$\mathsf{E}\,(h(X,Y)\,|\,\mathcal{A}) = \psi(X),$$

where for a rv X independent from the sub-σ-algebra $\mathcal{A} \subset \mathcal{F}$ and Y being $\mathcal{A}$-measurable we have a function $h : \mathbb{R}^2 \to \mathbb{R}$ s.t. $h(X,Y) \in L^1$, and then $\psi(x) = \mathsf{E}(h(x,Y))$ given by the expectation of the rv $h(x,Y)$, which is obtained by keeping X as a constant becomes unconditional thus we get we get $\psi(X) = \mathsf{E}(h(X,Y))$. In the current setting, we have $\mathcal{A} = \mathcal{F}_t$, $X = \tilde{B}_T - \tilde{B}_t$ and $Y = S_T$. With a further little effort we eventually arrive to the *Black-Scholes formula*

$$C_t = y F_Z(\mathrm{d}_1) - \mathrm{e}^{-r(T-t)} K F_Z(\mathrm{d}_2), \tag{10.20}$$

where

$$\mathrm{d}_1 = \frac{\ln\left(\frac{y}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$
$$\mathrm{d}_2 = \mathrm{d}_1 - \sigma\sqrt{T-t},$$

see Exercise . It is apparent that evaluating this formula at $t = 0$, for the known[12] $S_0 = y$, the known r and the (estimated) value of volatility σ we get a numerical result without the need of Monte Carlo simulation, see Sect. 8.4.

10.5 Exercises

10.1 Suppose the weekly random loss X € is discrete with masses

$$\mathsf{P}(X = -105) = 0.03, \quad \mathsf{P}(X = -57) = 0.015, \quad \mathsf{P}(X = -10) = 0.12,$$
$$\mathsf{P}(X = 39) = 0.3, \quad \mathsf{P}(X = 98) = 0.1, \quad \mathsf{P}(X = 226) = 0.435.$$

Compute the 5% VaR.

10.2 Prove that $\mathrm{VaR}_{c,T}(X) = -Q_X^+(c) = Q_{-X}^-(1-c)$, for any $c \in (0,1)$.

10.3 Show that the diffusive equation (10.17) has solution $S_t = S_0 \mathrm{e}^{(\alpha - \sigma^2/2)t + \sigma B_t}$.

10.4 Prove the Black-Scholes formula for $t = 0$.

[12] Obviously, the strike price K and the maturity are also known.

Appendix

Details on VaR and Quantiles

Given a set $\{x \in \mathbb{R} \mid P(x)\}$, where $P(x)$ is a property depending on x, we have that

$$\inf\{x \in \mathbb{R} \mid P(-x)\} = -\sup\{y \in \mathbb{R} \mid P(y)\}, \quad \text{where} \quad y = -x. \tag{10.21}$$

As an example $P(-x)$ given by the relation $3 < -x < 5$. Thus, rewrite the set as $\{x \in \mathbb{R} \mid -5 < x < -3\}$ and take the infimum to get -5. On the other hand, applying Eq. (10.21) yields $-\sup\{y \in \mathbb{R} \mid 3 < y < 5\} = -5$, with $y = -5$. Equation (10.21) holds true by interchanging the infimum with the supremum. More generally we have for $y = a + b\,x$:

- $\inf\{x \in \mathbb{R} \mid P(a + bx)\} = a + b \inf\{y \in \mathbb{R} \mid P(y)\}$, with $a \in \mathbb{R}$ and $b > 0$;
- $\inf\{x \in \mathbb{R} \mid P(a + bx)\} = a + b \sup\{y \in \mathbb{R} \mid P(y)\}$, with $a \in \mathbb{R}$ and $b < 0$.

The first property implies that for quantiles $Q_{a+bX}(c) = a + b\,Q_X(c)$, for $b > 0$.

Itô Stochastic Integral

It essentially in the treatment of stochastic integrals depending on Brownian motion that the filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathsf{P})$ refers to $\{\mathcal{F}_t\} = \sigma(B_s, s \leqslant t)$. Here we assume that if $A \in \mathcal{F}$ and $\mathsf{P}(A) = 0$ then $A \in \mathcal{F}_t$ for all $t \geqslant 0$. This technical condition ensures that each $\mathcal{F}_t$ contains all the P-null events of $\mathcal{F}$, which implies that for rvs $X \overset{\text{a.s.}}{=} Y$, with Y being $\mathcal{F}_t$-measurable we also have X as $\mathcal{F}_t$-measurable. We start from $\int_0^t Z_u \mathrm{d}B_u$, where $Z = (Z_t)_{t \geqslant 0}$ is a continuous-time process driven by the Brownian motion. The problem is that this stochastic integral $\int_0^t Z_u(\omega)\mathrm{d}B_u(\omega)$ cannot be defined as a Riemann-Stieltjes integral for each scenario (i.e. each path), because $\omega \mapsto B_t(\omega)$ is nowhere differentiable. This is linked to B having paths with unbounded variation on every finite interval $[0, T]$. Heuristically, taking

$$\sup_{\mathcal{P}_n} \sum_{i^1}^{n} |B_{t_i} - B_{t_{i-1}}|$$

for each $\omega \in \Omega$ and each partition

$$\mathcal{P}_n = \{0 = t_0, t_1, \ldots, t_n\} \subset [0, T],$$

it can be showed that it goes to ∞ as $n \to \infty$ and $\sup |t_i - t_{i-1}| \to 0$, see [9] or [12]. Instead, the *quadratic variation*

$$\sum_{i=1}^{n} (B_{t_i} - B_{t_{i-1}})^2$$

is finite on $[0, T]$, and to see this intuitively we let $t_{i-1} = \frac{(i-1)t}{2^n}$ and $t_i = \frac{it}{2^n}$, for a dyadic partition of any $[0, t] \subset [0, T]$, then rewrite the quadratic variation rv as

$$H_n := \sum_{i=1}^{2^n-1} (B_{t_i} - B_{t_{i-1}})^2$$

for which[13] $H_n \xrightarrow{\text{a.s.}} t$. Observe that

$$\mathsf{E}(H_n) = \mathsf{E}\left(\sum_{i=1}^{2^n-1} (B_{t_i} - B_{t_{i-1}})^2 \right) = \sum_{i=1}^{2^n-1} \left(\frac{it}{2^n} - \frac{(i-1)t}{2^n} \right) = t.$$

This can be written $(\mathrm{d}B_t)^2 = \mathrm{d}t$ or $\langle B \rangle_t$.

Remark 10.8 Let $\Delta B_t = B_{t+\Delta t} - B_t = Z\sqrt{\Delta t}$, where $Z \sim \mathrm{N}(0, 1)$, apply Chebyshev's inequality:

$$\begin{aligned} \mathsf{P}\left(\left| (\Delta B_t)^2 - \Delta t \right| > \epsilon \right) &= \mathsf{P}\left(\Delta t \left| Z^2 - t \right| > \epsilon \right) \\ &\leqslant \frac{(\Delta t)^2 \mathsf{E}(Z^2 - 1)}{\epsilon^2}. \end{aligned}$$

Then, the right-hand side of the inequality tends to zero as $\Delta t \to 0$.

Knowing the behavior of B for $t \geqslant 0$, we define the *Itô stochastic integral* of a *simple process*:

- partition $[0, t]$ into subintervals $[t_{i-1}, t_i]$, through

$$\mathcal{P}_n = \{0 = t_0, t_1, \ldots, t_n = t\};$$

- define $C_s^{(n)} = Z_{t_n}$, if $s = t_n$, and $C_s^{(n)} = Z_{t_{i-1}}$ if $t_{i-1} \leqslant s < t_i$, for $i = 1, \ldots, n$, where $Z = (Z_t)_{t \geqslant 0}$ is a continuous time process;

[13] Actually, the convergence is $H_n \xrightarrow{\text{ms}} t$ and then $H_n \xrightarrow{\mathsf{P}} t$, from which we can extract an a.s. convergent subsequence.

- define Z adapted to the Brownian natural filtration, s.t. $Z_{t_{i-1}}$ is $\mathcal{F}_{t_{i-1}}$ measurable, and interpret this as a gamble where at each t_{i-1} we stake the amount $Z_{t_{i-1}}$ to be repaid by $Z_{t_{i-1}}(B_{t_i} - B_{t_{i-1}})$ at t_i in a non anticipative way;
- assume $\mathsf{E}(Z_{t_i}^2) < \infty$, each t_i.

The total 'amount stacked' is the desired stochastic integral

$$I_t(C^{(n)}) = \sum_{i=1}^{n} C_{t_{i-1}}^{(n)}(B_{t_i} - B_{t_{i-1}}),$$

where obviously $C_{t_{i-1}}^{(n)} = Z_{t_{i-1}}$, of the simple process $C^{(n)} = (C_t^{(n)})_{t \geqslant 0}$. It can be shown that:

- $\mathsf{E}(I_t(C^{(n)})) = 0$;
- $\mathsf{E}\left((I_t(C^{(n)}))^2\right)$ is equal to $\mathsf{E}\left(\sum_{i=1}^{n} Z_{t_{i-1}}^2(t_i - t_{i-1})\right)$, by the independence of increments $B_{t_i} - B_{t_{i-1}}$ from $\mathcal{F}_{t_{i-1}}$ and the fact that $\mathsf{V}(B_{t_i} - B_{t_{i-1}}) = t_i - t_{i-1}$.

If

$$\int_0^t \mathsf{E}\left(Z_s - C_s^{(n)}\mathrm{d}s\right) = \mathsf{E}\left(\int_0^t Z_s - C_s^{(n)}\mathrm{d}s\right) \to 0$$

as $n \to \infty$ and $\mathcal{P}_n$ becomes finer and finer, then $I_t(C^{(n)}) \xrightarrow{\text{ms}} I_t(Z)$.

Remark 10.9 Actually, the stronger condition

$$\mathsf{E}\left(\sup_{0 \leqslant s \leqslant t} \left|I_s(Z) - I_s(C^{(n)})\right|^2\right) \to 0,$$

as $n \to \infty$ implies mean square convergence.

Thus, $I = (I_t(Z))_{t \geqslant 0}$ is $\mathcal{F}_t$-adapted with $\mathsf{E}(I_t(Z)) = 0$ and variance:

$$\begin{aligned}\mathsf{V}(I_t(Z)) &= \mathsf{E}\left((I_t(Z))^2\right) = \mathsf{E}\left(\left(\int_0^t Z_s \mathrm{d}B_s\right)^2\right) \\ &= \mathsf{E}\left(\int_0^t Z_s^2 \mathrm{d}B_s\right) = \int_0^t \mathsf{E}(Z_s^2)\mathrm{d}s.\end{aligned}$$

The latter property is called *Itô isometry*. As an alternative notation,[14] we can write

[14] It is only s typographical convention, since what has been defined is the stochastic integral.

$$\mathrm{d}I_t(Z) = Z_t \mathrm{d}B_t \iff I_t(Z) = I_0(Z) + \int_0^t Z_s \mathrm{d}B_s.$$

For example, we can write $\int_0^t \mathrm{d}B_s = B_t - B_0 = B_t$.[15] A further property of the Itô integral is linearity:

$$\int_0^t \left\{ a\, Z_s^1 + b\, Z_s^2 \right\} \mathrm{d}B_s = a \int_0^t Z_s^1 \mathrm{d}B_s + b \int_0^t Z_s^2 \mathrm{d}B_s$$

for all $a, b \in \mathbb{R}$ and a pair of Itô-integrable processes Z^1, Z^2. The Itô integral is also linear with respect to intervals where it is computed:

$$\int_0^t Z_s \mathrm{d}B_s = \int_0^u Zs \mathrm{d}B_s + \int_u^t Z_s \mathrm{d}B_s,$$

for $0 \leqslant u < t$. Moreover, we can define $I_{s,t}(Z) = \int_s^t Z_u \mathrm{d}B_u$ and it can showed that (see [14]) that

$$\mathsf{E}\left(I_{s,t}(Z) \,|\, \mathcal{F}_s\right) = 0, \quad \text{for } 0 \leqslant s < t. \tag{10.22}$$

This property enable us to show that an Itô integral is a martingale: write $I_t(Z) = I_u(Z) + I_{u,t}(CZ)$ and apply the general conditional expectation

$$\mathsf{E}(I_t(Z) \,|\, \mathcal{F}_u) = \mathsf{E}(I_u(Z) \,|\, \mathcal{F}_u) + \mathsf{E}(I_{u,t}(Z) \,|\, \mathcal{F}_u),$$

and since $I_u(C)$ is $\mathcal{F}_u$-measurable we have $\mathsf{E}(I_u(Z) \,|\, \mathcal{F}_u) = I_u(Z)$ then by (10.22) it follows $\mathsf{E}(I_{u,t}(Z) \,|\, \mathcal{F}_u) = 0$, which entails $\mathsf{E}(I_t(C) \,|\, \mathcal{F}_u) = I_u(C)$ Formally, we can mix and write the SDE

$$\mathrm{d}Y_t = X_t \mathrm{d}t + Z_t \mathrm{d}B_t$$

to be meant

$$Y_t = Y_0 + \int_0^t X_s \mathrm{d}s + \int_0^t Z_s \mathrm{d}B_s,$$

for an additional $\mathcal{F}_t$-adapted process X_t. Such a SDE is called a *diffusion equation*. For $x \in \mathbb{R}$, it can be shown that:

$$X_t = \lim_{h \downarrow 0} \frac{1}{h} \mathsf{E}\left(Y_{t+h} - Y_t \,|\, Y_t = x\right),$$

[15] Under suitable conditions, $\int_0^t Z_s \mathrm{d}B_s$ can be understood in the Riemann sense, for each path.

and

$$Z_t^2 = \lim_{h \downarrow 0} \frac{1}{h} \mathsf{E}\left(|Y_{t+h} - Y_t|^2 \,|\, Y_t = x \right).$$

Moreover, $X_t = \alpha(t, Y_t)$, called the drift, and $Z_t = \sigma(y, Y_t)$, called the diffusive term, are suitable functions α, σ. For example, the Brownian motion is a diffusive process since $Y_t = B_t$ with $\alpha(t, Y_t) = 0$ and $\sigma(y, Y_t) = 1$. An Itô stochastic integral, as a process itself, is a continuous times martingale:

$$\mathsf{E}\left(I_t(Z) \,|\, \mathcal{F}_s\right) = I_s(Z), \quad \text{for } 0 \leqslant s < t.$$

In general, a diffusive SDE does not provides a martingale, unless the drift is zero. The following is crucial in our contingent claim pricing problem.

Theorem 10.4 (Itô's Formula) *Suppose $(Y_t)_{t \geqslant 0}$ is a diffusive process and that $f(t, y)$ is a function with continuous second partial derivatives. Then, the stochastic process $f(t, Y_t)$ satisfies the diffusion equation*

$$\mathrm{d}f(t, Y_t) = \left(X_t \frac{\partial f}{\partial y} + \frac{\partial f}{\partial t} + \frac{1}{2} Z_t^2 \frac{\partial^2 f}{\partial y^2} \right) \mathrm{d}t + Z_t \frac{\partial f}{\partial y} \mathrm{d}B_t,$$

where the partial derivatives are evaluated at $(t, Y_t) = (t, y)$.

The above theorem make implicitly use of the following multiplication rules: $\mathrm{d}t\mathrm{d}t = 0$, $(\mathrm{d}B_t)^2 = \mathrm{d}B_t \mathrm{d}B_t = \mathrm{d}t$ and $\mathrm{d}t\mathrm{d}B_t = 0$. In fact, to prove Itô's Formula heuristically we can proceed as follows. Write the Taylor expansion of $f(t, Y_t)$ up to second order:

$$\begin{aligned} \mathrm{d}f(t, Y_t) = & \frac{\partial f}{\partial t} \mathrm{d}t + \frac{\partial f}{\partial Y_t} \mathrm{d}Y_t + \frac{1}{2} \frac{\partial^2 f}{\partial t^2} (\mathrm{d}t)^2 \\ & + \frac{1}{2} \frac{\partial^2 f}{\partial Y_t^2} (\mathrm{d}Y_t)^2 + \frac{\partial^2 f}{\partial t \partial Y_t} \mathrm{d}t \mathrm{d}Y_t. \end{aligned}$$

Now, from SDE $\mathrm{d}Y_t = X_t \mathrm{d}t + Z_t \mathrm{d}B_t$ and using the above multiplication rules we compute:

$$\begin{aligned} (\mathrm{d}Y_t)^2 &= X_t^2 (\mathrm{d}t)^2 + Z_t^2 (\mathrm{d}B_t)^2 + 2 X_t Z_t \mathrm{d}B_t \\ &= Z_t^2 \mathrm{d}B_t \mathrm{d}B_t = Z_t^2 \mathrm{d}t. \end{aligned}$$

Plugging the above into the Taylor formula we get Itô's Formula. Indeed, the latter can be written

$$\mathrm{d}f(t, Y_t) = \frac{\partial f}{\partial t} \mathrm{d}t + \frac{\partial f}{\partial Y_t} \mathrm{d}Y_t + \frac{1}{2} \frac{\partial^2 f}{\partial Y_t^2} (\mathrm{d}Y_t)^2 \tag{10.23}$$

which we used in the expression for $\tilde{V}_t$.

Solutions to Exercises

Exercises of Chap. 1

1.1 We have to collect those $\omega \in \Omega$ s.t. $S_1(\omega)$ has value less than or equal to 3.5:

$$\{\omega \mid S_1(\omega) \leqslant 3.5\} = \{\text{uud, udu, duu, udd, dud, ddu, ddd}\}.$$

1.2 First we note that $\{S_1 > 3.5\} = \{S_1 \leqslant 3.5\}^c$, where for a subset A we write A^c the collection containing those outcomes $\omega \in \Omega$ s.t. $\omega \notin A$, given Ω as the universal set. Therefore, we have the singleton:

$$\{\omega \mid S_1(\omega) > 3.5\} = \{\text{uuu}\}.$$

Moreover $\{S_1 \geqslant 3.5\} = \{S_1 < 3.5\}^c = \{\text{uuu, uud, udu, duu}\}$ while $\{S_1 < 3.5\} = \{\text{udd, dud, ddu, ddd}\}$ and $\{S_1 = 3.5\} = \{\text{uud, udu, duu}\}$.

1.3 It is easily seen that

$$\{\omega \mid X(\omega) \leqslant a\} \cup \{\omega \mid a < X(\omega) \leqslant b\}$$

is the event containing those outcomes ω for which at the same time $X(\omega) \leqslant b$, $X(\omega) > a$ and $X(\omega) \leqslant a$, thus we have the desired event $\{X \leqslant b\}$. Observe that $\{X \leqslant a\} \cap \{a < X \leqslant b\} = \varnothing$, i.e. they are incompatible events: the rv cannot maps the same outcomes ω for which $X(\omega) > a$ and $X(\omega) \leqslant a$ hold. Since X is a function, it maps each $\omega \in \Omega$ into one real number; it is ruled out that the same ω is mapped into the number $X(\omega) = a$ and into a different number $X(\omega) = c > a$.

1.4 We have

$$\{\omega \mid X(\omega) > a\} \cup \{\omega \mid X(\omega) = a\}$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

is the event containing those ω for which either $X(\omega) > a$, or $X(\omega) = a$ (but not both), thus we have the desired event $\{X \geqslant a\}$.

1.5 The assumption prevent us from the need to encode outcomes ω as we did in previous examples (using u,d). Now, the event A contains those ω s.t. $1 < S_1(\omega) < 3$, or $X(\omega) \in B = (1, 3)$. Then, $S_1(\omega) \leqslant 3 - \frac{1}{i}$ for some $i = 1, 2, \ldots$. As byproduct, $S_1(\omega) \in \cup_{i=1}^{\infty}(1, 3 - \frac{1}{i}]$, i.e. at least one of the events $A_i = \{1 < S_1 \leqslant 3 - \frac{1}{i}\}$ occur and we conclude $A = \cup_{i=1}^{\infty} A_i$. Conversely, if $S_1(\omega) \in \cup_{i=1}^{\infty}(1, 3 - \frac{1}{i}]$ then $1 < S_1(\omega) \leqslant 3 - \frac{1}{i}$, for some $i = 1, 2, \ldots$. Thus $S_1(\omega) \in (1, 3)$. If $A_i \in \mathcal{F}$ then their union A is also a member of the σ-algebra $\mathcal{F}$: we can assign probabilities to each A_i and to A.

1.6 The event A contains those ω s.t. $1 \leqslant S_1(\omega) \leqslant 3$, or $S_1(\omega) \in B = [1, 3]$. Then, $1 - \frac{1}{i} < S_1(\omega)$ for all $i = 1, 2, \ldots$. As byproduct, $S_1(\omega) \in \cap_{i=1}^{\infty}(1 - \frac{1}{i}, 3]$, i.e. all events $A_i = \{1 - \frac{1}{i} < S_1 \leqslant 3\}$ occur and we conclude $A = \cap_{i=1}^{\infty} A_i$. Conversely, if $S_1(\omega) \in \cap_{i=1}^{\infty}(1 - \frac{1}{i}, 3]$ then $1 - \frac{1}{i} < S_1(\omega) \leqslant 3$, for all $i = 1, 2, \ldots$. Thus $S_1(\omega) \in [1, 3]$.

1.7 Since every event comes from $\mathcal{F}$, by property S2 we have $A_i^c \in \mathcal{F}$ for all $i = 1, 2, \ldots$. Thus, by property S3 $\cup_{i=1}^{\infty} A_i^c \in \mathcal{F}$ and by DeMorgan's Laws $\left(\cup_{i=1}^{\infty} A_i^c\right)^c = \cap_{i=1}^{\infty} A_i \in \mathcal{F}$. Moreover, recall that $A \setminus B = A \cap B^c$ and observe that since $B^c \in \mathcal{F}$ by property S2, then $A \cap B^c \in \mathcal{F}$ by the above reasoning.

1.8 We establish both $A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$, and $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$. First, $\omega \in A \cap (B \cup C) \Rightarrow \omega \in A$ and $\omega \in B \cup C$, thus we have $\omega \in A$ and $\omega \in B$ or $\omega \in C$. Now we distinguish 2 cases.

$\underline{\omega \in B}$. Thus $\omega \in A$ and $\omega \in B$, i.e. $\omega \in A \cap B$
$\underline{\omega \in C}$. Thus $\omega \in A$ and $\omega \in C$, i.e. $\omega \in A \cap C$.

Both yield $\omega \in (A \cap B) \cup (A \cap C)$. Summing up: $\omega \in A \cap (B \cup C)$ does imply $\omega \in (A \cap B) \cup (A \cap C)$, which means that $A \cap (B \cup C)$ is a subset of $(A \cap B) \cup (A \cap C)$. Second, we prove the converse $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$. Let $\omega \in (A \cap B) \cup (A \cap C)$. As a consequence, $\omega \in A \cap B$ or $\omega \in A \cap C$ and we again distinguish two cases.

$\underline{\omega \in A \cap B}$. We have $\omega \in B$ and so $\omega \in B \cup C$, i.e. $\omega \in A \cap (B \cup C)$.
$\underline{\omega \in A \cap C}$. We have $\omega \in C$ and so $\omega \in B \cup C$, i.e. $\omega \in A \cap (B \cup C)$.

Summing up: $\omega \in (A \cap B) \cup (A \cap C)$ does imply $\omega \in A \cap (B \cup C)$, i.e. $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$. This together with the reverse inclusion (we proved previously) entail the desired equality $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

1.9 We have:

$$\begin{aligned} \omega \in \left(\cup_{i=1}^{\infty} A_i\right)^c &\iff \omega \notin \cup_{i=1}^{\infty} A_i \\ &\iff \omega \notin A_i \quad \text{for all } i = 1, 2, \ldots \end{aligned}$$

$$\iff \omega \in A_i^c \quad \text{for all } i = 1, 2, \ldots$$
$$\iff \omega \in \cap_{i=1}^{\infty} A_i^c.$$

In a similar way, we can prove DeMorgan's Law $\left(\cap_{i=1}^{\infty} A_i\right)^c = \cup_{i=1}^{\infty} A_i^c$.

1.10

- First, we rewrite $A \cup (A \cap B) = (A \cap \Omega) \cup (A \cap B)$. Then we use distributivity and compute $A \cap (\Omega \cup B) = A \cap \Omega = A$.
- First, we use distributivity when intersecting $(A \cup B)$ with the union $A \cup C$:

$$\begin{aligned}(A \cup B) \cap (A \cup C) &= [(A \cup B) \cap A] \cup [(A \cup B) \cap C] \\ &\overset{\text{distrib. again}}{=} (A \cap A) \cup (A \cap B) \cup (A \cap C) \cup (B \cap C) \\ &\overset{\text{distrib. again}}{=} A \cap (\Omega \cup B \cup C) \cup (B \cap C) \\ &= (A \cap \Omega) \cup (B \cap C) = A \cup (B \cap C).\end{aligned}$$

1.11 Assume $\mathsf{P}(A \cap B) = \mathsf{P}(B)$ and $\mathsf{P}(A^c) = 1$ to find a contradiction. We have

$$\mathsf{P}(B) = \mathsf{P}(A \cap B) + \mathsf{P}(A \cap B^c),$$

thus by the first hypotheses $\mathsf{P}(A \cap B^c) = 0$. This implies $\mathsf{P}((A \cap B^c)^c)$ is 1 and we can write:

$$\begin{aligned}1 &= \mathsf{P}((A \cap B^c)^c) = \mathsf{P}(A^c \cup B) \\ &\overset{\text{subadd.}}{\leqslant} \mathsf{P}(A^c) + \mathsf{P}(B).\end{aligned}$$

From $\mathsf{P}(A^c) = 1$ we have $\mathsf{P}(A) = 0$ and then by the above it holds $\mathsf{P}(B^c) \geqslant 1$ which is false. As byproduct, it must be true $\mathsf{P}(A \cap B) = \mathsf{P}(B)$ implies $\mathsf{P}(A) = 0$.

1.12

- Checking Pr1. Decompose Ω as the disjoint union $A \cup A^c$. Then write $\mathsf{P}(\Omega) = \frac{|\Omega|}{n} = 1 = \frac{|A \cup A^c|}{n}$, and recall that $|A \cup A^c| = |A| + |A^c|$. Thus

$$1 = \frac{|A|}{n} + \frac{|A^c|}{n} \iff \frac{|A^c|}{n} = 1 - \frac{|A|}{n} \iff \mathsf{P}(A^c) = 1 - \mathsf{P}(A).$$

- Checking Pr2. For $A, B \in \mathcal{F}$ s.t. $A \subset B$ we have $|A| \leqslant |B|$, and $\mathsf{P}(A) = \frac{|A|}{n} \leqslant \frac{|B|}{n} = \mathsf{P}(B)$.
- Checking Pr3. For A, B as above we have $|B \setminus A| = |B| - |A|$ and the result follows easily.
- Checking Pr4. For events $A, B \in \mathcal{F}$ we have $|A \cup B| = |A| + |B| - |A \cap B|$ and the result follows easily.

1.13 It suffices to show that $\mathcal{H}$ contains $\varnothing$ and it is closed under complements and countable unions. Indeed, $\varnothing$ belong to $\mathcal{H}$ as $\varnothing \in \mathcal{S}_i$ for all $i \in I$. Assume $A \in \mathcal{H}$, then every $\mathcal{S}_i \in \mathcal{G}$ contains A, then it contains its complement A^c too, which in turn belong to $\mathcal{H}$. Now suppose $(A_i)_{i\in\mathbb{N}}$ is a sequence of measurable sets in $\mathcal{H}$. This implies $A_i \in \mathcal{G}$ for every $i \in I$. Then $\cup_i A_i$ belongs to $\mathcal{S}_i \in \mathcal{G}$ and consequently to $\mathcal{H}$.

1.14 Let $\mathcal{G}$ be the collection of all σ-algebras containing $\mathcal{C}$. Then $\mathcal{G}$ is nonempty as it contains 2^Ω. Furthermore, by Exercise 1.13 and its solution the intersection of all σ-algebras in $\mathcal{G}$ is itself a σ-algebra which contains the class $\mathcal{C}$.

1.15 Given $A, B \in \mathcal{F}$ we have $A^c \in \mathcal{F}$ by (S2) and $B \setminus A = B \cap A^c \in \mathcal{F}$, thus we have proved the first claim regardless the additional hypotheses $A \subset B$. Furthermore, any sequence $(A_i)_{i\in\mathbb{N}} \subset \mathcal{F}$ of events in the σ-algebra satisfies (S3), $\cup_i A_i \in \mathcal{F}$, which is obviously holds when, in addition, the sequence is monotone increasing.

1.16 We need to check properties S1–S3 of Definition 1.2. S1 is verified, $\varnothing \in \mathcal{F}_E$, since $A = \varnothing$ gives $A \cap E = \varnothing$ or alternatively $E^c \cap E = \varnothing \in \mathcal{F}_E$, given that $E^c \in \mathcal{F}$. To show S2, for every $A \cap E \in \mathcal{F}_E$ we must have $(A \cap E)^c \in \mathcal{F}_E$, where E is the new reference space. Indeed, taking complements has to be redefined as $(A \cap B)^c := E \setminus (A \cap E)$. The term on the right-hand side reduces to $E \cap (A \cap E)^c = E \cap (A^c \cup E^c)$. But $A^c \cup E^c \in \mathcal{F}$, then the last term on the right-hand side belongs to $\mathcal{F}_E$ and we are done. Finally, for a sequence $(A_i)_{i\in\mathbb{N}} \subset \mathcal{F}$ we have $A_i \cap E \in \mathcal{F}_E$ for all i, and we have

$$(A_1 \cap E) \cup (A_2 \cap E) \cup \cdots = \cup_i (A_i \cap E) = (\cup_i A_i) \cap E,$$

since $\cup_i A_i \in \mathcal{F}$. Then, $\cup_i (A_i \cap E) \in \mathcal{F}_E$, and S3 is verified.

Exercises of Chap. 2

2.1 To show (e) we simply consider $\{X > a\} = \{X \leqslant a\}^c$ and by property Pr1 of $\mathsf{P}(\cdot)$ we have

$$\mathsf{P}(X > a) = 1 - \mathsf{P}(X \leqslant a) = 1 - F_X(a).$$

To show (f), first consider the decreasing sequence of events $\{a - \frac{1}{i} < X \leqslant a\}$ for $i \in \mathbb{N}$, with intersection $\cap_{i=1}^{\infty} \{a - \frac{1}{i} < X \leqslant a\} = \{X = a\}$. Thus, by continuity of $\mathsf{P}(\cdot)$ we have

$$\mathsf{P}(X = a) \quad = \quad \lim_{i\to\infty} \mathsf{P}(a - \tfrac{1}{i} < X \leqslant a)$$

$$\begin{aligned} &\overset{\text{Theorem 2.1(d)}}{=} \lim_{i\to\infty}\left[F_X(a) - F_X(a-\tfrac{1}{i})\right] \\ &= F_X(a) - F_X(a^-). \end{aligned}$$

2.2 The density function is continuous on the closed interval $[x-\frac{h}{2}, x+\frac{h}{2}]$, whose total length is h. By the mean value theorem we find $c \in (x-\frac{h}{2}, x+\frac{h}{2})$ s.t.

$$hf_X(c) = \int_{x-\frac{h}{2}}^{x+\frac{h}{2}} f_X(x)\mathrm{d}x,$$

which is the required probability $\mathsf{P}\big(x-\frac{h}{2} \leqslant X \leqslant x+\frac{h}{2}\big)$. Evaluating the density at a different point $x \neq c$ inside the above closed interval we have $hf_X(x) \approx \mathsf{P}(x-\frac{h}{2} \leqslant X \leqslant x+\frac{h}{2})$, i.e. geometrically the area of a rectangle. When the base becomes smaller and smaller, we end with

$$\mathsf{P}(X=x) = \lim_{h\to 0}\int_{x-\frac{h}{2}}^{x+\frac{h}{2}} f_X(x)\mathrm{d}x = \int_x^x f_X(u)\mathrm{d}u = 0,$$

confirming that the density alone is not a probability.

2.3 Take a suitable $B \subset \mathbb{R}$. Then, $\{\mathbf{I}_A \in B\}$ can be:

- the impossible event $\varnothing$, provided $0 \notin B$ and $1 \notin B$;
- the certain event Ω, whenever both 1 and 0 are in B;
- the event A, if $0 \notin B$ and $1 \in B$;
- the contrary event A^c, when $0 \in B$ and $1 \notin B$.

2.4 The diagram $\Omega \xrightarrow{X} \mathbb{R} \xrightarrow{\mathbf{I}_B} \mathbb{R}$ shows that $\mathbf{I}_B \circ X$ is itself an indicator function, since it assume value 1 whenever $X(\omega) \in B$ and 0 otherwise, but this means $\omega \in \{X \in B\}$ thus we have $\mathbf{I}_B(X) = \mathbf{I}_{\{X\in B\}}$.

Remark 1 From Appendix in Chap. 2, we can use the concept of Borel-measurable function and the fact that a composition of two measurable functions is again measurable. Thus, $\mathbf{I}_B$ is a Borel-measurable function and the composed map $\mathbf{I}_B \circ X$ is also measurable because X is a rv.

2.5 The union of left-open intervals $(a, b-\frac{1}{i}]$ contains those real numbers x such that $a < x \leqslant b-\frac{1}{i}$ for at least one $i \in \mathbb{N}$. On the other hand, the condition $x \leqslant b-\frac{1}{i}$ for at least one i is equivalent to $x < b$. Eventually, $(a, b-1] \cup (a, b-\frac{1}{2}] \cup (a, b-\frac{1}{3}] \cup \cdots$ will be (a, b). Since every left-open interval is a Borel set, then their countable union is Borel too. Now, the intersection of left-open intervals $(a-\frac{1}{i}, b]$ contains those real numbers x such that $a-\frac{1}{i} < x \leqslant b$ for all $i \in \mathbb{N}$. The condition $a-\frac{1}{i} < x$ for all i is equivalent to $a \leqslant x$. For i large enough

$(a-1,b]\cap(a-\frac{1}{2},b]\cap(a-\frac{1}{3},b]\cap\cdots$ will be $[a,b]$. The same reasoning as above provides this closed interval as a Borel set.

2.6

(1) By our convention, the class $\mathcal{G}$ contains finite disjoint unions of left-open intervals $(a,b]$ with $-\infty \leqslant a < b < \infty$ and such that $(a,\infty) = (a,\infty]$. Thus, it suffices to choose $a_1 = -\infty, b_1 = a_2, b_2 = a_3$ and then take the finite disjoint union

$$(a_1,b_1]\cup(a_2,b_2]\cup(a_3,\infty] = (-\infty,\infty)\in\mathcal{G}.$$

But clearly $(-\infty,\infty) = \mathbb{R}$. (2) By Formula $\mathsf{P}_0(A) := \sum_{i=1}^{n} F(b_i) - F(a_i)$ and the previous exercise, with $\mathbb{R} = (a_1,b_1]\cup(a_2,b_2]\cup(a_3,\infty]$ we have:

$$\begin{aligned}\mathsf{P}_0(\mathbb{R}) &= F(b_1) - F(-\infty) + F(b_2) - F(a_2) + F(\infty) - F(a_3)\\ &= F(b_1) - 0 + F(b_2) - F(b_1) + 1 - F(b_2) = 1,\end{aligned}$$

since F is a distribution function with the additional properties $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$, and moreover $b_1 = a_2, b_2 = a_3$. (3) Formula (2.26) is shown to be finite additive on finite disjoint unions of sets in $\mathcal{G}$. Since

$$\mathsf{P}_0(\mathbb{R}) = \mathsf{P}_0(\mathbb{R}\cup\varnothing) = \mathsf{P}_0(\mathbb{R}) + \mathsf{P}_0(\varnothing)$$

and $\mathsf{P}_0(\mathbb{R}) = 1$ we must have $\mathsf{P}_0(\varnothing) = 0$.

2.7 It suffices to check properties PA1,PA2 of Definition 1.3 in Sect. 1.3 hold.

- To verify PA1 we observe that:

$$\mathsf{P}_X(\varnothing) := \sum_{i=1}^{\infty} p_i\delta_{x_i}(\varnothing) = 0,$$

since each $\delta_{x_i}(\varnothing)$ is zero by definition; similarly

$$\mathsf{P}_X(\mathbb{R}) := \sum_{i=1}^{\infty} p_i\delta_{x_i}(\mathbb{R}) = \sum_{i=1}^{\infty} p_i = 1,$$

because each $\delta_{x_i}(\mathbb{R})$ is one by definition.

- To check countable additivity PA2 for pairwise disjoint Borel sets $(B_j)_{j\in\mathbb{N}}\subset\mathcal{B}$ we first note that

$$\delta_{x_i}(\cup_{j=1}^{\infty} B_j) = \sum_{j=1}^{\infty}\delta_{x_i}(B_j),$$

since by disjointedness either $x_i \in B_j$ for only one B_j (so $\delta_{x_i}(B_j) = 1$ and the Dirac deltas for all indexes different from j are zero) or x_i does not belong to any B_j (all $\delta_{x_i}(B_j)$ are zero). Then we compute:

$$\begin{aligned}\mathsf{P}_X(\cup_{j=1} B_j) &:= \sum_{i=1}^{\infty} p_i \delta_{x_i}(\cup_{j=1}^{\infty} B_j) \\ &= \sum_{i=1}^{\infty} p_i \sum_{j=1}^{\infty} \delta_{x_i}(B_j) \\ &= \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} p_i \delta_{x_i}(B_j) = \sum_{j=1}^{\infty} \mathsf{P}_X(B_j).\end{aligned}$$

2.8 We have to transform the given rv by taking its absolute value, $|X|$. We also have:

$$|X| = X^+ + X^-, \quad \text{and} \quad X = X^+ - X^-.$$

Eventually, X^+ and X^- are both $\leqslant |X|$.

2.9 It suffices to draw the graphs of the corresponding functions.

2.10 Recall that by definition of quantile function (see Sect. 2.8, Definition 2.8, and Proposition 2.2, items 1 and 2) it holds $Q_X(F_X(x)) \leqslant x$, for all $x \in \mathbb{R}$, and $F_X(Q_X(c)) \geqslant c$, for all $c \in (0, 1)$. Therefore, assuming $X \leqslant_{\text{st}} Y$ we have $F_Y(x) \leqslant F_X(x)$ for every x and then

$$Q_X(F_Y(x)) \leqslant Q_X(F_X(x)) \leqslant x, \quad \text{for all } x \in \mathbb{R}.$$

Conversely, assuming $x \geqslant Q_X(F_Y(x))$ for every x we have

$$F_X(x) \geqslant F_X(Q_X(F_Y(c))) \geqslant F_Y(x), \quad \text{for all } x \in \mathbb{R},$$

and the equivalence is proved.

2.11 For a discrete X, we first evaluate formula (2.14) at $x^+ = \lim_{h \downarrow 0} x + h$ for $h > 0$:

$$\begin{aligned}F_X(a - x^+) + F_X(a + (\lim_{h \downarrow 0} x + h)^-) &= F_X(a - x^+) \\ &\quad + F_X(a + (\lim_{h \downarrow 0}(\lim_{h \downarrow 0} x + h) - h)) \\ &= F_X(a - x^+) + F_X(a + x) = 1.\end{aligned}$$

Now, subtracting the latter expression from (2.14) and rearranging we get

$$\begin{aligned} 1-1 &= \big[F_X(a-x) - F_X(a-x^+)\big] - \big[F_X(a+x) - F_X(a+x^-)\big] \\ &= \mathsf{P}(X=a-x) - \mathsf{P}(X=a+x), \end{aligned}$$

and the above masses must be equal.[1] If X is absolutely continuous we simply differentiate the CDF and get

$$\tfrac{\mathrm{d}}{\mathrm{d}x} F_X(a-x) + \tfrac{\mathrm{d}}{\mathrm{d}x} F_X(a+x^-) = -f_X(a-x) + f_X(a+x) = 0$$

i.e. the above densities are equal.[2]

2.12 First, by Theorem 2.3, $Y=-X$ is a rv. By definition, $\mathsf{P}(X=-x) = \mathsf{P}(X=x)$ and the event $\{X=-x\}$ is the same as the collection of those scenarios $\omega\in\Omega$ s.t. $\{-X=x\}$. Now, the events $\{X=x\}$ and $\{-X=x\}$ have the same probability for all $x\in\mathbb{R}$, which implies that the rvs X and $-X$ share the same discrete distribution fully characterized by the corresponding masses. More generally, by (2.14) in Definition 2.7 we have for all $x\in\mathbb{R}$ that $\mathsf{P}(X\leqslant -x) = \mathsf{P}(X\geqslant x)$ is equivalent to $\mathsf{P}(-X\geqslant x) = \mathsf{P}(X\geqslant x)$ which in turn equals

$$\underbrace{1-\mathsf{P}(-X\geqslant x)}_{=\mathsf{P}(-X<x)} = \underbrace{1-\mathsf{P}(X\geqslant x)}_{=\mathsf{P}(X<x)}$$

If X is continuous the last two probabilities do not change by adding the endpoints, i.e. $\mathsf{P}(-X\leqslant x) = \mathsf{P}(X\leqslant x)$ which means $F_{-X}(x) = F_X(x)$ so the CDFs are the same and $-X$ is distributed as X. In case of absolute continuity, we can differentiate both CDFs and get $f_{-X}(x) = f_X(x)$.

2.13 We have to check properties (a) to (d) of Theorem 2.1 in Sect. 2.1. Indeed, $F(x)$ is the CDF of some rv Z defined on the same probability space.

(a) Pick any $x_1 < x_2$, then

$$\begin{array}{rcl} F(x_1) & := & \alpha F_X(x_1) + (1-\alpha)F_Y(x_1) \\ \leqslant & \text{since} & \leqslant \qquad\qquad \leqslant \\ F(x_2) & := & \alpha F_X(x_2) + (1-\alpha)F_Y(x_2). \end{array}$$

[1] Note that $(a-x)^- = \lim_{h\downarrow 0}(a-x)-h = \lim_{h\downarrow 0} a-(x+h) = a-x^+$.

[2] Observe that $\frac{\mathrm{d}}{\mathrm{d}x} F_X(a+x^-)$ is the same as $\frac{\mathrm{d}}{\mathrm{d}x} F_X(a+x)$, because the CDF is continuous so $F_X(a+x^-) = F_X(a+x)$.

(b) For any $h > 0$:

$$\begin{aligned}\lim_{h\downarrow 0} F(x+h) &:= \lim_{h\downarrow 0} \big\{\alpha F_X(x+h) + (1-\alpha)F_Y(x+h)\big\} \\ &= \alpha \lim_{h\downarrow 0} F_X(x+h) + (1-\alpha) \lim_{h\downarrow 0} F_Y(x+h) \\ &= \alpha F_X(x) + (1-\alpha)F_Y(x) \ \ [\text{the two CDFs are yet right-cont.}] \\ &= F(x).\end{aligned}$$

(c) We have:

$$\begin{aligned}\lim_{x\to-\infty} F(x) &:= \lim_{x\to-\infty} \big\{\alpha F_X(x) + (1-\alpha)F_Y(x)\big\} \\ &= \alpha \underbrace{\lim_{x\to-\infty} F_X(x)}_{=0} + (1-\alpha) \underbrace{\lim_{x\to-\infty} F_Y(x)}_{=0} = 0.\end{aligned}$$

Similarly:

$$\begin{aligned}\lim_{x\to\infty} F(x) &:= \lim_{x\to\infty} \big\{\alpha F_X(x) + (1-\alpha)F_Y(x)\big\} \\ &= \alpha \underbrace{\lim_{x\to\infty} F_X(x)}_{=1} + (1-\alpha) \underbrace{\lim_{x\to\infty} F_Y(x)}_{=1} = 1.\end{aligned}$$

(d) To compute $\mathsf{P}(a < Z \leqslant b)$ for some rv Z we observe that:

$$\begin{aligned}&\big\{\alpha F_X(b) + (1-\alpha)F_Y(b)\big\} - \big\{\alpha F_X(a) + (1-\alpha)F_Y(a)\big\} \\ &\qquad = \alpha\big\{F_X(b) - F_X(a)\big\} + (1-\alpha)\big\{F_Y(b) - F_Y(a)\big\} \\ &\qquad = \alpha\,\mathsf{P}(a < X \leqslant b) + (1-\alpha)\,\mathsf{P}(a < Y \leqslant b).\end{aligned}$$

The last expression is the convex combination of two probabilities of the rvs X, Y lying into the interval $(a, b]$, which can be taken as the probability of a third rv Z lying in the same interval.

Exercises of Chap. 3

3.1 We have

$$(-\infty, \mathbf{b}] \setminus (-\infty, \mathbf{a}_1] \cup (-\infty, \mathbf{a}_2] = (-\infty, \mathbf{b}] \cap \Big[(-\infty, \mathbf{a}_1] \cup (-\infty, \mathbf{a}_2]\Big]^c,$$

where the complement equals $(-\infty, \mathbf{a}_1]^c \cap (-\infty, \mathbf{a}_2]^c$. The last intersection equals

$$\left\{(x, y) \in \mathbb{R}^2 \,|\, x > a_1 \text{ or } y > b_2\right\} \cap \left\{(x, y) \in \mathbb{R}^2 \,|\, x > b_1 \text{ or } y > a_2\right\},$$

but all its points must also belong to $(-\infty, \mathbf{b}]$, i.e. $x \leqslant b_1$ and $y \leqslant b_2$. Putting all these conditions together we finally obtain the set $(\mathbf{a}, \mathbf{b}]$.

3.2 We compute:

$$\begin{aligned}\underbrace{\mathsf{P}(a_1 < X_1 \leqslant b_1, a_2 < X_2 \leqslant b_2)}_{=\mathsf{P}(\mathbf{a}<\mathbf{X}\leqslant\mathbf{b})} = \underbrace{\mathsf{P}(X_1 \leqslant b_1, X_2 \leqslant b_2)}_{=\mathsf{P}(\mathbf{X}\leqslant\mathbf{b})} \\ - \Big[\underbrace{\mathsf{P}(X_1 \leqslant a_1, X_2 \leqslant b_2)}_{=\mathsf{P}(\mathbf{X}\leqslant\mathbf{a}_1)} \\ + \underbrace{\mathsf{P}(X_1 \leqslant b_1, X_2 \leqslant a_2)}_{=\mathsf{P}(\mathbf{X}\leqslant\mathbf{a}_2)} \\ - \underbrace{\mathsf{P}(X_1 \leqslant a_1, X_2 \leqslant a_2)\Big]}_{=\mathsf{P}(\mathbf{X}\leqslant\mathbf{a})}\end{aligned}$$

The result now follows from the definition of joint CDF. It is worth noting that

$$(-\infty, a_1] \times (-\infty, a_2] =: (-\infty, \mathbf{a}] = (-\infty, \mathbf{a}_1] \cap (-\infty, \mathbf{a}_2] \neq \varnothing$$

and

$$(-\infty, \mathbf{a}_1] \cap (-\infty, \mathbf{a}_2] \subset (-\infty, \mathbf{b}].$$

This together with formula (3.22) yields:

$$\{\mathbf{a} < \mathbf{X} \leqslant \mathbf{b}\} = \{\mathbf{X} \leqslant \mathbf{b}\} \setminus \{\mathbf{X} \leqslant \mathbf{a}_1\} \cap \{\mathbf{X} \leqslant \mathbf{a}_2\},$$

where the event $\{\mathbf{X} \leqslant \mathbf{b}\}$ is bigger than the joint event $\{\mathbf{X} \leqslant \mathbf{a}_1\} \cap \{\mathbf{X} \leqslant \mathbf{a}_2\} \neq \varnothing$. To compute the probability of the event in the right-hand side of the above equation we used properties Pr2,Pr3 and Pr4 of Proposition 1.1.

3.3 Figure 1 illustrates three regions in the plane containing points $(x, y) = (X_1(\omega), X_2(\omega))$ whose outcomes ω induce the events A_1, A_2 and A_3. Indeed, all scenarios in A_1 yield points (x_1, x_2) s.t. $x_1 + x_2 = 1$, i.e. the equation of a straight line passing through the points $(0, 1)$ and $(1, 0)$. Next, the condition for A_2 is that of a circle with radius $r = 1$ centered at the origin $(0, 0)$, i.e. all the scenarios in A_2 entail points (x_1, x_2) s.t. $\sqrt{x_1^2 + x_2^2} \leqslant 1$ and $\sqrt{x_1^2 + x_2^2}$ is the Euclidean distance from the origin. Finally, those scenarios in A_3 ensure $\max\{X_1, X_2\}$ is less than or equal 1: if for some scenarios $\max\{X_1, X_2\} = X_1$ then $X_2 < X_1 \leqslant 1$, while for

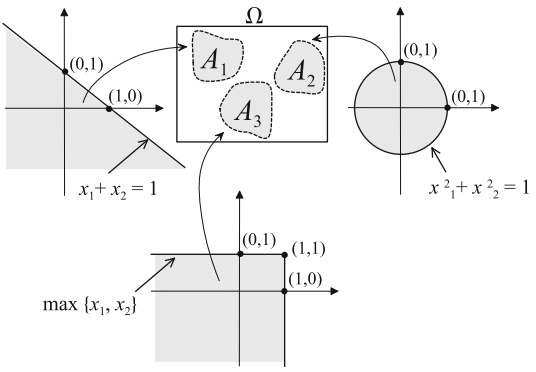


Fig. 1 Events in Ω are mapped into three regions in the Cartesian plane

some of the remaining scenarios $\max\{X_1, X_2\} = X_2$ and $X_1 < X_2 \leqslant 1$. There can be residual scenarios s.t. $X_1 = X_2 \leqslant 1$. Hence, all these scenarios verify the three conditions and we have $\{\max\{X_1, X_2\} \leqslant 1\} = \{X_1 \leqslant 1\} \cap \{X_2 \leqslant 1\}$.

3.4 In general for $a_1 \leqslant a_2$ and $b_1 \leqslant b_2$

$$F_{\mathbf{X}}(a_1, b_1) \leqslant F_{\mathbf{X}}(a_2, b_1) \leqslant F_{\mathbf{X}}(a_2, b_2)$$
$$F_{\mathbf{X}}(a_1, b_1) \leqslant F_{\mathbf{X}}(a_1, b_2) \leqslant F_{\mathbf{X}}(a_2, b_2).$$

Now, to verify the first equality above we note that

$$\{X \leqslant b_1, Y \leqslant y\} = \{X \leqslant a_1, Y \leqslant y\} \cup \{a_1 < X \leqslant b_1, Y \leqslant y\},$$

and the union is disjoint. Thus applying the probability function and rearranging we have

$$\mathsf{P}(a_1 < X \leqslant b_1, Y \leqslant y) = \mathsf{P}(X \leqslant b_1, Y \leqslant y) - \mathsf{P}(X \leqslant a_1, Y \leqslant y) \tag{1}$$

and we are done. The second equality is verified similarly.

3.5 Firstly, we have that

$$\{a_1 < X \leqslant b_1, Y \leqslant b_2\} = \{a_1 < X \leqslant b_1, Y \leqslant a_2\} \cup \{a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2\},$$

and the union is disjoint. Thus applying the probability function we have

$$\mathsf{P}(a_1 < X \leqslant b_1, Y \leqslant b_2) = \mathsf{P}(a_1 < X \leqslant b_1, Y \leqslant a_2) + \mathsf{P}(a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2).$$

Now applying (1) to the term on the left-hand side and to the first term on the right-hand side of the latter equation, eventually rearranging, we get:

$$\begin{aligned}\mathsf{P}(a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2) &= \mathsf{P}(a_1 < X \leqslant b_1, Y \leqslant b_2) \\ &\quad - \mathsf{P}(a_1 < X \leqslant b_1, Y \leqslant a_2) \\ &= F_{\mathbf{X}}(b_1, b_2) - F_{\mathbf{X}}(a_1, b_2) \\ &\quad - [F_{\mathbf{X}}(b_1, a_2) - F_{\mathbf{X}}(a_1, a_2)]\,. \end{aligned} \tag{2}$$

This yields the desired result since $\mathsf{P}(a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2) \geqslant 0$.

3.6 First we have $\frac{\partial F_{\mathbf{X}}}{\partial x}(x, y) = \lim_{\Delta x \to 0} \frac{F_{\mathbf{X}}(x+\Delta x, y) - F_{\mathbf{X}}(x, y)}{\Delta x}$. Second we also have:

$$\begin{aligned}\frac{\partial}{\partial y}\left[\frac{\partial F_{\mathbf{X}}}{\partial x}\right](x, y) = \lim_{\Delta y \to 0}\Bigg[\lim_{\Delta x \to 0} \frac{F_{\mathbf{X}}(x+\Delta x, y+\Delta y) - F_{\mathbf{X}}(x, y+\Delta y)}{\Delta x}\frac{1}{\Delta y} \\ - \lim_{\Delta x \to 0} \frac{F_{\mathbf{X}}(x+\Delta x, y) - F_{\mathbf{X}}(x, y)}{\Delta x}\frac{1}{\Delta y}\Bigg].\end{aligned} \tag{3}$$

By definition $f_{\mathbf{X}}(x, y) = \frac{\partial^2 F_{\mathbf{X}}}{\partial x \partial y}(x, y)$, so by the continuity assumption on the second order partial[3] and by (3) we can write:

$$\lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{F_{\mathbf{X}}(x + \Delta x, y + \Delta y) - F_{\mathbf{X}}(x, y + \Delta y) - F_{\mathbf{X}}(x + \Delta x, y) + F_{\mathbf{X}}(x, y)}{\Delta x \Delta y}.$$

Since by (2) the numerator in the above expression is $\geqslant 0$ we have proved item (a). To prove item (b) we interpret the second order differential of the CDF as

$$\mathrm{d}^2 F_{\mathbf{X}}(x, y) = \partial^2 F_{\mathbf{X}} = \partial x \partial y f_{\mathbf{X}}(x, y) =: f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y.$$

This acts as $\mathrm{d}F_X(x)$ in the one-dimensional case. Thus, we integrate it and by Fubini's Theorem get:

$$\begin{aligned}\int_{a_1}^{b_1}\int_{a_2}^{b_2} f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y = \int_{a_1}^{b_1}\int_{a_2}^{b_2} \mathrm{d}^2 F_{\mathbf{X}}(x, y) = \int_{(a_1,b_1]\times(a_2,b_2]} \mathrm{d}^2 F_{\mathbf{X}}(x, y) \\ = \mathsf{P}(a_1 < X \leqslant b_1, a_2 < Y \leqslant b_2).\end{aligned}$$

[3] Namely, the limit exists and is finite.

To show item (c) we apply item (b) as follows:[4]

$$\begin{aligned} F_{\mathbf{X}}(x, y) &= \lim_{n\to\infty} \mathsf{P}(-n < X \leqslant x, -n < Y \leqslant y) \\ &= \lim_{n\to\infty} \int_{-n}^{x}\int_{-n}^{y} f_{\mathbf{X}}(u, v)\mathrm{d}u\mathrm{d}v = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{\mathbf{X}}(u, v)\mathrm{d}u\mathrm{d}v. \end{aligned}$$

Now item (d) easily follows from item (c) by sending x, y to ∞.

3.7 We show that $\mathsf{P}(\,\cdot \mid B)$ satisfies all the axioms of a probability measure. $\mathsf{P}(\Omega \mid B) = \frac{\mathsf{P}(\Omega\cap B)}{\mathsf{P}(B)} = \frac{\mathsf{P}(B)}{\mathsf{P}(B)} = 1$. If events $(A_i)_{i\in\mathbb{N}}$ are pairwise disjoint then:

$$\mathsf{P}(\cup_{i=1}^{\infty} A_i \mid B) = \frac{\mathsf{P}\left(\left(\cup_{i=1}^{\infty} A_i\right) \cap B\right)}{\mathsf{P}(B)} = \frac{\mathsf{P}\left(\cup_{i=1}^{\infty}(A_i \cap B)\right)}{\mathsf{P}(B)}.$$

The events $(A_i \cap B)_{i\in\mathbb{N}}$ are pairwise disjoint and the expression on the right-hand side rewrites: $\sum_{i=1}^{\infty} \frac{\mathsf{P}(A_n\cap B)}{\mathsf{P}(B)} = \sum_{i=1}^{\infty} \mathsf{P}(A_i \mid B)$. Thus, $\mathsf{P}(\,\cdot \mid B)$ is countably additive and we are done. The 'fraction' of event B including A is measured by $\mathsf{P}(A \cap B)$, as the *area* of $A \cap B \subset B$. Assuming B is occurred yield this event behaves as a *new sample* space $\Omega' = B$ whose *area* is $\mathsf{P}(B)$ which is not necessarily $= 1$. The original events $A \subset \Omega$ are then replaced by subsets $A \cap B$ of Ω' and the new σ-algebra for them can be safely be assumed to be $\{A \cap B \mid A \in \mathcal{F}\}$. The probability $\mathsf{P}(A)$ is replaced by $\frac{\mathsf{P}(A\cap B)}{\mathsf{P}(B)}$, i.e. the *area* $\mathsf{P}(A \cap B)$ is normalized by the reference *area* $\mathsf{P}(B)$.

...Take a Look at This Example...
Let $\Omega = \{\omega = (i, j) \mid i, j \in \{1, \ldots, 6\}\}$ be the sample space for rolling a fair dice twice. Assume a uniform probability space such that $\mathsf{P}(\{\omega\}) = \frac{1}{36}$ for every $\{\omega\} \in \mathcal{F} = 2^{\Omega}$, i.e. event are *equally likely*. Next, assume $B = \{\text{the first roll shows 3}\}$ is occurred. Thus, Ω reduces to $\Omega' = \{\omega = (3, j) \mid j = 1, \ldots, 6\} = B$. Only 6 outcomes are possible with new probability $\mathsf{P}(\{\omega\} \mid B) = \frac{\mathsf{P}(\{\omega\}\cap B)}{\mathsf{P}(B)} = \frac{\mathsf{P}(\{\omega\})}{\mathsf{P}(B)} = \frac{6}{36} = \frac{1}{6}$, where of course $\{\omega\} \cap B = \{\omega\}$, for every $\omega \in \Omega'$. We may think of a *new* uniform probability space with sample space

$$\Omega' = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\} = B$$

and σ-algebra given by events $A \cap B$ and $A \in \mathcal{F}$. It is easily seen that it coincides with $2^{\Omega'}$.

[4] We also apply continuity of $\mathsf{P}(\,\cdot\,)$.

3.8 With all the information available, we can assume a uniform probability space with 9 scenarios. Note that $\mathsf{P}(\{\omega\}) > 0$ whether $\omega \notin \{\text{du}, \text{us}\}$. To compute the required probability observe that

$$A = \text{'price \#1 is fallen'}, \qquad B = \text{'price \#2 is the same or is raised'}.$$

(Do not forget to decompose A, B and $A \cap B$ into their singletons and then add probabilities). We have $\mathsf{P}(B) = 0.32 > 0$ then $\mathsf{P}(A \mid B) := \frac{\mathsf{P}(A \cap B)}{\mathsf{P}(B)} = \frac{0.1}{0.32} \approx 0.31$, which gives the required probability of A conditional on B. The re-weighting of event A has now produced a greater evaluation of its likelihood.

3.9 Let the bi-variate random vector $\mathbf{Z} = (U, V)$ has a joint density function $f_{\mathbf{Z}}(u, v)$ and suppose it equals $h(u)k(v)$, the product of two real-valued functions. Thus, let

$$\int h(u)\mathrm{d}u = a, \quad \text{and} \quad \int k(v)\mathrm{d}v = b.$$

Then we have

$$\begin{aligned} a\,b &= \left(\int h(u)\mathrm{d}u\right)\left(\int k(v)\mathrm{d}v\right) \\ &\overset{\text{Fubini's Th.}}{=} \iint h(u)k(v)\mathrm{d}u\mathrm{d}v \\ &= \iint f_{\mathbf{Z}}(u, v)\mathrm{d}u\mathrm{d}v = 1, \end{aligned}$$

i.e., $a\,b = 1$. Now, computing the marginals

$$f_U(u) = \int h(u)k(v)\mathrm{d}v = h(u)b, \quad f_V(v) = \int h(u)k(v)\mathrm{d}u = k(v)a,$$

and putting things together we have

$$f_{\mathbf{Z}}(u, v) = h(u)k(v)a\,b = f_U(u)\,f_V(v),$$

i.e., we find a necessary condition for independence between U and Y: their joint density factors by two functions $h(u), k(v)$ of u alone and of v alone. New, let $A(u) = \{x \mid h(x) \leqslant u\}$ and $B(v) = \{y \mid k(y) \leqslant v\}$, then passing to the joint CDF we have:

$$\begin{aligned} F_{\mathbf{Z}}(u, v) &= \mathsf{P}(U \leqslant u, V \leqslant v) \\ &= \mathsf{P}(X \in A(u), Y \in B(v)) \\ &\overset{\text{Indep. of } X \text{ and } Y}{=} \mathsf{P}(X, \in A(u))\mathsf{P}(Y \in B(v)). \end{aligned}$$

Since we can differentiate to get the joint density,

$$f_{\mathbf{Z}}(u,v) = \frac{\partial^2}{\partial u \partial v} F_{\mathbf{Z}}(u,v)$$
$$= \frac{\mathrm{d}}{\mathrm{d}u}\mathsf{P}(X \in A(u)) \frac{\mathrm{d}}{\mathrm{d}v}\mathsf{P}(Y \in B(v)),$$

we end with the product of two functions, of u alone and of v alone, which together with the above necessary condition entails $U = f(X)$ and $V = g(Y)$ as independent.

Exercises of Chap. 4

4.1 Linearity of expectation for discrete rvs is:

$$\mathsf{E}(a + b\,X) = \sum_i (a + b\,x_i)\,\mathsf{P}(X = x_i)$$
$$= a \sum_i \mathsf{P}(X = x_i) + b \sum_i x_i \mathsf{P}(X = x_i) = a + b\,\mathsf{E}(X).$$

4.2 Taking $g(x) = a\,x + b$ we have:

$$\mathsf{E}(a + b\,X) = \int_{-\infty}^{\infty} (a + b\,x)\, f_X(x)\,\mathrm{d}x$$
$$= a \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x + b \int_{-\infty}^{\infty} x\, f_X(x)\,\mathrm{d}x = a + b\,\mathsf{E}(X).$$

4.3 By definition we have:

$$\mathsf{E}(X) - \mathsf{E}(Y) = \int_0^{\infty} [1 - F_X(x)]\mathrm{d}x - \int_{-\infty}^{0} F_X(x)\mathrm{d}x$$
$$- \int_0^{\infty} [1 - F_Y(x)]\mathrm{d}x + \int_{-\infty}^{0} F_Y(x)\mathrm{d}x$$
$$= \int_0^{\infty} [F_Y(x) - F_X(x)]\mathrm{d}x + \int_{-\infty}^{0} [F_Y(x) - F_X(x)]\mathrm{d}x$$
$$= \int_{-\infty}^{\infty} [F_Y(x) - F_X(x)]\mathrm{d}x \leqslant 0$$

which yields $\mathsf{E}(X) \leqslant \mathsf{E}(Y)$. Observe that $X \preccurlyeq_{\text{FSD}} Y$ is also equivalent to

$$1 - \mathsf{P}(X > x) \geqslant 1 - \mathsf{P}(Y > x) \Longleftrightarrow \mathsf{P}(Y > x) \geqslant \mathsf{P}(X > x).$$

4.4 By assumption $X - a \overset{\text{d}}{=} a - X$ then:

$$\mathsf{P}(X - a \leqslant 0) = \mathsf{P}(a - X \leqslant 0) \Longleftrightarrow \mathsf{P}(X \leqslant a) = \mathsf{P}(X \geqslant a).$$

Thus we have:

$$\mathsf{P}(X \leqslant a) = 1 - \mathsf{P}(X > a) \geqslant 1 - \mathsf{P}(X \geqslant a) = 1 - \mathsf{P}(X \leqslant a)$$

which implies at once $\mathsf{P}(X \leqslant a) \geqslant \frac{1}{2}$ and $\mathsf{P}(X \geqslant a) \geqslant \frac{1}{2}$, i.e. $a = \mathsf{med}(X)$. Eventually, by symmetry again we have $\mathsf{E}(X - a) = -\mathsf{E}(X - a)$ iff $\mathsf{E}(X - a) = 0$ i.e. $\mathsf{E}(X) = a$.

4.5 First, since $f_X(x) \geqslant 0$ for all $x \in \mathbb{R}$ then $\frac{1}{b} f_X\left(\frac{x-a}{b}\right) \geqslant 0$, for every $x, a \in \mathbb{R}$ and $b > 0$. Eventually:

$$\begin{aligned} \int \tfrac{1}{b} f_X\left(\frac{x-a}{b}\right) \mathrm{d}x &= \int f_X(y) \mathrm{d}y \qquad \left[\text{let } y = \tfrac{x-a}{b} \text{ then } \mathrm{d}y = \tfrac{1}{b}\mathrm{d}x\right] \\ &= 1 \qquad \left[\text{because } f_X(y) \text{ is a density}\right]. \end{aligned}$$

4.6 First, assume $d < m$ without loss of generality. We have $|X - \alpha| = (X - \alpha)\mathbf{I}_{\{X \geqslant \alpha\}} - (X - \alpha)\mathbf{I}_{\{X < \alpha\}}$, for $\alpha = d, m$. Write the difference between absolute values and add:

$$\begin{aligned} |X - d| - |X - m| = {} & X\mathbf{I}_{\{X \geqslant d\}} - d\mathbf{I}_{\{X \geqslant d\}} - X\mathbf{I}_{\{X < d\}} + d\mathbf{I}_{\{X < d\}} \\ & - X\mathbf{I}_{\{X \geqslant m\}} + m\mathbf{I}_{\{X \geqslant d\}} + X\mathbf{I}_{\{X < d\}} - m\mathbf{I}_{\{X < d\}} + X - X. \end{aligned}$$

Arrange terms in X,

$$X\left(\mathbf{I}_{\{X \geqslant d\}} + 1 - \mathbf{I}_{\{X < d\}}\right) - X\left(\mathbf{I}_{\{X \geqslant m\}} + 1 - \mathbf{I}_{\{X < m\}}\right),$$

and rewrite them as $2X\mathbf{I}_{\{X \geqslant d\}} - 2X\mathbf{I}_{\{X \geqslant m\}}$, where $\mathbf{I}_{\{X \geqslant \alpha\}} = 1 - \mathbf{I}_{\{X < \alpha\}}$. Arrange terms in d and write

$$\begin{aligned} d\left(\mathbf{I}_{\{X < d\}} - \mathbf{I}_{\{X \geqslant d\}}\right) &= d\left(1 - \mathbf{I}_{\{X \geqslant d\}} - \mathbf{I}_{\{X \geqslant d\}}\right) \\ &= d\left(1 - 2\mathbf{I}_{\{X \geqslant d\}}\right) \\ &= d - 2d\mathbf{I}_{\{X \geqslant d\}} \\ &= d\left(\mathbf{I}_{\{X \geqslant m\}} + \mathbf{I}_{\{X < m\}}\right) - 2d\mathbf{I}_{\{X \geqslant d\}} \\ &= 2d\mathbf{I}_{\{X \geqslant m\}} - d\mathbf{I}_{\{X \geqslant m\}} + d\mathbf{I}_{\{X < m\}} - 2d\mathbf{I}_{\{X \geqslant d\}}, \end{aligned}$$

where in the fourth equality $1 = \mathbf{I}_{\{X<m\}} + \mathbf{I}_{\{X\geqslant m\}}$. Now, arrange all the above terms to get:

$$(m-q)\left(\mathbf{I}_{\{X\geqslant m\}} - \mathbf{I}_{\{X<m\}}\right) + 2(X-d)\left(\mathbf{I}_{\{X\geqslant d\}} - \mathbf{I}_{\{X\geqslant m\}}\right).$$

Note that $\mathbf{I}_{\{X\geqslant d\}} = 1 - \mathbf{I}_{\{X<d\}}$ and events $\{X < d\}$ and $\{X \geqslant m\}$ are incompatible (recall that $d < m$) so $\mathbf{I}_{\{X<d\}} + \mathbf{I}_{\{X\geqslant m\}} = \mathbf{I}_{\{X<d\}\cup\{X\geqslant m\}} = \mathbf{I}_{\{d\leqslant X<m\}}$. We finally obtain:

$$|X-d| - |X-m| = (m-d)(2\mathbf{I}_{\{X\geqslant m\}} - 1) + 2(X-d)\mathbf{I}_{\{d\leqslant X<m\}}.$$

Taking expectation we have:

$$\begin{aligned}\mathsf{E}(|X-d|) - \mathsf{E}(|X-m|) &= (m-d)\,(2\mathsf{P}(X\geqslant m) - 1)\\ &\quad + 2\mathsf{E}\left[(X-d)\mathbf{I}_{\{d\leqslant X<m\}}\right].\end{aligned}$$

The second term on the right-hand side is $\geqslant 0$ as well the first, since $2\mathsf{P}(X \geqslant m) \geqslant 1$ for m is the median of F_X. This proves the claim.

4.7 If $|X| \geqslant 1$ pointwise then $|X|^n \leqslant |X|^m$, otherwise for $|X| < 1$ we have $|X|^n \leqslant 1$. Taking expectation we get

$$\mathsf{E}(|X|^n) \leqslant 1 + \mathsf{E}(|X|^m) < \infty$$

and we are done. We might write $|X|^n \leqslant \max\{1, |X|^m\} \leqslant 1 + |X|^m < \infty$ at once.

4.8 Let us write the expression of $(g(a+bX))^2$:

$$\begin{aligned}\left(\frac{a+bX-\mathsf{E}(a+bX)}{\mathsf{SD}(a+bX)}\right)^2 &= \frac{(a-a+b(X-\mathsf{E}(X)))^2}{(|b|\mathsf{SD}(X))^2}\\ &= \frac{|b|((X-\mathsf{E}(X)))^2}{(|b|\mathsf{SD}(X))^2} = \frac{(X-\mathsf{E}(X))^2}{(\mathsf{SD}(X))^2}\\ &= (g(X))^2.\end{aligned}$$

4.9 We have $|XY| \leqslant \frac{X^2}{2} + \frac{Y^2}{2}$ and taking expectation of both sides $\mathsf{E}(|XY|) \leqslant \frac{1}{2}\mathsf{E}(X^2) + \frac{1}{2}\mathsf{E}(Y^2)$ and for $X, Y \in L^2$ means $\mathsf{E}(X^2) < \infty$ and $\mathsf{E}(Y^2) < \infty$, then $\mathsf{E}(XY)$ is finite, i.e. $XY \in L^1$.

4.10 We first compute $\int_0^1 g(x)\mathrm{d}F(x)$ by definition, observing that $g(x) = 1$ for $x \in (0, 1]$:

$$\lim_{n\to\infty}\sum_{i=1}^{n}[F(x_i) - F(x_{i-1})] \quad = \quad \lim_{n\to\infty}\big\{[F(x_1) - F(x_0)] + [F(x_2) - F(x_1)]$$

$$+\cdots+[F(x_n)-F(x_{n-1})]\big\} = \lim_{n\to\infty}\big\{F(1)-F(0)\big\}$$
$$= \quad F(1)-F(0) = 1 = \int_0^1 \mathrm{d}F(x),$$

where the limit is meant for finer partitions $0 = x_0 < x_1 < \cdots < x_n = 1$ of $(0, 1]$ and the larger $x_i - x_{i-1}$ goes to zero.[5] The second integral is $\int_1^2 g(x)\mathrm{d}F(x) = 0$ because $g(x) = 0$ for $x \in (1, 2]$. Nevertheless, the integral $\int_0^2$ does not exists: the Stieltjes sum

$$\sum_{i=1}^{n} g(c_i)[F(x_i) - F(x_{i-1})]$$

has different values depending on the choice $c_i = 1$, when $g(c_i) = 1$, or $c_i = \lim_{h\downarrow 0}(1+h) = 1^+$ for $h > 0$, when $g(c_i) = 0$. Despite the two integrals $\int_0^1$ and $\int_1^2$ exist the whole integral $\int_0^2$ does not.

4.11 First, observe that $n\mathsf{P}(X > n) \leqslant \mathsf{E}(X\mathbf{I}_{\{X>n\}})$, since the term on the left-hand side of the inequality equals $\mathsf{E}(n\mathbf{I}_{\{X>n\}})$. Now, write the term on the right-hand side as $\mathsf{E}\left(X\left[1-\mathbf{I}_{\{X\leqslant n\}}\right]\right)$, which in turns is $\mathsf{E}(X) - \mathsf{E}(X\mathbf{I}_{\{X\leqslant n\}})$. The sequence of nonnegative rvs $X\mathbf{I}_{\{X\leqslant n\}}$ for $n = 1, 2, \ldots$ is increasing and converges pointwise to X, indeed $\lim_{n\to\infty}\mathbf{I}_{\{X\leqslant n\}} = \mathbf{I}_\Omega = 1$. Thus applying the MCT we have

$$\lim_{n\to\infty}\mathsf{E}(n\mathbf{I}_{\{X>n\}}) = \mathsf{E}(X) - \mathsf{E}\left(\lim_{n\to\infty} X\mathbf{I}_{\{X\leqslant n\}}\right) = 0.$$

Since $n\mathsf{P}(X > n)$ is dominated by $\mathsf{E}(X\mathbf{I}_{\{X>n\}})$ for every $n \in \mathbb{N}$, then we get the desired result.[6]

4.12 Take $A = \{X = \infty\}$ and consider its truncated version $X\mathbf{I}_A$. Thus, using the abstract definition of the expectation and $\mathsf{P}(A) > 0$ we get:

$$\begin{aligned}\mathsf{E}(X\mathbf{I}_A) &= \int_\Omega X\mathbf{I}_A \mathrm{d}\mathsf{P}\\ &= \int_A X\mathrm{d}\mathsf{P} = \infty\int_A \mathrm{d}\mathsf{P}\\ &= \infty\times\mathsf{P}(A) = \infty.\end{aligned}$$

The latter equality contradicts the hypotheses of finite expectation, since for every $A \in \mathcal{F}$ we always have $\mathsf{E}(X\mathbf{I}_A) \leqslant \mathsf{E}(X)$. The remedy is $\mathsf{P}(A) = 0$ and consequently

[5] This can be written $n \to \infty$ and $\sup\{x_i - x_{i-1} \mid i = 1, \ldots, n\} \to 0$.

[6] Recall that given two sequences of nonnegative real numbers $(x_n)_{n\in\mathbb{N}}$, $(y_n)_{n\in\mathbb{N}}$ s.t. $y_n \to 0$ and $0 \leqslant x_n \leqslant y_n$, each n, then $x_n \to 0$ too.

$\mathsf{P}(X < \infty) = 1$, using the multiplication rule $\infty \times 0 = 0$ valid for every probability measure.

4.13 We start from discrete nonnegative rvs X, Y. It can be shown that the rv $X+Y$ has representation $\sum_{i=1}^{n}\sum_{j=1}^{m}(x_i + y_j)\mathbf{I}_{A_i \cap B_j}$, where events $A_i = \{X = x_i\}$ and $B_j = \{Y = y_j\}$ form two different partitions of Ω, thus it is discrete too. Hence,

$$\sum_i \sum_j x_i \mathsf{P}(A_i \cap B_j) + \sum_i \sum_j y_j \mathsf{P}(A_i \cap B_j)$$
$$= \sum_i x_i \sum_j \mathsf{P}(A_i \cap B_j) + \sum_j y_j \sum_i \mathsf{P}(A_i \cap B_j).$$

But events A_i e B_j decompose Ω and then $\sum_j \mathsf{P}(A_i \cap B_j) = \mathsf{P}(\cup_j (A_i \cap B_j)) = \mathsf{P}(A_i)$ and $\sum_i \mathsf{P}(A_i \cap B_j) = \mathsf{P}(\cup_i (A_i \cap B_j)) = \mathsf{P}(B_j)$. Thus, the last equality above is just $\mathsf{E}(X) + \mathsf{E}(Y)$. Applying this reasoning[7] to increasing sequences $(X_n)_{n\in\mathbb{N}}$, $(Y_n)_{n\in\mathbb{N}}$ of discrete rvs converging pointwise to two nonnegative rvs X and Y, so $X_n + Y_n \to X + Y$, and applying MCT we get linearity again, which can be extended to general rvs using positive and negative parts.

4.14 First, we have $X = \sum_{i=1}^{n} x_i \mathbf{I}_{A_i}$ and $Y = \sum_{i=1}^{m} y_j \mathbf{I}_{B_j}$ since both the rvs are discrete, where the events A_i and B_j are two different partition of Ω. Summing we get $\sum_{i=1}^{n} x_i \mathbf{I}_{A_i} + \sum_{i=1}^{m} y_i \mathbf{I}_{B_j}$, but the summation can be arranged to give values $x_i + y_j$ for those scenarios $\omega \in A_i \cap B_j$. In fact we also have $A_i \cap B_j = \{X + Y = x_i + y_j\}$. Eventually,

$$X + Y = \sum_{i=1}^{n}\sum_{j=1}^{m}(x_i + y_j)\,\mathbf{I}_{A_i \cap B_j},$$

where some $A_i \cap B_j$ can be the impossible event, and all still form a partition of the sample space. It is worth noting that kX is a discrete rv for $k \in \mathbb{R}$ and X being discrete. Also, the product of two discrete rvs XY is discrete too because it can be represented as $\sum_{i=1}^{n}\sum_{j=1}^{m}(x_i y_j)\,\mathbf{I}_{A_i \cap B_j}$.

4.15 We use the standard machine, starting from discrete rvs X, Y and then extending to general rvs via MCT. If X, Y are both discrete and $X(\omega) \leqslant Y(\omega)$ for each $\omega \in \Omega$, then $x_i \leqslant y_j$ in their respective representations over $A_i \cap B_j$. As byproduct:

$$\sum_i \sum_j x_i \mathsf{P}(A_i \cap B_j) \leqslant \sum_i \sum_j y_j \mathsf{P}(A_i \cap B_j) \iff \mathsf{E}(X) \leqslant \mathsf{E}(Y).$$

[7] It is sometime called the 'standard machine' [16].

4.16 Solution 1. Recall that C, the X's support, is the set of points $x \in \mathbb{R}$ where $F_X(x)$ is strictly increasing. By definition $Q_X(c)$ is the smallest value in C where $F_X(Q_X(c)) \geqslant c$, for every $0 < c < 1$. In fact, we also have $Q_X(F_X(x)) \leqslant x$ and for each *finite* $x \in C$ there is some probability level $c \in (0, 1)$ s.t. $Q_X(c) = u$, for otherwise $F_X(x)$ does not increases at x and we may find a smaller value $x' < x$ where $Q_X(F_X(x')) \leqslant x'$ and thus $x' = Q_X(c)$ is the correct quantile.[8] As byproduct, $Q_X(c) = x$ and we can write:

$$\mathsf{E}(X) = \int_{-\infty}^{\infty} x \, \mathrm{d}F_X(x) = \int_C Q_X(c) \mathrm{d}F_X(Q_X(c)).$$

Now we consider two cases.

1. If the CDF has not jump discontinuities in the X's support we safely let $F_X(Q_X(c)) = c$.
2. If the CDF has jump discontinuities at the points $x \in C$, then here we have that $x\mathsf{P}(X = x) = \int_{F_X(x^-)}^{F_X(x)} Q_X(c)\mathrm{d}c$. Indeed, for c in the open interval $\big(F_X(x^-), F_X(x)\big)$ it holds $Q_X(c) = x$ and the above integral equals $x \int_{F_X(x^-)}^{F_X(x)} \mathrm{d}c = x[F_X(x) - F_X(x^-)]$.

Putting things together and considering the (at most countable) sum $\sum_x \int_{F_X(x^-)}^{F_X(x)} Q_X(c)\mathrm{d}c$ we eventually get the desired result. Solution 2. Recall the quantile transform: for a rv X with CDF F_X and quantile function Q_X we always have $X \overset{\mathrm{d}}{=} Q_X(U)$, where $U \sim \mathrm{U}(0, 1)$. We can write

$$\mathsf{E}(X) = \int_\Omega X \mathrm{d}\mathsf{P} = \int_\Omega \underbrace{Q_X(U)}_{=g(U)} \mathrm{d}\mathsf{P} = \int_0^1 Q_X(c)\mathrm{d}c,$$

where we used the abstract definition of expectation and the change of variable formula with $g(c) = g(U(\omega))$ and $c = U(\omega)$ for every scenario $\omega \in \Omega$ s.t. $U(\omega)$ varies in the unit interval $(0, 1)$. In fact, under these conditions we have

$$\int_\Omega g(U)\mathrm{d}\mathsf{P} = \int_{-\infty}^{\infty} g(c)\mathrm{d}F_U(c) = \int_{-\infty}^{\infty} g(c) \underbrace{f_U(c)}_{=\mathbf{I}_{\{0 \leqslant c \leqslant 1\}}} \mathrm{d}c = \int_0^1 Q_X(c)\mathrm{d}c.$$

4.17

- For a discrete $X = \sum_{i=1}^n x_i \mathbf{I}_{A_i}$ with $x_i > 0$ then $\mathsf{P}(A_i) = \mathsf{P}(X = x_i) = 0$ since $X \overset{\text{a.s.}}{=} 0$ by hypotheses, thus $\mathsf{E}(X) = 0$. Passing to $X \geqslant 0$ with $\mathsf{P}(X > 0) = 0$, we consider a discrete rv Z s.t. $0 \leqslant Z \leqslant X$, then it holds $Z \overset{\text{a.s.}}{=} 0$. We can write

[8] This happens at all the points in C^c.

$\mathsf{E}(X) = \sup_{\{0 \leqslant Z \leqslant X,\ Z \text{ discrete}\}} \mathsf{E}(Z) = 0$, since each $\mathsf{E}(Z) = 0$. The general case follows by the standard machine.

- Start from discrete rvs X, Y s.t. $X \overset{\text{a.s.}}{=} Y$, i.e. $\mathsf{P}(X \neq Y) = 0$. Then:

$$\begin{aligned} \mathsf{E}(Y) &= \mathsf{E}(Y\mathbf{I}_{\{X \neq Y\}}) + \mathsf{E}(Y\mathbf{I}_{\{X=Y\}}) \\ &\overset{\text{equality a.s.}}{=} \mathsf{E}(Y\mathbf{I}_{\{X \neq Y\}}) + \mathsf{E}(X\mathbf{I}_{\{X=Y\}}). \end{aligned}$$

Pick a discrete sequence $Y_1, Y_2, \ldots$ s.t. $Y_n \to Y$ pointwise. Hence, $Y_n\mathbf{I}_{\{X \neq Y\}}$ increases pointwise to $Y\mathbf{I}_{\{X \neq Y\}}$. Moreover $Y_n\mathbf{I}_{\{X \neq Y\}} \leqslant \delta$ each n, since any simple rv is bounded above by some $\delta > 0$. Then

$$0 \leqslant \mathsf{E}(Y_n\mathbf{I}_{\{X \neq Y\}}) \leqslant \mathsf{E}(\delta\,\mathbf{I}_{\{X \neq Y\}}) = \delta\,\mathsf{P}(X \neq Y) = 0,$$

so $\mathsf{E}(Y_n\mathbf{I}_{\{X \neq Y\}}) = 0$. Since $Y_n\mathbf{I}_{\{X \neq Y\}} \to Y\mathbf{I}_{\{X \neq Y\}}$ pointwise, MCT gives $0 = \lim_n \mathsf{E}(Y_n\mathbf{I}_{\{X \neq Y\}}) = \mathsf{E}(Y\mathbf{I}_{\{X \neq Y\}})$. Similarly, $\mathsf{E}(X\mathbf{I}_{\{X \neq Y\}}) = 0$. Putting things together:

$$\begin{aligned} \mathsf{E}(Y) &= \mathsf{E}(Y\mathbf{I}_{\{X \neq Y\}}) + \mathsf{E}(X\mathbf{I}_{\{X=Y\}}) \\ &= 0 + \mathsf{E}(X\mathbf{I}_{\{X=Y\}}) \\ &= \mathsf{E}(X\mathbf{I}_{\{X \neq Y\}}) + \mathsf{E}(X\mathbf{I}_{\{X=Y\}}) = \mathsf{E}(X). \end{aligned}$$

For general rvs, $X \overset{\text{a.s.}}{=} Y$ implies $X^+ \overset{\text{a.s.}}{=} Y^+$ and $X^- \overset{\text{a.s.}}{=} Y^-$. Applying the above separately to X^+ and X^- then adding we are done.

- Let $A = \{X > 0\}$ and $A_n = \{X \geqslant \frac{1}{n}\}$. Clearly $A_1 \subset A_2 \subset \cdots$ and $A = \cup_i A_i$, i.e. $A_n \uparrow A$ which implies

$$0 \leqslant X\mathbf{I}_{A_n} \leqslant X\mathbf{I}_A.$$

By monotonicity of expectation we also have $0 \leqslant \mathsf{E}(X\mathbf{I}_{A_n}) \leqslant \mathsf{E}(X\mathbf{I}_A) \leqslant \mathsf{E}(X) = 0$. As byproduct,

$$0 = \mathsf{E}(X\mathbf{I}_{A_n}) \geqslant \mathsf{E}(\tfrac{1}{n}\mathbf{I}_{A_n}) = \tfrac{1}{n}\mathsf{P}(A_n)$$

which further implies $\mathsf{P}(A_n) = 0$ for all $n \geqslant 1$. But $\mathsf{P}(A) = \lim_{n\to\infty} \mathsf{P}(A_n)$ and we conclude $\mathsf{P}(A) = 0$.

Remark 2 Combining items 1 and 2 we show that for $X \overset{\text{a.s.}}{=} Y$ the condition $|X - Y| \overset{\text{a.s.}}{=} 0$ also implies $\mathsf{E}(X) = \mathsf{E}(Y)$. In fact, we have

$$|\mathsf{E}(X) - \mathsf{E}(Y)| = |\mathsf{E}(X - Y)| \leqslant \mathsf{E}(|X - Y|).$$

If we assume $X \overset{\text{a.s.}}{=} Y$ then $|X - Y| \overset{\text{a.s.}}{=} 0$ because

$$1 = \mathsf{P}(X = Y) = \mathsf{P}(X - Y = 0) = \mathsf{P}(|X - Y| = 0),$$

and obviously $|x - y| = 0$ iff $x - y = 0$. But this implies $\mathsf{E}(|X - Y|) = 0$ by item 1 and then

$$|\mathsf{E}(X) - \mathsf{E}(Y)| = 0 \Longleftrightarrow \mathsf{E}(X) - \mathsf{E}(Y) = 0,$$

thus we get $\mathsf{E}(X) = \mathsf{E}(Y)$.

Remark 3 Observe that for a nonnegative rv $X \geqslant 0$ with $A^c = \{X = 0\}$ and $\mathsf{P}(A) = 0$ then every nonnegative discrete rv Z satisfying $0 \leqslant Z \leqslant X$ must be zero on the same set A^c. Indeed, for $Z = z_1\mathbf{I}_{A_1} + \cdots + z_n\mathbf{I}_{A_n}$ we have $Z\mathbf{I}_A = \sum_{i=1}^n z_i\mathbf{I}_{A\cap A_i}$. The expectation of $Z\mathbf{I}_A$ is clearly $\sum_{i=1}^n z_i\mathsf{P}(A \cap A_i)$, where $\mathsf{P}(A \cap A_i)$ must be zero because $A \cap A_i \subset A$, and by monotonicity of P it holds $\mathsf{P}(A \cap A_i) \leqslant \mathsf{P}(A) = 0$. This implies that $\mathsf{E}(Z\mathbf{I}_A) = 0$ if and only if $\mathsf{P}(A) = 0$.

4.18 The rv

$$Y_n(\omega) = \begin{cases} X(\omega) & \text{if } X(\omega) \in \{x_1, x_2, \ldots, x_n\} \\ 0 & \text{if } X(\omega) \in \{x_{n+1}, x_{n+2}, \ldots\}, \end{cases}$$

is such that $Y_n(\omega) < X(\omega)$, for every $\omega \in \Omega$ and some $n \in \mathbb{N}$. Since $x_i \geqslant 0$ we have Y_n is discrete. Taking expectation: $\mathsf{E}(Y_n) = \sum_{i=1}^n x_i\mathsf{P}(X = x_i)$. The sequence $(Y_n)_{n\in\mathbb{N}}$ converges pointwise to the nonnegative discrete rv X. Then, by MCT we have:

$$\lim_n \mathsf{E}(Y_n) = \lim_n \sum_{i=1}^n x_i\mathsf{P}(X = x_i) = \sum_{i=1}^\infty x_i\mathsf{P}(X = x_i) = \mathsf{E}(X).$$

4.19 Write the finite sum $S_n = \sum_{i=1}^n X_i$. Of course $S_n \leqslant S_{n+1}$ for every $n \in \mathbb{N}$, and further

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{i=1}^n X_i = \sum_{i=1}^\infty X_i$$

pointwise. MCT now yields

$$\lim_{n\to\infty} \mathsf{E}(S_n) = \mathsf{E}\Big(\lim_{n\to\infty} S_n\Big) = \mathsf{E}\left(\sum_{i=1}^\infty X_i\right).$$

4.20 Assume $X_n \to X$ pointwise. Then $\lim_{n\to\infty}(X_n - X) = X - X = 0$. By definition $(X - X_n)^+ = (X - X_n)\mathbf{I}_{\{X-X_n\geqslant 0\}}$ which is either zero or $X - X_n$ if

$0 \leqslant X \leqslant X_n$. This obviously implies $(X - X_n)^+ \leqslant X_n$. Analogously, $(X_n - X)^- = -(X_n - X)\mathbf{I}_{\{X_n - X \leqslant 0\}}$ is either zero or $X - X_n$ whenever $0 \leqslant X_n \leqslant X$ so that $(X_n - X)^- \leqslant X$. Verification in the case P-a.s. only require to add $\mathbf{I}_A$, where $A = \{\lim_{n\to\infty} X_n = X\}$.

4.21 Let $\mathbf{g}(\mathbb{R}^n) = \{\mathbf{y} \in \mathbb{R}^n \mid \exists\, \mathbf{x} \in \mathbb{R}^n \text{ with } \mathbf{y} = \mathbf{g}(\mathbf{x})\}$. Since $\mathbf{g}$ is a diffeomorphism we have that $\mathbf{g}(\mathbb{R}^n)$ is an open subset of $\mathbb{R}^n$ and the inverse function $\mathbf{g}^{-1}$ is well defined on this subset, continuously differentiable with non-zero Jacobian. Let $B \in \mathcal{B}^n$ and $\tilde{B} = \mathbf{g}^{-1}(B)$. We have:

$$\begin{aligned}\mathsf{P}(\mathbf{X} \in \tilde{B}) &= \int_{\tilde{B}} f_{\mathbf{X}}(\mathbf{x})\mathrm{d}\mathbf{x} = \int_{\mathbf{g}^{-1}(B)} f_{\mathbf{X}}(\mathbf{x})\mathrm{d}\mathbf{x} \\ &= \int_B f_{\mathbf{X}}(\mathbf{g}^{-1}(\mathbf{y}))|\det \mathbf{J}_{\mathbf{g}^{-1}}(\mathbf{y})|\mathrm{d}\mathbf{y},\end{aligned}$$

where the third equality is due to Theorem 4.9 with $\mathbf{g}^{-1}$. Furthermore, events $\{Y \in B\}$ and $\{X \in \tilde{B}\}$ coincide, then their probabilities are equal,

$$\mathsf{P}(X \in \tilde{B}) = \int_B f_{\mathbf{Y}}(\mathbf{y})\mathrm{d}\mathbf{y}.$$

Since B is any Borel set in $\mathbb{R}^n$ we get formula (4.28).

4.22 We have:

$$\mathbf{g}(x_1, x_2) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix} \in \mathbb{R}^2.$$

Here the function $\mathbf{g}$ is a diffeomorphism and the inverse function

$$\mathbf{g}^{-1}(u, v) = \begin{bmatrix} \frac{u+v}{2} \\ \frac{u-v}{2} \end{bmatrix}$$

is well defined, continuously differentiable with constant Jacobian matrix

$$\mathbf{J}_{\mathbf{g}^{-1}}(u, v) = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2}. \end{bmatrix}$$

Its Jacobian does not depends on variables u, v :

$$\det \mathbf{J}_{\mathbf{g}^{-1}} = \tfrac{1}{2}(-\tfrac{1}{2}) - \tfrac{1}{2}\tfrac{1}{2} = -\tfrac{1}{2}.$$

Note that the new random vector $\mathbf{Y} = (U, V)$ has $U = X_1 + X_2$ and $V = X_1 - X_2$. By formula (4.28) its joint density is

$$\begin{aligned} f_{\mathbf{Y}}(u, v) &= f_{\mathbf{X}}\left(\frac{u+v}{2}, \frac{u-v}{2}\right) |\det \mathbf{J}_{\mathbf{g}^{-1}}| \\ &= f_{X_1}(\tfrac{u+v}{2}) f_{X_2}(\tfrac{u-v}{2}) \, \tfrac{1}{2} \\ &= \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(\frac{u+v}{2})^2} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(\frac{u-v}{2})^2} \, \frac{1}{2} \\ &= \frac{1}{\sqrt{4\pi}} \mathrm{e}^{-\frac{u^2}{4}} \frac{1}{\sqrt{4\pi}} \mathrm{e}^{-\frac{v^2}{4}}, \end{aligned}$$

for $u, v \in \mathbb{R}$, where f_{X_i} denotes the density of X_i, each $i = 1, 2$. By visual inspection, we recognize the product of two Gaussian densities referring to $U, V \sim \mathrm{N}(0, 2)$. Thus $\mathbf{Y}$ is a Gaussian vector with independent Gaussian component.

4.23 Regardless the expectation may be infinite, we use an extension of formula (4.1) for discrete rvs. Therefore we have:

$$\mathsf{E}(X) = \sum_{i=1}^{\infty} i \mathsf{P}(X = i) = \sum_{i=1}^{\infty} \mathsf{P}(X \geqslant i).$$

The second equality follows from the fact that:

- for $i = 1$ we have $\mathsf{P}(X \geqslant 1) = \mathsf{P}(X = 1) + \mathsf{P}(X > 1)$, and $\mathsf{P}(X > 1) = \mathsf{P}(X = 2) + \mathsf{P}(X = 3) + \mathsf{P}(X = 4) + \cdots$;
- for $i = 2$ we have $\mathsf{P}(X \geqslant 2) = \mathsf{P}(X = 2) + \mathsf{P}(X > 2)$, and $\mathsf{P}(X > 2) = \mathsf{P}(X = 3) + \mathsf{P}(X = 4) + \cdots$;
- $\vdots$

Collecting the terms we found $i = 1$ term $\mathsf{P}(X = 1)$ and $i = 2$ terms $\mathsf{P}(X = 2)$ and $i = 3$ terms $\mathsf{P}(X = 3)$, etc. Summing up we obtain $\sum_{i=1}^{\infty} i \, \mathsf{P}(X = i) = \sum_{i=1}^{\infty} \mathsf{P}(X \geqslant i)$.

Remark 4 The integer valued rv $X \geqslant 0$ is discrete with image $X(\Omega) = \mathbb{N}$. By definition, for $x \in [i - 1, i)$ and $i = 2, 3, \ldots$ we have

$$\mathsf{P}(X \geqslant i) = \mathsf{P}(X > x) = 1 - F_X(x).$$

In fact, if $x = i - 1$ then $\mathsf{P}(X > x) = \mathsf{P}(X > i - 1) = \mathsf{P}(X \geqslant i)$, and the same is true for $i - 1 < x < i$ since X is integer valued and the probability between two points $i - 1, i \in C$ in its support (where the CDF increases) is zero. Now we integrate the survival function $1 - F_X(x)$ over $(0, \infty)$ by dividing it into subintervals

$[i-1, i)$, i.e. we compute the Riemann integrals $\int_{i-1}^{i}[1-F_X(x)]\mathrm{d}x$ and then sum up. Eventually:

$$\int_0^\infty [1-F_X(x)]\mathrm{d}x = \sum_{i=1}^{\infty}\int_{i-1}^{i}[1-F_X(x)]\mathrm{d}x = \sum_{i=1}^{\infty}\mathsf{P}(X \geqslant i).$$

We used $\int_{i-1}^{i}[1-F_X(x)]\mathrm{d}x = \int_{i-1}^{i}\mathsf{P}(X > x)\mathrm{d}x = \int_{i-1}^{i}\mathsf{P}(X \geqslant i)\mathrm{d}x = \mathsf{P}(X \geqslant i)$, for $x \in [i-1, i)$.

4.24 First we recall that $\mathsf{E}(X) = \int_{-\infty}^{\infty} x\,\mathrm{d}F_X(x)$. Then, we split the Stieltjes integral as

$$\int_{-\infty}^{0} x\,\mathrm{d}F_X(x) + \int_0^\infty x\,\mathrm{d}F_X(x), \quad \text{where } x \text{ is continuous at zero.}$$

Next, we compute the first integral via integration by parts:

$$\int_{-\infty}^{0} x\,\mathrm{d}F_X(x) = 0\,F_X(0) - \lim_{x\to-\infty}(x\,F_X(x)) - \int_{-\infty}^{0} F_X(x)\mathrm{d}x = -\int_{-\infty}^{0} F_X(x)\mathrm{d}x,$$

On the other hand, $\int_0^\infty x\,\mathrm{d}F_X(x) = \int_0^\infty [1-F_X(x)]\mathrm{d}x$ by formula (4.5). Since (by the finiteness of the expectation) we also have

$$0 \leqslant \lim_{x\to-\infty}|x|F_X(x) = \lim_{x\to-\infty}|x|\int_{-\infty}^{x}\mathrm{d}F_X(u) \leqslant \lim_{x\to-\infty}\int_{-\infty}^{x}|u|\mathrm{d}F_X(u) = 0,$$

the above limit is zero and we conclude

$$\mathsf{E}(X) = \int_0^\infty [1-F_X(x)]\mathrm{d}x - \int_{-\infty}^{0} F_X(x)\mathrm{d}x. \tag{4}$$

4.25 First, by the results of the previous exercises let $\mathbf{I}_B(\omega, x)$ with $B := \big\{(\omega, x)\,|\,0 \leqslant x - X(\omega)\big\}$. Thus, the rv $(x-X)^+$ may be written as

$$\int_{-\infty}^{x}\mathbf{I}_B(\omega, u)\mathrm{d}u = \int_0^{x-X(\omega)}\mathrm{d}u = x - X(\omega), \quad \text{for those } \omega \text{ s.t. } 0 \leqslant x - X(\omega),$$

and this equals $\max\{0, x - X(\omega)\}$. Next, we have the chain of equivalence:

$$\begin{aligned}
\mathsf{E}\big((x-X)^+\big) &= \mathsf{E}\left(\int_{-\infty}^{x} \mathbf{I}_{\{X\leqslant u\}}\mathrm{d}u\right) = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{x}\mathbf{I}_{\{X\leqslant u\}}\mathrm{d}u\right]\mathrm{d}F_X(u) \\
&\overset{\text{Fubini's Th}}{=} \int_{-\infty}^{x}\left[\int_{-\infty}^{\infty}\mathbf{I}_{\{X\leqslant u\}}\mathrm{d}F_X(u)\right]\mathrm{d}u = \int_{-\infty}^{x}\left[\int_{-\infty}^{u}\mathrm{d}F_X(u)\right]\mathrm{d}u \\
&= \int_{-\infty}^{x}[F_X(u) - F_X(-\infty)]\,\mathrm{d}u = \int_{-\infty}^{x} F_X(u)\mathrm{d}u.
\end{aligned}$$

4.26 We can apply the results of the previous exercises and compute

$$\begin{aligned}
\mathsf{E}\big((X-s)^+\big) &= \int_{-\infty}^{\infty}\left[\int_{s}^{\infty}\mathbf{I}_{\{X\geqslant x\}}\mathrm{d}x\right]\mathrm{d}F_X(x) \\
&= \int_{s}^{\infty}\left[\int_{x}^{\infty}\mathrm{d}F_X(u)\right]\mathrm{d}x = \int_{s}^{\infty}[F_X(\infty) - F_X(x)]\mathrm{d}x.
\end{aligned}$$

Exercises of Chap. 5

5.1 Let $C = \{0, 1, 2, \ldots, n\}$ be the support of $X \sim \mathrm{Bin}(n, p)$ and let $x_i = i$. Defining $Y = n - X$ via linear transformation, we have:

$$\mathsf{P}(Y = i) = \mathsf{P}(X = n - i) = \binom{n}{n-i}p^{n-i}q^i = \binom{n}{i}q^{n-i}p^i,$$

where the last line follows by the symmetry of the binomial coefficient.

5.2 We work out the mass function of $X + Y$ by conditioning on[9] X and then using the law of total probability (6.11), see Theorem 6.1 in Sect. 6.1 we have:

$$\begin{aligned}
\mathsf{P}(X+Y=i) &= \sum_{j=0}^{i}\mathsf{P}(X+Y=i\mid X=j)\mathsf{P}(X=j) \\
&\overset{\text{Indep. and } X=j}{=} \sum_{j=0}^{i}\mathsf{P}(X+Y=i-j)\mathsf{P}(X=j) \\
&= \sum_{j=0}^{i}\binom{n_2}{i-j}p^{i-j}q^{n_2-i+j}\binom{n_1}{j}p^j q^{n_1-j}
\end{aligned}$$

[9] The same is true by conditioning on Y.

$$= \quad p^i q^{n_1+n_2-i} \sum_{j=0}^{i} \binom{n_2}{i-j}\binom{n_1}{j}$$

$$\overset{\text{Vandermonde's identity}}{=} \quad p^i q^{n_1+n_2-i} \binom{n_1+n_2}{i}.$$

5.3 It is a simply matter of direct computation using a little combinatorial identities and manipulation of binomial coefficients:

$$\begin{aligned}
\mathsf{E}(X) = \sum_{i=1}^{n} i \binom{n}{i} p^i (1-p)^{n-i} &= \sum_{i=0}^{n} i \, \frac{n!}{i!(n-i)!} p^i (1-p)^{n-i} \\
&= np \sum_{i=1}^{n} \frac{(n-1)!}{(i-1)!(n-i)!} p^{i-1} (1-p)^{n-i} \\
&= np \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-1-k} \\
&= np(p+(1-p))^{n-1} = np.
\end{aligned}$$

5.4 First, we write the second moment of $X \sim \mathrm{Po}(\lambda)$:

$$\mathsf{E}(X^2) = \sum_{i=0}^{\infty} i^2 \mathsf{P}(X=i) = \mathrm{e}^{-\lambda} \sum_{i=0}^{\infty} i^2 \frac{\lambda^i}{i!}. \tag{5}$$

Next, we need to differentiate term by term the series $\sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$ with respect to λ getting

$$\begin{aligned}
\sum_{i=1}^{\infty} i \, \frac{\lambda^{i-1}}{i!} &= \frac{\mathrm{d}}{\mathrm{d}\lambda} \mathrm{e}^{\lambda} = \mathrm{e}^{\lambda} \\
&\Longleftrightarrow \\
\sum_{i=1}^{\infty} i \, \frac{\lambda^{i}}{i!} &= \lambda \mathrm{e}^{\lambda}.
\end{aligned}$$

Differentiating the left-hand side of the expression after the equivalence we have:

$$\begin{aligned}
\sum_{i=1}^{\infty} i^2 \frac{\lambda^{i-1}}{i!} &= \frac{\mathrm{d}}{\mathrm{d}\lambda} \lambda \mathrm{e}^{\lambda} = \mathrm{e}^{\lambda} + \lambda \mathrm{e}^{\lambda} = \mathrm{e}^{\lambda}(1+\lambda) \\
&\Longleftrightarrow \\
\sum_{i=1}^{\infty} i^2 \frac{\lambda^{i}}{i!} &= \lambda \mathrm{e}^{\lambda}(1+\lambda).
\end{aligned}$$

Using the last line of the latter equivalence we can rewrite

$$\mathsf{E}(X^2) = \mathrm{e}^{-\lambda}\lambda\mathrm{e}^{\lambda}(1+\lambda) = \lambda(1+\lambda),$$

and eventually we get

$$\mathsf{V}(X) = \mathsf{E}(X^2) - (\mathsf{E}(X))^2 = \lambda(1+\lambda) - \lambda^2 = \lambda,$$

and we are done.

5.5 As we did for the sum of two independent binomial rvs, we condition on X and use the law of total probability (6.11), see Chap. 6, for $x_i = i \in C = \{0, 1, 2, \ldots\}$:

$$\begin{aligned}
\mathsf{P}(X+Y=i) &= \sum_{j=0}^{i} \mathsf{P}(X+Y=i \mid X=j)\mathsf{P}(X=j) \\
&\overset{\text{Indep. and } X=j}{=} \sum_{j=0}^{i} \mathsf{P}(X+Y=i-j)\mathsf{P}(X=j) \\
&= \sum_{j=0}^{i} \frac{\mathrm{e}^{-\lambda_2}\lambda_2^{i-j}}{(i-j)!}\frac{\mathrm{e}^{-\lambda_1}\lambda_1^{j}}{j!} \\
&= \frac{\mathrm{e}^{-(\lambda_1+\lambda_2)}}{i!}\sum_{j=0}^{i}\binom{i}{j}\lambda_1^{j}\lambda_2^{i-j} \\
&\overset{\text{Bin. Th.}}{=} \frac{\mathrm{e}^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^{i}}{i!}.
\end{aligned}$$

5.6 We mainly apply the Bayes' formula (6.12), see Chap. 6, and independence to write down the probability of X conditioned on $X+Y=n$:

$$\begin{aligned}
\mathsf{P}(X=i \mid X+Y=n) &\overset{\text{Bayes' form.}}{=} \frac{\mathsf{P}(X+Y=n \mid X=i)\mathsf{P}(X=i)}{\mathsf{P}(X+Y=n)} \\
&\overset{\text{Indep. and } X=i}{=} \frac{\mathsf{P}(Y=n-i)\mathsf{P}(X=i)}{\mathsf{P}(X+Y=n)}.
\end{aligned}$$

The expression in the last line can be rewritten using the masses of X, Y and $X+Y$,

$$\mathsf{P}(X=i \mid X+Y=n) = \frac{\left(\frac{\mathrm{e}^{-\lambda_2}\lambda_2^{n-i}}{(n-i)!}\right)\left(\frac{\mathrm{e}^{-\lambda_1}\lambda_1^{i}}{i!}\right)}{\frac{\mathrm{e}^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^{n}}{n!}}$$

$$= \binom{n}{i} \frac{\lambda_1^i \lambda_2^{n-i}}{(\lambda_1+\lambda_2)^n}$$
$$= \binom{n}{i} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^i \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-i},$$

which yields the required binomial distribution for $i = 0, 1, 2, \ldots, n$.

5.7 Writing down the MGFs of the binomial X_n and the Poisson X and passing to the limit we get

$$\lim_{n\to\infty} M_{X_n}(u) = \lim_{n\to\infty} \mathsf{E}(\mathrm{e}^{uX_n}) = \lim_{n\to\infty} (p_n \mathrm{e}^u + (1-p_n))^n$$
$$= \lim_{n\to\infty} \left(1 + \frac{\lambda(\mathrm{e}^u - 1)^n}{n}\right)$$
$$= \mathrm{e}^{\lambda(\mathrm{e}^u-1)} = \mathsf{E}(\mathrm{e}^{uX}) = M_X(u).$$

5.8 Let $u \in (c, d)$ and write

$$\mathsf{P}(U \leqslant u \,|\, U \in (c,d)) = \frac{\mathsf{P}(U \leqslant u, c < U < d)}{\mathsf{P}(c < U < d)}$$
$$= \frac{\mathsf{P}(c < U \leqslant u)}{\mathsf{P}(c < U < d)} = \frac{u-c}{d-c}.$$

Then, the CDF of U conditional on $U \in (c, d)$ is 0 for $u \leqslant c$ and 1 for $u \geqslant d$. This entails the desired distribution $U \sim \mathrm{U}(c, d)$.

5.9 First, use the change of variable $z = \frac{x-\mu}{\sqrt{2}\sigma}$:

$$\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = \tfrac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\mathrm{d}x$$
$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \mathrm{e}^{-z^2}\mathrm{d}z,$$

where $\mathrm{d}x = \sqrt{2}\sigma\,\mathrm{d}z$ and $\exp(\bullet)$ is the exponential function. Now, let $I = \int_{-\infty}^{\infty} \mathrm{e}^{-z^2}\mathrm{d}z$. Since the volume under the surface $w = \mathrm{e}^{-(z^2+y^2)}$ can be given by the product of two-dimensional areas I, by the Fubini's theorem we may write it as an iterated integral:

$$I^2 = \left(\int_{-\infty}^{\infty} \mathrm{e}^{-z^2}\mathrm{d}z\right)\left(\int_{-\infty}^{\infty} \mathrm{e}^{-y^2}\mathrm{d}y\right)$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-(z^2+y^2)}\mathrm{d}z\mathrm{d}y.$$

Now we can use the bijective transformation $\mathbf{T} : \mathbb{R}^2 \to \mathbb{R}^2$, where $(z, y) = \mathbf{T}(r, \theta)$ is defined to be the vector function which maps the polar coordinates $(r, \theta) \in \mathbb{R}^2$ to the rectilinear coordinates $(z, y) \in \mathbb{R}^2$ by the functions $f, g : \mathbb{R}^2 \to \mathbb{R}$ as follows:

$$z = f(r, \theta) = r \cos\theta,$$
$$y = g(r, \theta) = r \sin\theta,$$

where $r > 0,\ \theta \in [0, 2\pi]$. At this point we need an appeal to the change of variable formula for multiple integrals, see Sect. 4.9, We have:

$$\mathrm{d}z\mathrm{d}y = \left|\det \begin{pmatrix} f_r & g_r \\ f_\theta & g_\theta \end{pmatrix}\right| \mathrm{d}r\mathrm{d}\theta,$$

where $f_r = \cos\theta$, $f_\theta = r\sin\theta$ and $g_r = \sin\theta$, $g_\theta = -r\cos\theta$ are partial derivatives. Recall that $\sin^2 + \cos^2\theta = 1$. Then passing to polar coordinates we get the new integral $2\int_0^c \mathrm{e}^{-r^2} r\,\mathrm{d}r$. Write $-\int_0^c -2\mathrm{e}^{-r^2} r\,\mathrm{d}r$ and observe that $\frac{\mathrm{d}}{\mathrm{d}r}\left(\mathrm{e}^{-r^2}\right) = -2r\mathrm{e}^{-r^2}\,\mathrm{d}r$. Hence, the last integral is $-\left(\mathrm{e}^{-a^2} - \mathrm{e}^0\right)$ and as $a \to \infty$ we obtain: $2\int_0^\infty \mathrm{e}^{-r^2} r\,\mathrm{d}r = 1$. Eventually, we rewrite

$$I^2 = \int_0^{2\pi}\int_0^\infty \mathrm{e}^{-r^2} r\,\mathrm{d}r\mathrm{d}\theta$$
$$= 2\pi \int_0^\infty \mathrm{e}^{-r^2} r\,\mathrm{d}r = \pi,$$

and find $I = \sqrt{\pi}$, that is $\frac{1}{\sqrt{\pi}} I = 1$.

5.10 We compute the MGF as:

$$M_{a+bX} = \mathrm{e}^{ua}\mathsf{E}(\mathrm{e}^{u\,bX}) = \mathrm{e}^{ua}\mathrm{e}^{u\,b\mu + \frac{u^2 b^2 \sigma^2}{2}}$$
$$= \mathrm{e}^{(a + b\,\mu) + \frac{u^2(a^2\sigma^2)}{2}},$$

where the second equality follows from the expression of $M_X(u)$. A linear transformation of a Gaussian rv provides the MGF of a new Gaussian variable $Y \sim \mathrm{N}(a + b\mu, a^2\sigma^2)$ where the localization parameter is a linear combination with the same coefficients and the scale parameter is itself scaled by a factor b^2.

5.11 Let $f_{\mathbf{X}}(x_1, x_2)$ the bi-variate Gaussian density. Then, the marginal density of X_1 is:

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, x_2)\mathrm{d}x_2$$
$$= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2(1-\rho^2)}(u^2 + v^2 - 2\rho uv)\right)\mathrm{d}v,$$

where we set $u = \frac{x_1-\mu_1}{\sigma_1}$ and $v = \frac{x_2-\mu_2}{\sigma_2}$, so that $\mathrm{d}x_2 = \sigma_2\mathrm{d}v$. Moreover, $u^2 + v^2 - 2\rho uv$ equals $(v-\rho u)^2 + u^2(1-\rho^2)$ whence the above becomes

$$f_{X_1}(x_1) = \frac{1}{\sigma_1\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right)$$
$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}(v-\rho u)^2\mathrm{d}v\right).$$

We recognize (viz. the integrand) the density $\mathrm{N}(\rho u, 1-\rho^2)$. Bringing back $u = \frac{x_1-\mu_1}{\sigma_1}$ entails $f_{X_1}(x_1)$ as the density of $\mathrm{N}(\mu_1, \sigma_1^2)$. The same reasoning can be applied to X_2.

5.12 Let $Y = \mathrm{e}^X$, then we firstly have

$$\mathsf{E}(Y^n) = \mathsf{E}(\mathrm{e}^{nX}) =: M_X(n) = \exp\left(n\mu + \frac{n^2\sigma^2}{2}\right), \quad n = 1, 2, \ldots,$$

where the first two equalities hold for any rv X, and the third is for $X \sim \mathrm{N}(\mu, \sigma^2)$ and its MGF, i.e., the nth moment of the log-normal Y is the MGF of the Gaussian X evaluated at $u = n$. For $n = 1$ Eq. (5.17) follows. Now, for $n = 2$ we have $\mathsf{E}(Y^2) = \exp\left(2\mu + 2\sigma^2\right)$ and then

$$\mathsf{V}(Y) = \mathsf{E}(Y^2) - (\mathsf{E}(Y))^2 = \exp\left(2\mu + 2\sigma^2\right) - \left(\exp\left(n\mu + n^2\sigma^2/2\right)\right)^2,$$

which after little algebra yields (5.18).

5.13 We compute the expectation over the interval $(0, \infty)$:

$$\mathsf{E}(X) = \int_0^{\infty} x\lambda\mathrm{e}^{-\lambda x}\mathrm{d}x.$$

Integrating by parts

$$\int_a^b f(x)g'(x)\mathrm{d}x = [f(x)g(x)]_a^b - \int_a^b g(x)f'(x)\mathrm{d}x$$

with $f(x) = x$, $g'(x) = \lambda \mathrm{e}^{-\lambda x}$ we get:

$$\begin{aligned}
\mathsf{E}(X) &= \left[-x\mathrm{e}^{-\lambda x}\right]_0^\infty + \int_0^\infty \mathrm{e}^{-\lambda x}\mathrm{d}x \\
&= \lim_{x\to\infty}\left(-x\mathrm{e}^{-\lambda x}\right) - 0 - \frac{1}{\lambda}\int_0^\infty -\lambda \mathrm{e}^{-\lambda x}\mathrm{d}x \\
&= -\frac{\lim_{x\to\infty} x}{\mathrm{e}^{\lambda \lim_{x\to\infty} x}} - \frac{1}{\lambda}\left[\mathrm{e}^{-\lambda x}\right]_0^\infty \\
&= -\frac{1}{\lambda \mathrm{e}^{\lambda \lim_{x\to\infty} x}} + \frac{1}{\lambda} = \frac{1}{\lambda},
\end{aligned}$$

where in the fourth equality we use l'Hôpital's rule. Using the same type of integration by parts and assuming $Y \sim \mathrm{Exp}(1)$ we also get

$$\mathsf{E}(Y^2) = \int_0^\infty x^2 \mathrm{e}^{-x}\mathrm{d}x = 2,$$

so $\mathsf{V}(Y) = \mathsf{E}(Y^2) - (\mathsf{E}(Y))^2 = 1$, using the result above with $\lambda = 1$. Therefore, we have

$$\mathsf{V}\left(\frac{1}{\lambda}X\right) = \frac{1}{\lambda^2}\mathsf{E}(Y) = \frac{1}{\lambda^2},$$

and the result follows.

5.14 First, we note that $\mathsf{E}((a+bX-a-b\mu)^p) = b^p\mathsf{E}((X-\mu)^p)$, for every a, $p \in \mathbb{R}$ and $b \neq 0$. Using this and the standard normal $Z = \frac{X-\mu}{\sigma}$ we have[10] $\mathsf{E}((X)^p) = \frac{1}{\sigma^p}\mathsf{E}(X^p)$, so we can instead find the pth moment of Z and then multiply by σ^2 to get the pth moment of the original Gaussian. Next, we use the MGF of Z through its Taylor series expansion:

$$M_Z(u) = \mathrm{e}^{u^2/2} = \sum_{p=0}^\infty \frac{(u^2/2)^p}{p!} = \sum_{p=0}^\infty \frac{(2p)!}{2^p\, p!}\frac{u^{2p}}{(2p)!}.$$

[10] Sine the standard normal has expectation zero, its pth central moment is equal to the pth moment.

Since $\mathsf{E}(Z^p)$ equals the coefficient of the term involving u^p in the Taylor series expansion of $M_Z(u)$ we have $\mathsf{E}(Z^{2p}) = \frac{(2p)!}{2^p p!}$, i.e. the even moments of Z. Hence, we eventually have

$$\mathsf{E}((X-\mu)^p) = \frac{(2p)!}{2^p p!}\sigma^{2p},$$

which is zero for odd $p = 1, 3, 5, \ldots$ by the symmetry of Z.

5.15 The density $g(x)$ is zero for $x \leqslant 0$, then integrating with the substitution $\lambda x = z$ we get:

$$\begin{aligned} 1 &= k\int_0^\infty x^{\alpha-1}\mathrm{e}^{-x\lambda}\mathrm{d}x \\ &= \frac{k}{\lambda^\alpha}\int_0^\infty z^{\alpha-1}\mathrm{e}^{-z}\mathrm{d}z = k\frac{\Gamma(\alpha)}{\lambda^\alpha} \\ \iff k &= \frac{\lambda^\alpha}{\Gamma(\alpha)}. \end{aligned}$$

5.16 First, from $x = h(y) = \lambda\, y$ we compute $h'(y) = \lambda$ then we write

$$\begin{aligned} f_Y(y) = f_X(x)\lambda &= \lambda\frac{1}{\Gamma(\alpha)}(\lambda\, y)^{\alpha-1}\mathrm{e}^{-\lambda\, y}y^{-1} \\ &= \lambda\frac{1}{\Gamma(\alpha)}(\lambda\, y)^{\alpha}(\lambda\, y)^{-1}\mathrm{e}^{-\lambda\, y}y^{-1}, \end{aligned}$$

which is the desired density for $y > 0$.

5.17 We first compute the expectation:

$$\begin{aligned} \mathsf{E}(X) &= \int_0^\infty x\frac{1}{\Gamma(\alpha)}x^{\alpha-1}\mathrm{e}^{-x}\mathrm{d}x \\ &= \frac{1}{\Gamma(\alpha)}\int_0^\infty \underbrace{x^\alpha \mathrm{e}^{-x}}_{=\Gamma(\alpha+1)}\,\mathrm{d}x \\ &= \frac{\alpha\Gamma(\alpha)}{\Gamma(\alpha)} = \alpha. \end{aligned}$$

Next, we compute the second moment:

$$\begin{aligned}\mathsf{E}(X^2) &= \int_0^\infty x^2 \frac{1}{\Gamma(\alpha)} x^{\alpha-1}\mathrm{e}^{-x}\mathrm{d}x \\ &= \frac{1}{\Gamma(\alpha)} \int_0^\infty \underbrace{x^{\alpha+1}\mathrm{e}^{-x}}_{=\Gamma(\alpha+2)}\mathrm{d}x \\ &= \frac{(\alpha+1)\alpha\Gamma(\alpha)}{\Gamma(\alpha)} = (\alpha+1)\alpha.\end{aligned}$$

As a result, $\mathsf{V}(X) = \mathsf{E}(X^2) - (\mathsf{E}(X))^2 = (\alpha+1)\alpha - \alpha^2 = \alpha$.

5.18 We first set[11] $A = \{x \in \mathbb{R} \mid g(v) \leqslant y\} = \{v^2 \leqslant y\} = [-\sqrt{y}, \sqrt{y}]$ so that

$$\mathsf{P}(W \leqslant y) = \mathsf{P}(V \in A) = F_V(\sqrt{y}) - F_V(-\sqrt{y}),$$

where we used the fact that V is a continuous rv. Differentiating we get:

$$F'_W(y) = f_W(y) = \frac{1}{2\sqrt{y}} \left(f_V(\sqrt{y}) - f_V(-\sqrt{y})\right).$$

Then, using $V \sim \mathrm{N}(0, \sigma^2)$ we get the expression for the transformed density:

$$\begin{aligned}f_W(y) &= \frac{1}{2\sqrt{y}} \frac{1}{\sqrt{2\pi\sigma^2}} \left(-\mathrm{e}^{-y/2\sigma^2} + \mathrm{e}^{-y/2\sigma^2}\right) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} y^{-1+1/2}\mathrm{e}^{-y/2\sigma^2}, \quad y > 0.\end{aligned}$$

By comparing this density with that of a gamma rv we are done.

Exercises of Chap. 6

6.1 First, we check that this function is increasing in the variable x. Indeed, for $x_1 < x_2$ it holds $F_X(x_1 \mid A) \leqslant F_X(x_2 \mid A)$ because in formula (6.2) we have $\{X \leqslant x_1\} \subset \{X \leqslant x_2\}$ for fixed $A \in \mathcal{F}$. Second, right-continuity writes:

$$\begin{aligned}\lim_{h \downarrow 0} F_X(x+h \mid A) &= \lim_{n\to\infty} \frac{\mathsf{P}\big(\{X \leqslant x + \frac{1}{n}\} \cap A\big)}{\mathsf{P}(A)} \\ &\overset{\text{cont. of } \mathsf{P}}{=} \frac{\mathsf{P}\big(\lim_{n\to\infty}\{X \leqslant x + \frac{1}{n}\} \cap A\big)}{\mathsf{P}(A)} = F_X(x \mid A).\end{aligned}$$

[11] We cannot use formula (4.11) in Sect. 4.5, since $W = V^2$ is not a monotone transform.

The verification that $\lim_{x\to-\infty} F_X(x \mid A) = 0$ and $\lim_{x\to\infty} F_X(x \mid A) = 1$ is done similarly. Eventually, $\mathsf{P}(a < X \leqslant b \mid A) = F_X(b \mid A) - F_X(a \mid A)$ because $\{X \leqslant b\} \cap A$ equals the disjoint union of $\{X \leqslant a\} \cap A$ and $\{a < X \leqslant b\} \cap A$.

6.2 It is easily to see $f_{X\mid Y}(x) \geqslant 0$ for every $x \in \mathbb{R}$ and any $y \in \mathbb{R}$ s.t. the marginal $f_Y(y) \neq 0$. Moreover:

$$\begin{aligned}\int_{-\infty}^{\infty} f_{X\mid Y}(x)\mathrm{d}x &= \int_{-\infty}^{\infty} \frac{f_{\mathbf{X}}(x,y)}{f_Y(y)}\mathrm{d}x \\ &= \tfrac{1}{f_Y(y)} \underbrace{\int_{-\infty}^{\infty} f_{\mathbf{X}}(x,y)\mathrm{d}x}_{=f_Y(y)} = 1.\end{aligned}$$

6.3 By formula (6.19) we have

$$\mathsf{E}(X \mid A) = \mathsf{E}(\mathbf{I}_B \mid A) = \frac{\mathsf{E}(\mathbf{I}_B\mathbf{I}_A)}{\mathsf{P}(A)} = \frac{\mathsf{E}(\mathbf{I}_{A\cap B})}{\mathsf{P}(A)} = \frac{\mathsf{P}(A\cap B)}{\mathsf{P}(A)},$$

which entails conditional probability $\mathsf{P}(B \mid A)$.

6.4 First, we observe that $\mathsf{P}(X = x_i) = \mathsf{P}(X = x_i \mid A)\mathsf{P}(A) + \mathsf{P}(X = x_i \mid A^c)\mathsf{P}(A^c)$, see the law of total probability. Thus,

$$\begin{aligned}\mathsf{E}(X) = \sum_{i=1}^{\infty} x_i \mathsf{P}(X = x_i) &= \sum_{i=1}^{\infty} x_i \mathsf{P}(X = x_i \mid A)\mathsf{P}(A) \\ &\quad + \sum_{i=1}^{\infty} x_i \mathsf{P}(X = x_i \mid A^c)\mathsf{P}(A^c) \\ &= \mathsf{E}(X \mid A)\mathsf{P}(A) + \mathsf{E}(X \mid A^c)\mathsf{P}(A^c),\end{aligned}$$

since $\mathsf{P}(X = x_i \mid A)\mathsf{P}(A) = p(x_i \mid A)$ and similarly for A^c.

6.5 Let X has finite expectation so that also $X\mathbf{I}_A \in L^1$. Assume $\mathsf{P}(A) > 0$. Thus choose $X = \mathbf{I}_D$ for any event $D \in \mathcal{F}$. We have:

$$\begin{aligned}\mathsf{P}(D \mid A) &= \tfrac{\mathsf{P}(D\cap A)}{\mathsf{P}(A)} = \tfrac{\mathsf{E}(\mathbf{I}_{D\cap A})}{\mathsf{P}(A)} \\ &= \tfrac{\mathsf{E}(\mathbf{I}_D\mathbf{I}_A)}{\mathsf{P}(A)} \\ &= \tfrac{\mathsf{E}(X\mathbf{I}_A)}{\mathsf{P}(A)} = \mathsf{E}(\mathbf{I}_D \mid A)\end{aligned}$$

and formula (6.19) is proved for indicator rvs. Now, take a discrete rv $X = \sum_{i=1}^{n} x_i \mathbf{I}_{D_i}$ with pairwise disjoint events $D_i \in \mathcal{F}$ partitioning Ω. Hence:

$$\begin{aligned}\mathsf{E}(X \mid A) &= \sum_{i=1}^{n} x_i \mathsf{P}(D_i \mid A) = \sum_{i=1}^{n} \frac{x_i \mathsf{E}(\mathbf{I}_{D_i}\mathbf{I}_A)}{\mathsf{P}(A)} \\ &= \frac{\mathsf{E}\Big((\sum_{i=1}^{n} x_i \mathbf{I}_{D_i})\mathbf{I}_A\Big)}{\mathsf{P}(A)} = \frac{\mathsf{E}(X\mathbf{I}_A)}{\mathsf{P}(A)}.\end{aligned}$$

Here we used the conditional distribution of X given by masses $\mathsf{P}(D_i \mid A) = \frac{\mathsf{P}(D_i \cap A)}{\mathsf{P}(A)}$, together with the previous definition $\mathsf{E}(\mathbf{I}_{D_i} \mid A) = \mathsf{P}(D_i \mid A)$. Observe that we can take $D_i = \{X = x_i\}$ for $x_i \in C = \{x_1, \ldots, x_n\}$. Eventually, let $X = X^+ - X^-$ and take two sequences $(X_n^+)_{n\in\mathbb{N}}$,$(X_n^-)_{n\in\mathbb{N}}$ converging pointwise to X^+ and X^- respectively. Then, we have $X_n^+\mathbf{I}_A \to X^+\mathbf{I}_A$ and $X_n^-\mathbf{I}_A \to X^-\mathbf{I}_A$, pointwise and

$$\begin{aligned}\mathsf{E}(X \mid A) &= \mathsf{E}(X^+ \mid A) - \mathsf{E}(X^- \mid A) \\ &= \lim_n \mathsf{E}(X_n^+ \mid A) - \lim_n \mathsf{E}(X_n^- \mid A) \\ &= \frac{\mathsf{E}(\lim_n X_n^+\mathbf{I}_A)}{\mathsf{P}(A)} - \frac{\mathsf{E}(\lim_n X_n^-\mathbf{I}_A)}{\mathsf{P}(A)} \\ &= \frac{\mathsf{E}(X^+\mathbf{I}_A)}{\mathsf{P}(A)} - \frac{\mathsf{E}(X^-\mathbf{I}_A)}{\mathsf{P}(A)}.\end{aligned}$$

The second equality follows from the MCT.

6.6 By definition of A_u we have:

$$0 \leqslant u\mathbf{I}_{A_u} \leqslant X\mathbf{I}_{A_u} \leqslant X.$$

Hence, applying expectation to all terms in the above chain of inequalities we get

$$\mathsf{E}(u\mathbf{I}_{A_u}) = u\mathsf{P}(A_u) \leqslant \mathsf{E}(X\mathbf{I}_{A_u}) \leqslant \mathsf{E}(X),$$

and after rearranging w.r.t. u we are done.

6.7 Since Y is constant it holds $\{Y = c\} = \Omega$. Hence $\mathsf{E}(X \mid Y)$ equals $\mathsf{E}(X \mid \{Y = c\}) = \frac{\mathsf{E}(X\mathbf{I}_\Omega)}{\mathsf{P}(\Omega)}$ which is just $\mathsf{E}(X)$. Next, $\mathsf{E}(\mathbf{I}_A \mid \mathbf{I}_B) = \mathsf{E}(\mathbf{I}_A \mid \{\mathbf{I}_B = 1\})$ for scenarios $\omega \in B$. As byproduct, the last conditional expectation equals $\mathsf{E}(\mathbf{I}_A \mid B) = \frac{\mathsf{E}(\mathbf{I}_A\mathbf{I}_B)}{\mathsf{P}(B)}$, where $\mathsf{E}(\mathbf{I}_A\mathbf{I}_B) = \mathsf{E}(\mathbf{I}_{A\cap B})$, which is $\mathsf{P}(A \cap B)$ and we get the desired result. Observe that taking $\omega \in B^c$ we also get $\mathsf{E}(\mathbf{I}_A \mid \mathbf{I}_B) = \mathsf{E}(\mathbf{I}_A \mid \{\mathbf{I}_B = 0\})$ which equals $\mathsf{P}(A \mid B^c)$.

6.8 Using conditional masses (6.10), we have:

$$\mathsf{E}(X \mid Y = y_j) = \sum_{i=1}^{\infty} x_i \, p(x_i \mid y_j) = \sum_{i=1}^{\infty} x_i \frac{\mathsf{P}(X = x_i, Y = y_j)}{\mathsf{P}(Y = y_j)}.$$

Similarly, the conditional expectation of X given that a *discrete* random vector $\mathbf{Y} = (Y_1, \ldots, Y_n)$ takes on values $(c_1, \ldots, c_n) \in \mathbb{R}^n$ is the rv $\mathsf{E}(X \mid Y_1, \ldots, Y_n) = f(Y_1, \ldots, Y_n)$ which is constant over events $\{Y_1 = c_1, \ldots, Y_n = c_n\}$ with values

$$\mathsf{E}(X \mid Y_1 = c_1, \ldots, Y_n = c_n) = f(c_1, \ldots, c_n).$$

6.9 By change of variable formula, see 4.3 in Sect. 4.2, and using the distribution of Y we compute:

$$\begin{aligned}
\mathsf{E}\left(\mathsf{E}(X \mid Y)\right) &= \sum_{j=1}^{\infty} \mathsf{E}(X \mid Y = y_j)\mathsf{P}(Y = y_j) \\
&= \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} x_i \mathsf{P}(X = x_i, Y = y_j) \\
&= \sum_{i=1}^{\infty} x_i \mathsf{P}(X = x_i) = \mathsf{E}(X),
\end{aligned}$$

thus $\mathsf{E}(\mathsf{E}(X \mid Y))$ coincides with expectation of X. Observe that we tacitly assumed that $\mathbf{X} = (X, Y)$ is a discrete random vector with joint distribution given by masses.

6.10 Since $\mathbf{X} = (X, Y)$ is a continuous random vector having continuous joint density $f_{\mathbf{X}}(x, y)$, with a similar reasoning as in the case X, Y are both discrete we write

$$\mathsf{E}(X \mid Y = y) = \int_{-\infty}^{\infty} x f_{X \mid Y}(x \mid y) \mathrm{d}x,$$

as the value of $\mathsf{E}(X \mid Y)$ at y. Recall that $f_{X \mid Y}(x \mid y) = \frac{f_{\mathbf{X}}(x,y)}{f_Y(y)}$ provided that $f_Y(y) > 0$. Now, $f(Y) = \mathsf{E}(X \mid Y)$ is a continuous rv. Furthermore we have

$$\begin{aligned}
\mathsf{E}\left(\mathsf{E}(X \mid Y)\right) &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x f_{X \mid Y}(x) \mathrm{d}x \right) f_Y(y) \mathrm{d}y \\
&= \int_{-\infty}^{\infty} x f_X(x) \mathrm{d}x = \mathsf{E}(X),
\end{aligned}$$

since $f_{X \mid Y}(x) f_Y(y) = f_{\mathbf{X}}(x, y)$ and the integral of the joint density for all y is just the marginal of X.

6.11 Items 1 and 2 are easy: take $p(x_i \mid y_j) = \mathsf{P}(X = x_i)$ and $f_{X\mid Y}(x \mid y) = f_X(x)$ in the calculations showed in the solutions to Exercises 6.9 and 6.10. For item 3 we have:

$$\begin{aligned}\mathsf{E}(X \mid Y = y_j) &= \frac{\mathsf{E}(X\mathbf{I}_{\{Y=y_j\}})}{\mathsf{P}(Y = y_j)} \\ &= \frac{\mathsf{E}(X)\mathsf{E}(\mathbf{I}_{\{Y=y_j\}})}{\mathsf{P}(Y = y_j)} = \mathsf{E}(X).\end{aligned}$$

6.12 First we note that if Y is discrete, then $\sigma(Y)$ is the smallest σ-algebra containing events $\{Y = y_j\}$, thus any other event can be represented by countable or even finite unions $\cup_j\{Y = y_j\}$. E.g., $\{a < Y \leqslant b\}$ for $a, b \in \mathbb{R}$ can be given by the countable union $\cup_{i\in I}\{Y = y_i\}$ where the index set is $I = \{j \in \mathbb{N} \mid a < y_j \leqslant b\}$. It follows $\{a < Y \leqslant b\} \in \sigma(Y)$. Moreover, pairwise disjoint events form a partition of Ω, and since $\mathsf{E}(X \mid Y)$ evaluated at y_i is constant over these events its $\sigma(Y)$-measurability follows. Next, pick $A = \cup_{i\in I}\{Y = y_i\} \in \sigma(Y)$ and observe that $\mathsf{E}(X\mathbf{I}_A) = \mathsf{E}\Big(X\sum_{j\in I}\mathbf{I}_{\{Y=y_j\}}\Big)$, because the disjoint union of events $\{Y = y_j\}$ in the indicator $\mathbf{I}_A$ yields the countable sum of $\mathbf{I}_{\{Y=y_j\}}$. The last expectation equals $\sum_{j\in I}\mathsf{E}(X\mathbf{I}_{\{Y=y_j\}})$. But

$$\mathsf{E}\left(\mathsf{E}(X \mid Y)\mathbf{I}_A\right) = \sum_{j\in I}\mathsf{E}(X \mid Y = y_j)\mathsf{P}(Y = y_j) = \sum_{j\in I}\mathsf{E}(X\mathbf{I}_{\{Y=y_j\}}),$$

since by definition $\mathsf{E}(X \mid Y = y_j) = \frac{\mathsf{E}(X\mathbf{I}_{\{Y=y_j\}})}{\mathsf{P}(Y=y_j)}$. This implies that $\mathsf{E}(X\mathbf{I}_A) = \mathsf{E}\left(\mathsf{E}(X \mid Y)\mathbf{I}_A\right)$.

6.13 Consider $A = \{X \geqslant c\} \in \mathcal{A}$ for any $c > 0$. We have

$$0 \leqslant c\,\mathsf{P}(A) = c\,\mathsf{E}(\mathbf{I}_A) \leqslant \mathsf{E}(X\mathbf{I}_A) = 0,$$

where the last equality holds by assumption. Then we must have $\mathsf{P}(A) = 0$. If we take $A = \{X \leqslant -c\}$ the probability is again zero. It follows that $\mathsf{P}(-c < X < c) = 1$ for every $c > 0$. We choose c small enough, $\mathsf{P}(-\frac{1}{n} < X < \frac{1}{n}) = 1$ for $n \in \mathbb{N}$. The event $\{X = 0\}$ is the countable intersection $\cap_{n=1}^{\infty}\{-\frac{1}{n} < X < \frac{1}{n}\}$ of the decreasing sequence of events $\{-\frac{1}{n} < X < \frac{1}{n}\} \supset \{-\frac{1}{n+1} < X < \frac{1}{n+1}\}$ and continuity of $\mathsf{P}(\,\cdot\,)$ yields

$$\mathsf{P}(X = 0) = \lim_{n\to\infty}\mathsf{P}(-\tfrac{1}{n} < X < \tfrac{1}{n}) = \lim_{n\to\infty} 1 = 1,$$

which means $X \overset{\text{a.s.}}{=} 0$ as required.

6.14 By Exercise 6.13 and its solution, we simply have to show that if $X \overset{\text{a.s.}}{=} X'$, then $\mathsf{E}(X \mid Y) \overset{\text{a.s.}}{=} \mathsf{E}(X' \mid Y)$. In fact, suppose X and X' are $\sigma(Y)$-measurable

versions of the X's expectation conditioned on Y. Then, we have $X - X' \overset{\text{a.s.}}{=} 0$ and

$$\mathsf{E}((X - X')\mathbf{I}_A) = 0 \iff \mathsf{E}(X\mathbf{I}_A) = \mathsf{E}(X'\mathbf{I}_A) \quad \text{for all} \quad A \in \mathcal{A} \subset \mathcal{F}. \tag{6.6}$$

This yields $\mathsf{E}(X \mid Y) \overset{\text{a.s.}}{=} \mathsf{E}(X' \mid Y)$.

6.15 Since $Z = \mathsf{E}(X \mid Y) \equiv \mathsf{E}(X \mid \sigma(Y))$, we must exhibit a Borel measurable function $f : \mathbb{R} \to \mathbb{R}$ s.t. $Z = f(Y)$, and for every event $A \in \sigma(Y)$ the unconditional expectations of $X\mathbf{I}_A$ and $Z\mathbf{I}_A$ are equal. We choose $A = \{Y \in B\}$ which is surely in $\sigma(Y)$, then recall that $\mathbf{X} = (X, Y)$ has a joint density $f_{\mathbf{X}}(x, y)$ and

$$\mathsf{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\mathrm{d}x = \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y)\mathrm{d}y\right]\mathrm{d}x,$$

since X has univariate density given by integrating the joint density w.r.t. Y. The same applies to $\mathsf{E}(Y)$. Therefore, integrating over a subset and interchanging the order of integration we can compute:

$$\begin{aligned}\mathsf{E}(X\mathbf{I}_{\{Y\in B\}}) &= \int_B \int_{-\infty}^{\infty} x f_{\mathbf{X}}(x, y)\mathrm{d}y\mathrm{d}x \\ &= \int_B \left[\int_{-\infty}^{\infty} x f_{\mathbf{X}}(x, y)\mathrm{d}x\right]\mathrm{d}y.\end{aligned}$$

Moreover, using the change of variable formula we compute:

$$\begin{aligned}\mathsf{E}(f(Y)\mathbf{I}_{\{Y\in B\}}) &= \int_B \int_{-\infty}^{\infty} f(y) f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y \\ &= \int_B f(y) \left[\int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y)\mathrm{d}x\right]\mathrm{d}y.\end{aligned}$$

By definition of conditional expectation given a rv, for every Borel set B, both unconditional expectations must be equal which implies

$$f(y) = \frac{\int_{-\infty}^{\infty} x f_{\mathbf{X}}(x, y)\mathrm{d}x}{\int_{-\infty}^{\infty} f_{\mathbf{X}}(x, y)\mathrm{d}x}.$$

Evaluating $f(\cdot)$ at the rv Y the proof is complete.

6.16 For $A = \varnothing$ formula (6.22),

$$\tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A), \quad \text{with } \mathsf{E}_{\mathsf{P}}(X) = 1,$$

yields $\tilde{\mathsf{P}}(\varnothing) = \mathsf{E}_{\mathsf{P}}(X\mathbf{I}_{\varnothing}) = 0$. The same is true for the certain event Ω. It remains to show this holds for any other event $A \in \mathcal{F}$, and additionally to prove countable additivity of $\tilde{\mathsf{P}}$. Let $X = \sum_{j=1}^{n} x_j \mathbf{I}_{G_j}$ be a nonnegative discrete rv, and for a pairwise disjoint sequence $(A_i)_{i\in\mathbb{N}} \subset \mathcal{F}$ take $\mathbf{I}_A = \mathbf{I}_{\cup_{i=1}^{m} A_i} = \sum_{i=1}^{m} \mathbf{I}_{A_i}$. Recall that $X\mathbf{I}_A$ is also nonnegative discrete, and $X\mathbf{I}_A = \sum_{j=1}^{n} x_j \mathbf{I}_{G_j \cap A}$. Whence we have:

$$\begin{aligned}\tilde{\mathsf{P}}(A) &= \sum_j x_j \mathsf{P}(G_j \cap (\cup_i A_i)) = \sum_j x_j \sum_i \mathsf{P}(G_j \cap A_i) \\ &= \sum_i \sum_j x_j \mathsf{P}(G_j \cap A_i) = \sum_i \mathsf{E}(X\mathbf{I}_{A_i}) = \sum_i \tilde{\mathsf{P}}(A_i).\end{aligned}$$

If $X \geqslant 0$ we pick a pairwise disjoint sequence $(B_k)_{k\in\mathbb{N}} \subset \mathcal{F}$ such that $A_m = \cup_{k=1}^{m} B_k$ and $A_1 \subset A_2 \subset \cdots$ is an increasing sequence in $\mathcal{F}$ with limit $A_m \uparrow A = \cup_m A_m$. This implies that $0 \leqslant \mathbf{I}_{A_1} \leqslant \mathbf{I}_{A_2} \leqslant \cdots$ converges pointwise, $\lim_{m\to\infty} \mathbf{I}_{A_m} = \mathbf{I}_A$. Then: $\tilde{\mathsf{P}}(A) = \mathsf{E}(X\mathbf{I}_A) = \lim_{m\to\infty} \mathsf{E}(X\mathbf{I}_{A_m}) = \lim_{m\to\infty} \sum_{k=1}^{m} \mathsf{E}(X\mathbf{I}_{B_k}) = \lim_{m\to\infty} \sum_{k=1}^{m} \tilde{\mathsf{P}}(B_k) = \sum_{k=1}^{\infty} \tilde{\mathsf{P}}(B_k) = \tilde{\mathsf{P}}(A)$, where we used MCT in the second equality, the identity $A = \cup_{m=1}^{\infty} A_m = \cup_{k=1}^{\infty} B_k$ and the fact that $\mathbf{I}_{A_m} = \sum_{k=1}^{m} \mathbf{I}_{B_k}$. Note how moving from A_m to $A_{m+1} = A_m \cup B_{m+1}$ if $\omega \in B_{m+1}$ then $\mathbf{I}_{A_{m+1}} = 1$ and $\mathbf{I}_{A_m} = 0$, for otherwise both indicators are 1 or zero. On the other hand, to show that $\tilde{\mathsf{P}} \ll \mathsf{P}$ we proceed as follows. Start from a nonnegative discrete $X = \sum_{i=1}^{n} x_i \mathbf{I}_{G_i}$ and pairwise disjoint events $A_1, \ldots, A_n$ in $\mathcal{F}$, thus assume $\mathsf{P}(A) = 0$ for $A \in \mathcal{F}$. Since $\mathbf{I}_A = \mathbf{I}_{\cup_{j=1}^{m} A_j} = \sum_{j=1}^{m} \mathbf{I}_{A_j}$ then

$$X\mathbf{I}_A = \sum_{i=1}^{n} x_i \mathbf{I}_{G_i \cap A} = \sum_{i=1}^{n} x_i \mathbf{I}_{\cup_{i=1}^{m} (G_i \cap A_j)} = \sum_{i=1}^{n} x_i \sum_{j=1}^{m} \mathbf{I}_{G_i \cap A_j},$$

and the latter equals $\sum_{i=1}^{n} \sum_{j=1}^{m} x_i \mathbf{I}_{G_i \cap A_j}$. Now, we have

$$\mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A) = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i \mathsf{P}(G_i \cap A_j),$$

but since $A_j \subset A$ each j and $0 \leqslant \mathsf{P}(A_j) \leqslant \mathsf{P}(A) = 0$ we also have $0 \leqslant \mathsf{P}(G_i \cap A_j) \leqslant \mathsf{P}(A) = 0$, i.e. $\mathsf{P}(G_i \cap A_j) = 0$. Thus $\tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A) = 0$. For general $X \geqslant 0$ we write

$$\tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A) = \sup_{\{0 \leqslant Z \leqslant X,\ Z \text{ discrete}\}} \mathsf{E}_{\mathsf{P}}(Z\mathbf{I}_A),$$

but the supremum is zero because each $\mathsf{E}_{\mathsf{P}}(Z\mathbf{I}_A) = 0$.

6.17 First, assume $X \geqslant 0$ and recall that $\tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A)$ is a measure such that $\tilde{\mathsf{P}} \ll \mathsf{P}$, see Exercise 6.16. Now, we have $\mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A) = \tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(\frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}}\mathbf{I}_A)$ which is item 3 of Theorem 6.2. Taking $A = \Omega$ and assuming $\mathsf{E}_{\mathsf{P}}(X) = 1$ we have that $X\mathbf{I}_A \in L^1$, then $\frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}}$ has also finite expectation and it is a version of $\mathsf{E}_{\mathsf{P}}(X \mid \mathcal{A})$. For the general case, we apply the previous result to the positive and negative parts of X. Hence, $Z_1 = \mathsf{E}_{\mathsf{P}}(X^+ \mid \mathcal{A})$ and $Z_2 = \mathsf{E}_{\mathsf{P}}(X^- \mid \mathcal{A})$ are well defined. Now, write $Z_1 - Z_2$ and observe this is a rv with finite expectation E_{P}. Then, for every $A \in \mathcal{A}$ we have

$$\begin{aligned}\mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A) &= \mathsf{E}_{\mathsf{P}}(X^+\mathbf{I}_A) - \mathsf{E}_{\mathsf{P}}(X^-\mathbf{I}_A)\\ &= \mathsf{E}_{\mathsf{P}}(Z_1\mathbf{I}_A) - \mathsf{E}_{\mathsf{P}}(Z_2\mathbf{I}_A) = \mathsf{E}_{\mathsf{P}}((Z_1 - Z_2)\mathbf{I}_A).\end{aligned}$$

We conclude $Z_1 - Z_2 \in L^1$ satisfies property 3 of Theorem 6.2 and is a version of the expectation of X conditional on the sub-σ-algebra $\mathcal{A}$.

6.18 For the case $X = \mathbf{I}_A$ we easily have $\mathsf{E}_{\tilde{\mathsf{P}}}(X) = \tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(\mathbf{I}_A Y)$, for every $A \in \mathcal{F}$. Passing to a nonnegative discrete $X = \sum_{i=1}^m x_i \mathbf{I}_{A_i}$, where events A_i partition the sample space, we additionally have:

$$\begin{aligned}\mathsf{E}_{\tilde{\mathsf{P}}}(X) &= \sum_{i=1}^m x_i \tilde{\mathsf{P}}(A_i) = \sum_{i=1}^m x_i \mathsf{E}_{\tilde{\mathsf{P}}}(\mathbf{I}_{A_i})\\ &= \sum_{i=1}^m x_i \mathsf{E}_{\mathsf{P}}(\mathbf{I}_{A_i} Y) \overset{\text{linear. of exp.}}{=} \mathsf{E}_{\mathsf{P}}\left(\sum_{i=1}^m x_i \mathbf{I}_{A_i} Y\right)\\ &= \mathsf{E}_{\mathsf{P}}(XY).\end{aligned}$$

For nonnegative $X \geqslant 0$, pick a sequence of nonnegative discrete rvs $0 \leqslant X_1 \leqslant X_2 \leqslant \cdots$ increasing pointwise to X. Since we showed $\mathsf{E}_{\tilde{\mathsf{P}}}(X_n) = \mathsf{E}_{\mathsf{P}}(X_n Y)$ now it follows:

$$\begin{aligned}\mathsf{E}_{\tilde{\mathsf{P}}}(X) = \mathsf{E}_{\tilde{\mathsf{P}}}(\lim_n X_n) &\overset{\text{MCT}}{=} \lim_n \mathsf{E}_{\tilde{\mathsf{P}}}(X_n)\\ &= \lim_n \mathsf{E}_{\mathsf{P}}(X_n Y) \overset{\text{MCT}}{=} \mathsf{E}_{\mathsf{P}}(XY).\end{aligned}$$

For a general rv X we have $\mathsf{E}_{\tilde{\mathsf{P}}}(X^+) = \mathsf{E}_{\mathsf{P}}(X^+Y)$ and $\mathsf{E}_{\tilde{\mathsf{P}}}(X^-) = \mathsf{E}_{\mathsf{P}}(X^-Y)$. Thus we get:

$$\mathsf{E}_{\tilde{\mathsf{P}}}(X) = \mathsf{E}_{\tilde{\mathsf{P}}}(X^+) - \mathsf{E}_{\tilde{\mathsf{P}}}(X^-) = \mathsf{E}_{\mathsf{P}}(X^+Y) - \mathsf{E}_{\mathsf{P}}(X^-Y) = \mathsf{E}_{\mathsf{P}}(XY),$$

provided that both sides are finite.

6.19 By assumption we have $\frac{X}{Y} \geqslant 0$. By Exercise 6.18, formula (6.25) can be written with X replaced by $\frac{X}{Y}$ and now formula (6.26) easily follows.

6.20 We have $\mathsf{E}_{\mathsf{P}}(X) = \sup\{\mathsf{E}_{\mathsf{P}}(Z) \mid 0 \leqslant Z \leqslant X\}$, for a simple rv $Z = \sum_{i=1}^{k} z_i \mathbf{I}_{A_i} \geqslant 0$, where events $A_1, \ldots, A_k$ form a partition of Ω. By assumption $\mathsf{P}(A) = 0$, it follows $\mathsf{P}(A \cap A_i) = 0$ each i and then

$$\mathsf{E}_{\mathsf{P}}(Z\mathbf{I}_A) = \sum_{i=1}^{k} z_k \mathsf{P}(A \cap A_i) = 0.$$

The supremum of $\{\mathsf{E}_{\mathsf{P}}(Z) \mid 0 \leqslant Z \leqslant X\}$ is also zero. We conclude that $\tilde{\mathsf{P}}(A) = \mathsf{E}_{\mathsf{P}}(X\mathbf{I}_A) = 0$ so $\tilde{\mathsf{P}} \ll \mathsf{P}$.

6.21 If $\mathcal{A} = \{\varnothing, \Omega\}$ only the constant rvs induce events in $\mathcal{A}$, i.e., they are $\mathcal{A}$-measurable, and trivially admit finite expectation. Then, items 1 and 2 of Theorem 6.2 are proved. We check item 3 for $A = \varnothing, \Omega$:

- $\mathsf{E}(X\mathbf{I}_{\varnothing}) = 0 = \mathsf{E}(\mathsf{E}(X)\mathbf{I}_{\varnothing})$,
- $\mathsf{E}(X\mathbf{I}_{\Omega}) = \mathsf{E}(X) = \mathsf{E}(\mathsf{E}(X)\mathbf{I}_{\Omega})$.

As a consequence we have $\mathsf{E}(X \,|\, \mathcal{A}) \overset{\text{a.s.}}{=} \mathsf{E}(X)$. If $\mathcal{A} = \mathcal{F}$, then $\mathsf{E}(X \,|\, \mathcal{F})$ induces events in $\mathcal{F}$ and item 3 holds for any $A \in \mathcal{F}$. This implies that $\mathsf{E}(X \,|\, \mathcal{F})$ must be equal to X or any other rv $Y \overset{\text{a.s.}}{=} X$ and we are done.

6.22 Let $Z = \mathsf{E}(X \,|\, \mathcal{A})$ and $G = \{Z > 0\} \in \mathcal{A}$, since Z is $\mathcal{A}$-measurable. Write condition 3 of Theorem 6.2 for G and G^c,

$$\mathsf{E}(Z\mathbf{I}_G) = \mathsf{E}(X\mathbf{I}_G) \leqslant \mathsf{E}(|X|\mathbf{I}_G), \quad \mathsf{E}(-Z\mathbf{I}_{G^c}) = \mathsf{E}(-X\mathbf{I}_{G^c}) \leqslant \mathsf{E}(|X|\mathbf{I}_{G^c}),$$

where the two inequalities above are a consequence of Jensen's inequality, while $\mathsf{E}(Z\mathbf{I}_G) > 0$ and $\mathsf{E}(-Z\mathbf{I}_{G^c}) \geqslant$ by the choice of G (note that $G^c = \{Y \leqslant 0\}$). Since $X \in L^1$ by assumption, we have $\mathsf{E}(|X|\mathbf{I}_G)$ and $\mathsf{E}(|X|\mathbf{I}_{G^c})$ are both finite. Adding each other we finally have $\mathsf{E}(|Z|) \leqslant \mathsf{E}(|X|) < \infty$, then $Z \in L^1$ as required.

6.23 We show first $\mathsf{E}\big(\mathsf{E}(X \,|\, \mathcal{A}) \,|\, \mathcal{H}\big) \overset{\text{a.s.}}{=} \mathsf{E}(X \,|\, \mathcal{H})$. Fix an event $A \in \mathcal{H} \subset \mathcal{A}$, then use item 4 of Proposition 6.1 with $Y = \mathbf{I}_A$,

$$\mathsf{E}\big(\mathsf{E}(X \,|\, \mathcal{A}) \,|\, \mathcal{H}\big)\mathbf{I}_A = \mathsf{E}\big(\mathbf{I}_A \mathsf{E}(X \,|\, \mathcal{A}) \,|\, \mathcal{H}\big),$$

since $\sigma(\mathbf{I}_A) \subset \mathcal{H}$ and $\mathbf{I}_A$ is $\mathcal{H}$-measurable. But $\mathbf{I}_A$ is also $\mathcal{A}$-measurable, thus the last term on the right-hand side above becomes $\mathsf{E}\,(\mathsf{E}(X\mathbf{I}_A \,|\, \mathcal{A}) \,|\, \mathcal{H})$. Let $Z' = \mathsf{E}(X\mathbf{I}_A \,|\, \mathcal{A})$ and $Z'' = \mathsf{E}\big(\mathsf{E}(X \,|\, \mathcal{A}) \,|\, \mathcal{H}\big)$, using the above and taking expectation:

$$\mathsf{E}(Z''\mathbf{I}_A) = \mathsf{E}\big(\mathsf{E}(Z' \,|\, \mathcal{H})\big). \tag{**}$$

Applying twice the property $\mathsf{E}(\mathsf{E}(X \,|\, \mathcal{A})) = \mathsf{E}(X)$, the right-hand term in (**) is now $\mathsf{E}(X\mathbf{I}_A)$. Then, by property (iii) of theorem 6.2 with $Z = \mathsf{E}(X \,|\, \mathcal{H})$ we have $Z'' = Z$ by a.s. uniqueness. For the other equality in item 3 of Proposition 6.1,

consider property 4 of the same Proposition and write:

$$\mathsf{E}\big(\mathsf{E}(X \mid \mathcal{H}) \mid \mathcal{A}\big) = \mathsf{E}(X \mid \mathcal{H})\mathsf{E}(1 \mid \mathcal{A}) = \mathsf{E}(X \mid \mathcal{H}).$$

Indeed, the random variable $Z = \mathsf{E}(X \mid \mathcal{H})$ has generated σ-algebra $\sigma(Z) \subset \mathcal{H} \subset \mathcal{A}$. Then, Z is $\mathcal{A}$-measurable and we can treat it as a constant.

Exercises of Chap. 7

7.1 Take the expectation of both sides of (7.2) and use property (6) of conditional expectation:

$$\mathsf{E}(\mathsf{V}(Y \mid X)) = \mathsf{E}(Y^2) - \mathsf{E}\left(\{\mathsf{E}(Y \mid X)\}^2\right). \tag{6}$$

Now, write the variance of the rv $\mathsf{E}(Y \mid X)$:

$$\begin{aligned}\mathsf{V}\left(\mathsf{E}(Y \mid X)\right) &= \mathsf{E}\left(\{\mathsf{E}(Y \mid X)\}^2\right) - \{\mathsf{E}(\mathsf{E}(Y \mid X))\}^2 \\ &= \mathsf{E}\left(\{\mathsf{E}(Y \mid X)\}^2\right) - \{\mathsf{E}(Y)\}^2, \end{aligned} \tag{7}$$

where we used property (6) again. Eventually, adding all sides of Eqs. (6) and (7) will complete the proof.

7.2 Let $W = \mathsf{E}(Y \mid \mathcal{A})$ we have:

$$\begin{aligned}\mathsf{E}((Y - Z)^2) &= \mathsf{E}((Y - W + W - Z)^2) \\ &= \mathsf{E}((Y - W)^2) + 2\mathsf{E}((Y - W)(W - Z)) \\ &+ \mathsf{E}((W - Z)^2).\end{aligned}$$

Now, $\mathsf{E}((Y - W)(W - Z))$ equals $\mathsf{E}\left(\mathsf{E}((Y - W)(W - Z) \mid \mathcal{A})\right)$ by the tower property, and this in turn equals $\mathsf{E}\left((W - Z)\mathsf{E}((Y - W) \mid \mathcal{A})\right)$ taking out what is known (i.e. the $\mathcal{A}$-measurable rv $W - Z$). Further, write $\mathsf{E}((Y - W) \mid \mathcal{A})$ as $\mathsf{E}(Y \mid \mathcal{A}) - W\mathsf{E}(1 \mid \mathcal{A})$ which is $W - W = 0$. We conclude the minimum of $\mathsf{E}((Y - Z)^2)$ is at $Z = W$.

7.3 Assume $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$. Observe that $\mathsf{V}(X) = \mathsf{E}(X^2)$, $\mathsf{V}(Y) = \mathsf{E}(Y^2)$, and both are equal to 1. Moreover, $\frac{\mathsf{cov}(X,Y)}{\sigma_X \sigma_Y} = \mathsf{corr}(X, Y) = \rho$. Now, let

$$g(a, b) = \mathsf{E}\left(\{Y - (a + bX)\}^2\right)$$

and compute

$$\begin{aligned} g(a,b) &= \mathsf{E}(Y^2) + \mathsf{E}(-2aY - 2bXY) + \mathsf{E}(a^2 + 2abX + b^2X^2) \\ &= \mathsf{V}(Y) - 2b\rho + a^2 + b^2\mathsf{V}(X) \\ &= 1 + a^2 + b(b - 2\rho). \end{aligned}$$

Hence, to find the minimum we also compute the partials and set them to zero:

$$\frac{\partial}{\partial a} = 2a = 0 \iff a = 0$$
$$\frac{\partial}{\partial b} = 2b - 2\rho = 0 \iff b = \rho.$$

Thus, the minimizer is $1 - \rho^2$, and the linear predictor is $Y = f(X) = \rho X$ and we are done. For the general case, we just consider the standardized versions of X, Y, then set

$$\frac{X - \mu_X}{\sigma_X} = \rho \frac{Y - \mu_Y}{\sigma_Y},$$

and eventually rearrange for Y to get Eq. (7.5).

7.4 Replace X with $X - \mathsf{E}(X)$ and Y with $Y - \mathsf{E}(Y)$. Next, observe that

$$\mathsf{V}(X) = \mathsf{E}\big((X - \mathsf{E}(X))^2\big) = \mathsf{E}\big(X^2 - 2X\mathsf{E}(X) + (\mathsf{E}(X))^2\big),$$

and the same is true for Y. Thus, applying the Cauchy-Schwarz inequality with the above change of variable we get

$$\left| \frac{\mathsf{E}\Big((X - \mathsf{E}(X))(Y - \mathsf{E}(Y))\Big)}{\sqrt{\mathsf{E}\big((X - \mathsf{E}(X))^2\big)\mathsf{E}\big((Y - \mathsf{E}(Y))^2\big)}} \right| \leqslant 1.$$

In fact $\mathsf{E}\big((X - \mathsf{E}(X))(Y - \mathsf{E}(Y))\big) = \mathsf{cov}(X, Y)$. Next, observe that $\mathsf{V}(a + bX) = b^2\mathsf{V}(X)$ and the same is true for Y. Use formula $\mathsf{cov}(X, Y) = \mathsf{E}(XY) - \mathsf{E}(X)\mathsf{E}(Y)$ to get $\mathsf{cov}(a + bX, c + dY) = b\,d\,\mathsf{cov}(X, Y)$ after some tedious algebra, and finally $\mathsf{corr}(a + bX, c + dY) = \frac{b\,d\,\mathsf{cov}(X,Y)}{\sqrt{b^2d^2\mathsf{V}(X)\mathsf{V}(Y)}} = \frac{|b\,d|\,\mathsf{cov}(X,Y)}{b\,d\,\sqrt{\mathsf{V}(X)\mathsf{V}(Y)}}$.

7.5 For the joint event $\{Y \leqslant y\} \cap \{Z \leqslant z\}$ we have two cases:

1. $\{Y \leqslant y\} := \{\omega \,|\, Y(\omega) = a \leqslant y\}$ is just Ω if $y \geqslant a$;
2. $\{Y \leqslant y\} = \varnothing$ if $y < a$.

In case 1 the joint event reduces to $\{Z \leqslant z\}$, and in case 2 it reduces to $\varnothing$. Applying $\mathsf{P}(\,\cdot\,)$ to the joint event we see that its value always equals $\mathsf{P}(Y \leqslant y)\mathsf{P}(Z \leqslant z)$. We know that for two independent rvs X, Y and two suitable functions $g, h : \mathbb{R} \to$ it holds $\mathsf{E}(g(X)h(Y)) = \mathsf{E}(g(X))\mathsf{E}(h(Y))$. Then choosing $g(x) = h(x) = x$ we

additionally get that the expectation of XY is equal to the product of the expectations of X and Y alone. Since we have proved that the degenerate rv $Y = a$ and the rv $Z = b\,X$ are independent then:

$$\begin{aligned}\mathsf{V}(Y+Z) &= \mathsf{E}\left((Y+Z)^2\right) - (\mathsf{E}(Y+Z))^2 \\ &= \mathsf{E}\left(Y^2 + 2YZ + Z^2\right) - (\mathsf{E}(Y))^2 - (\mathsf{E}(Z))^2 - 2\mathsf{E}(Y)\mathsf{E}(Z) \\ &= \mathsf{E}(Y^2) - (\mathsf{E}(Y))^2 + \mathsf{E}(Z^2) - (\mathsf{E}(Z))^2 = \mathsf{V}(Y) + \mathsf{V}(Z).\end{aligned}$$

But $\mathsf{V}(Y) = \mathsf{V}(a) = 0$ and $\mathsf{V}(Z) = \mathsf{V}(aX) = a^2\mathsf{V}(X)$ and the proof is complete.

7.6 First we have

$$\mathsf{E}(XY) = \mathsf{E}\,(X(a+bX)) = a\,\mathsf{E}(X) + b\,\mathsf{E}(X^2),$$

and clearly $\mathsf{E}(a + bX) = a + b\,\mathsf{E}(X)$. Thus formula $\mathsf{cov}(X, Y) = \mathsf{E}(XY) - \mathsf{E}(X)\mathsf{E}(Y)$ yields

$$\begin{aligned}\mathsf{cov}(X,Y) &= a\,\mathsf{E}(X) + b\,\mathsf{E}(X^2) - \mathsf{E}(X)\,(a + b\,\mathsf{E}(X)) \\ &= b\left[\mathsf{E}(X^2) - (\mathsf{E}(X))^2\right] = b\,\mathsf{V}(X).\end{aligned}$$

Now, computing the correlation we get:

$$\begin{aligned}\mathsf{corr}(X, a + b\,X) &= \frac{b\,\mathsf{V}(X)}{\sqrt{\mathsf{V}(X)b^2\,\mathsf{V}(X)}} \\ &= \frac{b\,\mathsf{V}(X)}{|b|\,\sqrt{\mathsf{V}(X)\mathsf{V}(X)}} = \mathrm{sign}(b),\end{aligned}$$

where $\mathrm{sign}(b) = 1$ if $b > 0$ and $\mathrm{sign}(b) = -1$ if $b < 0$.

7.7 Performing the product inside expectation we get

$$\begin{aligned}\mathsf{E}\left(X_1X_2 - X_1\tilde{X}_2 - \tilde{X}_1X_2 + \tilde{X}_1\tilde{X}_2\right) &= 2\,\mathsf{E}(X_1X_2) - 2\,\mathsf{E}(X_1)\mathsf{E}(X_2) \\ &= 2\,\mathsf{cov}(X_1, X_2).\end{aligned}$$

We used equality in distribution which implies $\mathsf{E}(g(X_1, X_2)) = \mathsf{E}(g(\tilde{X}_1, \tilde{X}_2))$ with $g(x, y) = xy$, and also independence which implies that the expectation of products factors into the product of expectations, but since block independence additionally implies $F_{X_i} \equiv F_{\tilde{X}_i}$ for $i = 1, 2$ then $\mathsf{E}(X_i) = \mathsf{E}(\tilde{X}_i)$.

7.8 Since $Y \overset{\text{a.s.}}{=} X$ means $\mathsf{P}(Y = X) = 1$, take $A = \{\omega \in \Omega \,|\, Y(\omega) = X(\omega)\}$. Next, considering $B = \{X \leqslant x\}$ we have that $A \cap B$ implies $\{X \leqslant x\} = \{Y \leqslant y\}$.

Moreover, from $A^c = \{X \neq Y\}$ and $A^c \cap B \subset A^c$ we get $\mathsf{P}(X \leqslant x) \leqslant \mathsf{P}(A^c) = 0$ which together with $A^c \cap \{Y \leqslant y\} \subset A^c$ imply $\mathsf{P}(X \leqslant x) = \mathsf{P}(Y \leqslant y) = 0$ on those scenarios s.t. the two rvs are different. Thus we have proved $Y \overset{\text{a.s.}}{=} X \Rightarrow Y \overset{\text{d}}{=} X$. It is worth noting how this further implies $\mathsf{E}(g(X)) = \mathsf{E}(g(Y))$ since expectation is based on the distribution.

7.9 Recall that equality a.s. implies $\mathsf{E}(g(X)) = \mathsf{E}(g(Y))$ for every bounded or continuous function $g : \mathbb{R} \to \mathbb{R}$. Now, the hypotheses implies $XY \overset{\text{a.s.}}{=} X^2$ since $\mathsf{P}(XY = XX) = \mathsf{P}(Y = X) = 1$. Thus, choosing $g(x) = x$ we have:

$$\begin{aligned}\mathsf{corr}(X, Y) &= \frac{\mathsf{E}(XY) - \mathsf{E}(X)\mathsf{E}(Y)}{\sqrt{\mathsf{V}(X)\mathsf{V}(Y)}} \\ &= \frac{\mathsf{E}(XX) - \mathsf{E}(X)\mathsf{E}(X)}{\sqrt{\mathsf{V}(X)\mathsf{V}(X)}} = \frac{\mathsf{V}(X)}{\mathsf{V}(X)} = 1.\end{aligned}$$

Note that $\mathsf{V}(Y) = \mathsf{E}(X^2) - (\mathsf{E}(X))^2$ by the hypotheses. For the case $Y \overset{\text{a.s.}}{=} -X$ we readily obtain $\mathsf{corr}(X, Y) = -1$.

7.10 We have to check $G(x, y) = \max\{0, F_X(x) + F_Y(y) - 1\}$ fulfills conditions (a)–(d) seen in Sect. 3.1, Chap. 3. First we clearly have $G : \mathbb{R}^2 \to [0, 1]$, by construction. To show (a), i.e., G is nondecreasing, we choose $x_1 \leqslant x_2$ and $y_1 \leqslant y_2$ and since marginals F_X, F_Y are CDFs (then are nondecreasing) we get:

$$\max\{0, F_X(x_1) + F_Y(y_1) - 1\} \leqslant \max\{0, F_X(x_2) + F_Y(y_2) - 1\}.$$

Right-continuity (b) is easily recovered by right-continuity of the marginals:

$$\lim_{h \downarrow 0} \left(\lim_{k \downarrow 0} \max\{0, F_X(x + h) + F_Y(y + k) - 1\} \right) = \max\{0, F_X(x) + F_Y(y) - 1\}.$$

To check (c) we compute:

$$\begin{aligned}\max\{0, F_X(-\infty) + F_Y(y) - 1\} &= 0 \\ \max\{0, F_X(x) + F_Y(-\infty) - 1\} &= 0 \\ \max\{0, F_X(\infty) + F_Y(\infty) - 1\} &= 1.\end{aligned}$$

Eventually, we verify G fulfills (d) for the same pairs (x_1, x_2) and (y_1, y_2) as above:

$$\begin{aligned}&\max\{0, F_X(x_2) + F_Y(y_2) - 1\} - \max\{0, F_X(x_2) + F_Y(y_1) - 1\} \\ &- \max\{0, F_X(x_1) + F_Y(y_2) - 1\} + \max\{0, F_X(x_1) + F_Y(y_1) - 1\} \geqslant 0.\end{aligned}$$

The verification that $H(x, y) = \min\{F_X(x), F_Y(y)\}$ is a CDF follows easily by the same reasoning above.

Exercises of Chap. 8

8.1 Take $\epsilon > 0$ and let $A_n(\epsilon) = \{|X_n - X| > \epsilon\}$, for every $n \in \mathbb{N}$. Then, by monotonicity, subadditivity and continuity of P we have

$$\lim_{n\to\infty} \mathsf{P}(A_n(\epsilon)) \leqslant \lim_{n\to\infty} \mathsf{P}(\cup_{i\geqslant n} A_i(\epsilon)) = \mathsf{P}(A_i(\epsilon) \text{ i.o.}) = 0,$$

where we used the hypotheses $X_n \xrightarrow{\text{a.s.}} X$. See Sect. 1.5 for the notation $\{A_i(\epsilon) \text{ i.o.}\}$.

Remark 5 We give some details on the last part of the solution above. The event $\{\lim_{n\to\infty} X_n = X\}$ has complimentary event comprising those scenario ω where $X_n(\omega)$ fails to converge to $X(\omega)$ iff there is some $\epsilon > 0$ s.t. $|X_n(\omega) - X(\omega)| \geqslant \epsilon$ for infinitely many n. As byproduct,

$$\left\{\lim_{n\to\infty} X_n = X\right\}^c = \cup_\epsilon \{|X_n - X| \geqslant \epsilon \text{ i.o.}\} = \cup_\epsilon \{A_n(\epsilon) \text{ i.o.}\}.$$

Note the right-hand side increases as ϵ decreases. The 'convergence' event $\{\lim_{n\to\infty} X_n = X\}$ has probability 1 iff its complement has probability zero, $\mathsf{P}(A_n(\epsilon) \text{ i.o.}) = 0$. Moreover, the 'convergence' event is $A_n(\epsilon) = \limsup_n \{A_n(\epsilon) \text{ i.o.}\}$ and by [2, Th 4.1] and the above zero probability condition it follows $\lim_{n\to\infty} \mathsf{P}(|X_n - X| > \epsilon) = 0$. It is worth noting that assuming $\sum_{n=1}^{\infty} \mathsf{P}(|X_n - X| > \epsilon) < \infty$ we have another sufficient condition for $X_n \xrightarrow{\text{a.s.}} X$, since by Borel-Cantelli lemma 1.1, part (i) it follows $\mathsf{P}(A_n(\epsilon) \text{ i.o.}) = 0$.

8.2 Assume $X_n \xrightarrow{\text{ms}} X$ and let $\epsilon > 0$. Then, by Markov's inequality (see Proposition 4.2, Sect. 4.6) we have:

$$\begin{aligned}\mathsf{P}(|X_n - X| > \epsilon) &= \mathsf{P}(|X_n - X|^2 > \epsilon^2) \\ &\leqslant \frac{\mathsf{E}((X_n - X)^2)}{\epsilon^2}.\end{aligned}$$

By hypotheses, the numerator on the right-hand side of the inequality tends to zero and we are done.

8.3 Assume $X_n \xrightarrow{\text{ms}} X$. By Cauchy-Schwarz inequality (4.3), Sect. 4.6, we have:

$$\mathsf{E}(|X_n - X|) \leqslant \sqrt{\mathsf{E}((X_n - X)^2)}.$$

By hypotheses, $\mathsf{E}((X_n - X)^2) \to 0$, as $n \to \infty$ and thus also $\mathsf{E}(|X_n - X|) \to 0$.

8.4 By Chebyshev's inequality (see Proposition 4.2, Sect. 4.6) we easily compute:

$$\mathsf{P}(|X_n - X| > \epsilon) \leqslant \frac{\mathsf{E}(|X_n - X|)}{\epsilon}.$$

Then, the hypotheses $\mathsf{E}(|X_n - X|) \to 0$ does imply $\mathsf{P}(|X_n - X| > \epsilon) \to 0$.

8.5 First, we observe that the constant $\frac{1}{\sqrt{2\pi}}$ is each sample density can be ignored without loss of generality. Then we write:

$$\begin{aligned}\prod_{i=1}^{n} f(R_i, \theta) &= \prod_{i=1}^{n} \sigma^{-1} \exp\left(-\frac{1}{2\sigma^2}(R_i - \mu)^2\right) \\ &= \sigma^{-1} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(R_i - \mu)^2\right).\end{aligned}$$

The latter expression can be rewritten

$$\sigma^{-1} \exp\left(-\frac{n\bar{V}_n}{2\sigma^2}\right) \exp\left(-\frac{n(\bar{R}_n - \mu)^2}{2\sigma^2}\right),$$

where $\bar{R}_n = n^{-1}\sum_{i=1}^{n} R_i$ is the sample mean and $\bar{V}_n = n^{-1}\sum_{i=1}^{n}(R_i - \bar{R}_n)^2$ is the sample variance. Indeed, $\sum_{i=1}^{n}(R_i - \mu)^2 = n\bar{V}_n + n(\bar{R}_n - \mu)^2$, since $\sum_{i=1}^{n}(R_i - \mu)^2$ is equal to $\sum_{i=1}^{n}(R_i - \bar{R}_n + \bar{R}_n - \mu)^2$. Passing to the log-likelihood,

$$h(\mu, \sigma) = -n \ln \sigma - \frac{n\bar{V}_n}{2\sigma^2} - \frac{n(\bar{R}_n - \mu)^2}{2\sigma^2},$$

setting the first-order conditions

$$\frac{\partial}{\partial \mu} h(\mu, \sigma) = 0, \quad \frac{\partial}{\partial \sigma} h(\mu, \sigma) = 0$$

we finally get the desired maximizers $\bar{R}_n$ and $\sqrt{\bar{V}_n}$.

8.6 We let $\alpha = \mathsf{E}(T_n)$ and compute:

$$\begin{aligned}\mathsf{E}((T_n - \theta))^2 &= \mathsf{E}((T_n - \theta + \alpha - \alpha))^2 \\ &= (\alpha - \theta)^2 + \mathsf{E}((T_n - \alpha))^2 \\ &= (\text{bias}(T_n))^2 + \mathsf{V}(T_n).\end{aligned}$$

Next, assuming $(\text{bias}(T_n))^2 \to 0$ and $\sqrt{\mathsf{V}(T_n)} \to 0$, as $n \to \infty$, then by the above result we also have MSE $\to 0$. As a consequence, $T_n \xrightarrow{\text{ms}} \theta$ and by Theorem 8.1, part (iii), convergence in probability follows.

8.7 We know that $X \overset{\text{a.s.}}{=} Y$ is equivalent to $X - Y \overset{\text{a.s.}}{=} 0$. Now, let $\mathcal{N}^p$ be the subset of $\mathcal{L}^p$ containing all rvs Z s.t. $\mathsf{E}(|Z|^p)^{1/p} = 0$ for $p \in [1, \infty)$, or equivalently $\mathsf{E}(|Z|^p) = 0$. Since $|Z| \geqslant 0$, by Exercise 4.17, part 3, we also have $Z \overset{\text{a.s.}}{=} 0$. Taking $Z = X - Y$ the relation $X \overset{\text{a.s.}}{=} Y$ now reads $X - Y \in \mathcal{N}^p$. Clearly, $X \overset{\text{a.s.}}{=} X$ is $X - X = 0 \in \mathcal{N}^p$ which trivially implies reflexivity. Since $X \overset{\text{a.s.}}{=} Y$ is the same as $Y \overset{\text{a.s.}}{=} X$ we have $Y - X \in \mathcal{N}^p$ and the relation is symmetric. To show transitivity, use the triangle inequality

$$\begin{aligned} |X - U| &= |X - Y + Y - U| \\ &\leqslant |X - Y| + |Y - U|, \end{aligned}$$

which is equivalent to

$$\mathsf{E}(|X - U|^p)^{1/p} \leqslant \mathsf{E}\left(|X - Y| + |Y - U|^p\right)^{1/p}.$$

Applying Minkowski's inequality (see Theorem 4.5) to the right-hand side we get

$$\mathsf{E}\left(|X - Y| + |Y - U|^p\right)^{1/p} \leqslant \mathsf{E}(|X - Y|^p)^{1/p} + \mathsf{E}(|Y - U|^p)^{1/p}.$$

Assume $X \overset{\text{a.s.}}{=} Y$ and $Y \overset{\text{a.s.}}{=} U$, i.e. $X - Y \in \mathcal{N}^p$ and $Y - U \in \mathcal{N}^p$. Expectations on the right-hand side of this last inequality are zero by assumption, so putting things together we have $\mathsf{E}(|X - U|^p)^{1/p} = 0$ or equivalently $\mathsf{E}(|X - U|^p) = 0$. Hence, $X - U \in \mathcal{N}^p$ which means $X \overset{\text{a.s.}}{=} U$ and transitivity is proved. Since $\overset{\text{a.s.}}{=}$ is an equivalence relation on $\mathcal{L}^p$ we can take its quotient space $\mathcal{L}^p \backslash_{\overset{\text{a.s.}}{=}}$ as the set denoted by L^p containing all the equivalence classes induced by $\overset{\text{a.s.}}{=}$. Thus, $X \in L^p$ if and only if $|X|^p \in \mathcal{L}^p$ and there are rvs W in $\mathcal{L}^p$ with $|W|^p \in \mathcal{L}^p$ and such that $X \overset{\text{a.s.}}{=} W$.

Exercises of Chap. 9

9.1 We have $B_{c\,t} \sim \mathrm{N}(0, c\,t)$ by the definition of a Brownian motion. Also, $\sqrt{c}B_t \sim \mathrm{N}(0, c\,t)$, since $\mathsf{E}(\sqrt{c}B_t) = \sqrt{c}\mathsf{E}(B_t)$, which is zero, and $\mathsf{V}(\sqrt{c}B_t) = (\sqrt{c})^2\mathsf{V}(B_t) = c\,t$.

9.2 We compute:

$$\begin{aligned} \mathsf{P}(B_4 < b \,|\, B_1 = a) &= \mathsf{P}(B_4 - B_1 < b - a \,|\, B_1 = a) \\ &\overset{\text{indep.}}{=} \mathsf{P}(B_4 - B_1 < b - a) \end{aligned}$$

$$\overset{\text{stat. incr.}}{=} \mathsf{P}(B_{b-a} < b-a)$$
$$\overset{\text{scaling}}{=} \mathsf{P}\left(\sqrt{b-a}B_1 < b-a\right) = \mathsf{P}\left(B_1 < \sqrt{b-a}\right).$$

The latter is the CDF of N(0, 1) evaluated at $\sqrt{b-a}$.

9.3 For the Brownian motion with drift we have:

$$\mathsf{E}(\tilde{B}_t) = \mu t + \sigma \mathsf{E}(B_t) = \mu t,$$
$$\mathsf{V}(\tilde{B}_t) = \mathsf{V}(\mu t + \sigma B_t) = \sigma^2 \mathsf{V}(B_t) = \sigma^2 t.$$

Observe that, for $s, t > 0$, we have

$$\begin{aligned}\mathsf{cov}(\tilde{B}_t, \tilde{B}_s) &= \mathsf{E}(\tilde{B}_t \tilde{B}_s) - \mathsf{E}(\tilde{B}_t)\mathsf{E}(\tilde{B}_s) \\ &= \mu t \mu s + \sigma^2 \mathsf{E}(B_t B_s) - \mu t \mu s = \sigma^2 \min\{s, t\}.\end{aligned}$$

For the geometric Brownian motion we firstly have, for $x > 0$:

$$\begin{aligned}\mathsf{P}(S_t \leqslant x) &= \mathsf{P}(S_0 \mathrm{e}^{\mu t + \sigma B_t} \leqslant x) \\ &= \mathsf{P}\left(\mu t + \sigma B_t \leqslant \ln\left(\frac{x}{S_0}\right)\right) \\ &= \mathsf{P}\left(B_t \leqslant \frac{\ln x - \ln S_0 - \mu t}{\sigma}\right) \\ &= \mathsf{P}\left(\sqrt{t} B_1 \leqslant \frac{\ln x - \ln S_0 - \mu t}{\sigma}\right) \\ &= \mathsf{P}\left(B_1 \leqslant \frac{\ln x - \ln S_0 - \mu t}{\sqrt{t}\sigma}\right).\end{aligned}$$

Eventually, the MGF $M_{\mu+\sigma B_t}(u) = M_{\mu+\sqrt{t}\sigma B_1}(u)$ is equal to $\exp(\mu t + \frac{t\sigma^2 u^2}{2})$, from which we get $\mathsf{E}(\mathrm{e}^{\sqrt{t} B_1}) = \mathrm{e}^{t\sigma^2 u^2/2}$ and further $\mathsf{E}(\mathrm{e}^{\sigma B_t}) = \mathrm{e}^{\sigma^2 t/2}$. From this and from Eqs. (5.17) and (5.18) in Sect. 5.7, the expressions for the expectation and variance of S_t follow.

9.4 We compute the probability that B_t is at most $a > 0$:

$$\begin{aligned}\mathsf{P}(B_t \geqslant a) \overset{\text{law of tot. pr.}}{=} &\ \mathsf{P}(B_t \geqslant a \mid T_a \leqslant t)\mathsf{P}(T_a \leqslant t) \\ &+ \mathsf{P}(B_t \geqslant a \mid T_a > t)\mathsf{P}(T_a > t).\end{aligned}$$

Assuming $T_a > t$ we have that B_t has not hit the barrier $y = a$, then $\{B_t \geqslant a\}$ has probability zero, and the second term in the right-hand side above vanishes.

Moreover, by the reflection principle we also have

$$\mathsf{P}(B_t \geqslant a \,|\, T_a \leqslant t) = \mathsf{P}(B_t \leqslant a \,|\, T_a \leqslant t) = \frac{1}{2},$$

thus $\mathsf{P}(B_t \geqslant a) = \frac{1}{2}\mathsf{P}(T_a \leqslant t)$ and eventually $\mathsf{P}(T_a \leqslant t) = 2\mathsf{P}(B_t \geqslant a)$. Observe that the latter is recovered by the normal standard $Z \sim \mathrm{N}(0, 1)$ as

$$2\mathsf{P}(B_t \geqslant a) = 2\left(1 - F_Z\left(\frac{a}{\sqrt{t}}\right)\right).$$

9.5 For the running maximum of a standard Brownian motion $\bar{B}_t := \sup_{u\in[0,t]} B_u$ we note that $\{\bar{B}_t > a\}$ and $\{T_a < t\}$, are equivalent, for positive a. Therefore,

$$\mathsf{P}(\bar{B}_t \leqslant a) = 1 - \mathsf{P}(\bar{B}_t > a) = 1 - \mathsf{P}(T_a \leqslant t),$$

where in the latter equality we used that T_a has a continuous distribution with CDF[12] $2(1 - F_Z(\frac{a}{\sqrt{t}}))$, for $Z \sim \mathrm{N}(0, 1)$, see Exercise 9.4. Eventually, the desired CDF is $2F_Z(\frac{a}{\sqrt{t}}) - 1$.

Exercises of Chap. 10

10.1 We compute the corresponding quantile $Q_X(0.05) = -10$ which is just $-\mathrm{VaR}_{c,T}(X)$, because

$$F_X(-10) = \mathsf{P}(X = -105) + \mathsf{P}(X = -57) + \mathsf{P}(X = -10) = 0.165.$$

We take the *smallest* value in the F_X's domain s.t. it crosses the given probability level $c = 0.05$, indeed $0.045 = F_X(-57) < 0.05 < F_X(-10)$. Here the support of the P&L rv is $C = \{-105, -57, -10, 39, 98, 226\}$, and we should be confident at $100(1 - c)\% = 95\%$ to have only revenues.

10.2 Write the following chain of equivalences:

$$\begin{aligned} Q^-_{-X}(1-c) &= \inf\left\{x \in \mathbb{R} \,\middle|\, \mathsf{P}(-X \leqslant x) \geqslant 1 - c\right\} \\ &= \inf\left\{x \in \mathbb{R} \,\middle|\, 1 - \mathsf{P}(X < -x) \geqslant 1 - c\right\} \\ &= \inf\left\{x \in \mathbb{R} \,\middle|\, \mathsf{P}(X < -x) \leqslant c\right\} \\ &= -\sup\left\{y \in \mathbb{R} \,\middle|\, \mathsf{P}(X < y) \leqslant c\right\} = -Q^+_X(c). \end{aligned}$$

[12] The event $\{T_a = t\}$ has zero probability.

10.3 To prove that $S_t = S_0 \mathrm{e}^{(\alpha-\sigma^2/2)t+\sigma B_t}$ is a solution to the SDE $\mathrm{d}S_t = \alpha S_t \mathrm{d}t + \sigma S_t \mathrm{d}B_t$, we apply Itô's Formula with $S_t = f(t, Y_t) = S_0 \mathrm{e}^{\mu t+\sigma Y_t}$ and considering the diffusion equation $\mathrm{d}Y_t = X_t \mathrm{d}t + Z_t \mathrm{d}B_t$, with $X_t = 0$ and $Z_t = 1$, i.e., $\mathrm{d}Y_t = \mathrm{d}B_t$ and $y = B_t$. Thus:

$$\begin{aligned} \mathrm{d}f(t, B_t) = \mathrm{d}S_t &= \left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial y^2}\right)\mathrm{d}t + \frac{\partial f}{\partial y}\mathrm{d}B_t \\ &= S_t\left((\mu + \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma \mathrm{d}B_t\right). \end{aligned}$$

Eventually, it suffices to let $\alpha = \mu + \frac{1}{2}\sigma^2$, i.e., $\mu = \alpha - \frac{1}{2}\sigma^2$. Observe that we used:

- $\frac{\partial f}{\partial y} = \sigma S_0 \mathrm{e}^{\mu t+\sigma B_t} = \sigma S_t$
- $\frac{\partial f}{\partial t} = \mu S_0 \mathrm{e}^{\mu t+\sigma B_t} = \mu S_t$
- $\frac{\partial^2 f}{\partial y^2} = \sigma^2 S_0 \mathrm{e}^{\mu t+\sigma B_t} = \sigma^2 S_t$.

Thus, we arrived to

$$\frac{\mathrm{d}S_t}{S_t} = \mathrm{d}\ln S_t = (\alpha - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma \mathrm{d}B_t,$$

and integrating both sides we get[13]

$$\begin{aligned} \int_0^t \mathrm{d}\ln S_t = \ln S_t - \ln S_0 &= \ln\left(\frac{S_t}{S_0}\right) \\ &= (\alpha - \frac{1}{2}\sigma^2)\int_0^t \mathrm{d}u + \sigma\int_0^t \mathrm{d}B_u \\ &= (\alpha - \frac{1}{2}\sigma^2)t + \sigma(B_t - B_0) \\ &\iff \\ S_t &= S_0 \mathrm{e}^{(\alpha-\frac{1}{2}\sigma^2)t+\sigma B_t}. \end{aligned}$$

10.4 Let $t = 0$ and $S_0 = y$ we have the following chain of equivalences:

$$\begin{aligned} S_T > K &\iff \mathrm{e}^{(r-\sigma^2/2)T+\sigma\sqrt{T}z} > K/y \\ &\iff z > \frac{\ln(K/y) - (r - \sigma^2/2)T}{\sigma\sqrt{T}} \\ &\iff z > \sigma\sqrt{T} - \mathrm{d}_1, \end{aligned} \tag{8}$$

[13] We also used $\frac{\mathrm{d}f}{f} = \frac{f'}{f} = \mathrm{d}\ln f$, as for a deterministic real valued function f.

where $\ln(y/K) = -\ln(K/y)$. Moreover we have:

$$\begin{aligned}
\mathsf{E}_{\tilde{\mathsf{P}}}(\mathbf{I}_{\{S_T>K\}}) &= \tilde{\mathsf{P}}(S_T > K) \\
&= \tilde{\mathsf{P}}(z > \sigma\sqrt{T} - \mathrm{d}_1) \\
&= \tilde{\mathsf{P}}(z < \mathrm{d}_1 - \sigma\sqrt{T}) \\
&= F_Z(\mathrm{d}_1 - \sigma\sqrt{T}) = F_Z(\mathrm{d}_2). \qquad (9)
\end{aligned}$$

Letting $c = \sigma\sqrt{T} - \mathrm{d}_1$ and using (8) we also have:

$$\begin{aligned}
\mathsf{E}_{\tilde{\mathsf{P}}}(\mathbf{I}_{\{S_T>K\}}\, S_T) &= \int_c^\infty y\, \mathrm{e}^{(r-\sigma^2/2)T+\sigma\sqrt{T}z}\, \tfrac{1}{\sqrt{2\pi}}\, \mathrm{e}^{-z^2/2}\mathrm{d}z \\
&= \tfrac{1}{\sqrt{2\pi}}\, y\, \mathrm{e}^{(r-\sigma^2/2)T} \int_c^\infty \mathrm{e}^{-(x^2-2\sigma\sqrt{T}\,z)/2}\mathrm{d}z \\
&= \tfrac{1}{\sqrt{2\pi}}\, y\mathrm{e}^{rT} \int_c^\infty \mathrm{e}^{-(z-\sigma\sqrt{T})^2/2}\mathrm{d}z \\
&= y\mathrm{e}^{rT}\, \tfrac{1}{\sqrt{2\pi}} \int_{-\mathrm{d}_1}^\infty \mathrm{e}^{-w^2/2}\mathrm{d}w \qquad \text{(letting } w = z - \sigma\sqrt{T}) \\
&= y\mathrm{e}^{rT}\, \mathsf{P}(z > -\mathrm{d}_1) = y\mathrm{e}^{rT}\, F_Z(\mathrm{d}_1). \qquad (10)
\end{aligned}$$

The European call option payoff is $(S_T - K)^+ = (S_T - K)\mathbf{I}_{\{S_T>K\}}$, thus using (9) and (10) in the expression

$$\begin{aligned}
C_0 &= \mathsf{E}(\mathbf{I}_{\{S_T>K\}}(S_T - K)) \\
&= \mathrm{e}^{-rt}\mathsf{E}(\mathbf{I}_{\{S_T>K\}}S_T) - K\,\mathrm{e}^{-rt}\mathsf{E}(\mathbf{I}_{\{S_T>K\}}),
\end{aligned}$$

we get the desired result.

Appendix A
Binary Expansions

For every real number $a \in \mathbb{R}$ we may write a **binary expansion**, that is an expression like

$$b_0.b_1b_2\ldots b_n\ldots = \left(b_0 + \tfrac{b_1}{2^1} + \tfrac{b_2}{2^2} + \cdots + \tfrac{b_n}{2^n} + \cdots\right), \tag{A.1}$$

where now the digits b_i are taken from the set $\{0, 1\}$, while b_0 can be expressed as $c_n 2^n + \cdots + c_1 2^1 + c_0 2^0$ and again $c_i \in \{0, 1\}$. We are sure that the infinite series $\sum_{i=1}^{\infty} \frac{b_i}{2^i}$ converges since its partial sums are increasing and bounded above, see Appendix B. We show that to each real number a it corresponds a binary expansion whose sum is indeed a. We restrict ourselves to the case $0 \leqslant a < 1$ (with $b_0 = 0$) since binary expansions for lager numbers are build analogously. Then, choose $a \in [0, 1)$ and observe that a lies in exactly one of the two intervals $[0, \frac{1}{2})$ and $[\frac{1}{2}, 1)$. This first choice is equivalent to split $[0, 1)$ into two subintervals, each of length $\frac{1}{2}$. If a belong to $[0, \frac{1}{2})$ we let $b_1 = 0$, otherwise we let $b_1 = 1$. Hence, as $b_0 = 0$, we can write the subinterval containing a as $[b_0 + \frac{b_1}{2}, b_0 + \frac{b_1+1}{2})$, indeed it is equivalent to consider the two intervals $[b_0, b_0 + \frac{1}{2})$ and $[b_0 + \frac{1}{2}, b_0 + \frac{2}{2})$. Obviously, the left-hand point $b_0 + \frac{b_1}{2}$ is the $\frac{\text{largest integer}}{2}$ less than or equal to a. By further subdividing $[b_0 + \frac{b_1}{2}, b_0 + \frac{b_1+1}{2})$ into two intervals of type $[\ ,\ )$ and equal length $\frac{1}{2}$ we let $b_2 = 0$ if a belong to the first subinterval, otherwise $b_2 = 1$. We can write the subinterval containing a as $[b_0 + \frac{b_1}{2} + \frac{b_2}{2^2}, b_0 + \frac{b_1}{2} + \frac{b_2+1}{2^2})$. Again, the left-hand point of this last interval is the $\frac{\text{largest integer}}{2^2}$ less than or equal to a. Of course, we carry on in this manner and find $b_0.b_1b_2\ldots b_n$ that by construction differs from a less than $\frac{1}{2^n}$, indeed we can write

$$b_0.b_1b_2\ldots b_n \leqslant a < b_0.b_1b_2\ldots b_n + \tfrac{1}{2^n},$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

since at the nth stage we have subdivided the interval containing a into n subintervals of equal length $\frac{1}{2^n}$. If we continue this procedure infinitely many times, then $n \to \infty$ and the equality $a = b_0.b_1b_2b_3, \ldots$ is proved by a squeezing argument. The same construction can be adapted to expand any real number a by a decimal representation. We start with the interval $[q_0, q_0 + 1)$ containing a, where q_0 is the largest natural number less than or equal to a. If we divide this interval in 10 subintervals of type [,) and equal length $\frac{1}{10}$, then a lies in exactly one of these and we may denote its left-hand point by $q_0 + \frac{q_1}{10}$ where $q_1 \in \{0, 1, 2, \ldots, 9\}$. Now $b_0 + \frac{q_1}{10}$ is the $\frac{\text{largest integer}}{10}$ which is less than or equal to a. By repeating this subdivision n times we get $q_0.q_1q_2 \ldots q_n$ that differs from a less than $\frac{1}{10^n}$. Then, the same limiting procedure as above entails the desired equality $a = q_0.q_1q_2q_3 \ldots$.

Appendix B
Sequences of Real Numbers

A function $f : \{1, 2, \ldots, n\} \to \mathbb{R}$ has image $f(\{1, 2, \ldots, n\}) = \{x_1, x_2, \ldots, x_n\} \subset \mathbb{R}$. We usually write its value at $i \in \{1, 2, \ldots, n\}$ as $f(i) = x_i$ and call it a **finite sequence**. It is possible to define an **infinite sequence** simply by switching the domain $\{1, 2, \ldots, n\}$ into the whole set of natural numbers $\mathbb{N}$. An infinite sequence is written $x_1, x_2, \ldots$ or briefly $(x_n)_n$ where the number x_n is called the ***n*th term** of the sequence and indeed $(x_n)_n$ is an indexed family of reals.

Example B.1 For every $n \in \mathbb{N}$ let define:

$$\begin{aligned} x_n &= (-1)^n, & x_1, x_2, x_3, x_4 &= -1, 1, -1, 1 \\ y_n &= \tfrac{1}{n+1}, & y_1, y_2, y_3, y_4 &= \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{4}, \tfrac{1}{5} \\ z_n &= 2(n-1), & z_1, z_2, z_3, z_4 &= 0, 2, 4, 6. \end{aligned}$$

They are all infinite sequences, whose first four terms are displayed on the right.

The previous example illustrates some important features of sequences. It may happen that $x_{n_1} \neq x_{n_2}$ for every $n_1 \neq n_2$ in the domain of the sequence, i.e. the terms are all distinct. But the sequence with nth term $x_n = (-1)^n$ in Example B.1 shows that recurring terms are admitted. Furthermore, it may happen that $x_n \leqslant x_{n+1}$ or $x_n \geqslant x_{n+1}$ for every $n \in \mathbb{N}$ and in the former case we call such a sequence **increasing**, while in the latter case we call it **decreasing**. In both cases we have a **monotonic sequence**. Thus for larger values n:

(a) the terms x_n are far each other as n goes to infinity,
(b) the terms x_n are never close each other,
(c) the terms x_n are closer each other as n goes to infinity.

A sequence $(x_n)_n$ of type (a) is said to be **divergent**. If $(x_n)_n$ is in addition increasing we write $\lim_{n\to\infty} x_n = \infty$ or $x_n \to \infty$. In case (c) the sequence is said to be **convergent** to a number $x \in \mathbb{R}$ which we call **limit** and write $\lim_{n\to\infty} x_n = x$ or

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https://doi.org/10.1007/978-3-031-54688-4

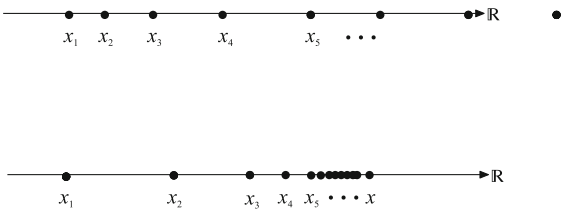


Fig. B.1 Top: increasing sequence that is divergent. Bottom: increasing sequence converging to x

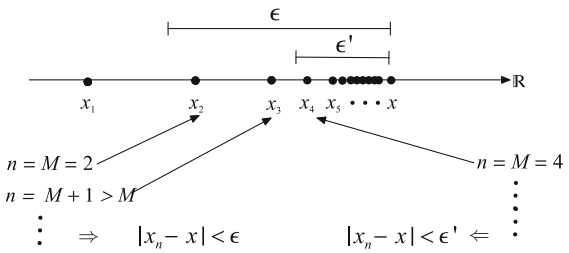


Fig. B.2 Given $\epsilon > 0$, we find $M = 2$ and from this value of the index on, $n \geqslant M$, the terms $x_2, x_3, \ldots$ are close each other and near x. Choosing $\epsilon' > 0$ we find $M = 4$ and $x_4, x_5, \ldots$ are closer and closer to x

$x_n \to x$. Figure B.1 displays the *tail behavior* of a sequence. For a more formal description we need the concept of **neighborhood**, which enable us to measure how close the terms x_n are each other in the tail of a sequence, for n large enough. If actually the sequence is convergent to x, then we need also to know the nearness of the terms x_n to the supposed limit x. This proximity can be captured by the open interval $(x-\epsilon, x+\epsilon)$ centered at x, for which we choose an arbitrary $\epsilon > 0$. Hence, if $x_n \to x$ we can find a positive integer M for every $\epsilon > 0$ such that $n \geqslant m$ implies $x_n \in (x-\epsilon, x+\epsilon)$. A similar reasoning applies to decreasing sequences converging to some x. A decreasing sequence that is divergent entails $\lim_{n\to\infty} x_n = -\infty$. Case (b) above refers to sequences that are neither convergent nor divergent as their terms instead oscillate. The formal definition of convergence of a sequence follows (see also Fig. B.2):

Definition B.1 A sequence $(x_n)_n$ is said to be **convergent** to the limit x if for every $\epsilon > 0$ there exists an integer $M \in \mathbb{N}$, which depends upon ϵ, such that $n \geqslant M \Rightarrow |x_n - x| < \epsilon$. We write either $\lim x_n = x$ or $x_n \to x$, for $n \to \infty$.

Example B.2 Let $(x_n)_n$ be a sequence with nth term $x_n = a^n$, where $a \in \mathbb{R}$. If $a = 1$ we have $1^n = 1$ for every integer n, then whenever $\epsilon > 0$ we find $N \in \mathbb{N}$ such that $n \geqslant N$ implies $|1 - 1| < \epsilon$ and the sequence is trivially convergent. Now, assume $|a| < 1$. Thus, for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $n \geqslant N$ implies $|a^n| < \epsilon$. In fact, $\epsilon^{1/n} \to 1$ while $|a| < 1$ and then we may find N such that $|a| < \epsilon^{1/N}$. From the latter relation we deduce that $|a^N| = |a|^N < \epsilon$ and this in turn gives $|a^n| < |a^N| < \epsilon$ for $n \geqslant N$. We conclude that $a_n \to 0$. If $|a| > 1$ the sequence diverges, $a^n \to \infty$. We call $a^1, a^2, a^3, \ldots$ a *geometric sequence*.

A sequence is bounded above (below) if the set $R = \{x_1, x_2, \ldots\}$ is such that $\sup R$ ($\inf R$) is finite. If a sequence $(x_n)_n$ is both bounded above and below, then is said to be **bounded**. We can write $\sup_n x_n$ and $\inf_n x_n$ to denote the supremum

and the infimum of the image set of the sequence. Clearly, for a bounded sequence we can find a number $U > 0$ such that $|x_n| < U$ for every $n \in \mathbb{N}$, that is $x_n \in (-U, U)$. Observe that the sequence $((-1)^n)_n$ in Example B.1 is bounded but is not convergent. We will see soon that, on the contrary, any convergent sequence is always bounded. But first, we analyze the behavior of monotonic sequences.

Theorem B.1 *Every increasing sequence (decreasing) which is bounded above (below) is convergent.*

Proof By assumption $(x_n)_n$ is bounded above then there exists $x = \sup\{x_n \mid n \in \mathbb{N}\}$. We need to show that $\lim_{n\to\infty} x_n = x$. Then, pick $\epsilon > 0$ and find $M \in \mathbb{N}$ such that $x - \epsilon < x_M \leqslant x$. We can write $x - x_m < \epsilon$. But $(x_n)_n$ is increasing, then for every $n > M$ we have $x_M \leqslant x_n$ and $x - x_n \leqslant x - x_M$. This implies $|x - x_n| = x - x_n \leqslant x - x_M < \epsilon$ and we are done. ■

If a sequence is increasing and bounded above we write $x_n \uparrow x$ and $x = \sup\{x_n\}$. Similarly, for decreasing sequence that are bounded below we write $x_n \downarrow x$ and $x = \inf\{x_n\}$. Hence, monotone sequences which are bounded always have a limit. The following properties are useful:

1. If $(x_n)_n$, $(y_n)_n$ and $(z_n)_n$ are such that $x_n \leqslant z_n \leqslant y_n$ for every n and $\lim x_n = \lim y_n = x$, then $z_n \to x$.
2. For every $a, b \in \mathbb{R}$ the sequence $(ax_n + by_n)_n$ converges to the limit $ax + by$ if and only if $\lim x_n = x$ and $\lim y_n = y$.
3. If $(x_n y_n)_n$ is obtained by multiplying two convergent sequences $x_n \to x$ and $y_n \to y$, then $\lim x_n y_n = xy$.
4. If $x_n \leqslant y_n$ for every n and $x_n \to x$, $y_n \to y$, then $x \leqslant y$ also known as **squeeze rule**. One can choose the **constant sequence** $y_n = a$ for each n. The choice $a = 0$ provides the **sign rule**.
5. If $\lim_n |x_n| = z$, then the original sequence also converges, $x_n \to z$ (**absolute convergence**).

Example B.3 Let $(x_n)_n$ be a sequence with nth term $x_n = \frac{(-1)^{n-1}}{n!}$. If we take the limit as $n \to \infty$ then $\frac{(-1)^\infty}{\infty}$. Thanks to property 5 above we instead consider the sequence of absolute values $|x_n| = \left|\frac{(-1)^{n-1}}{n!}\right| = \frac{|(-1)^{n-1}|}{|n!|}$ whose limit exists and equals zero. We conclude that $x_n \to 0$.

A convergent sequence has a unique limit. If $x_n \to x$ and $x_n \to y$ with $x \neq y$, then choosing $\epsilon = \frac{|y-x|}{4}$ we have:

$$(x - \epsilon, x + \epsilon) \cap (y - \epsilon, y + \epsilon) = \varnothing.$$

But by assumption we should have $M_1, M_2 \in \mathbb{N}$ such that implies

$$n \geqslant M_1 \Rightarrow x_n \in (x - \epsilon, x + \epsilon), \qquad \text{and} \qquad n \geqslant M_2 \Rightarrow x_n \in (y - \epsilon, y + \epsilon),$$

which is impossible. For a given sequence $(x_n)_n$ we can 'extract' a new sequence by a proper selection of a subset of indexes in $\mathbb{N}$. More precisely, let $i_1 < i_2 < i_3 < \cdots$ be positive integers and define the function $i_n \mapsto x_{i_n}$ for every $n \in \mathbb{N}$. This is nothing but a **subsequence** of $(x_n)_n$ written $(x_{i_n})_n$. For example, $(2(i_n - 1))_n$ for $i_n \in \{2, 4, 6, \ldots\} \subset \mathbb{N}$ is a subsequence of $(2(n-1))_n$ for $n \in \mathbb{N}$.

Appendix C
Limit Superior and Limit Inferior

Let $(x_n)_n$ be a bounded sequence. Thus, we can construct two monotonic sequences:

- $u_n = \sup\{x_n, x_{n+1}, \ldots\}$ such that $(u_n)_n$ is decreasing, $u_n \geqslant u_{n+1}$ for every $n \in \mathbb{N}$;
- $d_n = \inf\{x_n, x_{n+1}, \ldots\}$ such that $(d_n)_n$ is increasing, $d_n \leqslant d_{n+1}$ for every $n \in \mathbb{N}$.

For n lager enough the set $\{x_n, x_{n+1}, \ldots\}$ becomes smaller, then the set of its upper bounds is smaller too, while the set of its lower bounds is bigger. Clearly, $d_n \leqslant u_n$ for every $n \in \mathbb{N}$. This implies that for $m \leqslant n$:

$$d_m \leqslant d_n \leqslant u_n \leqslant u_m,$$

so $d_m \leqslant u_n$ for every pair of naturals m, n. We deduce that:

- $(u_n)_n$ is decreasing and bounded below by some d_m;
- $(d_n)_n$ increasing and bounded above by some u_n.

Theorem B.1 now implies $u_n \to u$ and $d_n \to d$. Furthermore, by the squeeze rule we also have $d \leqslant u$. We call **limit inferior** of $(x_n)_n$ the number

$$d = \liminf x_n = \sup_n \left\{ \inf_{k \geqslant n} x_k \right\},$$

and **limit superior** the number

$$u = \limsup x_n = \inf_n \left\{ \sup_{k \geqslant n} x_k \right\}.$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

In fact, $\inf_{k\geqslant n} x_k = d_n$ e $\sup_{k\geqslant n} x_k = u_n$. Eventually we have:

Proposition C.1 *Let $(x_n)_n$ be a bounded sequence. Then it is convergent to the limit x if and only if* $\liminf x_n = \limsup x_n = x$.

We rephrase Proposition C.1 by saying that any convergent sequence is always bounded, but the converse is not true.

Proof of Proposition C.1 ($\Leftarrow$) By definition, we have $d_n \leqslant x_n \leqslant u_n$, for every $n \in \mathbb{N}$. Now, assume $\liminf x_n = \limsup x_n = x$. We have that $\lim d_n = \lim u_n = x$, then by the squeeze rule $x_n \to x$. ($\Rightarrow$) Let us suppose that $\lim x_n = x$. By Definition B.1, for a given $\epsilon > 0$ we can find a positive integer M t.c. $n \geqslant M$ implies $|x_n - x| < \epsilon$. The sequence $\{s_M, s_{M+1}, \ldots\}$ has the lower bound $x - \epsilon$ and the upper bound $x + \epsilon$. Thus, it is bounded and we can construct two sequences $(d_n)_{n\geqslant M}$ e $(u_n)_{n\geqslant M}$ such that $x - \epsilon \leqslant d_M \leqslant u_M \leqslant x + \epsilon$. This implies:

$$x - \epsilon \leqslant \liminf x_n \leqslant \limsup x_n \leqslant x + \epsilon.$$

As $\epsilon > 0$ is an arbitrary number we obtain $x \leqslant \liminf x_n \leqslant \limsup x_n \leqslant x$, and it must hold $\liminf x_n = \limsup x_n = x$. ■

It is easily seen that $\inf\{x_n\} = -\sup\{-x_n\}$. We then have:

$$\begin{aligned}\sup_n \left\{ \inf_{k\geqslant n} x_k \right\} &= -\inf_n \left\{ -\left(-\sup_{k\geqslant n} -x_k \right) \right\} \\ &= -\inf_n \left\{ \sup_{k\geqslant n} -x_k \right\}.\end{aligned}$$

That is, $\liminf x_n = -\limsup(-x_n)$ and similarly $\limsup x_n = -\liminf(-x_n)$.

Example C.1 Let $(x_n)_n = ((-1)^n)_n$ and $(y_n)_n = ((-1)^n(2 + \frac{1}{n}) + 2)_n$ be two real sequences. We are going to compute their limits superior and inferior. For the first sequence, observe that for every $n \in \mathbb{N}$ we may find an integer k such that $k \geqslant n$ implies $(-1)^k = 1$. Clearly, $\sup\{x_n \mid k \geqslant n\} = 1$ and taking the limit

$$\lim_{n\to\infty} \sup\{x_n \mid k \geqslant n\} = \inf_n \{\sup\{x_n \mid k \geqslant n\}\} = \limsup x_n = 1.$$

In the same way we obtain $\liminf x_n = -1$. For the second sequence, with the same reasoning we find an integer k such that $k \geqslant n$ implies $\sup\{y_n \mid k \geqslant n\} = ((2 + \frac{1}{n}) + 2)$ and taking the limit:

$$\lim_{n\to\infty} \sup\{y_n : k \geqslant n\} = \inf_n \{\sup\{y_n : k \geqslant n\}\} = \limsup y_n = 4.$$

Similarly, we see that $\liminf y_n = 0$.

If $(x_n)_n$ and $(y_n)_n$ are two bounded sequences it is always true that:

$$\limsup(x_n + y_n) \leqslant \limsup x_n + \limsup y_n.$$

Moreover, since $\limsup((-x_n) + (-y_n)) \leqslant \limsup(-x_n) + \limsup(-y_n)$ we can multiply both sides by -1 and get:

$$\liminf(x_n + y_n) \geqslant \liminf x_n + \liminf y_n.$$

Thanks to Proposition C.1, every bounded sequence $(x_n)_n$ has at least one subsequence $(x_{i_n})_n \subset (x_n)_n$ which is convergent (Bolzano-Weierstrass theorem). An alternative characterization of convergence is the following:

Proposition C.2 *Let $(x_n)_n$ be a sequence. The set $\{n \in \mathbb{N} \mid |x_n - x| \geqslant \epsilon\}$ is finite if and only if $(x_n)_n$ converges to the limit x.*

Proof Suppose $x_n \to x$. Then, by Definition B.1 we have that the set $\{n \in \mathbb{N} : |x_n - x| \geqslant \epsilon\}$ contains at most $k - 1$ points. Conversely, if such is finite, then we would find an upper bound $U \in \mathbb{R}$. But choosing $k = U + 1$ we have $|x_n - x| < \epsilon$ whenever $n \geqslant k$ and so $x_n \to x$. ■

Recall that the product of a bounded sequence $(x_n)_n$ by a convergent sequence $(y_n)_n$ is always a convergent sequence in its own. For example, $((-\frac{1}{2})^n)_n$ can be written as the product between two sequence, $((-1)^n)_n$ and $((\frac{1}{2})^n)_n$, the former is bounded but not convergent while the latter is convergent to zero and we have $\lim_n (-1)^n \frac{1}{2})^n = 0$ too. We say that $(x_n)_n$ is a **Cauchy sequence** if for every $\epsilon > 0$ there exists a natural $k \in \mathbb{N}$ such that $i, j \geqslant k$ implies $|x_i - x_j| < \epsilon$. In this case by choosing $j = k$ we get:

$$x_k - \epsilon \leqslant x_i \leqslant x_k + \epsilon, \qquad \text{for every} \quad i \geqslant k.$$

It follows that $(x_n)_n$ bounded below the maximal element in $\{x_1, x_2, \ldots, x_{k-1}, x_k - \epsilon\}$, and is also bounded above by the minimal element in $\{x_1, x_2, \ldots, x_{k-1}, x_k + \epsilon\}$. Hence, every Cauchy sequence is bounded. Furthermore, it holds

$$x_k - \epsilon \leqslant d_k \leqslant u_k \leqslant x_k + \epsilon,$$

which implies:

$$x_k - \epsilon \leqslant d \leqslant u \leqslant x_k + \epsilon \iff 0 \leqslant u - d \leqslant 2\epsilon.$$

As the number $\epsilon > 0$ is arbitrary we conclude that $u = d$ and the limit inferior equals the limit superior, then the Cauchy sequence $(x_n)_n$ is also convergent. The converse is also true: any convergent sequence of real numbers is also a Cauchy sequence.

Example C.2 The sequence $((-1)^n)_n$ is bounded but is not convergent, then it is not a Cauchy sequence. In fact, its terms are never arbitrarily close each other. The sequence $((-\frac{1}{2})^n)_n$ has terms $-\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, \ldots$ that are closer each other as n is large enough and are also closer to zero. Then, the latter sequence is a Cauchy sequence.

Appendix D
Infinite Sums

Let $(x_n)_n$ be a sequence of real numbers. We can build **partial sums**:

$$s_1 = x_1, \quad s_2 = x_1 + x_2, \quad s_3 = x_1 + x_2 + x_3, \dots.$$

Clearly, $(s_n)_n$ is a new sequence for which it makes sense to verify its convergence:

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \sum_{k=1}^{n} x_k = \sum_{k=1}^{\infty} x_k.$$

The sequence of partial sums is said to be an **infinite series**, and is usually denoted by the symbol $\sum_{k=1}^{\infty} x_k$. If all terms x_n in the original sequence are all non-negative we have:

$$\text{either} \quad \sum_{k=1}^{\infty} x_k = x \qquad \text{or} \qquad \sum_{k=1}^{\infty} x_k = \infty, \qquad x_k \geqslant 0.$$

We can recover a sequence from a series by letting $x_1 = s_1$ and $x_n = s_n - s_{n-1}$, for $n \geqslant 2$. A series is said to be **absolutely convergent** if the series $\sum_k |x_k|$ is indeed convergent. This always implies that also $\sum_k x_k$ is convergent. Moreover, the convergence of the series, $\sum_k x_k = x$, implies that of the original sequence, $\lim_n x_n = 0$.

Example D.1 Let $\sum_{k=1}^{n} x_k = s_n$ be such that $x_k = \frac{1}{k}$. Define the infinite sum $\lim_n \sum_{k=1}^{n} x_k = \lim_n s_n = \sum_{k=1}^{\infty} x_k$. Observe that: $\sum_{k=1}^{\infty} x_k = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \dots + \frac{1}{8}\right) + \left(\frac{1}{9} + \dots + \frac{1}{16}\right) + \dots$, where

$$\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

$$\frac{1}{5} + \cdots + \frac{1}{8} > \frac{1}{8} + \cdots + \frac{1}{8} = \frac{1}{2}$$
$$\frac{1}{9} + \cdots + \frac{1}{16} > \frac{1}{16} + \cdots + \frac{1}{16} = \frac{1}{2}$$

and so on. The behavior of the partial sums is that of a divergent sequence:

$$\sum_{k=1}^{n} \frac{1}{k} > 1 + \underbrace{\left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right)}_{=2} + \underbrace{\left(\frac{1}{2} + \cdots + \frac{1}{2}\right)}_{=3} + \cdots$$

We conclude that $\lim_n \sum_{k=1}^{n} \frac{1}{k} = \infty$, i.e. the series diverges. The infinite sum $\sum_{k=1}^{\infty} \frac{1}{k}$ is also known as *harmonic series*.

Example D.2 Let $(x_n)_n$ be the sequence of Example B.2, with nth term $x_n = a^n$. Let us consider the infinite sum $\sum_{k=1}^{\infty} a^k$ and show that the nth partial sum is $a\,\frac{1-a^n}{1-a}$. Let

$$s_n = \sum_{k=1}^{n} a^k = a + a^2 + a^3 + \cdots + a^n$$

and write

$$a s_n = a^2 + a^3 + \cdots + a^{n+1}.$$

Subtracting terms by terms the above equalities we indeed get $s_n(1-a) = a(1-a^n)$. If we assume $|a| < 1$, then $\lim_n s_n = \frac{a}{1-a}$ and the series converges. The infinite sum $\sum_{k=1}^{\infty} a^k$ is also known as a *geometric series*. We may write $\sum_{k=0}^{\infty} a^k$ so that the series start from $a^0 = 1$. The previous computation of the nth term still applies and gives $\frac{1-a^n}{1-a}$ as the new partial sum, which converges to $\frac{1}{1-a}$ as $n \to \infty$ and $|a| < 1$.

A **double series** is defined by partial sums of type $(s_{nm})_{nm}$, where $n, m \in \mathbb{N}$. The terms $s_{11}, s_{12}, s_{21}, s_{13}, \ldots$ can be collected into a matrix with an infinite number of rows and columns. If we assume $s_{nm} \geqslant 0$ for every $n, m \in \mathbb{N}$, then we can always write:

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} s_{nm} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} s_{nm}. \tag{D.1}$$

Indeed, the previous equality holds whenever both sums are finite. On the other hand, we can write:

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} s_{nm} \geqslant \sum_{i=1}^{n} \sum_{j=1}^{m} s_{ij} = \sum_{j=1}^{m} \sum_{i=1}^{n} s_{ij},$$

and taking the limit as $n, m \to \infty$ we get

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} s_{nm} \geqslant \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} s_{nm}.$$

This reasoning applies as well by inverting the indexes of the sums, then equality (D.1) must hold.

Appendix E
Open Subsets of Real Numbers

A **topological space** is a non-empty set X together with a collection τ of its subsets such that:

T1 X and $\varnothing$ belong to τ,
T2 for arbitrary unions $\cup_i A_i$ of sets A_i in τ it holds $\cup_i A_i \in \tau$,
T2 finite intersections $\cap_{i=1}^n A_i$ of sets A_i in τ are again here, $\cap_{i=1}^n A_i \in \tau$.

The collection τ is said to contain **open sets**, actually subsets of X. We know that a given subset $O \subset \mathbb{R}$ sis open if for every $x \in O$ there exists an open interval (a, b) containing x and which is contained in O, i.e. $x \in (a, b) \subset O$. We can choose the **neighborhood** $O = (x - \delta, x + \delta)$ centered at x, where as usual $\delta > 0$. Recall that an open interval is the set $(a, b) := \{x \in \mathbb{R} \mid a < x < b\}$ and a closed interval is the set $[a, b] := \{x \in \mathbb{R} \mid a \leqslant x \leqslant b\}$. Half-open intervals $(a, b]$ and $[a, b)$ are defined analogously. In the former case it may happen $a = -\infty$ and in the latter $b = \infty$. A set C is said to be **closed** if its complement $C^c = \mathbb{R} \setminus C$ is indeed open. It can be shown that a set $O \in \tau$ is open if and only if its complement is closed. A function $f : A \to \mathbb{R}$ with domain $A \subset \mathbb{R}$ is said to be **continuous** at a point $c \in A$ if for every neighborhood $(f(c) - \epsilon, f(c) + \epsilon) \subset f(A)$ centered at $f(c)$ will correspond a neighborhood $(c - \delta, c + \delta) \subset A$ centered at c. we can write:

$$\forall \epsilon > 0,\ \exists \delta > 0 \,\text{t.c.}\, |x - c| < \delta \Rightarrow |f(x) - f(c)| < \epsilon, \tag{E.1}$$

or more compactly $\lim_{x \to c} f(x) = f(c)$. This is also the classical $\epsilon - \delta$ definition of limit. The function f is said to be simply continuous if it is continuous at every point of its domain A. If a function is not continuous at a point c on its domain we say that it is **discontinuous** at c. A function f whose domain contains an interval $[c, c + \delta)$ for some $\delta > 0$ is **continuous on the right** at c if:

$$\text{for each} \quad (x_n)_n \subset [c, c + \delta) \quad \text{such that} \quad x_n \to c, \quad \text{then} \quad f(x_n) \to f(c).$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

Similarly, f is said to be **continuous on the left** at c if the above limit exists with respect to a sequence $(x_n)_n \subset (c-\delta, c]$. Clearly, a function which is continuous at c must be both continuous on the right and on the left at c. The classical rephrasing of these sequential limits read: $\lim_{x\downarrow c} f(x) = f(c)$, where $x \downarrow c$ means $x \to c$ and $c < x$, and $\lim_{x\uparrow c} f(x) = f(c)$, where $x \uparrow c$ means $x \to c$ and $c > x$. A function has a *jump discontinuity* at c whenever the right-hand limit and the left-hand limit both exist (finite) but are different. Recall that the sum, the product and the quotient of two continuous functions is again a continuous function. In the latter case we must assume the function on the denominator being different from 0. There is an equivalent formulation of continuity for functions between topological spaces which is very useful. Here we specialize to functions from $A \subset \mathbb{R}$ to $\mathbb{R}$, where $(\mathbb{R}, \tau)$ denotes the real topological space.

Theorem E.1 *A function $f : A \to \mathbb{R}$ is said to be continuous if and only if for every open set $O \in \tau$ the inverse image $f^{-1}(O)$ is an open set in τ.*

Proof ($\Rightarrow$) By hypotheses f is continuous:

$$\exists\, \epsilon > 0 \text{ such that } x \in (c-\delta, c+\delta) \Rightarrow f(x) \in (f(c)-\epsilon, f(c)+\epsilon), \quad \text{for some } \delta > 0.$$

If $c \in f^{-1}(O)$ then $f(c) \in O$, but O is open so

$$\exists\, \epsilon > 0 \text{ such that } (f(c) - \epsilon, f(c) + \epsilon) \subset O.$$

This implies that $f(x) \in O$ if and only if $x \in (c-\delta, c+\delta)$, hence the open interval is a neighborhood of c and is contained in $f^{-1}(O)$, which is an open set. ($\Leftarrow$) Assume that O is open. Then, $f^{-1}(O)$ is open too. Pick $c \in A$ and $\epsilon > 0$ and consider the open neighborhood $(f(c) - \epsilon, f(c) + \epsilon)$. We can let $O = (f(c) - \epsilon, f(c) + \epsilon)$. The inverse image $f^{-1}(O)$ is also open and contains the point c. We can find $\epsilon > 0$ such that $(c - \delta, c + \delta) \subset f^{-1}(O)$, then it follows:

$$x \in (c - \delta, c + \delta) \subset f^{-1}\left((f(c) - \epsilon, f(c) + \epsilon)\right) \Rightarrow f(x) \in (f(c) - \epsilon, f(c) + \epsilon),$$

that is f is continuous. ■

The following result is fundamental:

Theorem E.2 *Let $\{O_i\}_{i\in I}$ be an indexed family of open sets in $(\mathbb{R}, \tau)$. Then:*

- *$\cup_{i\in I} O_i$ is open*
- *the finite intersection $\cap_{i=1}^n O_i$ is open.*

Proof For every $x \in \cup_{i\in I} O_i$ we can find $i \in I$ such that $x \in O_i$. Since O_i is open there exists an open neighborhood $(x - \delta, x + \delta) \subset O_i$, for some $\delta > 0$. But then $(x - \delta, x + \delta) \subset \cup_{i\in I} O_i$, and so the union is an open set. If $O_1, \ldots, O_n$ are open sets and x lies in $\cap_{i=1}^n O_i$, then $x \in O_i$ for every $i = 1, \ldots, n$. But O_i is open, thus there exists $\delta_i > 0$ such that $(x - \delta_i, x + \delta_i)$ is contained in O_i. Now, choose $0 < \delta = \min\{\delta_1, \ldots \delta_n\}$ and get $(x - \delta, x + \delta) \subset (x - \delta_i, x + \delta_i) \subset O_i$, for all

i. We conclude that $(x-\delta, x+\delta)$ is also contained in $\cap_{i=1}^{n} O_i$, which in turn is an open set. ■

Example E.1 Take the sequence of open intervals $O_i = (-\frac{1}{i}, b)$ with $b \in \mathbb{R}$. The countable intersection $\cap_{i=1}^{\infty} O_i = [0, b)$ is neither an open set nor a closed set in $(\mathbb{R}, \tau)$.

Eventually, it can be shown that any open set O in $\mathbb{R}$ can be covered by infinitely many open intervals $O = \cup_{i=1}^{\infty}(a_i, b_i), \qquad a_i \leqslant b_i, \qquad i = 1, 2, \ldots$.

Remark E.1 The continuity of a function in Theorem E.1 can be rephrased as follows. Let $O = \cup_{i=1}^{\infty}(a_i, b_i)$ be open and define $f : A \to \mathbb{R}$. The inverse image

$$f^{-1}(O) = f^{-1}\big(\cup_{i=1}^{\infty}(a_i, b_i)\big) = \cup_{i=1}^{\infty} f^{-1}((a_i, b_i))$$

is a countable union of open sets.

The Borel σ-algebra is generated by the open sets in $\mathbb{R}$, so let $\mathcal{O} = \{O \subset \mathbb{R} : O \text{ is open}\}$ and get $\mathcal{B} = \sigma(\mathcal{O})$.

Remark E.2 The length of a bounded half-open intervals $(a, b]$, $[a, b)$ is $b-a > 0$. This remains true for open and closed intervals with finite endpoints.

There is a sequential characterization of continuity for functions $f : A \subset \mathbb{R} \to \mathbb{R}$. Equation (E.1) implies that for every $x \in A$ there exists a sequence $(x_n)_n$ in A convergent to x, with $x_n \neq x$, such that:

$$\lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right) = f(x). \tag{E.2}$$

This definition of continuity by sequences is equivalent to that given in Theorem E.2.

Example E.2 Let $f(x) = \frac{\sin x}{x}$. Take a sequence $(x_n)_n$ in the domain of f such that $x_n \to 0$, with $x_n \neq 0$ and $n \to \infty$. Then, the $\epsilon - \delta$ definition $\lim_{x\to 0} f(x)$ is equivalent to:

$$\lim_{n\to\infty} \frac{\sin x_n}{x_n} = \frac{\sin\left(\lim_n x_n\right)}{\left(\lim_n x_n\right)} = 1.$$

The function f is continuous in zero and $f(0) = 1$ coincides with the limit.

Example E.3 The function $f(x) = \sin x$ has no limit for $x \to \infty$. To show this, let us find two sequences $(x_n)_n$ and $(y_n)_n$ in the domain of f such that $x_n \to \infty$ and $y_n \to \infty$, but $\sin(x_n)$ and $\sin(y_n)$ converge to two distinct finite limits. This clearly contradict the sequential characterization of a *unique* finite limit. Indeed, choosing $x_n = 2\pi n$ and $y_n = 2\pi n + \frac{\pi}{2}$, we have $x_n, y_n \to \infty$ for $n \to \infty$ but:

$$\lim_{n\to\infty} \sin x_n = \sin\left(\lim_{n\to\infty} x_n\right) = 0,$$

as $\sin(2\pi n) = 0$ for every $n \in \mathbb{N}$, and

$$\lim_{n\to\infty} \sin y_n = \sin\left(\lim_{n\to\infty} y_n\right) = 1,$$

as $\sin(2\pi n + \frac{\pi}{2}) = 1$ for every $n \in \mathbb{N}$.

The derivative of f at a point $c \in (a, b)$, where the open interval is contained in the domain of f, is given by

$$\lim_{x\to c} \tfrac{f(x)-f(c)}{x-c} = \lim_{h\to 0} \tfrac{f(c+h)-f(c)}{h} = \lim_{n\to\infty} \tfrac{f(x_n)-f(c)}{x_n-c},$$

and is denoted by $f'(c)$ provided the limit exists and is finite. Note that in the above definition, the third sequential version requires that $x_n \to c$ and $x_n \neq c$, for any sequence $(x_n)_n$ in the domain of f. A function is said to be **differentiable** in (a, b) if its derivative exists at every point $x \in (a, b)$. The derivative is itself a function $f'(x)$, and we say that f is of class C^1 if its derivative exists and is a continuous function on an open interval. Recall that for every continuous function $f : [a, b] \to \mathbb{R}$ the following hold.

1. The Weierstrass theorem: there exist points c and d in $[a, b]$ such that $f(c) \leqslant f(x) \leqslant f(d)$, for $x \in [a, b]$.
2. The Lagrange theorem or mean value theorem: if $f'(x)$ exists for every $x \in (a, b)$, then $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$.
3. If $f(a) < 0$ and $f(b) > 0$ (or viceversa), then there exists $c \in (a, b)$ such that $f(c) = 0$.
4. The rule sign: if $f(c) > 0$ for some $c \in (a, b)$ then there exists $\delta > 0$ such that $f(x) > 0$ for $x \in (c - \delta, c + \delta)$.
5. The intermediate value theorem: if $f(a) < m < f(b)$ then there exists some $c \in (a, b)$ such that $f(c) = m$.

A function g defined on an open interval $I \subset \mathbb{R}$ (bounded or unbounded) is said to be **convex** if

$$g(\lambda x_1 + (1 - \lambda)x_2) \leqslant \lambda g(x_1) + (1 - \lambda)g(x_2), \quad \forall\, x_1, x_2 \in I,\ \lambda \in [0, 1].$$

For $\lambda = \frac{1}{2}$ we have $g(\frac{x_1+x_2}{2}) \leqslant \frac{g(x_1)+g(x_2)}{2}$. For example, $g(x) = x^2$, $h(x) = \mathrm{e}^x$ and $k(x) = |x|$ are all convex with domain $I = \mathbb{R}$. The latter is not differentiable at $x = 0$. A convex function is continuous (in the interior of its domain), with finite left and right derivatives g'_- and g'_+. If it is twice differentiable its second derivative $g''(x) \geqslant 0$, a sufficient condition for convexity. The graph of a convex function g lies above its *tangent*, i.e. we may find some $a \in \mathbb{R}$ such that for a fixed $x_0 \in I$ it holds

$$g(x_0) + a(x - x_0) \leqslant g(x), \qquad \text{for every } x \in I \quad \text{(support line)}.$$

A function g defined on I is said to be **concave** if $-g(x)$ is convex for all $x \in I$.

Appendix F
Topology on n-Dimensional Euclidean Spaces

An open set can be defined through open neighborhoods, i.e. open intervals, and any open set is the union of all neighborhoods of its points.[1] This still holds in the n-dimensional Euclidean space $\mathbb{R}^n$, the set of all points/vectors $\mathbf{x} = (x_1, \ldots, x_n)$. We remind the definition of Euclidean distance or metric and its relation with the Euclidean norm:

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}. \tag{F.1}$$

An **open ball** around $\mathbf{x}$ with radius $\epsilon > 0$ is the set $B_\epsilon(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n \mid \|\mathbf{x}-\mathbf{y}\| < \epsilon\}$, see Fig. F.1. For $n = 1$ it is an open neighborhood; for $n = 2$ it is an open disk in the Cartesian plane; for $n = 3$ it equals the set of all points in $\mathbb{R}^3$ strictly inside a sphere with center $\mathbf{x}$ and radius ϵ. Open sets in $\mathbb{R}^n$ are generated by the family of open balls. More specifically, a subset $O \subset \mathbb{R}^n$ is said to be **open** if any $\mathbf{x} \in O$ is the center of an open ball entirely contained in the set, that is for every $\epsilon > 0$ and all $\mathbf{x} \in O$ we have $B_\epsilon(\mathbf{x}) \subset O$. For example, the Cartesian product of n open intervals in $\mathbb{R}$,

$$(a_1, b_1) \times \cdots \times (a_n, b_n), \qquad a_i \leqslant b_i,\ i = 1, \ldots, n,$$

is an open set by the above definition. We known that in the n-dimensional Euclidean space the Borel σ-algebra $\mathcal{B}^n$ is generated by the collection of all open sets. Indeed, for every $\mathbf{x} \in O \subset \mathbb{R}^n$ there exists an open ball with center $\mathbf{x}$ and rational radius $q \in \mathbb{Q}_+$, which is contained in the open set O. Thus O can be defined by the

[1] The class of all open neighborhoods in $\mathbb{R}$ is called a base for the *metric* topology. This is like the generator of a σ-algebra.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

countable union of all open balls centered at each of its points:

$$O = \cup_{q \in \mathbb{Q}_+} B_q(\mathbf{x}), \qquad \mathbf{x} \in O$$

This implies that for every $O \in \mathcal{B}^n$ we have $B_q(\mathbf{x}) \in \mathcal{B}^n$, then the Borel σ-algebra of $\mathbb{R}^n$ also contains the open balls. Conversely, the collection of all open balls with rational radius generates $\mathcal{B}^n$. Let F any subset of $\mathbb{R}^n$, we say that $\mathbf{x} \in F$ is an **interior point** of F if $\mathbf{x}$ is the center $\mathbf{x}$ of an open ball contained in F. The collection of all interior points of F is denoted by $\text{int}(F)$. If we take any $\mathbf{x} \in \text{int}(F)$ then there is some $\epsilon > 0$ such that $B_\epsilon(\mathbf{x}) \subset F$. Now, take any $\mathbf{y} \in B_\epsilon(\mathbf{x})$ and choose $\delta > 0$ small enough so that $B_\delta(\mathbf{x}) \subset B_\epsilon(\mathbf{x})$. Then, $B_\delta(\mathbf{x}) \subset F$ and eventually $\mathbf{y} \in \text{int}(F)$. This shows that the interior of F is itself an open set. Hence, we can rephrase the definition of an open set O by saying that all its points are interior points. Furthermore, any open set O is a neighborhood of a point $\mathbf{x}$ whenever O contains some open ball around $\mathbf{x}$. Note that in general $\text{int}(F) \subset F$. Using the triangle inequality and Fig. F.1 we can see that any open ball $B_\epsilon(\mathbf{x})$ is indeed an open set in $\mathbb{R}^n$. The set $F = \{\mathbf{x} = (x, y) \mid x < y\}$ is open in $\mathbb{R}^2$. A point $\mathbf{x} \in \mathbb{R}^n$ is said to be a **boundary point** of a set $F \subset \mathbb{R}^n$ if every open ball centered at $\mathbf{x}$ contains at least one point in F and at least one in F^c, that is $B_\epsilon(\mathbf{x}) \cap F \neq \varnothing$ and $B_\epsilon(\mathbf{x}) \cap F^c \neq \varnothing$. A set may includes none, some or all its boundary points. If $\mathbf{x} \in F$ but $\mathbf{x} \notin \text{int}(F)$ then $F \cap B_\epsilon(\mathbf{x}) \neq \varnothing$ for every $\epsilon > 0$, so $\mathbf{x}$ is a boundary point of F. The set of all boundary points of a set F is called the boundary of F and is denoted by ∂F. For example, the boundary of $B_\epsilon(\mathbf{y}) \subset \mathbb{R}^3$ is the sphere $S^3 := \{\mathbf{y} \in \mathbb{R}^3 \mid \|\mathbf{x} - \mathbf{y}\| = \epsilon\}$. A set is closed if it is the complement of an open set. Equivalently, a set F is closed if and only if it contains all its boundary points, and $F \cup \partial F =: \text{cl}(F)$ called the **closure** of F. It holds $F \subset \text{cl}(F)$. A point $\mathbf{x}$ lies in $\text{cl}(F)$ provided that every open ball centered at this point is such that $B_\epsilon(\mathbf{x}) \cap F \neq \varnothing$. For example, the Cartesian product

$$[a_1, b_1] \times \cdots \times [a_n, b_n], \qquad a_i \leqslant b_i, \ i = 1, \ldots, n$$

is a closed set. Observe that F and F^c have the same boundary points.[2] It can be shown that:

- $\text{int}(F)$ is the largest open set contained in F, that is $\text{int}(F) = \cup_{D \in \mathcal{D}} D$ with $\mathcal{D}$ being the collection of all open subsets of F;
- $\text{cl}(F)$ is the smallest closed set containing F, that is $\text{int}(F) = \cap_{G \in \mathcal{G}} G$ with $\mathcal{G}$ being the collection of all closed sets containing F.

Another useful characterization of closeness is the following:

Theorem F.1 *A set F in $\mathbb{R}^n$ is closed if and only if every convergent sequence of points in F has its limit in F.*

[2] Any open set contains none of its boundary points.

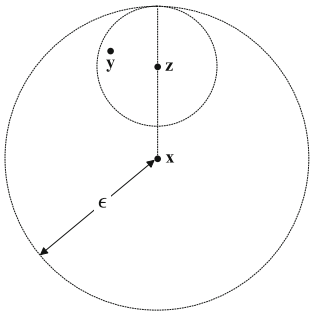


Fig. F.1 The open ball $B_\epsilon(\mathbf{x})$ is an open set

Recall that a sequence $(\mathbf{x}_n)_{n\in\mathbb{N}}$ in $\mathbb{R}^n$ converges to a point $\mathbf{x} \in \mathbb{R}^n$ provided that distances vanish, $\|\mathbf{x}_n - \mathbf{x}\| \to 0$ as $n \to \infty$, that is for each $\epsilon > 0$ there exists $k \in \mathbb{N}$ such that $\mathbf{x}_n \in B_\epsilon(\mathbf{x})$ for all $k > n$. Indeed, a point $\mathbf{x} \in \mathbb{R}^n$ lies in $\mathrm{cl}(F)$ if and only if $\mathbf{x}$ is the limit of a sequence $(\mathbf{x}_n)_n \subset F$. By Theorem F.1, the points that make F a closed set are also called **limit points**. They can be alternatively defined as those element $\mathbf{x}$ in $\mathbb{R}^n$ such that $B_\epsilon(\mathbf{x})$ contains infinitely many points of F for every $\epsilon > 0$, i.e. for each radius the corresponding open ball contains at least one point distinct from $\mathbf{x}$. We can summarize all the above by stating that $F \subset \mathbb{R}^n$ is open if and only if $F = \mathrm{int}(F)$, and closed if and only if $F = \mathrm{cl}(F)$. Some kind of open and closed sets in the n-dimensional Euclidean space are determined by continuous functions. A map $f : F \subset \mathbb{R}^n \to \mathbb{R}$ is said to be continuous at a point $\mathbf{x} \in F$ if for every $\delta > 0$ there exists a $\epsilon > 0$ such that the distance $|f(\mathbf{y}) - f(\mathbf{x})|$ is made strictly smaller that δ for all $\mathbf{y} \in F$ with $\|\mathbf{y} - \mathbf{x}\| < \epsilon$. We can equivalently write $f(B_\epsilon(\mathbf{x}) \cap F) \subset B_\delta(f(\mathbf{x}))$. In fact, $f^{-1}(O)$ is open whenever $O \subset \mathbb{R}^n$ is open and f is continuous. If f is continuous at very point in its domain F then we say that it is continuous on the whole F. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous function. we have:

- the sets $\{\mathbf{x} \in \mathbb{R}^n \mid f(\mathbf{x}) > 0\}$ and $\{\mathbf{x} \in \mathbb{R}^n \mid f(\mathbf{x}) < 0\}$ are open;
- the sets $\{\mathbf{x} \in \mathbb{R}^n \mid f(\mathbf{x}) \geqslant 0\}$ and $\{\mathbf{x} \in \mathbb{R}^n \mid f(\mathbf{x}) \leqslant 0\}$ are closed;
- the set $\{\mathbf{x} \in \mathbb{R}^n \mid f(\mathbf{x}) \neq 0\}$ is open and the set $\{\mathbf{x} \in \mathbb{R}^n \mid f(\mathbf{x}) = 0\}$ is closed,

see Fig. F.2. Recall the sequential definition of continuity: a function f is continuous at a point $\mathbf{x}$ in its domain if here there exists a sequence $(\mathbf{x}_n)_n \neq \mathbf{x}$ which has limit $\mathbf{x}$ and such that $\|f(\mathbf{x}_n) - f(\mathbf{x})\| \to 0$ as $n \to \infty$. For vector functions $\mathbf{f} : F \subset \mathbb{R}^n \to \mathbb{R}^m$, with $n \neq m$, all the above definitions apply ceteris paribus. Be warning that $\mathbf{f} = (f_1, \ldots, f_m)$ is indeed a vector of functions $f_i : F \subset \mathbb{R}^n \to \mathbb{R}$, each $i \in \{1, \ldots, m\}$. Thus, the continuity of $\mathbf{f}$ is understood in the appropriate norm/open ball. Notwithstanding, we have that $\mathbf{f}$ is continuous at $\mathbf{x} \in F$ if and only if each component function f_i is continuous at the same point. A set $F \subset \mathbb{R}^n$ is **bounded** if there exists $M > 0$ such that $\|\mathbf{x}\| \leqslant M$ for all $\mathbf{x} \in F$. Since for any convergent sequence the set $\{\mathbf{x}_1, \mathbf{x}_2 \ldots\} \subset \mathbb{R}^n$ is bounded in the sense of the aforementioned definition we have that a set $F \subset \mathbb{R}^n$ is bounded if and only if every sequence in

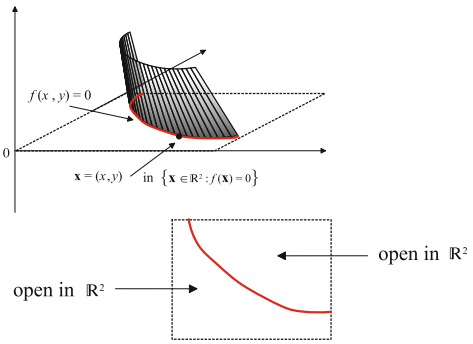


Fig. F.2 Bold line: those $\mathbf{x} \in \mathbb{R}^2$ such that $f(\mathbf{x}) = 0$ form a closed set. Its complement is the union of two open set, then is itself open

F has a subsequence $\{\mathbf{x}_{n_k}\} \subset F$ which converges to a limit. Closed and bounded subsets of $\mathbb{R}^n$ are said to be **compact**.

Theorem F.2 *A set $F \subset \mathbb{R}^n$ is compact if and only if every sequence of points in F has a subsequence that converges to a point belonging to F.*

Hence, boundedness implies convergence of subsequences in F, while closeness guarantees that the limit points are elements of F and together are equivalent to compactness. For example, The closure of the open ball $B_\epsilon(\mathbf{x})$ is a closed and bounded set in $\mathbb{R}^2$, then $\mathrm{cl}(B_\epsilon(\mathbf{x})) = B_\epsilon(\mathbf{x}) \cup \partial B_\epsilon(\mathbf{x})$ is a compact set. The above is the sequential characterization of compactness. Equivalently, let $\mathcal{G} = (G_i)_{i \in J}$ be a collection of open subsets of a given set $F \subset \mathbb{R}^n$, also called an **open cover** of F. A **subcover** of $\mathcal{G}$ is a subfamily of open sets that also cover F.

Theorem F.3 *A set $F \subset \mathbb{R}^n$ is compact (i.e. bounded and closed) if and only if every open cover of F has a finite subcover.*

A continuous function $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$ preserves compactness: for every compact set $K \subset \mathbb{R}^n$ the image $\mathbf{f}(K) \subset \mathbb{R}^m$ is compact too. For example let $n = m = 1$ and define a function $f : [a, b] \to \mathbb{R}$, then $f([a, b])$ is again a closed and bounded interval. This fact implies the Weierstrass theorem. So for $m = 1$ we have:

Theorem F.4 *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous function. Its restriction $f|K$ to a compact set $K \subset \mathbb{R}^n$ is such that $\{y \in \mathbb{R} \mid y = f(x),$ for some $\mathbf{x} \in K\}$ is bounded. Then, f attains a maximum and a minimum values.*

Appendix G
Vector Spaces and Normed Spaces

A **vector space** over the field of real numbers $\mathbb{R}$ is a a nonempty set V together with operations $(v, w) \mapsto v + w$ from $V \times V$ to $\mathbb{R}$ and $(\alpha, v) \mapsto \alpha v$ from $\mathbb{R} \times V$ to $\mathbb{R}$ such that:

(a1) $(v + w) + y = v + (w + y)$, for all $v, w, y \in V$;
(a2) $v + w = w + v$ for all $v, w \in V$;
(a3) there is an element $0 \in V$ with the property $v + 0 = v$, for all $v \in V$;
(a4) for each $v \in V$ there is an element $-x \in V$ with the property $v + (-v) = 0$
(s1) $1v = v$ holds for all $v \in V$;
(s2) $\alpha(\beta v) = (\alpha\beta)v$, for all $\alpha, \beta \in \mathbb{R}$ and all $v \in V$;
(d1) $(\alpha + \beta)v = \alpha v + \beta v$, for all $\alpha, \beta \in \mathbb{R}$ and all $v \in V$;
(d2) $\alpha(v + w) = \alpha v + \alpha w$, for all $\alpha \in \mathbb{R}$ and all $v, w \in V$.

Properties (a1)–(a4) refer to vector addition; properties (s1) and (s2) refer to scalar multiplication; properties (d1) and (d2) are distributive laws. Elements of V are called *vectors*.

Example G.1 The fundamental example of vector space is $\mathbb{R}^n$. For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, vector addition $\mathbf{x} + \mathbf{y}$ is meant componentwise, $x_i + y_i$ each $i = 1, \ldots, n$. Scalar multiplication $\alpha\mathbf{x}$ is done in the same fashion, αx_i for any $\alpha \in \mathbb{R}$ and and each $i = 1, \ldots, n$. The element $\mathbf{0} = (0, \ldots, 0) \in \mathbb{R}^n$ satisfies property (a3) and is called the *zero vector* or the *origin* of $\mathbb{R}^n$. We can identify elements $\mathbf{x} \in \mathbb{R}^n$ with row vectors, $1 \times n$, or column vectors $\mathbf{x}'$, after matrix transposition. If $n = 2$ we have the following geometric picture. Any element $\mathbf{x}$ is represented as an arrow in the Cartesian plane starting from the origin $\mathbf{0}$ and directed towards the point located at (x_1, x_2), the coordinates of $\mathbf{x}$. The vector $-\mathbf{x}$ points in the reverse direction. Addition between two vectors is called the **parallelogram law**. Scalar multiplication $\alpha\mathbf{x}$ modifies the length of the arrow by a factor α and also its direction if $\alpha < 0$. See Fig. G.1. Vectors $\mathbf{y}$ and $\mathbf{x}$ are parallel if $\mathbf{y} = \alpha\mathbf{x}$ for some scalar $\alpha \in \mathbb{R}$. All the above hold for the three-dimensional Euclidean space, see Appendix J.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

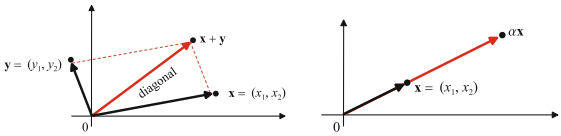


Fig. G.1 **y** is the displacement required to move **x** to **x** + **y**

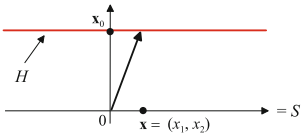


Fig. G.2 The hyperplane H as a shift of the set $S = \mathbb{R}$ by $\mathbf{x}_0$

A **subspace** of a vector space V is a subset $V' \subset V$ such that $\alpha v + \alpha w \in V'$ for all $\alpha, \beta \in \mathbb{R}$ and all $v \in V$, i.e. V' is a vector space in its own with respect to the same operations $+$ and $\cdot$ restricted to V'. If we let $v_0 \in V$ fixed and consider the set H of all vectors of the form $w = v + v_0$ where v range over the whole subspace V' while v_0 does not belong necessarily to V', it is called a **hyperplane** and is obtained by shifting the subspace V' by the vector v_0. Note that H in general is not itself a vector space (verify this).

Example G.2 In $V = \mathbb{R}^3$ the set $\mathbb{R}$ is a line and the set $\mathbb{R}^2$ is a plane. We can represents the hyperplane H as the set $\{\mathbf{y} \in \mathbb{R}^2 \mid \mathbf{y} = \mathbf{x} + \mathbf{x}_0,\ \mathbf{x} \in S,\ \mathbf{x}_0 \text{ fixed}\}$, see Fig. G.2, where S is the set of all those $x \in \mathbb{R}^2$ such that $\mathbf{x} = (x_1, 0)$ for every $x_1 \in \mathbb{R}$.

A linear combination of n vectors $v_1, \ldots, v_n \in V$ is $\sum_{i=1}^{n} \alpha_i v_i$, where $\alpha_i \in \mathbb{R}$ each i. A (finite) collection of vectors $\{v_1, \ldots, v_m\}$ is said to be **linearly independent** if we cannot find scalars $\alpha_1, \ldots, \alpha_m$, not all zero, such that their linear combination is the zero vector. An arbitrary set V' of vectors of V is said to be linearly independent if any nonempty finite subset of V' contains linearly independent vectors. A nonempty subset $V' \subset V$ of a vector space V can be used to form a third set $\mathcal{S}$ containing all linear combinations of elements in V'. The set $\mathcal{S}$ is a subspace called the **span** of V' denoted by $\text{span}(V')$. Since V' generate $\mathcal{S}$ we write $\mathcal{S} = \text{span}(V')$. If V contains a finite set of n linearly independent vectors and all other sets of $n + 1$ vectors is linear dependent, then we shall say that the **dimension** of V is n, written $\dim V = n$, and say that V is a **finite dimensional vector space** whatever V may contains an infinite number of vectors. If $S \subset V$ and S is a linearly independent set spanning V, then S is a **basis** of V.

Example G.3 The set $\mathbb{R}^n$ of all n-tuples $\mathbf{x} = (x_1, \ldots, x_n)$ is a finite dimensional vector space. Indeed the finite subset $\mathcal{S} = \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ of points $\mathbf{e}_i$ whose all coordinates are zero but the ith, which is equal to 1, is linearly independent and each $\mathbf{x} \in \mathbb{R}^n$ can be uniquely expressed as a linear combination $\mathbf{x} = \alpha_1 \mathbf{e}_1 + \cdots + \alpha_n \mathbf{e}_n$ for scalars α_i, all not zero. Thus, $\mathbb{R}^n = \text{span}(\mathcal{S})$, i.e. $\mathcal{S}$ spans $\mathbb{R}^n$, and $|\mathcal{S}| = n$ so that $\dim \mathbb{R}^n = n$.

A function $L : V_1 \to V_2$ defined between two vector spaces is said to be **linear** if $L(\alpha v + \alpha w) = \alpha L(v) + \alpha L(w)$, for all $\alpha, \beta \in \mathbb{R}$ and all $v \in V_1$. If $V_2 = \mathbb{R}$ we

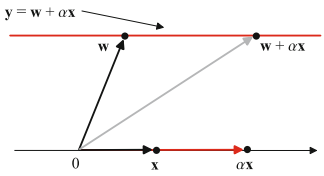


Fig. G.3 The line $\mathbf{y} = \mathbf{w} + \alpha\mathbf{x}$ with constant $\mathbf{w}$, $\mathbf{x}$ and $\mathbf{x} \neq \mathbf{0}$

call such an L a **linear functional**. Any bijective linear function has a linear inverse. A **line segment** connecting $v \in V$ and $w \in V$ is the set of points in V of the form $\lambda v + (1 - \lambda)w$ with $\lambda \in [0, 1]$. A subset $C \subset V$ of a vector space is **convex** if for each pair of $v, w \in C$ the line segment connecting them is included in C.

Example G.4 In $\mathbb{R}^3$ the equation $\mathbf{y} = \alpha\mathbf{x}$ represents all points on the line through the origin in the direction of $\mathbf{x}$ whenever α ranges over $\mathbb{R}$ and $\mathbf{x} \neq \mathbf{0}$. If $\mathbf{w}, \mathbf{x} \in \mathbb{R}^3$ and $\mathbf{x} \neq \mathbf{0}$ the equation $\mathbf{y} = \mathbf{w} + \alpha\mathbf{x}$ is a line through $\mathbf{w}$ in the direction of $\mathbf{x}$. See Fig. G.3.

A vector space V enables us to do algebraic operations with its elements. The linear structure of V can be enriched by introducing a function $\| \| : V \to \mathbb{R}$ such that

(1) $\|v\| \geqslant 0$ for each $v \in V$, and $\|v\| = 0$ if and only if $v = 0$;
(2) $\|\alpha v\| = |\alpha| \, \|v\|$ for all $\alpha \in \mathbb{R}$ and $v \in V$;
(3) $\|v + w\| \leqslant \|v\| + \|w\|$ for all $v, w \in V$.

The set V endowed with such a function is called a **normed space** and $\| \|$ is a **norm**. We sometimes shall write $(V, \| \|)$. Property (3) is called **triangle inequality**. The norm provides a vector space V with a topological structure. Indeed, $\|v - w\| = d(v, w)$ defines a metric for all $v, w \in V$. It turns out that concepts like open ball, open set, closed set and convergent sequences can be adapted to normed vector spaces.

Example G.5 Let $[a, b]$ be a closed bounded interval in $\mathbb{R}$ and let $C[a, b]$ be the set of all continuous real-valued functions f on $[a, b]$. It is easily verified that $C[a, b]$ is a vector space, provided that sums of functions and products between real numbers and functions are understood componentwise. The map $\| \| : C[a, b] \to \mathbb{R}$ defined by $\|f\| := \int_{[a,b]} |f| \mathrm{d}m$ is a norm on $C[a, b]$. Another norm can be defined via the formula $\|f\| := \sup\{|f(x)| \mid x \in [a, b]\}$, since any continuous function on the compact set $[a, b]$ is also bounded.

Example G.5 suggests that we can define different norms on the same vector space. Nonetheless, it can be shown that on a finite-dimensional normed vector space all norms are equivalent and define the same open sets. We shall provide another characterization of a vector space V. The **inner product** of two vectors $u, w \in V$, denoted by $\langle u, w \rangle$, is defined to be a function satisfying:

(I1) $\langle u, w \rangle = \langle w, u \rangle$
(I2) $\langle u + z, w \rangle = \langle u, w \rangle + \langle z, w \rangle$

(I3) $\alpha\langle u, w\rangle = \alpha\langle u, w\rangle$
(I4) $\langle u, u\rangle \geqslant 0$, with $\langle u, u\rangle = 0$ if and only if $u = 0$.

In any vector space V endowed with an inner product we can define a norm by $\|x\| = \sqrt{\langle u, u\rangle}$, and then a distance $d(u, w) = \|u - w\| = \sqrt{\langle u - w, u - w\rangle}$. Two vectors are said to be **orthogonal** if $\langle u, w\rangle = 0$, and we can write $u \perp w$. If a finite set $S = \{u_1, \ldots, u_n\}$ of vectors in V is such that $u_i \perp u_j$ for every $i \neq j \in \{1, \ldots, n\}$ we say that S is an **orthogonal set**. If, in addition, $u_i \in S$ has unit length $\|x\| = \langle u, u\rangle = 1$, each i, we call S an **orthonormal set**. The n-dimensional Euclidean space $\mathbb{R}^n$ is an inner product space with:

$$\langle \mathbf{x}, \mathbf{y}\rangle := \sum_{i=1}^{n} x_i y_i.$$

The induced norm is $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle}$. The usual basis of $\mathbb{R}^n$ is formed by vectors $\mathbf{e}_i$ with all entries zero but the ith which is 1. The set $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is indeed an orthonormal set. The Cauchy-Schwartz inequality holds:

$$|\langle \mathbf{x}, \mathbf{y}\rangle|^{1/2} \leqslant \|\mathbf{x}\| + \|\mathbf{y}\|.$$

The inner product gives additional geometric properties. In particular, since by the cosine rule

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$$

we can, after simplifying, write

$$\langle \mathbf{x}, \mathbf{y}\rangle = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta.$$

The following is the definition of the angle between two Euclidean vectors:

$$\cos\theta = \frac{\langle \mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

where $\theta \in [0, \pi]$. A hyperplane can now be defined in the following way. Consider two orthogonal vectors $\langle \mathbf{x}, \mathbf{y}\rangle = 0$ where $\mathbf{x}$ is kept fixed. In $\mathbb{R}^2$ this represents the equation of a line through the origin and in $\mathbb{R}^3$ it provides a plane through the origin. In both cases, $\mathbf{x}$ is orthogonal to the line or the plane, and is called the **normal**. Similarly, the equation $\langle \mathbf{x}, \mathbf{y} - \mathbf{w}\rangle = 0$ represents a plane through the fixed vector $\mathbf{w}$ with normal $\mathbf{x}$. Taking $\langle \mathbf{x}, \mathbf{w}\rangle = c$ the equation becomes $\langle \mathbf{x}, \mathbf{y}\rangle = c$. Note that the product $\mathbf{xA}$ between a vector $\mathbf{x} \in \mathbb{R}^n$ and a $n \times n$ matrix $\mathbf{A}$ is a vector in $\mathbb{R}^n$. Then we can always write $\langle \mathbf{xA}, \mathbf{y}\rangle$.

Appendix H
Sequence and Series of Functions

Let $(f_n)_n$ be a sequence of real-valued functions all defined on the interval $I \subset \mathbb{R}$. We say that $(f_n)_n$ **converge pointwise** to a function f defined on the same interval, if $f_n(x) \to f(x)$ as $n \to \infty$ for all $x \in I$. Observe that $f_1(x), f_2(x), \ldots$ is a sequence of real numbers which converges to the number $f(x)$, for each $x \in I$. If the convergence fails for some $x \in I$, then we can define

$$D = \left\{ x \in \mathbb{R} \mid \lim_{n \to \infty} f_n(x) \text{ is finite} \right\}$$

as the *region of convergence*. A stronger definition of convergence is the following:

Definition H.1 A sequence of functions $(f_n)_n$ is said to **converges uniformly** on an interval I if for each $\epsilon > 0$ there is an integer number M, depending on ϵ but not on x, such that $|f_n(x) - f(x)| < \epsilon$ for all $n > M$ and all $x \in I$.

It can be shown that this definition is equivalent to:

$$f_n \to f \text{ uniformly on } I \text{ if } \|f_n - f\| \to 0 \text{ as } n \to \infty,$$

where $\|f_n - f\| = \sup\{|f_n(x) - f(x)| \mid x \in I\}$. If $(f_n)_n$ is formed by continuous functions on I and $f_n \to f$ uniformly, then f is continuous on I. If each f_n has a continuous derivative on $[a, b]$, $f_n \to f$ pointwise and $(f_n')_n$ converges uniformly on $[a, b]$, then $f_n' \to f'$ on $[a, b]$. If each f_n is continuous and $f_n \to f$ uniformly on $[a, b]$, then $\int_a^b f_n(x)\mathrm{d}x \to \int_a^b f(x)\mathrm{d}x$. A **power series** in general form is $\sum_{i=0}^{\infty} a_i (x - a)^i$, which converges either for $x = a$, or all x, or for $|x - a| < R$, where $R = \sup_{x \in E} |x - a| > 0$ is the **radius of convergence**, where $E = \{x \in \mathbb{R} \mid \text{the series converges}\}$. Here a_i, a are constants. In fact, a power series is a partial sum of a sequence of functions $a_i (x - a)^i$ for $i = 0, 1, 2, \ldots$. More generally, given a sequence of functions $(f_n(x))_n$ the series $\sum_{i=0}^{\infty} f_i(x)$ converges to the sum $s(x)$ in D if given $\epsilon > 0$ there exists $N(x, \epsilon) \in \mathbb{N}$ such that $|s(x) - s_n(x)| < \epsilon$ whenever $n > N$, where $s_n(x) = \sum_{i=0}^{n} f_i(x)$. This is simply

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https://doi.org/10.1007/978-3-031-54688-4

the convergence of the sequence of partial sums $s_n(x)$ to the limiting function $s(x)$. Again, if $N(x, \epsilon) = N(\epsilon)$ does depend only on ϵ and not on x, uniform convergence comes into play. If $f_i(x)$ are continuous in $I = [a, b]$ and $\sum_{i=0}^{n} f_i(x)$ is uniformly convergent to $s(x)$ in I, then $s(x)$ is continuous in I. If $f_i(x)$ are integrable in I and $\sum_{i=0}^{n} f_i(x)$ is uniformly convergent to $s(x)$ in I, then

$$\int_a^b \sum_{i=0}^{n} f_i(x)\mathrm{d}x = \sum_{i=0}^{n} \int_a^b f_i(x)\mathrm{d}x,$$

i.e., we can exchange the definite integral with the infinite sum. Eventually, if $f_i(x)$ have continuous derivatives in I and $\sum_{i=0}^{n} f_i(x)$ converges to $s(x)$ while $\sum_{i=0}^{n} f_i'(x)$ is uniformly convergent in I, then

$$\frac{\mathrm{d}}{\mathrm{d}x} \sum_{i=0}^{n} f_i(x) = \sum_{i=0}^{n} \frac{\mathrm{d}}{\mathrm{d}x} f_i(x),$$

i.e., we can exchange the derivative with the infinite sum.

Appendix I
Review of Multivariate Differentiation and Integration

The partial derivative

$$\frac{\partial F_{\mathbf{X}}}{\partial x}(x, y) = \lim_{h \to 0} \frac{F_{\mathbf{X}}(x + h, y) - F_{\mathbf{X}}(x, y)}{h}$$

gives the change of the bi-variate CDF in the direction parallel to the x-axis. Similarly,

$$\frac{\partial F_{\mathbf{X}}}{\partial y}(x, y) = \lim_{k \to 0} \frac{F_{\mathbf{X}}(x, y + k) - F_{\mathbf{X}}(x, y)}{k}$$

measures the change of the bi-variate CDF in the direction parallel to the y-axis. If the limit

$$\lim_{h \to 0} \frac{F_{\mathbf{X}}(x + u_1 h, y + u_2 h) - F_{\mathbf{X}}(x, y)}{h} \tag{I.1}$$

exists[1] is called the **directional derivative** of $F_{\mathbf{X}}$ at (x, y), written[2] $\mathrm{d}_{\mathbf{u}} F_{\mathbf{X}}(x, y)$. Formula (I.1) gives the change of $F_{\mathbf{X}}$ in both the variables. In a more general setting we can write

$$\mathrm{d}_{\mathbf{u}} F_{\mathbf{X}}(\mathbf{x}) := \lim_{h \to 0} \frac{F_{\mathbf{X}}(\mathbf{x} + h\mathbf{u}) - F_{\mathbf{X}}(\mathbf{x})}{h}, \tag{I.2}$$

[1] Observe that $(x + u_1 h, y + u_2 h)$ is automatically in the domain $\mathbb{R}^2$ of the CDF.

[2] The direction of change is given by moving a distance $h\|\mathbf{u}\|$ from the vector (x, y) to the vector $(x + u_1 h, y + u_2 h)$ along the direction of the vector $\mathbf{u} = (u_1, u_2)$ for which $\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2}$.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

where $\mathbf{x}+h\mathbf{u} = (x_1+hu_1, \ldots, x_n+hu_n)$. It is worth noting that for $\mathbf{u} \equiv \mathbf{e}_i$ Eq. (I.2) entails $\frac{\partial F_\mathbf{X}}{\partial x_i}(\mathbf{x})$ for $i = 1, \ldots, n$. Recall that $\mathbf{e}_i$ is the n-dimensional vector whose entries are zeros except the ith which is one. Whenever the further limit

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{\left|F_\mathbf{X}(\mathbf{a}+\mathbf{h}) - F_\mathbf{X}(\mathbf{a}) - \mathbf{b}\cdot\mathbf{h}\right|}{\|\mathbf{h}\|} = 0 \tag{I.3}$$

exists we say that the CDF if **differentiable** at $\mathbf{a} \in \mathbb{R}^n$. Recall that the vector sum $\mathbf{a} + \mathbf{h} \in \mathbb{R}^n$ is understood componentwise, i.e., $a_i + h_i$ for $i = 1, \ldots, n$ are the entries of the resulting vector. Also, $\mathbf{h} \to \mathbf{0}$ is meant to holds componentwise, i.e. $h_i \to 0$ for $i = 1, \ldots, n$ and $\mathbf{0} \in \mathbb{R}^n$ being the vector of all zeros. Finally, the dot product $\mathbf{b}\cdot\mathbf{h}$ equals the *inner product* $\langle \mathbf{b}, \mathbf{h}\rangle$ of n-dimensional vectors defined as the scalar $\sum_{i=1}^n b_i h_i$, with $\|\mathbf{h}\| = \sqrt{\mathbf{h}\cdot\mathbf{h}}$. Now intuitively, if (I.3) holds true the change $F_\mathbf{X}(\mathbf{a}+\mathbf{h}) - F_\mathbf{X}(\mathbf{a})$ differs little from the linear function $g(\mathbf{h}) = \mathbf{b}\cdot\mathbf{h}$, and we can approximate it via

$$F_\mathbf{X}(\mathbf{a}+\mathbf{h}) - F_\mathbf{X}(\mathbf{a}) \approx \nabla F_\mathbf{X}(\mathbf{x})\cdot\mathbf{h}, \tag{I.4}$$

where $\nabla F_\mathbf{X}(\mathbf{x})$ is called the CDFs **gradient** i.e. the n-dimensional vector whose entries are the partial derivatives $\frac{\partial F_\mathbf{X}}{\partial x_i}(\mathbf{x})$. If this is the case, the multivariate CDF posses all the partial derivatives and all the directional derivatives[3] for $i = 1, \ldots, n$.

Example I.1 In the bi-variate case with CDF differentiable at $\mathbf{a} = (a_1, a_2)$ we have

$$\frac{\left|F_\mathbf{X}(x,y) - F_\mathbf{X}(a_1,a_2) - \frac{\partial F_\mathbf{X}}{\partial x}(a_1,a_2)(x-a_1) - \frac{\partial F_\mathbf{X}}{\partial y}(a_1,a_2)(x-a_2)\right|}{\|(x,y)-(a_1,a_2)\|} \to 0$$

as $(x, y) \to (a_1, a_2)$. We can say that $z = F_\mathbf{X}(a_1, a_2) + \frac{\partial F_\mathbf{X}}{\partial x}(a_1, a_2)(x - a_1) + \frac{\partial F_\mathbf{X}}{\partial y}(a_1, a_2)(y - a_2)$ is a good approximation of the CDF and this is the equation of the **tangent plane** (in $\mathbb{R}^3$) of its graph at the point $\mathbf{a} = (a_1, a_2)$. The analogous one-dimensional case is that of a real-valued function $F(x)$ differentiable at $a \in \mathbb{R}$ being in an open interval (neighborhood) contained in its domain, for which $y = f(a)+f'(a)(x-a)$ is the linear function whose graph is the tangent (in the Cartesian plane $\mathbb{R}^2$) of the F's graph at the point $(a, F(a))$: in fact, the tangent line given by the above equation is s.t. $\lim_{x\to a} \frac{F(x)-F(a)-F'(a)(x-a)}{x-a} = 0$.

In Eq. (I.3) we can let $\mathbf{x} = \mathbf{a} + \mathbf{h}$ so that $\mathbf{h} = \mathbf{x} - \mathbf{a}$ and the limit is equivalently taken as $\mathbf{x} \to \mathbf{a}$, componentwise. Recall that $\|\mathbf{x} - \mathbf{a}\| = \sqrt{\sum_{i=1}^n (x_i - a_i)^2}$ equals the Euclidean distance $d(\mathbf{x}, \mathbf{a})$. Thus the limit would imply that $d(\mathbf{x}, \mathbf{a}) \to 0$ as $\mathbf{x} \to \mathbf{a}$ (the CDF is evaluated in a neighborhood of $\mathbf{a}$).

[3] These can be given as $\mathrm{d}_\mathbf{u} F_\mathbf{X}(\mathbf{x}) = \nabla F_\mathbf{X}(\mathbf{x})\cdot\mathbf{u}$.

Remark I.1 As in the one-dimensional case if $F_{\mathbf{X}}$ is differentiable at $\mathbf{a}$ then it is continuous at this point. Moreover, if all the partial derivatives $\frac{\partial F_{\mathbf{X}}}{\partial x_i}$ exist and are continuous (in a neighborhood of $\mathbf{a}$) then the CDF is differentiable at this point. As by product we have the following hierarchy:

$$\text{continuous partials } \tfrac{\partial F_{\mathbf{X}}}{\partial x_i} \Longrightarrow F_{\mathbf{X}} \text{ differentiable} \Longrightarrow \text{partials exist.}$$

The second implication is due to the truly definition of partial derivative, since for the limit in (I.3) to exist is necessary that the gradient exists.

Here some remarks on multivariate differentiation.

- If a function $F(x_1, \ldots, x_n)$ admits second order partials $\frac{\partial^2 F}{\partial x_i \partial x_j}(\mathbf{a})$ at a point $\mathbf{a}$ in its domain,[4] for $i, j = 1, \ldots, n$, and they are continuous at that point then the order of partial differentiation does not matter (Schwarz's theorem or Young's theorem).
- We call $\sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 F}{\partial x_i \partial x_j}(\mathbf{a}) \mathrm{d}x_i \mathrm{d}x_j$ the *second order total differential*.

The very basics of multivariate integration is the **Riemann double integral**. Let $f : [a, b] \times [c, d] \to \mathbb{R}$ where the domain $\mathrm{dom}(f)$ is a compact subset of $\mathbb{R}^2$. We partition $[a, b]$ into $n \in \mathbb{N}$ subintervals $[x_{i-1}, x_i]$ for $i = 1, \ldots, n$ and $[c, d]$ into $m \in \mathbb{N}$ subintervals $[y_{j-1}, y_j]$ for $i = 1, \ldots, m$. Thus we compute the *Riemann sum*

$$S_{n,m} := \sum_{i=1}^{n} \sum_{j=1}^{m} f(c_i, c_j) \Delta x_i \Delta y_j, \quad (c_i, c_j) \in [x_{i-1}, x_i] \times [y_{j-1}, y_j], \tag{I.5}$$

where $\Delta x_i = x_i - x_{i-1}$ and $\Delta y_j = y_j - y_{j-1}$ for each i, j. Assuming further f is bounded then

$$\lim S_{n,m} = \int_a^b \left[\int_c^d f(x, y) \mathrm{d}y \right] \mathrm{d}x = \int_c^d \left[\int_a^b f(x, y) \mathrm{d}x \right] \mathrm{d}y, \tag{I.6}$$

provided the limit exists. The latter is taken for finer partitions of the two intervals above, as $n \to \infty$, $m \to \infty$ and s.t.

$$\sup \left\{ (x_i - x_{i-1})(y_j - y_{j-1}) \right\} \to 0.$$

[4] If the function has domain strictly included in $\mathbb{R}^3$, then all statements about continuity and differentiability are meant to hold at interior points and for small changes (so that all vectors involved do not move away from the domain).

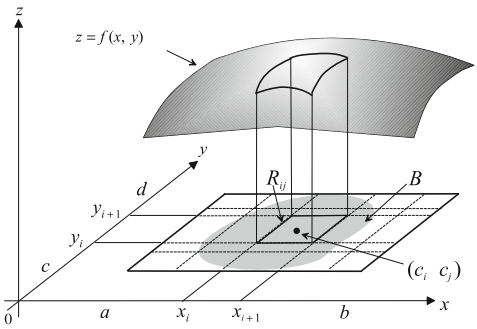


Fig. I.1 Surface as the graph of a function of two variable: the volume of a solid body over a sub-rectangle

The supremum means that we take the largest $(x_i - x_{i-1})(y_j - y_{j-1}) = \text{area}(R_{ij})$, i.e. the area of rectangles $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$. This is also called the *diameter* of[5] $[a, b] \times [c, d]$. The brackets in iterated integrals can be delete. Formula (I.6) especially holds if f is continuous on $\text{dom}(f) = [a, b] \times [c, d]$. If the function f is defined over a more general bounded region in the plane, say $\text{dom}(f) = B \subset \mathbb{R}^2$, then the above definition can be restated for the function

$$\tilde{f}(x, y) := \begin{cases} f(x, y), & \text{if } (x, y) \in B \\ 0, & \text{if } (x, y) \in [a, b] \times [c, d] \setminus B, \end{cases}$$

i.e. reconsidering the Riemann sum (I.5) for every (c_i, c_j) in $R_{ij} \cap B$. If the limit (I.6) exists we can write at once the double Riemann integral as

$$\iint_B \tilde{f}(x, y) \mathrm{d}x \mathrm{d}y,$$

and again it equals both the two iterated integrals in (I.6). In Fig. I.1, $R_{ij} \cap B = R_{ij}$ because R_{ij} is inside B. Assuming f is nonnegative, the double Riemann integral can be interpreted as the volume of a *solid body* given by every point $\mathbf{x} = (x, y, z)$ lying between the f's graph[6] and the rectangle $[a, b] \times [c, d]$ or the more general region B in the plane.

Remark I.2 (Cavalieri's Principle) Assume f is defined on $[a, b] \times [c, d]$. We can cut a slice of the solid by a plane for example parallel to the yz-plane (then perpendicular to the xz-plane), thus getting a cross-sectional area $A(x)$ at distance $x \in [a, b]$ from the reference plane. Now, incrementing x by Δx the volume of the

[5] One can define the diameter of any closed bounded set in $\mathbb{R}^2$ as the maximum distance between any two of its points.

[6] Recall it can be implicitly defined by the equation $g(x, y, z) = 0$, where $g(x, y, z) = z - f(x, y)$.

corresponding slab with this cross-sectional area and thickness Δx is approximately $V(x+\Delta x)-V(x) \approx A(x)\Delta x$. We obtain better approximation for smaller Δx and it is possible to get $V'(x) = A(x)$ in the limit, i.e. $V(b)-V(a) = \int_a^b A(x)\mathrm{d}x$ where we can let $V \equiv V(b)$ and $V(a) = 0$. In summary: Let $S \subset \mathbb{R}^3$ be a solid and let $(\Pi_x)_{a\leqslant x\leqslant b}$ a family of parallel planes s.t.

- S lies between Π_a and Π_b,
- $A(x)$ denotes the area of the S's slice cut by Π_x,

then $\mathrm{vol}(S) = V = \int_a^b A(x)\mathrm{d}x$. This approach can be restated by parallel planes $(\Pi_y)_{c\leqslant y\leqslant d}$, but the volume remained unchanged.

We have the following two-dimensional version of the mean value theorem, now for double integrals:

- $\iint_B f(x,y)\mathrm{d}x\mathrm{d}y = \mu \iint_B \mathrm{d}x\mathrm{d}y = \mu\mathrm{area}(B)$, for $\mu_* \leqslant \mu \leqslant \mu^*$ and bounded $f(x,y) \in [\mu_*, \mu^*]$ for every $(x,y) \in B$;
- if B is **connected**[7] then $\mu = f(c_1, c_2)$ for some $(c_1, c_2) \in B$.

The fact that $\iint_B \mathrm{d}x\mathrm{d}y = \mathrm{area}(B)$ can be justified by the following probabilistic example.

Example I.2 Let $f(x,y) = \frac{1}{\pi r^2}$ for $x^2+y^2 \leqslant r^2$ with $r > 0$, and zero otherwise. We show that this function is indeed a density of some bi-variate absolutely continuous random vector $\mathbf{X} = (X,Y)$, i.e. $f \equiv f_{\mathbf{X}}$. In fact, $B = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leqslant r^2\}$ is the subset of the plane given by a circle with radius r, so the double integral $\iint_B \mathrm{d}x\mathrm{d}y = \pi r^2$ is equivalent to $\mathrm{area}(B)$ and putting $f(x,y)$ inside the above integral gives 1. Then f is a joint density and can be the model for the experiment of choosing at random a point into the circle with radius r.

More generally, we can find a solid in $\mathbb{R}^3$ given by points $\mathbf{x} = (x,y,z)$ between the plane Π_z with equation $z = f(x,y)$ and the connected region $B \subset \mathbb{R}^2$, whose boundary ∂B is the closed line ℓ obtained by the equation $g(x,y) = 0$. Thus the volume of the cylinder is

$$V = \iint_B k\mathrm{d}x\mathrm{d}y = k\iint_B \mathrm{d}x\mathrm{d}y,$$

[7] It is the closure of a connected open subset in $\mathbb{R}^2$, i.e. it is not the union of two disjoint closed subsets in $\mathbb{R}^2$.

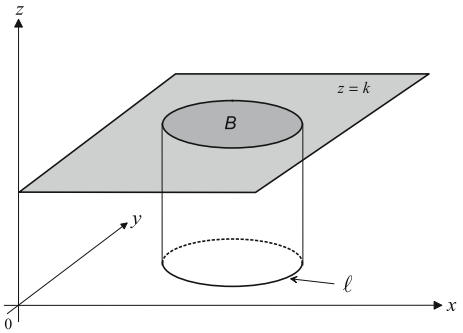


Fig. I.2 Cylinder with base given by the circle B. The B's boundary is the line ℓ given by the equation $g(x, y) = \sqrt{x + y} - r = 0$. A copy of B also lies in the plane Π_z identified by the equation $z = k$

where $k\mathrm{d}x\mathrm{d}y$ is the infinitesimal volume of a parallelepiped with height k w.r.t. to the xy-plane and base's area equal to $\mathrm{d}x\mathrm{d}y$, On the other hand, the volume also equals area$(B)k$ thus implying area$(B) = \iint_B \mathrm{d}x\mathrm{d}y$, see Fig. I.2.

Appendix J
A Glimpse to Vector Analysis in $\mathbb{R}^3$

This appendix is an application of results from Appendix G to the three-dimensional Euclidean space. Recall that for two points $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ we consider the concept of a vector as the directed line segment from $\mathbf{x}$ to $\mathbf{y}$. In particular, any point $\mathbf{x}$ can be identified with a vector starting from the origin $\mathbf{0} = (0, 0, 0)$ and ending at $\mathbf{x}$ itself. The *magnitude* of a vector starting from the origin is its Euclidean norm $\|\mathbf{x}\|$, and since $h\,\mathbf{x} = (h\,x_1, h\,x_2, h\,x_3)$ we have that, for $h > 0$, $\|h\,\mathbf{x}\| = h\|\mathbf{x}\|$, the second vector has the same direction of the first but h times its magnitude. Introducing the *unit vectors* $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$ and $\mathbf{e}_3 = (0, 0, 1)$ we can express any other vector $\mathbf{x} = (x_1, x_2, x_3)$ as $x_1\,\mathbf{e}_1 + x_2\,\mathbf{e}_2 + x_3\,\mathbf{e}_3$, where each $x_i\,\mathbf{e}_i$ is the *component* of X in the direction of the x_i-axes. These unit vectors are sometimes called *standard basis vectors* and denoted by $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$. Since the Euclidean norm satisfies (see also Appendix G)

1. $\|\mathbf{x}\| \geqslant 0$ for each $\mathbf{x} \in \mathbb{R}^n$ and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$,
2. $\|h\mathbf{x}\| = |h|\,\|\mathbf{x}\|$ for all $\mathbf{x} \in \mathbb{R}^n$ and $h \in \mathbb{R}$,
3. $\|\mathbf{x} + \mathbf{y}\| \leqslant \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ (triangle inequality),

by item 2 we have that for $h < 0$ the vector $h\,\mathbf{x}$ moves along the same straight line but in opposite direction than $\mathbf{x}$. For $h = -1$ the magnitude is the same, the direction is opposite. If $\mathbf{y} = h\,\mathbf{x}$ then $\frac{1}{h}\,\mathbf{y}$ is a unit vector with the same direction as $\mathbf{x}$ since $\|\frac{1}{h}\,\mathbf{y}\| = \frac{1}{|h|}\,\|h\,\mathbf{x}\| = \frac{1}{|h|}\,|h| = 1$. Writing the inner product $\langle \cdot, \cdot \rangle$ as the dot product, for the standard unit vectors we have:

$$\mathbf{e}_1 \cdot \mathbf{e}_1 = \mathbf{e}_2 \cdot \mathbf{e}_2 = \mathbf{e}_3 \cdot \mathbf{e}_3 = 1, \quad \mathbf{e}_1 \cdot \mathbf{e}_2 = \mathbf{e}_2 \cdot \mathbf{e}_3 = \mathbf{e}_3 \cdot \mathbf{e}_1 = 0.$$

We deduce:

- $\|\mathbf{e}_i\| = \mathbf{e}_i \cdot \mathbf{e}_i$, for $i = 1, 2, 3$;
- these vectors are *orthogonal*.

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These features can be extended to all the vectors, thus $\mathbf{x}, \mathbf{y}$ are orthogonal iff their inner product is zero (they are perpendicular). To get more geometric intuition, we first recall that the vector sum $\mathbf{x}+\mathbf{y}$ yields a new vector $\mathbf{z}$ representing the diagonal of a parallelogram with two sides given by $\mathbf{x}$ and $\mathbf{y}$ (starting both from the origin), a third side being the vector parallel to $\mathbf{y}$ and starting from the tip of $\mathbf{x}$, a fourth side being the vector parallel to $\mathbf{x}$ and starting from the tip of $\mathbf{y}$. More interestingly, we can identify any vector $\mathbf{w}$ not necessarily starting from $\mathbf{0}$ by simply considering the *vector difference* $\mathbf{y}+(-1)\mathbf{x} := \mathbf{y}-\mathbf{x}$. Thus, if $\mathbf{w}$ is just the difference then it moves from the tip of $\mathbf{x}$ to the tip of $\mathbf{y}$ and has coordinates $(y_1-x_1,\, y_2-x_2,\, y_3-x_3)$. Now we are ready to state the geometric version of the orthogonality condition between two vectors. Recall that

$$\mathbf{x}\cdot\mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\|\cos\theta, \quad \text{for } 0 \leqslant \theta \leqslant \pi.$$

Here θ is the angle between the two vectors. To prove the above we first note that a distributive law holds between the inner product and the vector sum:

$$\mathbf{x}\cdot(\mathbf{y}+\mathbf{z}) = \mathbf{x}\cdot\mathbf{y}+\mathbf{x}\cdot\mathbf{z}.$$

Then we apply the cosine law and get:

$$\|\mathbf{y}-\mathbf{x}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\|\,\|\mathbf{y}\|\cos\theta. \tag{J.1}$$

It suffices to rewrite $\|\mathbf{y}-\mathbf{x}\|^2 = (\mathbf{y}-\mathbf{x})\cdot(\mathbf{y}-\mathbf{x})$, $\mathbf{x}^2 = \mathbf{x}\cdot\mathbf{x}$ and $\mathbf{y}^2 = \mathbf{y}\cdot\mathbf{y}$ thus expand the inner product

$$\begin{aligned}(\mathbf{y}-\mathbf{x})\cdot(\mathbf{y}-\mathbf{x}) &= \mathbf{y}\cdot(\mathbf{y}-\mathbf{x}) - \mathbf{x}\cdot(\mathbf{y}-\mathbf{x})\\ &= \mathbf{y}\cdot\mathbf{y}-\mathbf{y}\cdot\mathbf{x}-\mathbf{x}\cdot\mathbf{y}+\mathbf{x}\cdot\mathbf{x}\\ &= \mathbf{x}\cdot\mathbf{x}+\mathbf{y}\cdot\mathbf{y}-2\mathbf{x}\cdot\mathbf{y},\end{aligned}$$

and eventually rearranging to get the result. Henceforth, two vector are orthogonal iff $\theta = \frac{\pi}{2}$, so that $\cos\theta = 0$ and the inner product is zero too. Equation J.1 is the Carnot's theorem: For any triangle with a side of length c and opposite angle θ between the two remaining adjacent sides of length a, b we have: $\boxed{c^2 = a^2+b^2-2a\,b\cos\theta}$. It is interesting to note how the gradient of a real-valued function $F(x_1, x_2, x_3)$ of three variables may be formally written using the vector calculus we have briefly developed. Let

$$\nabla := \mathbf{e}_1\tfrac{\partial}{\partial x_1} + \mathbf{e}_2\tfrac{\partial}{\partial x_2} + \mathbf{e}_3\tfrac{\partial}{\partial x_3}$$

we have (considering F as a scalar)

$$\nabla F := \nabla\cdot F = \mathbf{e}_1\tfrac{\partial F}{\partial x_1} + \mathbf{e}_2\tfrac{\partial F}{\partial x_2} + \mathbf{e}_3\tfrac{\partial F}{\partial x_3}.$$

Thus we can safely evaluate the above *operator* at a point $\mathbf{x} \in \mathbb{R}^3$ to get

$$\nabla F(\mathbf{x}) = \left(\frac{\partial F}{\partial x_1}(\mathbf{x}), \frac{\partial F}{\partial x_2}(\mathbf{x}), \frac{\partial F}{\partial x_3}(\mathbf{x}) \right).$$

Now it is easy to prove that $\nabla(F+G) = \nabla F + \nabla G$. Moreover, if $F(\mathbf{x}) = k$ defines a surface $\mathcal{S} \subset \mathbb{R}^3$, for $k \in \mathbb{R}$, then $\nabla F(\mathbf{x})$ is the vector orthogonal to this surface at a point lying in $\mathcal{S}$ and identified with the tip of a vector[1] $\mathbf{x}$. In fact, defining the **total differential** as

$$\mathrm{d}F(\mathbf{x}) = \frac{\partial F}{\partial x_1}(\mathbf{x})\,\mathrm{d}x_1 + \frac{\partial F}{\partial x_2}(\mathbf{x})\,\mathrm{d}x_2 + \frac{\partial F}{\partial x_3}(\mathbf{x})\,\mathrm{d}x_3,$$

i.e. the measure of how F changes by small variations in all its variables, then from the $\mathcal{S}$'s equation we deduce that $\mathrm{d}F(\mathbf{x}) = \mathrm{d}k = 0$. But observing how the total differential can be rewritten,

$$\underbrace{\left(\mathbf{e}_1 \frac{\partial F}{\partial x_1} + \mathbf{e}_2 \frac{\partial F}{\partial x_2} + \mathbf{e}_3 \frac{\partial F}{\partial x_3} \right)}_{=\nabla F} \cdot \underbrace{(\mathbf{e}_1 \mathrm{d}x_1 + \mathbf{e}_2 \mathrm{d}x_2 + \mathbf{e}_3 \mathrm{d}x_3)}_{=\mathrm{d}\mathbf{x}},$$

where $\mathrm{d}\mathbf{x}$ is a vector lying in the plane tangent to the surface, we finally get that the latter is the inner product $\nabla F \cdot \mathrm{d}\mathbf{x} = 0$ which entails the orthogonality condition claimed.

Example J.1 Consider a bi-variate CDF $F_{\mathbf{X}}$ and a point lying on the surface $\mathcal{S}$ defined implicitly by the equation giving the graph of the CDF, $g(x, y, z) = z - F_{\mathbf{X}}(x, y) = 0$. Taking $\mathbf{a} = (a_1, a_2, z)$ as its position vector and another point in $\mathcal{S}$ with position vector $\mathbf{x} = (x, y, z)$, we have that $\mathbf{x} - \mathbf{a}$ is a new vector which lies on the plane tangent to the surface at $\mathbf{a}$. Since we know that ∇g is orthogonal to $\mathcal{S}$ at the point with position vector $\mathbf{x}$, we deduce

$$(\mathbf{x} - \mathbf{a}) \cdot \nabla g(\mathbf{a}) = 0.$$

Observe that

$$\nabla g(\mathbf{a}) = \left(\frac{\partial g}{\partial x}(\mathbf{a}), \frac{\partial g}{\partial y}(\mathbf{a}), \frac{\partial g}{\partial z}(\mathbf{a}) \right) = \left(-\frac{\partial F_{\mathbf{X}}}{\partial x}(\mathbf{a}), -\frac{\partial F_{\mathbf{X}}}{\partial y}(\mathbf{a}), 0 \right).$$

Expanding the inner product we finally have:

$$\begin{aligned} (x - a_1, y - a_2, z - z) \cdot \left(-\frac{\partial F_{\mathbf{X}}}{\partial x}(\mathbf{a}), -\frac{\partial F_{\mathbf{X}}}{\partial y}(\mathbf{a}), 0 \right) \\ = -\frac{\partial F_{\mathbf{X}}}{\partial x}(\mathbf{a})(x - a_1) - \frac{\partial F_{\mathbf{X}}}{\partial y}(\mathbf{a})(y - a_2) + 0. \end{aligned}$$

[1] The vector $\mathbf{x}$ whose tip identifies a point lying in a surface is called *position vector*.

Example J.2 The last expression $-\frac{\partial F_{\mathbf{X}}}{\partial x}(\mathbf{a})(x-a_1)-\frac{\partial F_{\mathbf{X}}}{\partial y}(\mathbf{a})(y-a_2)$ is that of the inner product $(\mathbf{x}-\mathbf{a})\cdot\nabla g(\mathbf{a})$ which must be equal to zero. Moreover we started from $F_{\mathbf{X}}(x,y)-z=0$ and this is also valid at (a_1,a_2). Thus combining all these results we finally have the equation of the plane tangent at the point $\mathbf{a}$ to the surface representing the CDF's graph:

$$z=F_{\mathbf{X}}(a_1,a_2)\frac{\partial F_{\mathbf{X}}}{\partial x}(a_1,a_2)(x-a_1)+\frac{\partial F_{\mathbf{X}}}{\partial y}(a_1,a_2)(y-a_2).$$

Appendix K
Stieltjes Integral

$\int_0^\infty x \, \mathrm{d}F_X(x)$ is a particular instance of a Stieltjes integral. In fact, for a continuous function $g(x)$ over the interval $[a, b]$ it is defined through the following steps:

1. $[a, b]$ is partitioned by points $a = x_0 < x_1 < \cdots < x_n = b$;
2. take the 'representative value' $g(c_i)$, for $c_i \in (x_{i-1}, x_i]$;
3. use an increasing (right-continuous)[1] function $F(x)$ over the same interval to compute the weighted sum $S_n := \sum_{i=1}^n g(c_i)\,[F(x_i) - F(x_{i-1})]$;
4. take $\lim_{n\to\infty} S_n = \int_a^b g(x)\mathrm{d}F(x)$, sending the largest interval length (viz. mesh) to zero.

Since we may choose $g(x)$ as bounded over $[a, b]$, it is crucial for the limit to exist that $g(x)$ and $F(x)$ do not share discontinuities at the same points $x \in [a, b]$, indeed $F(x)$ must be continuous at the discontinuities of $g(x)$. Sufficient conditions for the existence of Stieltjes integrals: g continuous and F increasing, or F expressed as the difference of two increasing functions. In our applications to probability we consider:

- $g(x) = x$ and $F(x) \equiv F_X(x)$;
- $c_i = x_{i-1} = \frac{i-1}{2^n}$ in formula (4.37);
- $\mathsf{E}(X_n)$ is then the expectation of a discrete rv with these values and masses $\mathsf{P}(X_n = \frac{i-1}{2^n})$.

Remark K.1 For continuous $g(x)$ we have the Riemann-Stieltjes integral $\int_a^b g(x)\mathrm{d}F(x)$ exists. On the other hand, for $F(x) = x$ we have the classical Riemann integral. Be warned that in real analysis, for a *measurable* $g(x)$ and a function $F(x)$ corresponding to some *Lebesgue-Stieltjes measure* (see Appendix in Chap. 2) we have $\int_a^b g(x)\mathrm{d}F(x)$ as a Lebesgue-Stieltjes integral, which exists provided that $g(x)$ is continuous except for a countable set of points in its domain.

[1] Right-continuity is not necessary, but in our application to probability is obvious.

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https://doi.org/10.1007/978-3-031-54688-4

It is worth noting that any CDF F_X may be expressed as the difference of two increasing functions whenever it is of a **bounded variation** on [0, 1], i.e.,

$$\sup_{\mathcal{P}} \sum_{i=1}^{n} |F_X(x_i) - F_X(x_{i-1})| < \infty, \qquad \mathcal{P} \text{ is a partition of } [0, 1], \tag{K.1}$$

and we write $F_X \in \mathrm{BV}[0, 1]$, where $\mathrm{BV}[0, 1]$ denotes the set of all bounded variation functions with domain $[0, 1]$. More generally, the Riemann-Stieltjes integral $\int_0^1 g(x)\mathrm{d}F_X(x)$ exists if the following conditions hold:

- $g(x)$ and $F_X(x)$ do not have discontinuities at the same points $x \in [0, 1]$;
- $g(x)$ is of a bounded p-variation and $F_X(x)$ is of a bounded q-variation for some $p, q > 0$ such that $\frac{1}{p} + \frac{1}{q} = 1$.

The definition of p-variation is similar to Eq. (K.1), excepts that we must add the exponent $p > 0$ to each absolute differences, $|F_X(x_i) - F_X(x_{i-1})|^p$. The q-variation is defined similarly.

Appendix L
Leibniz's Formula

Let $f(x,t)$ be defined over $[a,b]\times[c,d]$, where it is continuous. Assuming further that the partial derivative $\frac{\partial f}{\partial t}(x,t)$ is also continuous over the same domain. Then, we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_c^d f(x,t)\mathrm{d}t = \int_c^d \frac{\partial f}{\partial x}(x,t)\mathrm{d}t. \tag{L.1}$$

When it is assumed that $f(x,t)$, $\frac{\partial f}{\partial x}(x,t)$ are continuous for every $x\in[a,b]$ and every $t\in\mathbb{R}$, that $\int_{-\infty}^{\infty} f(x,t)\mathrm{d}t$ is finite for every $x\in[a,b]$ and there exists another function $h(t)$ s.t. $\int_{-\infty}^{\infty} h(t)\mathrm{d}t$ is finite and $|\frac{\partial f}{\partial x}(x,t)|\leqslant h(t)$ for every $t\in\mathbb{R}$ and $x\in[a,b]$ then

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{-\infty}^{\infty} f(x,t)\mathrm{d}t = \int_{-\infty}^{\infty} \frac{\partial f}{\partial x}(x,t)\mathrm{d}t. \tag{L.2}$$

Minor changes yield the following.

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_c^{\infty} f(x,t)\mathrm{d}t = \int_c^{\infty} \frac{\partial f}{\partial x}(x,t)\mathrm{d}t, \quad \left|\frac{\partial f}{\partial x}(x,y)\right| \leqslant h(t) \text{ for every } t\geqslant c \tag{L.3}$$

and the same for $\frac{\mathrm{d}}{\mathrm{d}x}\int_{-\infty}^{d} f(x,y)\mathrm{d}t$ with $|\frac{\partial f}{\partial t}(x,y)|\leqslant h(t)$ for every $t\leqslant d$. A more general version of formula (L.1) is

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(x,t)\mathrm{d}t = f(x,v(x))v'(x) - f(x,u(x))u'(x) + \int_{u(x)}^{v(x)} \frac{\partial f}{\partial x}(x,t)\mathrm{d}t, \tag{L.4}$$

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https://doi.org/10.1007/978-3-031-54688-4

provided that $u(x), v(x)$ are differentiable functions (with continuous derivatives) defined over $[a, b]$ and their images are contained in $[c, d]$.

Example L.1 We give more details in the derivation of the marginal density $f_{X_1}(x_1)$ from a multivariate CDF $F_{\mathbf{X}}(x_1, x_2, \ldots, x_n)$ having a joint density $f_{\mathbf{X}}(x_1, x_2, \ldots, x_n)$. First we take $x_2, \ldots, x_n \to \infty$ in the CDF and write it using the density (via Fubini's theorem):

$$F_{\mathbf{X}}(x_1, \infty, \ldots, \infty) = \frac{\mathrm{d}}{\mathrm{d}x_1} \int_{-\infty}^{x_1} \left[\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(u_1, u_2, \ldots, u_n) \mathrm{d}u_2 \cdots \mathrm{d}u_n \right] \mathrm{d}u_1.$$

Next, we define $G(u_1) := \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(u_1, u_2, \ldots, u_n) \mathrm{d}u_2 \cdots \mathrm{d}u_n$ as a function of only u_1. Thus we need to differentiate the integral $\int_{-\infty}^{x_1} G(u_1) \mathrm{d}u_1$ w.r.t. to the variable x_1 and to this end we invoke the Leibniz's formula (L.4):

$$\frac{\mathrm{d}}{\mathrm{d}x_1} \int_{-\infty}^{x_1} G(u_1) \mathrm{d}u_1 = G(x_1) \frac{\mathrm{d}x_1}{\mathrm{d}x_1} - \int_{-\infty}^{x_1} \frac{\partial G}{\partial x_1}(u_1) \mathrm{d}u_1.$$

As byproduct $G(x_1)$ equals $\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, u_2, \ldots, u_n) \mathrm{d}u_2 \cdots \mathrm{d}u_n$, while $\frac{\mathrm{d}x_1}{\mathrm{d}x_1} = 1$ and the partial derivative is zero. Finally, the $n-1$ iterated integrals yield the desired marginal density $f_{X_1}(x_1)$. We used formula (L.4) with $f(x, u_1) = G(u_1)$, $v(x_1) = x_1$ and $u(x_1) = \infty$.

Appendix M
Lebesgue Measure

A measure on the measurable space $(\mathbb{R}, \mathcal{B})$ which plays a central role in the theory of probability distributions is described in the following:

Theorem M.1 *There exists a unique measure m on $(\mathbb{R}, \mathcal{B})$, called the **Lebesgue measure**, which assigns to each interval $(a, b] \in \mathcal{B}$ its length $m((a, b]) = b - a$ and is translation invariant.*

The Lebesgue measure on $\mathbb{R}$ is diffusive and attaches zero to singletons, finite sets and countable sets.

Remark M.1 We give a formulation of *Lebesgue null sets*, similar to that of P-null events. For each $\epsilon > 0$ there exists a sequence of intervals[1] $(I_i)_i \subset \mathcal{B}$ such that $A \subset \cup_i I_i$, i.e. they cover A, and their total length is arbitrary small, $\sum_i m(I_i) < \epsilon$. By Theorem M.1 we take $I_i = (a_i, b_i]$ with length given by $m(I_i) = b_i - a_i$. Each singleton $\{x\} \in \mathcal{B}$ is such that $\{x\} \subset \left(x - \frac{\epsilon}{4^i}, x + \frac{\epsilon}{4^i}\right]$ and can be covered by infinitely many intervals for $i = 1, 2, \ldots$. We have

$$\sum_i m(I_i) = \tfrac{\epsilon}{2} \sum_{i=1}^{\infty} \tfrac{1}{2^i} = \tfrac{\epsilon}{2},$$

then $\sum_i m(I_i) < \epsilon$. Thus by monotonicity, $\{x\} \leqslant \sum_i m(I_i)$, and the arbitrarily of ϵ we must have $m(\{x\}) = 0$.

Remark M.2 There are other Lebesgue null sets than the countable ones. A classic example of uncountable null set is due to George Cantor. Given $[0, 1]$, remove the middle third $(\frac{1}{3}, \frac{2}{3})$ to obtain $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Next, remove the middle third of each of these two intervals to obtain C_2 and continue the process n times. Now, C_n

[1] They may overlap.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

is the union of 2^n disjoint closed intervals each of length $\frac{1}{3^n}$, then $m(C_n) = \left(\frac{2}{3}\right)^n$. The *Cantor set* is $C = \cap_{n=1}^{\infty} C_n$. Let $\epsilon > 0$, pick a large $n \in \mathbb{N}$ such that $\left(\frac{2}{3}\right)^n < \epsilon$ and observe that $C \subset C_n$. Therefore, we find a sequence of intervals C_n covering C whose total length is less than an arbitrary ϵ and conclude $m(C) = 0$. But it can be shown that C has the same cardinality of $[0, 1]$, then it is an uncountable subset of $\mathbb{R}$ with Lebesgue measure zero.

Now, we come to translation invariance. Since the length of an interval does not change under a shifting, $m((a + x, b + x]) = b - a$ for all $x \in \mathbb{R}$, then it can be shown that this property still holds for general Borel sets of the form $A + x := \{a + x \mid a \in A\}$: they have Lebesgue measure $m(A + x) = m(A)$. A similar behavior hold under *dilation*: for every $x \in \mathbb{R}$ and a given $A \in \mathcal{B}$, the Borel set $xA := \{xa \mid a \in A\}$ has measure $m(xA) = |x|m(A)$. E.g. $A = (a, b)$ and $x < 0$ imply $m(xA) = |x|(b-a)$, so that length is not altered by negative dilation.[2] Interesting enough, if we restrict ourselves to the unit interval $\Omega = (0, 1]$ and take the smallest σ-algebra $\mathcal{B}_{(0,1]}$ generated by the collection

$$\left\{ \cup_{i=1}^{n} I_i \mid I_i = (a_i, b_i] \text{ are disjoint, } 0 < a_i < b_i \leqslant 1 \right\},$$

then the Lebesgue measure on $\left(\Omega, \mathcal{B}_{(0,1]}\right)$ is a probability measure. The Lebesgue measure can be extended to Borel subsets of the n-dimensional Euclidean space. First, Theorem M.1 is valid in $\mathbb{R}^n$ so that the pair $(\mathbb{R}^n, \mathcal{B}^n)$ can be equipped with the n-dimensional Lebesgue measure m^n which assigns a volume to half-open rectangles simply by the product of lengths of each side in the Cartesian product:

$$m^n((\mathbf{a}, \mathbf{b}]) = m((a_1, b_1]) \times \cdots \times m((a_n, b_n]) = \prod_{i=1}^{n} (b_i - a_i), \qquad \mathbf{a} < \mathbf{b} \in \mathbb{R}^n.$$

The volume of $[\mathbf{a}, \mathbf{b}]$, $(\mathbf{a}, \mathbf{b})$ and $[\mathbf{a}, \mathbf{b})$ is again $\prod_{i=1}^{n} (b_i - a_i)$. Therefore, the volume remains unchanged if we consider the Cartesian product $I_1 \times \cdots \times I_n$ of intervals $I_1, \ldots, I_n$ in $\mathbb{R}$ which are open, closed or neither open nor closed.

Example M.1 Let $\mathbb{R} \times \mathbb{R}$ be the two-dimensional Euclidean space and take $I_i = [a_i, b_i]$, each $i = 1, 2$. Then consider the sets $N_i = \{(x_1, x_2) \mid a_i \leqslant x_i \leqslant b_i, x_j = a_j\}$ with $i \neq j$ taken from $\{1, 2\}$. We can cover them by countable collections of open rectangles whose union has arbitrary small total volume, thus N_1 and N_2 are m^2-null and $I_1 \times I_2 \setminus (N_1 \cup N_2)$ has 2-dimensional Lebesgue measure $m^2(I_1 \times I_2) = (b_1 - a_1)(b_2 - a_2)$, which is equal to that of the Cartesian product $(a_1, b_1] \times (a_2, b_2]$ as the 2-dimensional version of Theorem M.1 requires.

[2] Note that $m((-\infty, a]) = \infty$ and $m((-\infty, \infty)) = \infty = m(\mathbb{R})$.

With the n-dimensional Lebesgue measure at hand, let $\frac{1}{m^n(E)}$ be attached to a Borel set $E \in \mathcal{B}^n$ such that $0 < m^n(E) < \infty$. Formula $\mathsf{P}(A) := \frac{m^n(A \cap E)}{m^n(E)}$ defines a probability measure on the measurable space $(E, \mathcal{F}_E)$, where $\mathcal{F}_E$ denotes the trace-σ-algebra of subset taken from $E \subset \mathbb{R}^n$, that is $\{A \cap E \mid A \in \mathcal{B}^n\}$, with $\mathsf{P}(E) = 1$. The case $n = 1$ with $E = (0, 1]$ and $m^1 = m$ has been treated above. Pay attention to the behavior of $A \cap E \subset E$: we have $0 < m^n(E) < \infty$ so that every subset has positive Lebesgue measure too. A final word. We know that $X = \frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}}$ is the Radon-Nikodým derivative of $\tilde{\mathsf{P}}$ w.r.t. the old probability measure P, provided the former is absolutely continuous (in the sense of measure) w.r.t. the latter. The term 'derivative' is analogous to consider the density $f_X(x) = \frac{\mathrm{d}F_X(x)}{\mathrm{d}x}$ of an absolutely continuous CDF. To see how this is intimately true, let us recall the continuous uniform probability measure on the sample space $\Omega = [0, 1)$ s.t. $\mathsf{P}(A) = b - a$ for every bounded interval $A = (a, b), (a, b], [a, b), [a, b]$. Now write $f(x) = \frac{1}{b-a}\mathbf{I}_{[a,b]}(x)$ and re-define the set function

$$\mathsf{P}(B) := \tfrac{1}{b-a} m([a, b] \cap B), \quad \text{for every } B \subset \mathbb{R},$$

where the Bs are Borel subsets of reals, and $m([a, b]) = b - a$ is the Lebesgue measure acting on bounded intervals $[a, b]$ as the uniform probability measure, where now $A \subset \mathbb{R}$. It follows that $m(B) = 0$ implies $\mathsf{P}(B) = 0$, indeed $[a, b] \cap B \subset B$ and then by monotonicity $0 \leqslant m([a, b] \cap B) \leqslant m(B) = 0$. The function $f(x)$ is actually the density of a rv X's probability law, i.e., we can identify $\mathsf{P} \equiv \mathsf{P}_X$ and then $\mathsf{P}_X(B) = \mathsf{P}(X \in B)$ is s.t. $\mathsf{P}_X \ll m$, and we also identify $f \equiv f_X$. In fact, it can be shown that for any probability distribution over $B \subset \mathbb{R}$ the existence of $f_X(x) \geqslant 0$ s.t. $\int_{\mathbb{R}} f_X(x)\mathrm{d}x = 1$ entails

$$\mathsf{P}_X(B) = \int_B f_X(x)\mathrm{d}x,$$

and $\mathsf{P}_X \ll m$. In fact, f_X is the density of P_X and $X = \frac{\mathrm{d}\tilde{\mathsf{P}}}{\mathrm{d}\mathsf{P}}$ may safely be called the *density* of $\tilde{\mathsf{P}}$ w.r.t. P. Since the latter is also the Radon-Nikodým derivative (of $\tilde{\mathsf{P}}$ w.r.t. P) the ultimate connection is the following: for an absolutely continuous CDF F_X it holds $F_X(x) = \int_{(-\infty,x]} f_X(u)\mathrm{d}u$ and then we have

$$\mathsf{P}_X(B) = \int_B \mathrm{d}F_X(x) = \int_B f_X(u)\mathrm{d}x = \int_{\mathbb{R}} f_X(u)\mathbf{I}_B(x)\mathrm{d}x,$$

with the obvious analogy on one hand between E_{P} and $\int_{\mathbb{R}}$, on the other hand between X and f_X.

Appendix N
Lebesgue Integral

We recall the following result about Riemann integration.

Theorem N.1 *$f : [a, b] \to \mathbb{R}$ is Riemann-integrable iff for all $\epsilon > 0$ there exists a partition $\mathcal{P}$ s.t. $U(\mathcal{P}, f) - L(\mathcal{P}, f) < \epsilon$.*

Here,

$$L(\mathcal{P}, f) := \sum_{i=1}^{n} m_i (x_i - x_{i-1}), \qquad m_i = \inf \left\{ f(x) \mid x \in [x_{i-1}, x_i] \right\}$$

are lower Riemann sums and

$$U(\mathcal{P}, f) := \sum_{i=1}^{n} M_i (x_i - x_{i-1}), \qquad M_i = \sup \left\{ f(x) \mid x \in [x_{i-1}, x_i] \right\}$$

are upper Riemann sums, for every partition $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$. Instead, we can refer to the common Riemann sum

$$S(\mathcal{P}, f) := \sum_{i=1}^{n} f(c_i)(x_i - x_{i-1}), \qquad \mathcal{P} \text{ partition of } [a, b],$$

where $c_i \in [x_{i-1}, x_i]$. We define the *mesh*

$$\|\mathcal{P}\| := \max \left\{ (x_i - x_{i-1}) \mid \mathcal{P} \text{ partition } [a, b] \right\} = \max_{1 \leqslant i \leqslant n} (x_i - x_{i-1}).$$

If the Riemann integral exists, then

$$\lim_{\|\mathcal{P}\| \to 0} S(\mathcal{P}, f) = R < \infty, \tag{N.1}$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

for every $\epsilon > 0$ s.t. there exists $\delta > 0$ and a partition of $[a, b]$ with mesh $\|\mathcal{P}\| < \delta$ s.t. $|S(\mathcal{P}, f) - R| < \epsilon$, for all $c_i \in [x_{i-1}, x_i]$.

Example N.1 The function $f(x) = \mathbf{I}_{\mathbb{Q}}(x)$ is not Riemann-integrable. Over the restricted domain [0, 1], this function has lower Riemann sums are 0, while all upper Riemann sums are 1. Then, f is not integrable in the Riemann sense.

Other pathological behaviors refer to sequences of functions $(f_n(x))_n$ defined over the same closed interval converging to a limit function $f(x)$: it is not always true that $\lim_n \int_a^b f_n(x)\mathrm{d}x = \int_a^b \lim_n f_n(x)\mathrm{d}x = \int_a^b f(x)\mathrm{d}x$, and interchanging limits with integral fails. To overcome this an other issues, the following procedure of integration comes into play. Given a bounded $f : [a, b] \to \mathbb{R}$ with image $G = f([a, b]) = \{f(x) \mid x \in [a, b]\}$,we instead partition $[\inf G, \sup G]$ via $\mathcal{P} = \{y_0, y_1, \ldots, y_n\}$ yielding[1] subintervals $J_i = [y_{i-1}, y_i)$. We require that f is Borel-measurable, then $f^{-1}(J_i)$ is again a Borel subset of $\mathbb{R}$ to which we attach its Lebesgue measure. Next, fora all i we define the Lebesgue sum

$$\mathrm{Le}(\mathcal{P}, f) = \sum_{i=1}^{n} c_i m(J_i), \tag{N.2}$$

where we can choose either $c_i \in J_i$ or $c_i = \inf J_i$. Therefore, f is Lebesgue-integrable if the limit

$$\lim_{\|\mathcal{P}\| \to 0} \mathrm{Le}(\mathcal{P}, f) = L = \int_{[a,b]} f(x)\mathrm{d}m, \tag{N.3}$$

exists and is finite, i.e., for every $\epsilon > 0$ exists $\delta > 0$ and a partition of $[\inf G, \sup G]$ with finite mesh $\|\mathcal{P}\| < \delta$ yielding $|\mathrm{Le}(\mathcal{P}, f) - L| < \epsilon$. Observe that $\mathcal{P}$ partitions $[\inf G, \sup G]$ through intervals $J_1, \ldots, J_n$, and this induces a partition of the f's domain via the Borel sets

$$A_i = \{x \in D \mid y_{i-1} \leqslant f(x) < y_i\} = f^{-1}(J_i),$$

and the domain is the disjoint union $\cup_{i=1}^n A_i$. Observe that J_i are mutually disjoint.

Remark N.1 It is worth noting that step functions $g : [a, b] \to \mathbb{R}$ defined as $g(x) = c_i$ for $x \in (x_{i-1}, x_i)$, for a partition $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ of the finite interval $[a, b]$, are Riemann-integrable with Riemann integral $\int_a^b g(x)\mathrm{d}x = \sum_{i=1}^n c_i(x_i - x_{i-1})$. Indeed, for a Riemann-integrable function $f : [a, b] \to \mathbb{R}$ it can be shown that

$$\int_a^b f(x)\mathrm{d}x = \sup\left\{\int_a^b g(x)\mathrm{d}x \;\middle|\; g \leqslant f,\ g \text{ step func.}\right\}.$$

[1] Here $y_0 = \inf G$ and $y_n = \sup G$.

We can set $g(x) = \sum_{i=1}^{n} c_i \mathbf{I}_{I_i}$ for intervals I_i stemming from a partition $\mathcal{P}$ of $[a, b]$ s.t. $[a, b] = \cup_{i=1}^{n} I_i$, with at most one common point, i.e., $I_i \cap I_j = \varnothing$ or $= \{x\}$ with $x \in [a, b]$ and $i \neq j$.

As showed by Remark N.1, the Lebesgue integral of a nonnegative step function $g(x) = \sum_{i=1}^{n} c_i \mathbf{I}_{A_i}$, where $A_i \subset [a, b]$ are pairwise disjoint and s.t. $\cup_{i=1}^{n} A_i = [a, b]$, is $\int_{[a,b]} g(x)\mathrm{d}m = \sum_{i=1}^{n} c_i m(A_i)$. Moreover, any nonnegative function f is Lebesgue-integrable iff there exist nonnegative step functions g_1, g_2 s.t. $g_1(x) \leqslant f(x) \leqslant g_2(x)$ for $x \in [a, b]$ and

$$\int_{[a,b]} g_2(x)\mathrm{d}m - \int_{[a,b]} g_1(x)\mathrm{d}m < \epsilon,$$

for any $\epsilon > 0$. For general functions we can always take $f^-, f^+ \geqslant 0$ s.t. $f = f^+ - f^-$ and

$$\int_{[a,b]} f(x)\mathrm{d}m = \int_{[a,b]} f^+(x)\mathrm{d}m - \int_{[a,b]} f^-(x)\mathrm{d}m.$$

All the usual properties of the Riemann integral transfers to the Lebesgue integral.[2] In particular:

1. If $f(x) = c$ for every x, then it is Lebesgue-integrable and $\int_{\mathbb{R}} f(x)\mathrm{d}m = c$.
2. If both f, g are Lebesgue-integrable and $a, b \in \mathbb{R}$, then $\int_{\mathbb{R}} \{a f(x) + b g(x)\}\mathrm{d}m = a \int_{\mathbb{R}} f(x)\mathrm{d}m + b \int_{\mathbb{R}} g(x)\mathrm{d}m$.
3. If $f(x) \leqslant g(x)$ for every x, and both functions are Lebesgue-integrable, then $\int_{\mathbb{R}} f(x)\mathrm{d}m \leqslant \int_{\mathbb{R}} g(x)\mathrm{d}m$.
4. $\int_{\mathbb{R}} f(x)\mathrm{d}m = \sum_{i=1}^{\infty} \int_{B_i} f(x)\mathrm{d}m$ provided that $B_1, B_2, \ldots$ form a partition of $\mathbb{R}$, i.e. $B_i \cap B_j = \varnothing$ for $i \neq j$ and $\cup_{i=1}^{\infty} B_i = \mathbb{R}$.
5. For every sequence of nonnegative Borel-measurable functions $f_1, f_2, \ldots$ it holds $\int_{\mathbb{R}} \sum_{n=1}^{\infty} f_n \mathrm{d}m = \sum_{n=1}^{\infty} \int_{\mathbb{R}} f_n \mathrm{d}m$.
6. $\left|\int_{\mathbb{R}} f(x)\mathrm{d}m\right| \leqslant \int_{\mathbb{R}} |f(x)|\mathrm{d}m$

Properties 1 and 3 can be stated in the a.e. sense, with the Lebesgue measure being zero over the sets $\{f \neq c\}$ and $\{f > g\}$. Moreover, the Lebesgue integral can be defined over general Borel subset of $\mathbb{R}$ just by indicators, $\int_B f(x)\mathrm{d}m = \int_{\mathbb{R}} \mathbf{I}_B f(x)\mathrm{d}m$. The function in Example N.1 is Lebesgue-integrable over every finite interval $[a, b]$, since

$$\int_{[a,b]} g(x)\mathrm{d}m = \int_{[a,b]} \mathbf{I}_{\mathbb{Q}\cap[a,b]}\mathrm{d}m = 1m(\mathbb{Q} \cap [a, b]) + 0m((\mathbb{Q} \cap [a, b])^c) = 0,$$

[2] Exchanging limits and integrals is permitted under weaker conditions than for Riemann integration.

for $\mathbb{Q} \cap [a, b]$ is countable and has Lebesgue measure zero, and so is $(\mathbb{Q} \cap [a, b])^c$. Riemann and Lebesgue integrals are linked:

Theorem N.2 *Let $f : [a, b] \to \mathbb{R}$ with $-\infty < a < b < \infty$.*

- *If f is Riemann-integrable over $[a, b]$, then f is Lebesgue-integrable and $\int_a^b f(x)\mathrm{d}x = \int_{[a,b]} f(x)\mathrm{d}m$.*
- *If f is bounded in $[a, b]$ then f is Riemann-integrable iff it is continuous m-almost everywhere (a.e.).*

The last statement means $A = \{x \in [a, b] \mid f(x) \text{ is not continuous}\}$ has Lebesgue measure zero, $m(A) = 0$. This can happen on a countable set of real numbers. Eventually, if f is Lebesgue-integrable over $[a, b]$ then

$$F(x) = \int_{[a,x]} f(u)\mathrm{d}m, \qquad \text{for every } x \in [a, b]$$

implies $F'(x) = f(x)$ m-a.e. in $[a, b]$, and if f is also differentiable we also have F' as Lebesgue-integrable and $F(x) - F(a) = \int_{[a,x]} F'(u)\mathrm{d}m$.

Appendix O
Double Lebesgue Integral

Double integrals as $\iint_B \mathrm{d}x\mathrm{d}y$ are related to the *Lebesgue measure* m^2 on $\mathbb{R}^2$. This can be defined as a **product measure**, i.e.:

- for any subset $B = B_1 \times B_2$ given by a Cartesian product with $B_1, B_2 \subset \mathbb{R}$, it holds $m^2(B) = m(B_1)m(B_2)$ where m is the one-dimensional Lebesgue measure on $\mathbb{R}$;
- for pairwise disjoint subsets $B, C \subset \mathbb{R}^2$ it holds $m^2(B \cup C) = m^2(B) + m^2(C)$;
- for subsets of the plane s.t. $B \subset C$ it holds $m^2(B) \leqslant m^2(C)$.

Formally m^2 can be written $m \times m$. Recall that

$$m((a,b]) = m((a,b)) = m([a,b)) = m([a,b]) = b - a$$

for reals $a < b$ and $m(\mathbb{R}) = \infty$. Hence, m^2 is just the product of the sides' length $m([a,b]) = b - a$ and $m([c,d]) = d - c$ of the rectangle $[a,b] \times [c,d]$. The area of any other figure in the plane like circles, which cannot be written as Cartesian products, may nevertheless be assigned by $m \times m$. This is because (as any probability measure) the Lebesgue product measure $m \times m$ is continuous in that we can find an decreasing sequence of rectangles $R_1 \supset R_2, \supset \ldots$ s.t. the circle B equals their union $B = \cup_n R_n$ and

$$m \times m(B) = \lim_{n\to\infty} m \times m(R_n) \iff \text{area}(B) = \lim_{n\to\infty} \text{area}(\square_n).$$

Remark O.1 The domain of the one-dimensional Lebesgue measure m as a set function is the smallest σ-algebra containing intervals like $(a,b]$ and (a,b). For the two-dimensional Lebesgue measure $m \times m$ the domain is the smallest σ-algebra containing Cartesian products $(a,b] \times (c,d]$, for reals $a < b$ and $c < d$. In both cases the endpoints of such intervals can be $-\infty$ or ∞. Observe that m evaluates finite or countable set $\{x_1, x_2, \ldots\}$ with zero as well as $m \times m$ evaluates countable set $\{\mathbf{x}_1, \mathbf{x}_2, \ldots\} \subset \mathbb{R}^2$, segments, lines and planes with zero.

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

Integrating functions of two variables w.r.t. to $m \times m$ starts from the choice $f(x, y) = \mathbf{I}_B(x, y)$ for a subset $B \subset \mathbb{R}^2$ picked from the appropriate σ-algebra:

$$\begin{aligned} m \times m(B) &= \int_{\mathbb{R}\times\mathbb{R}} \mathbf{I}_B(x, y)\mathrm{d}m \times m \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbf{I}_B(x, y)\mathrm{d}x \right] \mathrm{d}y \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbf{I}_B(x, y)\mathrm{d}y \right] \mathrm{d}x, \end{aligned} \tag{O.1}$$

where we used the identification $\mathrm{d}x \equiv \mathrm{d}m$ and $\mathrm{d}y \equiv \mathrm{d}m$. This is the first step in defining integration w.r.t. the Lebesgue product measure as *Fubini's theorem Round 1*. If $B = B_1 \times B_2$ we note that $\mathbf{I}_B(x, y) = \mathbf{I}_{B_1}(x)\mathbf{I}_{B_2}(y)$ and substituting above we get that the double integral $\iint_{\mathbb{R}^2} \mathbf{I}_B(x, y)\mathrm{d}x\mathrm{d}y = \iint_B \mathrm{d}x\mathrm{d}y$ equals

$$\int_{-\infty}^{\infty} \mathbf{I}_{B_1}(x)\mathrm{d}x \int_{-\infty}^{\infty} \mathbf{I}_{B_2}(y)\mathrm{d}y = \int_{B_1} \mathrm{d}x \int_{B_2} \mathrm{d}y,$$

so the product of two one-dimensional integrals yields just the 'area'

$$m \times m(B) = m(B_1)m(B_2).$$

The above double integral can be extended[1] to bounded, continuous or more general functions $f(x, y)$, instead of indicators $\mathbf{I}_B(x, y)$, s.t. the integral $\iint_B f(x, y)\mathrm{d}x\mathrm{d}y$ exists.

Remark O.2 In Fubini's theorem Round 1 given by formula (O.1) it is crucial that the two functions

$$y \mapsto \int_{\mathbb{R}} \mathbf{I}_B(x, y)\mathrm{d}x$$

and

$$x \mapsto \int_{\mathbb{R}} \mathbf{I}_B(x, y)\mathrm{d}y$$

are integrable in $\mathbb{R}$ w.r.t m and their one-dimensional integrals coincide.

The above approach to integration may be pushed further, by considering any pair of nonempty sets S_1, S_2 endowed with their own σ-algebras Σ_1, Σ_2. Thus one

[1] Using a standard machine similar to that used in defining expectations.

can show there exists a product measure $m_1 \times m_2$ s.t. $m_i : \Sigma_i \to [0, \infty]$ is a measure for $i = 1, 2$ and:

- $m_1 \times m_2(B) = m_1(B_1)m_2(B_2)$ for $B = B_1 \times B_2 \in \Sigma_1 \times \Sigma_2$;
- formula (O.1) holds, with generic integrable function $f(x, y)$ replacing $\mathbf{I}_B(x, y)$ and the one-dimensional Lebesgue measure replaced by the two m_1, m_2;
- the functions $y \mapsto \int_{S_1} f(x, y)\mathrm{d}m_1$ and $x \mapsto \int_{S_1} f(x, y)\mathrm{d}m_2$ are integrable w.r.t. m_2 and m_2, and their integrals coincide.

Here $\Sigma_1 \times \Sigma_2$ is called the *product σ-algebra* and is generated by Cartesian products of subsets taken individually from S_1 and S_2. But be warned that it is not the Cartesian product of the two σ-algebras, despite we used the same symbol.

Example O.1 We can consider $S_1 \times S_2 = \mathbb{R}^2$ and the (probabilistic) product measure $\mathsf{P}_{\mathbf{X}} = \mathsf{P}_X \times \mathsf{P}_Y$ corresponding to the laws of two independent rvs X, Y s.t.

$$\mathsf{P}_{\mathbf{X}}(B) = \mathsf{P}_X(B_1)\mathsf{P}_Y(B_1) = \int_{B_1} \mathrm{d}F_X(x) \int_{B_2} \mathrm{d}F_Y(y),$$

where $B = B_1 \times B_2$ for subsets $B_1, B_2 \subset \mathbb{R}$ for which $\mathsf{P}_X(B_1) = \mathsf{P}(X \in B_1)$ and $\mathsf{P}_Y(B_2) = \mathsf{P}(Y \in B_2)$. Here $\mathbf{X} = (X, Y)$. In fact the independence assumption entails $\mathsf{P}_{\mathbf{X}} = \mathsf{P}_X\mathsf{P}_Y$ and by the Fubini's theorem we have, for a $B \subset \mathbb{R}^2$ other than a Cartesian product,

$$\mathsf{P}(\mathbf{X} \in B) = \mathsf{P}_{\mathbf{X}}(B) = \iint_B \mathrm{d}F_X(x)\mathrm{d}F_Y(y),$$

But when the independence assumption is dropped the joint CDF $F_{\mathbf{X}}$ no longer corresponds to a product probability law $\mathsf{P}_{\mathbf{X}} = \mathsf{P}_X\mathsf{P}_Y$. For the Fubini's theorem to hold we need the further assumption that $\mathbf{X}$ is absolutely continuous with joint density s.t.

$$\mathsf{P}_{\mathbf{X}}(B) = \iint_B f_{\mathbf{X}}(x, y)\mathrm{d}x\mathrm{d}y.$$

This corresponds to an iterated integral provided B equals the Cartesian product of two subsets in $\mathbb{R}$, like intervals. Then, for absolutely continuous random vectors we have $\mathsf{P}_{\mathbf{X}} \ll m \times m$ with Radon-Nikodým derivative just the density $f_{\mathbf{X}}$, i.e. $m \times m(B) = 0$ implies $\mathsf{P}_{\mathbf{X}}(B) = 0$.

Example O.2 Another typical usage of product measures and Fubini's theorem is when $m_1 = \mathsf{P}$, $m_2 = m$ and $f(x, y) = f(\omega, y)$. Therefore, $S_1 \times S_2 = \Omega \times \mathbb{R}$, but in some applications the 'second set' in the Cartesian product can be reduced to a subset such as $[0, \infty)$. The product measure is $m \times \mathsf{P}$ and the double integral with

the corresponding iterated integrals (viz. Fubini's theorem) are:

$$\iint_{\Omega\times\mathbb{R}} f(\omega, y)\mathrm{d}\mathsf{P}\times m = \int_{\Omega}\left[\int_{-\infty}^{\infty} f(\omega, y)\mathrm{d}y\right]\mathrm{d}\mathsf{P}$$
$$= \int_{-\infty}^{\infty}\left[\int_{\Omega} f(\omega, y)\mathrm{d}\mathsf{P}\right]\mathrm{d}y.$$

For example letting $f(\omega, y) = \mathbf{I}_B(\omega, y)$ with the set $B = \{(\omega, y)\,|\,0 \leqslant y \leqslant X(\omega)\}$ and replacing $\mathbb{R}$ with $[0, \infty)$ we get

$$\iint_{\Omega\times[0,\infty)} \mathbf{I}_B(\omega, y)\mathrm{d}\mathsf{P}\times m = \int_{\Omega}\left[\int_{0}^{\infty} \mathbf{I}_B(\omega, y)\mathrm{d}y\right]\mathrm{d}\mathsf{P}$$
$$= \int_{0}^{\infty}\left[\int_{\Omega} \mathbf{I}_B(\omega, y)\mathrm{d}\mathsf{P}\right]\mathrm{d}y,$$

where $X(\omega)$ is a nonnegative rv.

Example O.3 Now

$$\int_0^{\infty} \mathbf{I}_B(\omega, y)\mathrm{d}y = \int_0^{X(\omega)} \mathrm{d}y = X(\omega)$$

and

$$\int_{\Omega} \mathbf{I}_B(\omega, y)\mathrm{d}\mathsf{P} = \int_{\{X\geqslant y\}} \mathrm{d}\mathsf{P} = \mathsf{P}(X \geqslant y).$$

Thus we get

$$\int_{\Omega} X(\omega)\mathrm{d}\mathsf{P} = \mathsf{E}(X) = \int_0^{\infty} \mathsf{P}(X \geqslant y)\mathrm{d}y,$$

an alternative way of deriving the formula for expectations of nonnegative rvs.

Remark O.3 Alternatively we can consider the product measure $\mathsf{P}_X \times m$ so that

$$\iint_{\mathbb{R}^2} f(x, y)\mathrm{d}\mathsf{P}_X\times m = \int_{-\infty}^{\infty}\left[\int_{0}^{\infty} \mathbf{I}_B(x, y)\mathrm{d}y\right]\mathrm{d}\mathsf{P}_X$$
$$= \int_{0}^{\infty}\left[\int_{-\infty}^{\infty} \mathbf{I}_B(x, y)\mathrm{d}\mathsf{P}_X\right]\mathrm{d}y,$$

where $B = \{(x, y)\,|\,0 \leqslant y \leqslant x\}$. Thus, the first bracket is $\int_0^x \mathrm{d}y = x$ while the second is $\int_y^{\infty} \mathrm{d}\mathsf{P}_X = \mathsf{P}(X \geqslant y)$ for a corresponding rv X with law P_X.

Given two probability spaces $(\Omega_1, \mathcal{F}_1, \mathsf{P}_1)$ and $(\Omega_2, \mathcal{F}_2, \mathsf{P}_2)$, how can we merge the σ-algebras to build a new probability measure over subsets of $\Omega_1 \times \Omega_2$, using both P_1 and P_2 (this is an example of compound experiment)? Here a useful device to get the right answer.

Theorem O.1 *Let $(\Omega_i, \mathcal{F}_i, \mathsf{P}_i)$ be probability spaces, each $i = 1, \ldots, n$. Then, there exists a unique probability measure P on the product σ-algebra $\mathcal{F} = \times_{i=1}^n \mathcal{F}_i$ such that its value for rectangles $A_1 \times A_2 \times \cdots \times A_n \subset \Omega_1 \times \Omega_2 \times \cdots \times \Omega_n$ is the product of single probabilities, $\mathsf{P}(A_1 \times \cdots \times A_n) = \mathsf{P}_1(A_1)\,\mathsf{P}_2(A_2) \cdots \mathsf{P}_n(A_n)$.*

The P above is the product of the P_i. For events $A \in \mathcal{F}$ that are not expressed as Cartesian products, $\mathsf{P}(A)$ does not factor. Sometimes, to emphasize the underlying product sample space Ω we write $\mathsf{P} = \mathsf{P}_{1,2,\ldots,n}$ or $\mathsf{P}_1 \times \cdots \times \mathsf{P}_n$. What is the σ-algebra $\mathcal{F} = \times_{i=1}^n \mathcal{F}_i$ introduced in Theorem O.1? To answer the question we specialize to the case $i = 1, 2$. Then, $\mathcal{F}_1 \times \mathcal{F}_2$ is the smallest σ-algebra containing rectangles $A_1 \times A_2$, for $A_i \in \mathcal{F}_i$ each i. It is the product σ-algebra. Be warned that $\mathcal{F}_1 \times \mathcal{F}_2$ is not the Cartesian product of $\mathcal{F}_1$ and $\mathcal{F}_2$, despite we use the same symbol $\times$. It is worth noting that $\mathcal{F}_1 \times \mathcal{F}_2$ can be equivalently generated by the collection containing sets of the form $A_1 \times \Omega_2$ or $\Omega_1 \times A_2$, for $A_i \in \mathcal{F}_i$, i.e. Cartesian products in which one factor is the whole ith sample space.

Example O.4 Recall that to a given joint CDF $F_{\mathbf{X}}(x_1, x_2)$ on $\mathbb{R}^2$ does correspond 1-to-1 a unique probability measure $\mathsf{P}_{\mathbf{X}}$ on $(\mathbb{R}^2, \mathcal{B}^2)$ via

$$\mathsf{P}_{\mathbf{X}}((a_1, b_1] \times (a_2, b_2]) = \Delta_{a_1,b_1}\Delta_{a_2,b_2} F_{\mathbf{X}}(x_1, x_2) \geqslant 0.$$

Now, to get $\mathsf{P}_{\mathbf{X}}$ as a product measure we first note that the two-dimensional Borel σ-algebra $\mathcal{B}^2$ is equal to the product σ-algebra $\mathcal{B} \times \mathcal{B}$ of two one-dimensional Borel σ-algebras on $\mathbb{R}$ (this is a non-trivial result). Please: do not confuse $\mathcal{B} \times \mathcal{B}$ with the Cartesian product of the one-dimensional Borel σ-algebra, and recall that we only use the same symbol $\times$. Next, by Theorem O.1 above we must have $\mathsf{P}_{\mathbf{X}}(B_1 \times B_2) = \mathsf{P}_{X_1}(B_1) \times \mathsf{P}_{X_2}(B_2)$ for any Borel set $B_i \in \mathcal{B}$ and $i = 1, 2$ and a random vector $\mathbf{X} = (X_1, X_2)$. Specifying the CDF $F_{\mathbf{X}}(x_1, x_2)$ as equal to $F_{X_1}(x_1)F_{X_2}(x_2)$, i.e. using independence of $F_{X_i}(x_i)$ CDFs on $\mathbb{R}$, we yield the desired result: starting from left-open two-dimensional intervals and then extending to the whole $\mathcal{B} \times \mathcal{B}$. Note that

$$\Delta_{a_1,b_1}\Delta_{a_2,b_2} F_{\mathbf{X}}(x_1, x_2) = (F_{X_1}(b_1) - F_{X_1}(a_1)) \cdot (F_{X_2}(b_2) - F_{X_2}(a_2)) \geqslant 0.$$

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D. Rossello, *A Cookbook with Probability One*, La Matematica per il 3+2 161,
https://doi.org/10.1007/978-3-031-54688-4

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https://doi.org/10.1007/978-3-031-54688-4

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